# AIRBUS A220

Edited: 24rd of October 2026

# Created by

**Note:** this guide is based on real Airbus A220 manuals and operational experience and may not accurately represent functionality of the Synaptic Airbus A220.

Material in this document is NOT for operational use.

# Change Log

| Date | Page(s) | Changes | Description |
|---|---|---|---|
| 08-January-2026 | ALL | - | First issue |
| 20-January-2026 | 19<br>26<br>119<br>143<br>157, 158<br>186<br>222<br>241<br>248<br>251 - 253<br>258, 275<br>259<br>276, 277<br>284<br>285 - 286<br>293 - 298<br>284 | Major | a) ACMP 2A – AUTO corrected to ACMP 2B – AUTO operation (typo).<br>b) VMINTRIM indication corrected (- 3 kts and "SPEED" inhibited < 30ft AGL).<br>c) "Operational Engineering Bulletin (OEB)" chapter added, OEBs relocated.<br>d) Note regarding CG and TRIM disappearing after EXEC prompt revised.<br>e) Approach guidance and matrix split, final course extension further clarified.<br>f) Page added explaining temperature compensation for arrival and approach.<br>g) "EICAS Message List" separated from non-normal into a separate chapter.<br>h) "Nuisance Guidance" chapter relocated after "EICAS Message List" chapter.<br>i) Minimum Equipment List / Configuration Deviation List chapter added.<br>j) Operation with systems inoperative relocated to the "MEL / CDL" chapter.<br>k) "Performance" and "Operational" renamed Performance dispatch / in-flight.<br>l) New performance table added to determine MTOW during pre-flight.<br>m) Detailed take-off perf tables moved to "Performance (in-flight)" section.<br>n) "Quick Reference" chapter added for easy access during flight.<br>o) Abbreviated normal procedure created for quick reference.<br>p) Holdover timetables relocated to "Quick Reference" chapter.<br>q) Various other tables / procedures added for quick access. |
| 27-January-2026 | 151<br>154<br>159, 160<br>285 | Minor | a) Slide added explaining high altitude flight (MACH, performance etc.)<br>b) VPA correction for gross weight rectified (5t above/below reversed).<br>c) 2000 ft and 1300 ft AGL corrected to AAE.<br>d) BATT 1 / 2 – ON corrected to BAT 1 / 2 – AUTO. |
| 09-February-2026 | -<br>151<br>46, 53, 78, 94, 98<br>146, 147, 164, 165, 157-160 | Major | a) All call-outs reviewed and aligned with new Airbus procedures.<br>b) A220 speed envelope made to scale, various clarifications added.<br>c) Various system related illustrations added, replaced or improved.<br>d) Airbus FOTs for TRIM AIR, APU BLEED and LOC guidance added. |
| 15-April-2026 | 273, 289 | Minor | Manual load & trim sheet with explanation added. |
| 11-May-2026 | 19, 71, 73, 107<br>254, 274, 288 | Minor | Part 1: Hydraulic pump logic in AUTO mode, automatic refuel sequence and FMS step climbs revised.<br>Part 2: Performance and stabilizer trim tables revised, Quick Reference Sheet low resolution fixed. |
| 17-June-2026 | 137 - 167 | Major | Normal procedures overhauled. |
| 15-July-2026 | 166, 280 | Minor | Slide with practical information added, idle descent table added. |
| 24-October-2026 | 16, 93, 95, 103, 105, 106, 114<br>138, 139, 146, 157, 160, 163, 164<br>197, 200, 202, 205-208, 211-216 | Major | a) Various clarifications added for systems, HOT revised for winter 2026-2027.<br>b) Normal procedures revised to reflect latest Airbus manuals, new FOT added.<br>c) Non-normal procedures updated to include Airbus "ECL Actions" procedure. |

# Page Navigation

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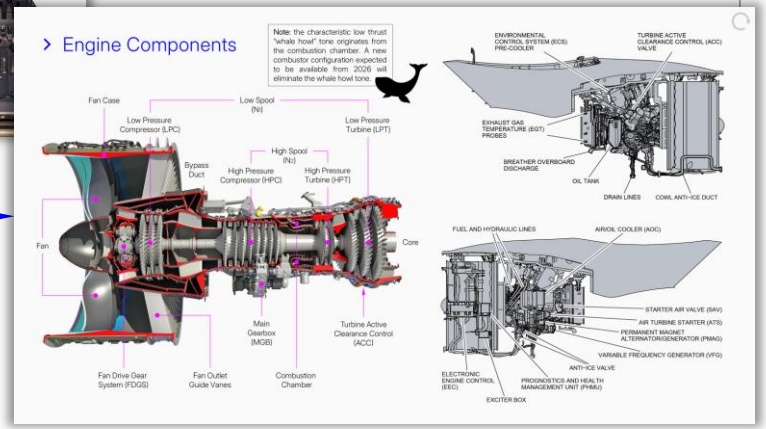


Disclaimer

⚠ Not for operational use

# 1. Aircraft Systems

# Part 1 – Aircraft Systems


- Introduction
- Air Conditioning, Bleed Air & Pressurization
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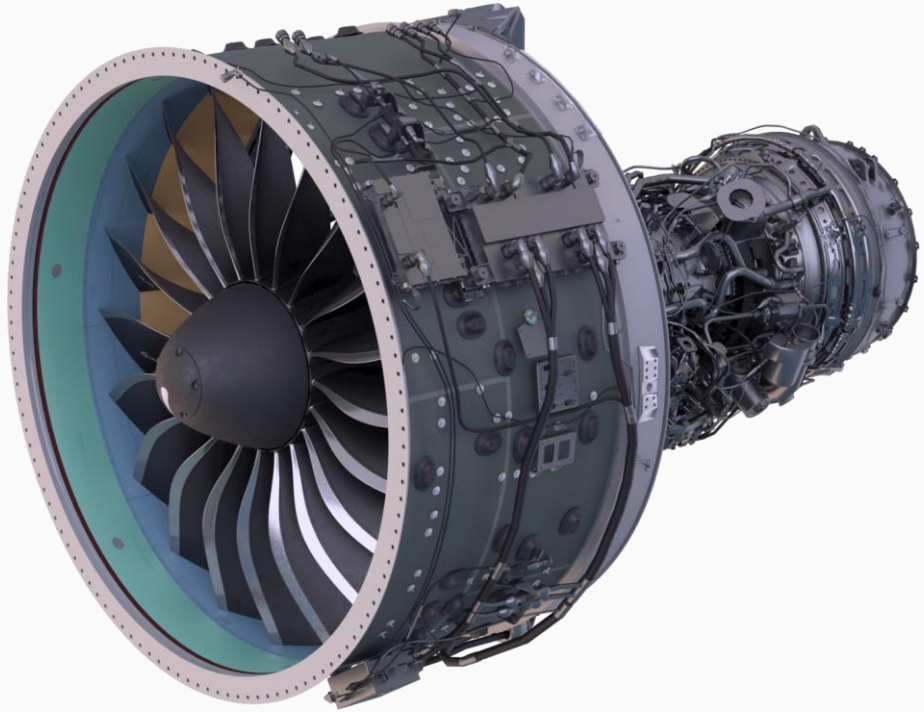
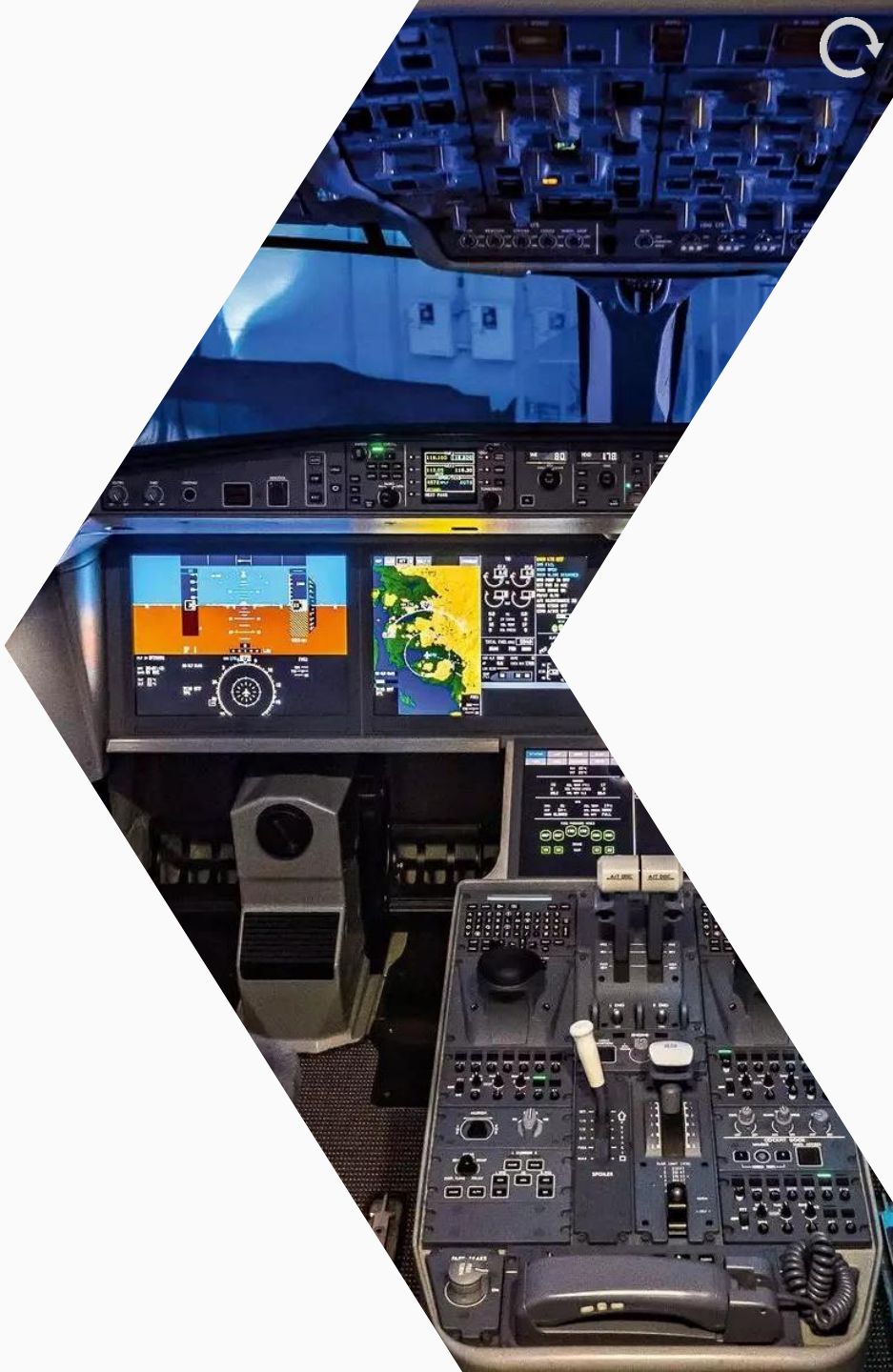


To Part 2 - Procedures

# Introduction



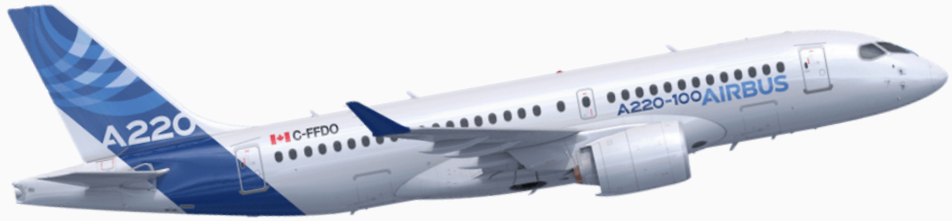
# Introduction

The Airbus A220 (former Bombardier C Series) is a narrow body, twin engine, medium range aircraft. Originally designed by Canadian manufacturer Bombardier Aerospace, the aircraft comes in 2 variants:

- A220-100 (former CS100 )
- A220-300 (former CS300)

The ICAO type designator is BCS1 and BCS3 respectively. The official type certificate designator for the aircraft is BD500.

The CS100 took off on its maiden flight in 2013 and entered service with Swiss in 2016. The CS300 first flew in 2015 and entered service in 2016 as well with airBaltic.

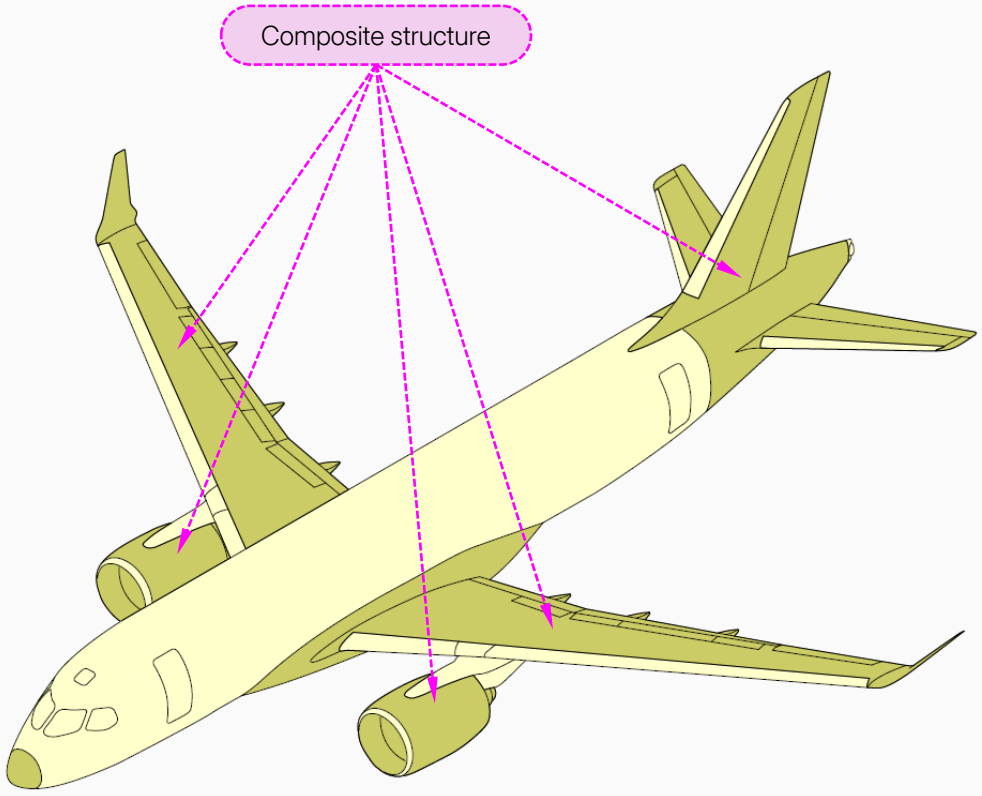


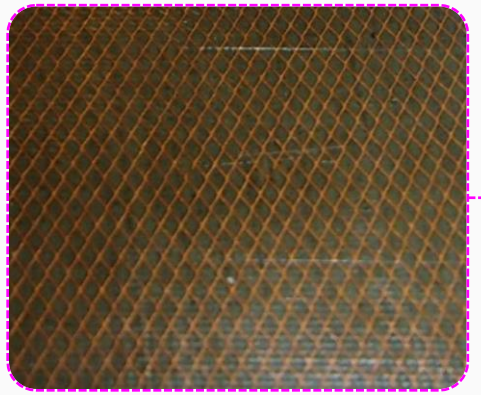
# Introduction

The Airbus A220 is certified to carry up to 130 (-100) or 150 passengers (-300) in a rather unusual 2-aisle-3 configuration. The aircraft includes two pressurized cargo compartments under the cabin floor.

Most of the fuselage is made of an aluminum-lithium alloy. The wings, center wing box, wing-to-body fairing, empennage, aft fuselage and engine nacelles are made of composite materials.

The aircraft has a Fly-By-Wire (FBW) flight control system and is powered by two Pratt & Whitney PW1500G, geared, ultra-high bypass, turbofan engines controlled by a Full Authority Digital Engine Control.



The composite structure is protected against lightning strikes by a copper foil wrap applied to the top layer.

# Dimensions

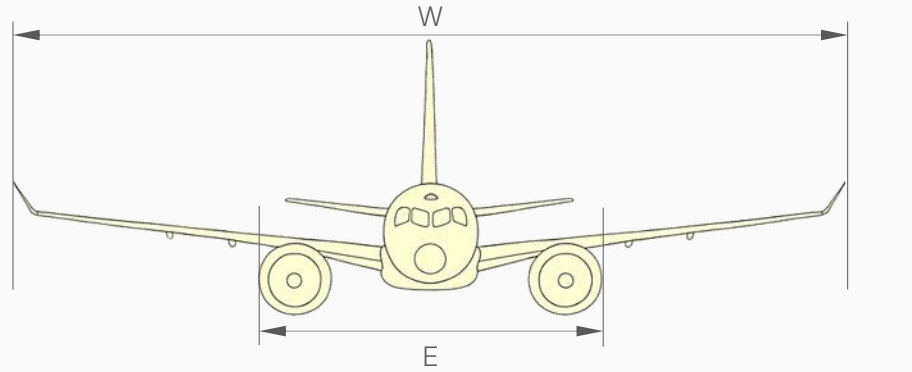
The A220-100 and A220-300 share the same wing and wing center box design, wingspan is therefore the same.

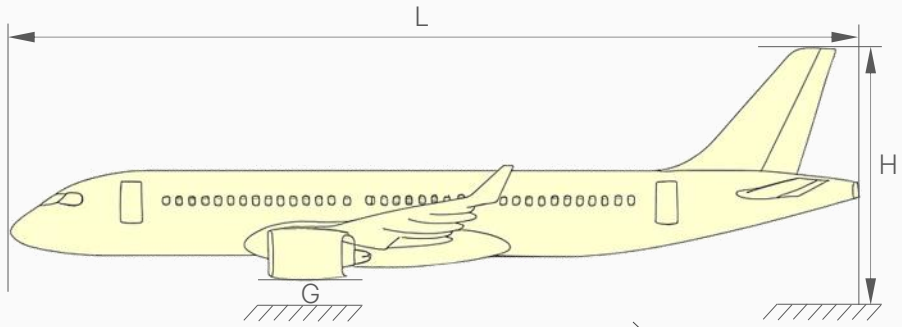
| Dimensions | A220-100 | A220-300 |
|---|---|---|
| L = Length | 34.9 m | 38.7 m |
| H = Height | 11.5 m | |
| W = Wingspan | 35.1 m | |
| G = Engine ground clearance | 0.51 m | |
| E = Engine outside diameter | 13.4 m<br>(minimum taxiway width 15 m) | |

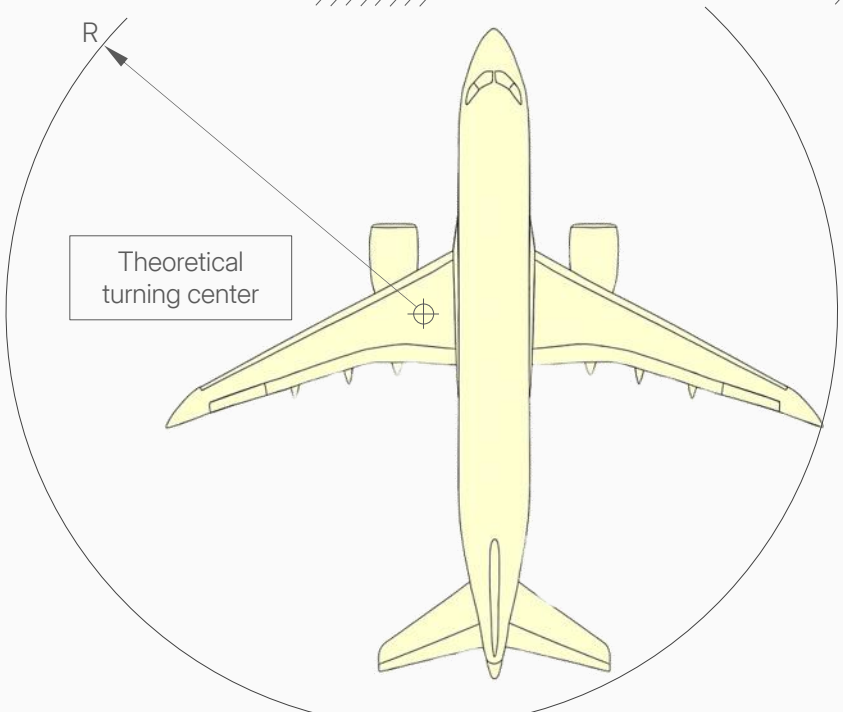
Although sharing the same wing, fuel capacity is slightly less on the A220-300 due to structural differences.

| Fuel Capacity (0.809 kg/l) | A220-100 | A220-300 |
|---|---|---|
| (a) Left wing tank | 3080 kgs | 3050 kgs |
| (b) Right wing tank | 3080 kgs | 3050 kgs |
| (c) Center tank | 11.480 kgs | 11.300 kgs |
| **Total (a + b + c) =** | **17.640 kgs** | **17.400 kgs** |

The A220-100 / -300 can make a 180-degree turn (R) under its own power (nosewheel deflected 80°, no differential thrust or braking) on a runway with a width of at least 20.82 m / 23.50 m respectively.

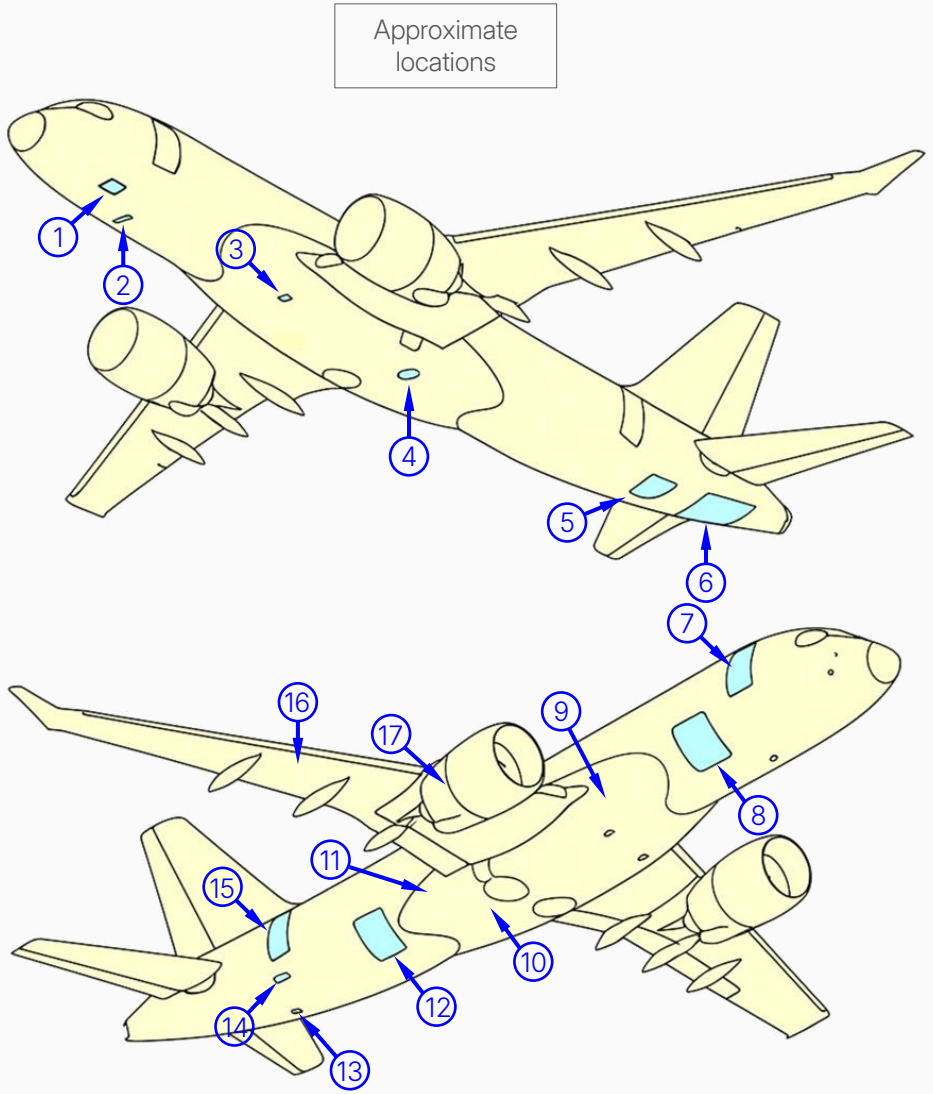


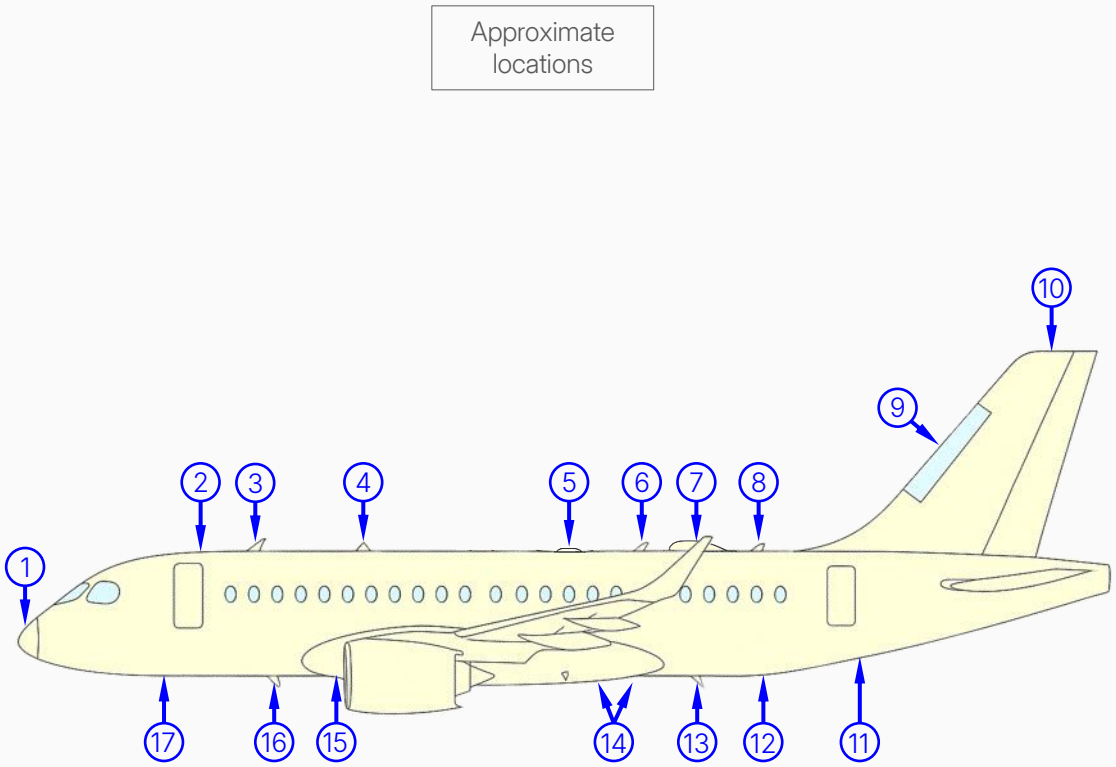
# Ground Servicing

1. Electrical / towing service panel including ground power connector. Crew oxygen system refill panel (forward of electrical / towing panel).
2. Forward avionic compartment door (pressurized section).
3. Low Pressure Ground Connection (LPGC) and High-Pressure Ground Connection (HPGC) doors (separate panels).
4. Mid avionic compartment door (pressurized section).
5. Aft equipment bay door (unpressurized section).
6. APU access, clamshell doors.
7. Forward service door (R1).
8. Forward cargo compartment door.
9. Refuel / defuel control panel.
10. Ram Air Turbine (RAT) panel location.
11. Battery compartment access panel.
12. Aft cargo compartment door.
13. Waste (lavatory) servicing panel.
14. Potable water servicing panel.
15. Aft service door (R2).
16. Refuel / defuel adapter (left side optional).
17. Engine oil servicing panel (within fan cowl).



# Antenna Locations

1. VHF LOC & GS, WXR
2. TCAS & Transponder
3. GNSS
4. VHF 1
5. SATCOM (optional)
6. VHF 3
7. K-BAND (internet, optional)
8. ELT
9. HF
10. VOR
11. Marker Beacon
12. DME 2
13. ADF (optional)
14. RAD ALT (2)
15. DME 1
16. VHF 2
17. TCAS & Transponder



# Flight Compartment

The flight compartment is equipped with a lockable cockpit door. Behind the pilot seats there is a foldable observer (jump) seat which slides into position in front of the flight deck door when in use.

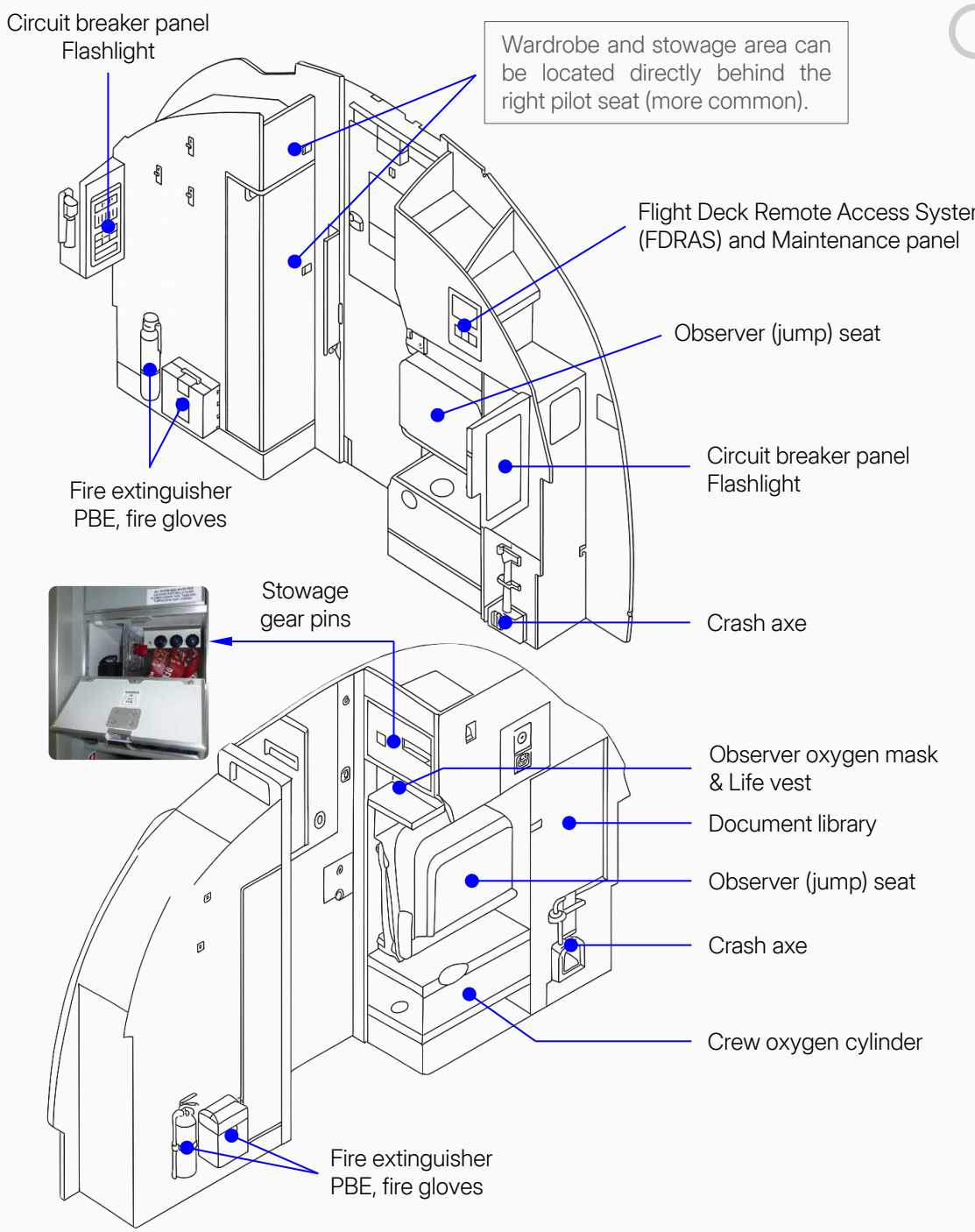
Located behind the left pilot seat:

- Circuit breaker panel and flashlight (to the side)
- Oxygen bottle (below observer seat in stowed position)
- Crash axe and document storage
- Flight Deck Remote Access System (FDRAS) & Onboard Maintenance System (OMS) panel
- Observer oxygen mask and life jacket
- Storage area that holds gear pins (3)

Located behind the right pilot seat:

- Circuit breaker panel and flashlight (to the side)
- Protective Breathing Equipment (PBE)
- Fire extinguisher and fire gloves
- Wardrobe, and storage area (more commonly behind the right seat)

Furthermore, the flight compartment is equipped with an emergency escape hatch fitted with an escape rope behind the overhead panel.



# Seating

## Flight Deck Seats

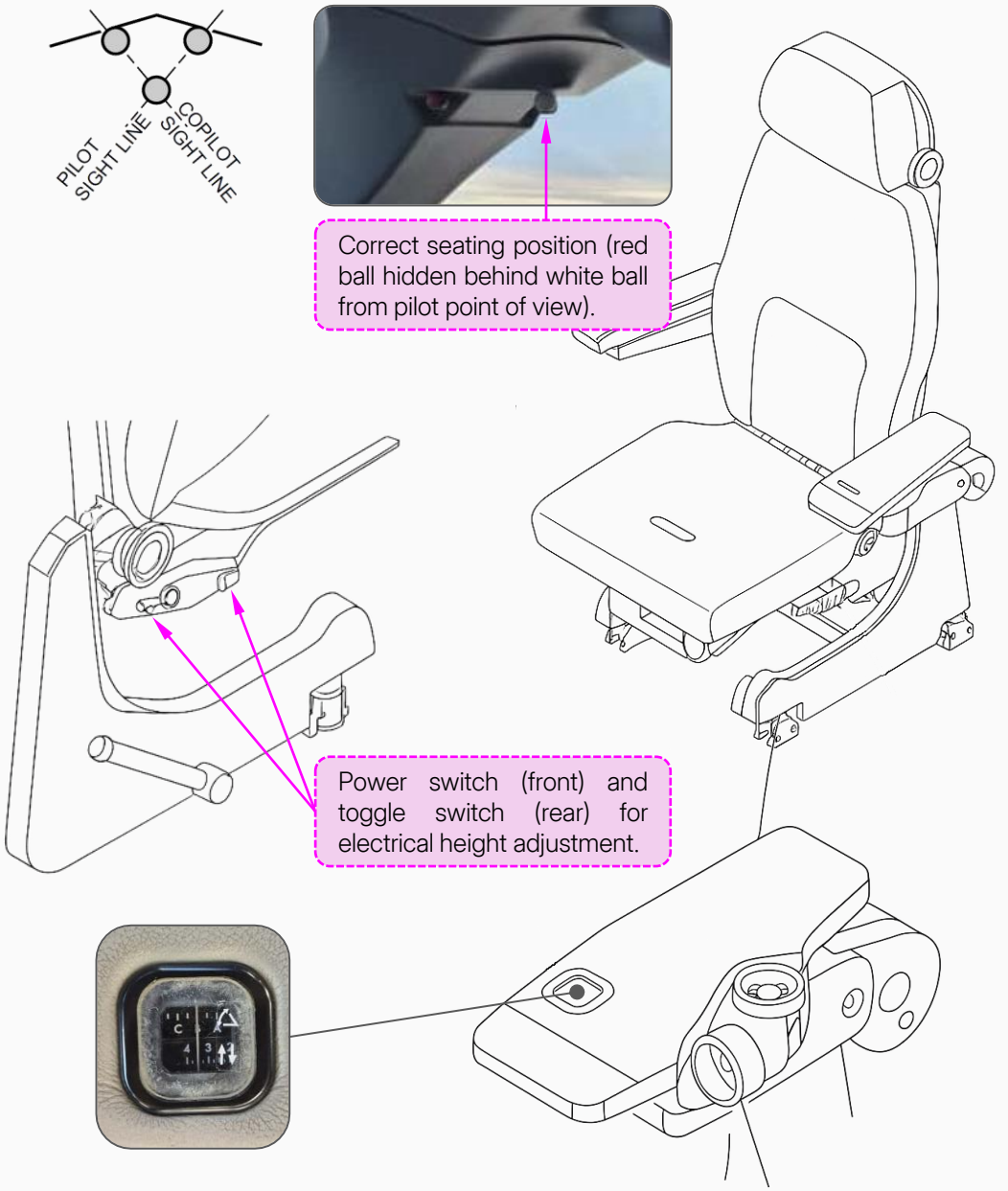
The flight deck seats feature adjustments for:

- Forward and aft movement with outboard movement at the end of the track for easy ingress and egress.
- Electric vertical adjustment with mechanical back-up.
- Adjustable armrests, the outboard armrest features height and tilt adjustment for comfortably holding the sidestick controller. The height is labelled in numbers (1 to 4), tilt in letters (A to E), making it easy to find the correct position during pre-flight (e.g. 3½ B).
- Adjustments for recline, seat pan and lumbar support.

## Flight deck Vision

The eye level indicator (alignment balls) on the center post are used to find the correct eye height and therefore seating position. The correct seating position is achieved when:

- Flight controls are unrestricted throughout full travel
- Flight instruments are visible and unobstructed
- Outside visibility is unobstructed
- Seating is comfortable and the same for VFR and IFR

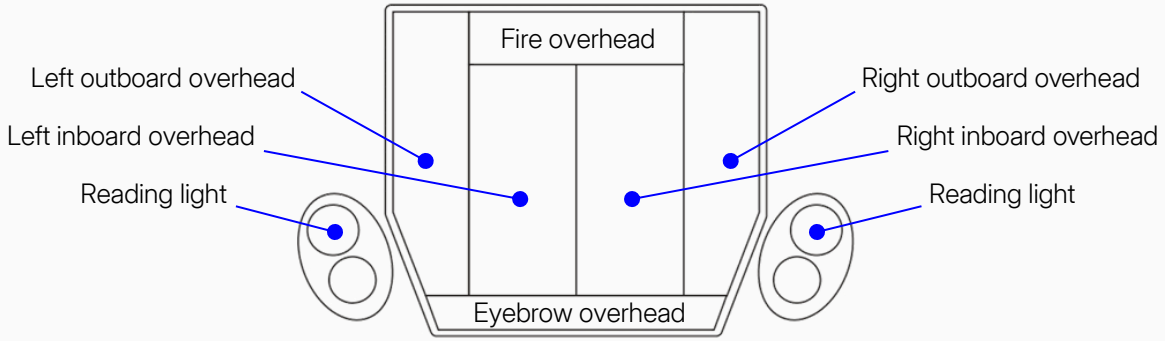


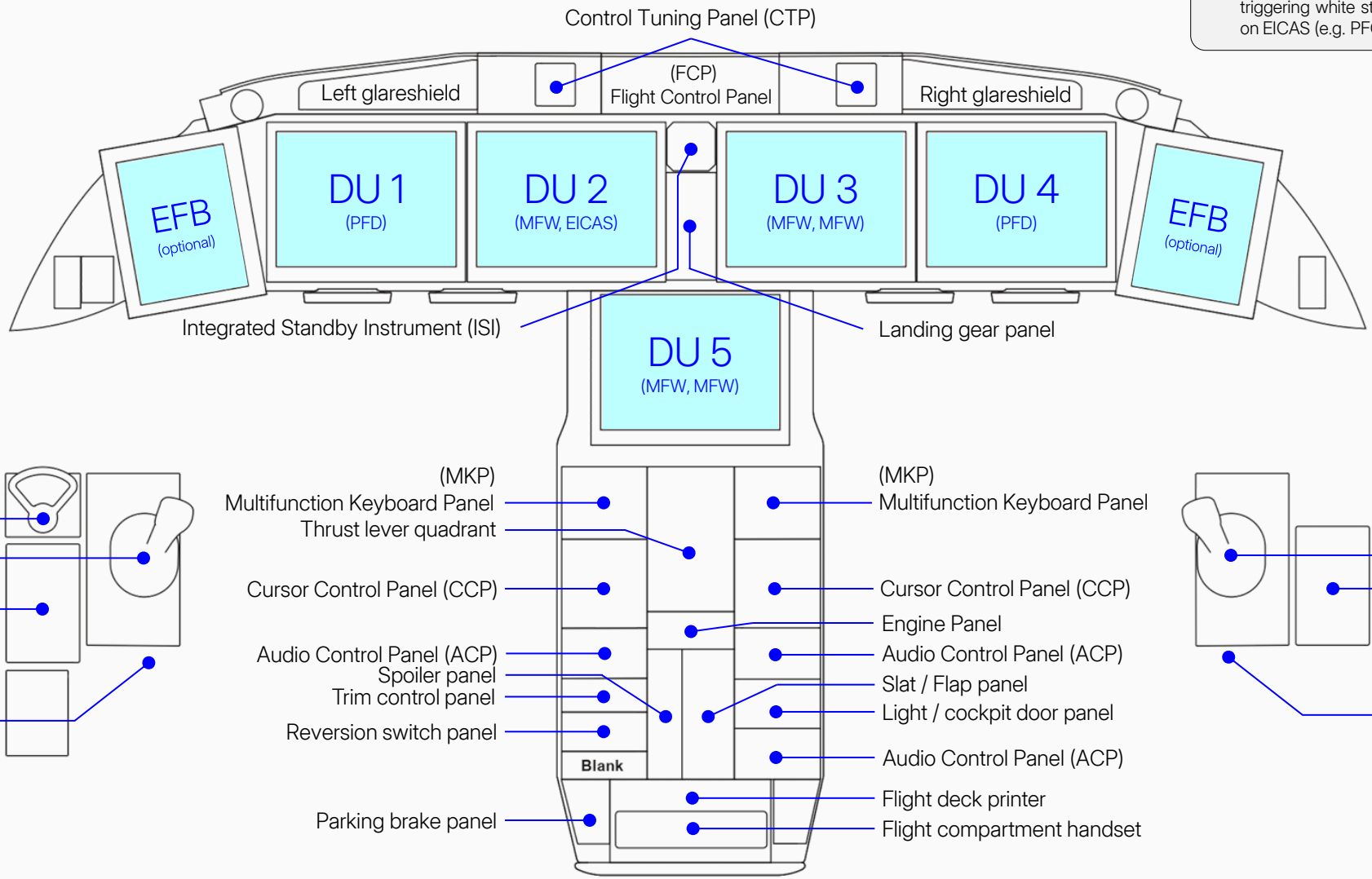
# Flightdeck

In normal flight, switches are in the twelve o'clock position, guards closed with all lights extinguished (dark cockpit philosophy).

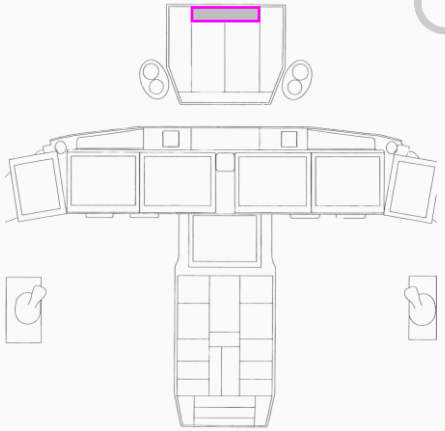
Guarded switches are color-coded based associated procedures:

- RED guard cover for switches associated with emergency procedures or triggering a red alert on EICAS (FIRE and RAT).
- YELLOW cover for switches associated with non-normal procedures, causing amber alert on EICAS (EMER DEPRESS).
- BLACK guard for switches triggering white status message on EICAS (e.g. PFCC, TAWS).

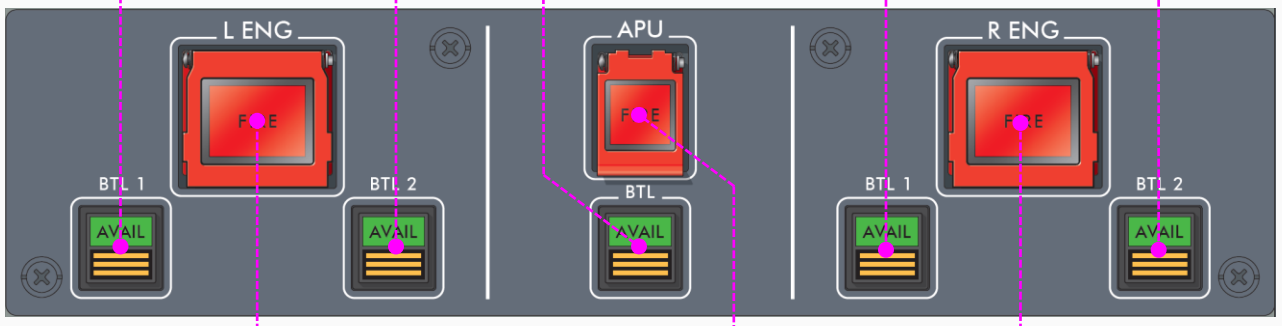


# Engine and APU Fire Panel



**Engine and APU fire-extinguishing bottle (BTL) switches**
- AVAIL when associated fire (guarded) switch is pressed (bottle is armed).
- Amber light bar when bottle is discharged or has failed (low pressure or squib failure).




| ISOLATION | L ENG FIRE or R ENG FIRE (Guarded) | APU FIRE (Guarded) |
|---|---|---|
| FIRE | Fire extinguishing bottles are armed | Fire extinguishing bottle is armed |
| FUEL | Fuel shut-off valve closes<br>FADEC commands shutdown | APU fuel shut-off valve closes<br>ECU commands APU shutdown |
| AIR | Fan air valve and bleed valve closes | APU bleed air valve closes |
| HYD | EDP hydraulic shut-off valve closes | - |
| ELEC | VFG deactivates | VFG deactivates |

**Note:** APU fire on ground will cause an automatic shutdown of the APU, 10 seconds later the APU fire extinguishing bottle discharges automatically.

The crew can override the automatic bottle discharge by pressing the APU FIRE guarded switch.

In-flight, detection of an APU fire will not cause an automatic shutdown or discharging of the APU fire bottle.

**Note:** the aircraft is equipped with two engine fire bottles installed in the wing-to-body fairing between the two wheel wells.

Engine fire bottles (BTL) can be discharged in either engine.

# Left Outboard Module

**Entrance Light**
Controls brightness of one ceiling and 2 lights at the rear of the pedestal, illuminating the floor. When turned halfway the floor lights will be at max brightness, past halfway the overhead light will come on at maximum brightness.

**Aural Warning Inhibit (Guarded)**
When pressed (INHIB), inhibits ALL aural warnings.

**Window Heat**
Switches windshield and side window heat between automatic mode (not selected) and OFF.

**PFCC - Primary Flight Control Computers (Guarded)**
Switches each of the three Primary Flight Control Computers between ON (not selected) and OFF.

**Panel Lights**
CB INTEG: controls the brightness of the left and right circuit breaker panel integral lighting.

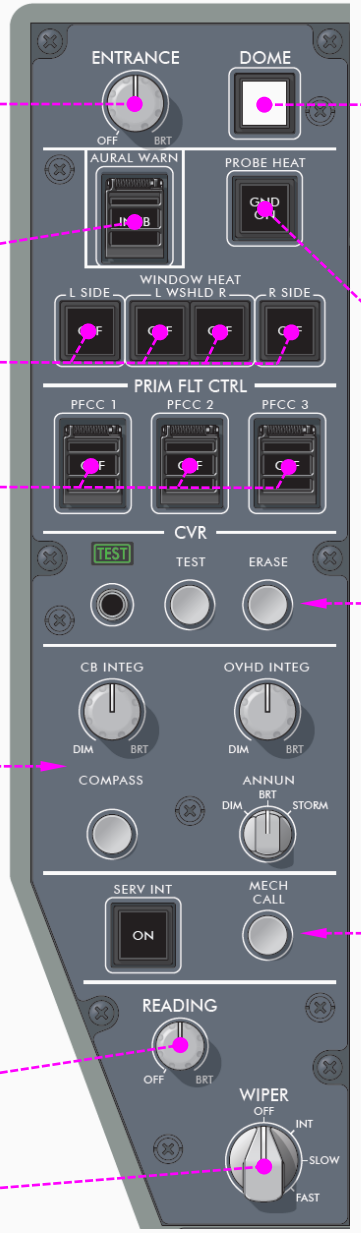
OVHD INTEG: controls the brightness of the overhead panel integral lighting.

COMPASS: toggles the magnetic compass light ON / OFF.

ANNUN:
- DIM – sets switches, lights and MKP to min brightness.
- BRT – sets switches / lights and MKP to full brightness.
- STORM – as BRT + sets integral lights of all panels to full brightness and turns on the dome lights.

**Left Pilot Reading light**
Controls brightness of the left pilot reading light.

**Left Windshield Wiper**
OFF – off and parked (vertical), INT – once every 5 sec, SLOW – 80 cycles per min, FAST 120 cycles per min.




**Dome Light**
Switches the 2 dome lights in the ceiling. The button (dome finder light) lights up for 20 min when ENTRY light is on, and aircraft is unpowered.

**Probe Heat**
Automatic mode (not selected): ADSP (4) and TAT (2) probes are heated when one engine is running (modulate below 55 kts, full heat above 55 kts or weight of wheels).

When selected (GND ON), heats the ADSP and TAT probes for 2 minutes while on ground with engines OFF.

AOA sensors are always heated when aircraft is powered.

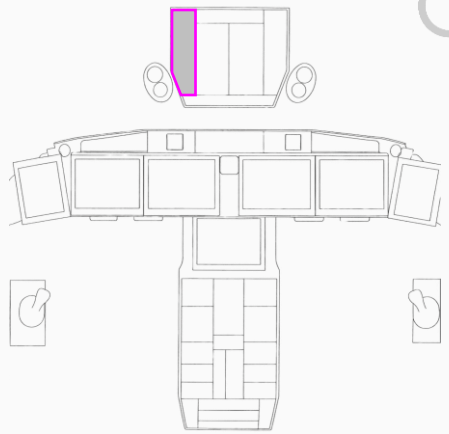
**CVR – Cockpit Voice Recorder**
TEST switch: when pressed and held performs an internal test. If successful, TEST illuminates.

ERASE switch: when pressed for 2 seconds with aircraft Weight-On-Wheels and parking brake ON, deletes all voice recordings. A tone is heard over the headphone jack.

**Communication**
SERV INT: if not pushed, the intercom is active between the three flight deck stations (left pilot, right pilot and observer), electrical / towing service panel and refuel / defuel panel.

When selected (ON), communication is opened to all maintenance and service panels (in addition to the above: forward, mid, aft equipment bays and the low-pressure ground connection panel).

MECH CALL: generates a horn to sound near the nose.



**Note:** a CVR test failure soon after aircraft is powered up might indicated that the CVR battery is not adequately charged, try again in 15 minutes.

**Note:** if left and right wipers are set to run at the same speed, their movement will be synchronized.

# Left Inboard Module

**Power Transfer Unit (PTU)**
Uses pressure from hydraulic system 2 to pressurize hydraulic system 1 (without exchanging fluid).
OFF – PTU does not operate. ON – PTU operates if hydraulic system 2 is pressurized and hydraulic fluid quantity is above 5%. AUTO – PTU will operate automatically when any of the conditions below exist:
- Take-off thrust applied until slats / flaps retracted
- In-flight with landing flaps / slats deployed (4 or 5) or ALTN FLAP selected (3) until slats / flaps are up
- Engine Driven Pump 1A or left engine has failed
- Single engine taxi (left engine off), park brake off

**Bus Isolation Switch**
Controls automatic reconfiguration of line and bus tie contactors (used during smoke isolation procedure).
AUTO – allows automatic reconfiguration.
MAIN – inhibits automatic reconfiguration of EPC 1/2.
ESS – inhibits automatic reconfiguration of EPC 3.

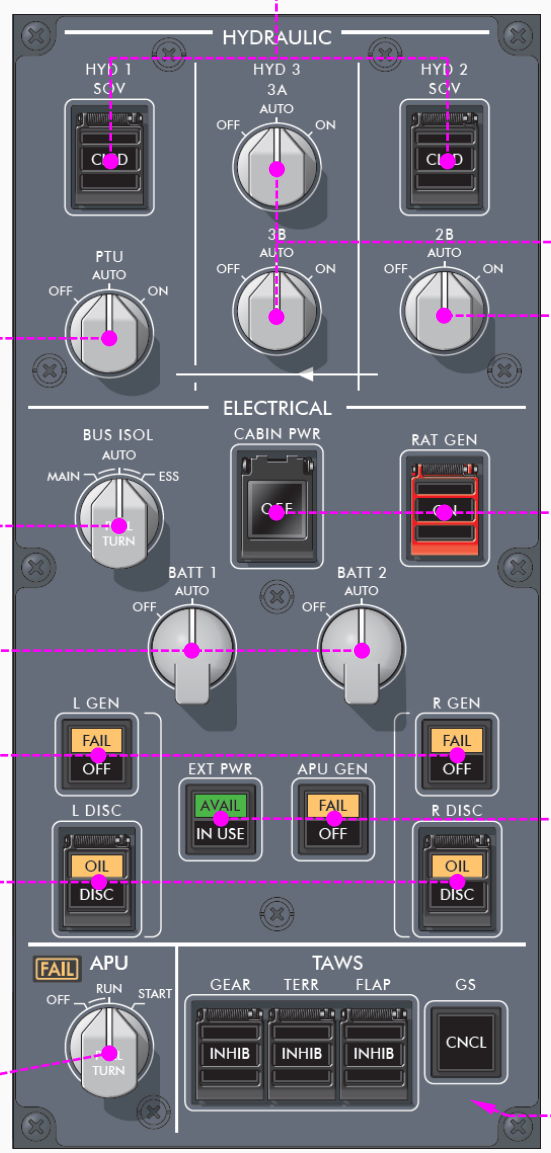
**Battery 1 and 2 Switches**
OFF – disconnects corresponding BAT from ESS bus.
AUTO – enables the BPCU to connect the battery.

**Left and Right Generator Switches**
Connect or disconnect (OFF) the Variable Frequency Generator (VFG), FAIL indicates a fault.

**Left and Right Disconnect Switches (Guarded)**
Disconnects (DISC) the VFG from the gearbox, automatic disconnect occurs when second high oil temperature threshold is reached (OIL).

**APU panel**
OFF – APU shuts down, RUN – APU is running, START (hold 3 sec), APU start sequence initiates.
FAIL indicates ECU BITE, fire detection or start fail.

**EDP 1A and 2A Shutoff Valves (Guarded)**
When pressed (CLSD) shuts off hydraulic fluids from the engine pylon to the Engine Driven Pump (EDP).
Also closes when L ENG or R ENG fire switch is pressed or automatically when hydraulic fluid temperature exceeds 125°C




**AC Motor Pump (ACMP) 3A and 3B**
OFF – ACMP 3A or 3B do not operate. ON – ACMP 3A or 3B operates if AC power is available and hydraulic fluid quantity is above 5%.
AUTO - ACMP 3A will operate automatically:
- Either left or right engine started or when airborne.

AUTO - ACMP 3B will operate automatically when any of the conditions below exist:
- Take-off thrust is applied
- Slats / flaps in transit while on ground
- Slat / flap lever or position out of zero in flight
- ACMP 3A has failed

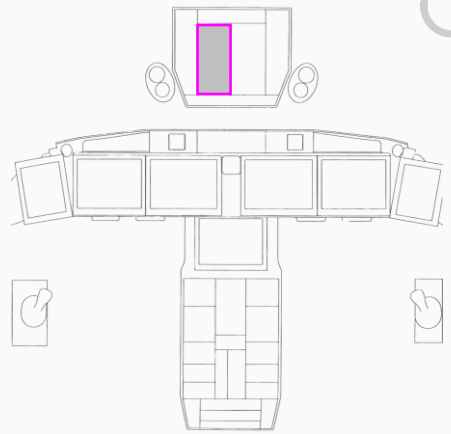
**AC Motor Pump (ACMP) 2B**
OFF – ACMP 2 does not operate. ON – ACMP 2 operates if AC power is available and hydraulic fluid quantity is above 5%. AUTO – ACMP 2B will operate automatically when any of the conditions below exist:
- Take-off thrust applied until slats / flaps retracted
- In-flight with landing flaps / slats deployed (4 or 5) or ALTN FLAP selected (3) until slats / flaps are up
- Engine Driven Pump 2A or right engine has failed
- PTU is operating (automatically or manually)
- Single engine taxi (right engine off), park brake off

**Cabin Power Switch (Guarded)**
When selected (OFF), deactivates flight compartment charging outlets and galley / entertainment systems.

**Ram Air Turbine (RAT) Switch (Guarded)**
RAT automatically deploys (ON), when AC power is lost in-flight. When pressed, manually deploys the RAT. If pressed with RAT deployed, RAT GCU resets.

**External Power Switch**
If connected (not used) AVAIL, when selected IN USE

**APU Generator Switch**
Connect or disconnect the APU VFG (OFF).
FAIL indicates a VFG fault.



**Note:** if AVAIL, external power connects automatically (IN USE) in case of engine or APU shutdown.

**Note:** a disconnected VFG cannot be reconnected in-flight.

**Gear Inhibit (Guarded)**
Inhibits (INHIB) the TAWS alerts associated with the landing gear - "TOO LOW GEAR".

**Terrain Inhibit (Guarded)**
Inhibits (INHIB) the TAWS predictive (forward looking) terrain alerts and terrain display - "TOO LOW TERRAIN"

**Flap Inhibit (Guarded)**
Inhibits (INHIB) the TAWS alerts associated with the flaps - "TOO LOW FLAPS".

**Glideslope Cancel**
Cancels (CNCL) the glideslope alert - "GLIDESLOPE".

# Right Inboard Module

### Manual Transfer
R or L – opposite boost pump runs if set to AUTO. 227 kgs transfers as selected, valves close and boost pump stops.
CTR – L or R boost pump must be set ON, 454 kgs transfers from L or R to center tank, then set boost pump back to AUTO.

### Gravity Transfer
When ON opens the gravity transfer valve, connecting the left and right main (wing) tanks. Side slipping will increase fuel transfer to the low wing.

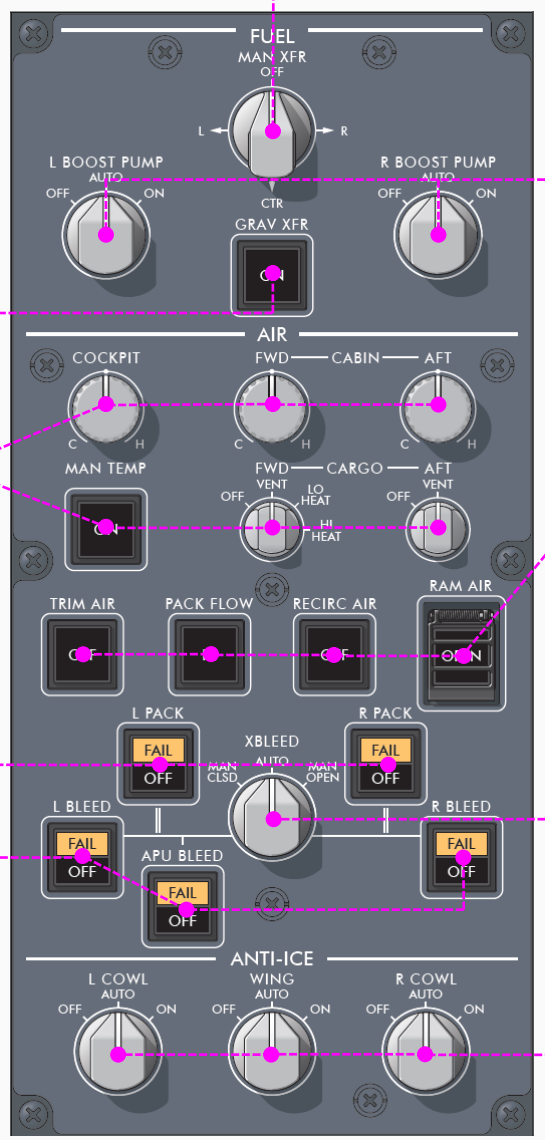
### Compartment Temperature Control
Crew can select the compartment temperature (or DUCT temp if MAN TEMP is ON) between 18°C and 30°C, center position is 24°C. The cabin crew can further adjust compartment temperature ±3°C.

### Forward Cargo Temperature Control
The forward cargo compartment can be set to:
- OFF (not ventilated)
- VENT (between +4°C and +25°C)
- LO HEAT (between 15°C and 20°C)
- HI HEAT (between 20°C and 25°C)

### Aft Cargo Temperature Control
Aft cargo compartment can be switched between OFF (not ventilated) and VENT (ventilated), the temperature in the aft cargo compartment is between -2°C and +25°C in VENT.

### L Pack and R Pack
Controls the Flow Control Valves (FCV). Normally in the auto (deselected position), can be switched OFF or illuminate FAIL in case of a fault. Switching the PACK off, then back to auto performs a reset.

### L, R and APU Bleed Switches
Normally in auto mode (deselected), can be set OFF. FAIL illuminates if bleed is not available while commanded by the IASC. Can be caused by Bleed Valve Fault, dual loss of Bleed Temperature Sensor, loss of Bleed Air Leak Detection or IASC 1 + 2 failure.




### Boost Pump Switches
OFF or ON – boost pump manually runs or turned off.
AUTO – boost pump runs automatically if low fuel pressure is sensed (ejector pump failure), APU is running without engines running (normally left pump) or fuel imbalance is sensed (automatic crossfeed):
- When center tank is empty, and
- Boost pumps are AUTO, and
- Imbalance exceeds 180 kgs,

The boost pump in the tank containing more fuel will run until tanks are within 45 kgs (for example in case of one engine inoperative or APU running of L tank).

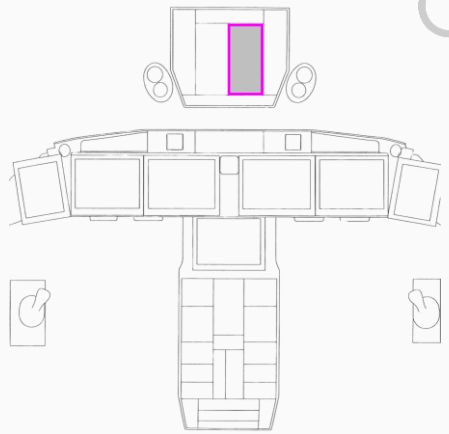
### Trim Air
Trim air is used to control compartment temperature of cockpit, forward and aft cabin zones and the forward cargo compartment. When switched OFF, valves will close, and zone temperature will decrease.

### Pack Flow
The amount of airflow is determined by the number of occupants entered in the FMS. When HI is selected, the high flow mode is activated, recirculation fans and packs will deliver maximum flow. Can be used for smoke or fumes evacuation.

### Recirculation Air
In auto mode (deselected), the recirculation fan is on, when OFF (selected), the recirculation fan is off.

### Ram Air (Guarded)
Controls the Emergency Ram Air Valve (ERAV) used to ventilate the flightdeck and cabin for flight without air-conditioning, OPEN displayed when selected on.

### Crossbleed Valve
AUTO – the IASC controls the crossbleed valve based on available bleed sources and demand. The valve can be set to MAN CLSD or MAN OPEN.



**Note:** Wing and Cowl anti-ice is inhibited until 1500 ft or 2 min after take-off (whichever occurs first) if not set ON before take-off.

**CAUTION:** cargo that is sensitive to temperature (e.g. live animals) shall be loaded in the forward cargo compartment only.

### Wing Anti-Ice System (WAIS)
OFF – wing anti-ice is off, and Wing Anti-ice Valves (WAIVs) are closed. AUTO – wing anti-ice is controlled by the IASCs and activated when ice is detected. ON – wing anti-ice is ON (in-flight) or armed (on ground) to activate above 60 kts.

### Cowl Anti-ice System (CAIS)
OFF – cowl anti-ice is off, and Cowl Anti-ice Valves (CAIVs) are closed. AUTO – cowl anti-ice is controlled by the FADEC and activated when ice is detected. ON – cowl anti-ice is ON.

# Right Outboard Module

**Emergency Locator Transmitter (ELT)**
ON – activates the ELT, ARM – the ELT is ready to activate on high impact, TEST / RESET – to test the ELT or reset after it has been activated. Newer models are equipped with Distress Tracking (ELT-DT).

**Equipment Cooling – Inlet**
Supplemental cooling installed: INLET switch control the heat exchanger fans for the forward and mid equipment bays and the ground valve. On other aircraft this switch only controls aft galley and lavatory air extraction.

- AUTO - controlled by the IASC or OFF.

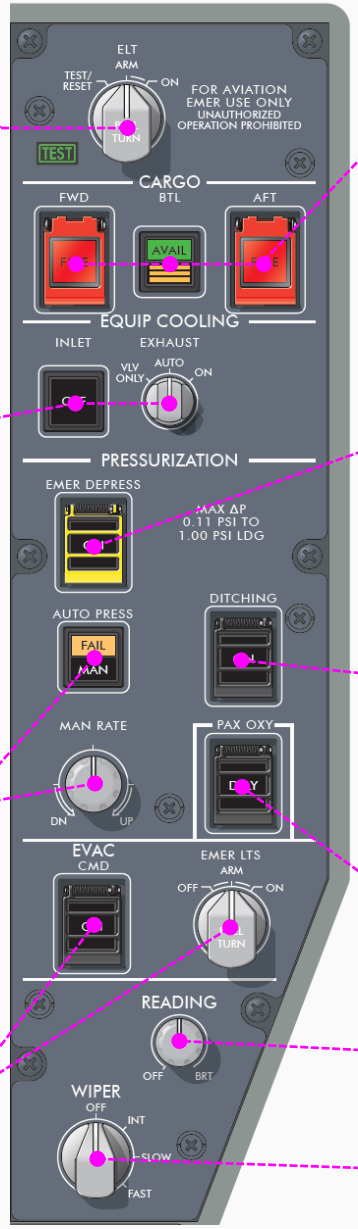
**Equipment Cooling – Exhaust**
Controls the Avionic Exhaust Valves (AEVs) & back-up fan.

- AUTO – the Avionic Cooling and heat Extraction System (ACES) is controlled by the IASCs.
- ON – both AEVs fully open and the back-up fan is on.
- VLV ONLY – AEVs are fully open, back-up fan is off.

**Cabin Pressurization System**
Deselected - pressurization is controlled by the IASCs. FAIL indicates a failure of the pressurization system. When pressed MAN illuminates, the pressurization system is in manual mode and is controlled with the MAN RATE knob. 12 o'clock position corresponds to 0 ft/min, 9 and 3 o'clock position corresponds to -1000 and +1000 ft/min respectively. Maximum rate is 2500 ft/min.

**Evacuation Command (Guarded)**
Activates (ON) or deactivates the evacuation command. Cabin crew can also activate or deactivate the EVAC horn.

**Emergency Lights**
ARM – lights will automatically come on when main power is lost. Cabin crew can also activate emergency lights.




**Cargo Fire Panel**
Fire switches illuminate if 2 out of 4 smoke detectors are detecting smoke. Pressing the guarded FWD or AFT fire switch will close cargo ventilation and arm the fire bottles.

Pressing the BTL switch when AVAIL, activates the bottle squibs. The High-Rate Discharge (HRD) bottle fires first to extinguish the fire, followed by the Low-Rate Discharge (LRD) bottle to supress the fire for 75 min (or longer as an option). Amber light bar indicates HRD bottle discharge or fault detected. There is no indication for the LRD bottle.

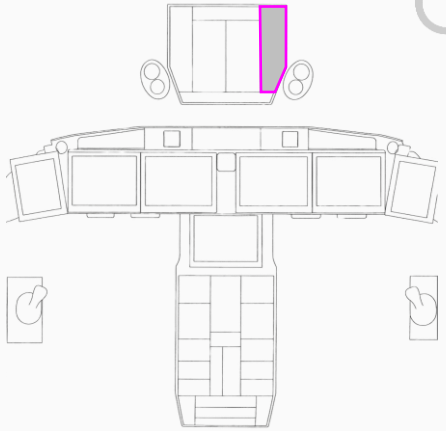
**Emergency Depressurization (Guarded)**
When selected (ON), the emergency depressurization mode is activated. If the pressurization is in auto mode, cabin rate is limited to 2500 ft/min, if pressurization is in manual mode, the cabin rate is limited to 3500 ft/min. The cabin will depressurize to 15.000 ft or current level if lower.

**Ditching switch (Guarded)**
When selected (ON), starts the ditching sequence by closing all valves below the water line (except RAM AIR): 1. Avionic Exhaust Valves (AEV), 2. Flow Control Valves (FCV), 3. Trim Air Pressure Regulating Valve (TAPRV), 4. Bulkhead check valve and 5. Outflow Valve (OFV) (last).

**Passenger Oxygen (Guarded)**
Oxygen masks will automatically deploy if cabin altitude exceeds 14.800 ft or can be manually deployment (DPLY), NO SMOKING / NO PED and FASTEN SEATBELT sign will come on and pre-recorded cabin announcement will play.

**Right Pilot Reading light**
Controls brightness of the right pilot reading light.

**Right Windshield Wiper**
OFF – off and parked (vertical), INT – once every 5 sec, SLOW – 80 cycles per min, FAST 120 cycles per min.

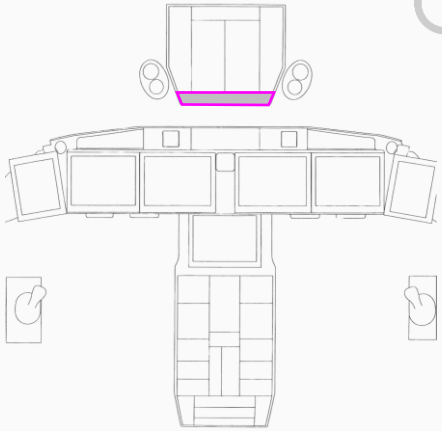


**Note:** some aircraft are not equipped with a supplemental avionic cooling system, therefore no ground valve installed (located behind right wing trailing edge above wing-to-body fairing).

Ground valve, opens on ground between 3 and 30°C OAT. Absent on some aircraft.

**Note:** if left and right wipers are set to run at the same speed, their movement will be synchronized.

# Eyebrow Panel



**Exterior Lights Panel**
- Navigation lights
- Beacon lights
- Strobe lights
- Logo lights
- Wing inspection lights

**Taxi lights**
NARROW only illuminates the nose taxi light, WIDE illuminates the nose and wing taxi lights (20° beam).

**Landing lights**
Separate switches for L, R and NOSE lights projecting a focused 3° beam.

**Passenger Signs**
SEATBELTS and NO PED or NO SMOKING. AUTO – signs go off above 10.000 ft, cabin crew can control signs

CVR Area Microphone

**Note:** toggle switch direction is optional and selected by operator. (Up = ON versus Down = ON)

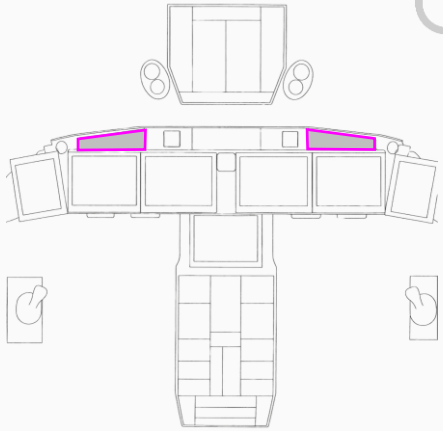
**Note:** nose taxi and nose landing light will only illuminate when nose landing gear is down and locked.

# Glareshield Panel

**Display Unit (DU) 1, 2 Brightness Control**
Brightness control for OUTBD (DU 1) and INBD (DU 2). Fully counter-clockwise turns off the DU. The right glareshield similarly controls DU 3 and 4. In case of switch failure, the display will automatically revert to 80% brightness.

**MASTER WARNING – CAUTION**
Flashes red in case of WARNING (top half) or amber in case of CAUTION (bottom half), press to cancel. Also stops flashing of the respective message on EICAS.

**Quick Response Panel (CPDLC)**
- ACPT – ATC is advised that the flight crew have accepted the uplinked message.
- RJCT – ATC is advised that the flight crew is unable to comply with the uplink request.
- STBY – ATC is advised that the message has been read but a further response is expected, response timer is paused.
- LOAD – loads the message into MKP scratchpad (not functional with Link 2000+).
- Refresh – refreshes the CPDLC window on the EICAS and shows the next message in case of more than one uplink message.



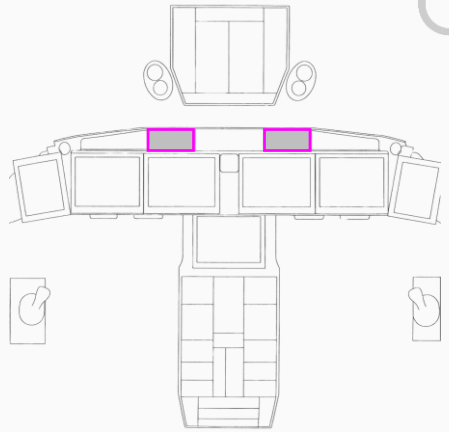
**Note:** right glareshield panel is a mirror of the left glareshield panel.

**Chrono Button**
Starts, stops and resets the timer (displayed on PFD). 1st press starts the timer, 2nd press stops the timer, 3rd press resets and removes the timer, repeat.

**Sidestick Priority Switch (Guarded)**
MOMENTARY priority: pressing and holding the AP/PTY (red) button on the sidestick:
- Disengages AP, disables the opposite sidestick
- A green PTY (priority) will flash on the priority side
- An aural "PRIORTY LEFT / RIGHT" will sound
- A red → or ← will appear on the disabled side
- If sidestick AP/PTY is released, priority is cancelled

LATCHED priority: lifting the guard and pressing the SIDESTICK priority switch on the glareshield.
- Disables the opposite sidestick, does not affect AP
- A green PTY will be steady on the priority side
- An aural "PRIORTY LEFT / RIGHT" will sound
- A red → or ← will appear on the disabled side
- Priority is canceled when switch is pressed again

# Control Tuning Panel (CTP)



**Note:** left and right CTP are identical in layout (not mirrored).

**Note:** navigation and data entry is performed with the 7 Line-Select-Keys (LSKs) and the TUNE / DATA rotary knob.

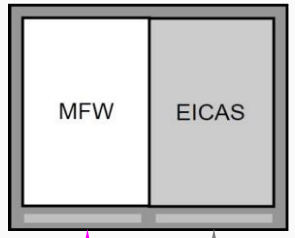
**MAP range selector**
Push for WXR ON / STBY is not functional on A220.

**Display Select Switches**
Allow selection of the left or right half of DU2 (left pilot) or DU3 (right pilot), called MFW - Multi Function Window. The selected MFW is highlighted with a green bar.

With the Quick Action Keys (QAKs) a format for the MFW is chosen. MAP is the only format that support full screen.

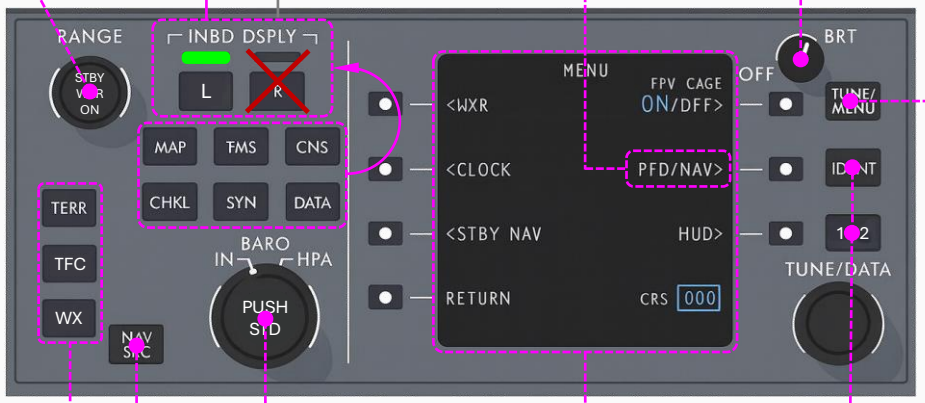
EICAS is displayed on the right MFW of DU2 (default) or left MFW on DU3 (closest to center), the display select switch for MFW currently displaying EICAS is inhibited.

DU 2



**PFD/NAV MENU page**
BARO / RAD minima bug, BRG pointers and manual LDG ELEV.

**Brightness Control**
Full CCW turns off the CTP display (but does not inhibit the buttons).




TUNE

ATC ID (callsign) is taken from the ACARS INIT page (optional) or is manually entered in the transponder. It is not coupled to the FMS.

**MAP overlay buttons NAV SRC (cycle)**
- FMS 1
- NAV 1
- FMS 2
- NAV2

**BARO Selector**
Rotate outer for IN or HPA, rotate inner to change pressure. Press for standard.

**MENU Page**
Access to WXR controls, clock, STBY NAV, HUD (if installed), PFD / NAV, and HSI course.

**Onside / Cross-side**
Swap between VHF 1, NAV 1, etc. and VHF 2, NAV 2, etc.

**Transponder IDENT**

**TUNE Page**
Tuning of VHF, NAV, XPDR, TCAS, ADF SELCAL and HF. (as installed).

# Flight Control Panel (FCP)

- Active modes in green, armed modes are white.
- CAP(ture) indicates an altitude capture mode.
- V indicates VNAV active for example VS → VVS.
- (V)TO, (V)GA or WSHR guidance activated by pressing either TOGA switch on thrust levers.

**Speed Rotary Switch –** Outer knob switches between FMS controlled speed target (magenta speed bug) and manually selected speed target (cyan speed bug).

Inner knob select manual target speed. When pushed, selects between IAS and MACH (automatic passing 27.500 ft).

**FD Switch**
Switches OFF Flight Directors (both) if Flight Guidance is coupled to the left (◄ XFR ►).

**Autoland Modes**
- Lateral at 150 ft – ALIGN
- Vertical at 65 ft – FLARE
- Throttle at 20 ft – RETARD
- Ground +2 sec – ROLLOUT

**HDG** – activates HDG mode. HDG is the default lateral mode.

**HDG Rotary Selector** – select heading target, push to synchronize heading. If HDG is not the active mode, push selects the heading to auto (remains automatically synched with current heading).

**NAV –** arms NAV mode based on active navigation source (FMS / VOR / LOC).

**APPR** – arms approach based on approach selected in FMS (LOC + GS or FMS + VGP).

**1/2 BANK –** automatic passing 31.500 ft, when pressed below, 1/2 BANK is only active in HDG.

**AP** – activates the autopilot if sidestick is neutral. When activated, sidestick centering force is increased. The AP can be switched OFF by applying force to the sidestick (non-standard), pressing AP on the FCP (inhibited below 1500 ft on ILS approach), or the sidestick AP / PTY button (standard).

**AT** – activates / deactivates the autothrottle in-flight, arms AT on ground to activate when TLA is above 23° (approx. 68% $N_1$).

**XFR** – switches Flight Guidance between left pilot (Channel A) and right pilot (Channel B).

**EDM (Guarded)** – activates the Emergency Descent Mode.

**FLC –** autothrottle engages in THRUST mode (idle for descent or CLB thrust for climb), aircraft follows target speed with pitch.

**ALT –** altitude hold mode, when activated, aircraft maintains the altitude at the time it is pressed.

**VNAV** – activates VNAV mode, ("V" is added on FMA before the vertical mode) VNAV will level off at programmed constraints and intercepts descent PATH. It will not climb above or descent below the preselected altitude.

**ALT Rotary Selector** – used to preselect altitudes in 1000 ft intervals, push and turn for 100 ft intervals. Outer knob (ring) switches between FT and M.

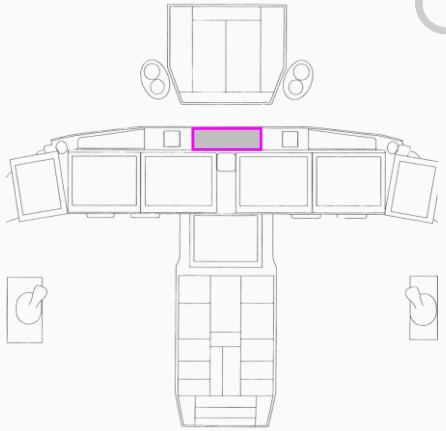
**VS** – engages the vertical speed mode, vertical speed is selected with the thumb wheel.

**FPA** – engages the Flight Path Angle mode, the angle is selected with the thumb wheel. FPA is the default vertical mode.

**OSPD / USPD** – overspeed and under speed protection, AT adjusts thrust, flight guidance appears to recover speed, FLC mode activates after recovery. AT and AP do not auto engage!

**WSHR –** windshear guidance.

**FD Switch**
Switches OFF Flight Directors (both) if Flight Guidance is coupled to the right (◄ XFR ►).



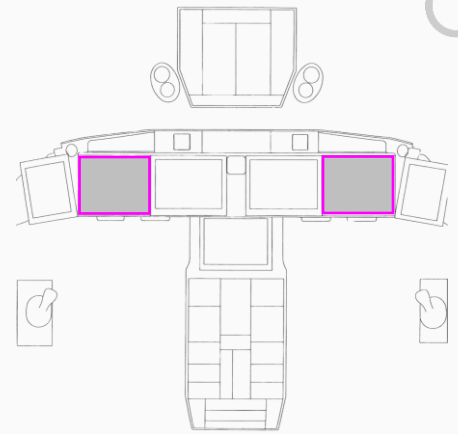
**Emergency Descent Mode (EDM)**

The Emergency Descent Mode activates automatically if above 25.000 ft when the cabin altitude exceeds 14.500 ft. It can also be manually selected above 25.000 ft. EDM provides the following functions:

- "EMERGENCY DESCENT" aural alert sounds.
- AP and AT engage (if not already engaged), thrust idle.
- Altitude is preselected to 15.000 ft (can be adjusted).
- HDG mode engages, and present heading is maintained (can be adjusted).
- Descent speed is set to $V_{MO}$ - 10 kts or $M_{MO}$ – 0.02M.
- Transponder is set to 7700 and seatbelts come on (only for automatic EDM activation).

However, speed brakes do NOT automatically extend.

# Primary Flight Display



Active IAS / MACH target, magenta for FMS, cyan for manual speed.

Speed tape, numbered every 10 kts below 200 kts, every 20 kts above 200 kts. Speed trend vector (green line) shows projected speed in next 10 sec with current rate.

Manual speed target bug – cyan (not active speed target in this example).

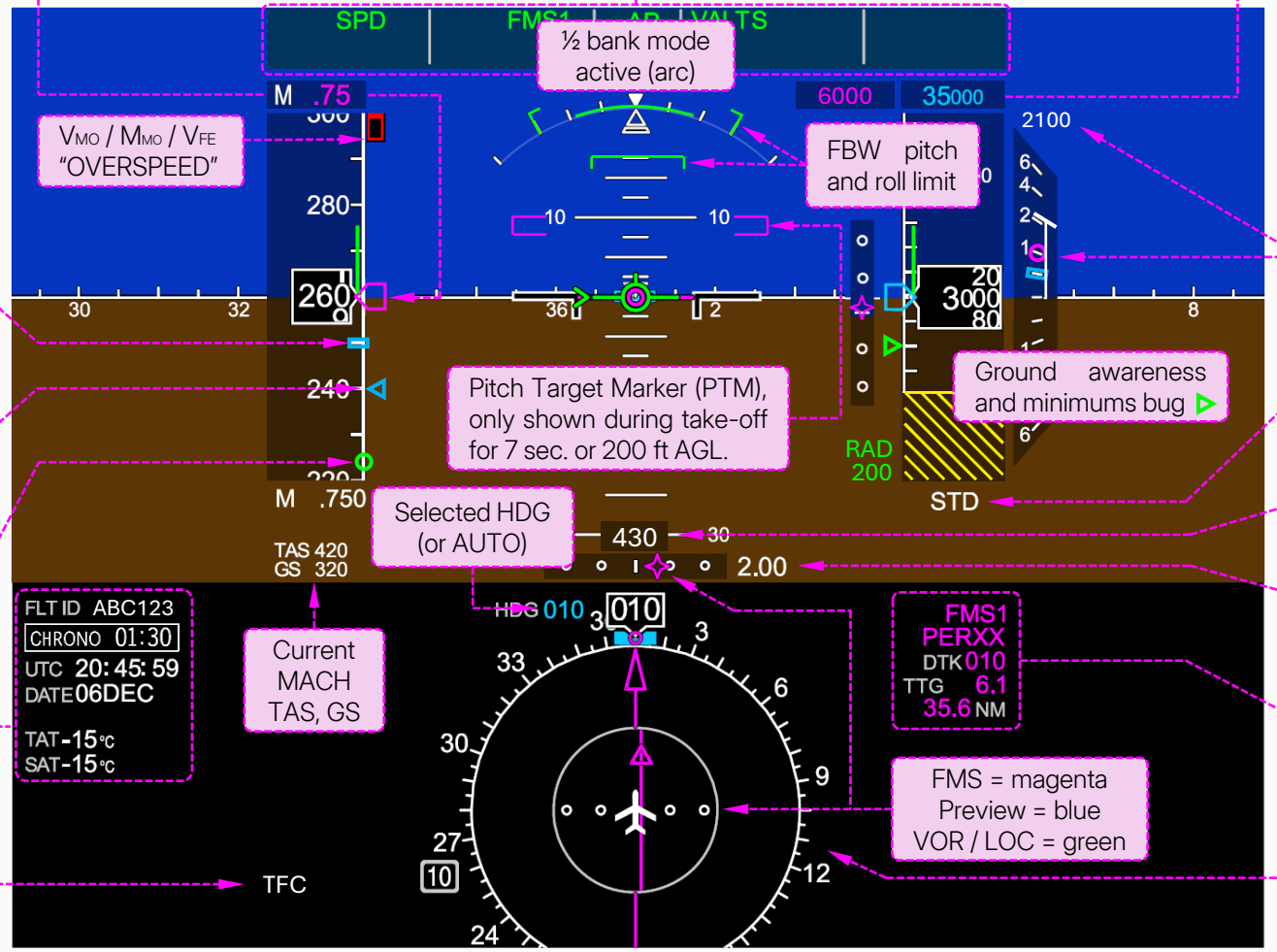
Speed trim bug (only shown in manual flight), speed the aircraft will fly without sidestick input. Adjusted with trim switch.

Best lift / drag ratio speed (green dot speed)

Flight number, timer (activated with CHRONO switch), UTC time and date, TAT and SAT.

Overlays displayed on HSI (under normal conditions only traffic).

**Flight Mode Annunciator (FMA)**
Speed | Lateral | AP/AT | Vertical | Source reversion (IRS / ADS)




Preselected FCP altitude (cyan), FMS constraint (magenta), also shown on the altitude tape as a magenta line.

Altitude tape is labelled every 500 ft. An altitude trend vector (green line) shows projected altitude in the next 10 seconds.

Current vertical rate (white), selected target (cyan), VNAV target (○).

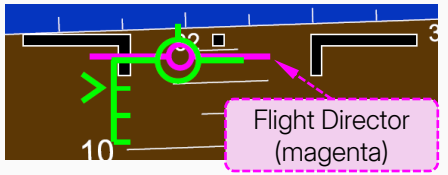
BARO, amber boxed if left and right different or passing transition altitude

Radio altimeter reading below 2500 ft AGL.

Cross track error, current RNP if nav source is FMS

Active waypoint box. Waypoint name, Desired Track (DTK), Time To Go (TTG) and distance.

HSI with active nav source CDI and bearing pointers (if selected). Range at bottom left.




**Flight Path Vector** (FPV) – (green) shows the actual path the aircraft is flying, top of the vertical tail on the horizon line equals a 3° path.
**Acceleration cue** showing energy trend (negative in the example).
**Speed error tape** with tick marks each 5 kts (in this example aircraft is 15 kts or more below target).

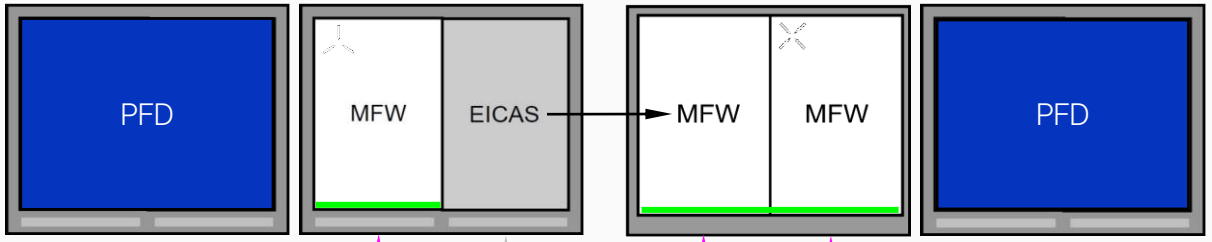
Below $V_{MIN\ TRIM}$ -3 kts
- Speed amber
- Single "SPEED" (inhibited < 30 ft AGL)

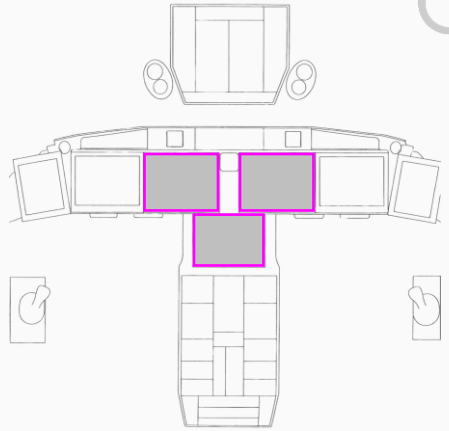
Below $V_{AOA\ SOFT}$
- Speed red
- "SPEED, SPEED, SPEED"
- Stick shaker activates

Below $V_{AOA\ HARD}$ - 5 kts
- "STALL, STALL, STALL"
- STALL on PFD
- Speed flashing red (inhibited < 30 ft AGL)

# Multifunction Displays (MFDs)

CCP Cursors cannot access PFDs and EICAS.

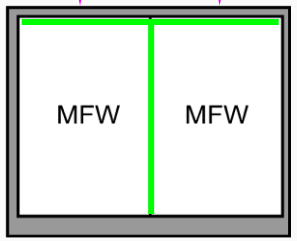


**Left Pilot Cursor**
Can be used on DU 2 and left side of DU 5.

**Right Pilot Cursor**
Can be used on DU 3, and right side of DU 5.

Left or right cursor can move over to the offside MFW on DU 5 but this will "bump" the onside cursor away (both cursors can never be on the same MFW). Right CCP can control DU2 in case of electrical emergency (only DU 1 and 2 available).




**Scratchpad, FMS quick access keys** (MSG, ROUTE, Direct-to, DEP / ARR) and **CNCL**, **EXEC**

**Arrow keys** can be used to move cursor or cycle between data entry fields.

**CAS** cycles through EICAS messages

**Double Stack Knob (DSK)**
Outer and inner wheel for data entry, push to enter.

Track ball

Data enter button (2)

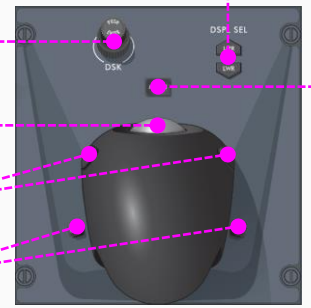
Push-to-talk button (2)

**Cursor Control Panel**
The CCP is used to move the cursor on the MFWs. Cursor is moved between DU 2/3 and DU 5 with the UPR and LWR button, or by moving the cursor to the top or bottom of the display, pause for 1 sec, then continue in the direction required (marked green here).

**Note:** EICAS is normally displayed on the right side of DU2 but can be swapped to left side of DU3 (explained later). EICAS cannot be displayed on DU5 except in case of multiple DU failure.

**CCP MENU**
Instead of using the CTP or MKP Quick Action Keys (QAKs), the cursor can be placed onto the MFW, pressing the MENU button on the CCP will open the MFW menu options (similar to QAKs).

Cursor Control Panel



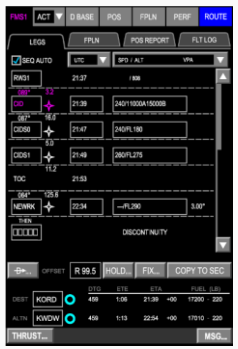
# Multifunction Windows (MFWs)

Database

Position

Flight Plan

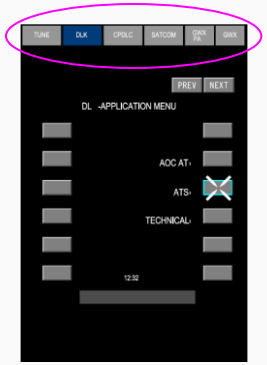
Performance

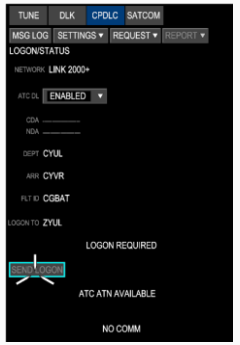

Route

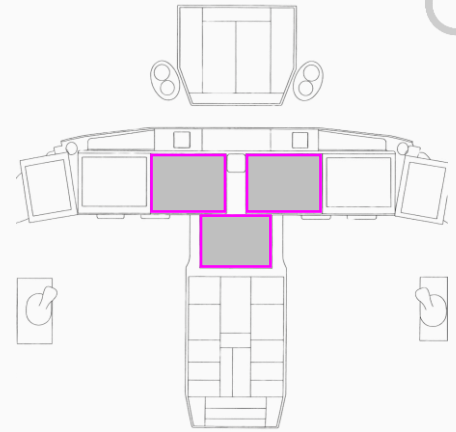
Optional

Tune

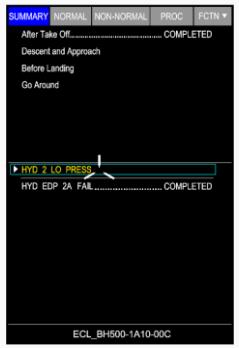
Tiles


ACARS


CPDLC



**Note:** to switch between MFW tiles, either select the tile with the cursor, or press QAK successively to scroll through different tiles. For example, pressing the SYN quick action key repeatedly cycles through aircraft synoptic pages.

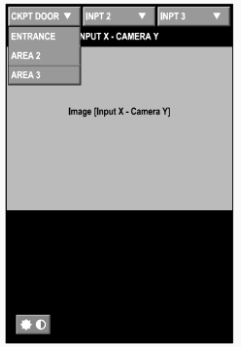


Checklist

MAP

MAP can be selected on two adjacent MFWs to display in full screen.

MAP | FMS | CNS
CHKL | SYN | DATA



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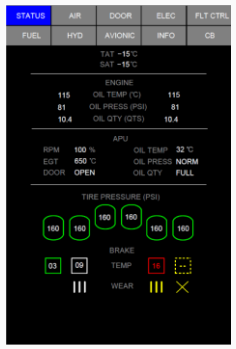
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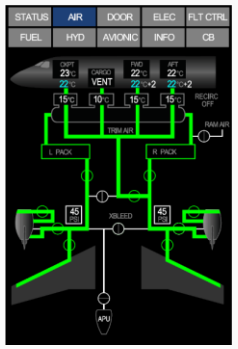
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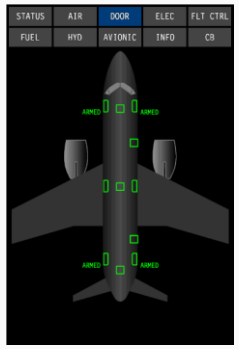
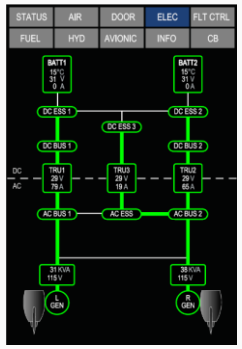

Doors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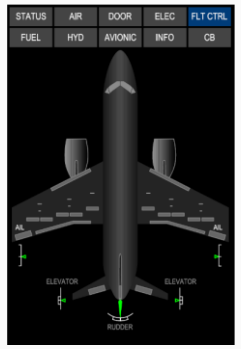
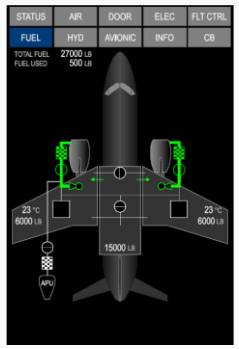
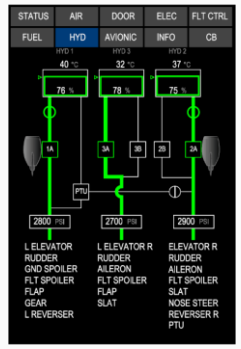
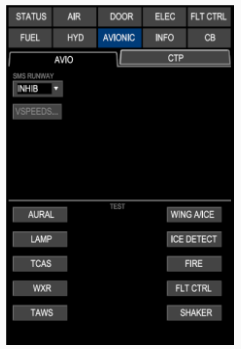
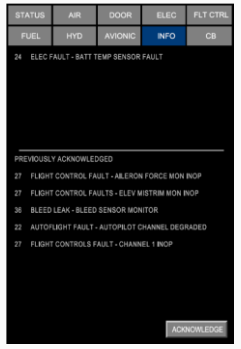
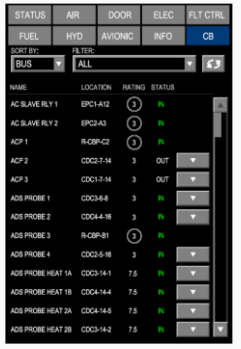

Flight Controls


Fuel


Hydraulics


Avionics


Info

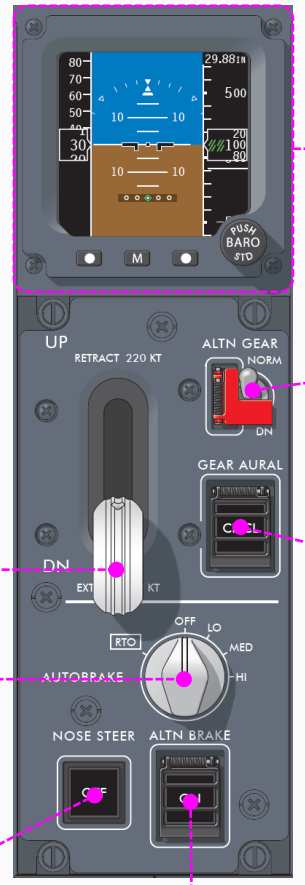

Circuit Breakers

# Center Panel

**Landing Gear Lever**
Extend (max 250 kts) and retract (max 220 kts). The lever must be pulled out before operating. On the ground, gear retraction is inhibited. Landing gear indications are removed from EICAS when gear is up for 30 sec (declutter).

**Autobrake Switch**
- OFF, autobrake is deactivated (can also be deactivated by pressing brake pedals during rejected take-off or landing rollout).
- RTO (Rejected Take-off), in case of rejected take-off above 70 kts, maximum deceleration is applied, below 70 kts, deceleration is reduced (2.44 m/s²). In case take-off is rejected below 60 kts, autobrake does not activate.
- LO – low rate of deceleration selected (1.83 m/s²).
- MED – medium rate of deceleration selected (2.74 m/s²).
- HI – maximum rate of deceleration.

**Nose Wheel Steering**
When selected OFF, disables the nosewheel steering system. Both the nose wheel steering tiller and rudder pedal steering is deactivated.




**Alternate Brake (Guarded)**
When pushed (ON), the BDCUs are bypassed and the brakes are controlled directly by the brake pedals (without brake protections such as anti-skid).

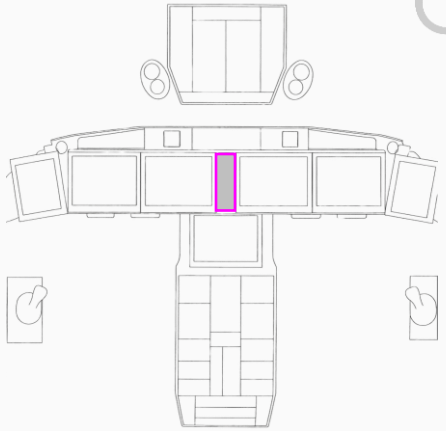
**Integrated Standby Instrument (ISI)**
Provides necessary information to fly the aircraft during emergency. It includes course guidance for approach LOC / GS (NAV 1 only).

**Alternate Gear Extension (guarded)**
NORM – alternate extension is off.
DN – the landing gear free-falls to the down and locked position.

**Gear Alert Cancel Switch (guarded)**
Allows the flight crew to mute (CNCL) the continuous "GEAR" aural alert if above 15.000 ft and all radio altimeters have failed.

It cannot be muted if:

- Any landing gear if not down and locked
- RAD ALT is below 1000 ft
- Rate of descent > 250 ft/min for 3 seconds
- SLAT / FLAP lever in landing position, and
- Both thrust levers below 17°, or
- Single thrust lever below 30° in case of one engine inoperative



**Note:** every 550 flight hours, the landing gear performs a dormancy check and uses the alternate gear extension system to lower the landing gear. Indicated by ALTN GEAR DN on EICAS, if a fault is detected the advisory GEAR FAULT will show. Fault can be reset on ground by selecting ALTN GEAR to DN for a few seconds, then return NORM.

# Pedestal

**Engine Run Switches**
ON - when START switch is in AUTO, command to initiate engine start is given to the EEC:
1. Starter Air Valve (SAV) opens
2. Igniters are energized
3. Fuel is injected at 18% - 20% $N_2$
4. SAV and igniters off at 51% $N_2$

OFF – engine shutdown command is given to the EEC.

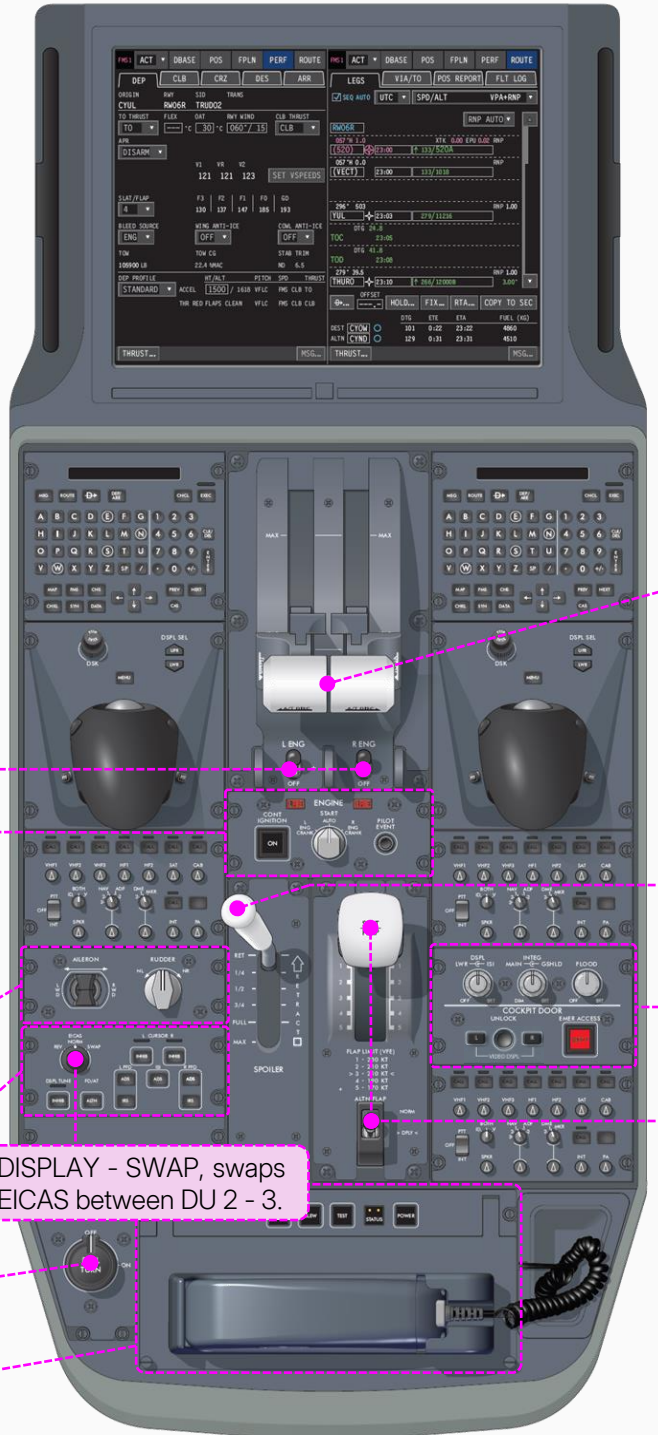
**Engine Panel**
CONTINUOUS IGNITION – provides continuous ignition if engines are running. If engines are not running, signals manual start request to EEC.
L-R FIRE lights – illuminates below corresponding engine switch if fire.
START switch – AUTO for auto start sequence, L-R ENG CRANK allows dry cranking after failed start attempt or manual start (CONT IGNITION ON).
PILOT EVENT – FDR data is marked for maintenance investigation.

AILERON and RUDDER trim. Aileron trim only available in DIRECT mode. Trim stops after 3 sec (double beep).

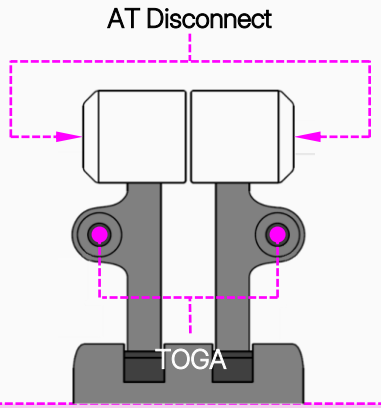
REVERSION panel – change FD/AT, ADS / IRS source, inhibit cursors and display tuning. Display reversion.

Parking brake switch (pull and turn).

Flightdeck printer and Interphone Handset



DISPLAY - SWAP, swaps EICAS between DU 2 - 3.

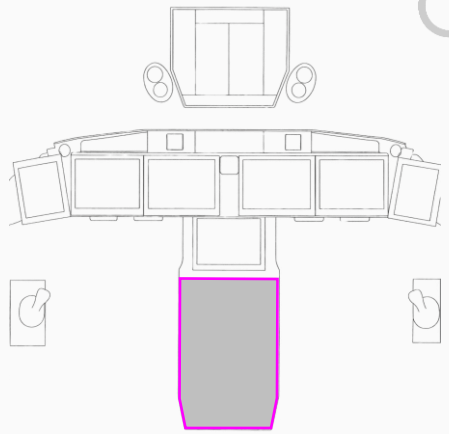



**Thrust Levers**
Stops at MAX thrust, IDLE and MAX REV. Thrust reverse is activated by raising the finger lifts and moving the thrust levers back. The thrust levers are mechanically held in the reverse idle position until reversers are deployed, similarly, when stowing, thrust levers are held at forward IDLE until reversers are stowed. Each thrust lever has an AT disconnect switch and a TOGA switch.

**Spoiler Lever**
Must be lifted to move from RET to FULL with friction detents at ¼, ½ and ¾. From FULL must be lifted again to move to MAX (only for emergency use). Ground spoilers deploy and retract automatically during rejected take-off and landing, there is no arm position.

**SLAT / FLAP Lever**
Six positions from 0 (UP) to 5. A release lever must be lifted before the flap lever can be moved. There are gates at the F4 position (go-around slats / flaps for a F5 landing), and the F2 position (go-around slats / flaps for a F4 landing).

**Alternate Flap Deploy (Guarded)**
If moved from NORM to DPLY, will bypass the SLAT/FLAP lever and deploy slats / flaps to position 3 (used if slat / flap lever jam or failed).



**Note:** Multipurpose Keyboard Panel (MKP) and Cursor Control Panel (CCP) previously discussed.

**Note:** three identical Audio Control Panels (ACPs) are installed (L, R and observer), see next page.

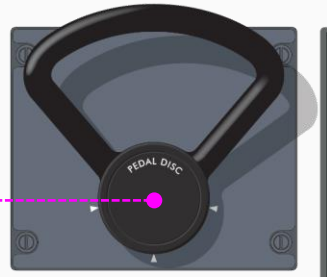
**Lighting Control Panel (top)**
1. DU 5 and ISI brightness.
2. Main panel and Glareshield.
3. Flood lighting brightness.

**Cockpit Door Panel (bottom)**
1. L – R will display cockpit door camera on either the left or right side of DU 5.
2. UNLOCK, door unlocks for 5 seconds.
3. DENY will deny access and cancel the "COCKPIT DOOR" aural alert after an emergency access request. If ignored, cockpit door will unlock after a time delay determined by the operator for 5 seconds.

# Side Consoles

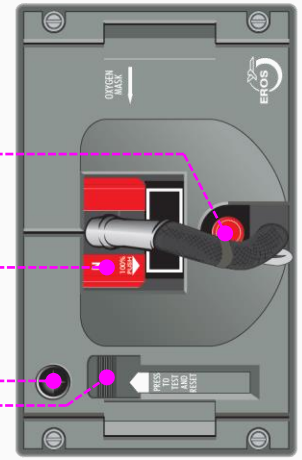
### Nosewheel Steering Tiller

Installed on the left side console (both sides optional). The steering tiller commands the nosewheel up to 80° left / right, at higher speeds maximum deflection is reduced. Rudder pedals command the nosewheel up to 9° left / right, when rudder pedals and steering tiller are moved simultaneously, the commands are algebraically added. The center PEDAL DISC button disconnects the pedal steering to allow for flight control checks without moving the nosewheel.




### Oxygen Mask

The oxygen mask has a built-in smoke goggle system and is standard in design. It has an oxygen flow indicator (blinker), TEST / RESET button to allow for checks without removing the mask from its container. A N / 100% rocker for selection of mixed or 100% oxygen and an emergency flow knob to supply 100% oxygen under pressure in case of smoke or fumes. The mask has a built-in microphone that activates automatically. To deactivate the oxygen mask microphone after use, close the left container door and press TEST / RESET switch.



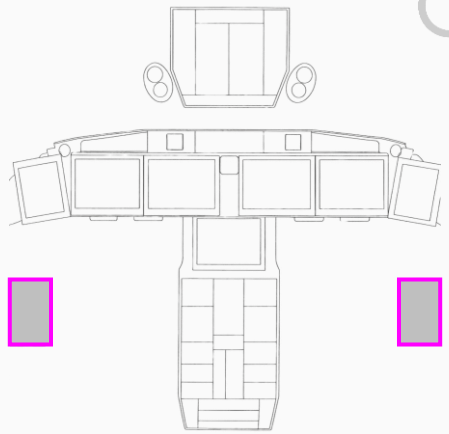
### Sidestick Controller (SSC)

Two separate sidesticks are used for manual pitch and roll control, they are not interconnected. Each sidestick has:

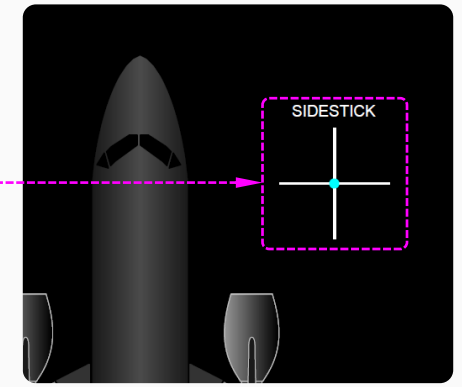
- A pitch trim rocker switch to control the speed trim bug in FBW normal mode, in direct mode the trim switch directly commands the stabilizer.
- An autopilot disconnect / priority switch (AP / PTY), pressed once disconnects the AP, pressed again cancels the AP disconnect warning. Press and hold for sidestick priority, the opposite sidestick inputs are ignored ("PRIORITY LEFT / RIGHT").
- An intercom and radio Push-To-Talk switch (INT / PTT) (trigger).

When both sidesticks are moved simultaneously (without taking priority) the signals are added algebraically, DUAL is shown on both PFDs and "DUAL INPUT" aural alert is heard.

On ground before engine start, sidestick input can be checked on the flight controls synoptic page.



**Note:** audio Control Panels (3) are located on the pedestal, added here due to lack of space.

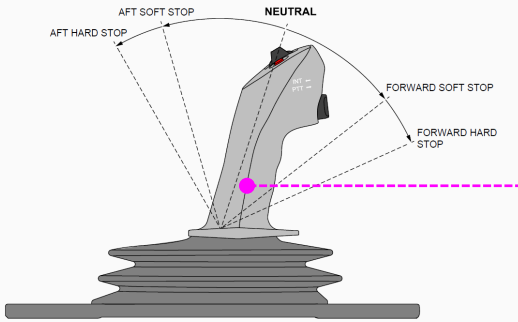


Hold 3 sec for EMER call

PA PTT

### Audio Control Panel (Pedestal)

- CALL buttons, when selected connects the microphone, a green bar illuminates (only one can be selected).
- For incoming calls, CALL button flashes. Pressing the CAB CALL button performs a normal call, pressing CALL for 3 sec performs emergency cabin call (three chimes).
- Volume knobs need to be unlatched (white band visible) for source to be audible, volume can then be adjusted.
- The button above PA is functioning as a PTT switch (hold to transmit), for PA using handheld mic, press both PTTs.

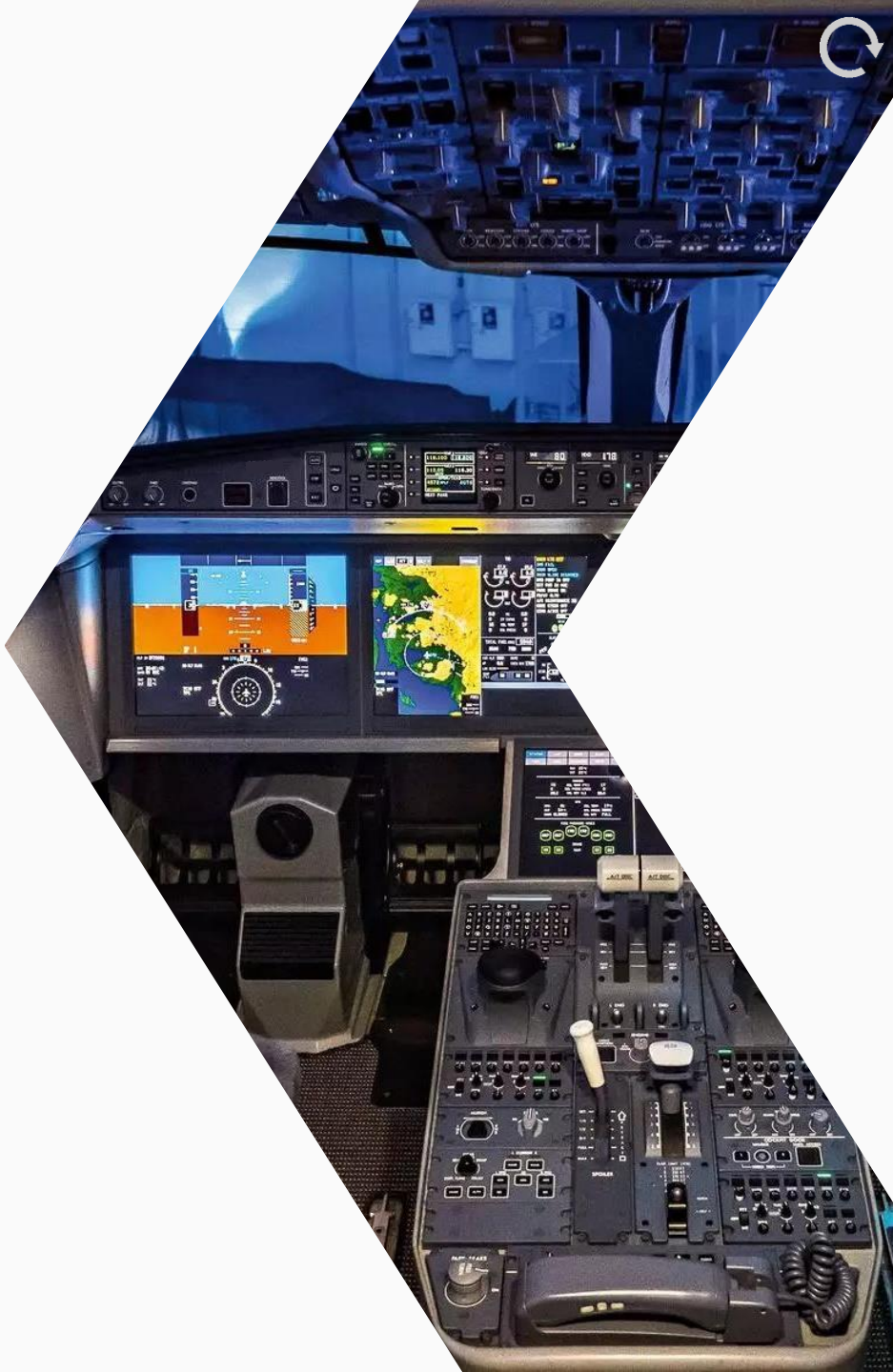



### Soft-stop and Hard-stop

The sidestick has a "soft stop" position (only on pitch axis) that can be overruled with some force to the mechanical "hard stop".

This allows for limited (safe) excursions outside of the normal flight envelope in an emergency.

# Air Conditioning
# Bleed Air and Pressurization



# Bleed Air System

Bleed air for air conditioning and pressurization is controlled by the Integrated Air Management System (IAMS) consisting of two dual channel Integrated Air System Controllers (IASCs). They manage, control and monitor the following:

- Bleed air system
- Bleed Air Leak and Overheat Detection System (BALODS)
- Airconditioning system
- Avionics cooling and heat extraction system
- Pressurization system
- Wing anti-ice system

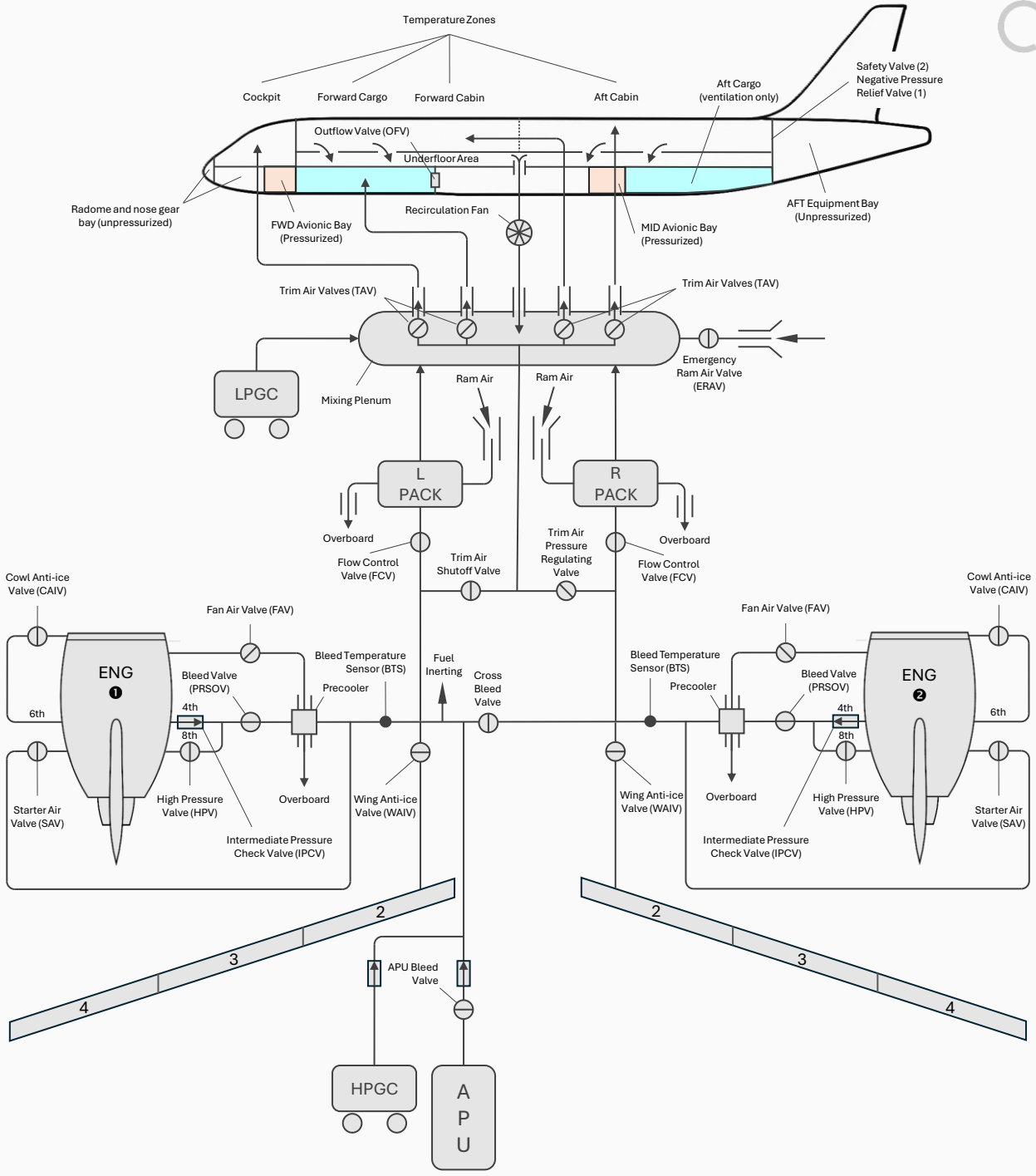
Additionally, bleed air is used for engine cowl anti-ice, engine start and fuel inerting but these functions are not controlled by the IASCs.

Bleed air can be provided by the following (kept between 38 and 45 psi):

- Engine bleed air from 4th (intermediate pressure) or 8th (high pressure) compressor stage. Cowl anti-ice uses 6th compressor stage bleed air.
- External air source – high pressure (unconditioned) air via the High-Pressure Ground Connection (HPGC) or low pressure (conditioned) air via the Low-Pressure Ground Connection (LPGC) directly into the manifold.
- Auxiliary Power Unit.

When APU and engine bleed sources are both available, the IASC will prioritize engine bleed if:

- Wing anti-ice is ON, or
- Aircraft is in clean configuration, or
- Altitude is above 16.500 ft.



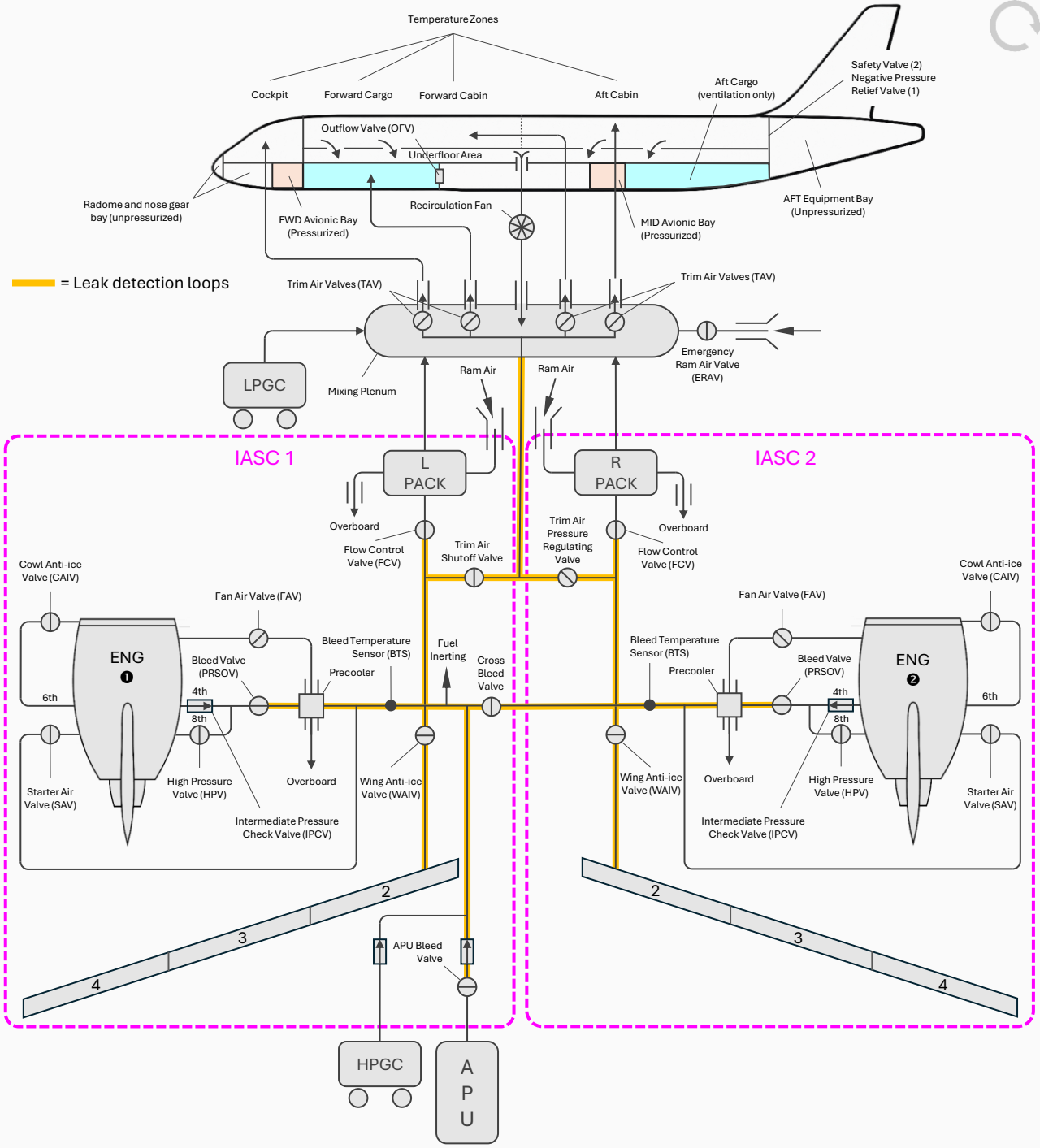
# Control and Leak Detection

Each IASC has 2 channels, Channel A, Channel B and a safety channel. Channel A and Channel B alternate control daily. In case of a complete IASC failure (both channels), the remaining IASC can take over offside IASC functions except the offside engine bleed and offside air-conditioning pack. The offside bleed valve will close together with the Fan Air Valve (FAV) and High-Pressure Valve (HPV). Also, the pack Flow Control Valve (FCV) will close.

Dual independent detection loops detect overheat and leaks. Loops are monitored by the B channels of each IASC. IASC 1B monitors Loop A, IASC 2B monitors Loop B. Both loops must detect a leak to trigger an alert (or one loop if the other is sensed as faulty). If both B channels fail, there is no leak detection, and the entire bleed system shuts down (emergency ram air available).

The following leak detection zones and isolation logic exists:

- L BLEED MANIFOLD (closes left bleed, APU bleed and crossbleed)
- R BLEED MANIFOLD (closes right bleed, APU bleed and crossbleed)
- APU MANIFOLD (closes APU bleed and crossbleed)
- L PACK MANIFOLD (closes left pack flow control valve)
- R PACK MANIFOLD (closes right pack flow control valve)
- TRIM AIR MANIFOLD (closes trim air pressure regulating and shut-off valves)
- L WING ANTI-ICE MANIFOLD (closes left wing anti-ice valve)
- R WING ANTI-ICE MANIFOLD (closes right wing anti-ice valve)



# Temperature Control

Pack airflow is regulated by the IASC based on the number of PAX entered into the FMS, if nothing is entered the IASC assumes cabin is full. High flow is automatically selected when APU is operating on ground depending on duct temperature.

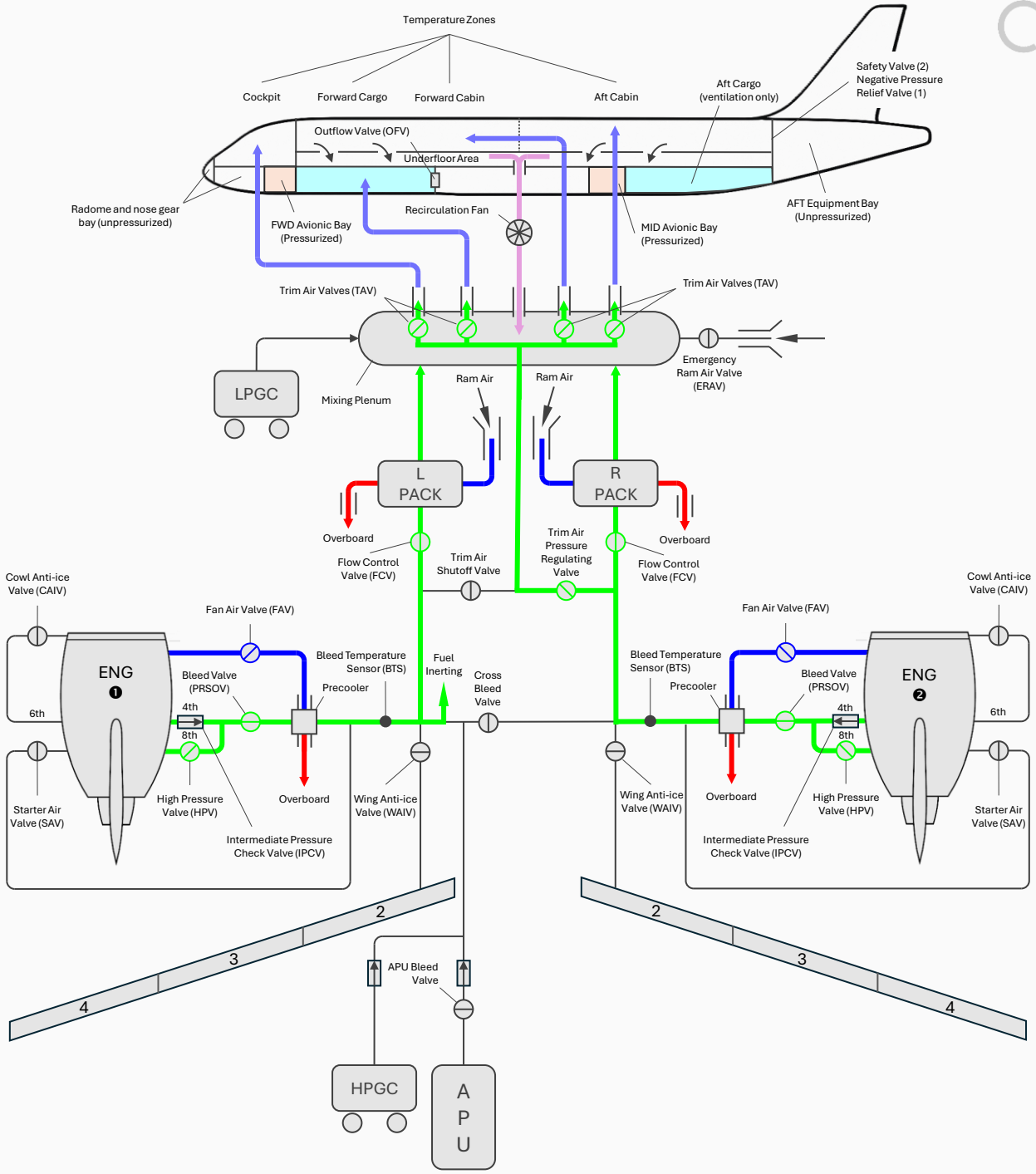
Trim air provides hot bleed air through 4 Trim Air Valves (TAVs) for: flight compartment, forward cargo, forward cabin and aft cabin. Aft cargo is not heated. In AUTO, zone temperature for cockpit and forward / aft cabin is selectable between 18ºC to 30ºC, center position is 24ºC, in MAN TEMP the temperature switches select duct temperature instead of zone temperature

Forward cargo can be set to LO HEAT (15ºC to 20ºC), HI HEAT (20ºC to 25ºC) or VENT only. Each cargo compartment has 2 shutoff valves, one inflow and one outflow which are closed with the respective CARGO FIRE switch.

For aircraft equipped with supplemental avionic cooling, the FWD avionic bay is cooled with air from underfloor and FWD galley via a filter, heat exchanger and supply fan. MID avionic bay is cooled via underfloor, AFT galley and a GROUND VALVE (on ground if temperature is between 3ºC and 30ºC until doors closed, reopens 1 min after landing). Avionic cooling is controlled by the INLET switch.

**Note:** some aircraft do not have a supplemental avionic cooling system (and therefore no GROUND VALVE). On those aircraft the INLET switch controls the aft galley and lavatory air extraction only.

Heat extraction from the FWD and MID equipment bays, flight deck displays, flight deck oxygen bottle and FWD / AFT galley is via 2 fans. On ground, the extracted air goes underfloor. In-flight or in case of fan failure, two Avionic Exhaust Valves (AEVs) open, cabin differential pressure forces air out. MID equipment bay has a back-up fan for ground use.



# Pressurization System

A single outflow valve is located on the aft pressure bulkhead of the forward cargo compartment. The OFV is controlled by IASC A channels in automatic mode (one active, one stand-by) or B channels in manual mode. To change the IACS channel, cycle AUTO PRESS switch from AUTO to MAN, back to AUTO.

Two safety valves and a negative pressure release valve are located on the cabin aft pressure bulkhead.

- Max positive differential pressure = 8.8 psid
- Max negative differential pressure = -0.5 psid
- Max differential pressure for take-off = 0.11 psid
- Max differential pressure for landing = 1.00 psid

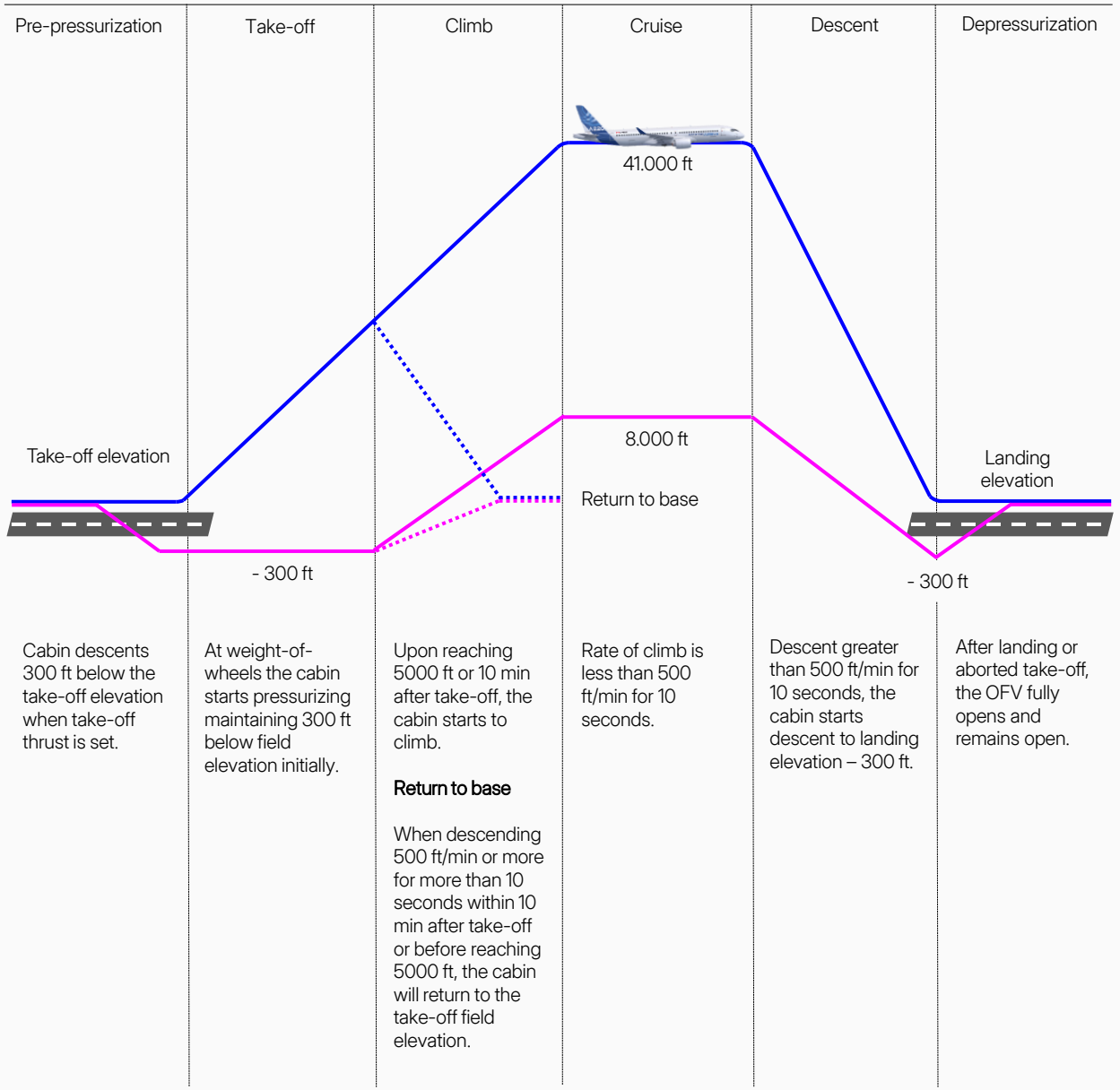
Four Pressure Equalizing Valves (PEVs) prevent pressure differences between the cabin and cargo compartments. Large PEVs prevent cabin pressurization when cargo door is not closed, small PEVs equalize pressure during fast climbs.

Emergency depressurization will depressurize the aircraft to 15.000 ft cabin altitude or aircraft altitude if lower. In automatic mode, the rate is limited to 2500 ft/min, in manual mode the rate is limited to 3500 ft/min.

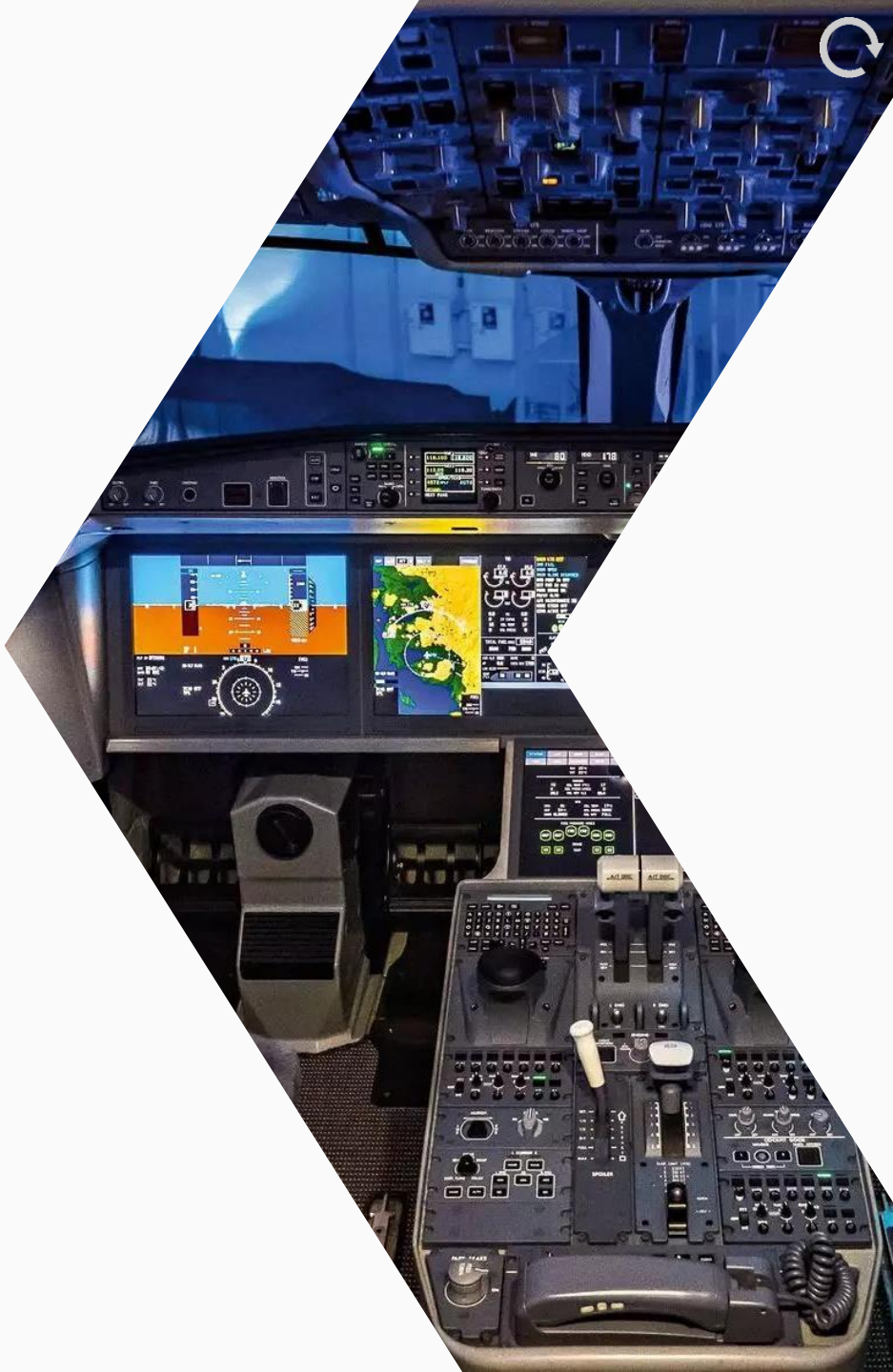
DITCHING closes all valves below the water line except for the RAM AIR valve:

- Avionics Exhaust Valves (AEVs)
- Flow Control Valves (FCVs)
- Trim Air Pressure Regulating Valve (TAPRV)
- Bulkhead check valve
- Outflow valve (closed last)

CABIN ALT (caution) will show when cabin altitude exceeds 8500 ft.
CABIN ALT (warning) will show when cabin altitude exceeds 10.000 ft.



# Automatic Flight



# Automatic Flight Control System

The Automatic Flight Control System (AFCS) consists of the following:

- Flight Guidance (FG)
- Autopilot (AP) – integrated within each PFCC, only available in normal mode
- Autoland (AL) – integrated within each PFCC, only available in normal mode
- Autothrottle (AT)

The aircraft has two AFCS systems hosted within the DCU Module Cabinets (DMC). DMC 1 hosts AFCS 1 and DMC 2 hosts AFCS2. One AFCS is in use based on an odd-even day cycle, in case of AFCS failure the other AFCS will automatically assume control. The flight crew can also manually change the AFCS in use by pressing the FD/AT – ALTN pushbutton on the reversion panel.

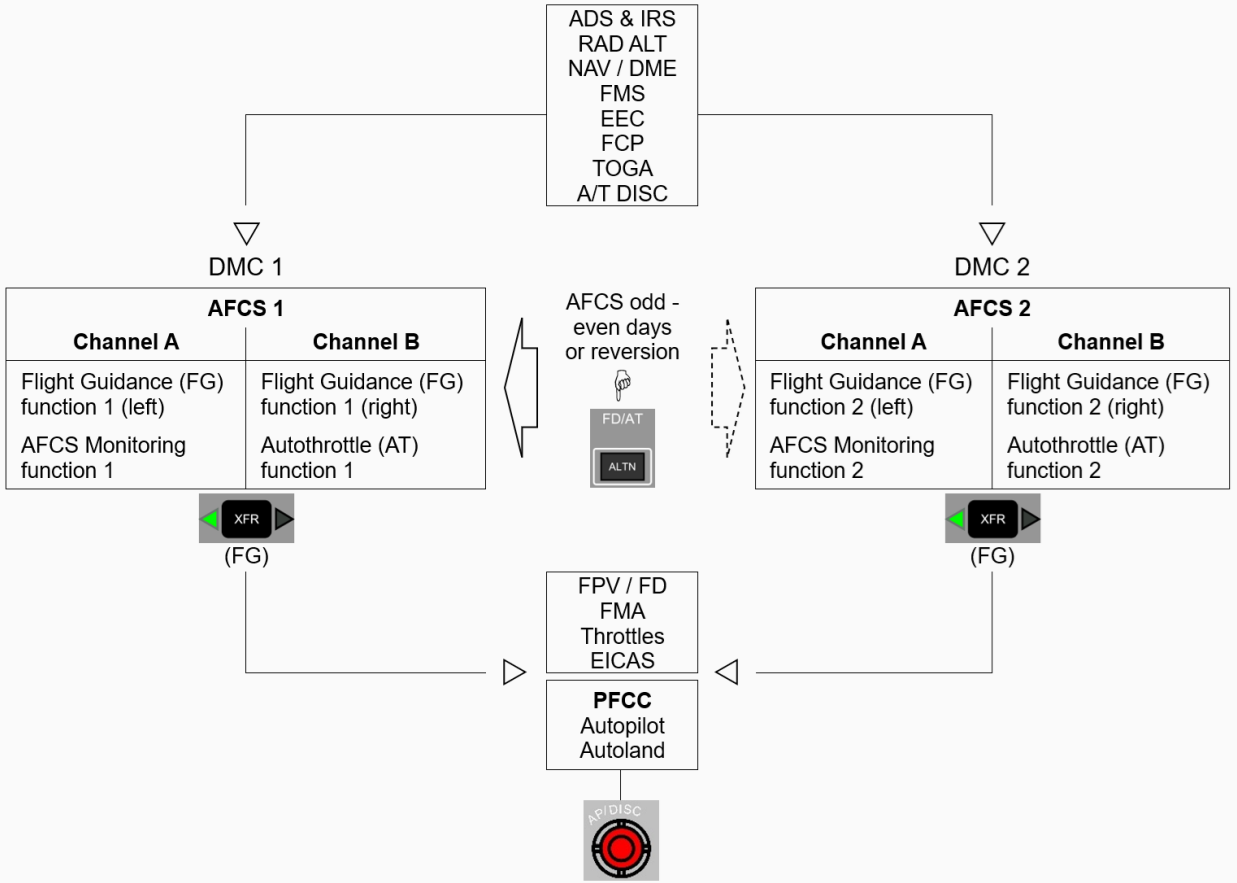
Each AFCS has 2 channels, Flight Guidance (FG) of Channel A is in use when the transfer switch is set to the left pilot, FG of Channel B is in use when the transfer switch is set to the right pilot. During Autoland, flight guidance is transferred from the AFCS to the PFCCs directly (AP switch on FCP inhibited).

Autopilot disengagement (automatic):

- Reversion of FBW to DIRECT mode
- Invalid data or AP monitor fault detected
- Stick shaker activation (except in WINDSHEAR guidance mode)
- Below 50 ft AGL in APPR 1 and APPR 2 (no Autoland)

Autopilot disengagement (manual):

- Pressing AP / PTY switch on the sidestick (press and hold for priority)
- Pressing the AP button on the FCP (inhibited during ILS approach)
- Moving sidestick (unless sidestick disabled due to latched priority)
- Moving rudder or steering tiller during ROLLOUT mode (autoland)



# Lateral and Vertical Modes

## Lateral Modes

HDG is the default lateral mode, ROLL mode is selected if PFCC cannot maintain heading or if below 200 ft. Heading synchronization is automatic when:

- In TO lateral mode and the aircraft is on ground, or
- Upon activation of the go-around (GA) mode, or
- Reverting to basic mode (e.g. end of FMS route).

Pressing NAV mode while FMS is the active NAV source, will command an intercept heading, even if the aircraft is moving away from but within 2.5 nm of the desired track.

Flight guidance automatically switches to half bank mode (15°) above 31.500 ft. Half bank will be active in both HDG and NAV modes. When manually selecting ½ bank mode below 31.500 ft, ½ bank will only be active in the HDG mode.

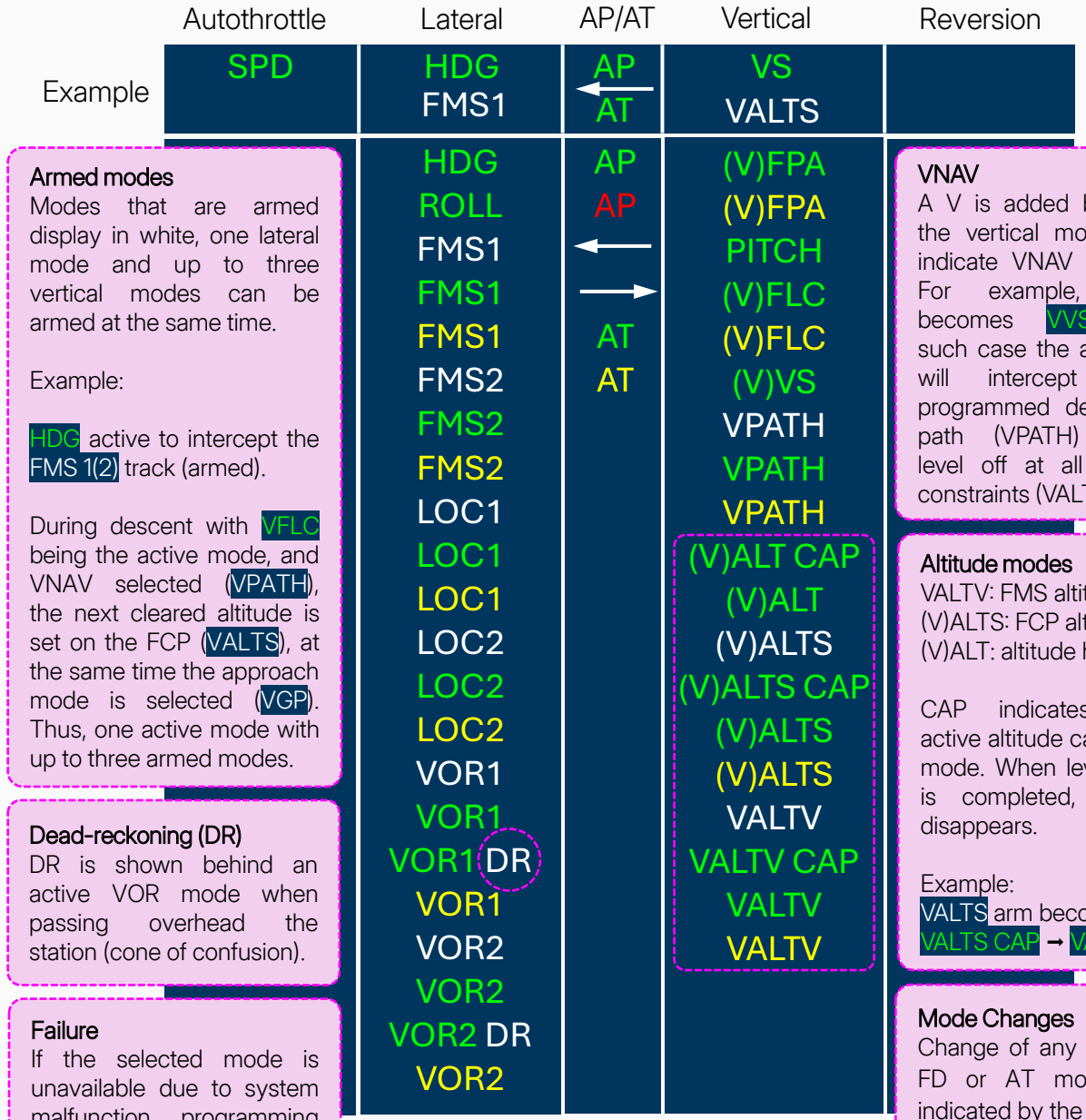
## Vertical Modes

FPA is the default vertical mode and can be selected from +9.9 to -9.9 (0.1 increment), the resulting pitch is limited to 15°.

A full-scale deflection on the VNAV vertical deviation scale represents 500 ft (250 ft each dot) or 150 ft (75 ft each dot) on approach.

Vertical Track Alerts (VTA) appear on the PFD, 60 seconds prior to a vertical change, then flash 5 seconds before with a double C-chord aural alert. FMS altitude constraints are indicated by VNAV FMS Altitude (VALTV) mode.

When track angle exceeds 75° or cross track error exceeds 10 nm, the route becomes undefined, VPATH (if active) changes to VFPA and re-arms with track error less than 75° and cross track error is less than 10 nm.



# Take-off and Go-around Mode

### Take-off (TO) and Go-around (GA)

When TOGA is pressed on ground, TO is the active lateral and vertical mode (or VTO using VNAV), heading synchronizes to current heading. A magenta Pitch Target Marker (PTM) is displayed, the PTM is based on selected V-speeds and weight. It is automatically adjusted in case of engine failure.

After take-off, TO mode maintains the runway heading, it does not follow the HDG bug, select any lateral mode (e.g. HDG) when passing 400 ft, if FMS was armed on ground, it will engage automatically passing 200 ft AGL.

For the FMS DEP PROFILE (automatic acceleration and thrust reduction) to function, NAV SRC and speed must be set to FMS and TOGA + VNAV must be selected before take-off. If not, no automatic change from TO to FLC mode will occur. In such case FLC mode must be manually selected at acceleration altitude, thrust reduction from TO to CLB thrust will occur when flaps are up.

In the go-around mode, heading at go-around is maintained to 400 ft RA, autopilot remains engaged, autothrottle engages if not already engaged. During autoland, pressing TOGA when on ground for less than 2 seconds will trigger an automatic go-around but the pilot will need to take control of the rudder. When on ground for more than 2 seconds, TOGA is inhibited.

Speed for go-around is $V_{GA}$ for AEO or $V_{AC}$ for OEI. NAV-to-NAV transfer back to FMS (in case of localizer-based approach) automatically occurs when passing 400 ft AGL to follow the programmed missed approach procedure.

One-engine Inoperative (OEI) guidance mode is active during take-off and go-around with an engine failure or opposite engine low thrust event until slats / flaps are up. Guidance (beta indicator) provides optimum rate of climb. When slats / flaps are up, the magenta slip indicator becomes white again, the green filled trapezoid displayed under the FPV remains displayed as a open trapezoid.

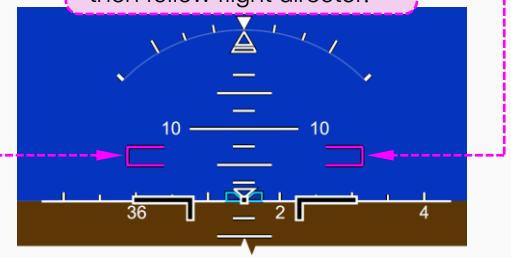
| | Autothrottle | Lateral | AP/AT | Vertical | Reversion |
|---|---|---|---|---|---|
| Example | SPD | HDG<br>FMS1 | AP<br>AT | VS<br>VALTS | |
| | THRUST<br>HOLD | TO<br>GA | | (V)TO<br>(V)GA | |

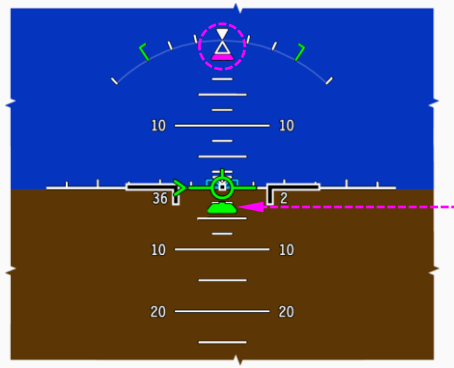
**HOLD Mode**
Autothrottle HOLD mode is active during take-off between 60 kts and 400 ft AGL. During this period, autothrottle will not command the thrust levers.

**Note:** Should windshear be encountered during take-off below 400 ft AGL, the autothrottle will NOT command full thrust while in HOLD mode.

**Take-off / Go-around**
Pressing either TOGA switch on the thrust levers while on ground activates TO mode, in-flight activates GA mode (or WSHR if windshear detected). Present HDG is synched and active. PTM will appear to indicate initial pitch required.

**Pitch Target Markers (PTM)**
Move airplane symbol (wings) into the brackets, then follow flight director.




**Beta indicator (flaps extended)**
Keep centered below FPV

# Overspeed, Underspeed, Windshear and Emergency Descent

**Overspeed & Underspeed**
Overspeed (OSPD) and underspeed (USPD) modes are available in the FBW normal and PFCC direct modes:

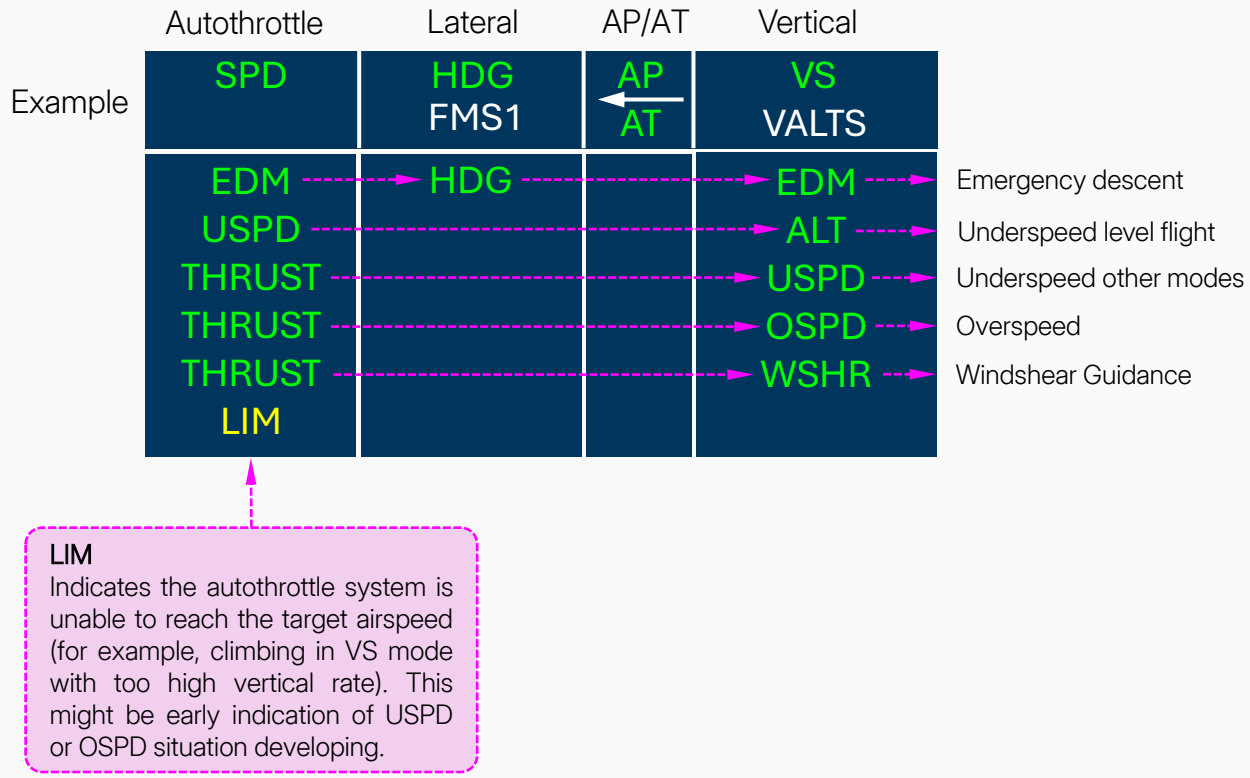
- OSPD: autothrottle (if engaged) reduces thrust until below $V_{MAX}$, flight director appears to provide pitch guidance. Below $V_{MAX}$ OSPD is replaced by FLC.
- USPD: in any altitude or altitude capture mode, the autothrottle (if engaged) increases thrust to max. In all other modes the flight director appears and provides pitch guidance. When above $V_{MIN\ TRIM}$ USPD is replaced by FLC.
- In both cases: The autopilot and autothrottle will not automatically engage.

**Pre-Avionic Built 6.5:** should an underspeed (USPD) occur on approach, VGP will no longer arm. First reselect VNAV, then deselect and reselect APPR mode.
**Post-Avionic Built 6.5:** should an underspeed (USPD) occur, reselect VNAV.

**Windshear Escape Guidance (WSHR)**
Windshear escape guidance (WSHR) provides vertical guidance to escape windshear detected by TAWS. Pressing TOGA while windshear alert is active activates WSHR escape guidance mode. The vertical guidance is a blend of airspeed, radio altitude and angle-of-attack. Once windshear is no longer encountered, change vertical mode, WSHR mode disregards selected altitudes.

**Emergency Descent Mode (EDM)**
When activated, initiates a high-speed descent to 15.000 ft using FLC and HDG mode. It is automatically activated should cabin altitude exceed 14.500 ft while the aircraft is above 25.000 ft. Above 25.000 ft it can also be manually selected.

- "EMERGENCY DESCENT" aural alert
- AP and AT engage (if not already), thrust flight idle
- Preselected altitude changes to 15.000 ft
- HDG mode and FLC mode active
- Descent speed $V_{MO}$ – 10 kts or $M_{MO}$ – 0.02M
- Transponder 7700, seatbelts on (auto EDM only)
- At 15.000 ft, changes to ALT and HDG mode, speed 250 kts
- EDM is cancelled with AP DISC or by pressing EDM switch
- Spoilers are not automatically deployed!



# Approach Modes

## Approach

NAV Preview is displayed when FMS is the active nav source within approximately 31 nm of the station and the FMS has autotuned the approach frequency. The navigation source automatically transfers (NAV-to-NAV transfer) from FMS to NAV when:

- In FMS lateral mode, and
- APPR mode selected, or
- Approach course captured.

Lateral mode automatically changes from APPR FMS1 (FMS2) to APPR LOC1 (LOC2).

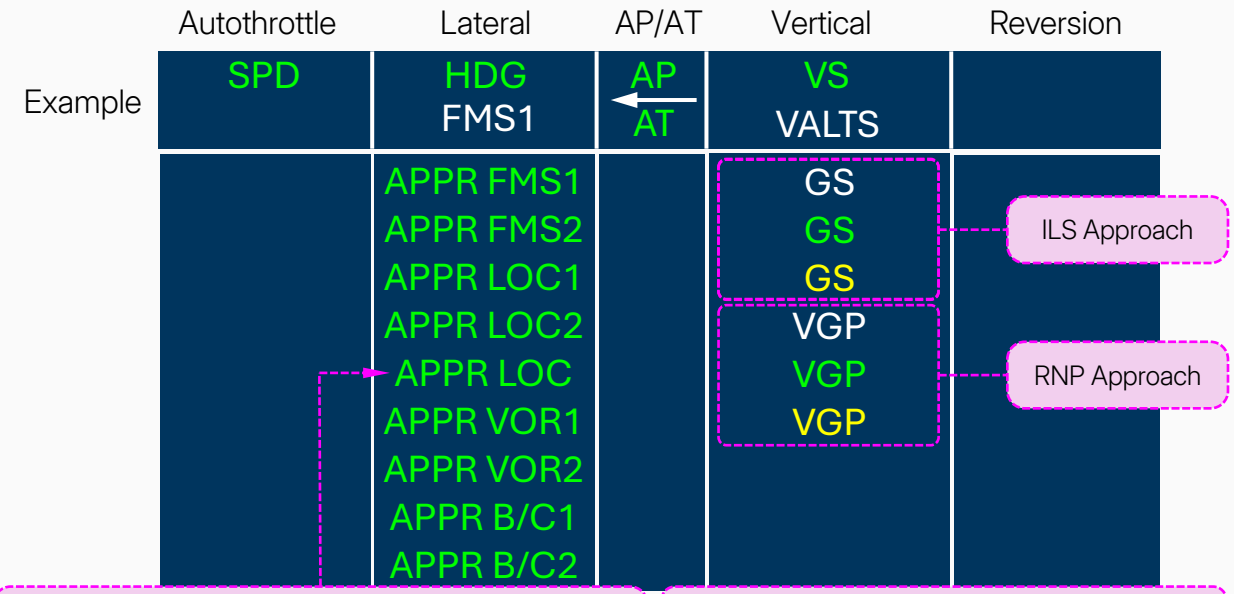
Approach Status Annunciator (ASA):

- ILS approach capability – APPR 1 (CAT I) or APPR 2 (CAT II) – no Autoland.
- Autoland capability – LAND 2 (fail passive) or LAND 3 (fail operational).
- Steep approach – STEEP (up to 5.5° available on A220-100 only).
- For all non-localizer-based approaches the ASA remains blank.

If slats / flaps are not in landing position (4 or 5) or flight spoilers are deployed, APPR 1 is displayed until correct configuration is achieved. If the correct landing configuration is not reached before 800 ft AAE, APPR 1 will remain displayed (no up-mode). Flying manually, APPR 2 is displayed with LAND 2/3 NOT AVAIL advisory until autopilot is ON.

Change of approach capability above 200 ft RA is indicated by a flashing amber ASA indication, e.g. NO APPR 2 in amber, then APPR 1 in green (or blank if all approach capability is lost). Below 200 ft RA this is indicated by a flashing red indication, e.g. NO APPR 2, then APPR 1 green (or blank if all approach capability is lost).

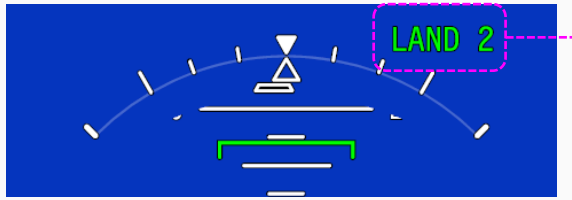
VGP mode is armed when APPR is pressed on the FCP, FMS is the active NAV source, a non-localizer-based approach is selected in the FMS, predictive RAIM is available, within the terminal area (30 nm) and FMS is not in Dead Reckoning (DR) mode. Amber VGP is displayed on FMA if criteria are not met. VGP behaves just like the ILS GS mode.




On an ILS approach, Flight Guidance is transferred from AFCS to PFCCs at 2000 ft above ground level, APPR LOC 1(2) changes to APPR LOC (without 1 or 2).

**Note:** All lateral approach modes have armed (white) and fail (amber) modes, not included here for clarity.

VOR and NDB approaches can be flown using FMS for lateral guidance if an FMS overlay is available. VNAV needs to be deselected latest at the FAF and FPA used as vertical mode. LNAV only approach can be flown like any other RNP approach (with VGP)




Approach Status Annunciator (ASA)

Within 31 nm from the airport, the FMS will autotune the approach frequency and course, a "preview" is shown in blue. When approach mode is active, the FMS will automatically transfer navigation source from FMS to NAV (magenta to green). Do not change NAV SRC manually as this will inhibit NAV-to-NAV transfer back to FMS in case of a go-around.

# Approach Modes

## Autoland

At the conclusion of an ILS approach, Autoland (AL) functions are:

- Approach tracking
- Runway alignment de-crabbing during crosswind conditions
- Landing flare and ground steering during landing rollout

Below 2000 ft AGL, flight guidance is transferred from DMC to the PFCC for precision approach guidance, indicated by a green APPR LOC (no 1 or 2) lateral mode on the FMA.

Between 1500 ft and 800 ft AAE, while in landing configuration with AP engaged, the highest autoland capability is shown on the ASA (LAND 2 or 3). The AL function is independent from the AP and embedded in each PFCC.

Approach Status Annunciator:

- NO AUTOLAND – loss of AL capability < 200 ft
- NO APPR 2 – loss of CAT II capability < 200 ft
- NO LAND 3 – loss of CAT IIIB capability > 200 ft
- NO AUTOLAND – loss of AL capability > 200 ft
- NO APPR 2 – loss of CAT II capability > 200 ft
- NO APPR 1 – loss of ILS capability > 200 ft
- STEEP – steep appr armed
- STEEP – steep appr active
- APPR 1 – CAT I capable
- APPR 2 – CAT II capable
- LAND 2 – CAT IIIA capable
- LAND 3 – CAT IIIB capable

| | | | |
|---|---|---|---|
| Above 2000 ft AGL | SPD \| APPR LOC2 \| AP AT \| VALTS GS | 200 ft 150 ft AAE | SPD RETARD \| ALIGN \| AP AT \| GS FLARE |
| Below 2000 ft AGL | SPD \| APPR LOC \| AP AT \| GS | Below 65 ft AGL | SPD RETARD \| ALIGN ROLLOUT \| AP AT \| FLARE |
| 1500 ft 800 ft AAE | LAND 2 \| LAND 3 | Below 20 ft AGL | RETARD \| ALIGN ROLLOUT \| AP AT \| FLARE |
| Below 400 ft AAE | SPD RETARD \| APPR LOC \| AP AT \| GS | 2s after WOW | ROLLOUT |
| 350 ft 250 ft AAE | SPD RETARD \| APPR LOC ALIGN \| AP AT \| GS FLARE | Below 30 kts GS | ROLLOUT command bar removed. NWS or rudder input disengages AP. |

# Autothrottle System

On ground, pressing AT on the FCP arms the autothrottle system, THRUST in white (armed) is shown on the FMA, the autothrottle engages and selects take-off thrust (THRUST green) when the thrust levers are moved beyond 23° TLA (≈ 68% $N_1$). Between 60 kts and 400 ft AGL, the AT will not command the thrust levers (HOLD).

The AT engages automatically during flight:

- During Go-around (pressing either TOGA switch), or
- Windshear escape (pressing TOGA with windshear detected) or,
- Emergency Descent Mode (automatic or manually selected).

The AT is disengaged by pressing A/T DISC switch outboard of either thrust lever, moving the thrust levers (requires some force) while autothrottle is engaged, pressing AT on the FCP or automatically when a system failure is detected.

An aural "AUTOTHROTTLE" will sound twice with a yellow AT on the FMA in case of manual disconnect. In case of automatic disconnect, the "AUTOTHROTTLE" alert will sound continuously until cancelled by pressing either A/T DISC switch.

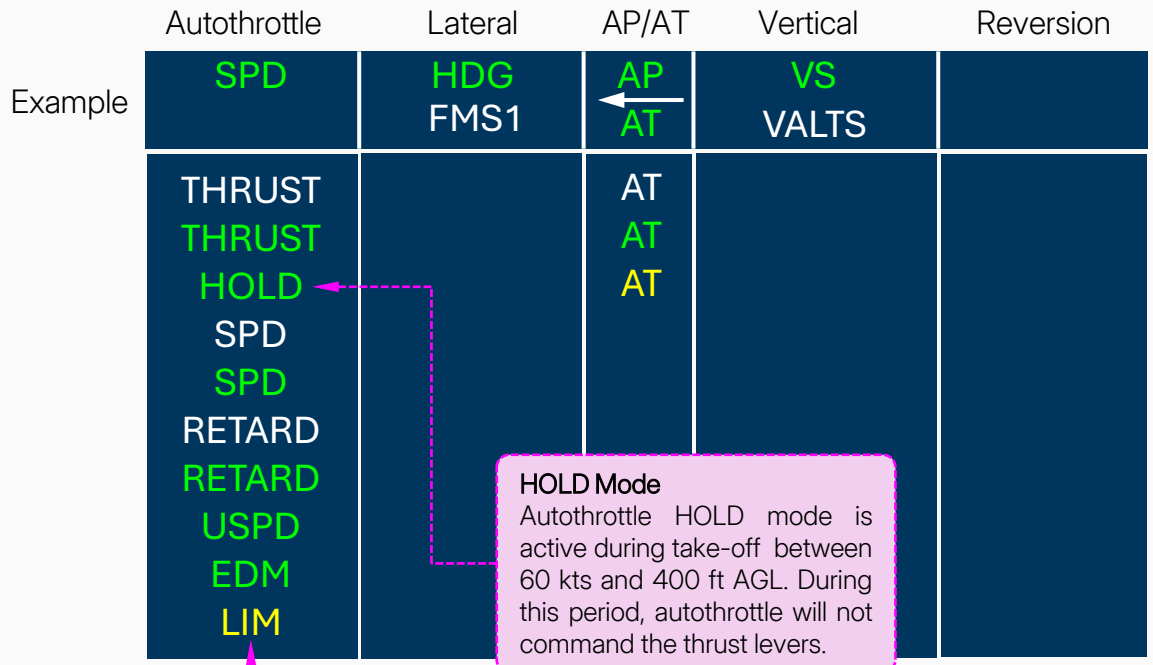
The AT operates in THRUST mode:

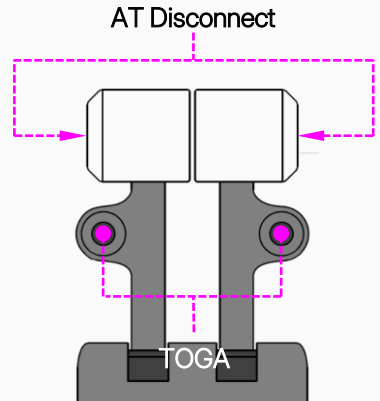
- Take-off (from 60 kts to 400 ft AGL HOLD mode) and go-around
- FLC (climb thrust for climb or flight idle for descent)
- LIM (speed target cannot be achieved by AT)
- RETARD (30 ft or between 20 ft - 15 ft during AL)
- Abnormal: WSHR, USPD, OSPD or EDM

During landing, the AT disconnects 2 seconds after weight on wheels (no aural alert), the ROLLOUT mode activates (in case of autoland), and TOGA switches are inhibited.

In all other vertical modes (ALT, ALTS, VS, VPATH, VGP, GS, FPA, etc.), the AT operates in SPD mode.

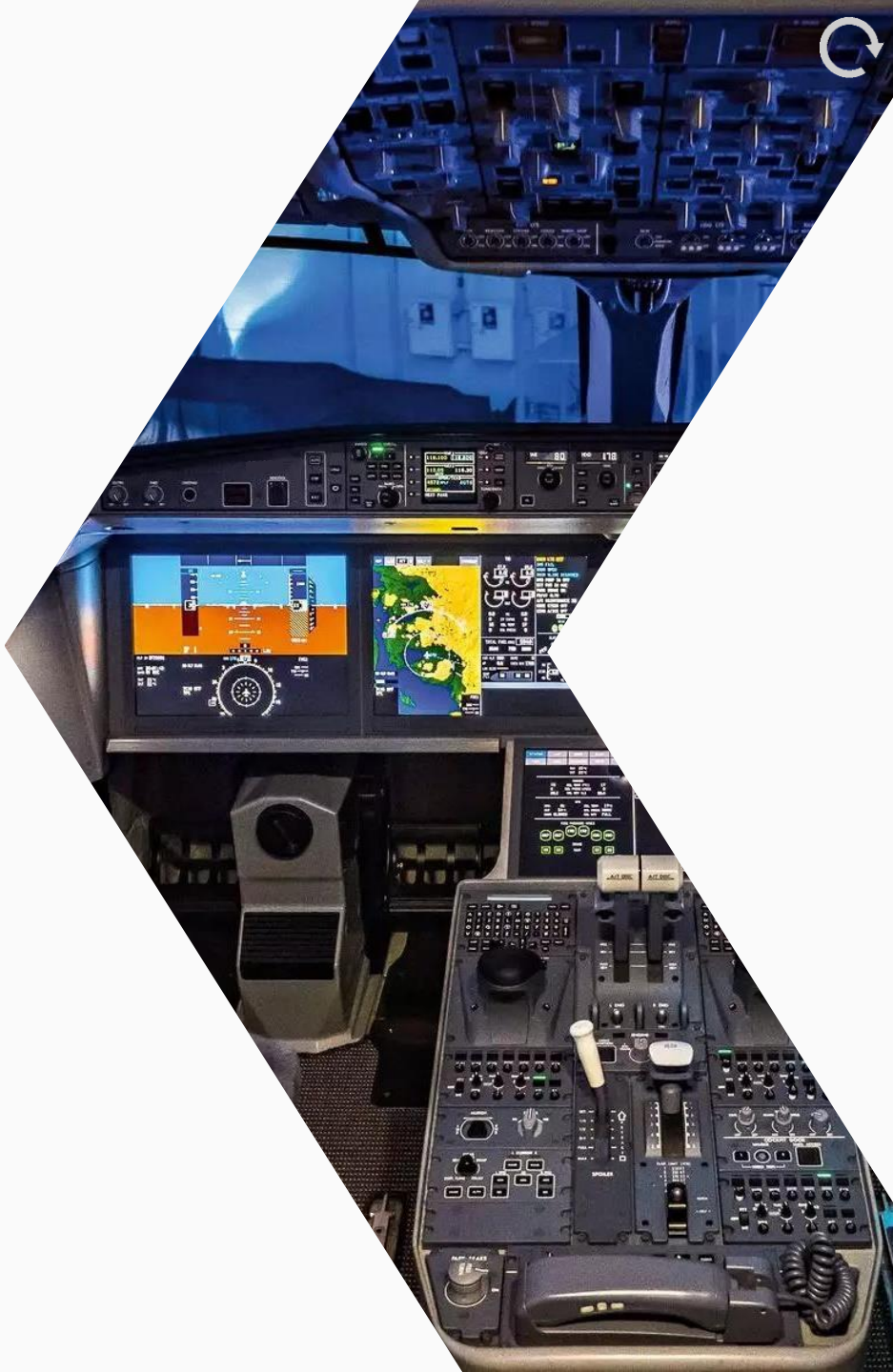
In case of OEI, the AT stops commanding the inoperative engine. After engine restart the AT must be cycled on the FCP for the AT to control both engine thrust levers again.




Contrary to the Airbus auto thrust philosophy the A220 thrust levers do move.

# Auxiliary Power Unit



# Auxiliary Power Unit (APU)

The APU provides 400 Hz, 115 VAC up to 75 kVA electrical power and bleed air. Based on pneumatic demand the ECU adjusts inlet guide vanes to control the volume of bleed air. APU related information such as RPM, EGT, oil temperature / pressure / quantity and inlet door position is found on the STATUS synoptic page.

The APU receives fuel from:

- Left main ejector pump if left engine is running
- Either tank AC boost pump (left is standard)
- Suction-feed from left main tank if no AC power

For start, the APU uses TRU 3 when AC power is available (and no other TRU inoperative) or otherwise both batteries. Battery 1 powers the ECU while battery 2 runs the APU starter. APU start sequence:

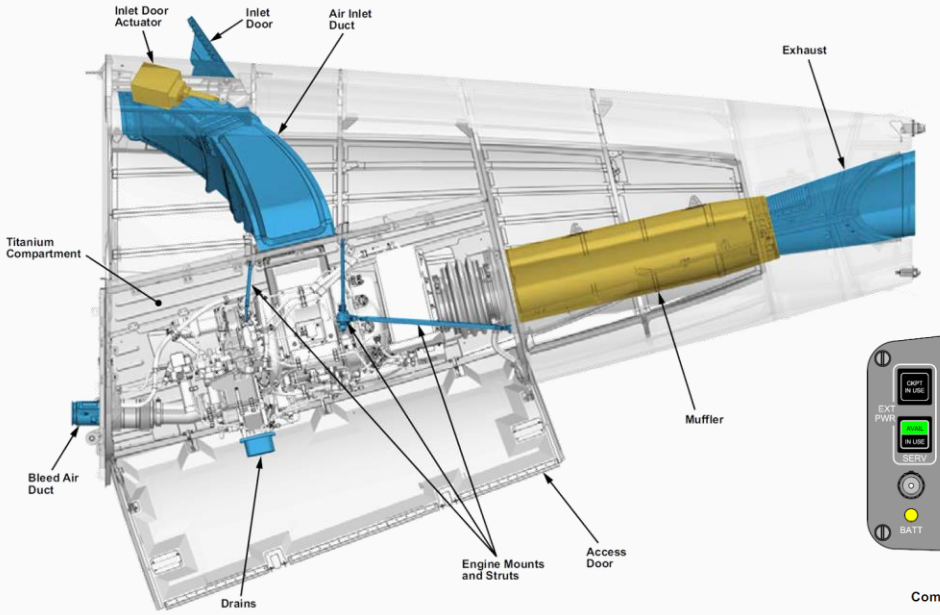
- APU switch must be held in START position for at least 3 seconds.
- APU BITE activates, APU inlet door opens and APU fuel shutoff valve opens.
- Left AC fuel boost pump runs (left engine not running and AC power available).
- APU start sequence begins and lasts until RPM reaches 70%.
- Upon reaching 95% RPM, APU is ready for electrical and pneumatic loading.

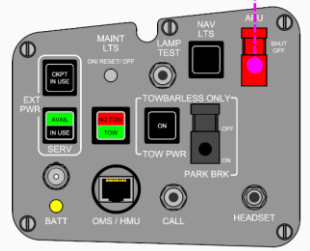
Maximum 3 start attempts within one hour with 2 min cooldown between attempts, the ECU prevents restart until RPM drops below 7%. APU can be started up to 41.000 ft.

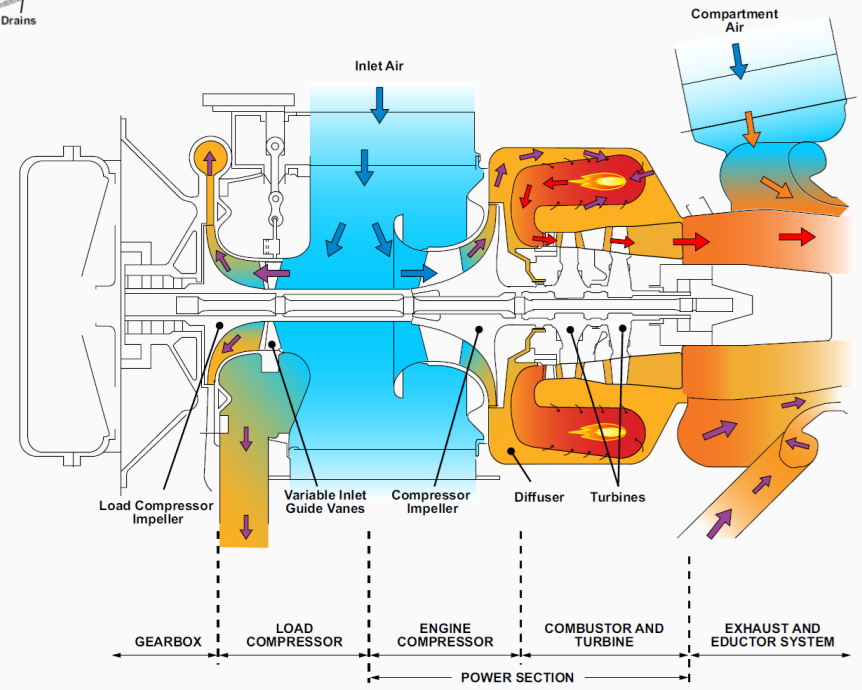
The APU provides 75 kVA up to 36.500 ft, then linearly reducing to 58 kVA at 41.000 ft. APU bleed air can be used up to 23.000 ft.

When APU is selected OFF, the ECU starts a 60 second cool down, then shuts down APU and closes the inlet door. APU can be shutdown from outside the aircraft through the external services panel. The system is certified to run in “unattended mode”.

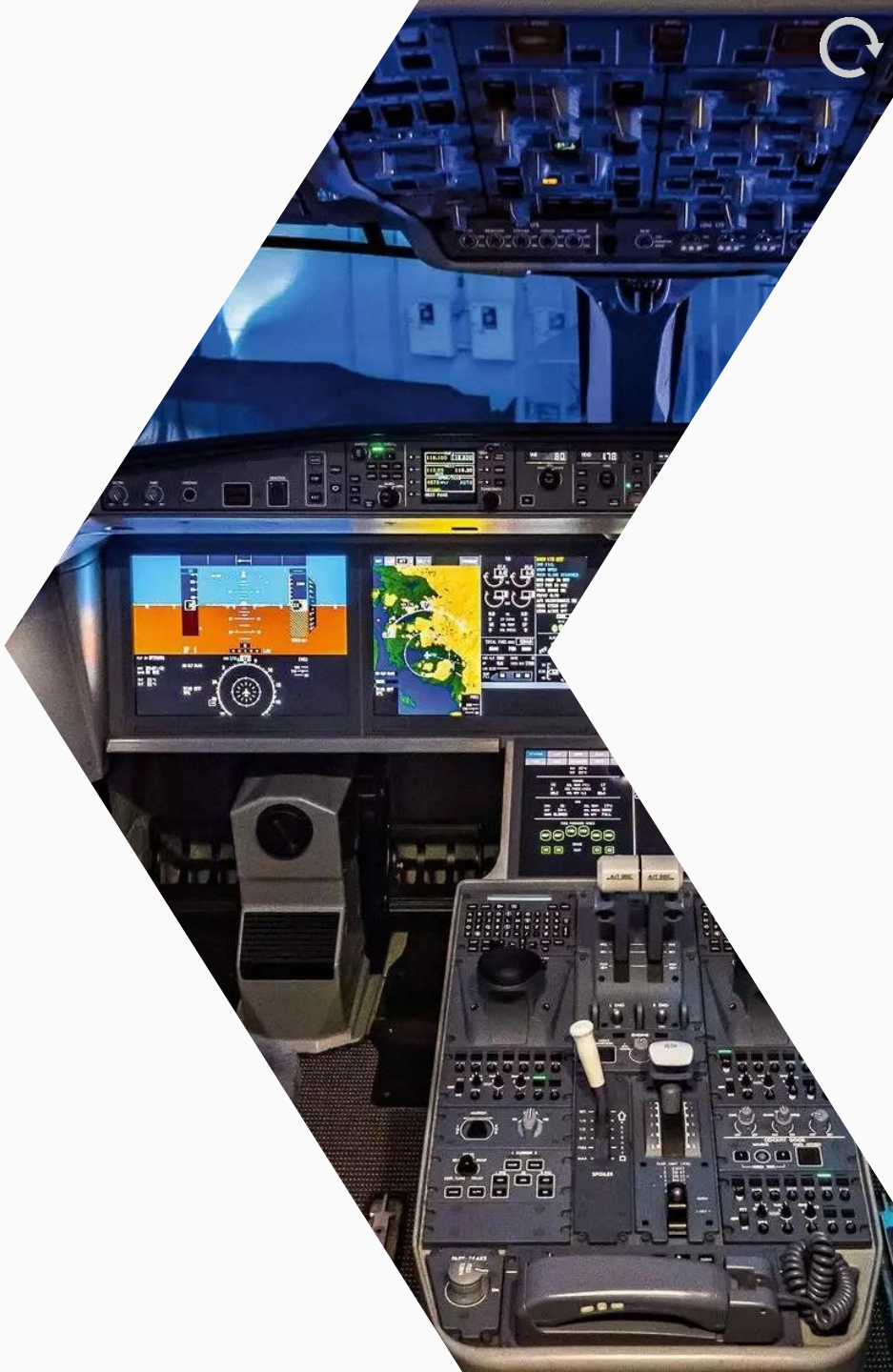
Automatic shutdown is available on ground in case of any malfunction and automatic fire extinguishing is provided. In-flight the APU will **only** shutdown automatically in case of start malfunction, RPM overspeed or loss of signal, critical ECU fault or loss of inputs to ECU. In-flight, the APU will NOT shutdown in case of fire, and there is no automatic extinguishing!



# Communication



# Communication

The system includes (* = optional):

- VHF 1, VHF 2, VHF 3 (data)
- HF 1 and HF 2*
- Selective calling (SELCAL)*
- Satellite Communication (SATCOM)*
- Data Link (ACARS) communication system
- Controller-Pilot Datalink Communications (CPDLC) – Link 2000+ or FANS 1/A+
- Internal communication (INTERCOM) (cockpit – cabin crew – ground crew)
- Passenger Address (PA) system

Communications is controlled by 2 independent Radio Interface Units (RIUs) each containing 2 channels. Only RIU 1 manages data link communications (DLK, CPDLC). Channel B of each RIU provides TCAS, EICAS and TAWS aural alerts. CTPs use Channel B, CNS (display tuning) uses Channel A.

VHF PTT switches have an open (stuck) microphone protection that becomes active after 30 seconds of continuous transmission. For HF, SATCOM, PA and cabin crew communications the open microphone protection activates after 2 minutes.

With the INT / PTT switch on the ACP or sidestick in the INT position, hot mic is activated and communication with the following stations is possible without having to press any Push-To-Talk (PTT) switch:

- Three pilot stations (left pilot, right pilot and observer)
- External service panel, refuel / defuel panel
- FWD, MID and AFT equipment bays*
- Low Pressure Ground Connection (LPGC) panel*

* SERV INT must be selected ON to connect these stations to the intercom system.

Emergency call to cabin is made by pressing CAB CALL button for more than 3 seconds. If the cabin crew makes a normal call to the flightdeck, the CALL button illuminates steady on the ACP with CABIN in cyan on EICAS, CALL is flashing on the ACP during an emergency call with yellow flashing CABIN on EICAS.

Control Tuning Panel (CTP)

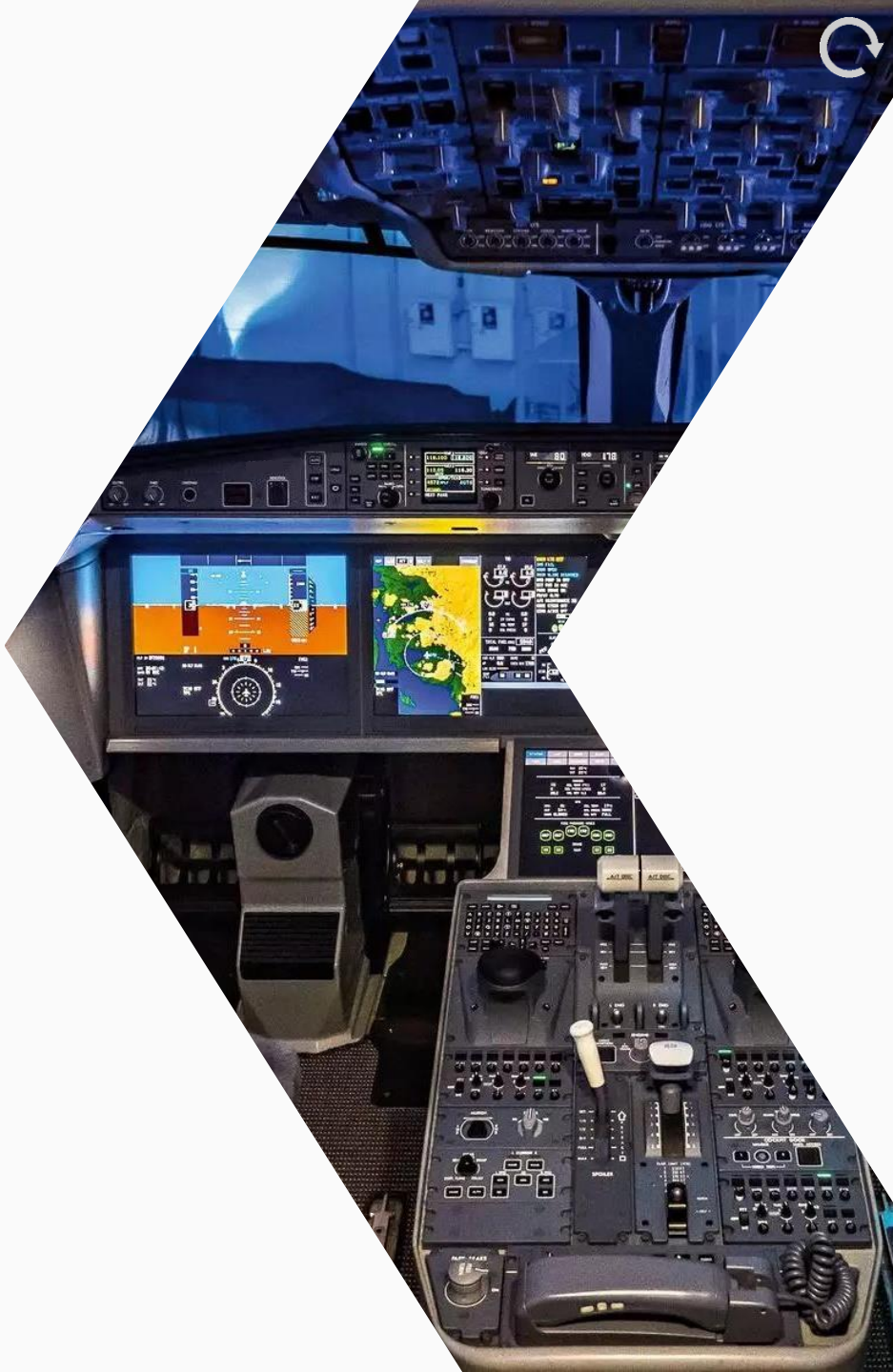
Display Tuning CNS – TUNE

Audio Control Panel (ACP)

Hold 3 sec for EMER call

PA PTT

# Doors



# Doors

The A220 has the following external doors:

- Forward and aft passenger doors (left side)
- Forward and aft service doors (right side, slightly smaller than passenger doors)
- Forward and aft cargo compartment doors (right side)
- Forward, mid and aft equipment bay doors (bottom)
- Two overwing emergency exit doors (4 optional)
- One flight crew emergency escape hatch with escape rope

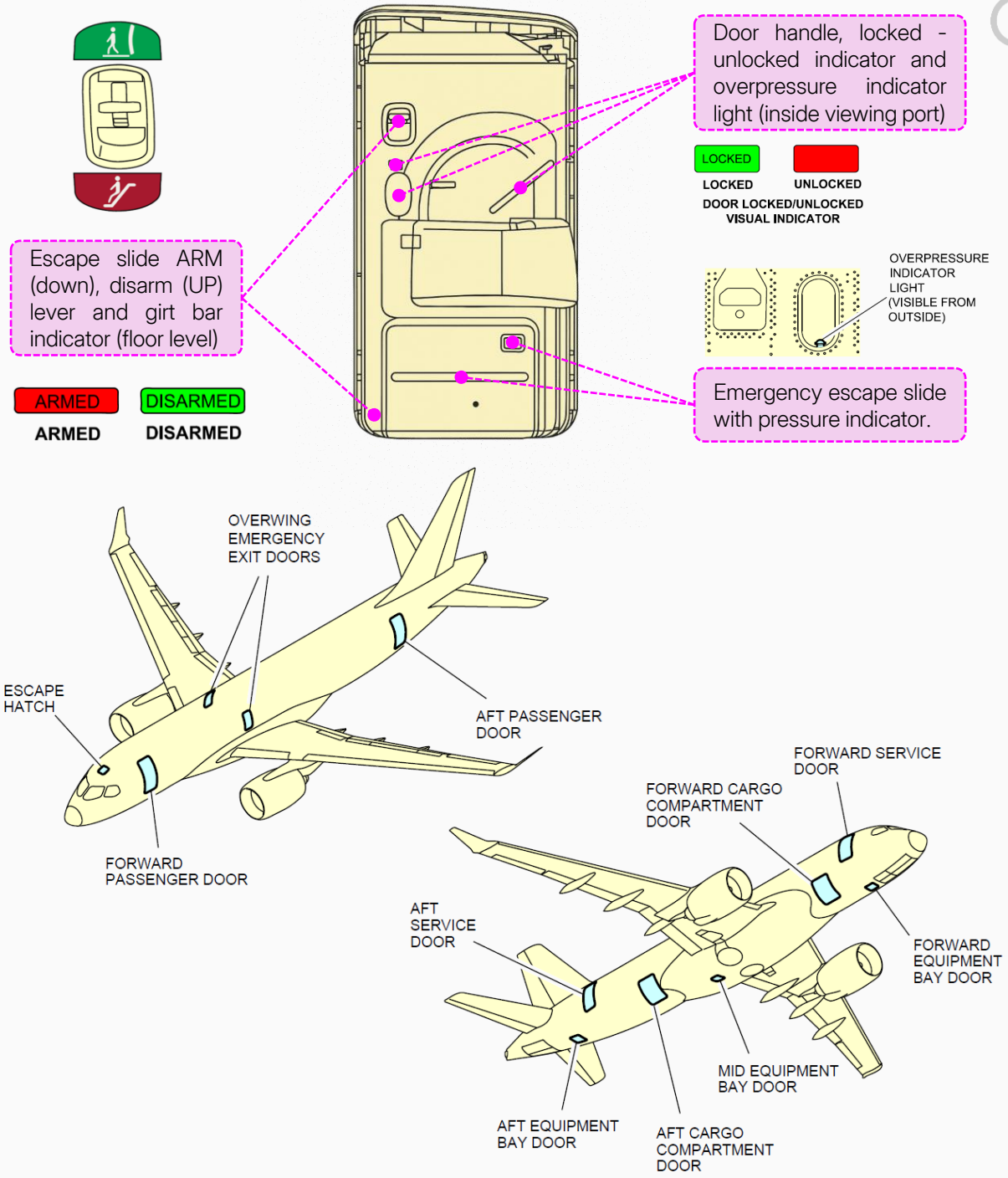
Door position is monitored by the Landing Gear Steering Control Units (LGSCUs) and shown on the DOORS synoptic page and CMS. The flight crew emergency escape hatch is not monitored (check the two mechanical locking indicators). Before opening any passenger or service door, the differential pressure indicator shall not be flashing.

Passenger, service and overwing exits are equipped with escape slides. The passenger and service door slides are disarmed with an internal handle, the overwing exit escape slides are always armed. When passenger or service doors are opened with an armed slide, a pneumatic actuator will assist to open the door and the slide will deploy. When any door is opened from the outside, the slide will automatically disarm.

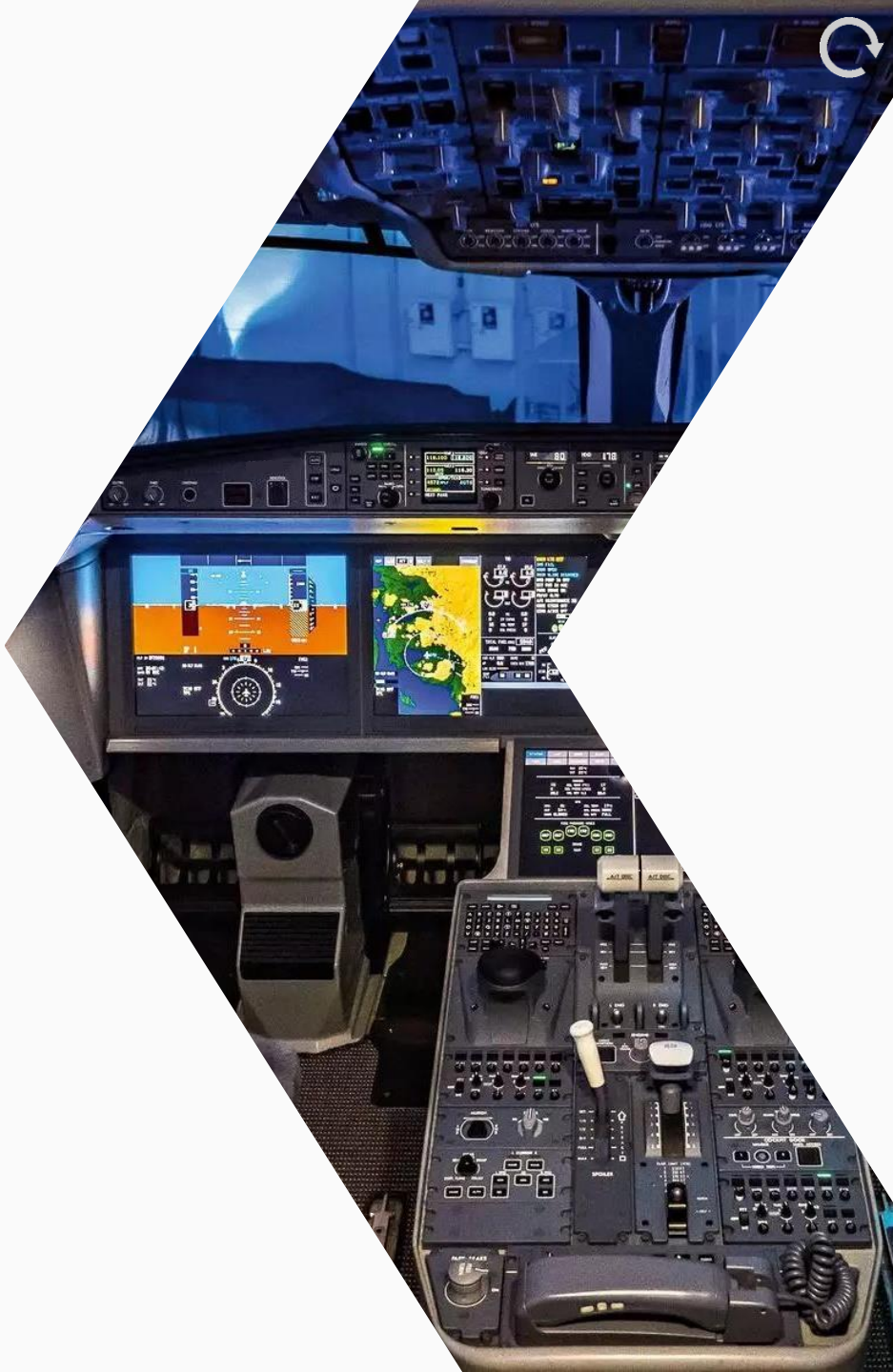
The flight deck escape hatch has 2 locking indicators, an escape rope is located behind a panel left of the escape hatch. The inboard top side of each flight crew seat is equipped with a step to reach the escape hatch.

Cargo doors are mechanically locked / unlocked, the door is then opened or closed with an electrical actuator powered by the BATT DIRECT BUS (also functions when aircraft is not electrically powered).

The forward and mid equipment bay doors are plug-type doors (compartment is pressurized). The mid equipment bay has two doors, the first is a service panel door, the second is a plug type pressure sealed door. The aft equipment bay is not pressurized.



# Electrical System



# Supply and Control

The aircraft uses AC and DC electrical power from the following suppliers:

### Alternating Current

- Main: (2) Variable Frequency Generators (VFGs), one per engine
- Auxiliary: (1) Variable Frequency Generator (VFG) powered by APU
- Emergency power: (1) Ram Air Turbine (RAT)
- External Power Unit

### Direct Current

- (3) Transformer Rectifier Units (TRU), transforming AC into DC
- (2) 24V Nickel Cadmium (NiCad) Batteries

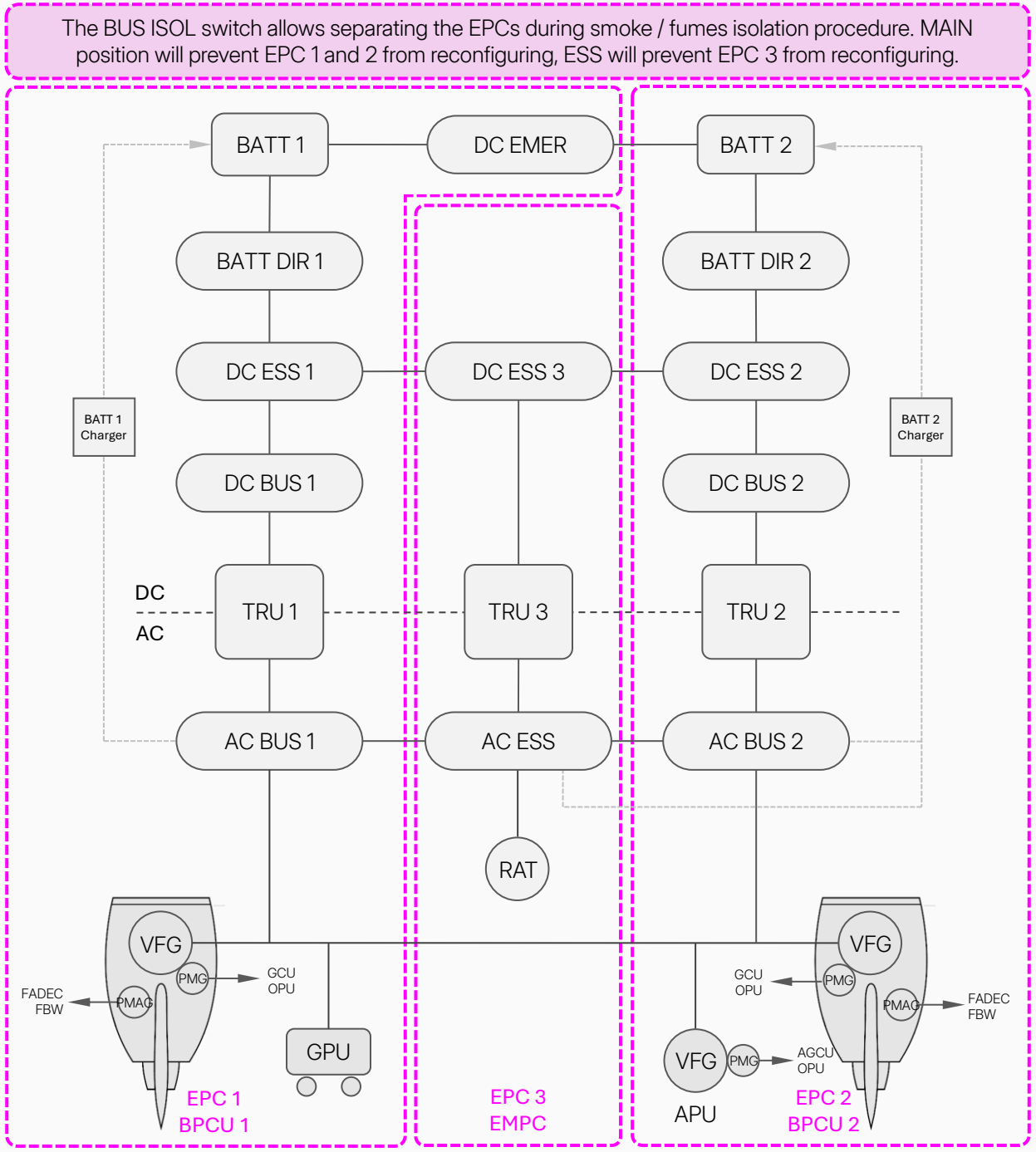
### Independent from the aircraft electrical system

- (2) Permanent Magnet Alternator / Generator (PMAGs), one in each engine to power the Fly-By-Wire system and engine FADEC (minimum 10% $N_2$).
- (3) Permanent Magnet Generators (PMGs), one in each VFG powering the Generator Control Unit (GCU) and Overvoltage Protection Unit (OPU).

The electrical control is divided into three Electrical Power Centers (EPCs) that are managed by the Bus Power Control Units (BPCUs) for EPC 1 and 2, and the Emergency Power Control (EMPC) for EPC 3. Power source priority:

| EPC | 1st | 2nd | 3rd | 4th |
|---|---|---|---|---|
| EPC 1 | L VFG (onside) | GPU | APU | R VFG (offside) |
| EPC 2 | R VFG (onside) | APU | GPU | L VFG (offside) |
| EPC 3 | AC BUS 2 | AC BUS 1 | RAT GEN | BATT |

The BUS ISOL switch allows separating the EPCs during smoke / fumes isolation procedure. MAIN position will prevent EPC 1 and 2 from reconfiguring, ESS will prevent EPC 3 from reconfiguring.



# Normal Power Distribution

## GPU

External power can be used to supply AC power in two different modes:

- Ground service mode: AC BUS 1, AC BUS 2 and DC BUS 1 are powered for ground servicing and battery charging. Selected via outside service panel.
- External power mode: GPU powers the electrical system (depending on other AC sources available – source priority). Selected from the flight deck.

GPU (if AVAIL) will automatically connect when engines or APU are shutdown.

## VFG

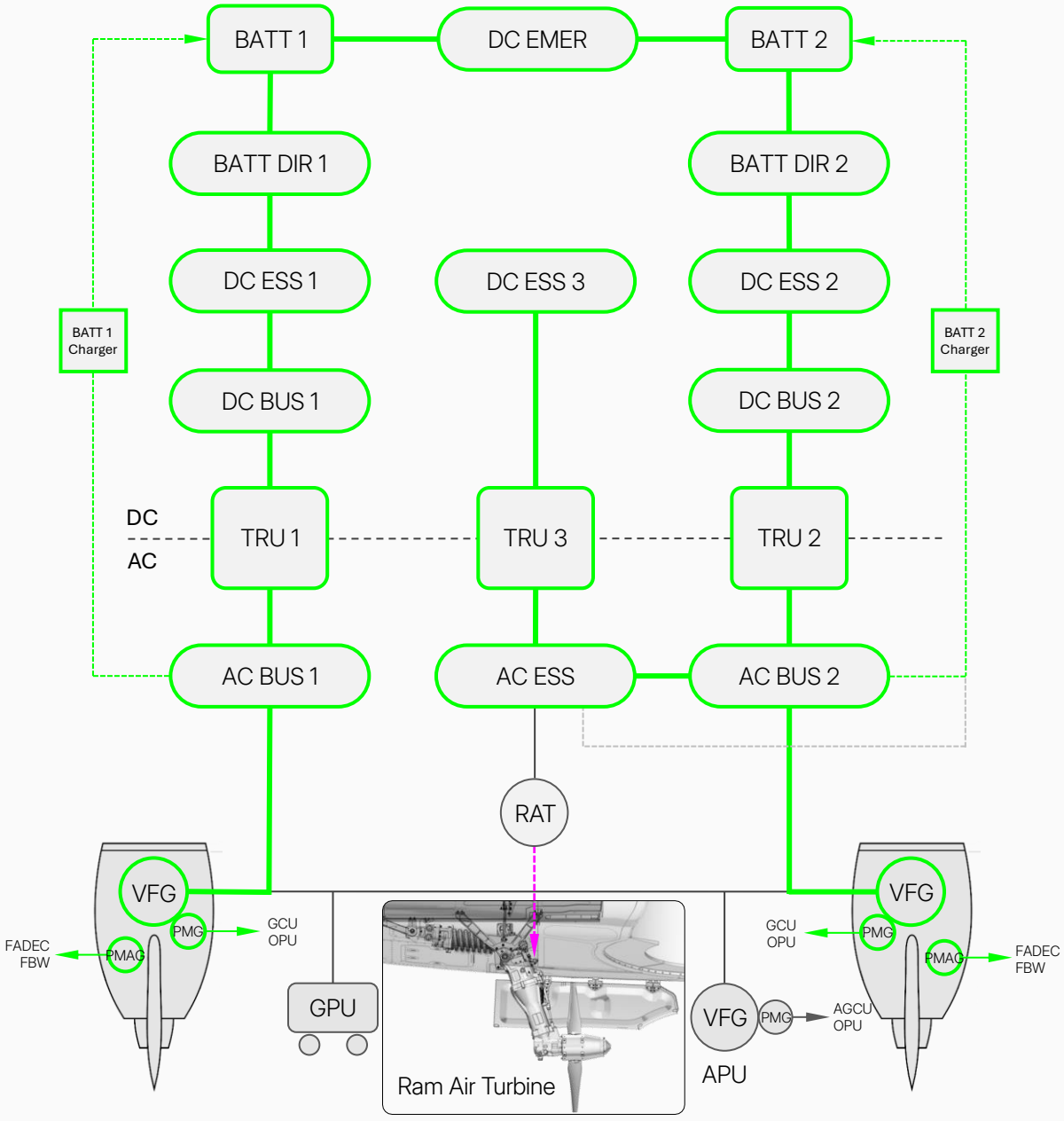
Each VFG produces 115 VAC at 75 kVA for continuous operation. Power fluctuations of 5 to 6 kVA are normal when windshield heat is on. Each VFG contains a Permanent Magnet Generator (PMG) that power the GCU and OPU.

When the L DISC or R DISC switch is pressed, the VFG physically disconnects from the gear box, the generator cannot be reconnected in flight, the engine will also automatically disconnect the VFG from the accessory gearbox in case of high VFG oil temperature.

## APU

APU electrical load reduces linearly from FL365 (75 kVA) to FL410 (58 kVA). APU ECU will request loadshedding through the CDCs to limit electrical load if required (especially above FL365), also during engine start to ensure sufficient bleed air available for the Air Turbine Starter (ATS).

APU (if running) will automatically connect when engines are being shutdown.



# Emergency Power Distribution

## RAT

The Ram Air Turbine (located between the two wheel wells) will automatically deploy when AC power is lost in flight, it can also be manually deployed. Pushing the RAT GEN switch while the RAT is already deployed will reset the RAT Generator Control (RGC).

Minimum speed is 148 kts for AC power and 115 kts for hydraulic power. After starting the APU or reestablishing VFG 1 / VFG 2, the RAT will continue to power the AC ESS bus and DC ESS 3 bus for the remainder of the flight.

## Batteries

DC EMER and BATT DIR busses are always powered. During ground operations, batteries are used for ground servicing (opening / closing cargo doors and refuelling operations) and to start the APU. Both batteries are needed to start the APU, battery 1 powers the APU ECU, battery 2 powers the starter.

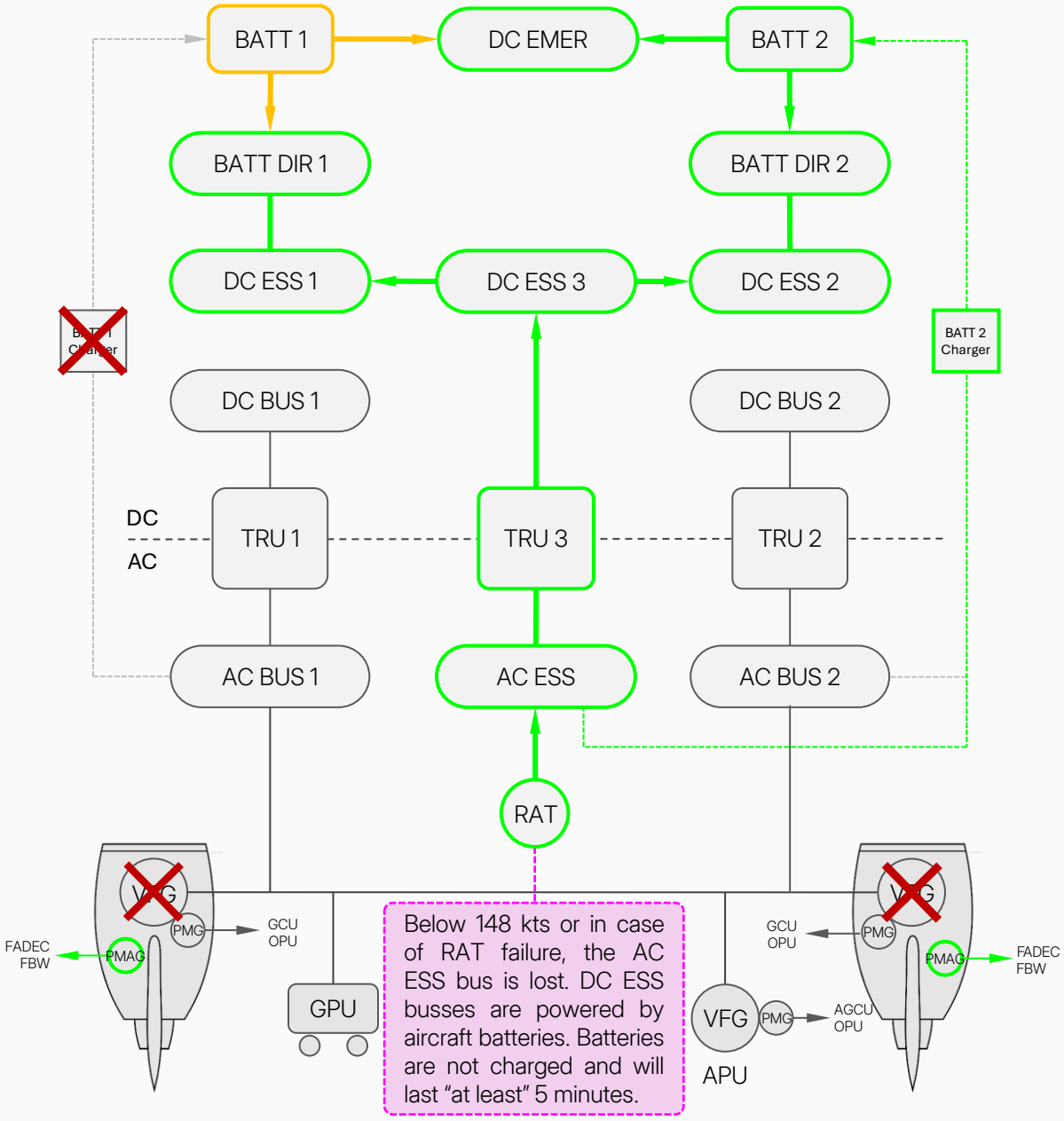
If TRUs are not working (fault or no AC power source available), batteries supply power to DC ESS busses and electric brakes. If batteries are supplying power on ground for more than 5 minutes, a horn will sound until AC power is connected or batteries are off.

Battery charger 1 is powered by AC BUS 1, battery charger 2 is powered by AC BUS 2 or AC ESS. Battery 1 is therefore not charged during an electrical emergency.

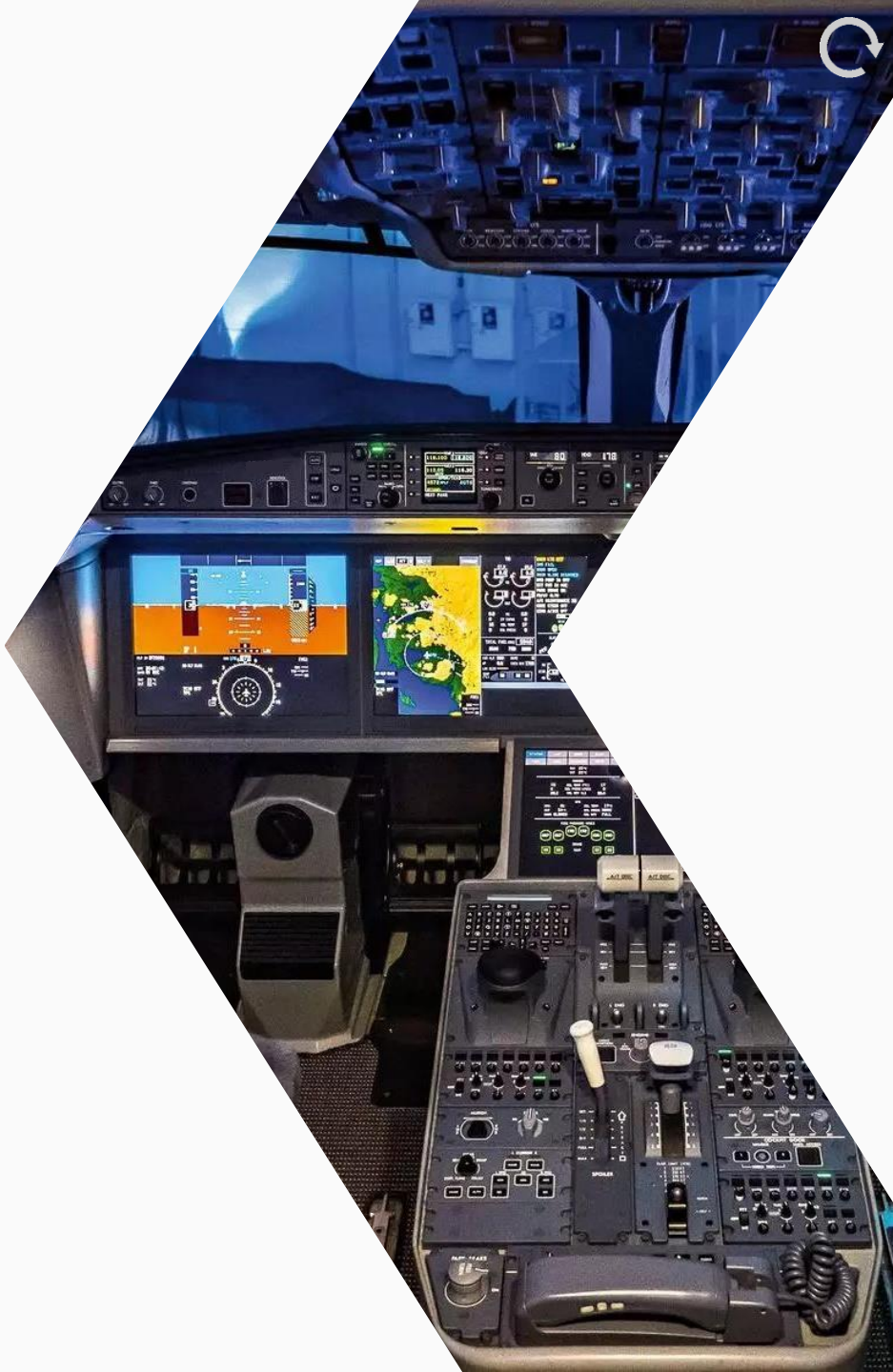
## Control and Distribution Cabinets (CDCs)

Five CDCs receive AC and DC power from the EPCs and redistribute electrical power to components that are not essential for flight. CDC 1 and 2 are masters, commanding the other CDCs (slaves). If excessive demand is placed on a VFG or TRU, the BPCU will command load shedding from the CDCs.

Circuit breakers are monitored and displayed on the CB synoptic page. CBs can be IN, TRIP (not acknowledged by crew), OUT (acknowledged), LOCK (by maintenance), SHED (turned off to limit electrical load) and INVALID (status unknown). Most CBs are electrical or solid state, very few thermal CBs are installed on the flight deck.



# Electronic Display



# Flight Instruments

On emergency power (RAT or batteries), only DU 1 and 2 are available. Both CCPs are able to control the DU2 MFW (but not simultaneously).

EICAS is normally displayed on the right half of DU 2, if there is a malfunction, EICAS is displayed on the left half of DU 3. Display reversion switch:

- SWAP – available when all DUs are working or only DU 5 failed, when selected, DU 2 displays 2 MFWs, DU 3 displays EICAS (left) and MFW (right).
- REV – manual reversion, only available when two or more DUs have failed, any DU displaying a full PFD will change to a half PFD to allow an extra MFW.
- NORM – automatic reversion in case of DU failure or DU switched OFF.

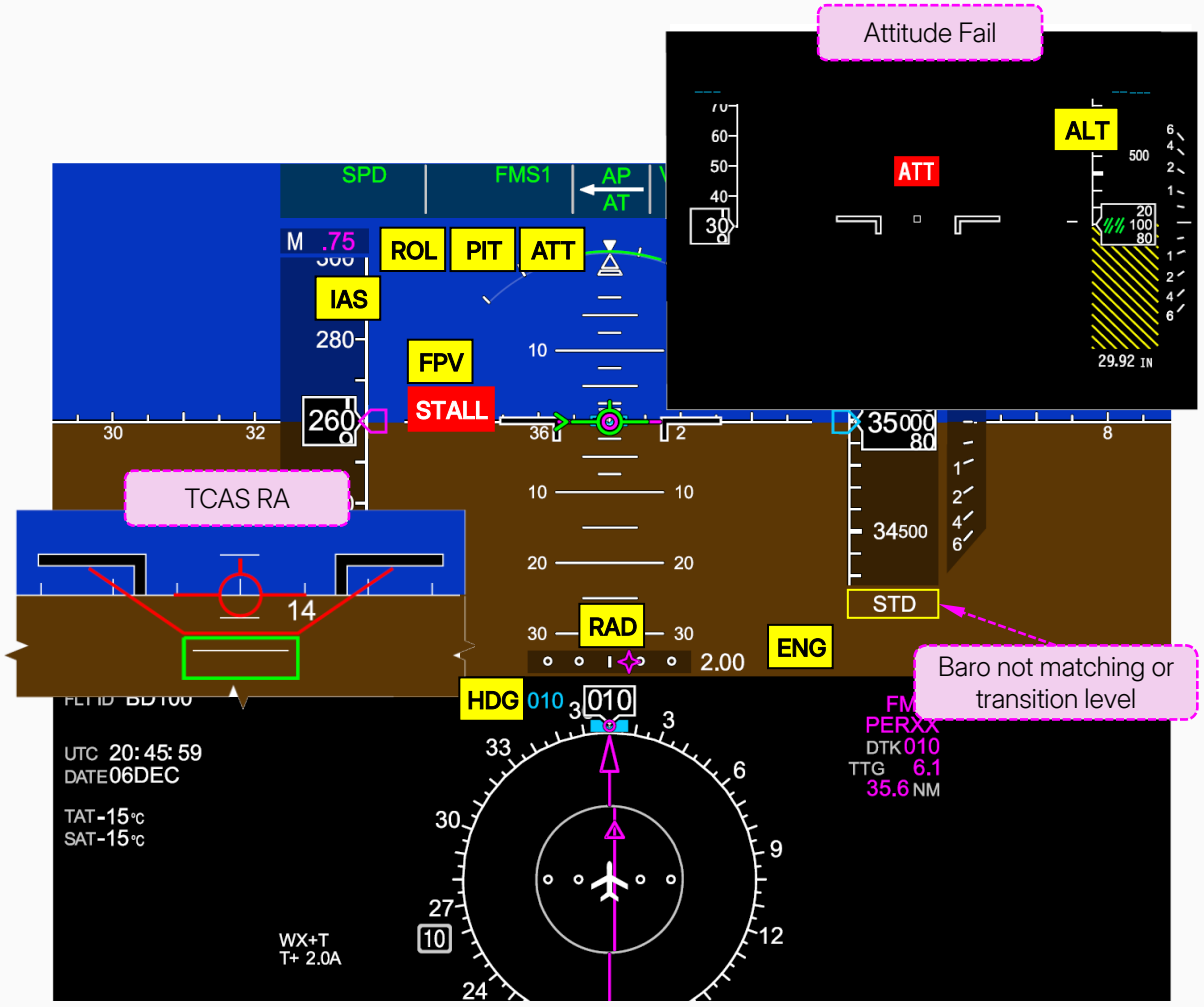
The FPV becomes laterally active at 80 kts and vertically active at weight off wheels. The flight director cue appears 3 seconds after weight off wheels.

Each "tick" on the FPV speed error tape represents 5 kts difference from the target speed up to maximum 15 kts (3 ticks).

The airspeed tape has speeds numerically marked every 10 kts when below 200 kts or 20 kts when above 200 kts. Mach is displayed above 0.45 M and removed when Mach is less than 0.40 M.

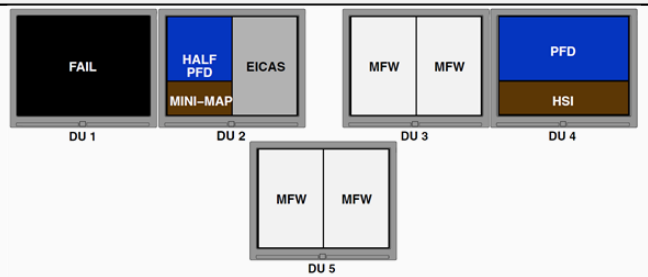
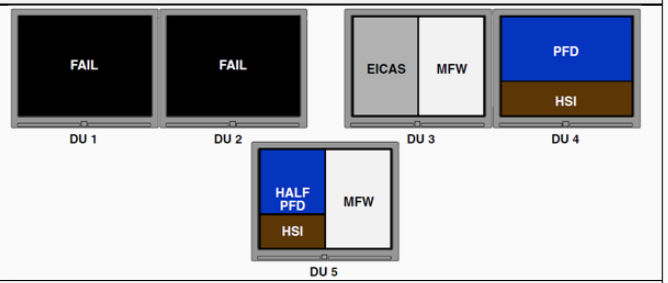
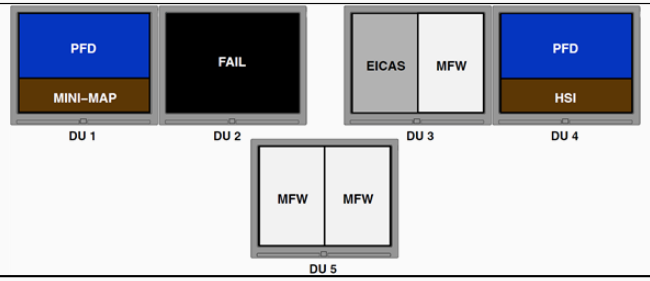
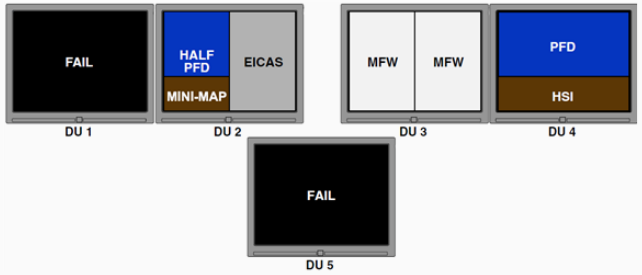
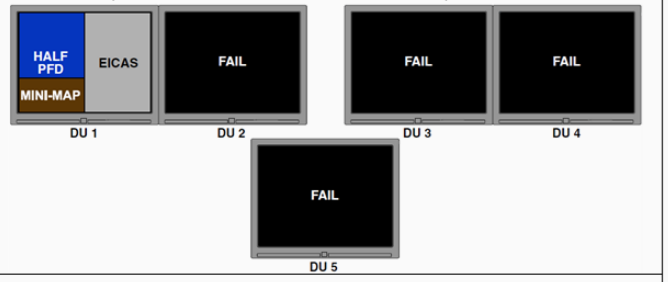
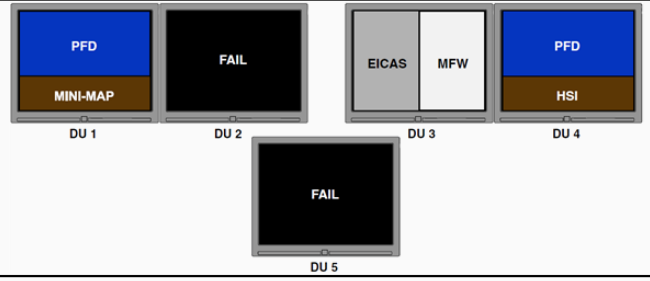
V speeds are removed when speed is greater than $V_2$ + 20 kts during take-off, on approach V speeds are removed after GA or on ground with airspeed below 50 kts for more than 5 seconds.

$V_{AC}$ (OEI) or $V_{GA}$ (AEO) is displayed on the speed tape when GA mode is active.




| | | | |
|---|---|---|---|
| PIT | More than 4 degrees disagree (3 degrees when GS captured) | ALT | Allowable difference increases with altitude. During RNP AR approach 75 ft, in RVSM 200 ft (between 29.000 ft and 41.000 ft) |
| ROL | More than 4 degrees disagree | | |
| ATT | More than 4 degrees disagree in both pitch and bank | RAD | Allowable difference depending on altitude |
| HDG | More than 6 degrees disagree | FPV | Difference between the FPVs |
| IAS | More than 10 kts disagree | ENG | Engine data differs between DUs |
| Red flags indicate complete loss of data. | | | |

# Automatic Reversion

| Failed DU | Reversion | Failed DU | Reversion |
|---|---|---|---|
| DU 1<br>or<br>DU 4 |  | DU 1<br>and<br>DU 2<br><br>or<br><br>DU 3<br>and<br>DU 4 |  |
| DU 2<br>or<br>DU 3 |  | 3 DUs<br>FAIL | When remaining DUs are paired, the full PFD and EICAS + MFW are shown. When remaining DUs are not paired, half PFD and EICAS is shown on one side, half PFD and MFW on the other side. Location depends on what DU has failed.<br> |
| DU 1<br>and<br>DU 5<br><br>or<br><br>DU 4<br>and<br>DU 5 |  | 4 DUs<br>FAIL | MFW (MAP, FMS, CNS, CHKL, SYN, DATA) are unavailable.<br>  |
| DU 2<br>and<br>DU 5<br><br>or<br><br>DU 3<br>and<br>DU 5 |  | Display<br>Reversion<br>Mode | The DISPLAY reversion mode (DISPLAY switch is set to REV) causes any DU that shows as full PFD to show a half PFD and either the EICAS or MFW. It is available when:<br><br>1. Automatic reversion is triggered and two DUs are inoperative<br>2. Only DU 1 and 2 available and one or both PFDs display in full<br><br>When DU 1 and DU 4 fail, manual reversion is not available. |

# EICAS

Engine Indication and Crew Alerting System (EICAS) is responsible for:

- Crew alerting
- Indication of engine and system states
- Take-off configuration warning
- Aural warning

To avoid distraction during take-off and landing phases, most caution and advisory messages are inhibited if no immediate crew action is required:

- Take-off inhibit: from 80 kts until 400 ft AGL.
- Landing inhibit: from 400 ft AGL until speed is less than 50 kts.

Take-off configuration warning is triggered when TLA is above 23° and indicated $N_1$ is above 55% on both engines in the following cases:

| | | | |
|---|---|---|---|
| CONFIG AP | Autopilot engaged on take-off | CONFIG RUDDER or STAB TRIM | Trim not in TO range (green arc) |
| CONFIG SIDESTICK | Sidestick priority latched at take-off | CONFIG FLAP | Slats / flaps not configured for TO |
| CONFIG BRAKE | Parking brake set | CONFIG SPOILER | Spoilers not stowed |

Aural alerts:

- A-chord: overspeed
- Double beep: trim running time limit (3 seconds)
- Tripple chime: warning message
- Single chime: caution message
- Cavalry charge: autopilot disconnect
- Single C-chord: altitude alert
- Double C-chord: VTA - Vertical Track Alert (VNAV)
- Voice message: EICAS warning, TAWS or TCAS

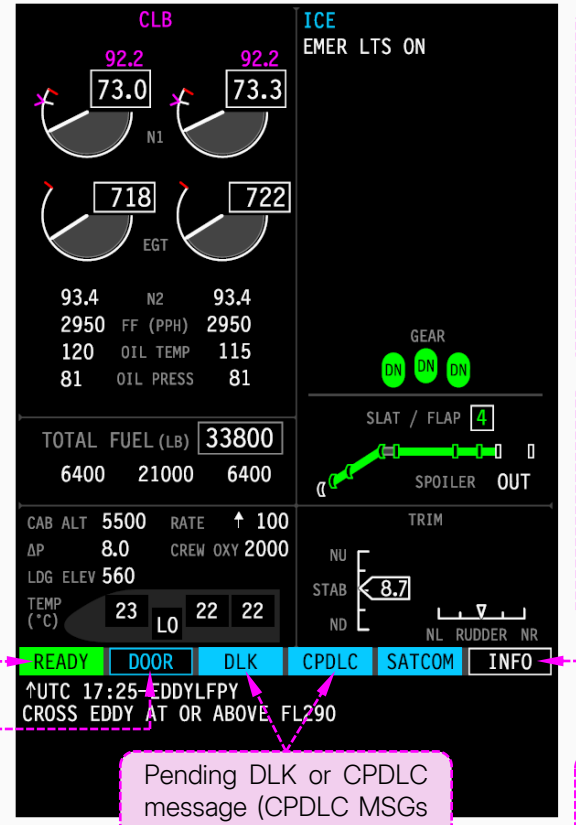
Priority of aural alerts is: STALL, TAWS, then TCAS.
System tests are available via the AVIONIC synoptic page.

Engine instruments:
- Active thrust mode and $N_1$ target >
- $N_1$ %
- EGT
- $N_2$ %
- Fuel flow
- Oil press / temp
- Fan vibration monitor (only if detected)

Permanently displayed:
- Fuel on board
- Cabin pressurization, crew oxygen and zone temperature
- Trims (aileron trim only in DIRECT mode)

Cabin crew signals ready for take-off or landing.

Doorbell – flight deck access requested.




Pending DLK or CPDLC message (CPDLC MSGs shown on lower EICAS).

Crew alerting section showing up to 18 EICAS messages (if gear and flaps are up and display decluttered), if not all messages can be shown (page full), the crew can cycle between pages, declutter and recall by using the CAS button.

Temporary displayed:
- Landing gear
- Flaps
- Spoilers

Removed when flaps and gear are up for 30 s and spoilers retracted.

Unconfirmed INFO msg, check and acknowledge on INFO synoptic page

| Level | Definition | Indication |
|---|---|---|
| Warning | Immediate awareness, action or crew decision required. | Master Warning + triple chime + flashing red message on EICAS + aural alert for some warnings (e.g. fire) |
| Caution | Immediate awareness, crew action required before end of the flight. | Master caution + single chime + flashing amber message on EICAS |
| Advisory | Crew awareness, no associated in-flight procedures. More info might be available on INFO synoptic page. | Cyan EICAS message flashing 5 seconds, then steady. |
| Status | Selected system states (e.g. SEATBELT ON). | White EICAS message |
| INFO messages are non-alerting-type messages with no in-flight associated procedures. INFO messages are used for dispatch and trouble shooting. They are displayed on the INFO synoptic page and require acknowledgement to remove the EICAS flag (will remain displayed on ground). | | |

EICAS message will stop flashing upon pressing MW / MC flasher.

# Air Data System (ADS)

The Air Data System provides:

- Static pressure
- Pitot pressure
- Angle-of-attack
- Outside temperature

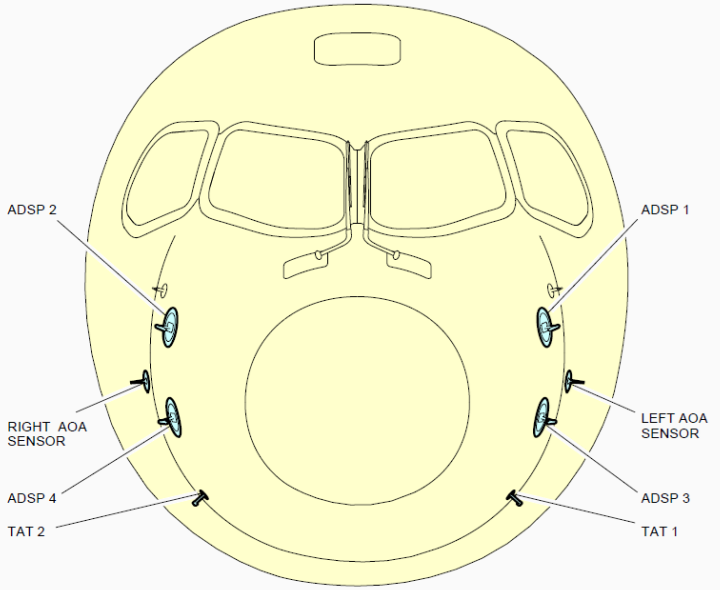
| | |
|---|---|
| Static pressure | 4 Air Data Smart Probes (ADSPs) measure static pressure, dynamic pressure and angle-of-attack. 2 Angle-of-attack sensors are used as back-up. |
| Pitot pressure | |
| Angle-of attack | |
| Outside temperature | 2 TAT probes (engine T2P2 sensor as back-up). |

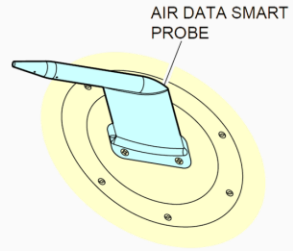
All probes are automatically heated as soon as engines are started, the angle-off attack probes are always heated when power is available.
In case of sensor fault, the system will automatically revert to an alternate source, this can also be achieved manually via the reversion panel.

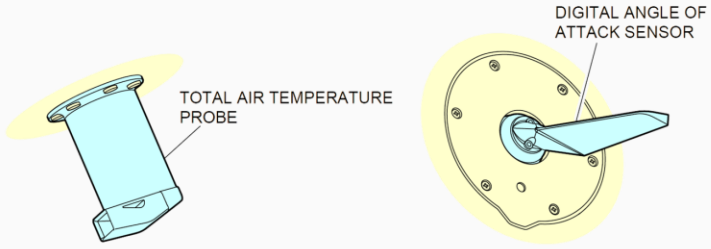
| Display | Normal | 1st Alternate | 2nd Alternate | 3rd Alternate |
|---|---|---|---|---|
| Left PFD | ADSP 1 | ADSP 4 | ADSP 2 | ADSP 3 |
| Right PFD | ADSP 2 | ADSP 4 | ADSP 1 | ADSP 3 |
| ISI | ADSP 3 | ADSP 4 | - | - |

Reversion to an alternate source is indicated on the right side of the FMA. An amber reversion indicates that both PFDs are using the same ADS or IRS source. On the ISI a reversion to ADS 4 is indicated by an amber ADSREF flag.

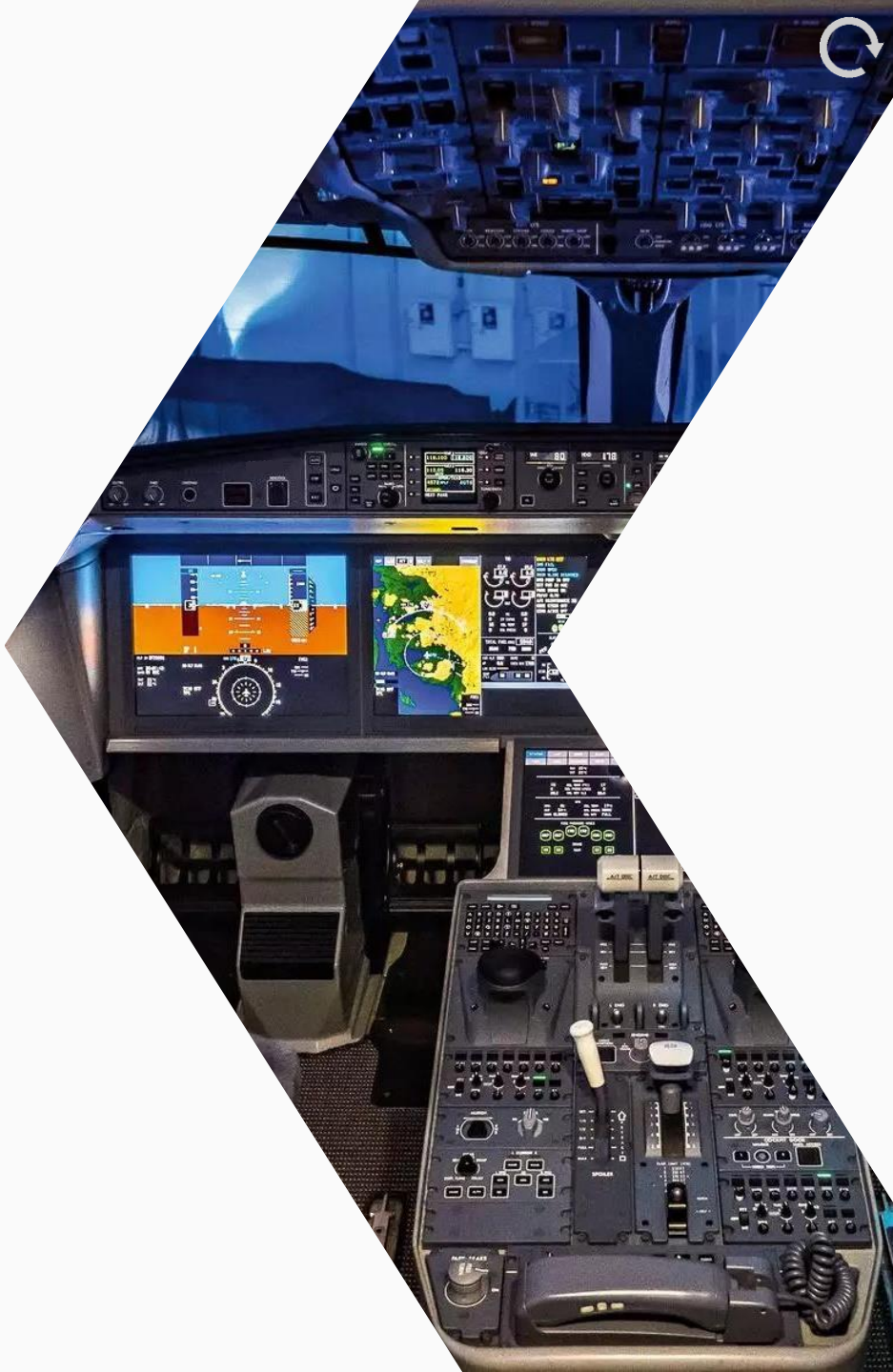
| Autothrottle | Lateral | AP/AT | Vertical | Reversion |
|---|---|---|---|---|
| SPD | HDG<br>FMS1 | ← | VS | ADS 4<br>IRS 2 |



# Fire Protection



# Fire and Overheat Protection System

The Fire Detection and Extinguishing (FIDEX) system provides monitoring, warning and fire-extinguishing capabilities for the following:

- Engine fire protection
- Auxiliary Power Unit (APU) fire protection
- Cargo compartment fire protection
- Equipment bay smoke detection (no extinguishing)
- Main landing gear bay overheat detection (no extinguishing)
- Lavatory smoke detection and fire protection (waste bins)

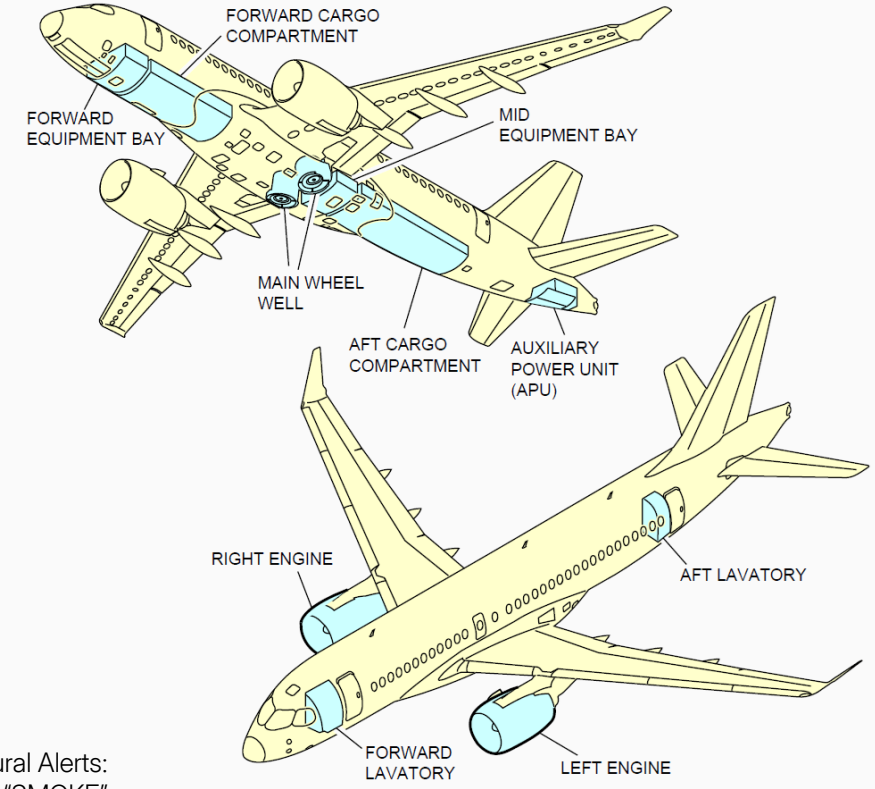
Engines, APU and main landing gear wheel wells have dual overheat / fire detection loops. Both loops must detect a fire or overheat for an alert to occur unless one loop is faulty. For engine fire extinguishing, two fire bottle are available (total) which can be discharged in either engine. One bottle is available for the APU. One high-rate and one low-rate bottle (2 optional) are available and can be used in either cargo compartment.

If an APU fire is detected during ground operations, the APU will automatically shutdown, 10 seconds later the fire bottle will discharge. Pressing the APU FIRE switch will override automatic discharge. In flight, APU fire warning will not cause an automatic shutdown. For L (R) ENG FIRE and APU FIRE switch functions, see table on the right.

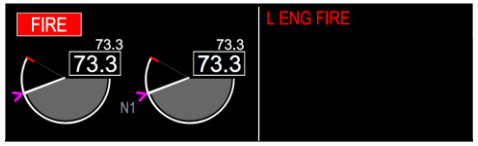
Each cargo compartment has 4 smoke detectors, at least 2 smoke detectors must indicate that smoke is present before an alarm is given. When CARGO FWD or AFT switch is pressed, cargo compartment ventilation and recirculation fans are shut off.

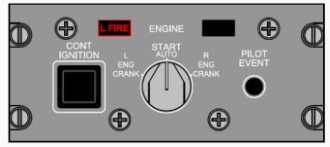
BTL indications only show discharge status of the HRD bottle. After the HRD bottle discharges the LRD bottle slowly discharges to supress further fires for 60 minutes + 15-minute hold (longer for ETOPS approved aircraft as an option).

Both FWD and MID avionic compartments have 2 smoke detectors in the exhaust ducts. Both detectors must sense smoke before a warning is issued unless one detector has failed.



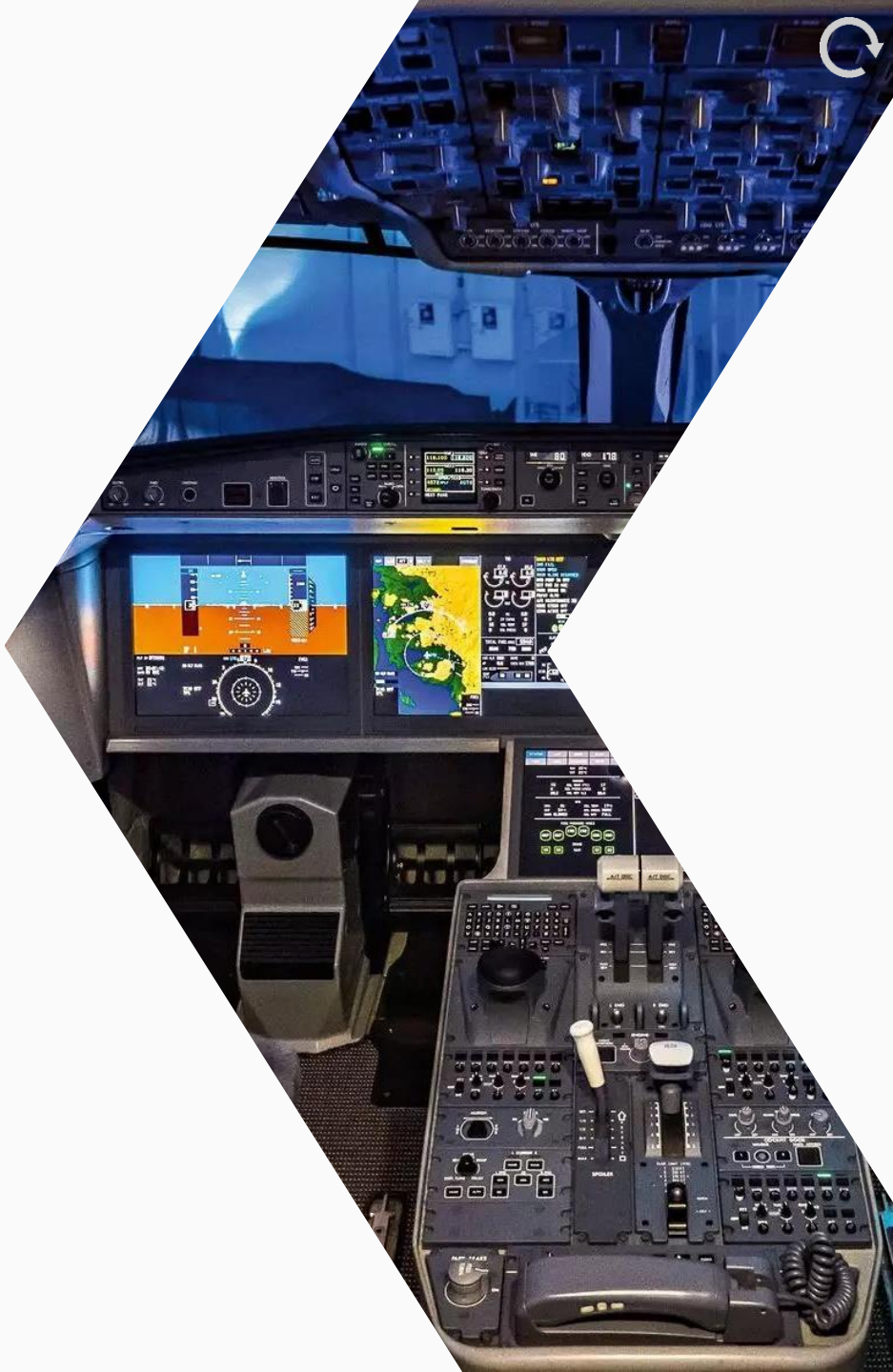
Aural Alerts:
"SMOKE"
"APU FIRE"
"CARGO FIRE"
"LEFT ENGINE FIRE"
"RIGHT ENGINE FIRE"
"GEAR BAY OVERHEAT"




| ISOLATION | L ENG FIRE or R ENG FIRE (Guarded) | APU FIRE (Guarded) |
|---|---|---|
| FIRE | Fire extinguishing bottles are armed | Fire extinguishing bottle is armed |
| FUEL | Fuel shut-off valve closes<br>FADEC commands shutdown | APU fuel shut-off valve closes<br>ECU commands APU shutdown |
| AIR | Fan Air Valve and bleed valve closes | APU bleed air valve closes |
| HYD | Hydraulic shut-off valve closes | - |
| ELEC | VFG deactivates | VFG deactivates |

# Flight Controls



# Fly-By-Wire (FBW)

The A220 flight control system consists of:

| Primary | Secondary |
|---|---|
| • Two ailerons<br>• Two elevators<br>• One rudder | • Multifunction Spoilers (MFS) – 4 per wing<br>• Ground Spoilers (GS) – 1 per wing<br>• Trimmable Horizontal Stabilizer (HSTAB)<br>• Slats and Flaps |

The Fly-By-Wire system controls and monitors all primary and secondary flight controls except high lift devices. Three PFCCs control the FBW system, any of three allows safe flight and landing. Only one PFCC is in control at a time, the PFCC in control is automatically determined at power-up, in case of failure, transfer is automatic.

Three Inceptor Interface Modules (IIMs) receive data from the flight deck controls and transmit those to the PFCCs. Each IIM is connected to all three PFCCs. The PFCCs issue commands to move flight control surfaces to the 10 Remote Electronic Units (REUs). The REUs control the Power Control Units (PCUs) and horizontal stabilizer Motor Control Electronics (MCEs) that move the flight control surfaces.

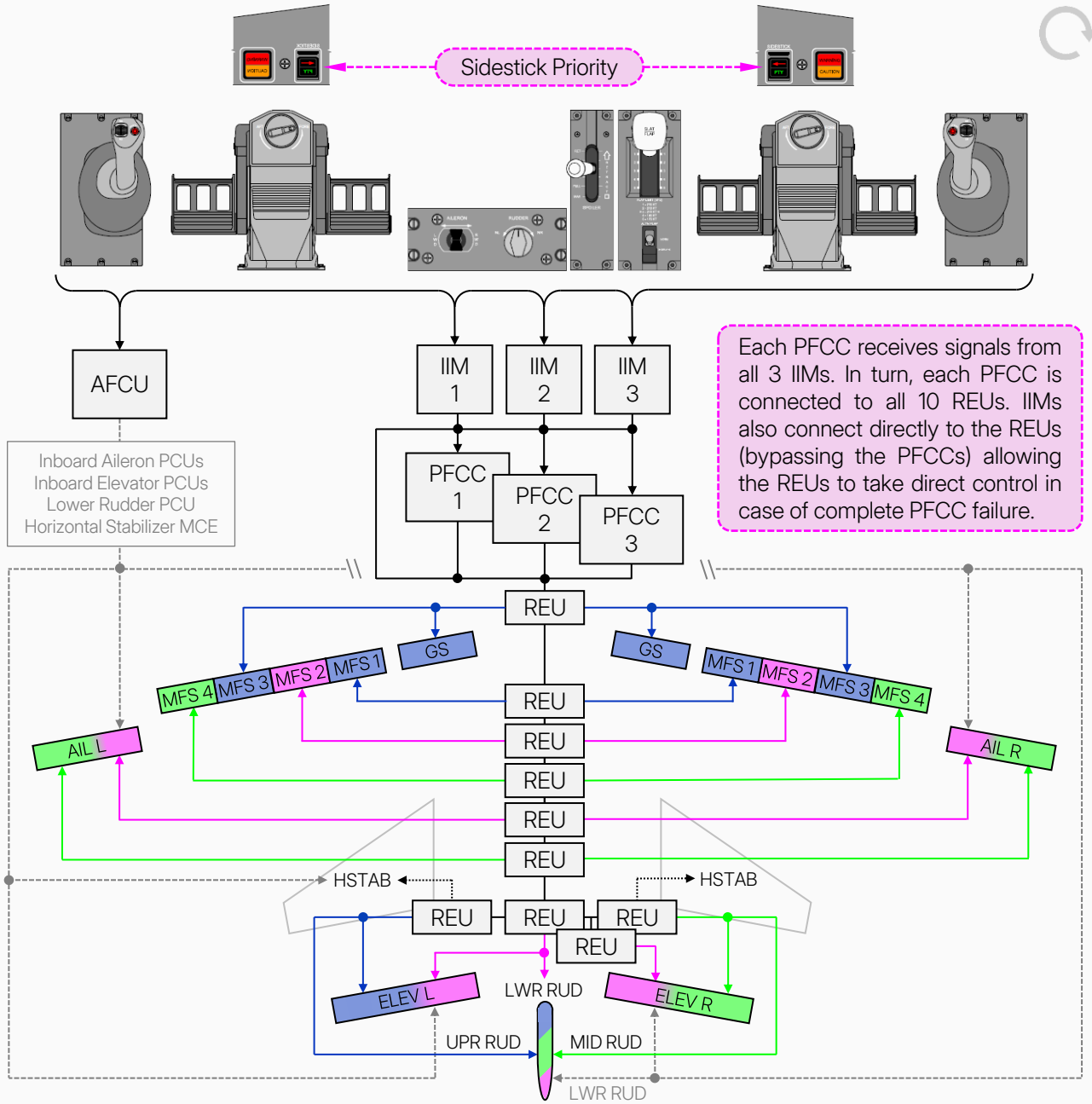
Flight control surfaces are divided equally between Hydraulic System 1, Hydraulic System 2 and Hydraulic System 3, except for ailerons (system 2 and 3 only). The horizontal stabilizer is electrically operated. The FBW system runs in two modes:

- Normal mode (autopilot and flight envelope protection available), or
- Direct mode – 3 sub modes: PFCC Direct, REU Direct, AFCU Direct.

If ADS or IRS data to the PFCC is invalid, the flight control system operates in the PFCC direct mode. Should all PFCCs fail, REUs will receive inputs directly from the cockpit controls through the IIMs (REU direct mode). The Alternate Flight Control Unit (AFCU) bypasses the REUs in case of a severe FBW system degradation. It provides direct interface from the flightdeck controls to the horizontal stabilizer MCE and one PCU for each flight control surface through hydraulic system 3:

- Left and right inboard aileron PCU
- Left and right inboard elevator PCU
- Horizontal stabilizer MCE (electrical)
- Lower rudder PCU

Ground spoilers are not available in REU Direct mode. Multifunction Spoilers and Ground Spoilers not available in AFCU Direct mode.




Each PFCC receives signals from all 3 IIMs. In turn, each PFCC is connected to all 10 REUs. IIMs also connect directly to the REUs (bypassing the PFCCs) allowing the REUs to take direct control in case of complete PFCC failure.

| Hydraulic System 1 | Hydraulic System 2 | Hydraulic System 3 |
|---|---|---|
| MFS 1 REU: MFS L1 / R1<br>MFS 3 REU: MFS L3 / R3, L AND R GS<br>AFT 1 REU: L ELEV OB, RUDDER UPR (HSTA) | MFS 4 REU: MFS L4 / R4<br>OB AIL REU: L AIL OB, R AIL OB<br>AFT 4 REU: R ELEV OB, RUDDER MID (HSTA) | MFS 2 REU: MFS L2 / R2<br>IB AIL REU: L AIL IB, R AIL IB<br>AFT 2 REU: L ELEV IB, RUDDER LWR<br>AFT 3 REU: R ELEV IB |

# Side Stick Controllers

Four position sensors are installed per sidestick, if there are fewer than 2 valid sensors per axis, the sidestick is declared faulty. Priority must be switched (latched) to the other sidestick. When all sensors are lost, the sidestick is failed.

When sidesticks are moved simultaneously, the system adds the signals algebraically. A red DUAL message is displayed with an aural “DUAL INPUT”. Momentary priority is assigned by pressing and holding the AP/PTY switch, continuous (latched) priority is assigned by lifting the guard and selecting the SIDESTICK guarded switch. This action deactivates the opposite sidestick.

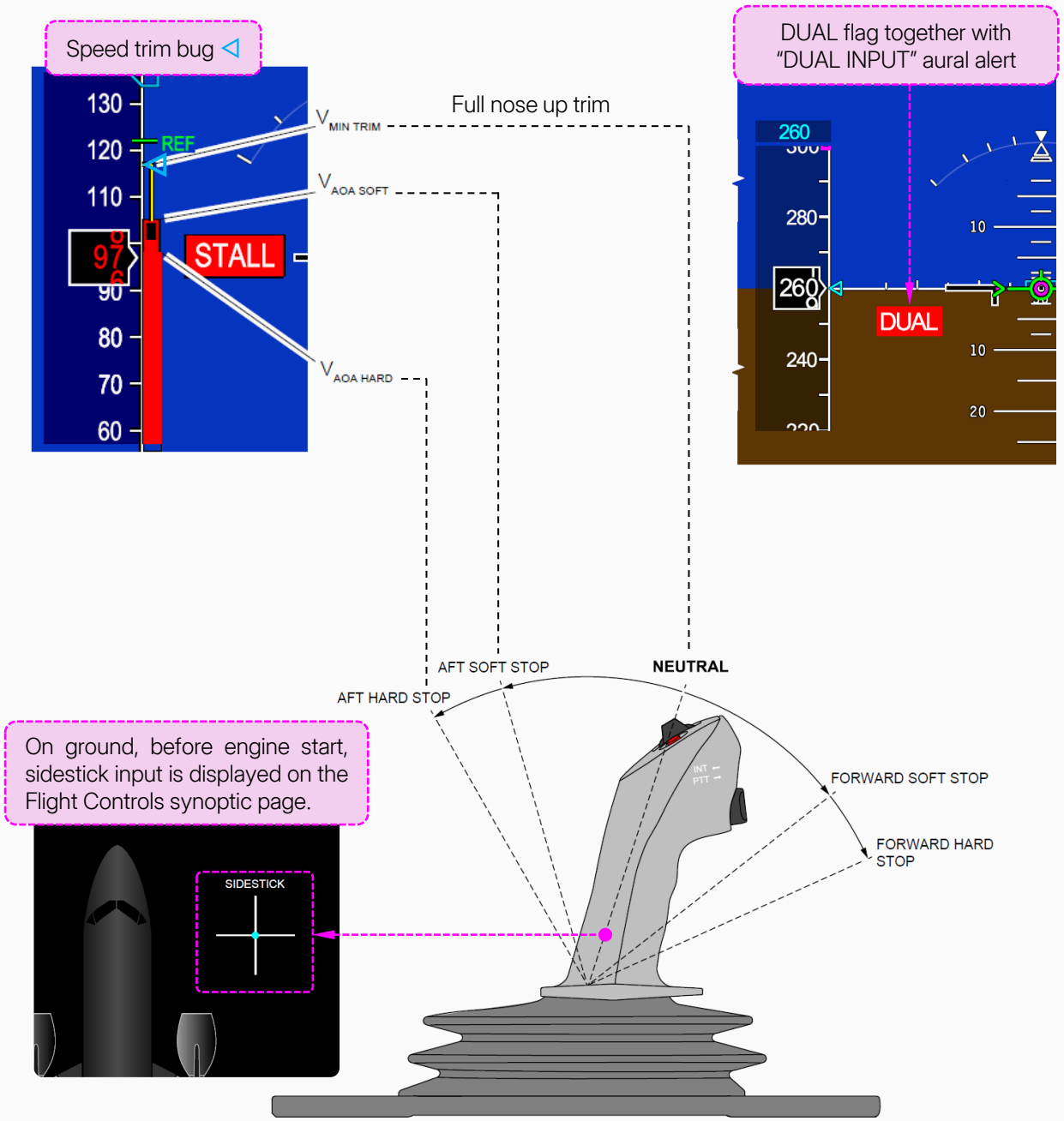
When the autopilot is engaged, the force maintaining the sidestick in neutral position is increased. If sufficient force is applied, the autopilot disengages and both sidesticks become free to move.

Sidesticks have a “soft stop” that can be overridden when sufficient force is applied to the mechanical “hard stop” (only on the pitch axis). Overriding the soft stop allows for limited (safe) excursions outside the operational envelope. When pulling the sidestick to the aft hard stop for example, the load factor is increased from 2.5 g to 2.75 g and speed is reduced to the stall marker (without stalling).

Minimum speed:

- Trimming full up, without sidestick input, the aircraft will target $V_{MIN\ TRIM}$.
- When pulling the sidestick to the aft soft stop, the aircraft targets $V_{AOA\ SOFT}$.
- When pulling the sidestick to the aft hard stop, the aircraft targets $V_{AOA\ HARD}$.

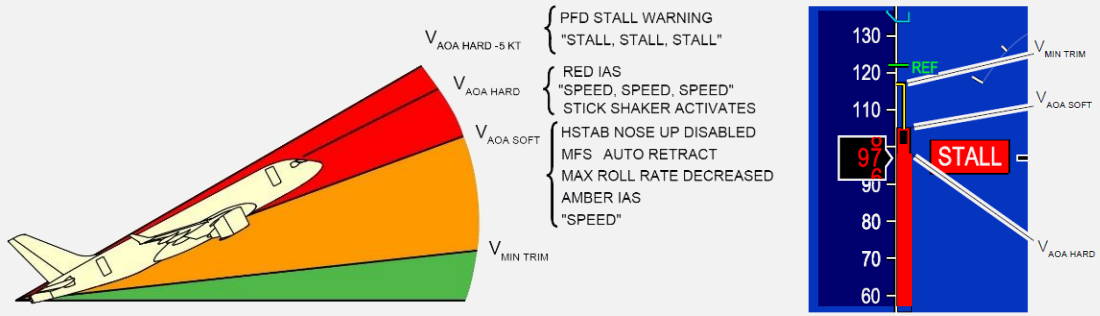
The angle of attack limit is automatically adjusted when in icing conditions if wing anti-ice is not activated or has failed (ALPHA LIMIT on EICAS), during abnormal slat / flap configurations, when side slip angle is greater than 5° or when speed is above MACH 0.65 (to allow for spoiler deployment during emergency descent without stick shaker activation).



# Normal Mode

## Pitch Control

- When the autopilot is disengaged, the **trim speed** is set manually using the pitch trim switches on each sidestick. Without sidestick input the aircraft will target trim speed.
- The FBW system inhibits **stabilizer movement** under certain conditions – during high AOA, the stabilizer is only allowed to trim down. During overspeed, the stabilizer is only allowed to trim up. Nose-down auto trim is disabled below 50 ft AGL.
- Automatic **nose-up compensation** is provided for bank angles up to 33°.
- When both thrust levers are set for take-off, elevators (at neutral sidestick) are moved 12° down until 80 kts to increase **nosewheel steering effectiveness**.
- The sidestick has a **soft stop** and **hard stop** on the pitch axis, allowing for limited (safe) excursions outside the normal load factor and AOA envelope.
- **HPP – High Pitch Protection:** nose-up pitch is limited based on airspeed, on ground nose-up pitch attitude is limited to 17°. Nose-down pitch is always limited to -20°.
- **HAP - High AOA Protection**:







- **High speed protection** applies nose-up trim and deploys multi-function spoilers to return to trim speed. Only active when slats / flaps are retracted.
- **Tail strike protection** reduces pitch rate during aggressive rotations, it compensates for CG position but cannot guarantee tail strike protection under all conditions.
- **Tailplane AOA protection** limits elevator commands as a function of airspeed and includes **elevator surface split limiting** protection.

## Roll / Yaw Control

- When the sidestick is released with bank angle less than 30°, bank angle is maintained. Maximum bank angle is 80°, further limited based on pitch and IAS.
- Above 31.500 ft, the half bank mode is automatically active (NAV and HDG modes).
- Turn coordination and aileron lift augmentation (ailerons droop when flaps selected).
- From 1.55 to 2.1 g, both ailerons move 10° up (wing maneuver load alleviation).
- Rudder travel limiter to limit structural loads based on airspeed and flap position.
- Partial compensation in case of engine failure or other failures causing sideslip. Sideslip angle is limited to 5°, bank angle is limited to 20°.
- Wing Drag Optimization – WDO: with flaps up and above ≈ 250 kts, both ailerons are commanded 2° trailing edge down to compensate for the ailerons "floating up".

## Ground-air-ground Transition

On ground:
- Direct, full authority of the elevator and horizontal stabilizer.
- Automatic setting of trim speed bug to $V_2$ + 10 kts ($V_2$ if engine failure detected).
- Sidestick shaker function enabled above 60 kts and yaw damper above 70 kts.
- Elevators move 12° down during take-off until 80 kts, reaching 0° by 90 kts.
- Nose-up pitch limited to 17°.

Take-off:
- Minimum speeds are shown on PFD speed tape 1 second after liftoff.
- Speed and auto trim enabled 1 second after lift-off, nose-up auto trim > 50 ft AGL.
- Transition is completed within 3 seconds after aircraft leaves the ground.
- Flight director is displayed 3 s after lift-off, PTM disappears 7 s after lift-off.
- Nose-up pitch limit is changed from 17° to in-flight limit (based on IAS).

Landing:
- Nose-up auto trim disabled < 50 ft AGL, speed and auto trim disabled.
- Direct, full authority of the elevators following weigh-on-wheels.
- Nose-down command when spoilers deploy after landing.
- Nose-up pitch limited to 17°, yaw damping disabled at 70 kts.

# Direct Mode

Autopilot and flight envelope protection are ONLY available in normal mode. Depending on the failure, the system can revert to one of three direct modes. Direct mode is axis specific, different modes might be applied on different axis. PFCC and REU Direct: FLT CTRL DIRECT, AFCU Direct: FLT CTRL DIRECT.

**PFCC Direct:**

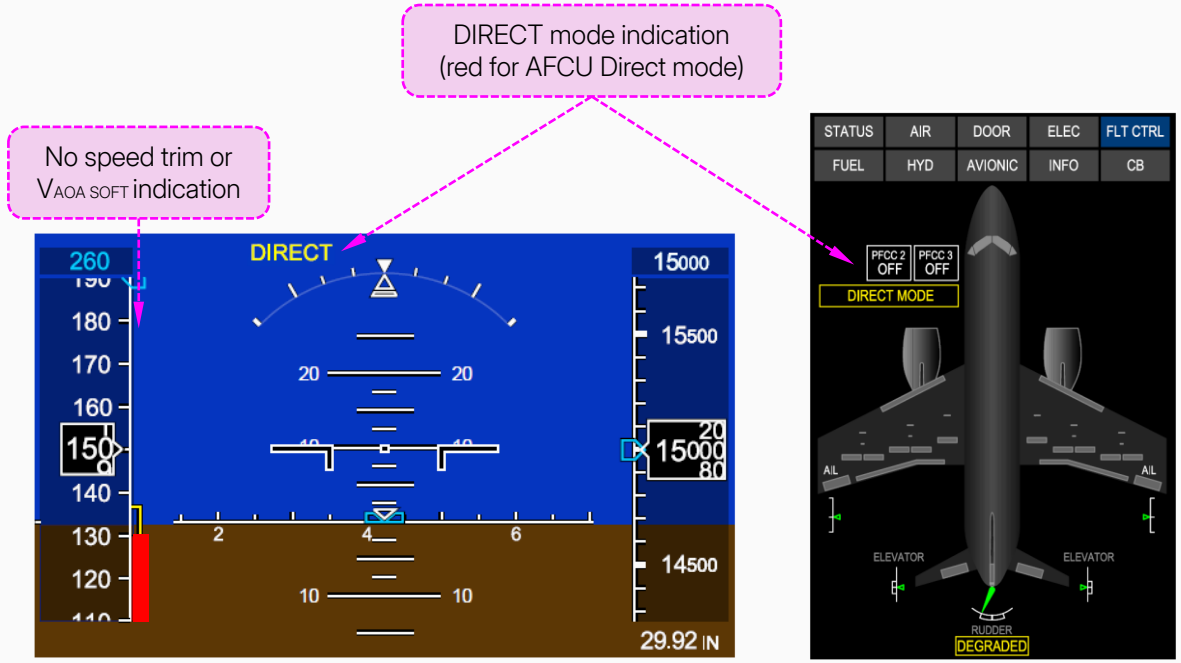
- Input to the PFCCs required for normal mode is degraded (e.g. IRS or ADS).
- MFS and ground spoilers available, MFS automatic deployment at landing is 5°.
- Multifunction spoilers, if extended, do not automatically retract (high AOA).
- Trim switches control horizontal stabilizer directly based of slat / flap position.
- $V_{AOA\ SOFT}$ is removed from the speed tape (amber for low speed, red for stall).

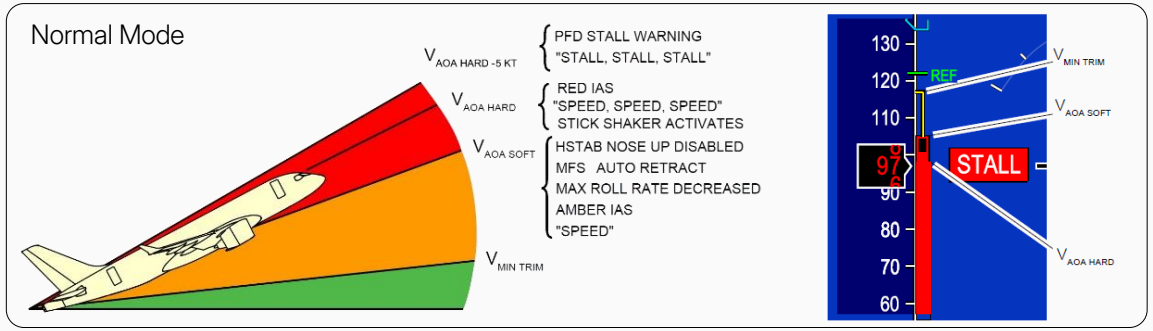
**REU Direct – as PFCC Direct, except:**

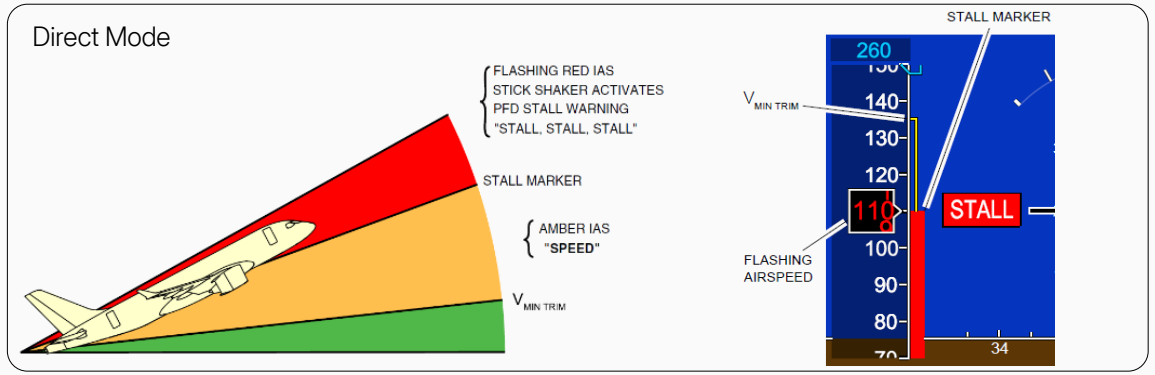
- All PFCCs have failed or all signals to the REUs are lost.
- No selection of sidestick priority.
- Ground spoilers not available.
- No automatic MFS deployment after landing.
- Pitch trim rate is a function of slat / flap position but slower.
- If a PFCC is recovered, it must be reset to enter PFCC direct.

**AFCU Direct – as REU Direct, except:**

- Control requirement is lost due to failure of multiple REUs.
- AFCU interfaces directly with flight controls and PCUs.
- Independent control path from flight controls to control surfaces.
- MFSs and ground spoilers not available (including roll assist).
- The rudder has no travel restrictions (full travel, use with care).
- Pitch trim rate regardless of slat / flap position.
- Automatically reverts to REU direct if REUs are recovered.



# Secondary Flight Controls

### Multifunction Spoilers

In normal mode, when left or right thrust lever TLA is above 14° or during high AOA conditions, the MFSs retract automatically. To redeploy the spoilers, the SPOILER lever must be returned to RET, then back to desired position.

In PFCC direct mode, the MFSs deploy to 5° to prevent excessive nose-up pitch after landing. The SPOILER lever must be manually set to the FULL or MAX position for additional MFS deployment.

### Ground Spoilers

Ground spoilers are only available in normal or PFCC direct mode. Inputs required for PFCC deployment of the ground and multi-function spoilers:

- Both main landing gear are weight on wheels
- Radio altimeter (less than 7 ft)
- Thrust lever position, engine running signal
- Wheel speed, airspeed
- Pitch attitude (for GS deployment only, less than 2.5°)

The spoilers are armed for deployment on ground when wheel speed is greater than 60 kts. Spoilers stow automatically if the thrust levers are advanced above forward idle.

### Horizontal Stabilizer

The horizontal stabilizer is controlled by a dual-channel Horizontal Stabilizer Trim Actuator (HSTA) which includes two electrical motors controlled by a Motor Control Electronics (MCE) unit, receiving signals from the respective REUs.

In normal mode, stabilizer trim is determined by the FBW system. Auto trim automatically offloads the elevators independent of sidestick trim switches (used to set the trim speed only). In direct mode, the trim switches control the horizontal stabilizer directly. The PFCC will disable trimming in-flight if trim is actuated for more than 3 seconds continuously (double beep sound), this is reset by releasing the trim switches.

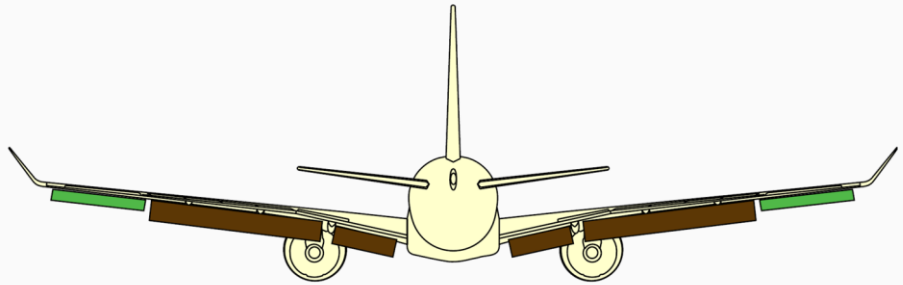
### High Lift System

Slats and flaps are controlled by 2 Slat / Flap Electronic Control Units (SFECUs). Consisting of 4 leading edge slat panels and 2 fowler-type flaps on each wing. They are actuated through 2 hydraulic motors, called Power Drive Units (PDUs) for the slats and 2 PDUs for the flaps (it can operate on 1 motor at half speed). The high lift system is not part of the FBW system but does provide inputs to it.

A hydraulic brake at each wingtip holds the slats and flaps in position between selections or stops the slats / flaps if an abnormal condition is detected (skew detection). Slats use hydraulic system 2 and 3 while flaps use hydraulic system 1 and 3.

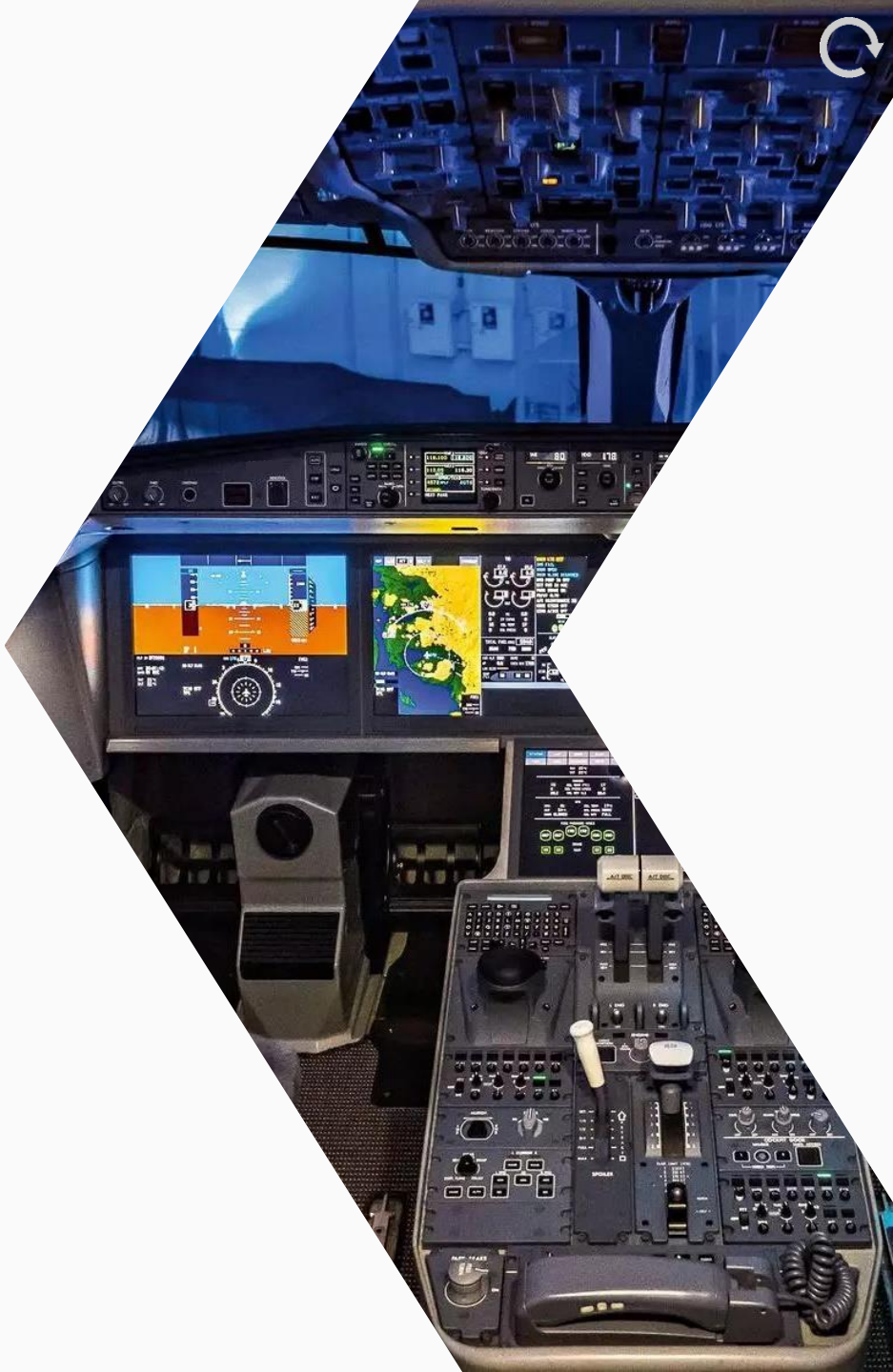
Slat / Flap lever position 2 (go-around for flap 4) and 4 (go-around for flap 5) are gated.

ALTN FLAP deploys the slats / flaps to position 3.



| Lever Detent | Use | Position (degrees) | | |
|---|---|---|---|---|
| | | Slats | Flaps | Aileron Droop |
| 0 | Cruise | UP | UP | 0° |
| 1 | Slats only | 21° | UP | 0° |
| 2 | TKOF (best climb) | 21° | 10° | 10° |
| 3 | TKOF | 21° | 15° | 10° |
| 4 | TKOF (short field) | 24° | 25° | 10° |
| | LDG | 24° | 25° | 0° |
| 5 | LDG (short field) | 27° | 37° | 0° |

# Fuel System



# Normal Fuel Supply

The A220 features 3 fuel tanks (left wing, right wing and center), a collector tank in each wing tank ensures fuel supply to each engine and APU. Fuel flows inboard by gravity, flappers prevent fuel from flowing out when maneuvering.

Fuel capacity at a fuel density of 0.809 kg / ltr for A220-300 (-100):

- Each wing tank: 3050 (3080) kgs
- Center tank: 11.300 (11.480) kgs
- Total fuel: 17.400 (17.640) kgs

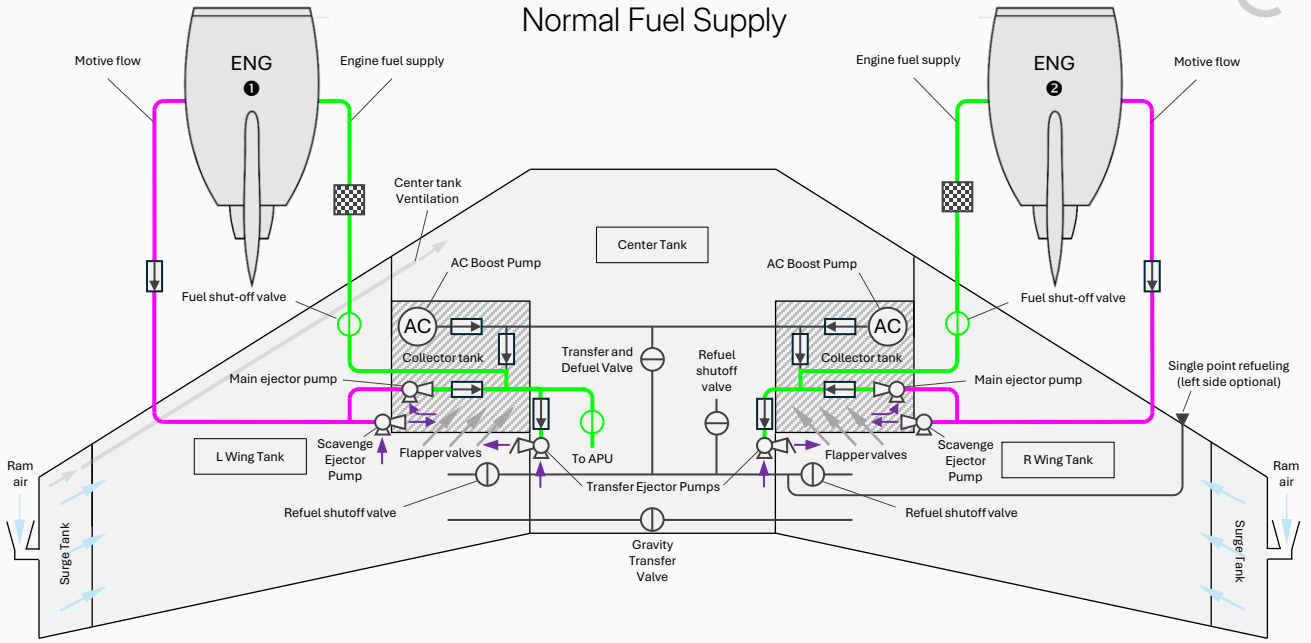
A fuel vent system provides positive pressure during flight (ram air via NACA scoops), left wing and center tank are vented through the left surge tank, right wing tank is vented through the right surge tank.

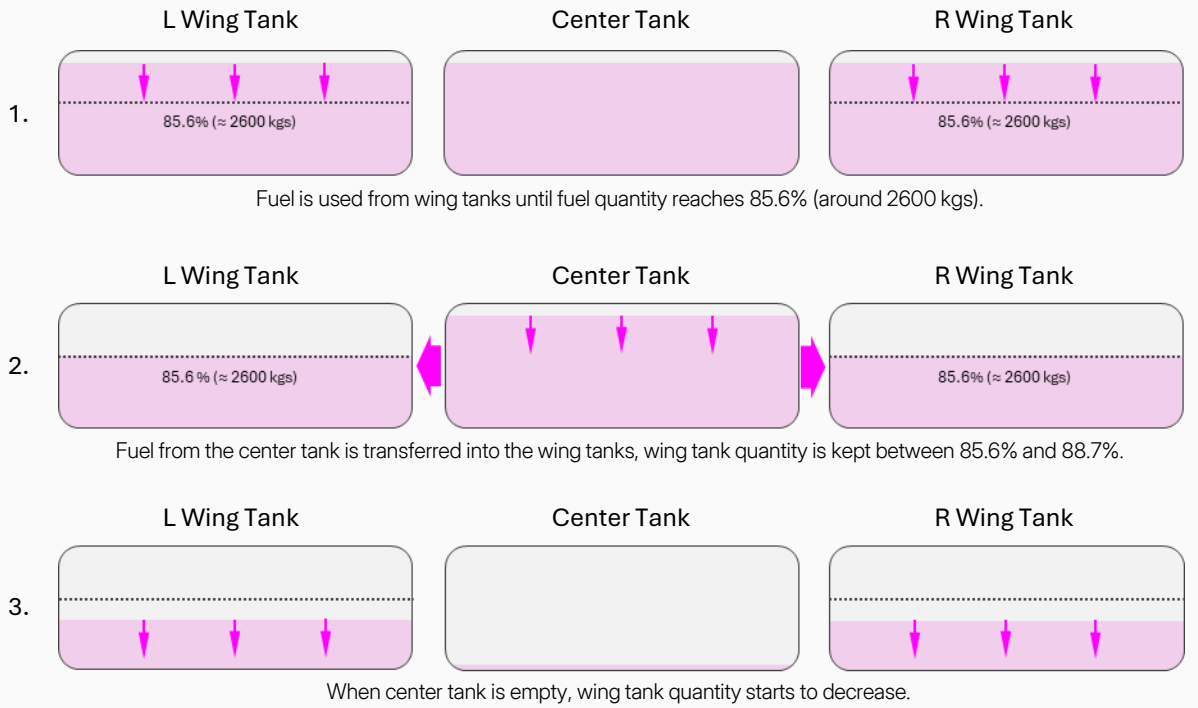
Motive flow (magenta) is the primary mean of fuel transfer. Part of the fuel is returned from the engine driven fuel pump to drive three ejector pumps. Ejector pumps have no moving parts, fuel flows through a venturi, the resulting drop in pressure draws more fuel into circulation. The fuel system uses motive flow for:

- Main Ejector Pump: supply fuel to the engines and APU.
- Scavenge Ejector Pump: transfer fuel from wing tanks to the collector tanks.
- Transfer Ejector Pump: transfer fuel from the center tank into the wing tanks when wing tank fuel quantity drops below 85.6% (≈ 2600 kgs).

An AC boost pump in each collector tank is used as a back-up for the main ejector pump and can also provide motive flow to the transfer ejector pumps.

APU receives fuel through the left main fuel ejector pump (left engine running), left or right boost pump (left boost pump primary) or suction feed if no AC source available and left engine is not running.



# Fuel Transfer

## Manual Transfer

MAN XFR L or R:
- The Fuel Quantity Computer allows 227 kgs to be transferred from wing to wing.
- With BOOST pumps AUTO the respective boost pump automatically activates.
- With both boost pumps manually set to ON, fuel transfer occurs but is reduced.
- FQC closes the transfer / defuel valve and the refuel valve when transfer complete.

MAX XFR CTR:
- The FQC allows 454 kgs (2 x 227 kgs) of fuel to be transferred from wing to center.
- Left or right boost pump must be manually selected to ON in flight (during ground operations both boost pumps can be in AUTO to transfer from both wings to center).
- FQC closes the transfer/defuel valve and the refuel valve when transfer complete.

GRAV XFR:
- Connects the left and right wing tanks, tanks will balance by gravity.

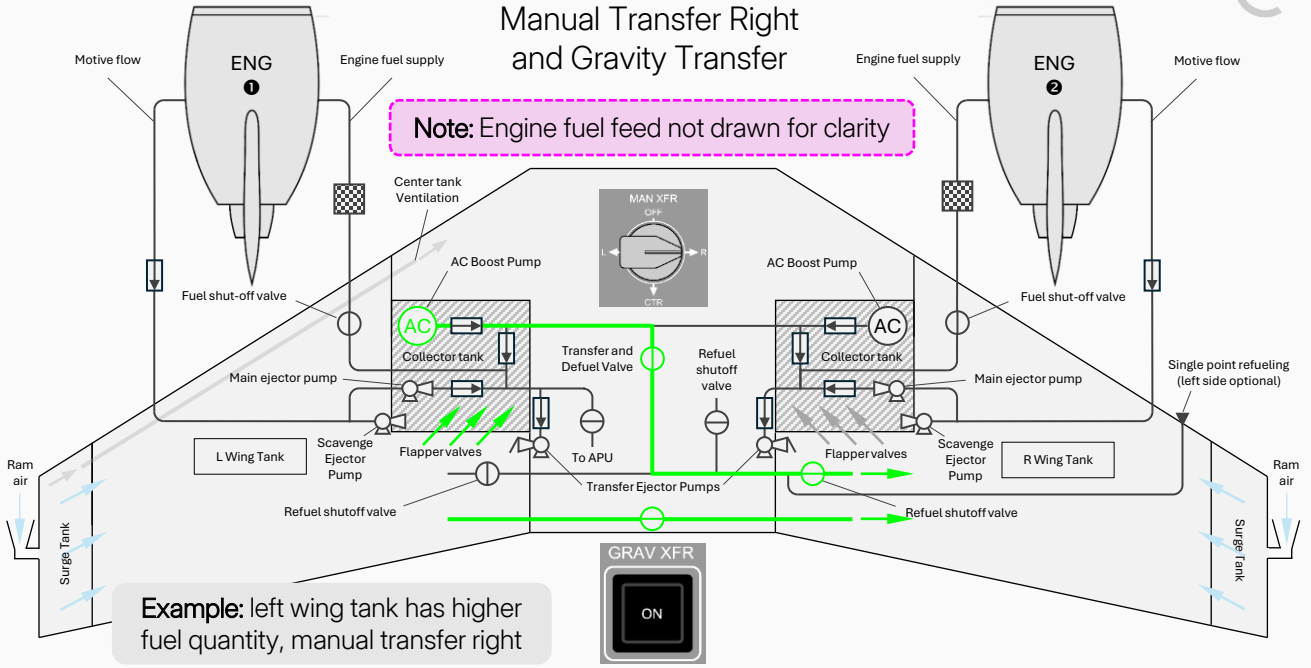
ENGINE FEED:
- In addition to transferring fuel, a fuel imbalance can also be burned off. Selecting either left or right boost pump to ON causes both engines (and APU) to be supplied from one wing tank.

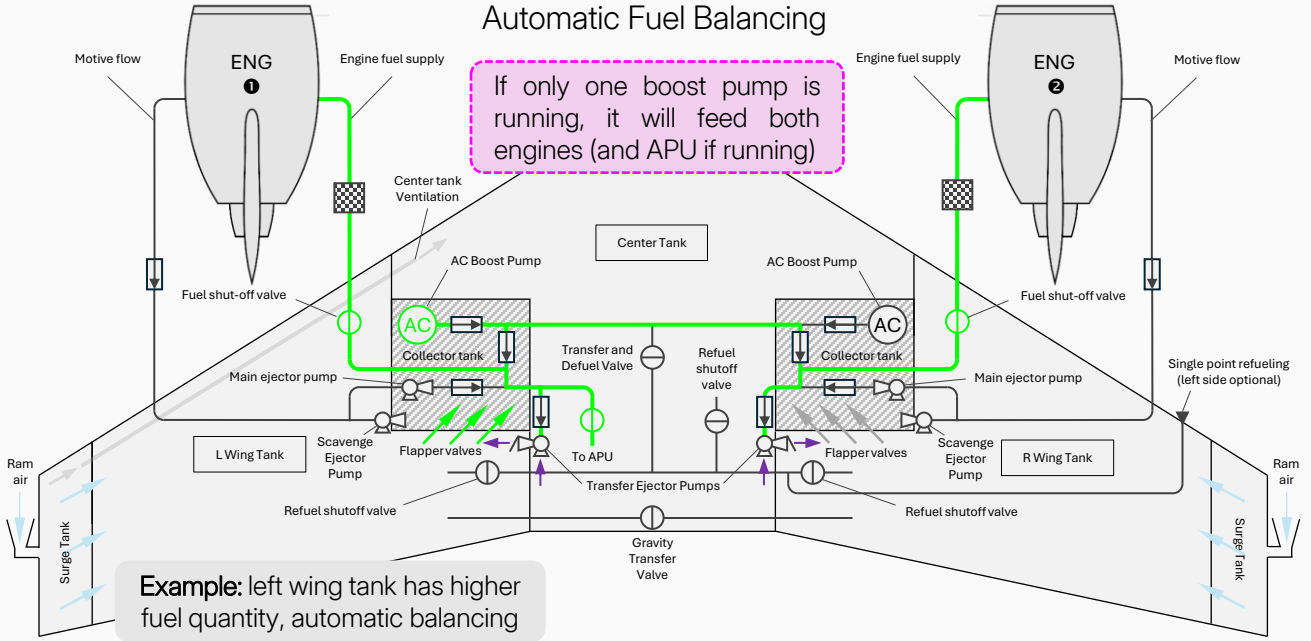
## Automatic fuel balancing

- Only active when center tank is empty (wing tanks kept 85.6% full until center tank empty).
- When boost pumps are in AUTO and the FQC senses an imbalance of more than 180 kgs between wing tanks, the AC boost pump of the heavier tank will automatically run.
- Both engines are fed from the heavier tank until tanks are within 45 kgs.
- If fuel pressure drops in the engine fuel feed line, both AC boost pumps will automatically run to restore engine fuel feed. With both pumps operating, automatic balancing is not available.

## Engine inoperative

- Left boost pump will supply the APU, boost pump on the operative engine side will supply the engine. When imbalance reaches 180 kgs, the FQC will attempt to balance the fuel.
- In case of right engine inoperative, the right boost pump is powered by the APU generator and cannot supply sufficient pressure to overcome the left main ejector pump if the $N_2$ is above 85%. The left tank keeps supplying left engine and APU, imbalance keeps worsening.
- Therefore, the imbalance needs to be corrected manually by the crew (manual transfer).



# Refueling and Defueling

## Fuel Quantity

Each tank contains 2 separate and independent capacitance-type fuel quantity probes, data is sent to the FQC which has 2 channels, one primary, and back-up. L or R FUEL LO QTY alert is generated when fuel quantity in either main tank is less than 442 kg.

A Fuel Tank Inerting System (FTIS) generates nitrogen-enriched (bleed) air to provide non-flammable air to backfill tanks as fuel is consumed.

## Fueling Panel

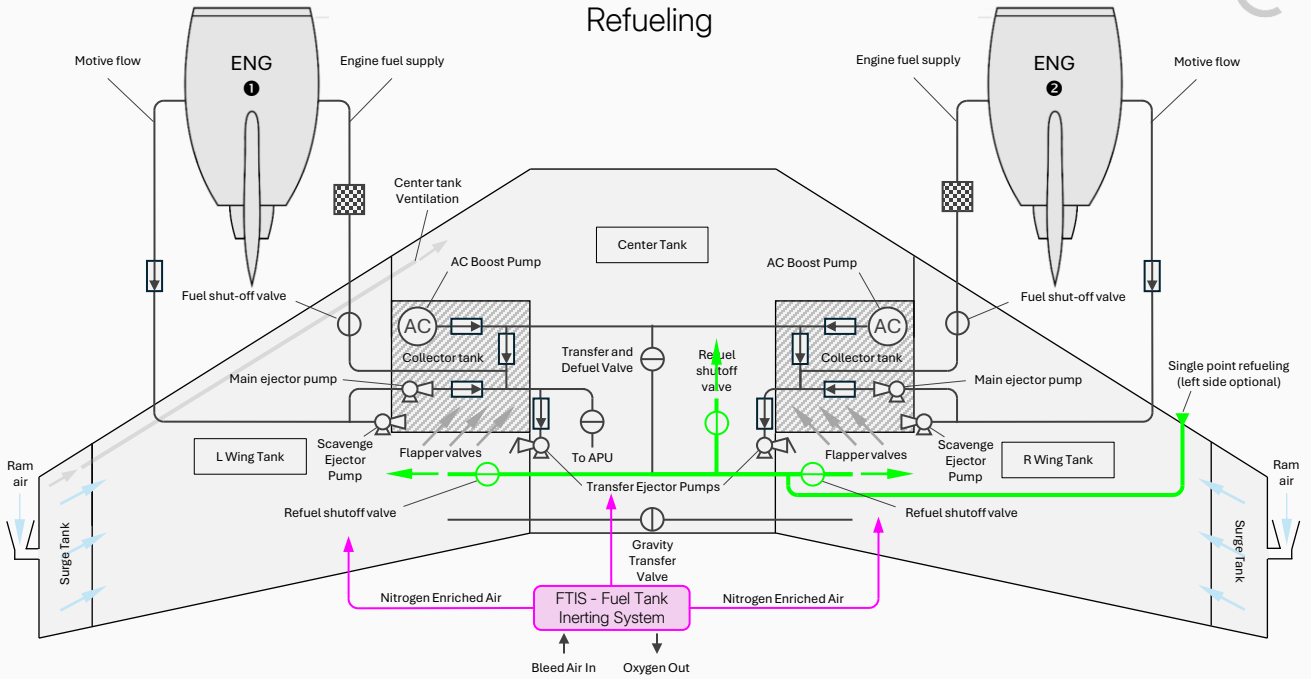
A refuel / defuel panel is located in front of the right-wing leading edge on the wing-to-body fairing. When opened, the FUEL FAULT advisory will display on EICAS with a FUELING DOOR OPEN message on the INFO synoptic page. The panel contains:

- ON / OFF switch – to power / depower the panel.
- PRESEL INCR / DECR spring loaded switch – to set the required quantity.
- REFUEL / DEFUEL switch – to select between refueling and defueling.
- AUTO / MANUAL switch – AUTO, refuelling stops when the preselected quantity is reached. MANUAL requires manual operation of the Left, Center, Right refuel valves.
- START / STOP – hold in spring-loaded START until a spinning animation is shown left of the PRESEL quantity to start refueling operations.

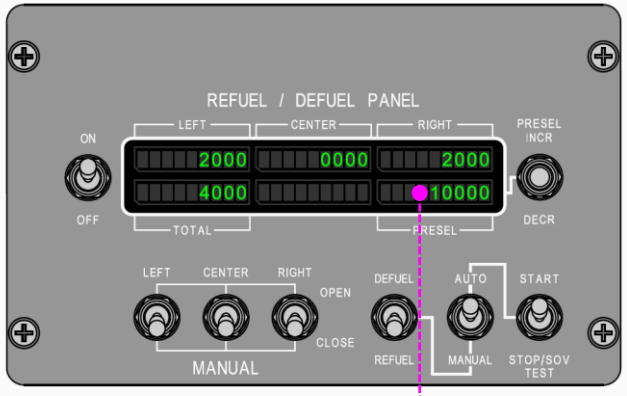
Refueling can be done with battery power only, but AC power is required for defueling. For defueling, the transfer/defuel valve connects the engine crossfeed manifold to the refuel manifold so that fuel can be pumped out of the collector tanks using the boost pumps. Center tank can only be defueled by transferring fuel first to the wing tanks. Refueling point on the left wing is optional. **Average refueling rate is approx. 600 kg/min.**

Automatic fuel distribution when refueling:

a. Up to 5200 kgs – fuel equally distributed between wing tanks (up to 85.6%).
b. 5200 to 6000 kgs – 2200 kgs in each wing (73.0%), remainder in center tank.
c. 6000 to 16.500 kgs – 2600 kgs in each wing (85.6%), remainder in center tank.
d. 16.500 to 17.400 kgs – center tank is full, remainder in wing tanks until full.

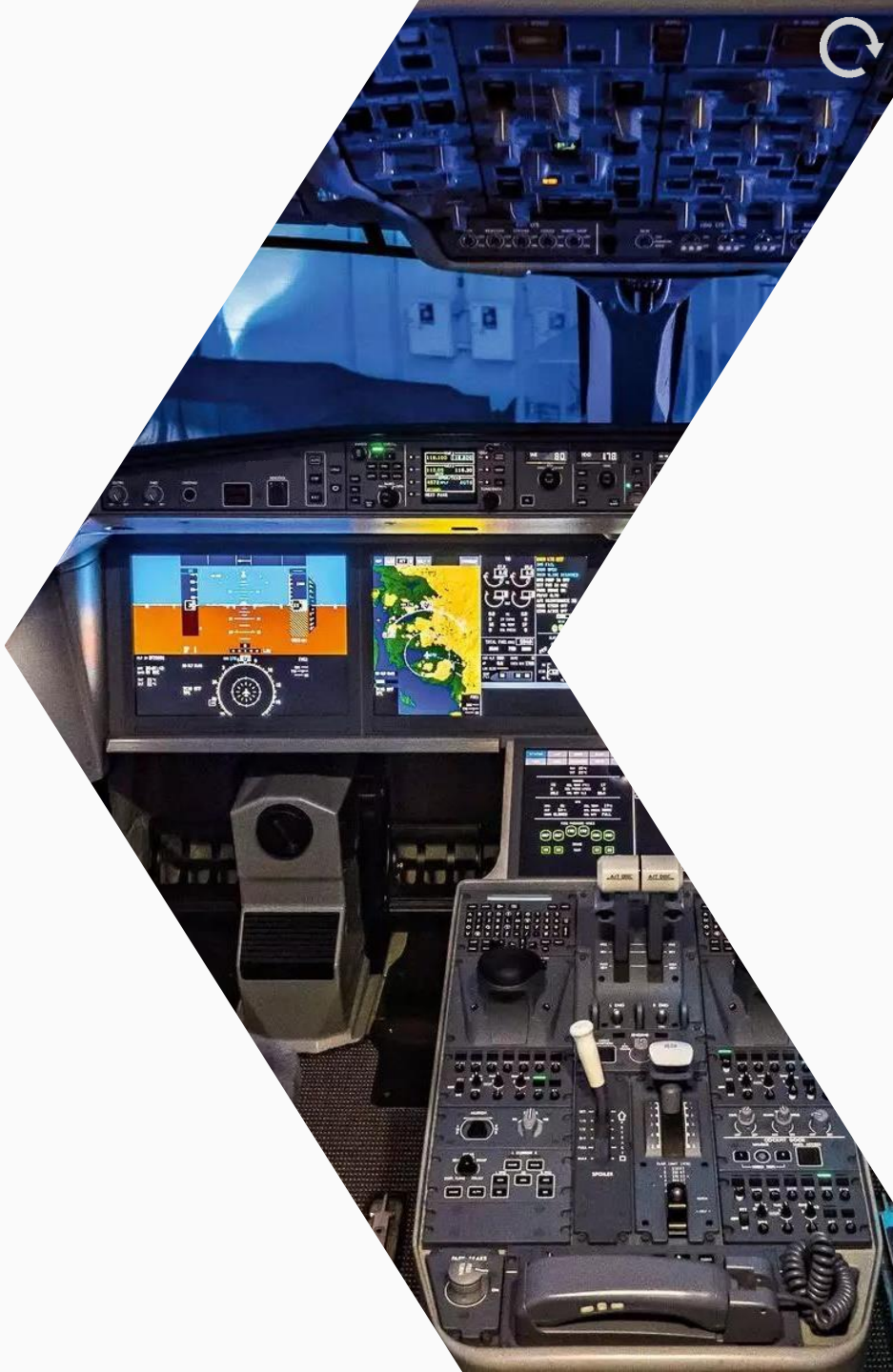



| Preselected fuel quantity | A220-300 Automatic Refuel Sequence - kgs | | |
|---|---|---|---|
| | Left Main | Center | Right Main |
| 2000 kg | 1000 | 0 (Empty) | 1000 |
| 3000 kg | 1500 | | 1500 |
| 4000 kg | 2000 | | 2000 |
| 5000 kg | 2500 | | 2500 |
| 5200 kg | 2600 (85.6%) | 0 | 2600 (85.6%) |
| 6000 kg | 2200 (73.0%) | 1600 | 2200 (73.0%) |
| 7000 kg | 2600 (85.6%) | 1800 | 2600 (85.6%) |
| 8000 kg | | 2800 | |
| 9000 kg | | 3800 | |
| 10.000 kg | | 4800 | |
| 11.000 kg | | 5800 | |
| 12.000 kg | | 6800 | |
| 13.000 kg | | 7800 | |
| 14.000 kg | | 8800 | |
| 15.000 kg | | 9800 | |
| 16.000 kg | | 10800 | |
| 16.500 kg | 2600 | 11300 | 2600 |
| 17.000 kg | 2850 | 11300 (Full) | 2850 |
| 17.400 kg | 3050 (Full) | | 3050 (Full) |




Automatic refueling might stop short of the preselected quantity due to slow fuel redistribution within the center tank. To prevent this from happening, add 50 kgs to the PRESEL quantity. FUEL CTR XFR FAIL illuminates if there is fuel in the center tank while wing tank quantity is below 2000 kgs. During refueling this may be displayed.

# Hydraulic System



# Hydraulic System 1 and 2

Hydraulic power is provided by 3 independent hydraulic systems. Each system delivers 3000 psi. The horizontal stabilizer and wheel brakes are electrically powered.

Pressure for system No. 1 and No. 2 is generated by:

| Hydraulic System 1 | Hydraulic System 2 |
|---|---|
| Engine Driven Pump – EDP 1A | Engine Driven Pump – EDP 2A |
| Power Transfer Unit – PTU (back-up) | AC Motor Pump – ACMP 2B (back-up) |

A Shut-off Valve (SOV) stops hydraulic supply to the EDP in case of:

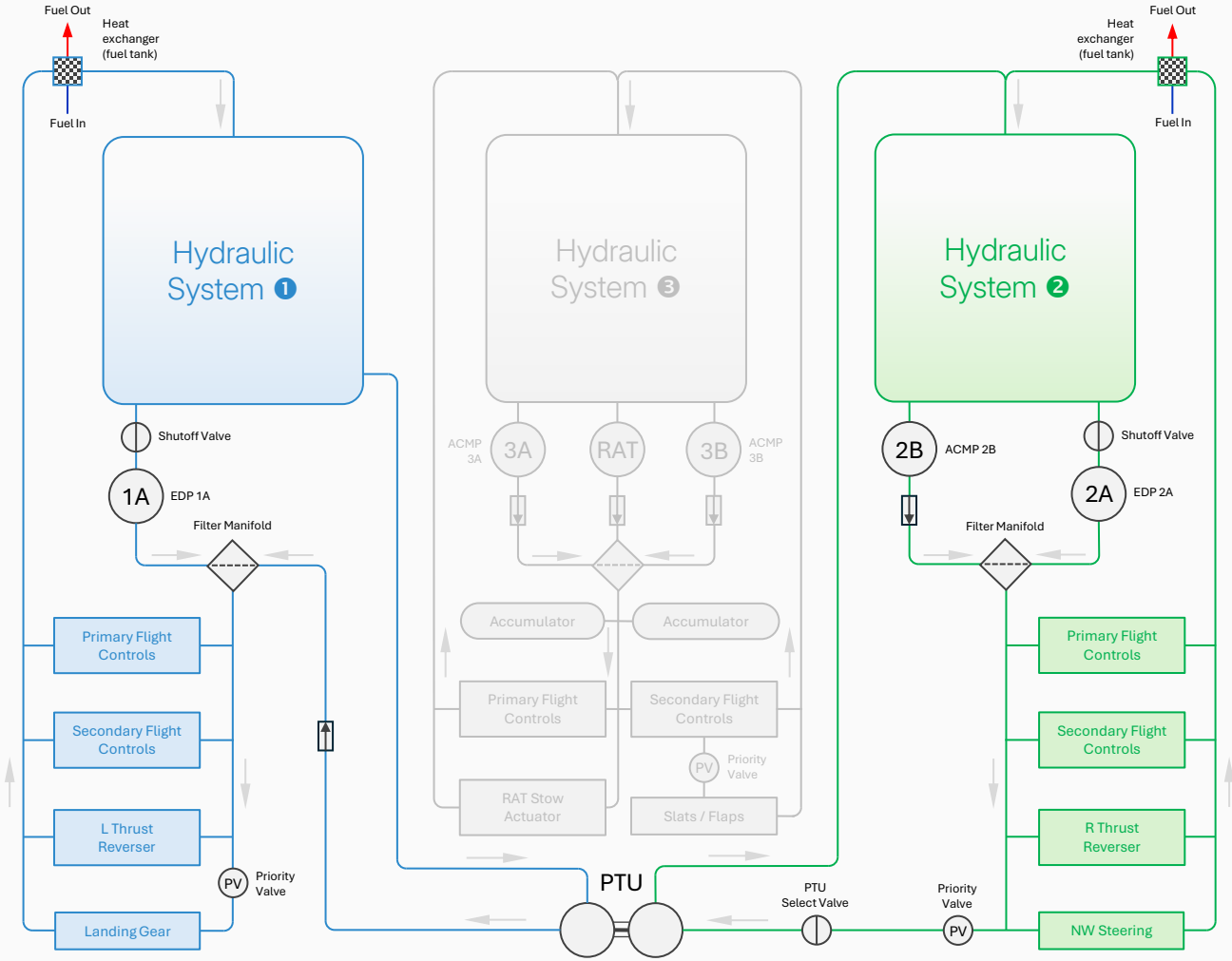
- Engine fire (pressing L or R ENG FIRE switch)
- Hydraulic fluid overtemperature (automatic)
- Manual selection

Both systems have a Priority Valve (PV) to preserve system pressure and flow capacity for high priority users if there is a pressure drop to 1700 psi or less. In system 1, the PV isolates the landing gear to maintain pressure for flight controls and left thrust reverser. In system 2, the PV isolates the PTU to maintain pressure for flight controls and right thrust reverser. PTU transfers power from HYD 2 to HYD 1 without exchanging fluid.

PTU and ACMP 2B will automatically operate in the following situations (if in AUTO):

| PTU | ACMP 2B |
|---|---|
| No overtemperature, fluid quantity more than 5%, | |
| Hydraulic system 2 pressurized, and: | AC electrical power available, and: |
| Take-off and landing phase: | |
| a. From take-off thrust applied until slats / flaps fully retracted<br>b. From slats / flaps deployed to F4 or F5 (landing position) or ALTN FLAP (F3) selected until slats / flaps are fully retracted | |
| Engine or hydraulic pump fail: | |
| Engine Driven Pump 1A or left engine failed | Engine Driven Pump 2A or right engine failed |
| Single engine taxi (PARK BRAKE – OFF): | |
| Taxi on right engine (left engine shutdown) | Taxi on left engine (right engine shutdown) |

Additionally, ACMP 2B runs whenever PTU is ON (automatic or manually selected on).




| Hydraulic System 1 | Hydraulic System 3 | Hydraulic System 2 |
|---|---|---|
| Primary flight controls | Primary flight controls | Primary flight controls |
| L ELEV OB, RUDDER UPR | L AIL IB, R AIL IB | L AIL OB, R AIL OB |
| Secondary flight controls | L ELEV IB, R ELEV IB | R ELEV OB, RUDDER MID |
| MFS L1 / R1 | RUDDER LWR | Secondary flight controls |
| MFS L3 / R3, L & R GS | Secondary flight controls | MFS L4 / R4 |
| FLAPS PDU | MFS L2 / R2 | SLATS PDU |
| Engine | SLATS PDU | Engine |
| LEFT THRUST REVERSER | FLAPS PDU | RIGHT THRUST REVERSER |
| Landing gear | Electrical system | Landing gear |
| LANDING GEAR EXTEND / RETRACT | RAT STOW ACTUATOR | NOSEWHEEL STEERING |

# Hydraulic System 3

Pressure for system No. 3 is generated by:

| Hydraulic System 3 |
| --- |
| AC Motor Pump – ACMP 3A |
| AC Motor Pump – ACMP 3B (back-up) |
| Ram Air Turbine – RAT Pump (emergency) |

System 3 includes a PV to preserve system pressure and flow capability for high priority users if there is a pressure drop in the system. The PV isolates the flap and slat Power Drive Units (PDUs) to maintain pressure for flight controls.

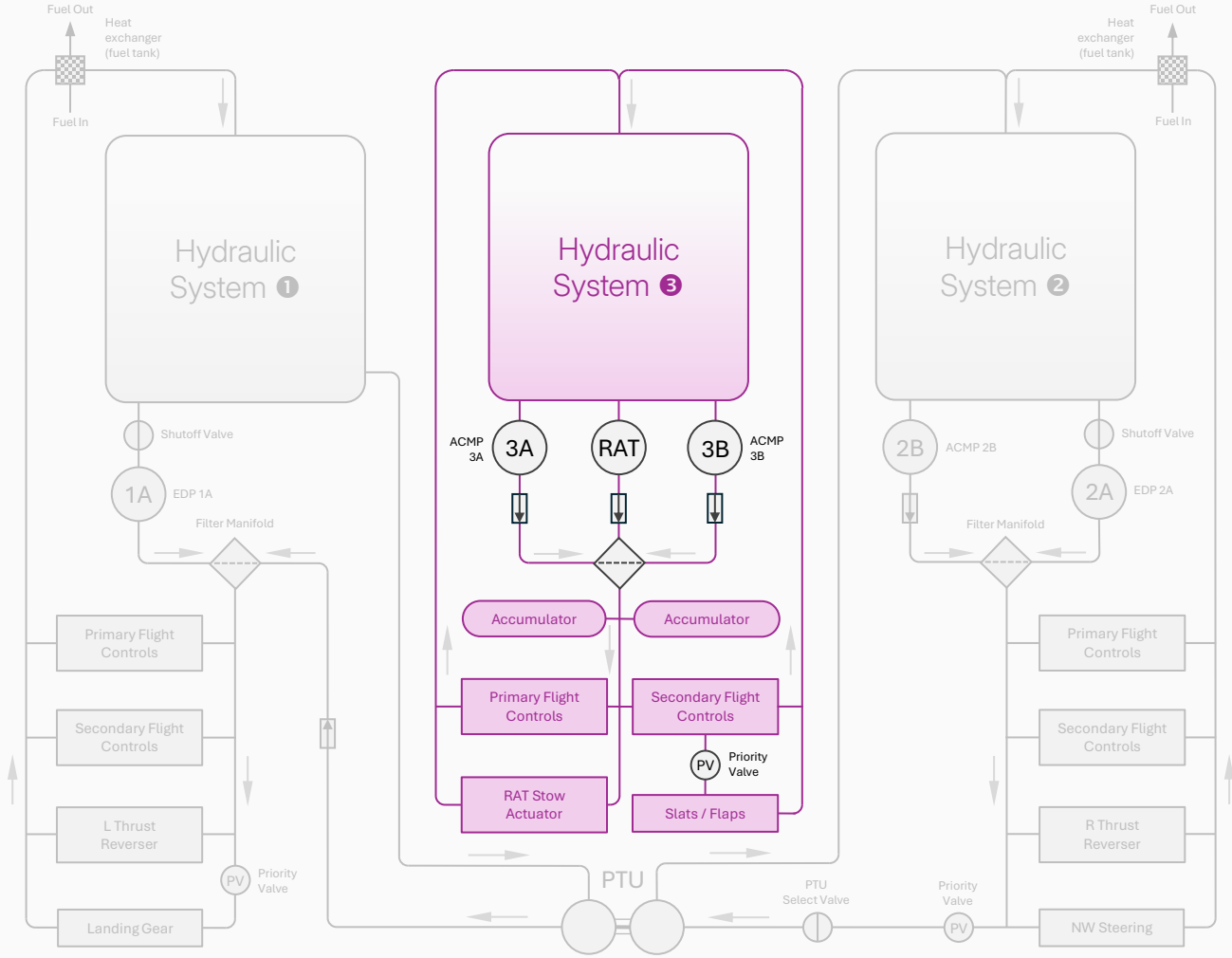
The RAT will automatically deploy and provide back-up hydraulic pressure to system 3 if there is a total loss of AC power in flight. During the deployment period, two accumulators maintain hydraulic pressure. The nominal hydraulic pressure is 2850 psi. The RAT minimum speed for hydraulics is 115 kts.

ACMP 3A will automatically operate in any the following conditions (if in AUTO):

- No overtemperature, AC power available and fluid quantity more than 5%, pressure less than 3350 psi, and
- Left or right engine started, or
- When airborne.

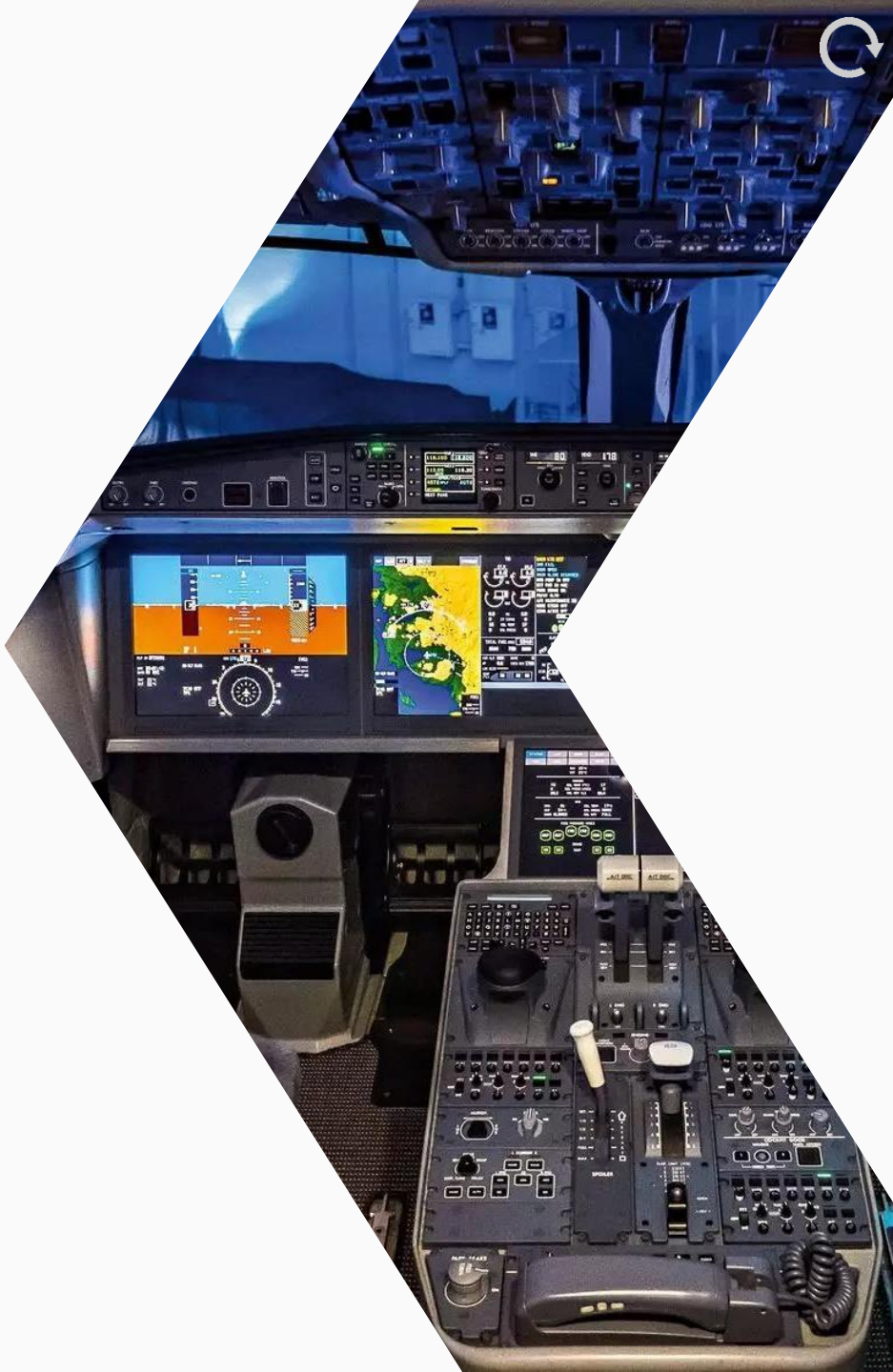
ACMP 3B will automatically operate in any the following conditions (if in AUTO):

- No overtemperature, AC power available and fluid quantity > 5%, and
- Take-off thrust applied, or
- The slats / flaps are in transit (on ground), or
- The slat / flap handle or position is out of zero (in flight), or
- ACMP 3A has failed.




| Hydraulic System 1 | Hydraulic System 3 | Hydraulic System 2 |
| --- | --- | --- |
| Primary flight controls | Primary flight controls | Primary flight controls |
| L ELEV OB, RUDDER UPR | L AIL IB, R AIL IB | L AIL OB, R AIL OB |
| Secondary flight controls | L ELEV IB, R ELEV IB | R ELEV OB, RUDDER MID |
| MFS L1 / R1 | RUDDER LWR | Secondary flight controls |
| MFS L3 / R3, L & R GS | Secondary flight controls | MFS L4 / R4 |
| FLAPS PDU | MFS L2 / R2 | SLATS PDU |
| Engine | SLATS PDU | Engine |
| LEFT THRUST REVERSER | FLAPS PDU | RIGHT THRUST REVERSER |
| Landing gear | Electrical system | Landing gear |
| LANDING GEAR EXTEND / RETRACT | RAT STOW ACTUATOR | NOSEWHEEL STEERING |

# Ice & Rain Protection



# Ice and Rain Protection

## Ice Detection

Ice detector will continue to indicate icing conditions for one minute after icing was last detected (ICE advisory on EICAS or ICE caution if anti-icing OFF). In AUTO, the anti-ice system will remain ON for an additional 2 minutes (3 minutes after last icing detected).

## Wing anti-ice

Slats inboard and directly outboard of the engine nacelles (slat 1 and part of slat 2) are unheated. Ice accumulation is not considered significant in these areas, for the same reason the vertical and horizontal stabilizers are unheated.

APU bleed cannot supply air for the WAIS system and the IASC will therefore change bleed source to engines prior to opening WAIVs.

## Cowl anti-ice

Two Cowl Anti-ice Valves (CAIVs) are mounted in series per engine and regulate 6th stage compressor bleed air, they are fail-safe to open position.

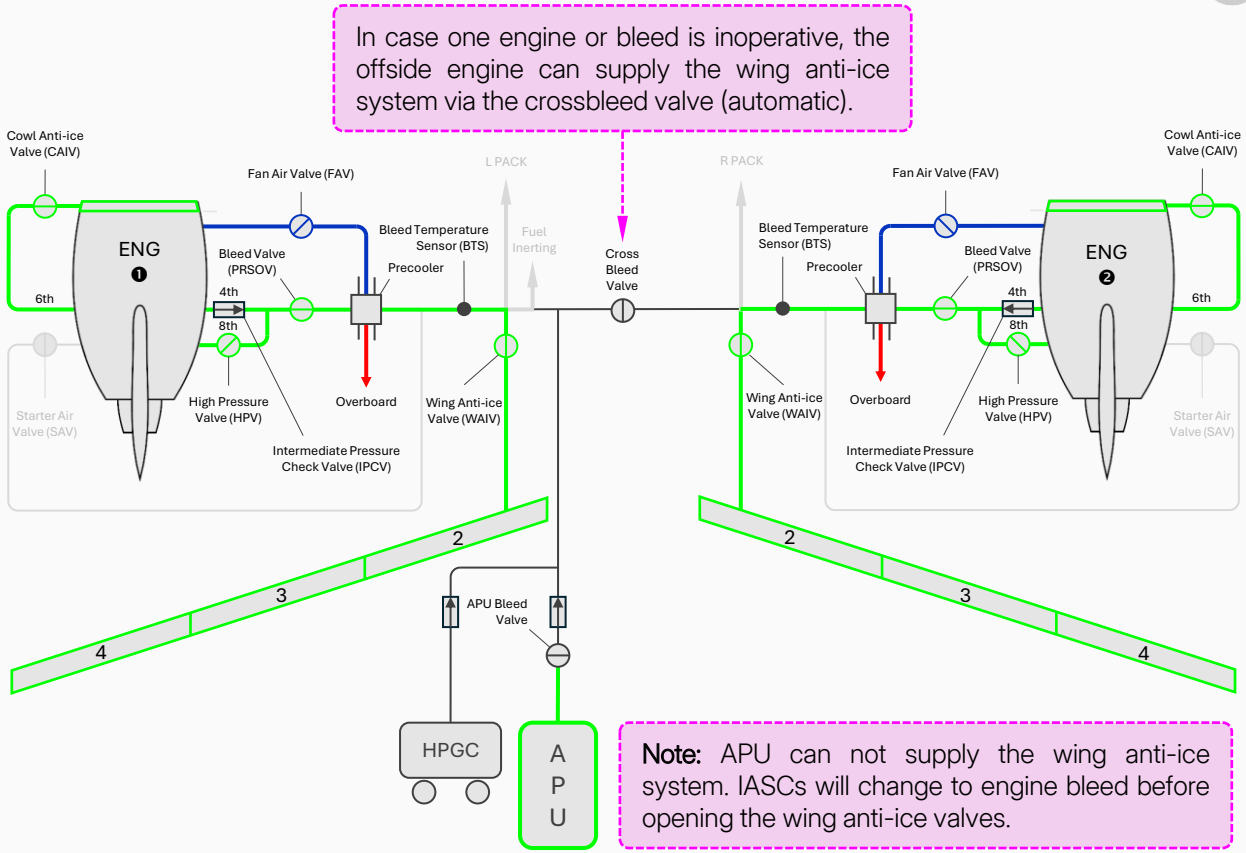
## Take-off

- Wing / Cowl Anti-ice AUTO: if icing conditions are encountered during take-off, WAIS and CAIS will not activate (inhibited) until 1500 ft or 2 min after take-off.
- Wing / Cowl Anti-ice ON: if icing conditions are anticipated during take-off, WAIS and CAIS are set ON before take-off. WAIS is inhibited on ground below 60 kts (armed). CAIS is active if engine is running, inhibited if outside temperature is > 15°C.

## ADS Probe and Sensor Ice Protection

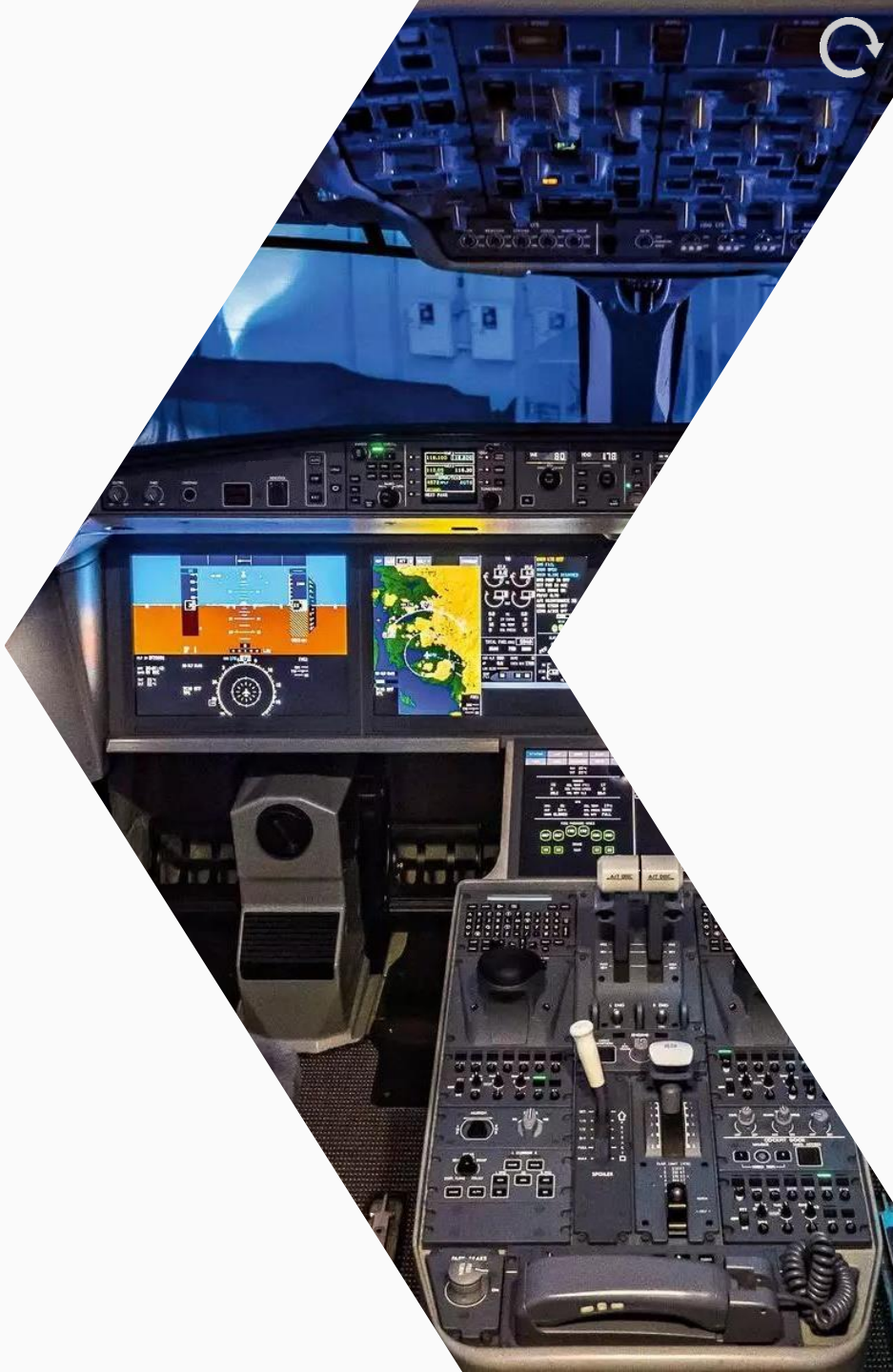
ADSP, AOA and TAT heating are automatically on when any engine is running, or weight-of-wheels detected (full heat above 55 kts or weight-off-wheels). On ground with engines off, PROBE HEAT set to GND ON will heat TAT and ADSP probes 2 min.

$P_2T_2$ probe is heated when engine is operating. AOA sensors are always heated.




| | | |
|---|---|---|
| Rain / Snow Removal | Windshield Wiper x 2 → synchronized if selected to same speed, maximum operating speed 250 kts. | |
| Ice Detection | Ice Detector x 2 → icing condition status to IASCs (WAIS), FADEC (CAIS) and FBW (stall warning parameters). | |
| Ice Protection | Engine Bleed Air:<br>• 4th or 8th compressor stage for WAI<br>• 6th compressor stage for CAI | Wing Anti-Ice System (WAIS), slats 2, 3 and 4 |
| | | Cowl Anti-ice System (CAIS) |
| | Electrical | Windshield heat (on whenever AC available) |
| | | Side window heat (on whenever AC available) |
| | | Air Data Smart Probes (ADSP) x 4 |
| | | Total Air Temperature Probes (TAT) x 2 |
| | | $P_2T_2$ probes (one per engine) |
| | | Angle of Attack Probes (AOA) x 2 (always on) |

# Landing Gear



# Landing Gear

Two Landing Gear and Steering Control Units (LGSCUs) operate in an active – standby configuration and change role each flight. They control the gear extension and retraction cycles, two Brake Data Concentrator Units (BDCUs) control the electric brakes. When up, the nose gear is fully enclosed within the nose gear bay, the main gear wheels are not fully covered when retracted (Boeing 737 style). For parking and towing purposes gear pins are installed in each landing gear leg.

Four proximity sensors (2 for up and 2 for down) are installed on each gear leg. The landing gear actuator remains pressurized in the extended position. In the retracted position, the gear is held up by mechanical up-locks, the actuators are not pressurized. 30 seconds after the gear is UP, landing gear indications are removed from the EICAS and will reappear when the landing gear is either selected down or in case of a fault. When on ground, the LGSCU inhibits landing gear retraction.

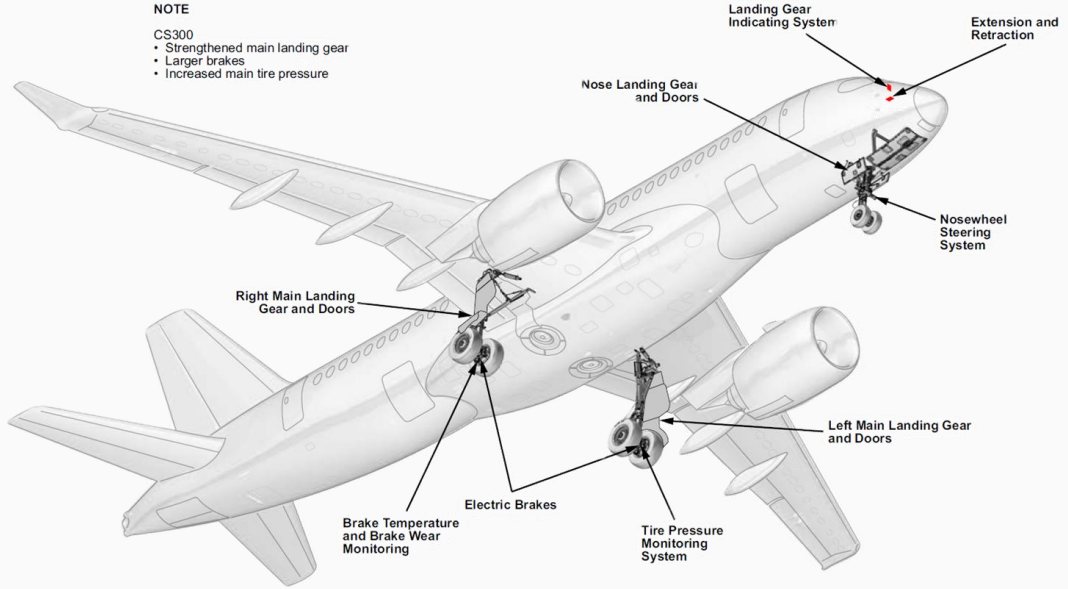
The "GEAR" aural warning can be muted if:

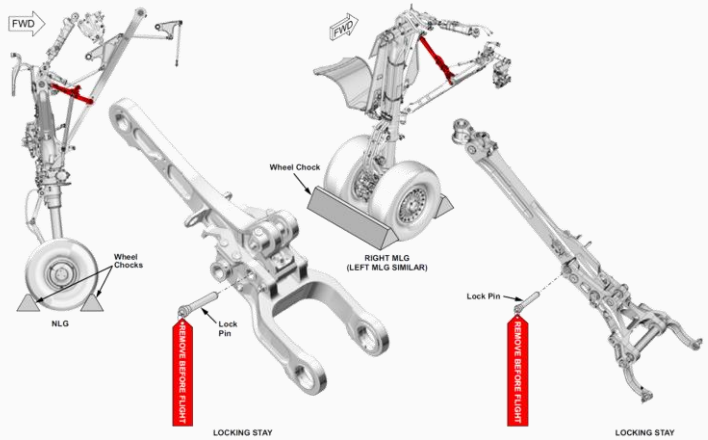
- The landing gear is not down and locked, and
- All radio-altimeters have failed, pressure altitude is below 15.000 ft, and
- Both thrust levers are at idle, and
- The SLAT / FLAP lever is in the landing position.

The "GEAR" aural warning cannot be muted if:

- The landing gear is not down and locked, and
- RAD ALT less than 1000 ft AGL with both thrust lever set to approach idle, or
- The SLAT / FLAP lever is in the landing position

The alternate gear extension allows for free-fall extension if normal extension fails. When ALTN GEAR switch is selected down, electro-mechanical actuators release the up-lock hooks and allow for a free-fall landing gear extension. Every 550 flight hours the LGSCU performs an alternate extension instead of normal extension to test alternate extension circuitry (ALTN GEAR DN displayed on EICAS).

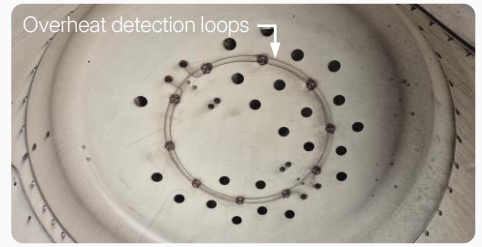



< 30 seconds

Gear indications are displayed on EICAS and removed 30 seconds after gear is up. White box is shown if gear in transit, white box with letters "UP" if gear is up and locked. Amber box if gear disagrees with gear lever (after time delay) or red for disagree during approach or landing.

Newer aircraft have holes drilled into the main gear wheel wells to get rid of sound resonance that older aircraft were suffering from. Also note the location of the main gear bay overheat detection loops.



# Nosewheel Steering

The Landing Gear and Steering Control Units (LGSCUs) command and monitor the steer-by-wire system. It receives inputs from the rudder pedals and nosewheel steering tiller (left console, right console optional).

Max nosewheel steering deflection:

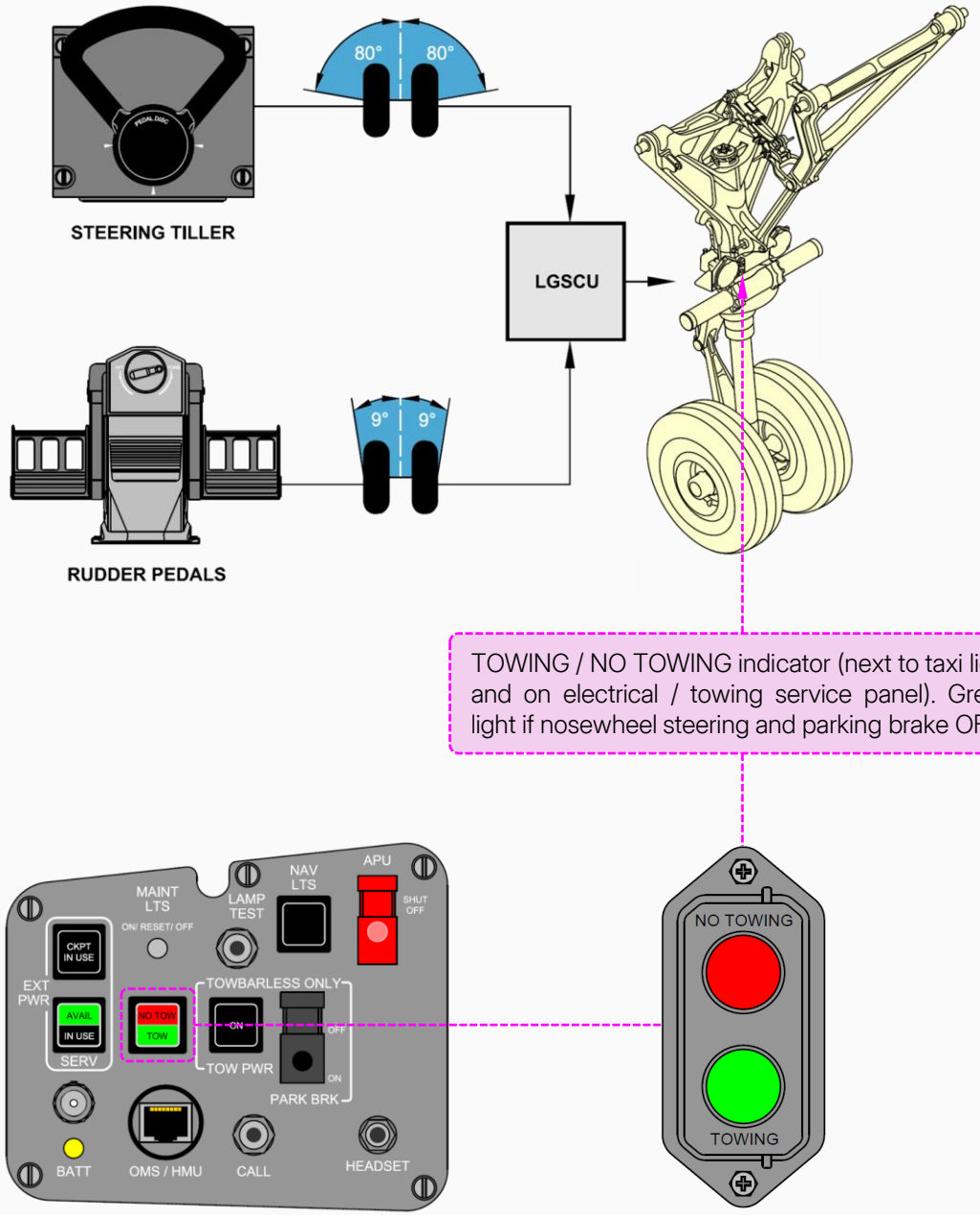
- Steering tiller: 80°
- Rudder pedals: 9°
- Towing (nosewheel steering OFF): 130°

Below 30 kts groundspeed the steering tiller controls the nosewheel up to 80°, between 30 kts and 100 kts the maximum nosewheel deflection automatically decreases from 80° to 9°. Above 100 kts the nosewheel deflection remains limited to 9°.

When rudder pedals and steering tiller are used at the same time (same or opposite direction), the LGSCU algebraically adds the two commands, resulting in an increase or decrease in steering angle.

For towbarless towing only, the parking brake (if aircraft not powered) and nosewheel steering can be switched off via the service panel. A towing (green), no towing (red) light on the service panel and nose landing gear leg indicates whether the steering is off with no brake pressure applied to start towing.

PEDAL DISC switch on the steering tiller (center) disconnects the rudder pedals for ground rudder checks without moving the nosewheel.



# Wheel Brakes

The A220 features electric brakes. Tires are equipped with overpressure protection plugs (400 psi) and fusible plugs (200°C).

Two Brake Data Concentrator Units (BDCUs) control and monitor the Electric Motor Actuators (EMAs), brake temperature, wheel speed and tire pressure. Each has a NORMAL and ALTERNATE module and work simultaneously in tandem. Should one fail, the other BDCU takes over the entire braking system.

Tire pressure, brake temperature and wear (only if < 5% of brake life remaining, equivalent to less than 100 landings) are displayed on the STATUS synoptic page and associated EICAS messages.

Each brake has 4 EMAs (2 pairs), each pair receives signals from a different Electric Motor Actuator Controller (EMAC) controlled by different Electric Motor Control Unit (EMCUs) for safety and redundancy.

ALTN BRAKE, when selected, bypasses the BDCU and provides direct control to the EMAs, autobraking and braking system protections are lost.

Parking brake locks the electric brakes mechanically, they then remain in a locked position without electrical power applied. Parking brake can be used to stop the aircraft when normal and alternate braking fails (no protections). If activated in the air, parking brake is failed until system is repowered on ground. **When holding position on ground for more than 1 minute, it is advisable to set parking brake, holding brakes manually for long periods can overheat the EMAs.**

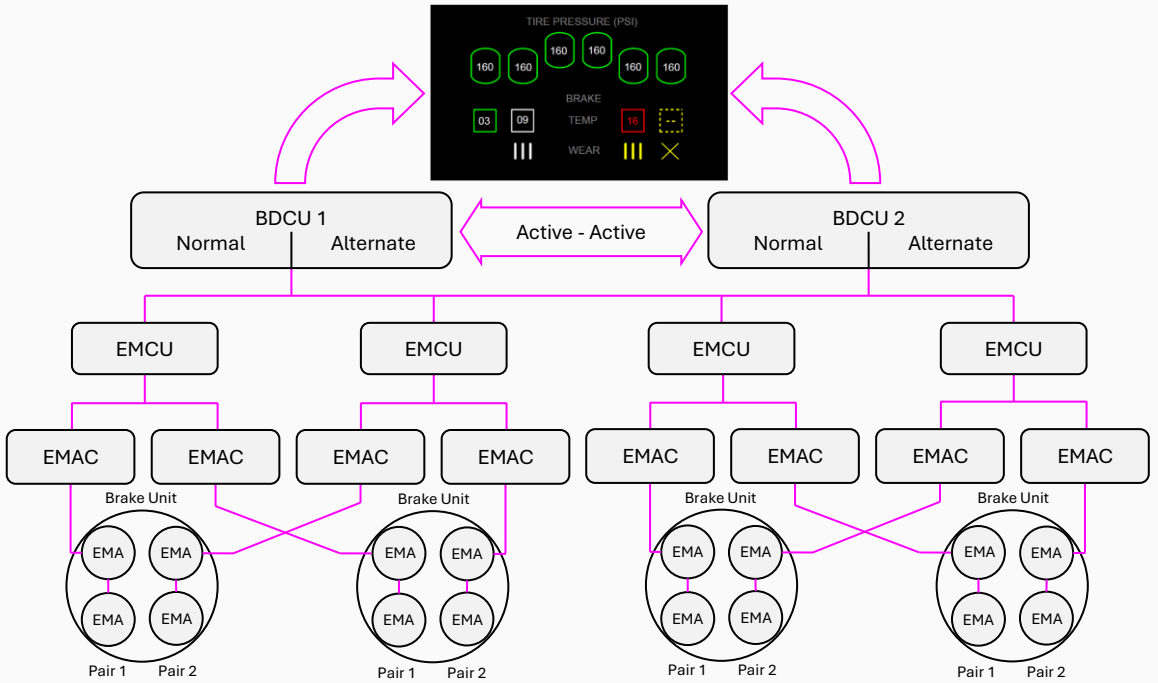
With autobrakes set to RTO and above 70 kts, should the thrust levers be set to idle, the autobrakes activate with maximum deceleration force. For landing autobrakes can be set to LO, MED or HI. Autobraking deactivates when manual braking is applied (> 20%) or autobrake switch is set to OFF.

Braking system protections:

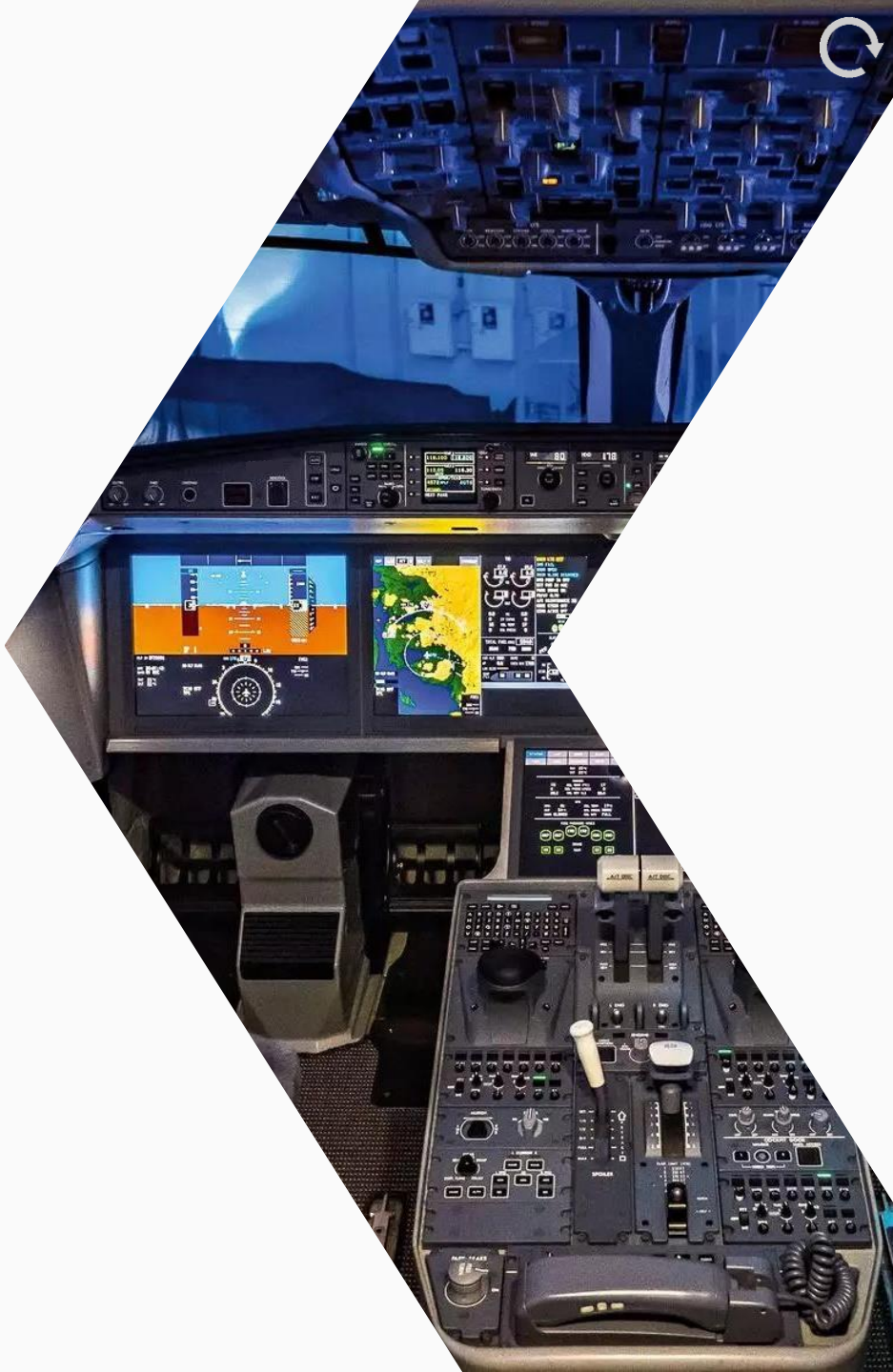
- Anti-skid protection – above 10 kts wheel speed.
- Landing gear retraction braking – brake signal for 5 seconds during retraction.
- Touchdown protection – prevents brake actuation before touchdown occurs.
- Locked wheel protection – releases brake if locked wheel detected > 30 kts.
- De-rotation – limits brake force with nose gear still in the air, it deactivates upon nose landing gear weight-on-wheel, nose-up pitch angle less than 1.5° or 5 seconds have elapsed since main wheel spin-up.

||| = approximately 100 landings left (5%)
||| = replace brakes immediately.

Brake temperature: below 6 is green, 7 to 14 is white. 15 and over is red (overheat, no dispatch).



# Lighting



# Lighting

FASTEN SEATBELT and NO SMOKING / NO PED (as installed) AUTO position: signs are ON below 10.000 ft. Cabin crew can control the signs above 10.000 ft via the CMS.

Entrance and “dome finder” lights, when activated, will be on for 20 minutes when aircraft is not electrically powered. Cabin advisory lights description (see below).

The nose taxi and nose landing lights are inhibited until the nose gear is in the down and locked position. Landing lights project a 3°, focused beam of light. Taxi lights project a wider, 20° pattern.

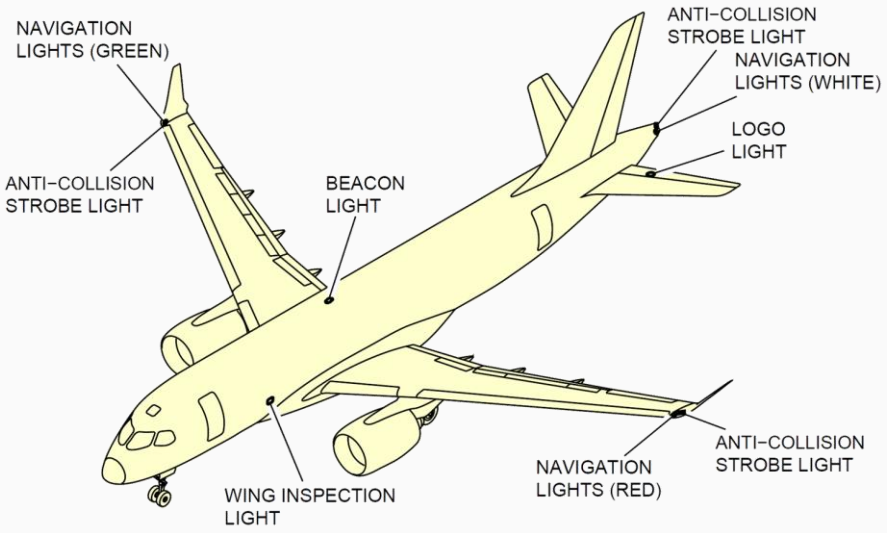
Taxi lights can be set to narrow (nose only) or wide (nose and wing taxi lights).

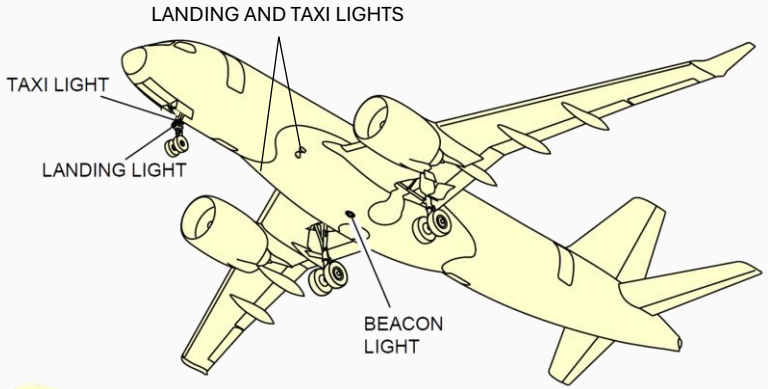
The cargo lights illuminate for 30 minutes when the switch is pressed and switch off when cargo door is closed. Service and maintenance lights stay ON for 20 minutes after switch is last pressed, after 15 minutes the lights will flash for 10 seconds signaling that only 5 minutes of illumination remain unless the switch is activated a second time.

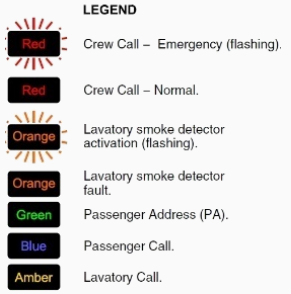
Six Emergency Power Supply Units (EPSUs) provide power to the emergency lights and exit signs. The EPSUs allow the emergency lighting system to operate for a minimum of 10 minutes after all other sources of power have been lost.

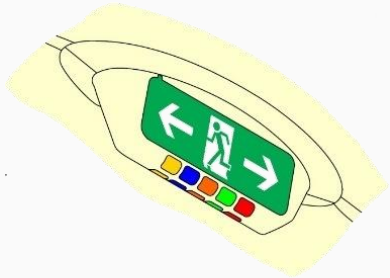
- Aisle lights
- Exit locator signs (3)
- Exit marker signs (6)
- Exit identifier signs (6)
- External forward emergency lights (2)
- External overwing emergency lights (two on each side) (4)
- External aft emergency lights (2)

The forward flight attendant can activate the emergency lights regardless of the cockpit EMER LTS switch position. The cockpit EMER LTS switch must be OFF before power is removed. The photo-luminescent tape along the cabin floor will provide up to 10 hours of illumination after a 15-minute charging period with cabin lights at full brightness.

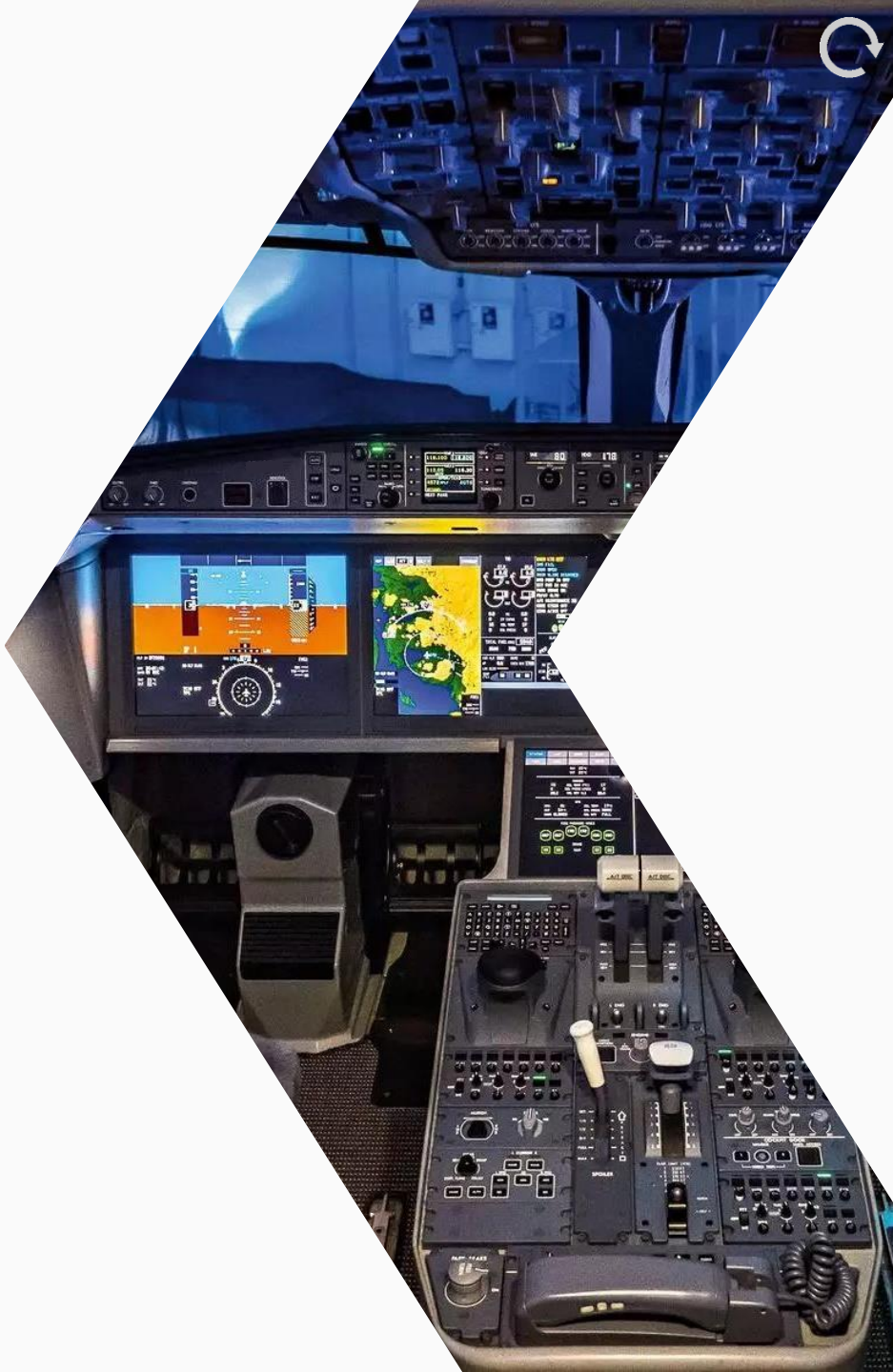


# Navigation



# Radio Navigation & GNSS

## Radio Navigation

The DME transceivers have three channels, the first channel is paired with the ILS or VOR frequency tuned (except when DME HOLD is used) the other two channels are tuned automatically by the FMS for position updating. When NAV tuning is set to AUTO, the FMS uses all three DME channels + VOR radials (if DME/DME unavailable) to determine aircraft position.

## Global Navigation Satellite System (GNSS)

The dual GNSS receivers provide position, speed and time data to the IRS, FMS and TAWS. All SBAS operations including LPV are supported. The GNSS is designed to operate with signals from any of the following SBAS systems:

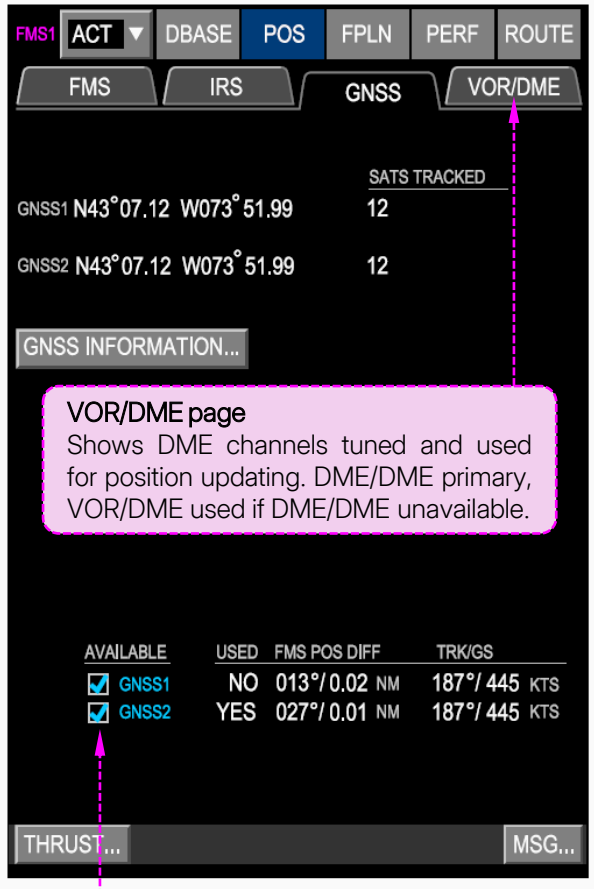
- EGNOS (Europe)
- WAAS (USA)
- GAGAN (India)
- MSAS (Japan)

The RAIM function of each GNSS compares the Horizontal Protection level (HPL) to the Horizontal Alert Limit (HAL). The HPL is equivalent to the ANP. HAL is automatically set and is equivalent to RNP as listed below:

- Oceanic / remote 4.0 nm
- Enroute 2.0 nm
- Terminal 1.0 nm
- Approach 0.3 nm (angular for LPV)

NO APPR or NO RNP is displayed on the PFD (HSI) when HPL exceeds the HAL (Actual Navigation Performance exceeding Required Navigation Performance).

Predicted RAIM availability at ETA +/- 15 min




**VOR/DME page**
Shows DME channels tuned and used for position updating. DME/DME primary, VOR/DME used if DME/DME unavailable.

GNSS sensor selected and used for navigation

| Independent Positioning | Dependent Positioning |
|---|---|
| Inertial Reference System (IRS) | Dual GNSS |
| Terrain Awareness and Warning System (TAWS) | NAV (VOR, LOC, ILS) |
| Traffic Surveillance System (TSS) including TCAS | DME |
| Weather Radar (WXR) | Marker Beacons |
| Radio Altimeters (RA) | ADF (optional) |

# Inertial Reference System (IRS)

The IRS includes 3 Inertial Reference Units (IRUs). IRU 1 and 3 are located within the left side-console while IRU 2 is located within the right side-console. They supply data to systems requiring the following parameters:

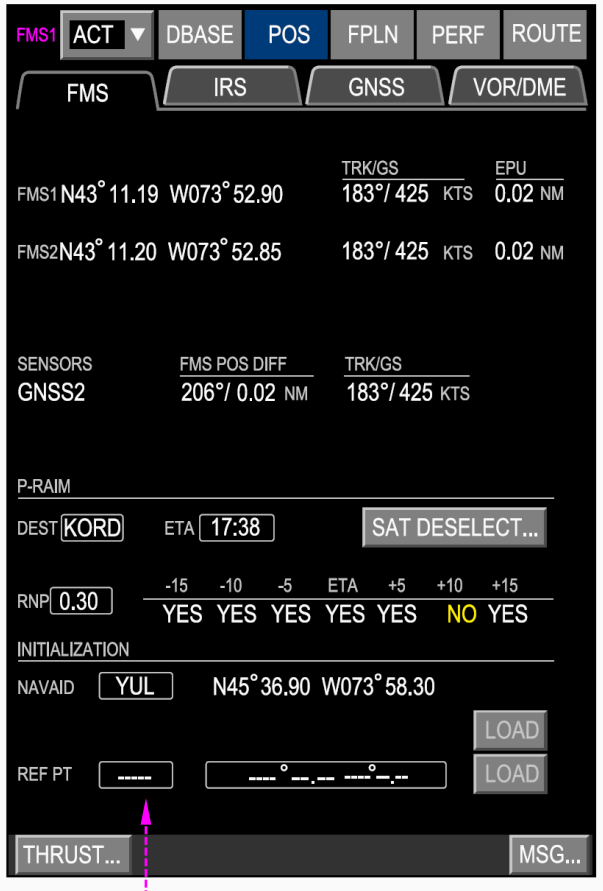
- Aircraft attitude (roll and pitch)
- Heading
- Wind speed and direction
- Aircraft flight path

Systems using IRU data include PFDs, FMS, FDR, AFCS, SATCOM, AIR DATA, WXR, FUEL SYSTEM, FBW SYSTEM, TCAS and TAWS. IRS 1 provides data to the left PFD, IRS 2 provides data to the right PFD, IRS 3 is an alternate source for either side. Pressing the left or right IRS switch on the reversion panel cycles from onside IRS to IRS3, then offside IRS.

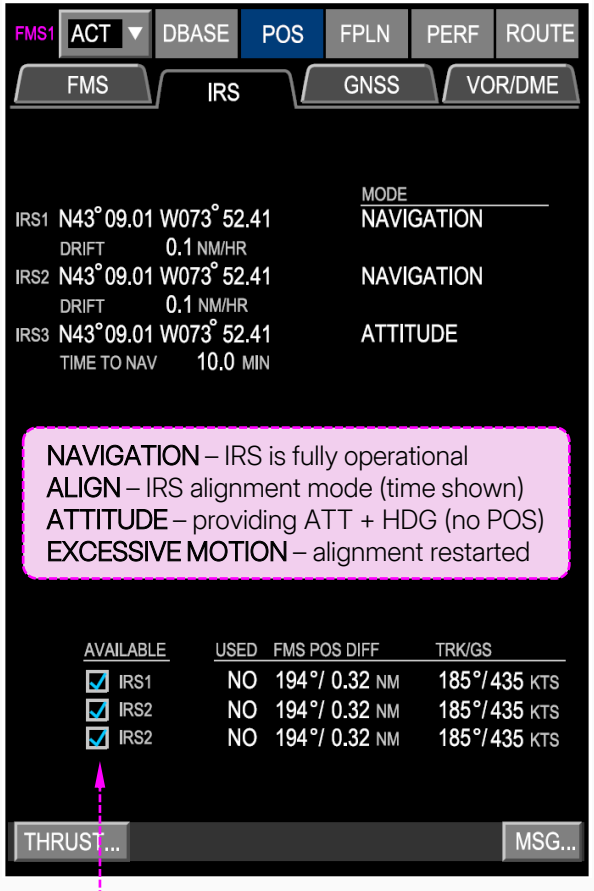
The IRUs receive GNSS and air data for alignment (position can also be entered by the flight crew), alignment at power-up or when IRS reset occurs is fully automatic with no crew action required. After power-up and platform levelling is done on each IRU, the IRUs enter "reversionary mode" to provide pitch, roll, heading, rotational rates and accelerations.

While in reversionary mode, the IRUs either enter Stationary Alignment (SA) or Alignment-in-Motion (AIM) depending on GNSS speed (below or above 30 kts). Stationary alignment is not affected by normal ground handling, but excessive motion such as push-back or taxiing may cause the IRS to restart the full alignment process.

Stationary alignment can take between 5 minutes at equator to 10 minutes between 60° and 70°N/S, extending to 17 minutes at higher latitude. In AIM the time varies from 15 to 25 minutes (flight with no heading or acceleration changes being the longest). When aligned the IRUs enter the Navigation Mode.




Enter position for IRS initialization in flight if required (normal alignment is automatic using GNSS position and ground speed).




IRS available and used for navigation, normally GNSS is preferred, then DME / DME position updating, IRS is used last (lowest accuracy).

| Side | Primary | 1st Alternate | 2nd Alternate |
|---|---|---|---|
| L PFD | IRS 1 (onside) | IRS 3 | IRS 2 (offside) |
| R PFD | IRS 2 (onside) | IRS 3 | IRS 1 (offside) |

# Traffic Surveillance System (TSS)

The aircraft is equipped with 2 transponders, XPDR 1 is a separate unit, XPDR 2 is embedded in the Traffic Surveillance System – TSS/TCAS computer. At power-up, XPDR / TCAS will be in AUTO (select STBY depending on local regulations). In AUTO the system controls transponder and TCAS automatically.

- Proximate Traffic (PT) – traffic within 6 nm horizontally and 1200 ft vertically.
- Traffic Advisory (TA) – 45 seconds from entering the collision area.
- Resolution Advisory (RA) – 30 seconds from entering the collision area.
- Other Traffic (OT) – other than above up to a range of 40 nm displayed.

Alerts are inhibited as a function of height, from high to low:

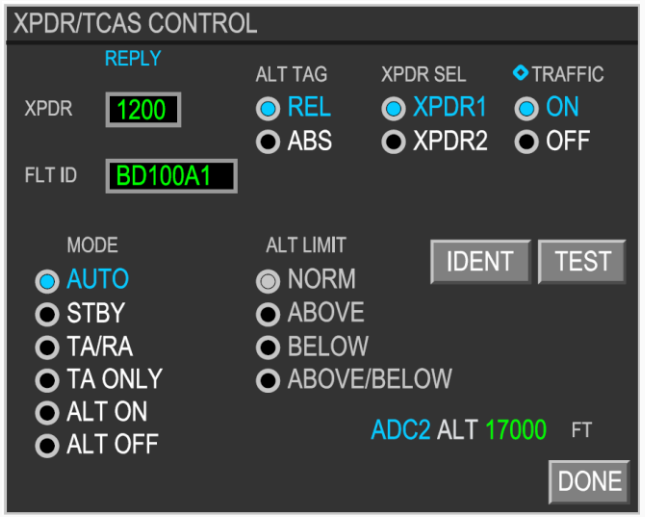
- "CLIMB" RAs are not issued above 40.000 ft.
- While descending, first "INCREASE DESCENT" RAs are inhibited
- Then all "DESCENT" RAs are inhibited.
- Then all RAs are inhibited (TA ONLY).
- Last, all alerts are inhibited (TAs still displayed, but no audio).

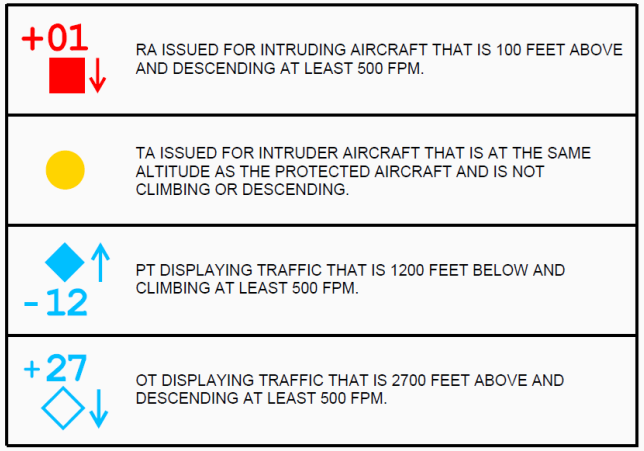
**Note:** during a TA or RA, MAP scale is automatically set to 10 nm and weather overlay removed. Weather need to be reselected by the pilots after a TA or RA.

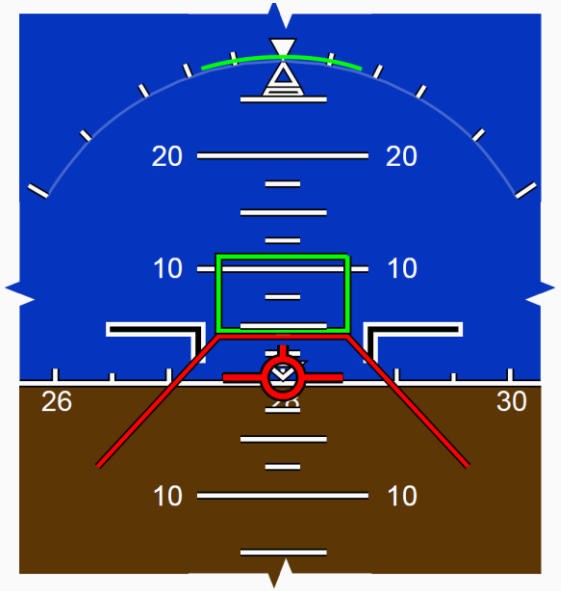
Transponder and TCAS is controlled via the CTP or TUNE CNS page. ALT LIMIT (only affects Other Traffic) can be manually set if not in AUTO:

- NORM (-2700 ft to +2700 ft)
- ABV (-2700 ft to +9900 ft)
- BLW (-9900 ft to +2700 ft)
- ABV and BLW (-9900 ft to +9900 ft)

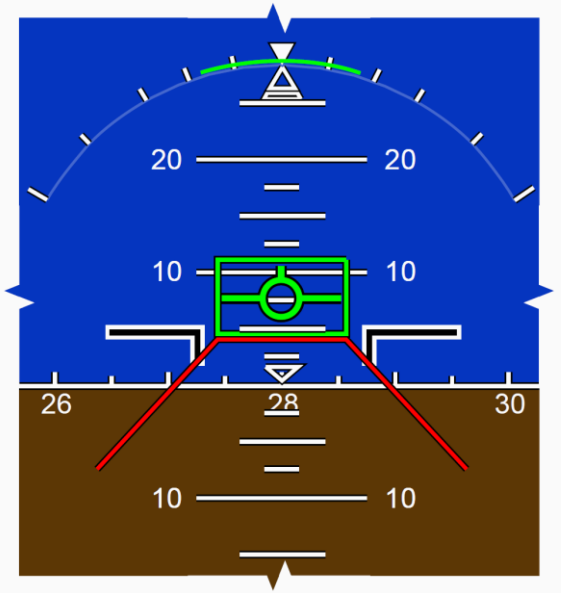
In AUTO mode NORM is selected if vertical speed is within 300 ft/min, ABOVE if rate of climb exceeds +300 or BELOW if rate of descent exceeds -300 ft/min.




Select FD off on the PF side (forces autothrottle speed mode), move FPV out of the red trapezoid, into the green box.



# Terrain Awareness and Warning System (TAWS)

TAWS is another term for EGPWS, it has the following modes:

- Mode 1: excessive descent rate (“SINK RATE”)
- Mode 2: excessive terrain closure rate (“TERRAIN TERRAIN, PULL UP”)
- Mode 3: altitude loss after take-off or go-around (“DON’T SINK”)
- Mode 4: unsafe terrain clearance (“TOO LOW TERRAIN / GEAR / FLAPS”)
- Mode 5: excessive deviation below glideslope (“GLIDESLOPE”)
- Mode 6: advisory callouts, bank angle and altitude (“BANK ANGLE”)
- Mode 7: windshear detection and alerting (“WINDSHEAR” x3)
- Premature Descent Alert (PDA) (“TOO LOW TERRAIN”)
- Forward Looking Terrain Avoidance (FLTA) (“TERRAIN AHEAD, PULL UP”)

The highest terrain within the half range ring is displayed on the map as a grey dot with elevation in feet if higher than 1000 ft above sea level.

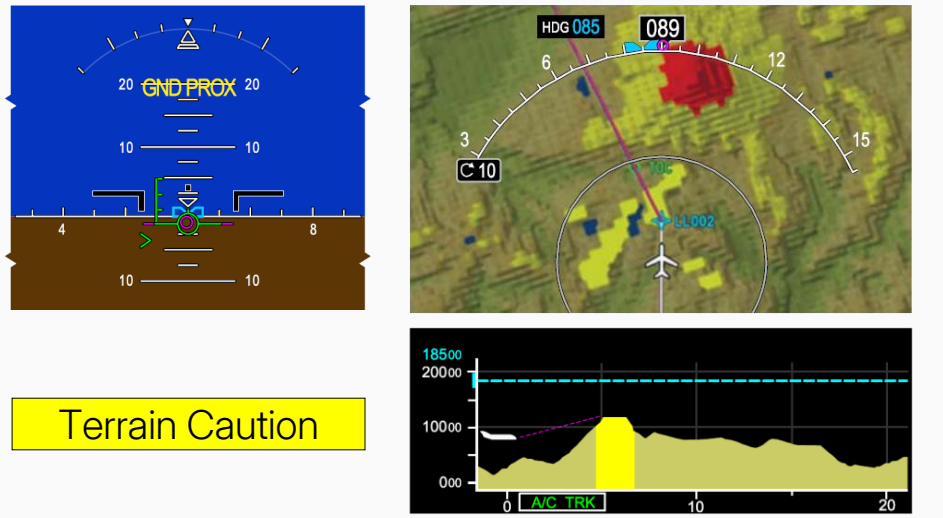
Terrain 500 ft below to 2000 ft above the aircraft is displayed as yellow. Terrain more than 2000 ft above the aircraft is displayed in red on the MAP display.

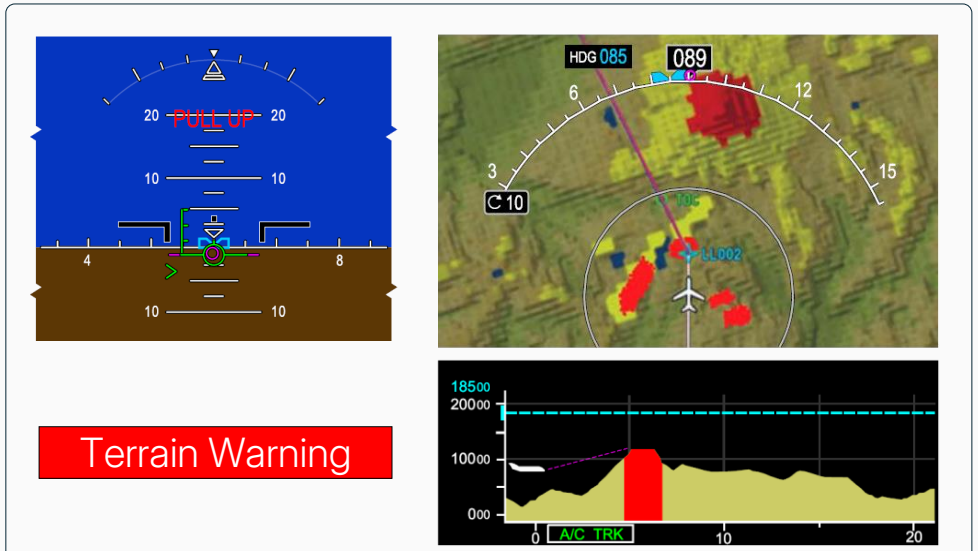
Terrain can also be shown on the Vertical Situation Display (VSD).

In case of terrain caution or warning, the map is automatically displayed on DU 2 and DU 3 at 10 nm range, WXR overlay is cancelled, and the conflicting terrain is shown in flashing yellow (caution) or red (warning).

Conflicting terrain is also shown on HSI and VSD (but not flashing).

Reactive windshear detection is active between 10 ft and 1500 ft AGL, reactive windshear alerts are provided for negative windshear such as decreasing headwind or increasing tailwind and severe down drafts. Windshear aural alerts have priority over all other ground proximity modes.




Reactive Windshear Alert

Peak elevation (highest elevation within half range scale if > 1000 ft AMSL).

# Weather Radar (WXR)

The weather radar system provides weather information up to a maximum range of 320 nm. The WXR is designed to work in automatic mode at all times, the system automatically scans and adjusts tilt angle and gain to provide an optimum weather picture. Operation in AUTO mode includes the following functions:

- Multi-scan
- Ground Clutter Suppression (GCS)
- Overflight (over scanning) protection
- Path Attenuation Correction (PAC)
- Temperature-based / geographic-based gain

The presented image is the result of multiple radar scans at different tilt angles. The system stores the scanned images in memory in order to construct and display a complete weather image, regardless of aircraft altitude.

The ground clutter suppression removes approximately 98% of the ground clutter. Above 22.000 ft, the overflight protection function activates to prevent detected storm cells from disappearing from the display as they approach during high altitude cruise.

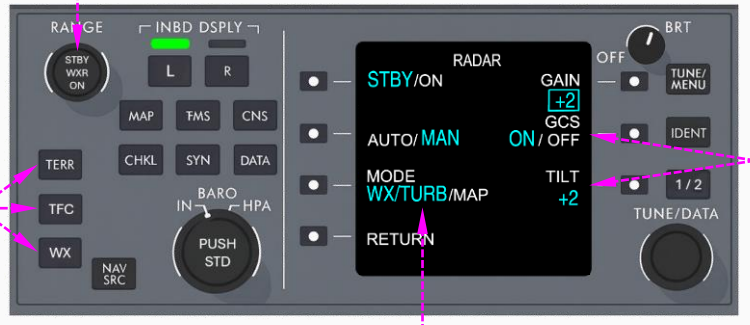
Path Attenuation Correction (PAC) shows an amber arc at the edge of the display to indicate the areas where the radar may not display precipitation correctly if at all (i.e. radar shadow). Temperature-based / geographic-based gain optimizes the gain setting for: variations in temperature, geographic location, time of day and altitude.

The WXR is automatically ON when the aircraft is weight-off-wheels. After landing, the WXR automatically switches to STBY mode. The windshear detection function operates below 2300 ft RA (inhibited during take-off between 80 kts and 50 ft), it also operates if WXR is set to standby or in TEST.

If the weather radar detects a windshear condition, "MONITOR RADAR DISPLAY" aural alert sounds (3 nm prior). Upon entering the warning area (1.5 nm prior) "WINDSHEAR AHEAD" x2 alert during take-off or "GO-AROUND, WINSHEAR AHEAD" on approach.

**Note:** European operators are encountering issues with the geographic-based gain, resulting in low weather detection sensitivity and late detection of weather. As a crosscheck to the AUTO mode, MAN mode can be used. For additional sensitivity and to improve weather detection range, GAIN can be increased as needed (table).

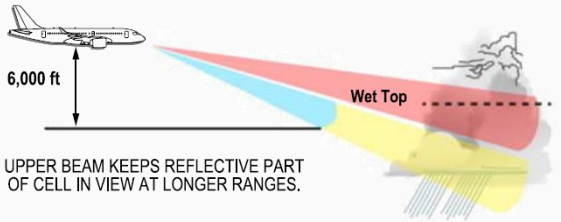
WXR STBY – ON button not functional on A220 aircraft

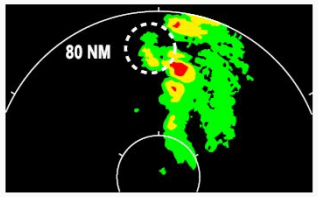



Map Overlays

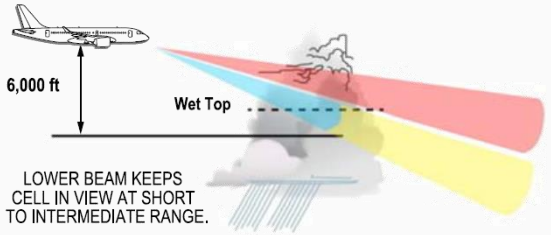
GCS only in auto mode, manual TILT control only available in MAN mode.

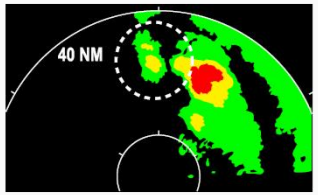
TURB shows turbulent areas detected within 40 nm in magenta



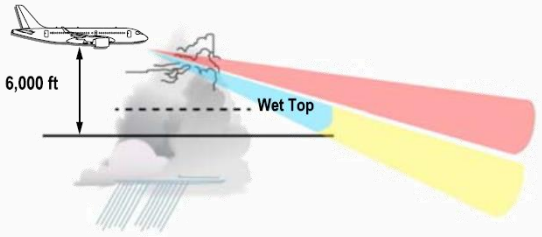



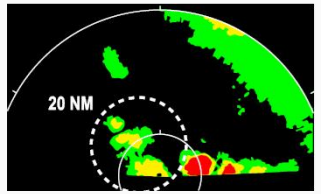

3 nm before: "MONITOR RADAR DISPLAY"



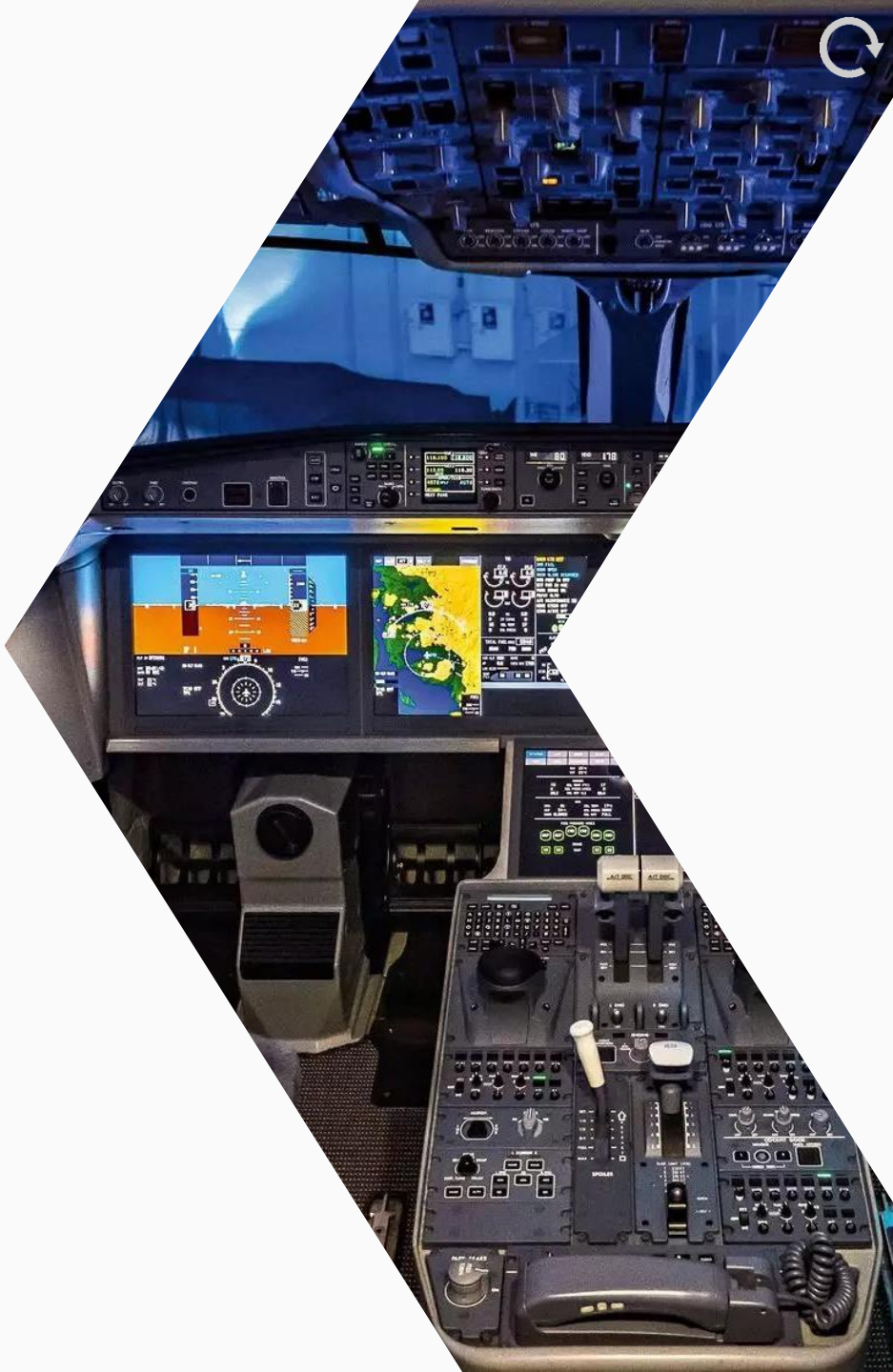




1½ nm prior: "(GO-AROUND) WINDSHEAR AHEAD"



# Oxygen & Emergency Equipment



# Oxygen Equipment

## Flight compartment oxygen system

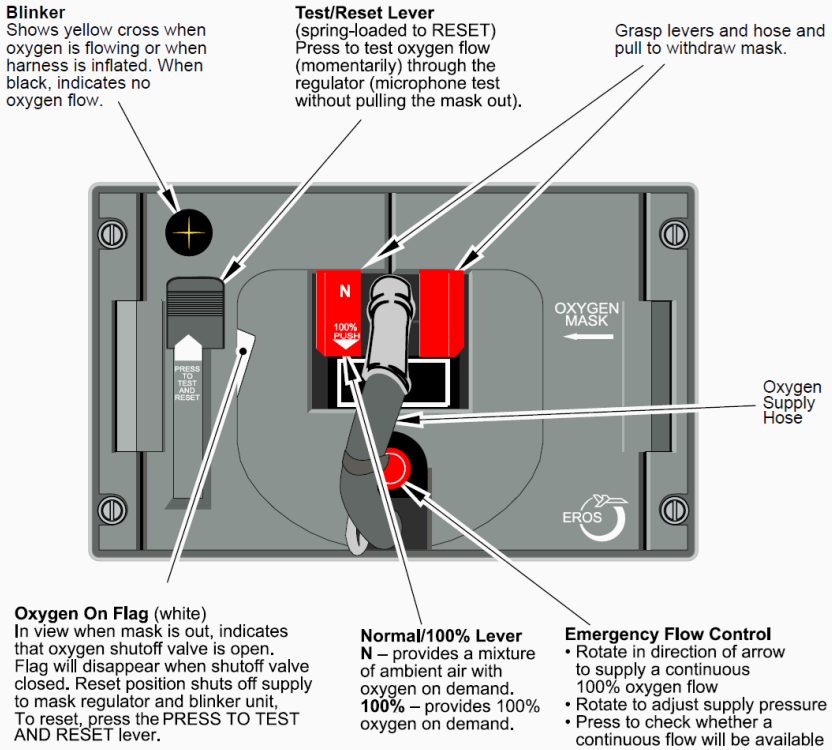
- Oxygen cylinder is stored in a ventilated compartment under the jump seat.
- Normal pressure is 1850 psi, below 1050 psi an alert is shown on EICAS.
- Oxygen masks are supplied with 70 psi, < 45 psi an alert is shown on EICAS.
- A green overboard blowout disc will discharge oxygen during overpressure.
- Oxygen mask is a EROS oxygen mask type with integrated smoke goggles.
- The microphone is automatically enabled, close left door of the stowage box, then press TEST / REST (flag removed) to return to headset microphone.
- Toggle for N / 100%, < 35.000 ft a mix is supplied in N, 100% oxygen above.
- Emergency (PRESS AND TURN) supplies 100% oxygen under positive pressure, used to prevent smoke / fumes entering the mask.

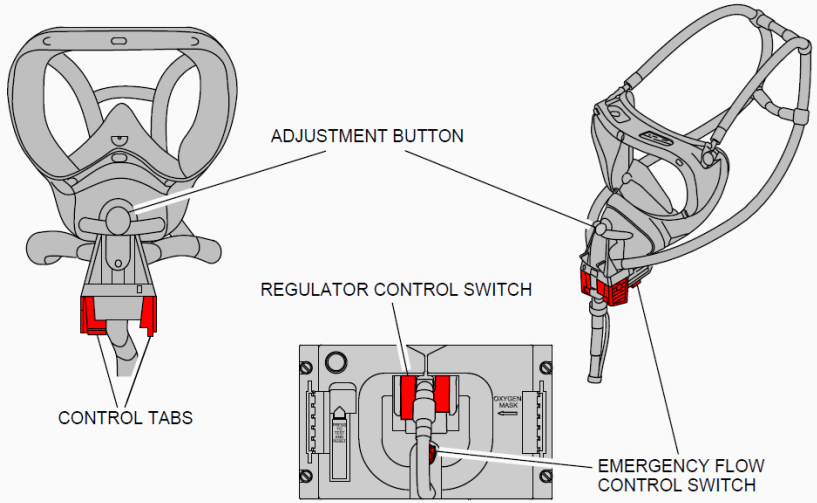
## Cabin oxygen system

- Passenger seats, flight attendant / galley stations and lavatories have masks.
- Chemical oxygen generators supplying oxygen for 13 min (22 min optional).
- When cabin altitude is > 14.800 ft the oxygen doors unlatch, and masks drop.
- The NO SMOKING (if installed), FASTEN SEATBELT and RETURN TO SEAT signs automatically turn ON and a video announcement will play.

## Portable oxygen system

- Used for first aid situations and to help during loss of cabin pressure.
- Number of cylinders depend on configuration. Charged to 1800 – 2000 psi.
- Each cylinder has 2 colour coded outlets for different flow rates.



# Emergency Equipment

### Fire-fighting equipment

- (4) clean-agent portable fire extinguishers, 1 in flightdeck, 2 FWD and 1 AFT cabin.
- Clean-agent portable fire extinguisher discharges in approximately 15-20 sec.
- (1) Portable water extinguisher, in the aft (optional) and discharges in 30-45 sec.
- (4) Protective Breathing Equipment (PBEs) installed near the fire extinguishers. PBEs contain a chemical oxygen generator to supply oxygen for approx. 15 min.
- (4) pairs of fire protective gloves are kept near each clean-agent fire extinguisher.

### Overwater emergency equipment

- Life vests for flight crew, observer, cabin crew and passengers (under each seat), life rafts (optional).

### Emergency exits

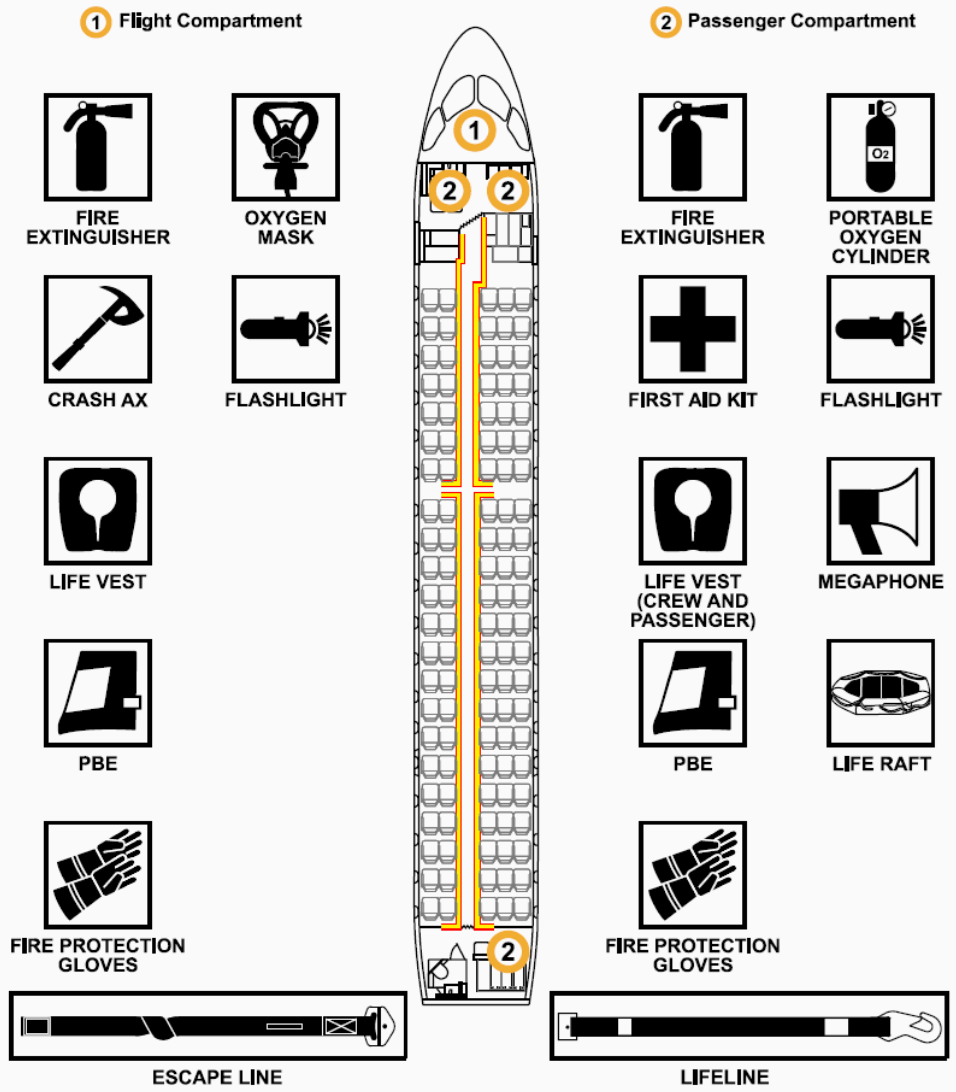
- (6) Escape slides for each passenger and service door (4) and overwing exits (2).
- (2) Lifelines at each overwing emergency exit door to hold on after landing on water.
- (1) Flight crew escape rope to be deployed through the emergency escape hatch.

### Emergency Locator Transmitter (ELT)

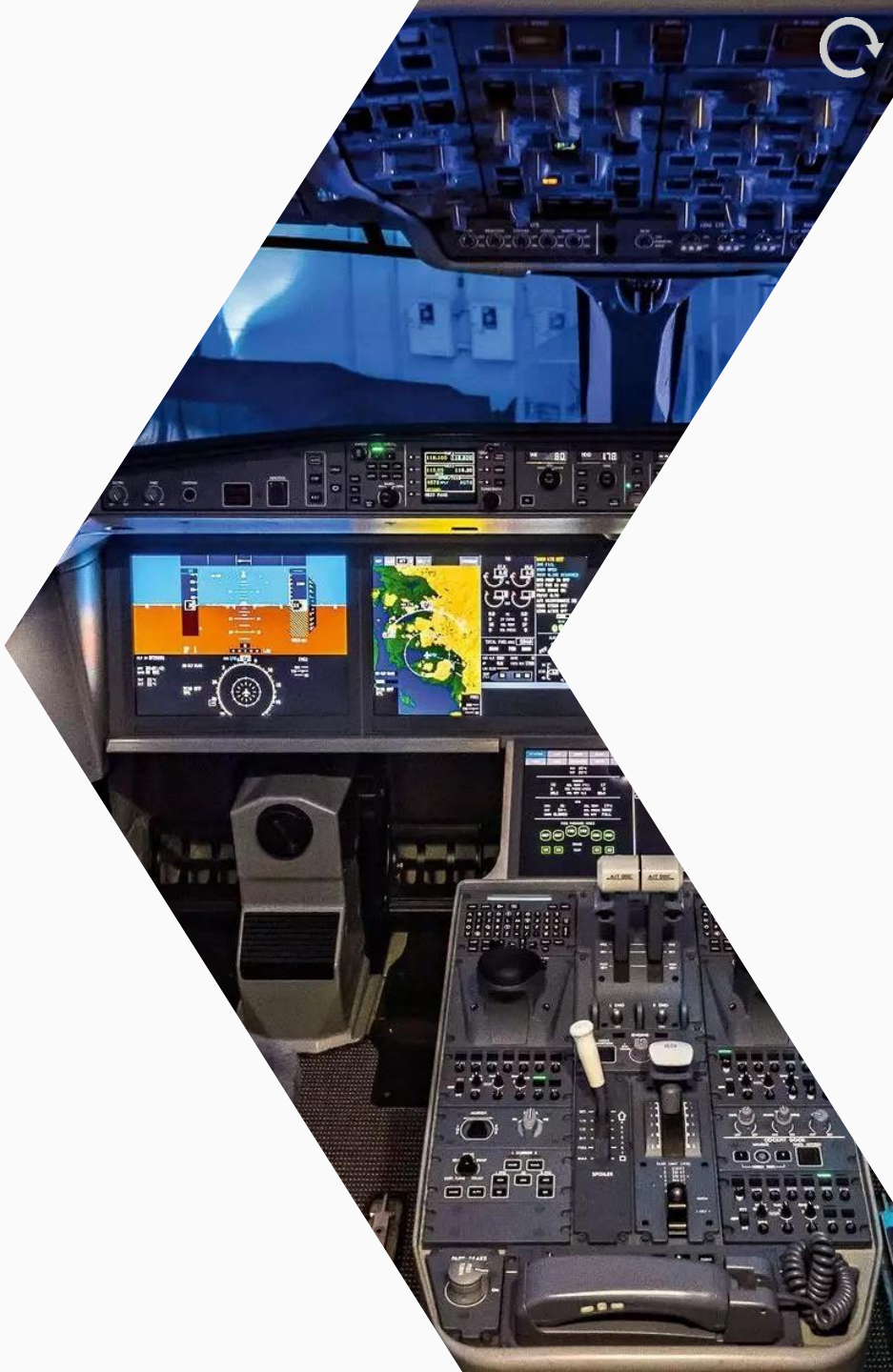
- (1) Fixed ELT transmitting on 121.5 MHz and 406 MHz, automatic type ELT beacon (with Distress Tracking (ELT-DT) on later models). (1) Portable ELT.

### Other emergency equipment

- (6) Standard or rechargeable (optional) flashlights, 1 at each crew station.
- (2) Megaphones, 1 forward attendant station and 1 aft attendant station.
- (2) First aid kits, 1 forward attendant station and 1 aft attendant station.
- (1) Crash axe installed in the flight compartment behind the left seat.
- Evacuation horn (EVAC CMD switch).



# Power Plant



# Introduction

The PW1500G is an ultra-high bypass, Geared Turbofan (GTF) engine with a bypass ratio of 12:1, around 80% of the thrust is provided by the bypass flow. The large diameter fan is driven at reduced speed by a Fan Drive Gear System (FDGS). The FDGS allows the fan to spin three times slower than the Low-Pressure Compressor. Each engine is controlled by a Full Authority Digital Engine Control (FADEC) system.

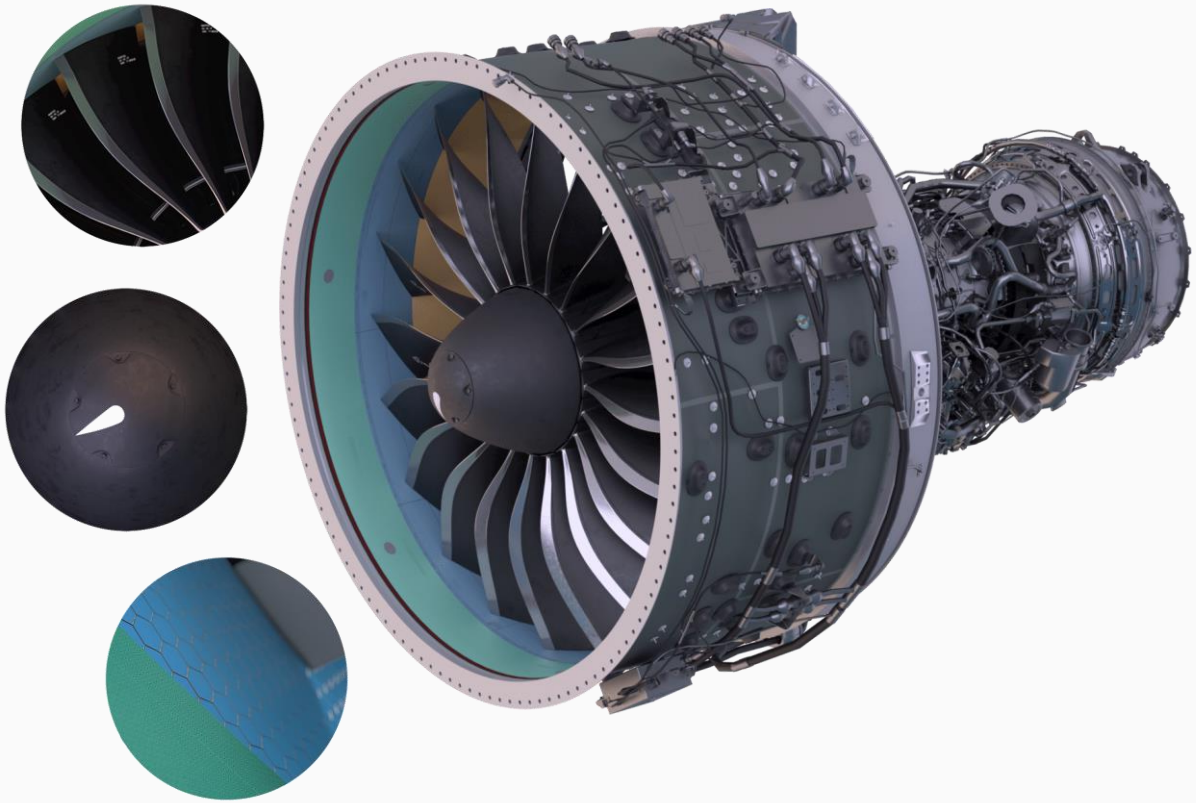
The PW1500G comes in 4 different thrust ratings. PW1519G is only used on the A220-100, the PW1525G has the same static thrust as the PW1524G but develops 5% more thrust growth at higher Mach numbers. These engines are physically the same and the thrust rating is merely a software configuration.

| TO Thrust Setting | (Installed) Thrust Rating (lbf) | | | |
|---|---|---|---|---|
| | % TO Thrust | 1519G | 1521G | 1524G / 1525G |
| TO | 100 | 18.900 | 21.000 | 23.300 |
| TO-1 | 90 | 17.000 | 18.900 | 21.000 |
| TO-2 | 80 | - | 17.000 | 18.900 |
| TO-3 | 70 | - | - | 17.000 |
| MAX FLEX | 75 | 75% of TO(-1,2,3) rating, minimum 14.000 lbf | | |

The Fan Drive Gear System (gearbox) allows the low-pressure compressor and fan to rotate at their optimum speeds, increasing efficiency. The relatively higher $N_1$ (low pressure compressor) speed reduces the number of stages required. The low spool has a three-stage Low Pressure Compressor (LPC) driven by a three-stage Low Pressure Turbine (LPT), the high spool consists of an eight-stage High Pressure Compressor (HPC) driven by a two-stage High Pressure Turbine (HPT). The Main Gearbox (MGB) is driven by the high spool and drives:

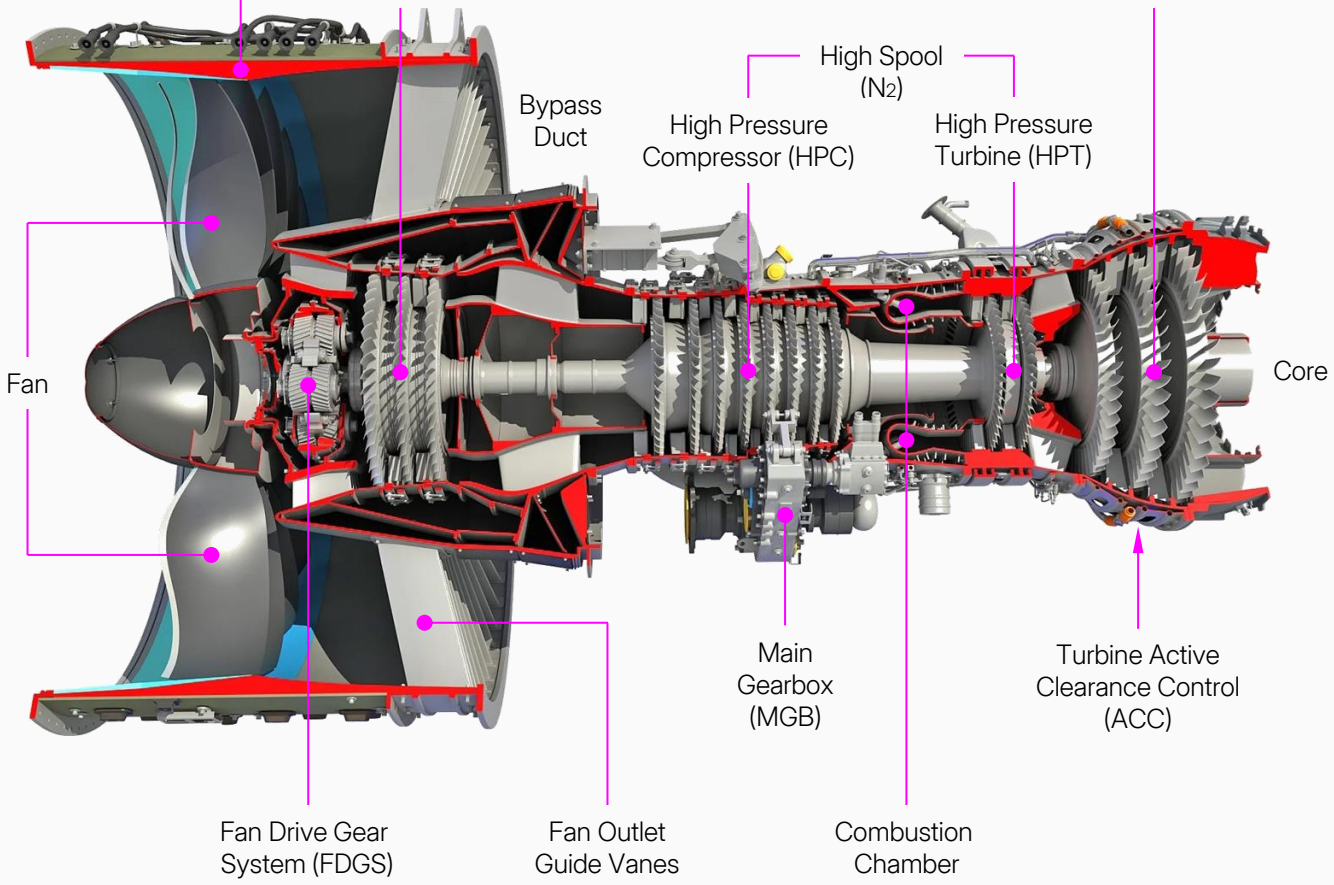
- Hydraulics: Engine Driven Pump (EDP) 1A and 2A
- Lubrication: Oil lubrication (1) and scavenge pumps (6), oil breather system
- Electrical: Variable Freq Gen (VFG), Permanent Magnet Alt - Gen (PMAG)
- Fuel: Integrated Fuel Pump and control (IFPC)
- Start: Air Turbine Starter (ATS)

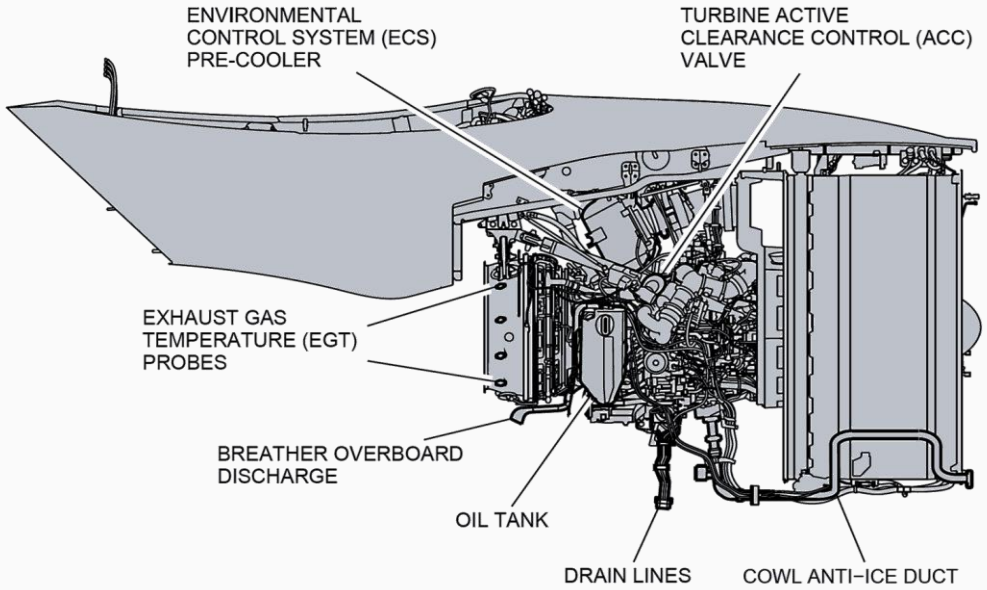
Below: PW1500 features large diameter fan blades made of high strength aluminum with titanium leading edge. The white droplet on spinner helps to identify whether an engine is running. Rub strips made of (blue) foam with metal mesh reduce the fan blade clearance.

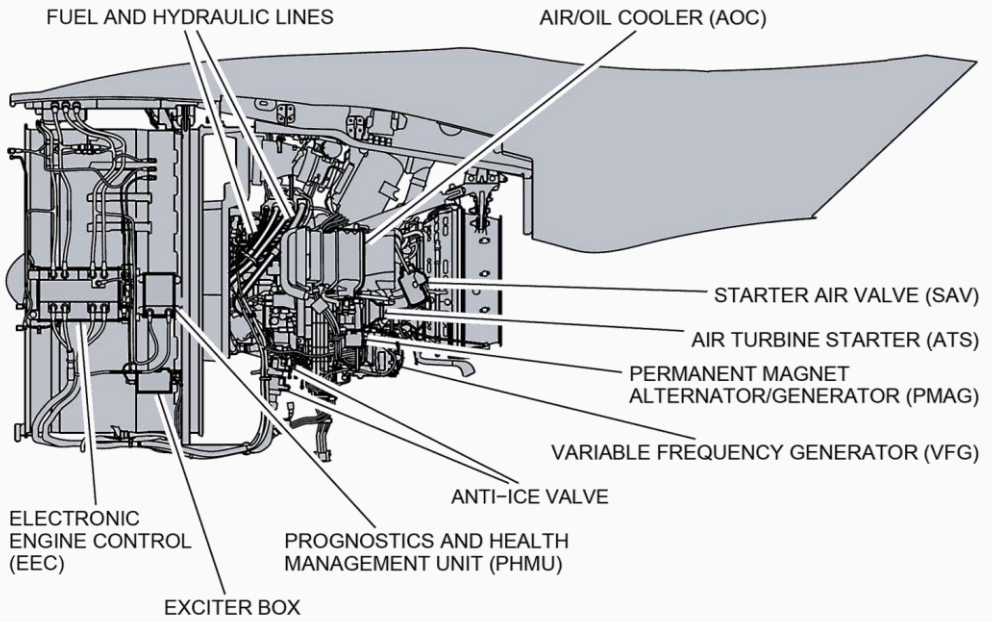


# Engine Components

**Note:** the characteristic low thrust "whale howl" tone originates from the combustion chamber. A new combustor configuration expected to be available from 2026 will eliminate the whale howl tone.



# Engine Start

Ensure that thrust lever is set to idle (FADEC will produce "THRUST LEVER" aural alert if TLA is not idle). The engine will start regardless of thrust lever position and thrust will quickly increase to the thrust lever setting after start which might be hazardous.

### Automatic start cycle

- Thrust lever at forward idle or above, and
- CONT IGNITION not selected, and
- START switch set to AUTO, and
- $N_2$ less than starter cut-out speed (51%), and
- Moving L or R ENG run switch to ON:
  - Starter used on ground and ≤ 250 kts
  - Windmilling start in-flight above 250 kts
- Motoring at 12% $N_2$ for 45 seconds (see note)
- Automatic start abort (up to engine stabilization):
  - $N_1$ locked rotor (no rotation), or
  - Hot or hung start, or
  - Loss of EGT data, or
  - Igniter failure (wet start), or
  - Fuel control failure (dry start)
- FADEC automatic restart after abort:
  - Motoring period of 30 seconds, then
  - Single re-start attempt below 20% $N_2$

Typical idle values after start:
- $N_1$ – 22%
- EGT – 550°C
- $N_2$ – 66%
- FF – 275 kg/hr/eng

### Manual start cycle

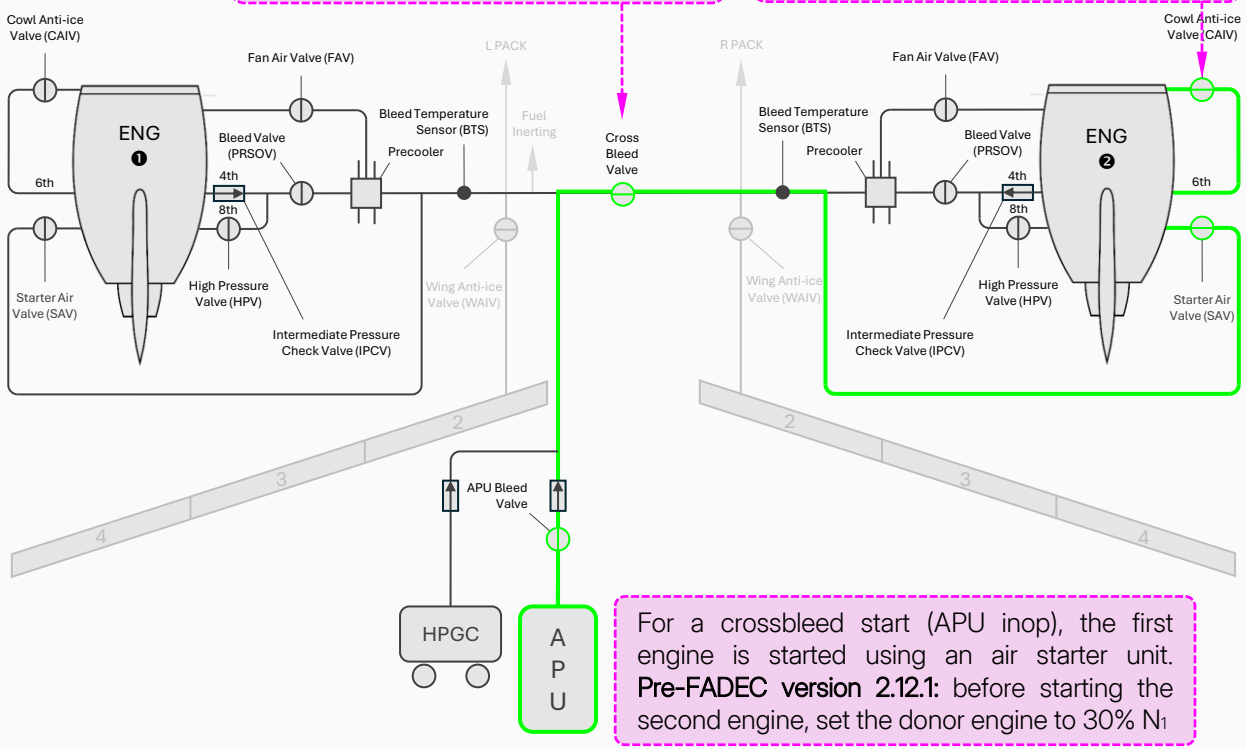
- CONT IGNITION set to ON (manual start command)
- START switch held in L or R ENG CRANK position
- Depending on conditions, start delay with motoring at 12% $N_2$
- L or R ENG run switch set to ON between 18% and 20% $N_2$
- Igniters and ATS automatically cut-off at 51% $N_2$ by the EEC

Always place the ENG RUN switch back to the OFF position after aborting the start sequence. If the ENG RUN switch is not set to the OFF position, the FADEC will not complete the ABORT. The engine might start and cause a hazardous situation.

**Note:** motoring will be longer if engine thrust has not been maintained at idle for 3 minutes before last engine shutdown or if OAT is lower than -17°C or above +43°C. Maximum motoring time is 120 seconds depending on altitude, ambient temperature and engine thermal state. ENG START DELAY advisory is displayed on EICAS if motoring time will exceed 55 seconds.

Example: starting right engine using APU bleed. Crossbleed valve is automatically configured by the IASCs.

Excess air during engine start is vented from the compressor via the HPC bleed valve and cowl anti-ice system.




For a crossbleed start (APU inop), the first engine is started using an air starter unit. **Pre-FADEC version 2.12.1:** before starting the second engine, set the donor engine to 30% $N_1$

# Engine Systems

## Engine Fuel System

Fuel scheduling is managed by the EEC. The Integrated Fuel Pump and Control (IFPC) includes a two-stage pump and a fuel control unit.

The first stage supplies fuel to the engine fuel distribution components, such as:

- Fuel nozzles
- Fuel-oil heat exchanger
- Fuel filter
- Fuel activated actuators and valves: Variable Inlet Guide Vanes (VIGVs), Variable Stator Vanes (VSVs), bleed valves, Active Clearance Control (ACC).

The second stage provides fuel to the ejector pumps to provide motive flow.

## Engine Ignition System

The ignition system provides engine start, relight and flameout protection and is controlled by the EEC. Each EEC channel controls one igniter. On ground, for engine start, the FADEC selects one igniter per engine. During subsequent start attempts the FADEC will alternate igniters to prolong igniter life. For manual starts, CAIS ON, engine out / surge detected, in-flight start attempts or continuous ignition, both igniters are used.

Continuous ignition is automatically selected for:

- Automatic re-start attempts
- Take-off and landing
- Cowl Anti-ice System (CAIS) ON
- Engine flame-out protection (> 60 kts)
- Compressor surge detected

## Engine Oil System

Oil flows from the oil tank to the lubrication pump (1), then passes though the Oil Control Module (OCM) where it is filtered before being directed to the heat exchangers, it is then distributed to engine bearings and gearboxes. The scavenge pumps (6) return the oil to the oil tank. Oil temperature, pressure, filtering and debris monitoring functions are accomplished by the OCM. Oil temperature is maintained by:

- Fuel-Oil Heat Exchanger (FOHE)
- Air-Oil Heat Exchanger (AOHE)
- VFG Oil-Oil Heat Exchanger

The Fan Drive Gear System (FDGS) is lubricated via an auxiliary distribution system which in certain situations separates from the main oil system. It consists of an auxiliary oil tank and fan driven oil pump. During normal operations, oil flows back to the (main) oil tank. During low power, windmilling, or low g events, oil is directed straight back to the FDGS by a shuttle valve in the OCM.

## Engine Bleed Air System

The engine bleed air system is used for engine airflow control, cooling, ingested debris removal and to provide bleed air to other aircraft systems (air conditioning, wing and cowl anti-ice, engine cross bleed start etc.). Engine airflow control removes excess air and optimizes airflow to improve engine stability (prevent compressor stalls).

The LPC bleed valve (ring) has a fuel-controlled actuator and removes excess air from the LPC, a spring-loaded bleed valve removes excess air from the HPC. During engine start, excess air is vented overboard via the HPC bleed valve and the Cowl Anti-ice System.

# Thrust Management

Thrust management is provided by the EECs and the FMS. Thrust mode changes are commanded by the EEC in accordance with the active flight plan. Available thrust settings are:

- Maximum Take-off thrust (TO)
- Maximum Continuous Thrust (MCT)
- Derated take-off thrust (TO-1, TO-2, TO-3)
- Reduced take-off thrust (FLEX)
- Climb thrust, including derates (CLB, CLB-1, CLB-2)
- Idle thrust (ground / flight idle, (steep) approach idle, reverse idle)
- Automatic Power Reserve (APR)

The thrust levers have stops at: IDLE, MAX and MAX REV. Maximum thrust is always available with thrust levers at the MAX position. The EEC resets redline limits to MCT after 5 minutes at TO thrust for All Engines Operating (AEO) or 10 minutes for One Engine Inoperative (OEI) or Opposite Engine Low-Thrust (OELT) event.

Thrust Limitation at Low Speed (TLLS) is active during take-off to ensure nose landing gear is sufficiently loaded. The FADEC initially limits the thrust to 60% of maximum take-off power, during acceleration, thrust increases and reaches full thrust by 80 kts (indicated by grey arc on the $N_1$ gauge).

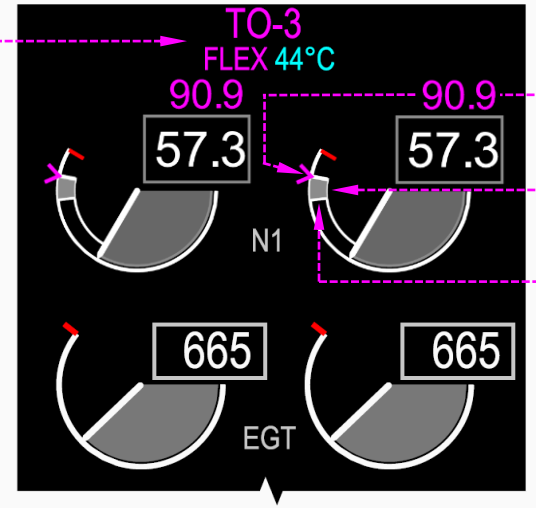
Derated take-off thrust is set via the FMS PERF page (TO-1, TO-2, TO-3). Derated take-off thrust can be used on wet or contaminated runways (except ice), derated take-off is forbidden in presence of windshear or thunderstorm activity.

**Pre-Avionic Built 6.5:** derated take-off used in combination with reduced take-off thrust (FLEX) is prohibited. **Post-Avionic Built 6.5:** derated thrust can be combined with FLEX.

Reduced take-off thrust (FLEX), the assumed temperature is entered on the FMS PERF page. The total thrust reduction is limited to 75% of the take-off thrust rating (but not below minimum take-off thrust of 13980 lbf) FLEX is allowed on wet runways, but prohibited on contaminated runways, in presence of windshear or thunderstorms.

**Active Thrust Mode**
Take-off thrust setting: TO, de-rate TO-1 / -2 / -3 or FLEX with assumed temperature. Climb thrust including de-rate, GA etc.

All values in Magenta for AUTO, cyan for MANUAL




Thrust Reference Value

TLLS segment below 80 kts

Commanded thrust

**FMS THRUST dialog box**
Select between AUTO (magenta) or MAN (cyan).

Change thrust mode:

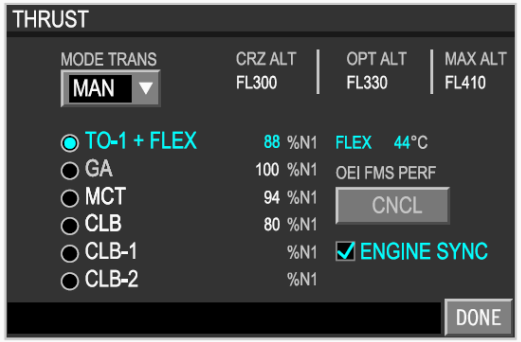
- TO, GA, MCT
- CLB, CLB-1 / -2

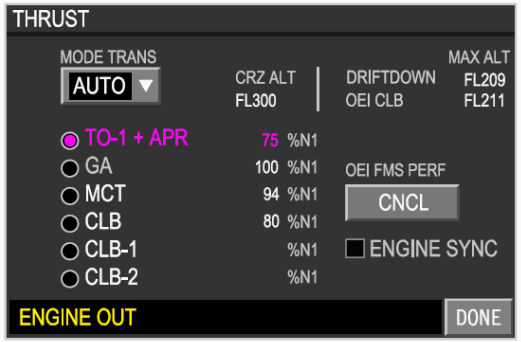
Thrust Reference Values (TRVs) shown.

Cruise altitude, optimum and maximum altitude (OEI climb ceiling and driftdown in case of engine failure).

ENGINE SYNC toggle to synchronize $N_1$ between engines to reduce harmonic noise (similar to prop sync).

Soft switch to cancel (CNCL) OEI FMS performance in case of engine restart.




**EICAS Engine Flags**
FIRE – fire indication
START – start sequence active
RELIGHT – automatic relight
START – start sequence failed
RELIGHT – auto relight failed
ATS – engine below idle (failed), within ATS envelope
WINDMILL – engine below idle (failed), windmill envelope
SYNC – engine synch active (to reduce harmonic noise)
REV – reverser deployed
REV - reverser in transit
REV – reverser failed
APR – APR is active
CAI – cowl-anti-ice is on
CAI – cowl anti-ice armed
CAI – cowl anti-ice failed
WAI – wing anti-ice on
WAI – wing anti-ice failed with caution condition or overheat
WAI – wing anti-ice failed with warning condition
IGN – ignition is active
IGN – ignition inhibited
> - FMS thrust reference value
> - manual thrust reference
VIB – high vibration, shown on $N_1$, $N_2$ or Fan vibration monitor (which is normally hidden).

# Thrust Management

If derated climb thrust is selected (CLB-1, CLB-2), a washout function automatically and gradually increases climb thrust to recover full CLB thrust at a pre-determined altitude by the EEC. When in cruise, the CLB $N_1$ reference will remain displayed.

Idle thrust is managed by the EEC, there are 5 idle settings based on engine stability, bleed demand and phase of flight:

- Ground idle – engine speed that reduces landing distance & brake wear during taxi.
- Flight idle – minimizes fuel consumption in clean config and increases with altitude.
- Approach idle – is intended to reduce spool-up time from idle to go-around thrust.
- Steep approach idle – lower than approach idle but will allow engines to achieve 80% of GA thrust within 8 seconds. Activated when:
    - Steep approach mode selected in the FMS
    - Approach mode engaged by the flight guidance system
    - No windshear has been detected
- Reverse idle – thrust reversers deployed, thrust levers in REV IDLE position.

Automatic Power Reserve (APR) provides an automatic additional thrust increase when an OELT condition exists during take-off. APR is automatically armed during a derated or reduced thrust take-off (can be disarmed in the FMS). When an OELT condition occurs, the FADEC will increase the $N_1$ to the next TO thrust level (e.g. TO-2 to TO-1).

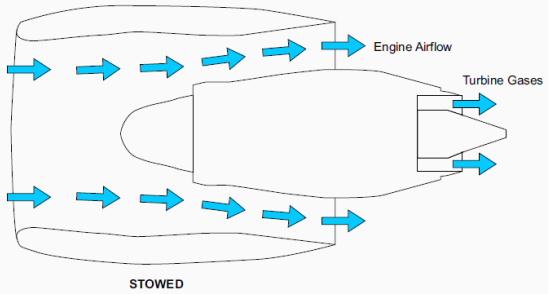
Activation:
- $N_1$ on opposite engine is 15% lower, and
- A derate or FLEX is selected, and
- Thrust Lever Angle (TLA) above 20°.

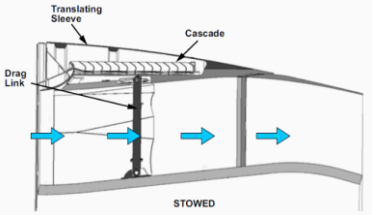
Deactivation:
- Above 400 ft radio altitude, or
- APR is set to DISARMED in the FMS, or
- Successful in-flight relight, or
- Another thrust mode is selected.

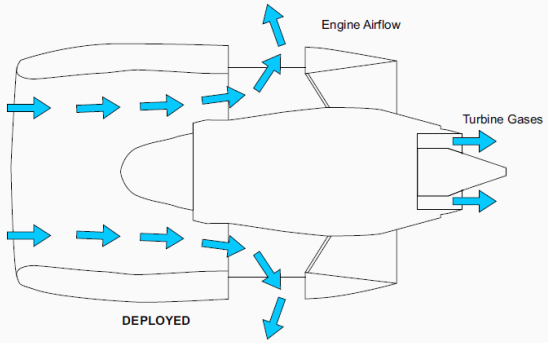
### Thrust Reverser System

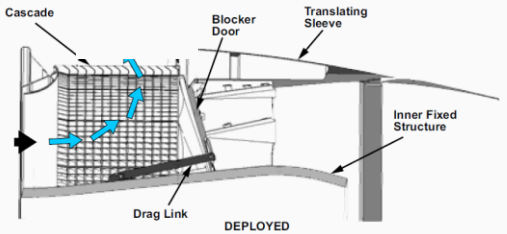
The thrust reverse system is of the translating sleeve type. When deploying, the hydraulic actuators move the translating sleeves aft, drag links will cause the blocker doors to close and redirect the fan airflow. Below 60 kts, reverse will automatically reduce to idle reverse (auto cut-back feature).

Tactile feedback is provided to the thrust levers during deployment, the mechanism blocks the thrust levers in the reverse idle position until thrust reverser deployment is confirmed, it can be overridden when a significant amount of force is applied. Similarly, the system blocks the thrust levers in forward idle thrust position until thrust reverser stowage is confirmed.

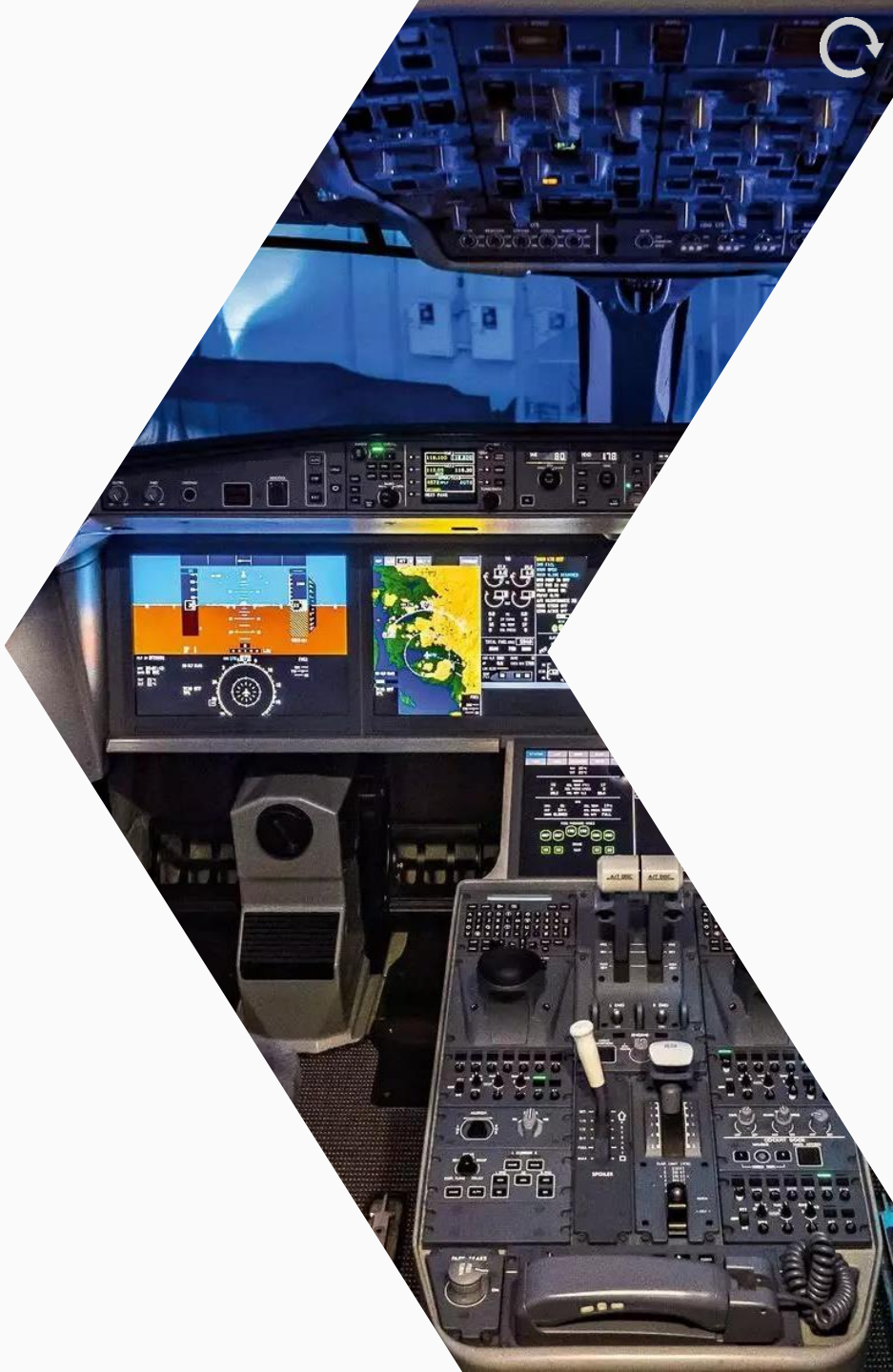


# Electronic Checklist



# Electronic Checklist (ECL)

Short description for each tile:

SUMMARY – is the ECL default page.

- Normal checklist: showing normal checklists for the current phase of flight, the three flight phases are: pre-flight, in-flight and post flight.
- Active subsection: displays non-normal checklists based on triggered EICAS messages, procedures selected by the crew (non-EICAS related) and follow-on checklists. A "priority pointer" ( ▶ ) indicates the procedure with highest priority, when completed it moves to the checklist with next priority (for EICAS triggered events only).
- History subsection displays completed non-normal checklists and procedures, completed with deferred and / or overridden items and associated carried limitations.

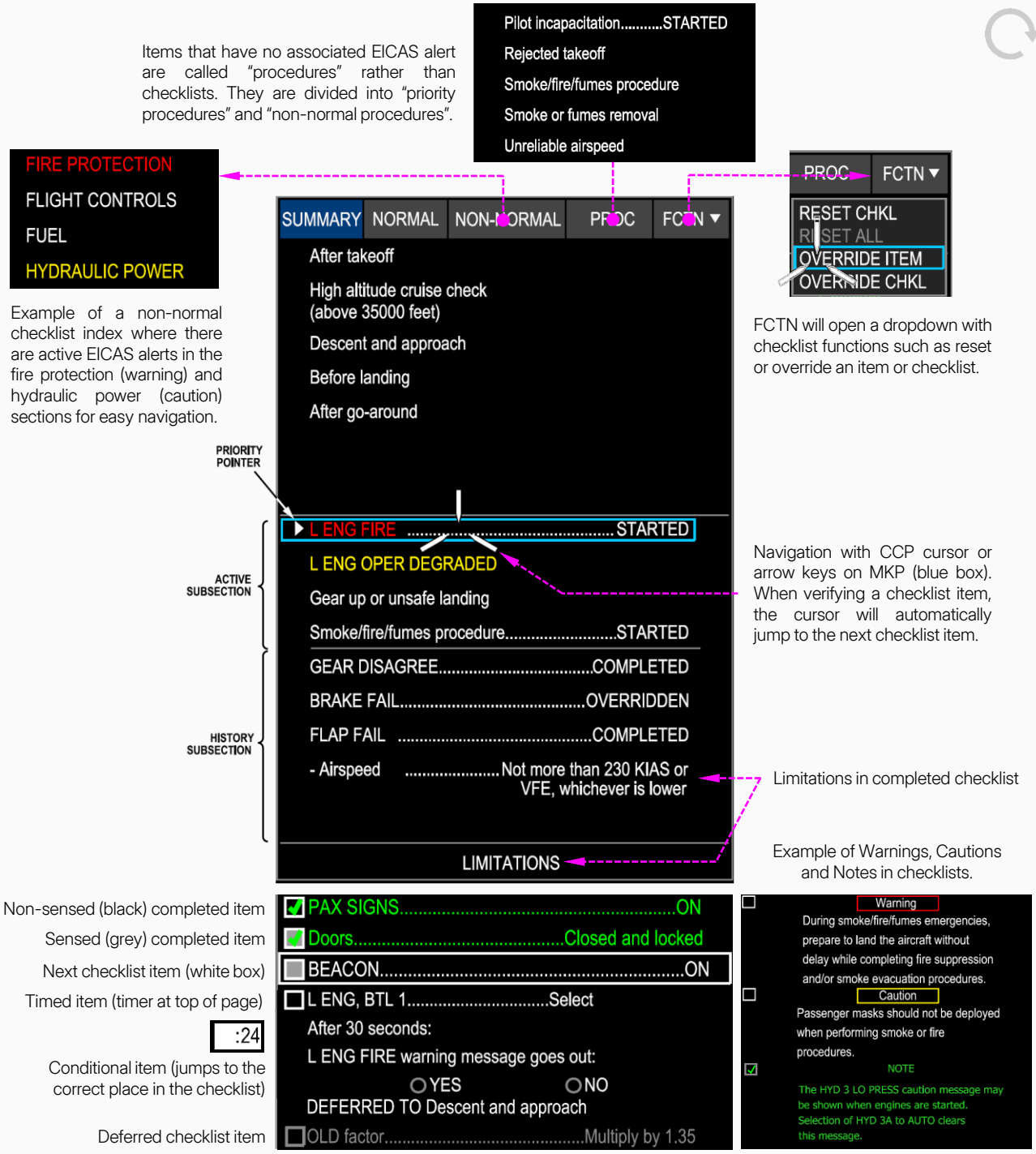
NORMAL – shows normal checklist index (all normal checklists regardless of flight phase).
NON-NORMAL – shows list of systems, clicking on a system displays subsystems. Systems and subsystems with active EICAS alerts are indicated in yellow (for caution messages) or red (for warning messages) for easy navigation.
PROC – shows the list of procedures not associated with an EICAS message (e.g. Pilot Incapacitation), so called "priority procedures" followed by, "non-normal procedures".
FCTN – function tile displays a drop-down to reset or override a checklist or item.

All checklists are auto reset on power-up or with the FCTN tile, reset all checklists is only available on ground, in flight only normal checklists can be reset.

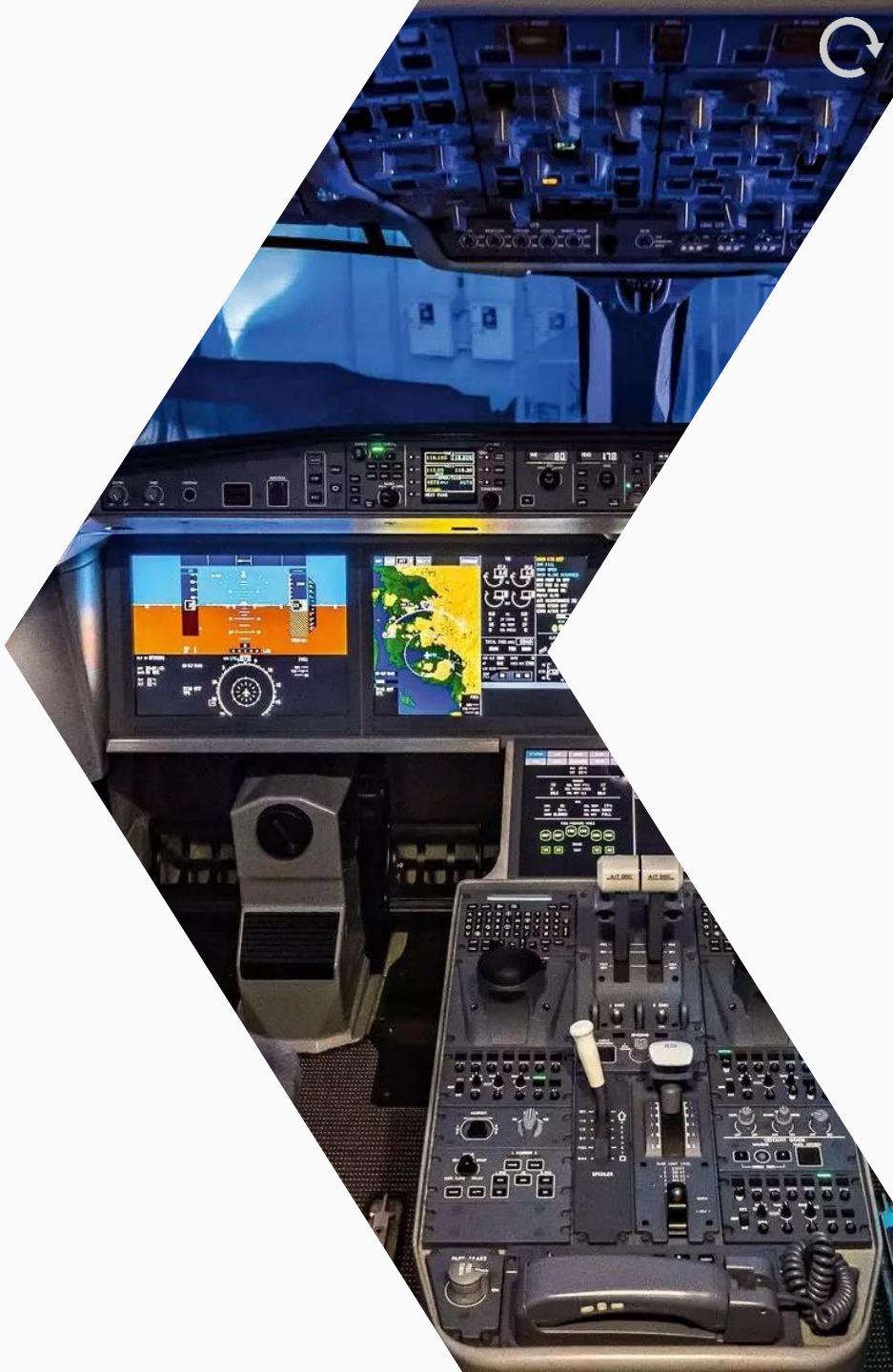
Checkboxes that are filled-in grey are "sensed items", they will automatically receive a green checkmark when the control is in the position requested. Sensed items can be overridden by using FCTN tile, then selecting "OVERRIDE ITEM".

When a checklist item has a time restriction, a countdown timer appears below the FCTN soft key, the ECL can only run one timer at a time. Deferred items (to be completed at a later stage during flight) will automatically show at the appropriate place in a later checklist.

**Note:** "Smoke / fire / fumes (priority) procedure" in ECL refers to the QRH (paper) checklist.



# Flight Management System



# Introduction

There are two Flight Management Systems (FMSs) installed in the aircraft, the FMS is accessed using the Quick Action Keys (QAKs) on the CTP (for DU 2 and 3) or the MKP QAKs (for DU 5). The left side (DU 2 + left half of DU 5) normally operates FMS 1, while the right side (DU 3 and right half of DU 5) normally operates FMS 2. The cross-side FMS can be accessed by pressing the NAV SRC button on the CTP to the offside FMS source.

The FMS can operate in three modes:

- Single mode – active when only one FMS is operational
- Synchronized mode – active when the master and slave FMS computers communicate (crosstalk) with each other. The pilot flying side is considered the master.
- Split mode – active when a fault prevents communication between the master and slave FMS computers.

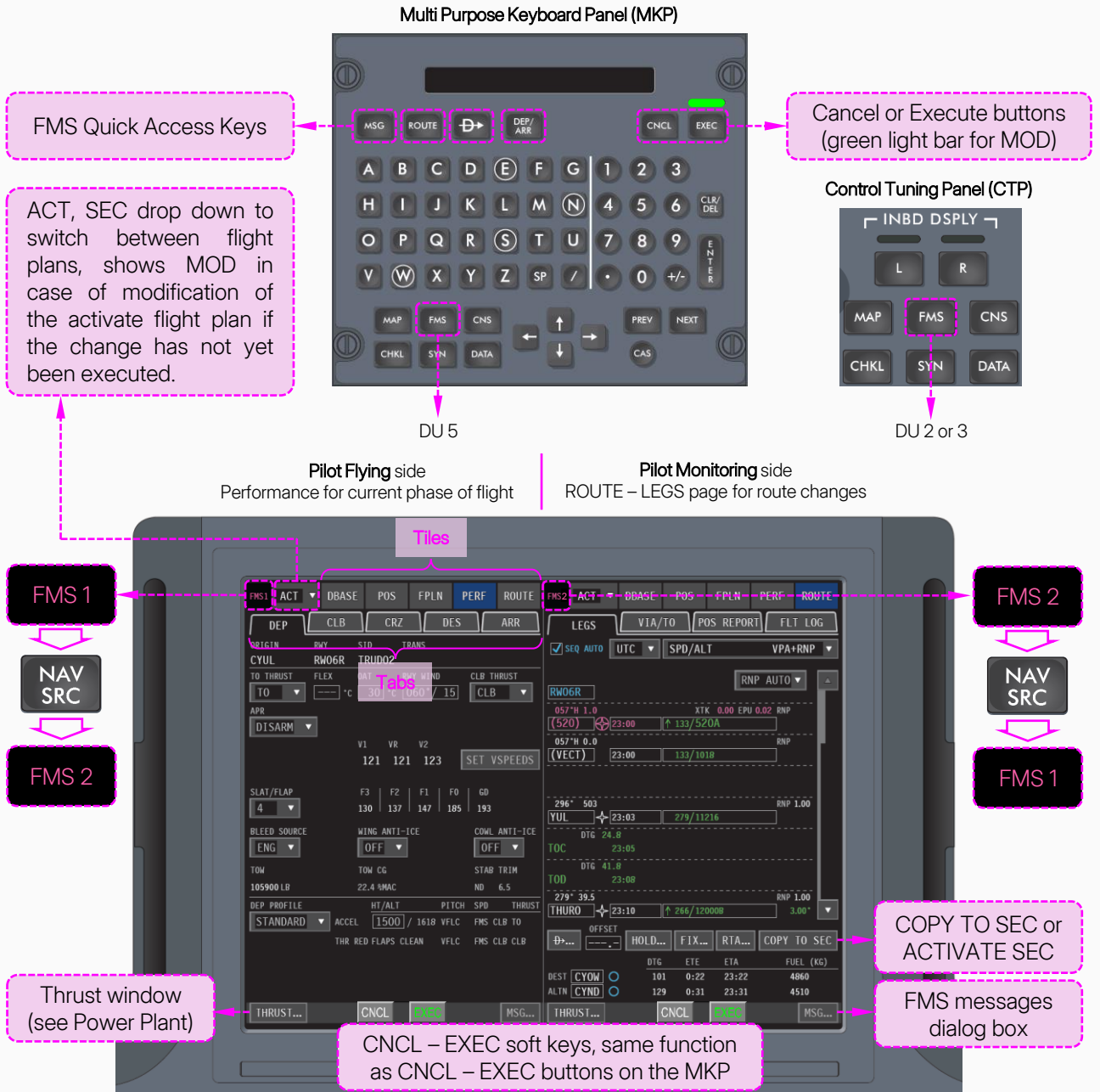
The FCP couple side decides which FMS is the master and which is the slave. FMS can be displayed on any MFW, however, it is standard to use DU 5 for FMS functions. ROUTE – LEGS page on Pilot Monitoring side, PERF for current phase of flight on Pilot Flying side.

The MKP has Quick Access Keys for:

- MSG opens the FMS messages dialog box.
- ROUTE to display the ROUTE tile, subsequent presses cycles tabs.
- DIRECT opens the direct-to dialog box.
- DEP / ARR opens departures / arrivals for the departure airport (within 50 nm from departure airport) or departure / arrivals for the destination (> 50 nm from origin).
- FMS QAK opens the last opened FMS page on the corresponding MFW, subsequent presses will cycle through FMS tiles (DBASE – POS – FPLN – PERF – ROUTE).

At the bottom row, the soft switches to access THRUST and MSG are always shown.
From the top left dropdown window:

- ACT – active flight plan. COPY TO SEC soft switch will copy active flight plan to SEC.
- SEC – secondary flight plan (cyan), ACTIVATE SEC to swap ACT and SEC flight plans.
- MOD - flight plan has been modified but change not yet executed. A green EXEC and white CNCL softswitch appears on FMS, green light bar above MKP EXEC illuminates.



# Database

DBASE – displays database information for navigation, performance, V-speeds and weight and balance.

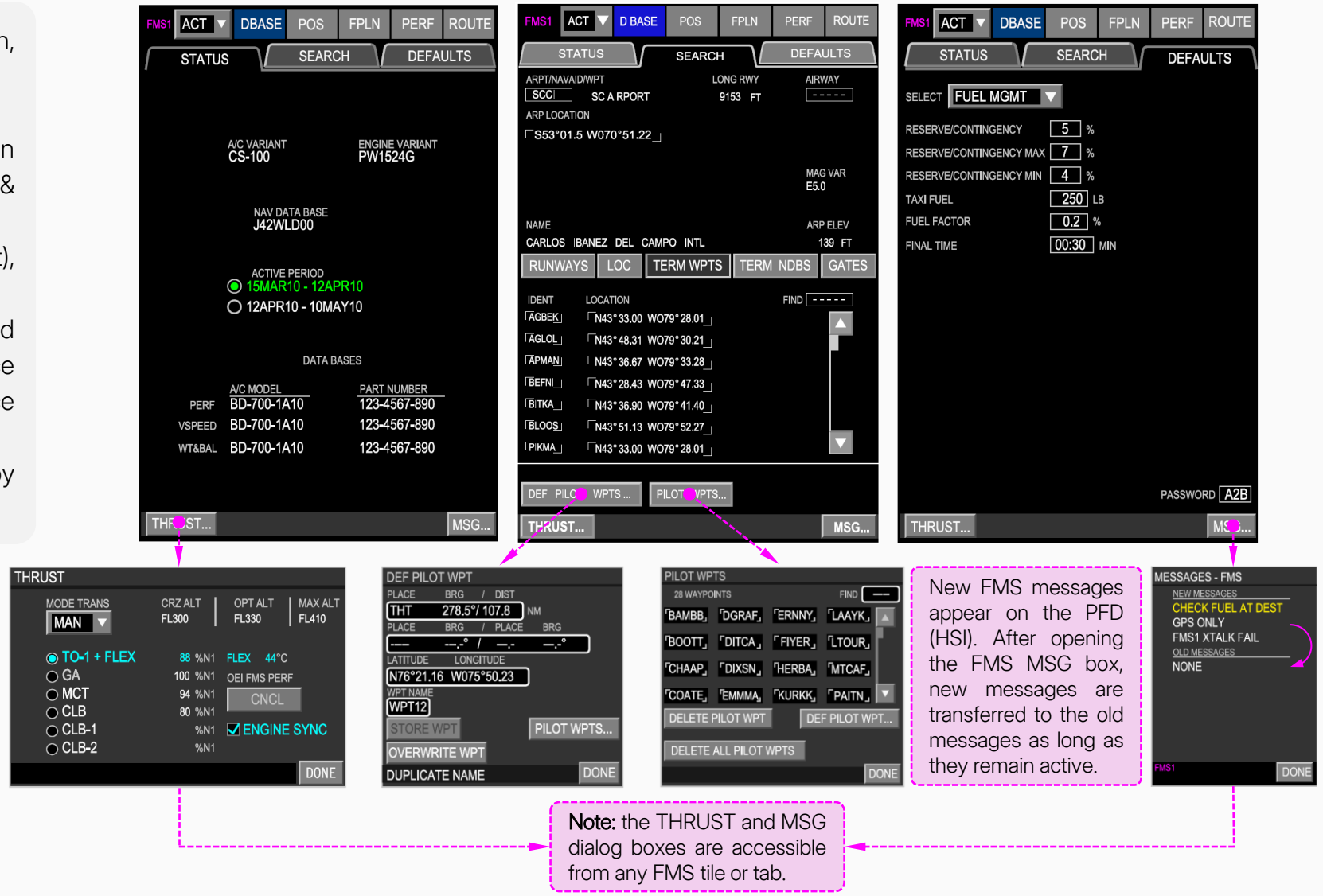
- STATUS – aircraft and engine variant, navigation database, performance, V-speeds, weight & balance database.
- SEARCH – airport, navaid, waypoint searches (left), airway data searches (right).
- PILOT derived waypoints can be created and accessed here. Waypoints are created from Place Bearing / Distance (PB/D), Place Bearing / Place Bearing (PB/PB) or latitude longitude (see below).
- DEFAULTS – modification to the default data by the operator, this page is password protected.

PB/D (Place Bearing/Distance) *("RW" for runway)*:
- **ABCDE 360/10** *(PB either with or without space)*

PB/PB (Place Bearing/Place Bearing):
- **ABC 360/DEF 090** *(PB with or without space)*

Coordinates (lat: DD MM.MM long: DDD MM.MM):
- **N50 30.15 E005 45.25** *(separated by spaces)*

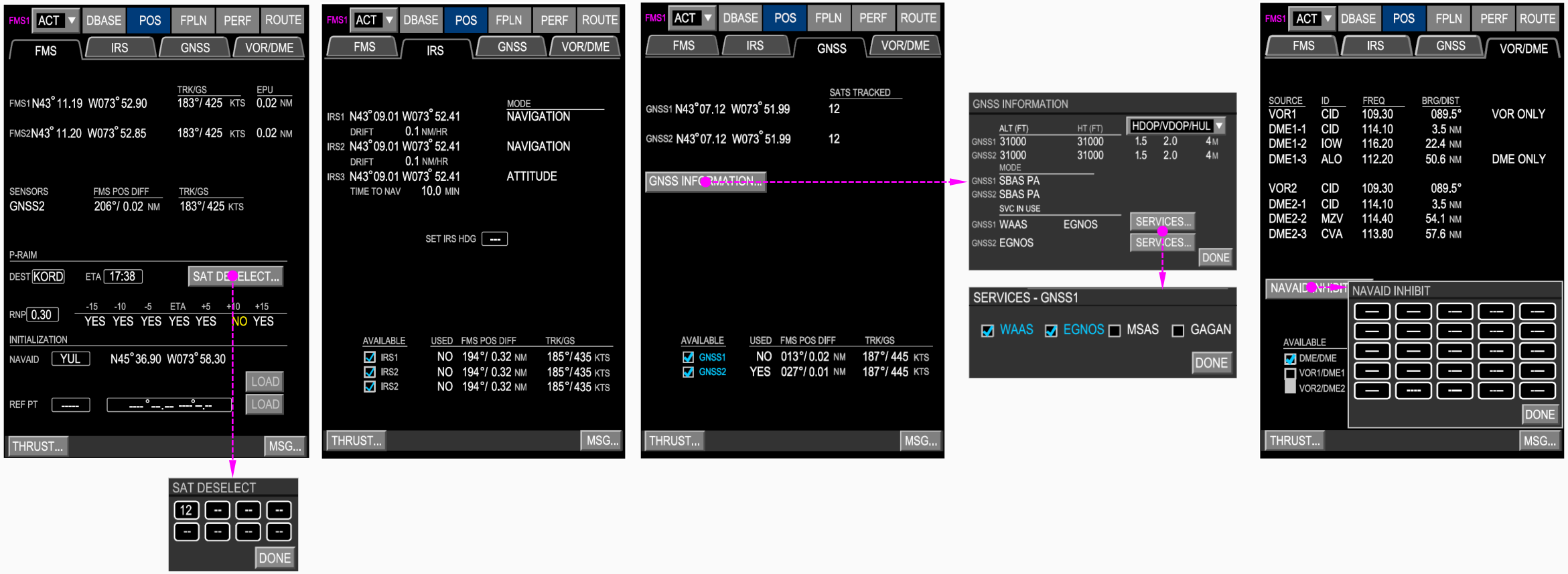
Waypoints can be created directly on the LEGS page by adding an extra "/" followed by a name.
- e.g. **ABCDE 360/10/TEST1** *(max. 5 characters)*



# Position

POS – the position tile has four tabs:

- FMS – current FMS position, sensor status, P-RAIM, SAT DESELECT dialog box and IRS position initialization (if required).
- IRS – position, mode, time to align, drift, SET IRS HDG (if required), select / deselect for navigation, FMS position difference.
- GNSS – position, satellites tracked, GNSS INFORMATION dialog box (showing mode, accuracy and SBAS), select / deselect.
- VOR / DME – navaids used for position updating, ident, frequencies, bearing / distance, NAVAID INHIBIT dialog box.



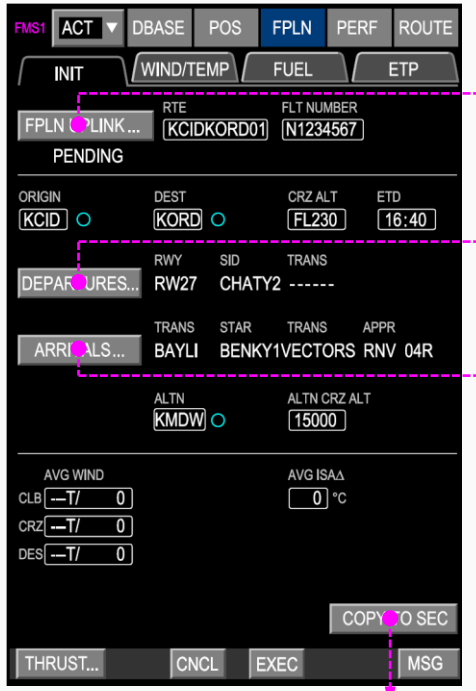
# Flight plan

FPLN – the flight plan tile has four tabs:

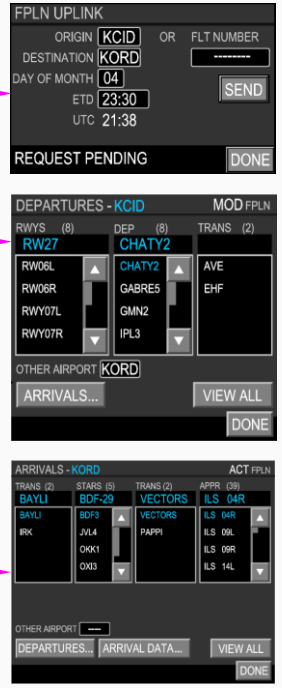
- INIT – flight plan ACARS uplink, origin and arrival airport, departures and arrivals, alternate, cruise levels, average winds and ISA deviation.
- WIND/TEMP – used to enter climb, en-route, descent winds and temperatures at various levels (or simply uplink winds aloft via ACARS).
- FUEL – used to enter total fuel on board, zero fuel weight, zero fuel weight CG, block / taxi / contingency fuel and number of passengers.
- ETP – nearest airports, runway lengths, bearing / distance, time and fuel, filter for MIN RWY LENGTH also filters the airports on the MAP.

**Note:** over the next 400 nm the FMS uses 100% measured data (wind, fuel flow etc.), beyond 400 nm the FMS uses 100% predicted data.

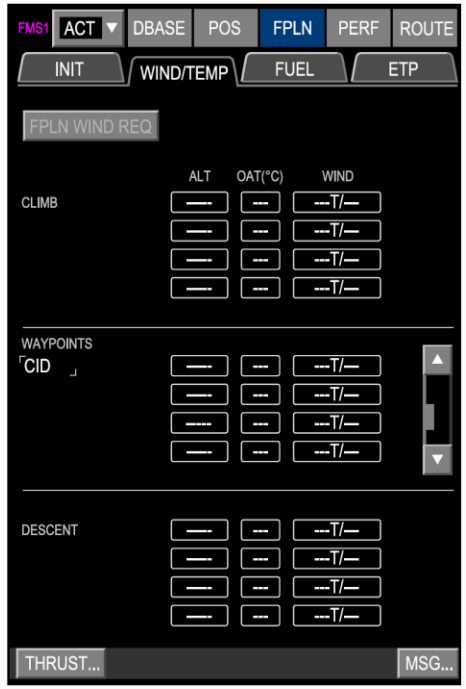
**Note:** the FLT NUMBER (IATA) on FMS FPLN – INIT page is fetched by in-flight entertainment systems. The ATC flight ID (ICAO) is set in the transponder and does not autofill from the FMS.



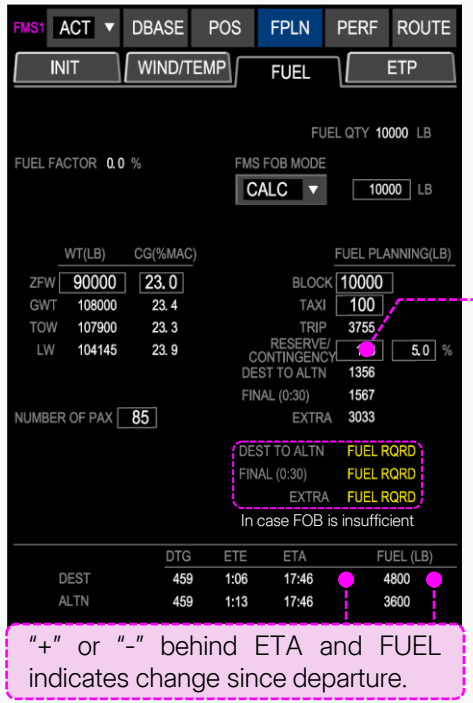

COPY TO SEC copies the active flight plan into SEC flight plan. While in the SEC flight plan, ACTIVATE SEC will be shown instead to swap the ACT and SEC flight plans.




The flight plan is uplinked as SEC flight plan as not to overwrite the ACT flight plan in flight.

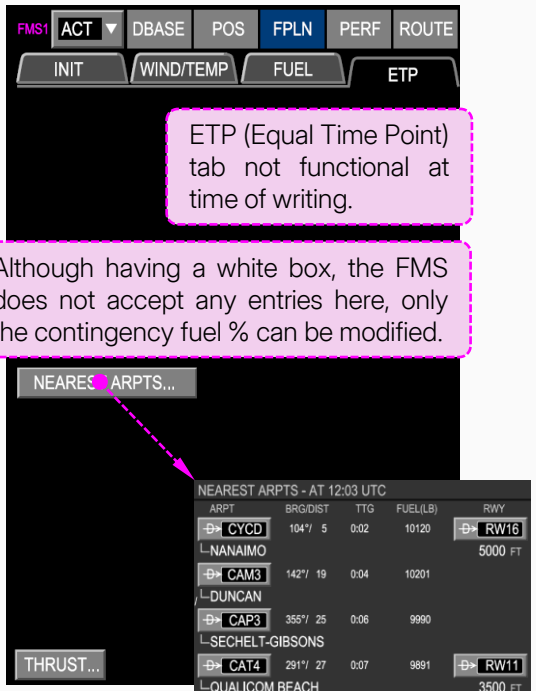



Flight plan wind and temperature can be uplinked (press FPLN WIND REQ), manually entered for climb, descent and en-route waypoints. Or use average wind + ISA on INIT tab.




"+" or "-" behind ETA and FUEL indicates change since departure.

The FMS FOB MODE is normally set to CALC, the FMS uses entered block fuel and subtracts the fuel used. SENSED directly uses FOB sensed by fuel quantity computer.




ETP (Equal Time Point) tab not functional at time of writing.

Although having a white box, the FMS does not accept any entries here, only the contingency fuel % can be modified.

Direct-to an airport or runway can be performed from NEAREST ARPTS box.

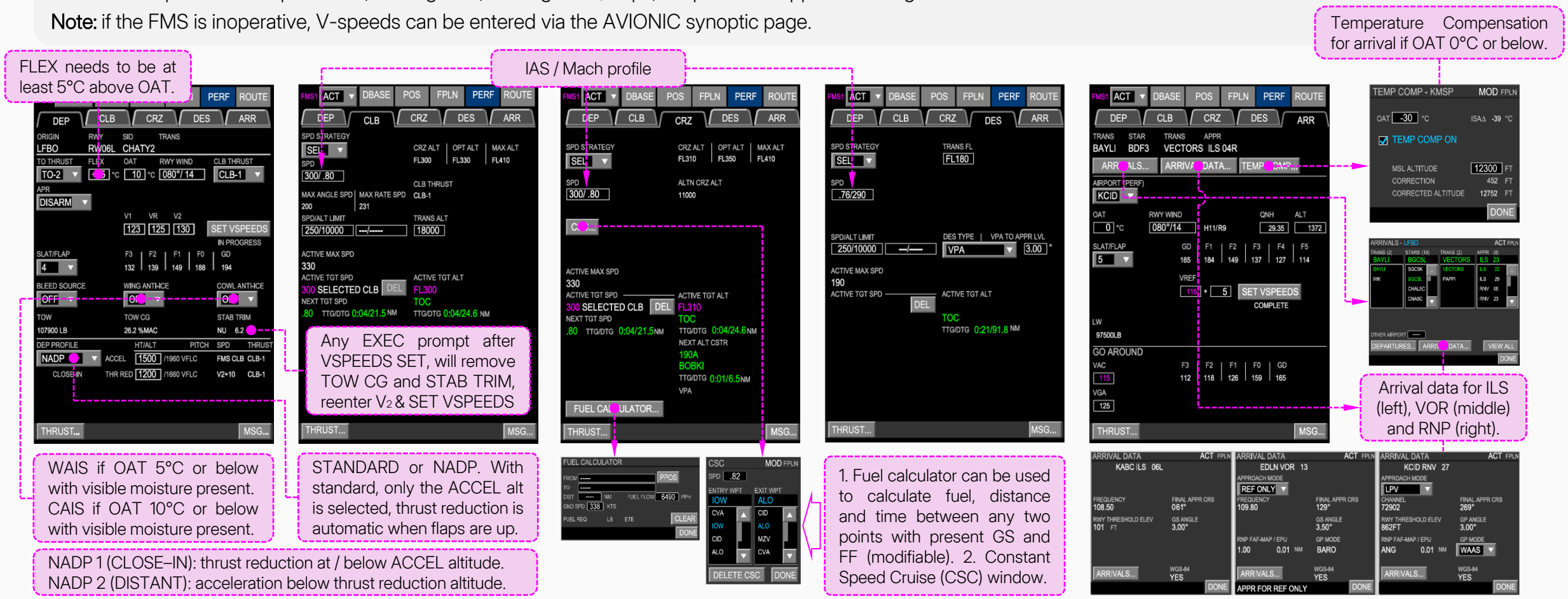
# Performance

PERF – the performance tile has five tabs:

- DEP – take-off and climb thrust selection, FLEX, APR, V-speeds, slats / flaps, bleed source, WAIS and CAIS and departure profile.
- CLB – speed strategy (not functional), speed / altitude limits and transition altitude, selected cruise level, optimum and maximum altitudes.
- CRZ – cruise, optimum, maximum and alternate altitudes, speed strategy (not functional), constant Speed Cruise (CSC) and fuel calculator.
- DES – speed strategy (not functional), speed and altitude limits, transition level, descent type (not functional) and Vertical Path Angle (VPA).
- ARR – temperature compensation, landing data, landing slats / flaps, V-speeds for approach and go-around.

**Note:** if the FMS is inoperative, V-speeds can be entered via the AVIONIC synoptic page.

**Pre-Avionic Built 6.5:**
Although the FMS may accept such input, FLEX in combination with derated thrust (TO-1, 2, 3) is not authorized. Select full rated thrust (TO) for using FLEX.

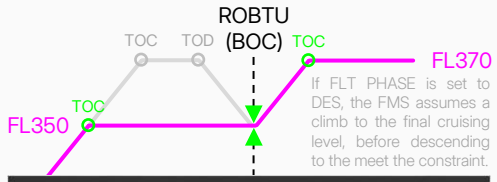


# Route

To program step climb(s):

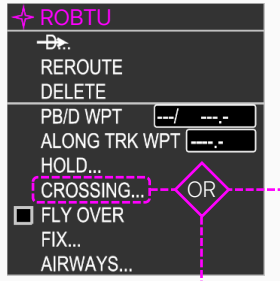
a. Set final cruising level on the FPLN – INIT page (e.g. FL370).
b. At the waypoint where further climb is expected (Bottom of Climb - BOC), select the initial cruising level (e.g. FL350).
c. On the CROSSING dialog box, ensure FLT PHASE is CLB, alternatively add "C" behind the flight level on the legs page.

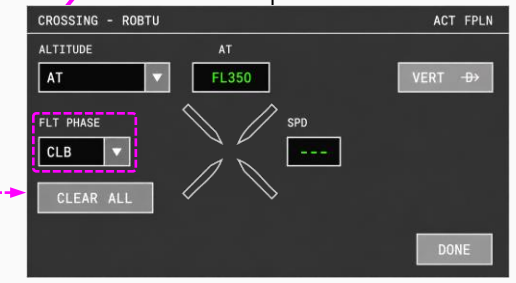
**Note:** the CROSSING dialog box only accepts "FL" as format.




ROUTE – the route tile has four tabs:

- LEGS – waypoints, dropdowns for route data, FIX, HOLD, DIRECT, flight plan summary for destination and alternate.
- VIA / TO – route legs (SID, STAR, transition, approach, airway) and waypoints, summary for destination and alternate.
- POS REPORT – last overflown waypoint, next three waypoints, ETA, time / distance to go, fuel, temperature & wind.
- FLT LOG – summary information for each waypoint, take-off / landing times, average speeds & remaining distance.



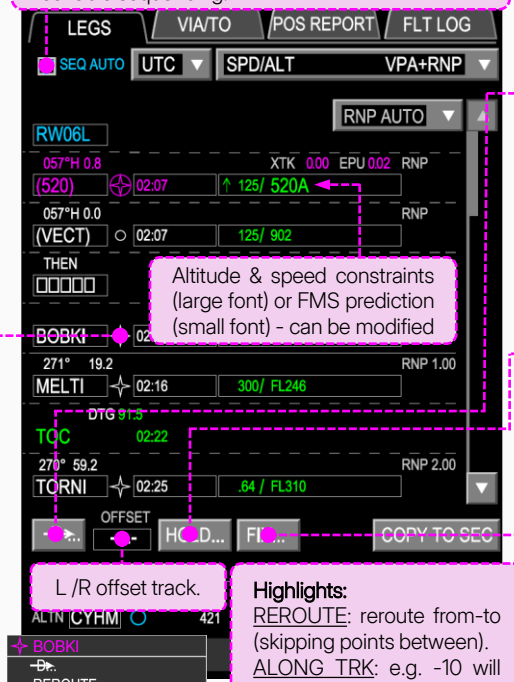




LEGS page

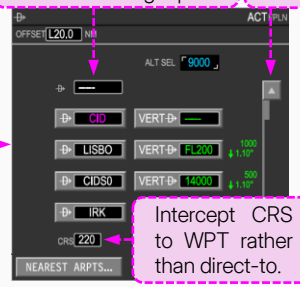
SEQ AUTO will prevent sequencing of the next waypoint if deselected. In case of missed approach, pressing TOGA will automatically reenable sequencing.

Direct-to a waypoint that is not in flight plan

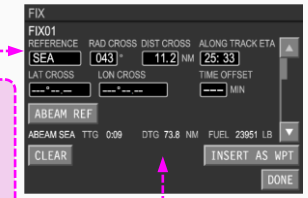
**Hidden feature:** scrolling all the way forward will show previous executed directs, enables rolling back previously executed changes.

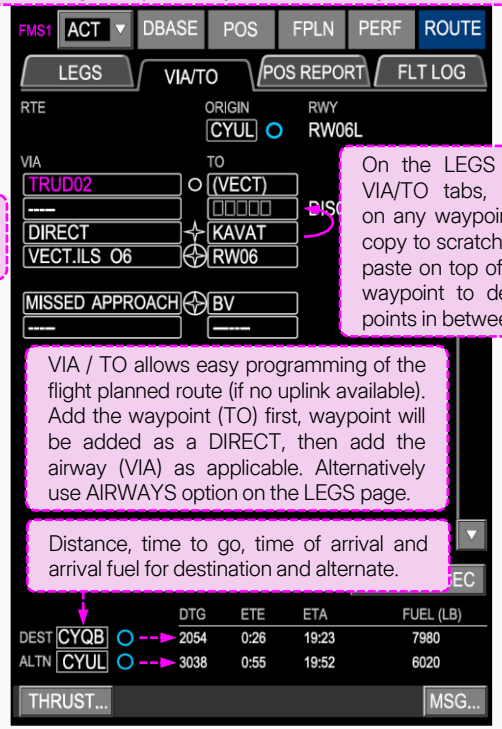



Altitude & speed constraints (large font) or FMS prediction (small font) - can be modified




Intercept CRS to WPT rather than direct-to.

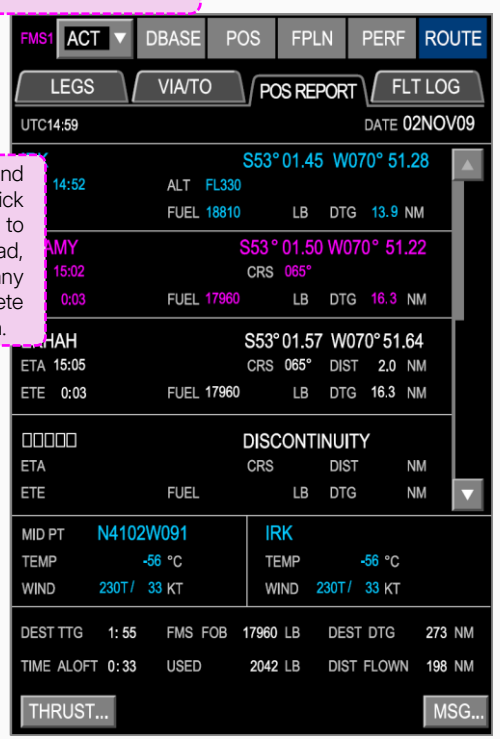


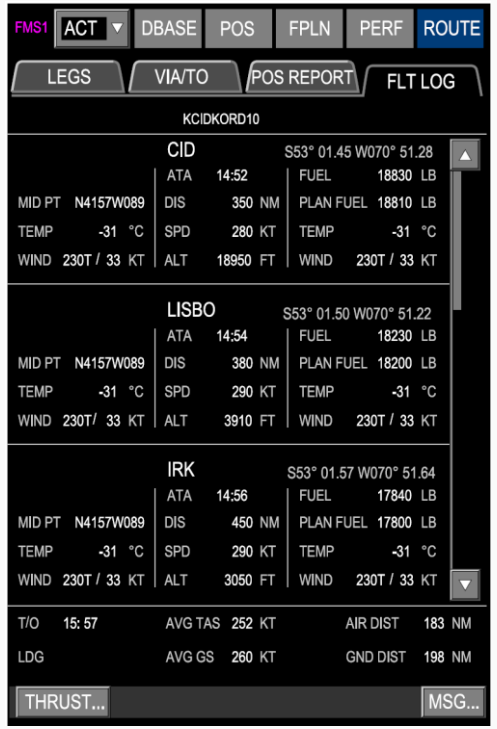




On the LEGS and VIA/TO tabs, click on any waypoint to copy to scratchpad, paste on top of any waypoint to delete points in between.

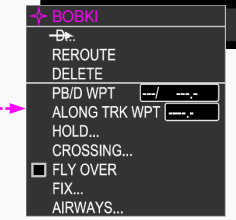
VIA / TO allows easy programming of the flight planned route (if no uplink available). Add the waypoint (TO) first, waypoint will be added as a DIRECT, then add the airway (VIA) as applicable. Alternatively use AIRWAYS option on the LEGS page.

Distance, time to go, time of arrival and arrival fuel for destination and alternate.




L /R offset track.




**Highlights:**
REROUTE: reroute from-to (skipping points between).
ALONG TRK: e.g. -10 will create a point 10 nm prior to the selected waypoint.
FLY OVER / BY checkbox
FIX: adds WPT to FIX page
AIRWAYS: use to manually enter flight plan (VIA - TO).

Top - Direct-to window for lateral or vertical directs. Direct to a waypoint, offset track or intercept a course to a waypoint. NEAREST ARPTS soft switch (as described before under FPLN – ETP tab).
Middle - HOLD prompt allows setting up holding at present position or at any waypoint. Select the INBD CRS and turn DIRECTION, maximum holding speed (select ICAO or FAA), best speed (green dot) versus selected holding speed, adjust leg timing or distance. EFC is the Expected Further Clearance (holding exit) time for fuel calculations (**Tip:** each lap in a standard holding takes 4 minutes).
Bottom - FIX window allows to create fixes on the map, this can be used to mark waypoints, draw radials or distance arcs (useful for drawing engine-out departure procedure or driftdown and depressurization procedures over high terrain). It is also possible to create ABEAM REF points and then INSERT AS WPT to the flight plan to conduct fuel checks (will be logged in FLT LOG at fly over).

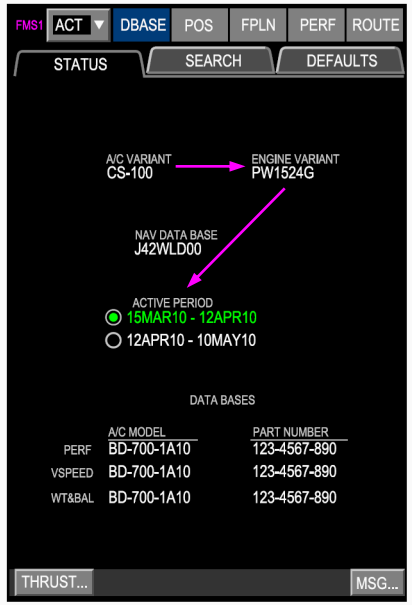
# FMS Preparation

1. DBASE
   - STATUS
2. POS
   - FMS
   - IRS
   - GNSS
   - VOR/DME
3. FPLN
   - INIT
   - WIND/TEMP*
   - FUEL
4. PERF
   - DEP
   - CLB
   - CRZ
   - DES
5. ROUTE
   - VIA/TO
   - LEGS (FIX*)

**Note:** if flight plan uplink unavailable, set-up the ROUTE before entering or uplinking wind and temperature.

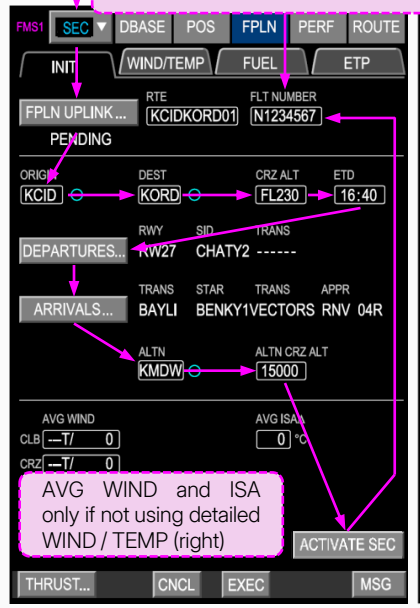
**Note:** on the LEGS tab (not shown here), add fixes as desired (e.g. engine out departure procedure for reference).

START

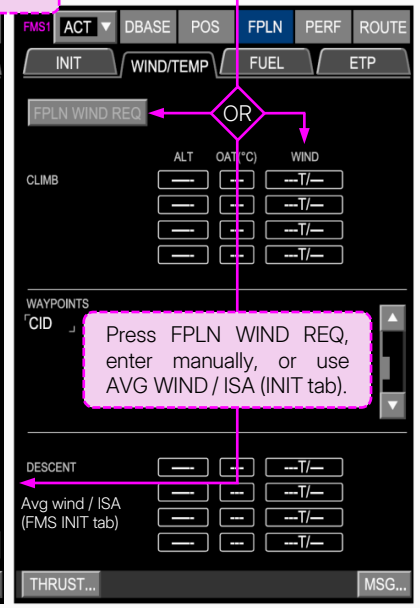


FPLN will be uplinked as SEC FPLN

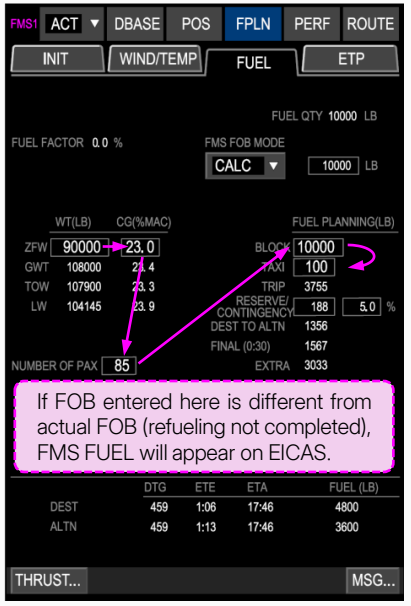
IATA FLT NO. after ACTIVATE SEC



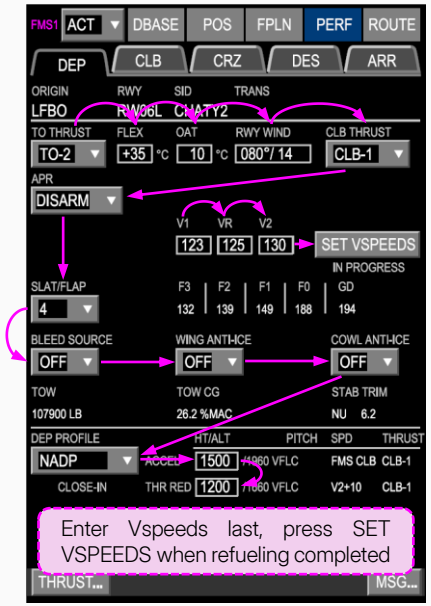

AVG WIND and ISA only if not using detailed WIND / TEMP (right)



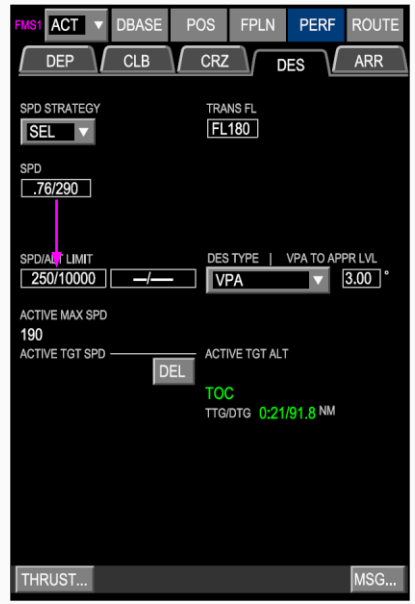

Press FPLN WIND REQ, enter manually, or use AVG WIND / ISA (INIT tab).

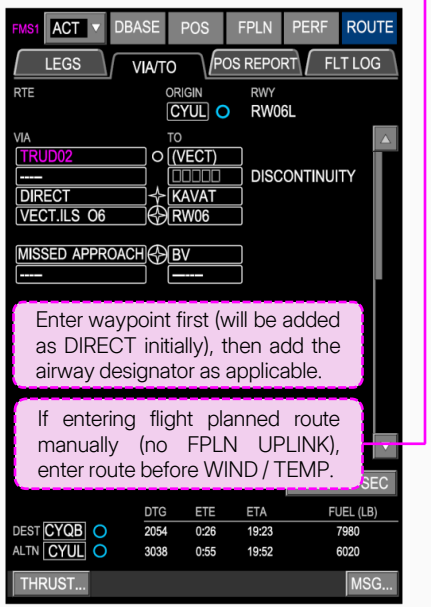



If FOB entered here is different from actual FOB (refueling not completed), FMS FUEL will appear on EICAS.




Enter Vspeeds last, press SET VSPEEDS when refueling completed




Enter waypoint first (will be added as DIRECT initially), then add the airway designator as applicable.

If entering flight planned route manually (no FPLN UPLINK), enter route before WIND / TEMP.

END

# Vertical Situation Display (VSD)

The Vertical Situation Display (VSD) gives a graphical picture of the vertical path relative to waypoints, constraints, terrain and the runway (for situational awareness purposes only). VSD normally displays vertical profile along the **flight plan** but automatically reverts to **aircraft track** when:

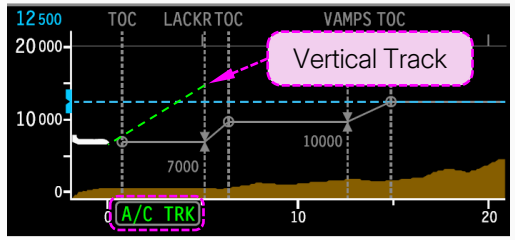
- Deviating from the lateral path by more than the RNP value
- Difference between current and desired track exceeds 90°
- Active lateral mode is not FMS or LOC
- Flight plan discontinuity
- Sequence inhibited
- Offset selected
- No valid flight plan
- Holding
- GNSS failure
- TAWS alert active

Symbology includes:

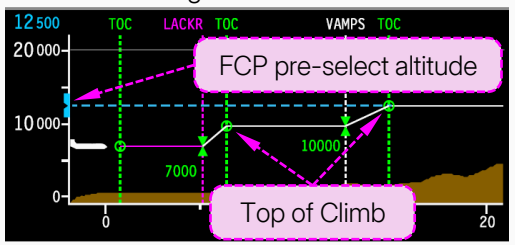
- Aircraft current vertical flight path / trend (green dashed line).
- Top of Climb (TOC), Top of Descent (TOD) and Final Approach Fix (FAF).
- FMS flight plan, active leg magenta, yellow if unable next altitude constraint, non-active legs are white, all legs are grey if A/C TRK mode active.
- FCP preselected altitude (blue bug with dashed blue line).
- Ground features: terrain, runways, distance markers. Terrain within a corridor of 2 times the RNP value (1 x RNP each side of track or flight plan) is shown.
- ABOVE, BELOW, AT, WINDOW constraints (green triangles).
- ILS GP feather (solid white), LPV feather (dashed white).

VSD is not available with MAP in the PLAN mode.
For VSD terrain symbology, see navigation chapter.

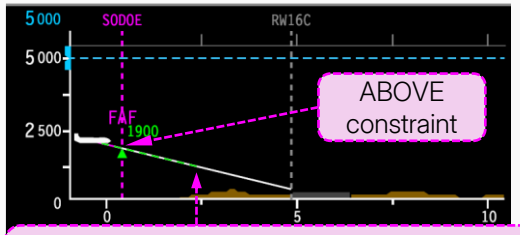
Aircraft Track mode



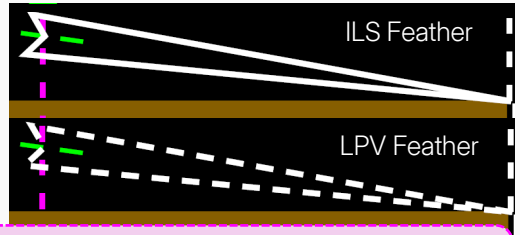

Flight Plan mode




APV or Non-precision approach




Precision approach (ILS or LPV)




**Note:** if for a non-precision approach, a path is shown from the FAF to the runway, then (advisory) VNAV is available for vertical guidance. Another place to check if vertical guidance exists is the ARRIVAL DATA window (GS ANGLE blank = no guidance).

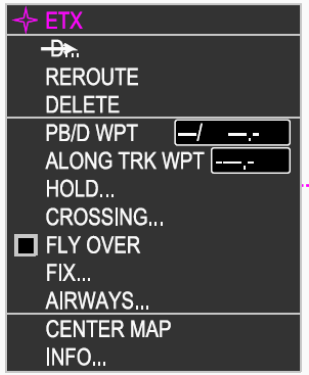
# Graphical Flight Planning

The FMS flight plan can be modified graphically on the MAP display. Selection of an object or symbol on the MAP will open a pop-up menu to make interactive modifications directly on the MAP display.

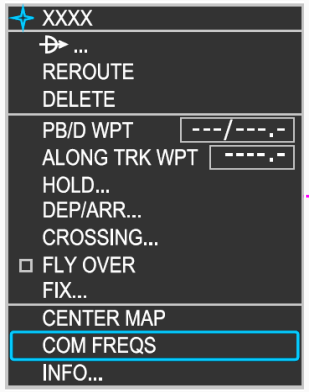
The Present Position (PPOS) menu can be accessed by selecting the aircraft symbol. This is also the place to find current wind components and ISA deviation for example.

Selecting a navaid opens the navaid or waypoint menu, tuning of NAV frequencies is possible from the MAP.

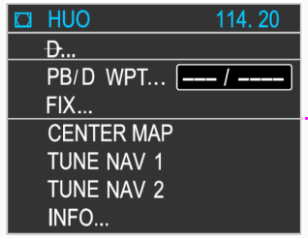
Selecting any blank space on the map allows creating any custom waypoint or opening en-route ATC frequency list. Although not recommended, VHF 1 and 2 can be tuned as well to any frequency directly from the MAP display.




Active waypoint menu




Airport menu



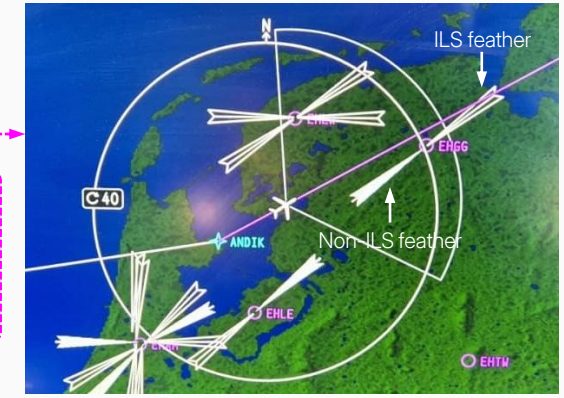

Navaid menu

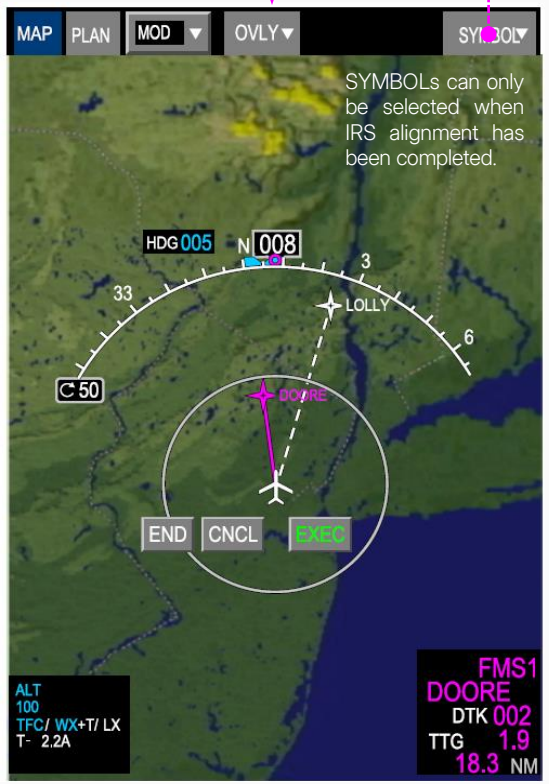
LAT / LONG menu

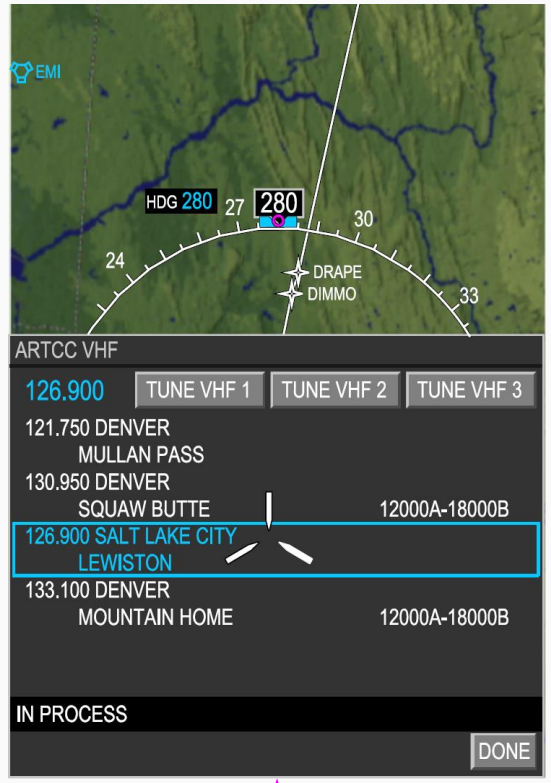
**Note:** direct-to and reroute can be done towards any symbol on the map (waypoint, airport, navaid, etc.) or any LAT / LONG by clicking any blank space on the map. Click any airport to access DEP / ARR window or any approach feather (decluttered at higher ranges) to access approach options (similar to NEAREST ARPTS).

Direct-to (modification white dashed before executing)

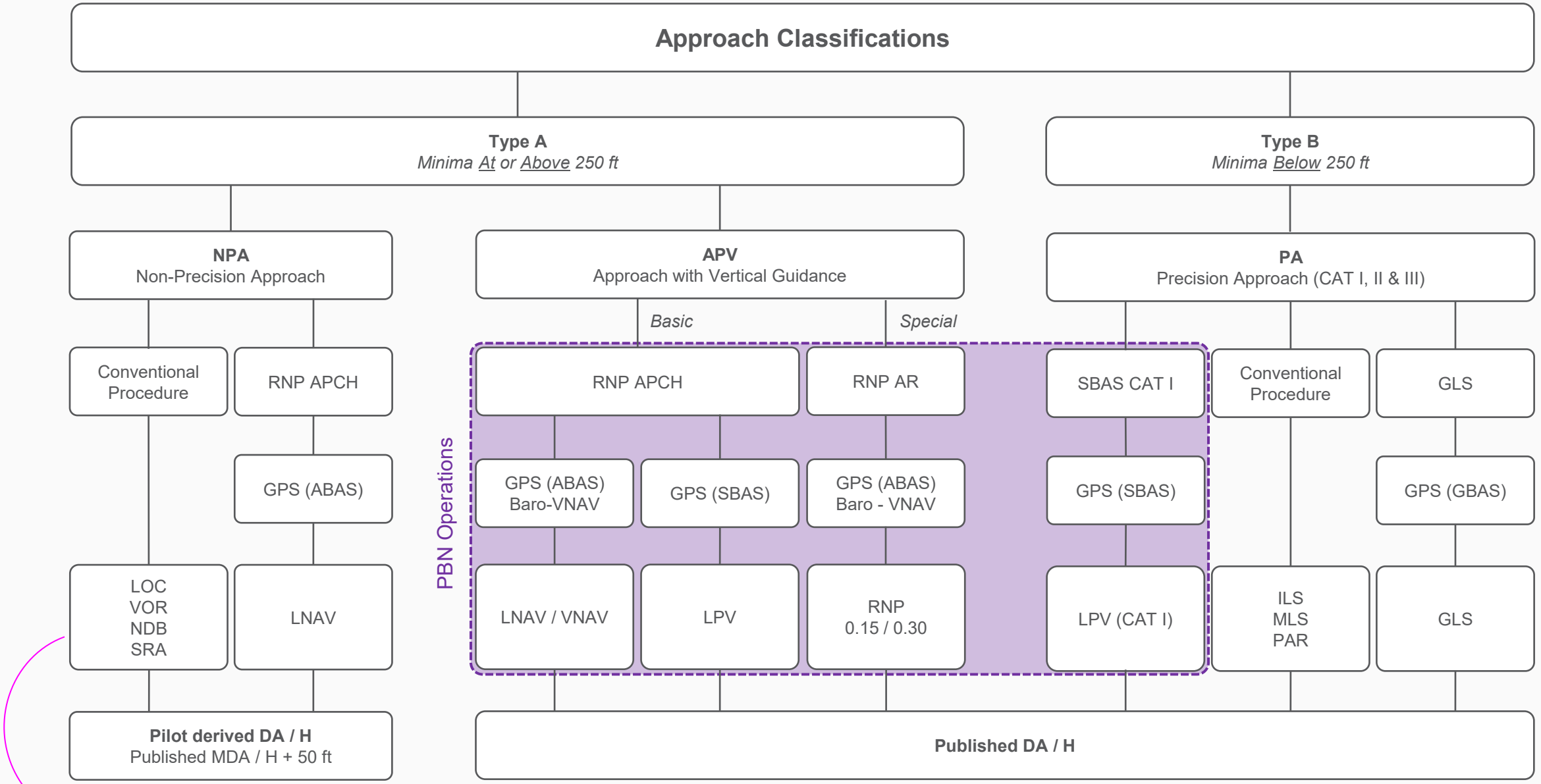
Select symbols to be displayed on the map, e.g. navaids, (nearest) airports, airways, waypoints.




ARTCC frequencies




**Note:** procedurally approaches are categorized as either 2D or 3D approach operations, most approaches can be flown as a 3D approach. Exceptions are LOC and VOR / NDB approaches without GNSS overlay. For VOR and NDB approaches without GNSS overlay, FMS overlay (REF ONLY APPROACH) with advisory Baro-VNAV might be available. No advisory VNAV is indicated by absence of a path from FAF to RWY on the VSD and GP ANGLE blank in ARRIVAL DATA. LOC approaches can only be flown conventionally.

# RNAV and RNP

RNP is similar to RNAV but with an onboard performance monitoring & alerting requirement. RNP-capable aircraft require a specific level of accuracy to be achieved 95% of the time and twice this value 99.999% of the time (containment limit). RNP is always relying on GNSS while RNAV can be flown using GNSS, IRS, DME/DME or combination thereof. The FMS automatically sets RNP as follows (RNP AUTO on ROUTE - LEGS page):

- **RNP 2.00** en-route or upon exit from terminal area.
- **RNP 1.00** standard instrument departures and arrivals and within terminal airspace
- **RNP 0.30** during FMS based approaches
- **Angular (ANG)** during LPV approach

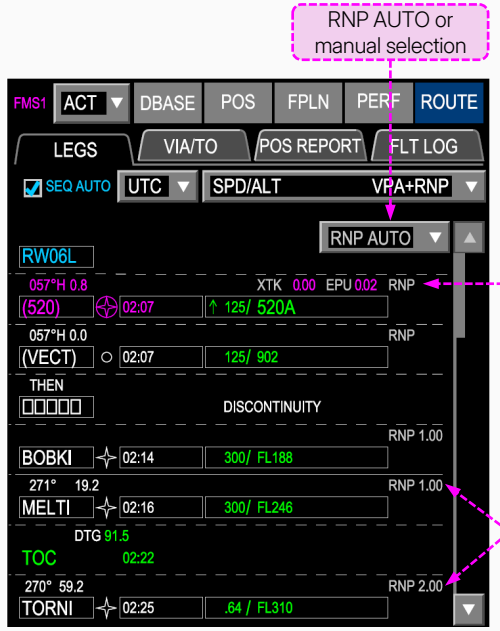
The Estimated Position of Uncertainty (EPU) is an error calculation made by the FMS based on sensors available for position updates. Full scale deflection represents a flight technical error equivalent to the RNP value. If the EPU would exceed the RNP value, a NO RNP message is shown (or NO APPR within 31 nm with RNP approach selected).

Procedurally, approach operations are very similar regardless of the type of approach flown, whether it is RNP (LNAV only, LNAV / VNAV), RNP-AR or LPV. HSI indications:

- RNP ARM appears 31 nm from the airport, replaced by RNP APPR 2 nm prior to FAF.
- LPV ARM appears 31 nm from the airport, replaced by LPV APPR 2 nm prior to FAF.
- RNP AR ARM 31 nm from the airport, RNP AR APPR at the first point of the approach.

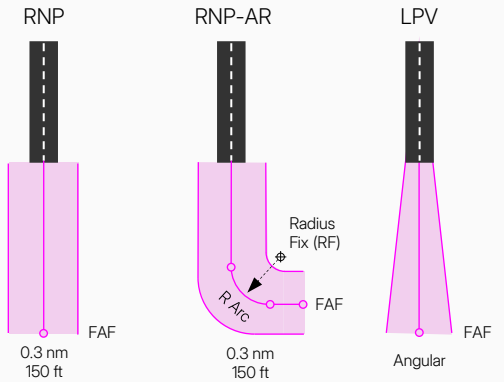
**RNP & RNP-AR APPR** – requires Aircraft Based Augmentation System (ABAS), for this RAIM is required (see POS – FMS page). For RF-legs (RNP AR) deselect ½ bank mode. Vertical guidance is BARO-VNAV, therefore GP angle is influenced by temperature.

**LPV APPR** – requires Satellite Based Augmentation System (SBAS) (WAAS in the US, EGNOS in Europe). Vertical guidance is based on geometric (GPS) altitude. Both lateral and vertical guidance are angular (like an ILS). The GP angle is not influenced by temperature, but temperature compensation can be used to ensure smooth transition from barometric to geometric altitude. A LPV approach can continue in case of single GNSS or SBAS failure within the IAF. If a single GNSS or SBAS receiver fails, while approach is not yet active (no LPV APPR), LPV will not be available on the failed side.

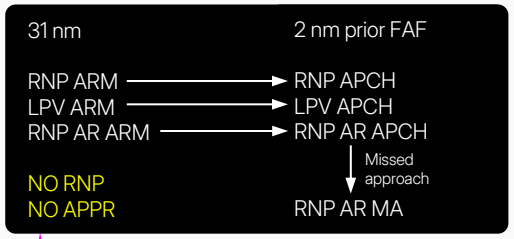


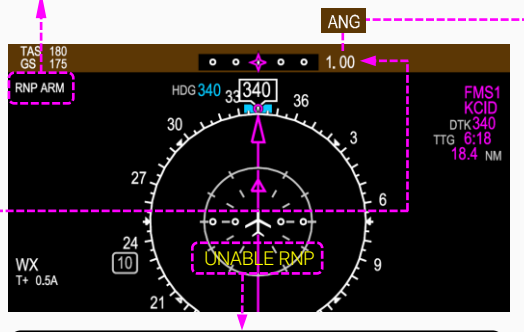

| Approach Type | RNP APPR LNAV only | RNP APPR LNAV VNAV | RNP AR APPR | LPV APPR |
|---|---|---|---|---|
| Lateral | Full scale = 0.3 nm (RNP 0.3) | | | ANG |
| Vertical | Full scale = 150 ft | | | |
| Glidepath | BARO | | | GPS |
| FMA | APPR FMS 1(2) VGP | | | |

Instrument approach charts are titled **RNP** or **RNP (AR)** for Authorization Required




**RNP-AR (Authorization Required)** is a type of RNP approach that requires special operator approval. Contrary to a "normal" straight-in approach, RNP-AR includes curved paths with RF (Radius-to-Fix) legs. A220 is certified for RNP-AR with a RNP of 0.3 for approach and RNP 1.0 for missed approach.




GNSS DISABLED
GNSS NOT AVAILABLE
GNSS REVERTED
INVALID SBAS CONFIG
LNAV AVAILABLE
SBAS DISABLE
SBAS NOT IN USE
SBAS PA NOT READY
UNABLE RNP

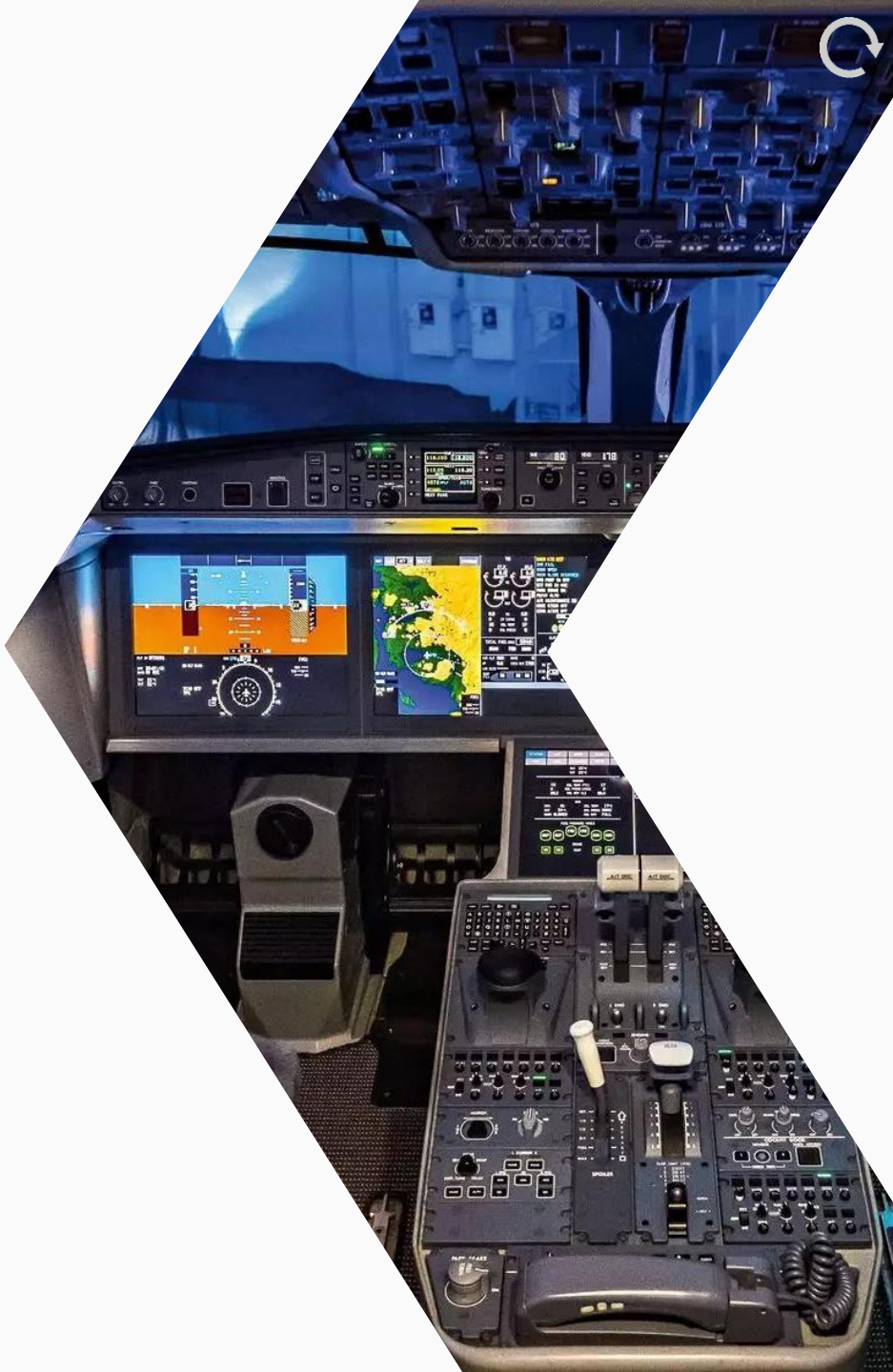
APPR NOT AVAILABLE
LPV NOT AVAILABLE
GNSS NOT AVAILABLE
UNABLE RNP

**LPV (Localizer Performance with Vertical guidance)** is an ILS like RNP approach. Both the lateral and vertical guidance are angular (ANG), they become more precise closer to the runway.

The vertical guidance is based on geometric (GPS) altitude and therefore not affected by temperature. There could be a jump in flight guidance when changing from barometric to geometric altitude depending on temperature.

All other RNP approaches are based on barometric altitude reference (BARO-VNAV).

# Surface Management System



# Surface Management System (SMS)

### Surface Management System (SMS) (if installed)

The Surface Management System (SMS) provides runway identification and visual / aural alerts to enhance safety during take-off, landing and ground operations. It consists of two functions:

- Airport Moving Map (AMM)
- Take-off and Landing Awareness Function (TLAF)

### Take-off and Landing Awareness Function (TLAF)

TLAF refers to the FMS active flight plan and operates automatically within the terminal area.

Mode 1 – not a runway during take-off (warning):

- Take-off thrust is applied (TLA > 23° and $N_1$ > 55% on both engines) for more than 3 seconds or ground speed exceeding 35 kts for more than 3 sec but not exceeding $V_1$, and
- The aircraft is not aligned (heading within 10 degrees of runway bearing) with any runway.
- A red RUNWAY flag will appear in the center of the PFD and an aural "NOT A RUNWAY, NOT A RUNWAY" is heard.

Mode 3 – runway mismatch during take-off (caution):

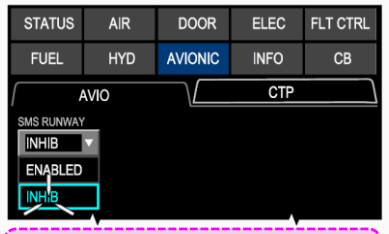
- Take-off thrust is applied (TLA > 23° and $N_1$ > 55% on both engines) for more than 3 seconds or ground speed exceeding 35 kts for more than 3 sec, but not exceeding $V_1$, and
- The aircraft is aligned (heading within 10 degrees of runway bearing) with a different runway than the FMS departure runway.
- A yellow RUNWAY flag will appear in the center of the PFD and an aural "RUNWAY DISAGREE, RUNWAY DISAGREE" is heard.

Mode 4 – not a runway during landing (warning):

- Active when aircraft track and position are within 10 nm of the FMS destination.
- Landing gear down and locked, aircraft is within 250 ft from runway elevation, the aircraft track does not intersect a box 150 ft wide on either side of the extended centerline (inhibited if within 1 dot lateral deviation on ILS or LPV approach).
- A red RUNWAY flag will appear in the center of the PFD and an aural "NOT A RUNWAY, NOT A RUNWAY" is heard, alert will cease if go-around initiated, upon landing or when the aircraft climbs 500 ft above the runway elevation.

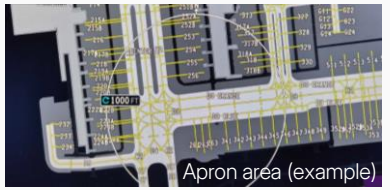
### Airport Moving Map (AMM)

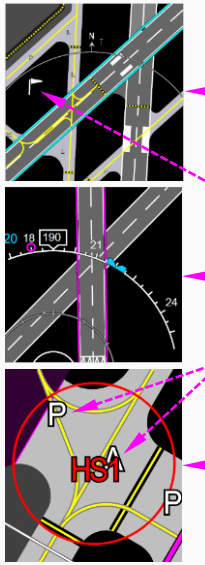
The Airport Moving Map (AMM) is accessed by using the MAP zoom rotary switch while on ground. Zooming in below 2 nm automatically displays the AMM. The AMM can be used in the MAP and PLAN modes. Positional information is provided by GNSS.




SMS can be inhibited on the AVIONIC synoptic page

Aerodrome Reference Point



Cyan FMS departure runway

Windsock location

Magenta FMS arrival runway

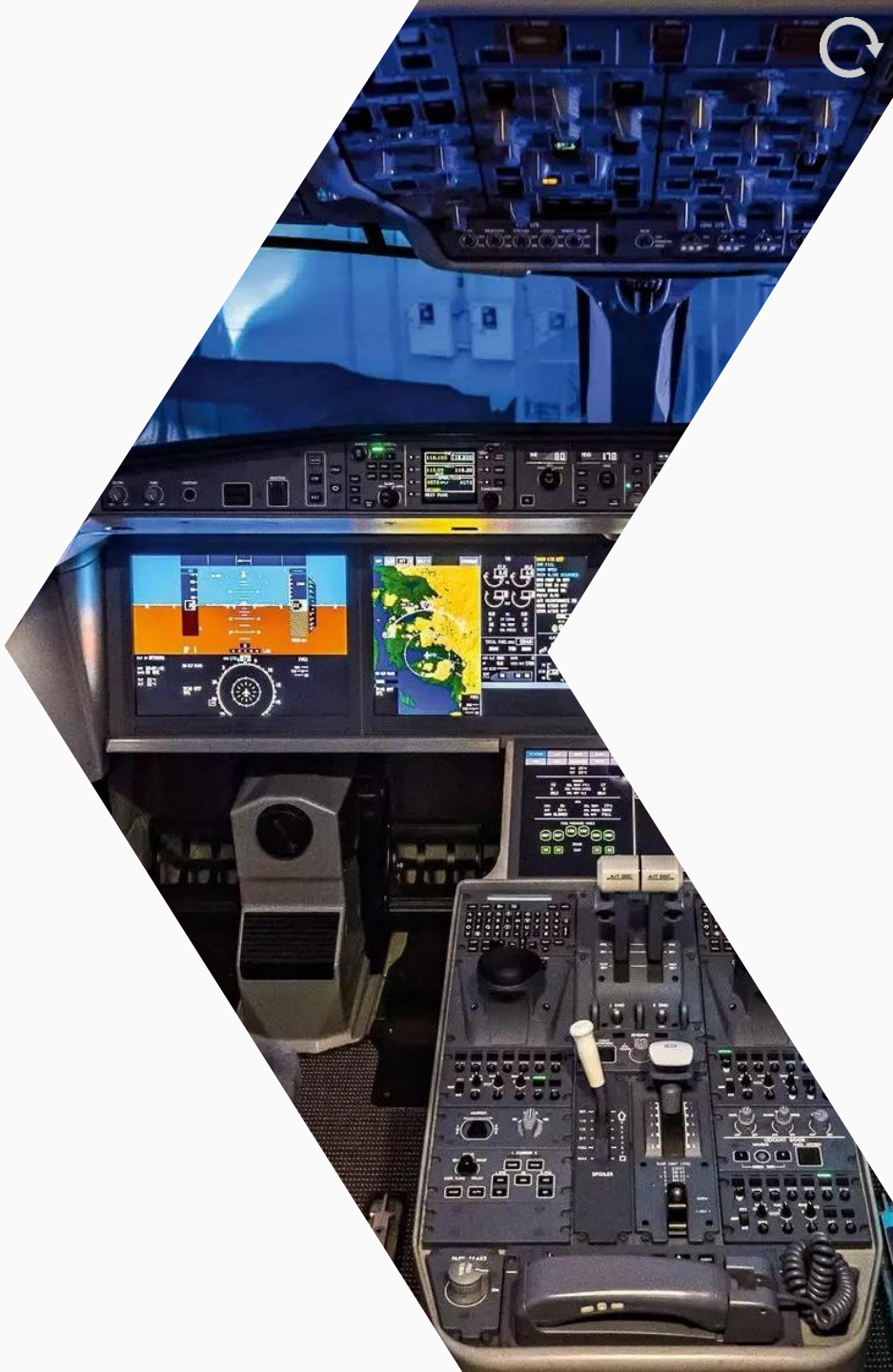
Taxiway designators

Runway Incursion Hotspots

TLAF Mode 1 and 4 "NOT A RUNWAY"

TLAF Mode 3 "RUNWAY DISAGREE"

# Miscellaneous



# Miscellaneous Systems

## Recording

Cockpit Voice Recorder & Flight Data Recorder.

Recording starts:

- Beacon or strobe lights ON, or
- Weigh-off-wheels (CVR) / first engine start (FDR)

Recording stops:

- On ground, and
- 10 minutes after beacon and strobe lights OFF (CVR) / last engine stopped (FDR)

Power removed (crash situation):

- No weight-on-wheels, and
- No airspeed and no engine oil pressure in at least one engine

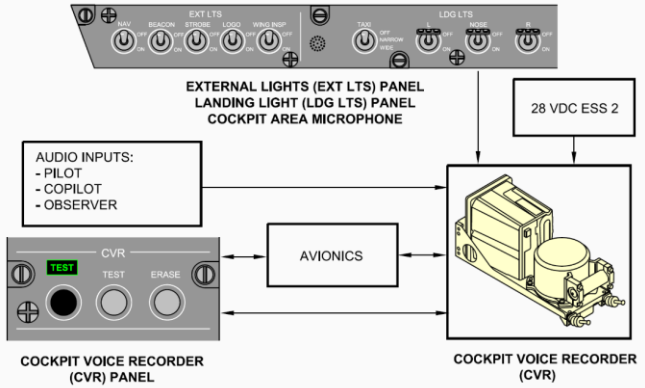
The high-load event indication function monitors gusts, turbulence and hard landing conditions, if a high load event is detected a “HIGH LOAD EVENT” advisory will appear on EICAS, requiring maintenance action.

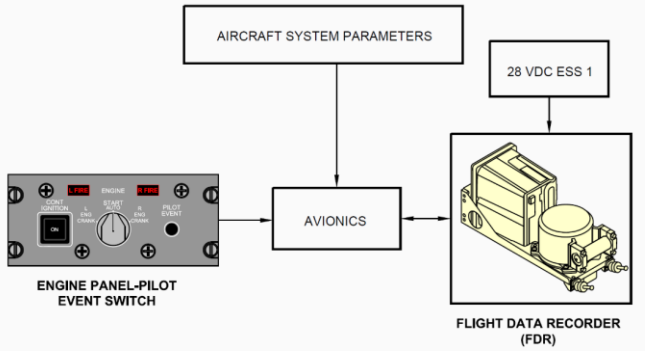
## Water and Waste

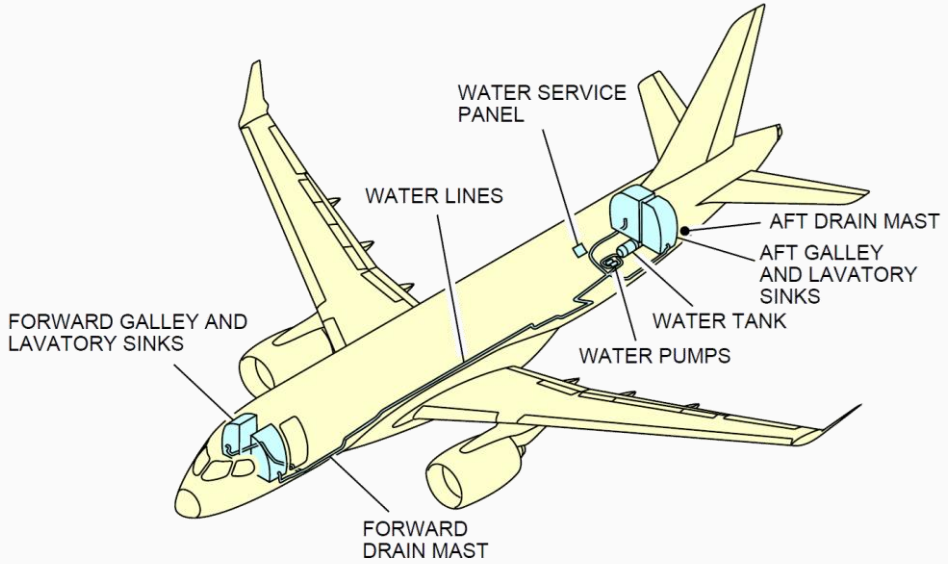
Potable water purging in flight:

- Landing gear up, and
- Drain mast heaters operative, and
- Aircraft altitude below 10.000 ft, and
- WATER TANK PURGE pressed on CMC

The waste system uses a vacuum generator below 16.000 ft, above 16.000 ft the system uses cabin differential pressure, waste tank is behind the rear bulkhead.



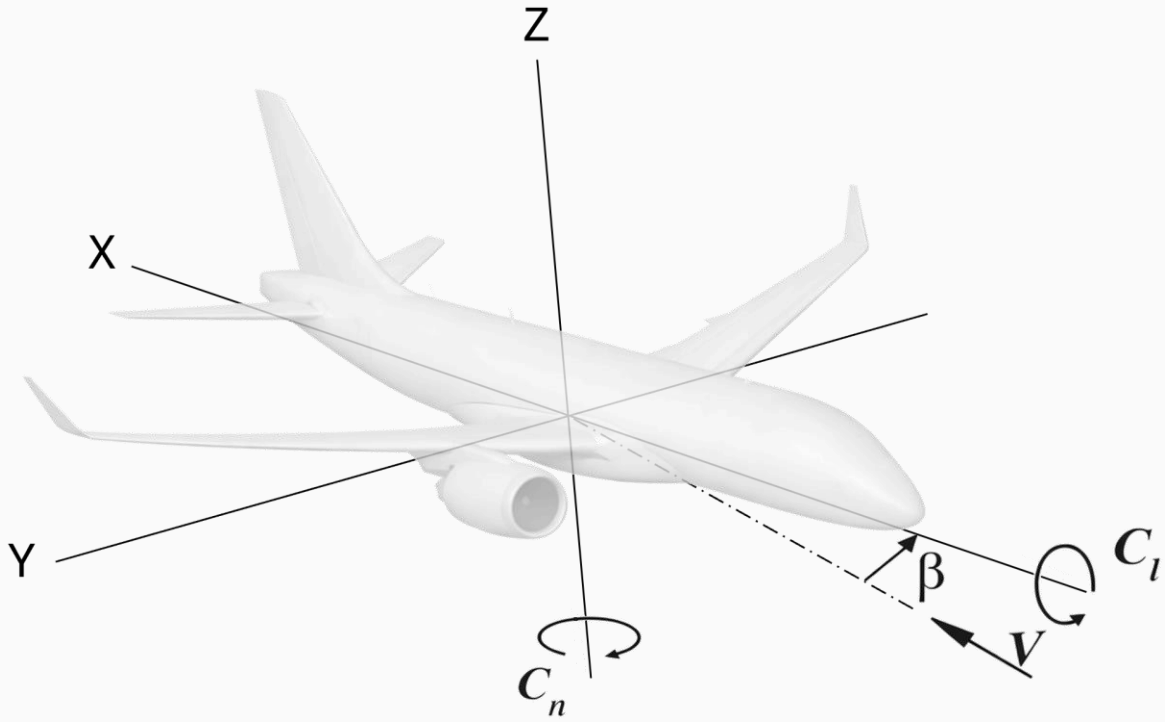



# 2. Procedures

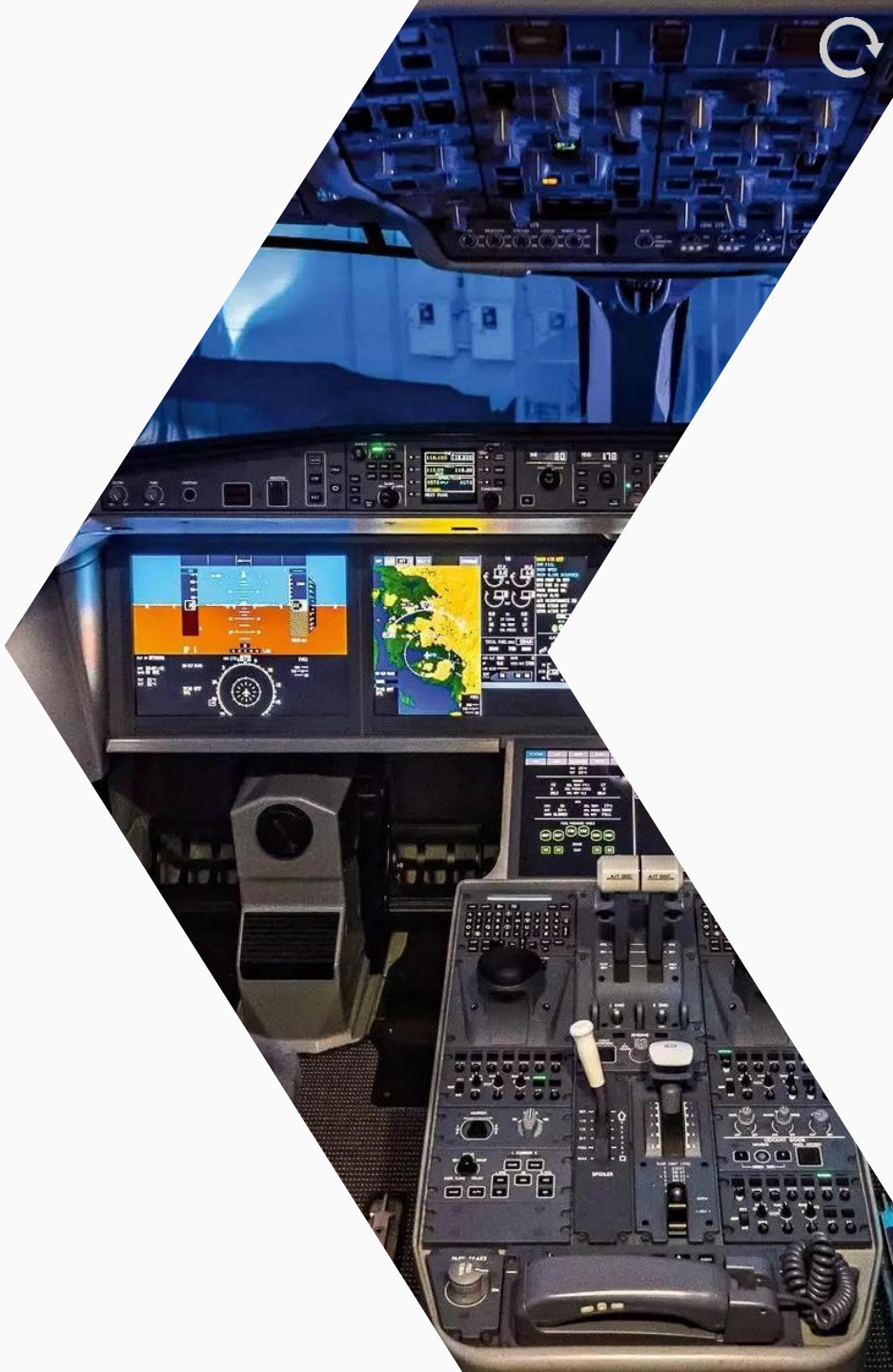
# Part 2 – Procedures

- OEBs
- Limitations
- Normal Procedures
- Supplemental Procedures
- Non-normal Procedures
- EICAS Message List
- Nuisance Guidance
- MEL / CDL
- Performance
- Quick Reference




To Part 1 – Aircraft Systems

# Operational Engineering Bulletin (OEB)



# Introduction

## General

The A220 is certified in the commercial air transport category (CS-25) in the following conditions, assuming all equipment and instruments are serviceable:

- VFR and IFR
- Extended overwater flight
- Flight in icing conditions up to 35.000 ft
- Landing with approach path angle up to and including 3.5°
  Special procedure for glide path exceeding 3.5° up to 3.8°
  A220-100 is certified for steep approach operations (5.5°)
- Maximum passenger capacity 130 (-100) or 150 (-300)

## Flight Operations Transmission (FOT)

An Airbus FOT (Flight Operations Transmission) is a rapid operational communication sent to airlines and flight crews to share time-sensitive safety, procedural, or technical information. It is typically used to provide interim guidance or awareness before formal documentation such as an OEB or manual revision is issued. FOTs are added with a light blue background in this guide.

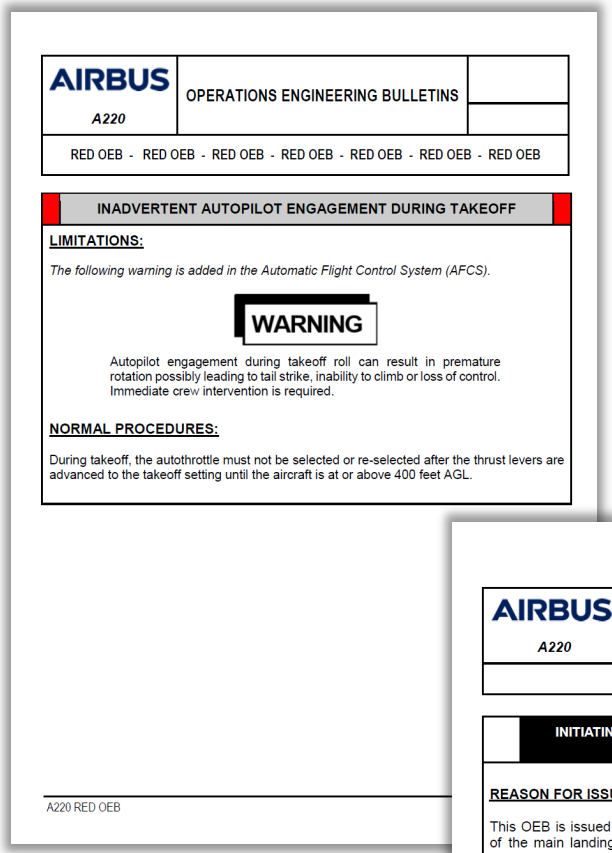
## Operational Engineering Bulletins (OEBs)

Airbus issues OEBs to rapidly inform operators of any deviations from initial design objectives that have a significant operational effect. An OEB provides technical information and temporary operational procedures. OEBs can be red or white:

Red OEBs are issued to indicate that non-compliance with the recommended procedures may have a significant impact on the safe operation of the aircraft, content must be made available to the flight crew without delay.

White OEBs are issued to indicate that non-compliance with the recommended procedures may have a significant impact on aircraft operation (e.g. diversion or delay), it is recommended to apply these OEBs within 30 days.

OEBs might include immediate actions that the flight crew must memorize in order to rapidly apply them without referring to the OEB. Before each flight, the flight crew must review all OEBs that are applicable to the aircraft.



AIRBUS A220 | OPERATIONS ENGINEERING BULLETINS

RED OEB - RED OEB - RED OEB - RED OEB - RED OEB - RED OEB - RED OEB

**INADVERTENT AUTOPILOT ENGAGEMENT DURING TAKEOFF**

**LIMITATIONS:**

*The following warning is added in the Automatic Flight Control System (AFCS).*

**WARNING**

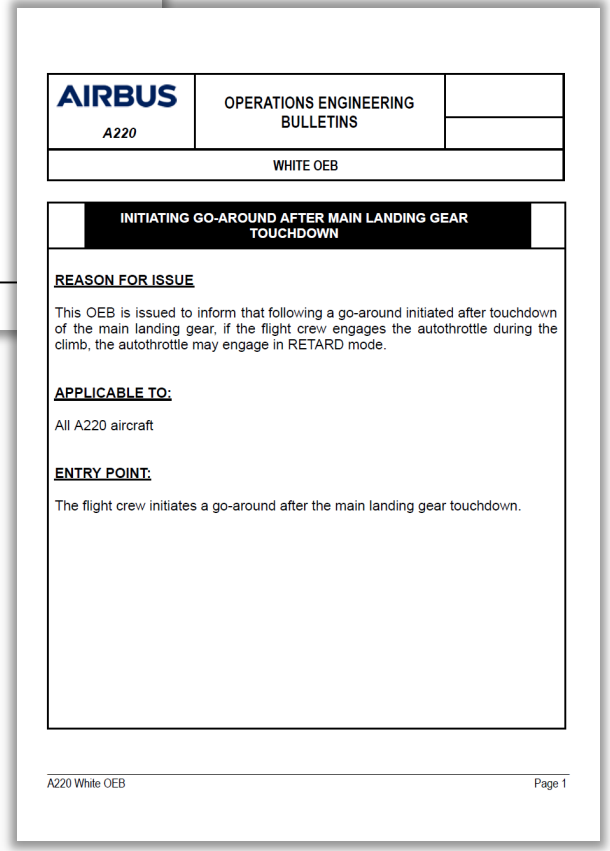
Autopilot engagement during takeoff roll can result in premature rotation possibly leading to tail strike, inability to climb or loss of control. Immediate crew intervention is required.

**NORMAL PROCEDURES:**

During takeoff, the autothrottle must not be selected or re-selected after the thrust levers are advanced to the takeoff setting until the aircraft is at or above 400 feet AGL.

A220 RED OEB

Example RED OEB



AIRBUS A220 | OPERATIONS ENGINEERING BULLETINS

WHITE OEB

**INITIATING GO-AROUND AFTER MAIN LANDING GEAR TOUCHDOWN**

**REASON FOR ISSUE**

This OEB is issued to inform that following a go-around initiated after touchdown of the main landing gear, if the flight crew engages the autothrottle during the climb, the autothrottle may engage in RETARD mode.

**APPLICABLE TO:**

All A220 aircraft

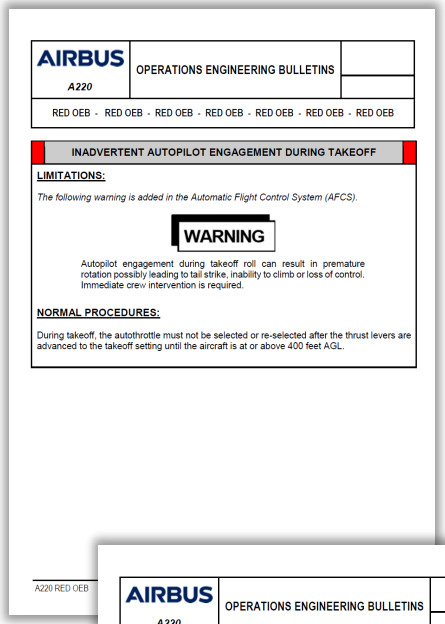
**ENTRY POINT:**

The flight crew initiates a go-around after the main landing gear touchdown.

A220 White OEB Page 1

Example White OEB

# (4) Active OEBs (depending on installed modifications)



AIRBUS A220 | OPERATIONS ENGINEERING BULLETINS

RED OEB - RED OEB - RED OEB - RED OEB - RED OEB - RED OEB - RED OEB

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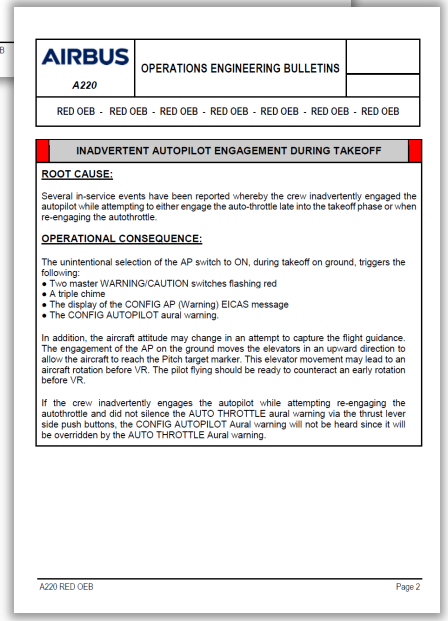
A220 RED OEB



AIRBUS A220 | OPERATIONS ENGINEERING BULLETINS

RED OEB - RED OEB - RED OEB - RED OEB - RED OEB - RED OEB - RED OEB

**INADVERTENT AUTOPILOT ENGAGEMENT DURING TAKEOFF**

**ROOT CAUSE:**

Several in-service events have been reported whereby the crew inadvertently engaged the autopilot while attempting to either engage the auto-throttle late into the takeoff phase or when re-engaging the autothrottle.

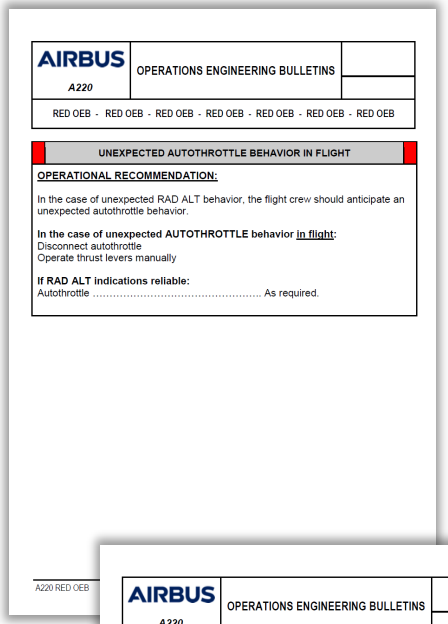
**OPERATIONAL CONSEQUENCE:**

The unintentional selection of the AP switch to ON, during takeoff on ground, triggers the following:
- Two master WARNING/CAUTION switches flashing red
- A triple chime
- The display of the CONFIG AP (Warning) EICAS message
- The CONFIG AUTOPILOT aural warning.

In addition, the aircraft attitude may change in an attempt to capture the flight guidance. The engagement of the AP on the ground moves the elevators in an upward direction to allow the aircraft to reach the Pitch target marker. This elevator movement may lead to an aircraft rotation before VR. The pilot flying should be ready to counteract an early rotation before VR.

If the crew inadvertently engages the autopilot while attempting re-engaging the autothrottle and did not silence the AUTO THROTTLE aural warning via the thrust lever side push buttons, the CONFIG AUTOPILOT Aural warning will not be heard since it will be overridden by the AUTO THROTTLE Aural warning.

A220 RED OEB Page 2



AIRBUS A220 | OPERATIONS ENGINEERING BULLETINS

RED OEB - RED OEB - RED OEB - RED OEB - RED OEB - RED OEB - RED OEB

**UNEXPECTED AUTOTHROTTLE BEHAVIOR IN FLIGHT**

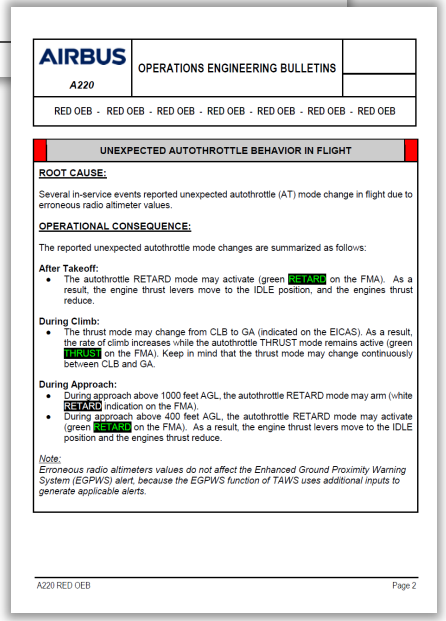
**OPERATIONAL RECOMMENDATION:**

In the case of unexpected RAD ALT behavior, the flight crew should anticipate an unexpected autothrottle behavior.

**In the case of unexpected AUTOTHROTTLE behavior in flight:**
Disconnect autothrottle
Operate thrust levers manually

**If RAD ALT indications reliable:**
Autothrottle ..... As required.

A220 RED OEB



AIRBUS A220 | OPERATIONS ENGINEERING BULLETINS

RED OEB - RED OEB - RED OEB - RED OEB - RED OEB - RED OEB - RED OEB

**UNEXPECTED AUTOTHROTTLE BEHAVIOR IN FLIGHT**

**ROOT CAUSE:**

Several in-service events reported unexpected autothrottle (AT) mode change in flight due to erroneous radio altimeter values.

**OPERATIONAL CONSEQUENCE:**

The reported unexpected autothrottle mode changes are summarized as follows:

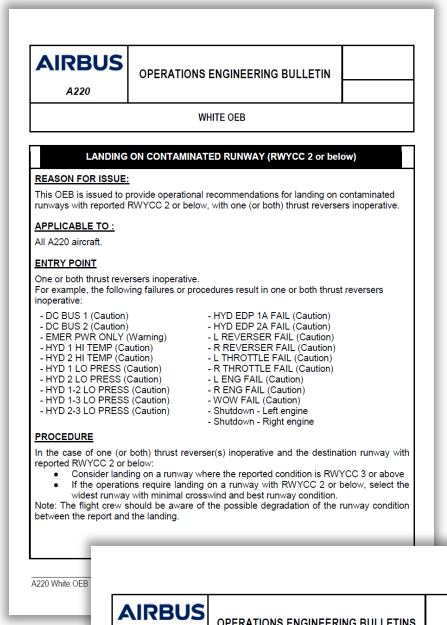
**After Takeoff:**
- The autothrottle RETARD mode may activate (green RETARD on the FMA). As a result, the engine thrust levers move to the IDLE position, and the engines thrust reduce.

**During Climb:**
- The thrust mode may change from CLB to GA (indicated on the EICAS). As a result, the rate of climb increases while the autothrottle THRUST mode remains active (green THRUST on the FMA). Keep in mind that the thrust mode may change continuously between CLB and GA.

**During Approach:**
- During approach above 1000 feet AGL, the autothrottle RETARD mode may arm (white RETARD indication on the FMA).
- During approach above 400 feet AGL, the autothrottle RETARD mode may activate (green RETARD on the FMA). As a result, the engine thrust levers move to the IDLE position and the engines thrust reduce.

*Note:*
*Erroneous radio altimeters values do not affect the Enhanced Ground Proximity Warning System (EGPWS) alert, because the EGPWS function of TAWS uses additional inputs to generate applicable alerts.*

A220 RED OEB Page 2



AIRBUS A220 | OPERATIONS ENGINEERING BULLETIN

WHITE OEB

**LANDING ON CONTAMINATED RUNWAY (RWYCC 2 or below)**

**REASON FOR ISSUE:**

This OEB is issued to provide operational recommendations for landing on contaminated runways with reported RWYCC 2 or below, with one (or both) thrust reversers inoperative.

**APPLICABLE TO :**

All A220 aircraft.

**ENTRY POINT**

One or both thrust reversers inoperative.
For example, the following failures or procedures result in one or both thrust reversers inoperative:

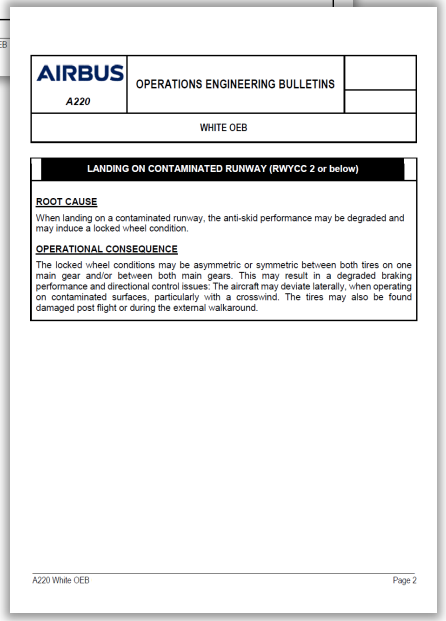
- DC BUS 1 (Caution)
- DC BUS 2 (Caution)
- EMER PWR ONLY (Warning)
- HYD 1 HI TEMP (Caution)
- HYD 2 HI TEMP (Caution)
- HYD 1 LO PRESS (Caution)
- HYD 2 LO PRESS (Caution)
- HYD 1-2 LO PRESS (Caution)
- HYD 1-3 LO PRESS (Caution)
- HYD 2-3 LO PRESS (Caution)
- HYD EDP 1A FAIL (Caution)
- HYD EDP 2A FAIL (Caution)
- L REVERSER FAIL (Caution)
- R REVERSER FAIL (Caution)
- L THROTTLE FAIL (Caution)
- R THROTTLE FAIL (Caution)
- L ENG FAIL (Caution)
- R ENG FAIL (Caution)
- WOW FAIL (Caution)
- Shutdown - Left engine
- Shutdown - Right engine

**PROCEDURE**

In the case of one (or both) thrust reverser(s) inoperative and the destination runway with reported RWYCC 2 or below:
- Consider landing on a runway where the reported condition is RWYCC 3 or above
- If the operations require landing on a runway with RWYCC 2 or below, select the widest runway with minimal crosswind and best runway condition.

Note: The flight crew should be aware of the possible degradation of the runway condition between the report and the landing.

A220 White OEB



AIRBUS A220 | OPERATIONS ENGINEERING BULLETINS

WHITE OEB

**LANDING ON CONTAMINATED RUNWAY (RWYCC 2 or below)**

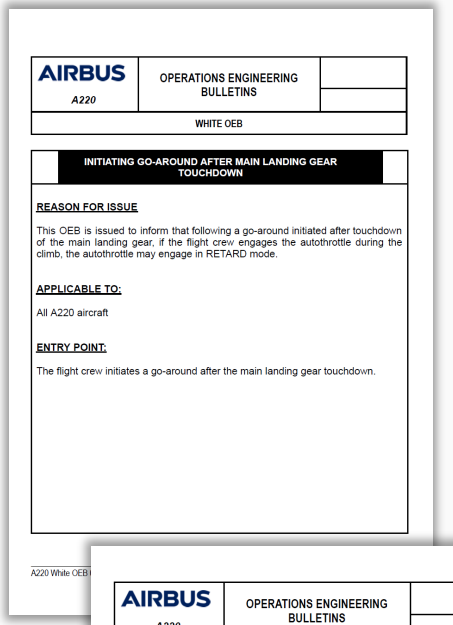
**ROOT CAUSE**

When landing on a contaminated runway, the anti-skid performance may be degraded and may induce a locked wheel condition.

**OPERATIONAL CONSEQUENCE**

The locked wheel conditions may be asymmetric or symmetric between both tires on one main gear and/or between both main gears. This may result in a degraded braking performance and directional control issues. The aircraft may deviate laterally, when operating on contaminated surfaces, particularly with a crosswind. The tires may also be found damaged post flight or during the external walkaround.

A220 White OEB Page 2



AIRBUS A220 | OPERATIONS ENGINEERING BULLETINS

WHITE OEB

**INITIATING GO-AROUND AFTER MAIN LANDING GEAR TOUCHDOWN**

**REASON FOR ISSUE**

This OEB is issued to inform that following a go-around initiated after touchdown of the main landing gear, if the flight crew engages the autothrottle during the climb, the autothrottle may engage in RETARD mode.

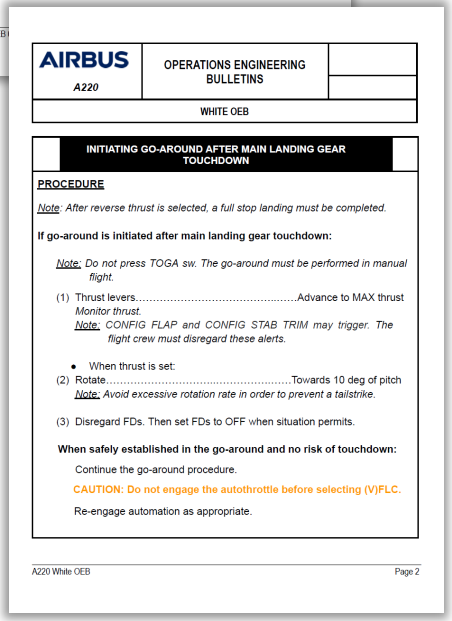
**APPLICABLE TO:**

All A220 aircraft

**ENTRY POINT:**

The flight crew initiates a go-around after the main landing gear touchdown.

A220 White OEB



AIRBUS A220 | OPERATIONS ENGINEERING BULLETINS

WHITE OEB

**INITIATING GO-AROUND AFTER MAIN LANDING GEAR TOUCHDOWN**

**PROCEDURE**

*Note: After reverse thrust is selected, a full stop landing must be completed.*

**If go-around is initiated after main landing gear touchdown:**

*Note: Do not press TOGA sw. The go-around must be performed in manual flight.*

(1) Thrust levers.....Advance to MAX thrust
*Monitor thrust.*
*Note: CONFIG FLAP and CONFIG STAB TRIM may trigger. The flight crew must disregard these alerts.*

- When thrust is set:

(2) Rotate.....Towards 10 deg of pitch
*Note: Avoid excessive rotation rate in order to prevent a tailstrike.*

(3) Disregard FDs. Then set FDs to OFF when situation permits.

**When safely established in the go-around and no risk of touchdown:**

Continue the go-around procedure.

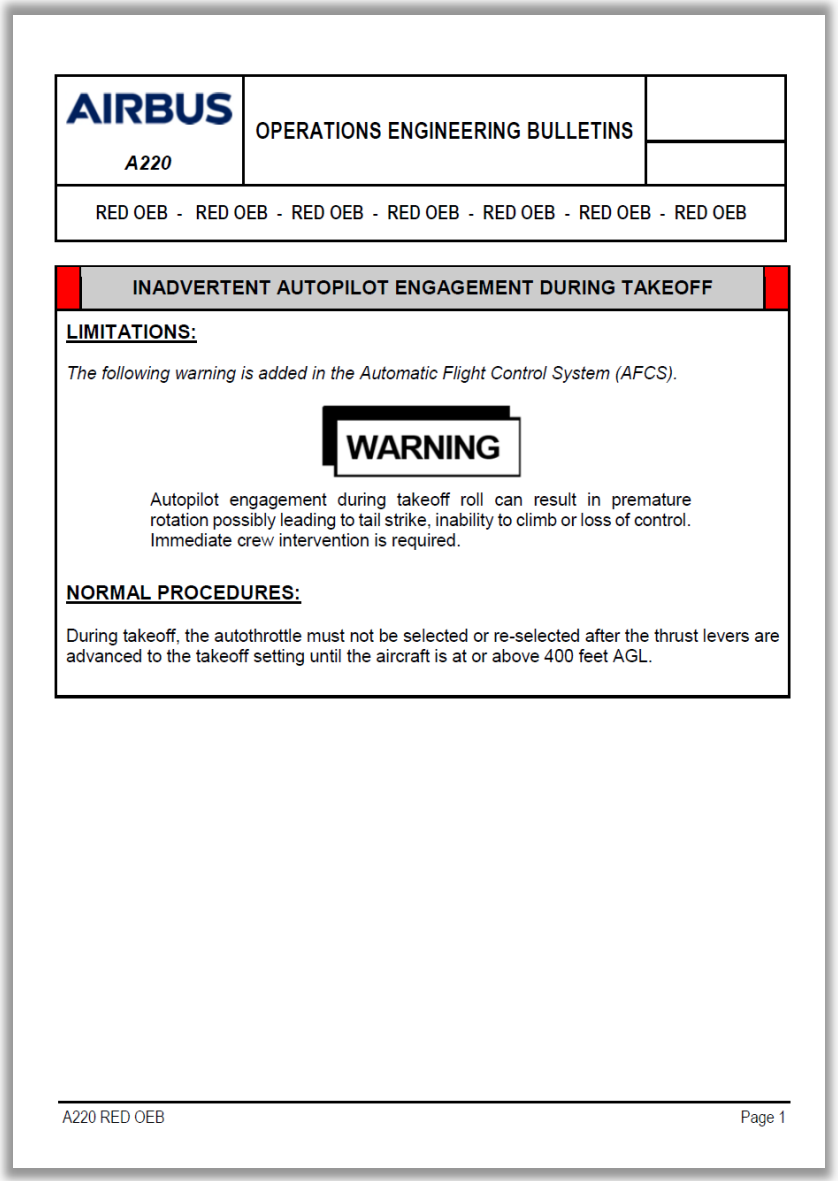
**CAUTION: Do not engage the autothrottle before selecting (V)FLC.**

Re-engage automation as appropriate.

A220 White OEB Page 2

# RED OEB 1

PFCC version 009 cancels this OEB.



AIRBUS A220 | OPERATIONS ENGINEERING BULLETINS

RED OEB - RED OEB - RED OEB - RED OEB - RED OEB - RED OEB - RED OEB

**INADVERTENT AUTOPILOT ENGAGEMENT DURING TAKEOFF**

**LIMITATIONS:**

*The following warning is added in the Automatic Flight Control System (AFCS).*

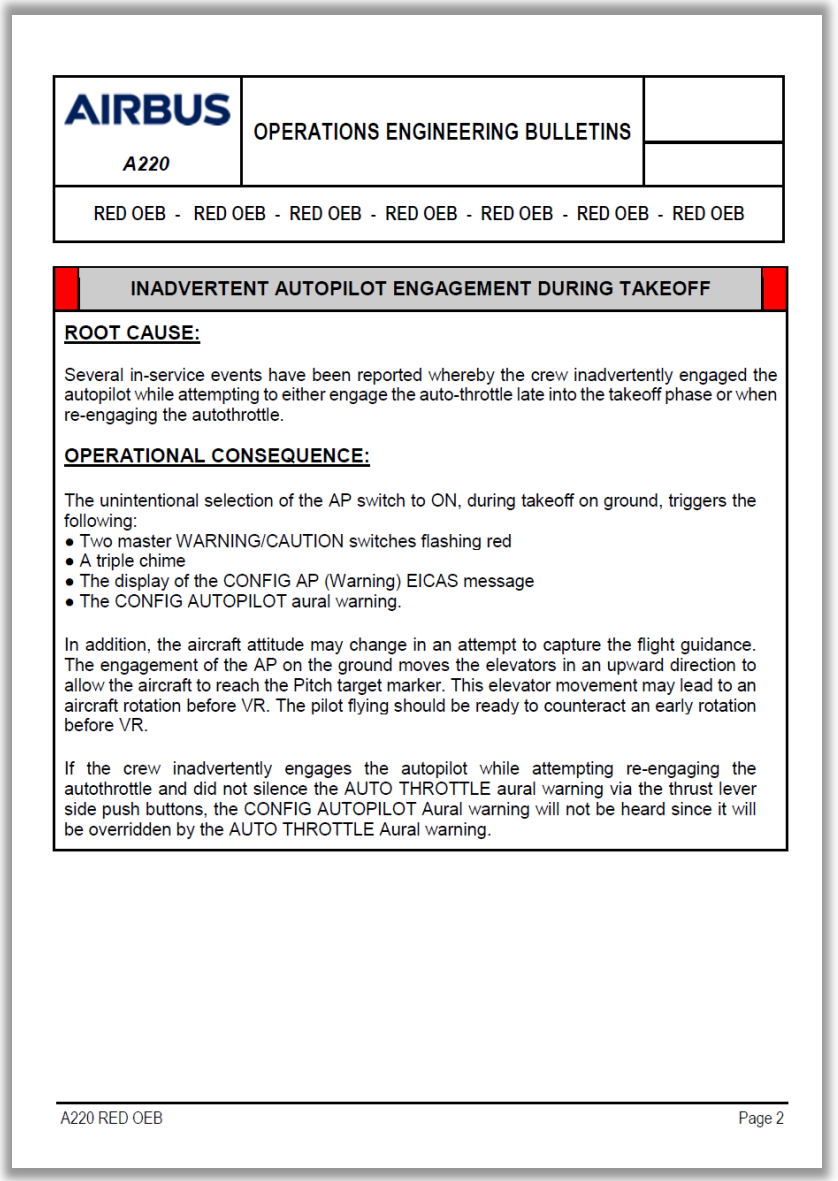
**WARNING**

Autopilot engagement during takeoff roll can result in premature rotation possibly leading to tail strike, inability to climb or loss of control. Immediate crew intervention is required.

**NORMAL PROCEDURES:**

During takeoff, the autothrottle must not be selected or re-selected after the thrust levers are advanced to the takeoff setting until the aircraft is at or above 400 feet AGL.

A220 RED OEB Page 1



AIRBUS A220 | OPERATIONS ENGINEERING BULLETINS

RED OEB - RED OEB - RED OEB - RED OEB - RED OEB - RED OEB - RED OEB

**INADVERTENT AUTOPILOT ENGAGEMENT DURING TAKEOFF**

**ROOT CAUSE:**

Several in-service events have been reported whereby the crew inadvertently engaged the autopilot while attempting to either engage the auto-throttle late into the takeoff phase or when re-engaging the autothrottle.

**OPERATIONAL CONSEQUENCE:**

The unintentional selection of the AP switch to ON, during takeoff on ground, triggers the following:
- Two master WARNING/CAUTION switches flashing red
- A triple chime
- The display of the CONFIG AP (Warning) EICAS message
- The CONFIG AUTOPILOT aural warning.

In addition, the aircraft attitude may change in an attempt to capture the flight guidance. The engagement of the AP on the ground moves the elevators in an upward direction to allow the aircraft to reach the Pitch target marker. This elevator movement may lead to an aircraft rotation before VR. The pilot flying should be ready to counteract an early rotation before VR.

If the crew inadvertently engages the autopilot while attempting re-engaging the autothrottle and did not silence the AUTO THROTTLE aural warning via the thrust lever side push buttons, the CONFIG AUTOPILOT Aural warning will not be heard since it will be overridden by the AUTO THROTTLE Aural warning.

A220 RED OEB Page 2

# RED OEB 2

Avionics built 8.0A3 together with a RAD ALT hardware improvement is expected to cancel this OEB in the future.



**UNEXPECTED AUTOTHROTTLE BEHAVIOR IN FLIGHT**

**OPERATIONAL RECOMMENDATION:**

In the case of unexpected RAD ALT behavior, the flight crew should anticipate an unexpected autothrottle behavior.

**In the case of unexpected AUTOTHROTTLE behavior in flight:**
Disconnect autothrottle
Operate thrust levers manually

**If RAD ALT indications reliable:**
Autothrottle ..... As required.



AIRBUS A220 | OPERATIONS ENGINEERING BULLETINS

RED OEB - RED OEB - RED OEB - RED OEB - RED OEB - RED OEB - RED OEB

**UNEXPECTED AUTOTHROTTLE BEHAVIOR IN FLIGHT**

**ROOT CAUSE:**

Several in-service events reported unexpected autothrottle (AT) mode change in flight due to erroneous radio altimeter values.

**OPERATIONAL CONSEQUENCE:**

The reported unexpected autothrottle mode changes are summarized as follows:

**After Takeoff:**
- The autothrottle RETARD mode may activate (green RETARD on the FMA). As a result, the engine thrust levers move to the IDLE position, and the engines thrust reduce.

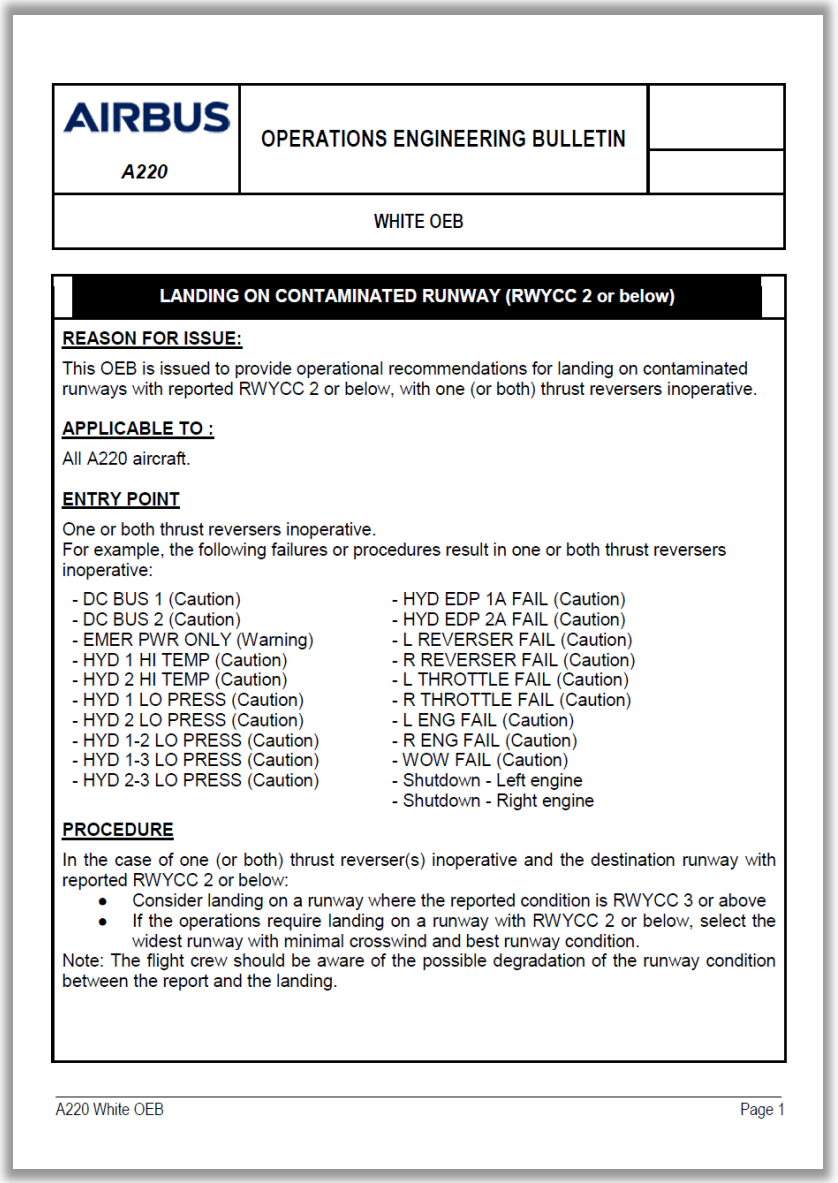
**During Climb:**
- The thrust mode may change from CLB to GA (indicated on the EICAS). As a result, the rate of climb increases while the autothrottle THRUST mode remains active (green THRUST on the FMA). Keep in mind that the thrust mode may change continuously between CLB and GA.

**During Approach:**
- During approach above 1000 feet AGL, the autothrottle RETARD mode may arm (white RETARD indication on the FMA).
- During approach above 400 feet AGL, the autothrottle RETARD mode may activate (green RETARD on the FMA). As a result, the engine thrust levers move to the IDLE position and the engines thrust reduce.

*Note:*
*Erroneous radio altimeters values do not affect the Enhanced Ground Proximity Warning System (EGPWS) alert, because the EGPWS function of TAWS uses additional inputs to generate applicable alerts.*

# White OEB 1

BCS version 4 cancels this OEB.



AIRBUS A220 | OPERATIONS ENGINEERING BULLETIN

WHITE OEB

**LANDING ON CONTAMINATED RUNWAY (RWYCC 2 or below)**

**REASON FOR ISSUE:**

This OEB is issued to provide operational recommendations for landing on contaminated runways with reported RWYCC 2 or below, with one (or both) thrust reversers inoperative.

**APPLICABLE TO :**

All A220 aircraft.

**ENTRY POINT**

One or both thrust reversers inoperative.
For example, the following failures or procedures result in one or both thrust reversers inoperative:

- DC BUS 1 (Caution)
- DC BUS 2 (Caution)
- EMER PWR ONLY (Warning)
- HYD 1 HI TEMP (Caution)
- HYD 2 HI TEMP (Caution)
- HYD 1 LO PRESS (Caution)
- HYD 2 LO PRESS (Caution)
- HYD 1-2 LO PRESS (Caution)
- HYD 1-3 LO PRESS (Caution)
- HYD 2-3 LO PRESS (Caution)
- HYD EDP 1A FAIL (Caution)
- HYD EDP 2A FAIL (Caution)
- L REVERSER FAIL (Caution)
- R REVERSER FAIL (Caution)
- L THROTTLE FAIL (Caution)
- R THROTTLE FAIL (Caution)
- L ENG FAIL (Caution)
- R ENG FAIL (Caution)
- WOW FAIL (Caution)
- Shutdown - Left engine
- Shutdown - Right engine

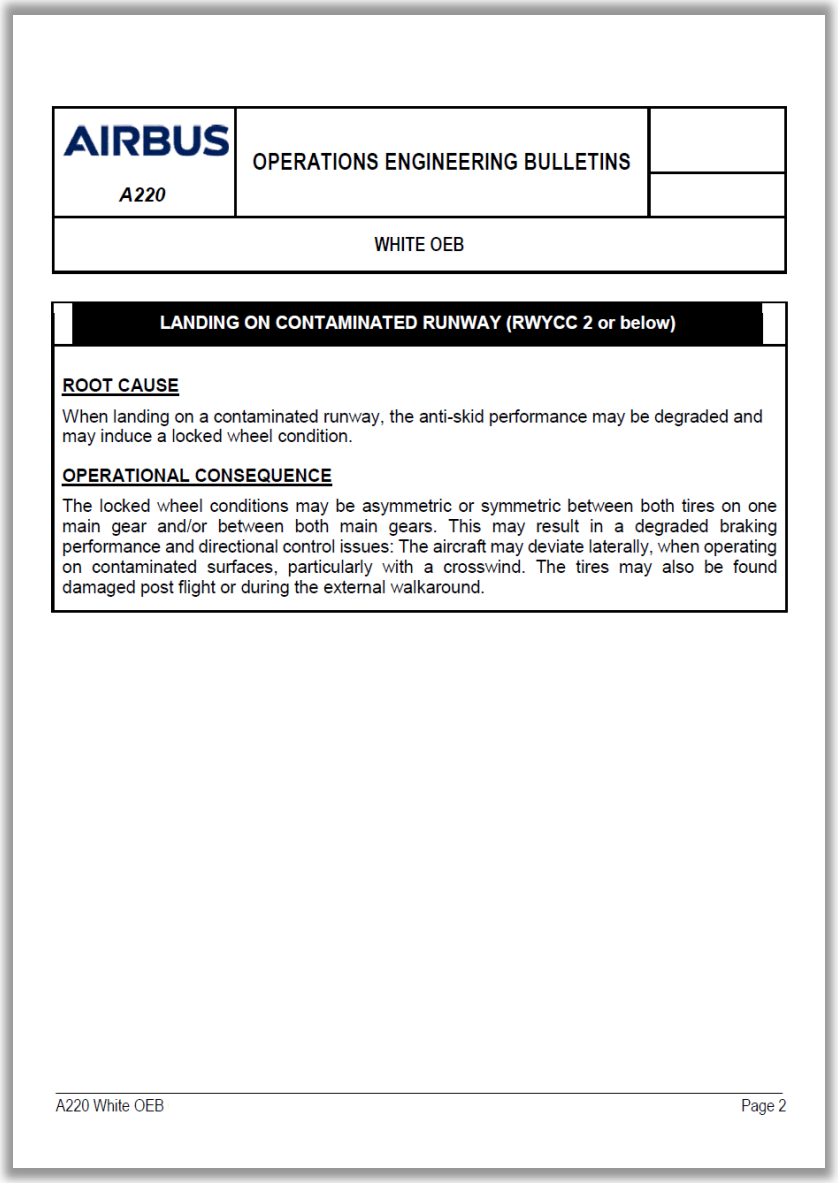
**PROCEDURE**

In the case of one (or both) thrust reverser(s) inoperative and the destination runway with reported RWYCC 2 or below:

- Consider landing on a runway where the reported condition is RWYCC 3 or above
- If the operations require landing on a runway with RWYCC 2 or below, select the widest runway with minimal crosswind and best runway condition.

Note: The flight crew should be aware of the possible degradation of the runway condition between the report and the landing.

A220 White OEB Page 1



AIRBUS A220 | OPERATIONS ENGINEERING BULLETINS

WHITE OEB

**LANDING ON CONTAMINATED RUNWAY (RWYCC 2 or below)**

**ROOT CAUSE**

When landing on a contaminated runway, the anti-skid performance may be degraded and may induce a locked wheel condition.

**OPERATIONAL CONSEQUENCE**

The locked wheel conditions may be asymmetric or symmetric between both tires on one main gear and/or between both main gears. This may result in a degraded braking performance and directional control issues: The aircraft may deviate laterally, when operating on contaminated surfaces, particularly with a crosswind. The tires may also be found damaged post flight or during the external walkaround.

A220 White OEB Page 2

# White OEB 2

An autothrottle fix is under investigation.



## INITIATING GO-AROUND AFTER MAIN LANDING GEAR TOUCHDOWN

**REASON FOR ISSUE**

This OEB is issued to inform that following a go-around initiated after touchdown of the main landing gear, if the flight crew engages the autothrottle during the climb, the autothrottle may engage in RETARD mode.

**APPLICABLE TO:**

All A220 aircraft

**ENTRY POINT:**

The flight crew initiates a go-around after the main landing gear touchdown.



AIRBUS A220 | OPERATIONS ENGINEERING BULLETINS
WHITE OEB

## INITIATING GO-AROUND AFTER MAIN LANDING GEAR TOUCHDOWN

**PROCEDURE**

*Note: After reverse thrust is selected, a full stop landing must be completed.*

**If go-around is initiated after main landing gear touchdown:**

*Note: Do not press TOGA sw. The go-around must be performed in manual flight.*

(1) Thrust levers.....Advance to MAX thrust
*Monitor thrust.*
*Note: CONFIG FLAP and CONFIG STAB TRIM may trigger. The flight crew must disregard these alerts.*

- When thrust is set:

(2) Rotate.....Towards 10 deg of pitch
*Note: Avoid excessive rotation rate in order to prevent a tailstrike.*

(3) Disregard FDs. Then set FDs to OFF when situation permits.

**When safely established in the go-around and no risk of touchdown:**

Continue the go-around procedure.

**CAUTION: Do not engage the autothrottle before selecting (V)FLC.**

Re-engage automation as appropriate.



AIRBUS A220 | OPERATIONS ENGINEERING BULLETINS
WHITE OEB

## INITIATING GO-AROUND AFTER MAIN LANDING GEAR TOUCHDOWN

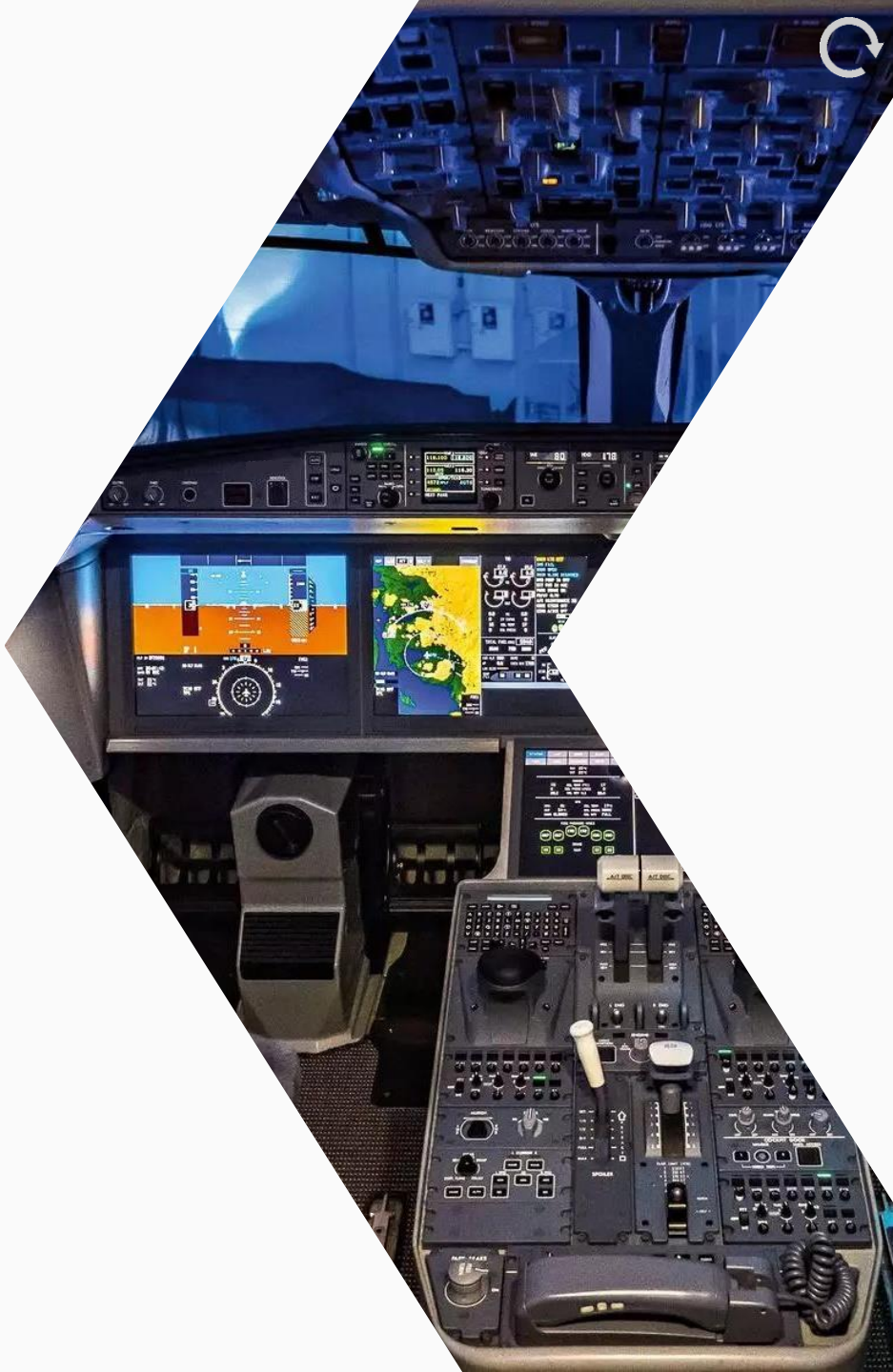
**ROOT CAUSE**

From 2 seconds after touchdown (confirmed by weight on the main landing gear), the avionic system prevents the autothrottle to engage when the flight crew presses the TOGA switch.
Then the autothrottle system may keep the "RETARD mode" as the latest memorized mode while a FG vertical mode is not set. As a result during the climb, when the flight crew engages the autothrottle before setting FG vertical modes, the autothrottle systems may engage the autothrottle in RETARD mode.

**OPERATIONAL CONSEQUENCE**

During a go around initiated 2 seconds (and more) after the aircraft is on the ground (confirmed by weight on wheels), and if the flight crew presses the TOGA switch, the autothrottle may not activate.
Then in climb, if the flight crew engages the autothrottle before selecting (V)FLC, the RETARD mode may activate. As a result, engines reduce the thrust to idle until the flight crew disconnects the autothrottle and manually adjusts the thrust.

# Limitations



# Weight & Balance Limitations

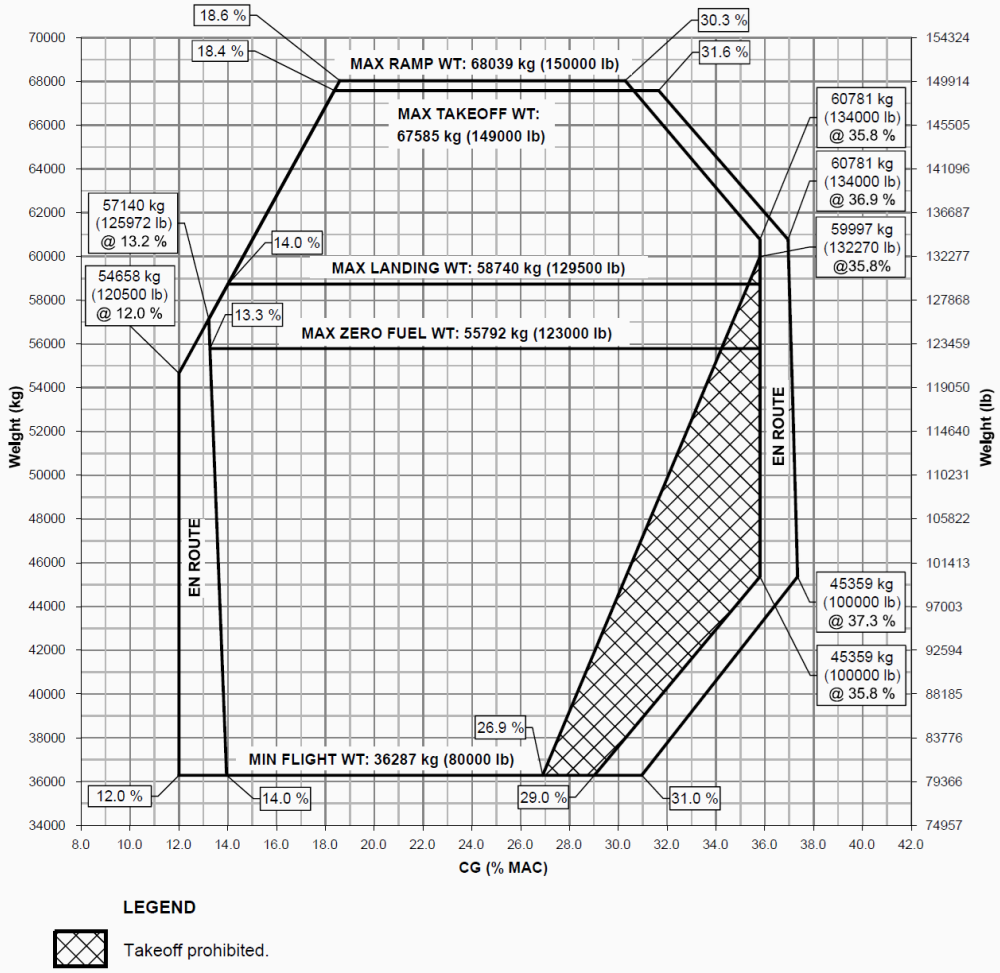
**Note:** in this guide, the 67.585 kg Maximum Structural Take-off Weight option is used for the purpose of A220-300 take-off performance and weight & balance.

| Weight | A220-100 | | A220-300 | |
|---|---|---|---|---|
| | kg | lbs | kg | lbs |
| Maximum Ramp Weight (MRW) | 64.183 | 141.500 | 71.214 | 157.000 |
| Maximum Take-off Weight (MTOW) | 63.730 | 140.500 | 70.896 | 156.300 |
| Maximum Landing Weight (MLW) | 54.658 | 120.500 | 58.740 | 129.500 |
| Maximum Zero Fuel Weight (MZFW) | 52.617 | 116.000 | 55.792 | 123.000 |

**Note:** additional approved (lower) weight variants are available and listed in the AFM, maximum structural weights in accordance with the approved Type Certificate listed above. Maximum Take-off Weight - MTOW and/or Maximum Landing Weight - MLW can be further limited by performance considerations.

| Forward Cargo Compartment | | Aft Cargo Compartment | |
|---|---|---|---|
| Section A | Section B | Section C | Section F |
| 800 kg | 1646 kg | 1858 kg | 1055 kg |
| Max compartment = 2446 kg | | Max compartment = 2606 kg | |
| Total = 5052 kg | | | |

Example



# Speed Limitations

## Definitions

- Take-off Decision Speed $V_1$ – the speed at which, due to failure or other causes, the pilot may elect to stop or continue the take-off. $V_1$ may be limited by maximum brake energy ($V_{1MBE}$), or minimum control speed ground ($V_{1MCG}$). ($V_{1MBE} \geq V_1 \geq V_{MCG}$).
- Rotation Speed $V_R$ – speed at which rotation is initiated.
- Take-off Safety Speed $V_2$ – target climb speed at or before 35 ft above the runway is reached during continued take-off following an engine failure.
- Final Take-off Speed $V_{FTO}$ – speed attained at the end of the flight path acceleration segment during continued take-off following engine failure = Green Dot.
- Reference Landing Speed $V_{REF}$ – the minimum speed at a height of 50 ft above the runway threshold. Corrected reference landing speed is the reference landing speed corrected for icing conditions or after a system failure ($\Delta V_{REF}$ as in the ECL). If applicable, $\Delta V_{REF}$ is applied to $V_{REF}$, $V_{AC}$ and $V_{GA}$.
- Approach speed $V_{APP}$ – speed to be maintained down to 50 ft over the threshold. The approach speed is $V_{REF}$ + increment for use of autothrottle or wind gust factor.
- Landing climb speed $V_{LC}$ – initial climb speed during balked landing (go-around during flare in landing configuration) = $V_{REF}$.
- Approach climb speed $V_{AC}$ – climb speed for go-around with one engine inoperative.
- Go-around Speed $V_{GA}$ – climb speed during a go-around with all engines operating.
- Minimum Control Speed Air ($V_{MCA}$), Minimum Control Speed Landing ($V_{MCL}$) – minimum flight speed at which the aircraft is controllable with a maximum of 5-degree bank when the critical engine becomes inoperative at TO or GA thrust.

| Bank Angle Limitations | | | | | |
|---|---|---|---|---|---|
| **Configuration** | **TKOF** | **CLB** | **Green Dot** | **APCH** | **GA** |
| All Engine Operating | 15° ($V_2$) | 25° (F)<br>30° (F + 10) | 30° < 31.500'<br>15° > 31.500' | 25° ($V_{REF}$) | 30° ($V_{GA}$) |
| One Engine Inoperative | 30° ($V_2$ + 10) | 15° (F) | 25° | 30° ($V_{REF}$ + 5) | 20° ($V_{AC}$) |

## Maximum Operating Speed / Mach ($V_{MO}$ – $M_{MO}$)

- 8000 ft and below – 300 kts (increasing linearly between 8000 ft to 10.000 ft)
- 10.000 ft and above – 330 kts
- 330 kts until reaching M0.82 (27.500 ft at ISA)

## Design Maneuvering Speed ($V_A$)

Avoid rapid and large control inputs as they may cause structural damage at any speed, including below $V_A$. In NORMAL mode, control laws provide automatic protection against full application of elevator, aileron and rudder. In DIRECT mode all maneuvers involving full application of controls must be conducted below $V_A$: 300 kts / M0.80.

## Maximum Slat / Flap Extended Speed ($V_{FE}$)

- FLAP 1: 230 kts
- FLAP 2 / 3: 210 kts
- FLAP 4: 190 kts
- FLAP 5: 170 kts

**Maneuvering Limit Load Factors**
- Flaps up: -1.0 g to +2.5 g
- Flaps down: 0.0 g to +2.0 g

## Maximum Landing Gear Operating ($V_{LO}$) & Extended Speeds ($V_{LE}$)

- Maximum speed for landing gear extension / extended: 250 kts
- Maximum speed for landing gear retraction: 220 kts

## Turbulent Penetration Speed ($V_{RA}$)

- Recommended airspeed for moderate to severe turbulence penetration.
- 10.000 ft and below: 265 kts, above 10.000 ft: 280 kts or M0.75 (whichever is lower).

**Windshield Wiper Operation:** windshield wipers must not be operated above 250 kts.

**Tire Limit Speed:** the tire limit speed is 195 kts (ground speed).

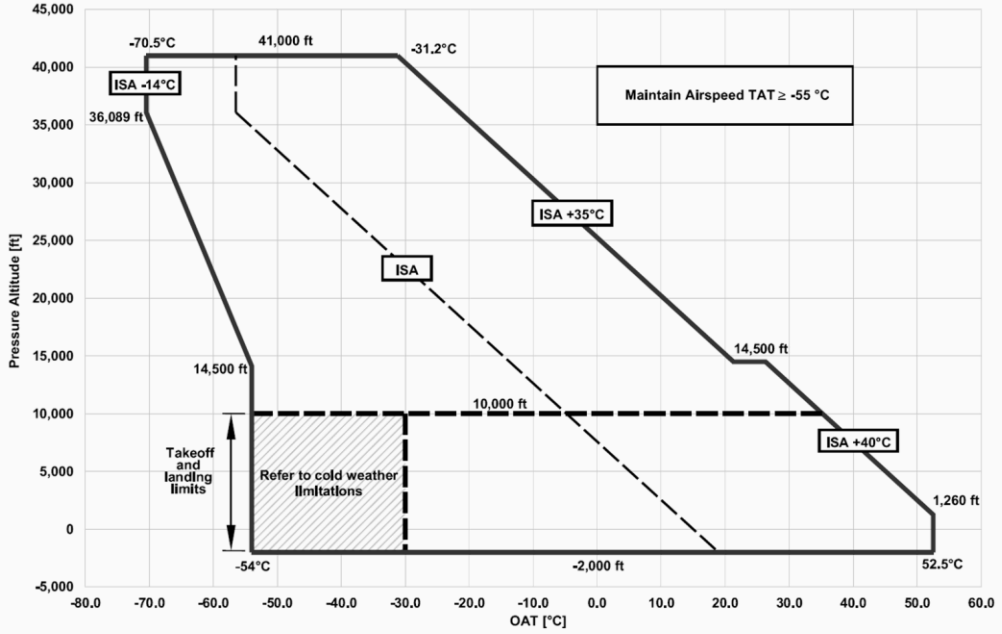
# Environmental Limitations

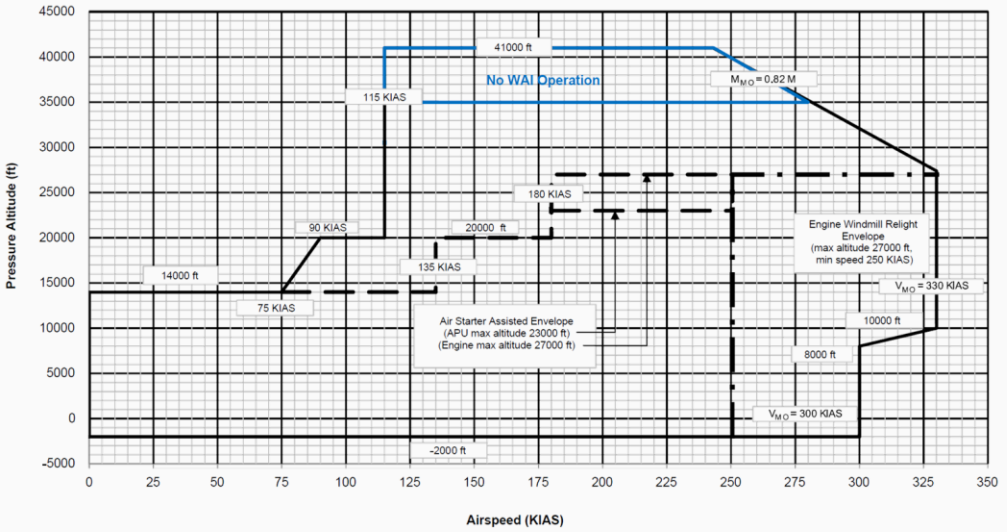
## Operating Altitude

- Maximum airport pressure altitude for take-off and landing is 8000 ft or 10.000 ft depending on installed modifications (autoland max 8466 ft).
- Maximum operating altitude is 41.000 ft.
- Flight in icing conditions up to 35.000 ft.
- Maximum altitude with slats / flaps extended:
    - A220-100: 15.000 ft
    - A220-300: 20.000 ft
- Air Starter Assisted Envelope:
    - Speed below 250 kts
    - APU bleed max 23.000 ft
    - Cross bleed max 27.000 ft
- Windmill Relight Envelope:
    - Speed above 250 kts
    - Max altitude 27.000 ft

## Temperature

- Maximum ambient temperature for take-off and landing is 52.5°C.
- Minimum ambient temperature for take-off and landing:
    - A220-100: -34°C
    - A220-300: -54°C
- Minimum temperature at the Tropopause: -70.5°C (ISA -14°C).



# Cold Weather Operations

## General

- Cold Weather operations refer to ground handling, take-off and landing conducted with Outside Air Temperature at or below 0°C.
- Take-off is prohibited with frost, ice, snow or slush adhering to any critical surfaces (wings, stabilizers, control surfaces and engine inlets).
- Engine operation in ground fog with OAT below -18°C is prohibited.
- Take-off is prohibited in freezing fog conditions when visibility is below 150 m.

## Icing Conditions:

- Ground: OAT is 5°C or below (wing anti-ice) or 10°C or below (cowl anti-ice) and visible moisture in any form is present (fog with visibility 1500 m or less, rain, snow, sleet or ice crystals), or when operating on runways, ramps or taxiways where surface is contaminated with snow, ice, standing water or slush.
- In-flight: TAT of 10°C or below and visible moisture in any form is present (clouds, rain, snow, sleet or ice crystals) except when SAT is -40°C or below.

## Cowl Anti-ice System

- On ground: the Cowl Anti-ice System must be ON when OAT is 10°C or below while ground icing conditions exist (as above).
- In flight: the Cowl Anti-ice System must be AUTO or ON in icing conditions (as above) or when ICE is annunciated by the ice detection system.

## Wing Anti-ice System

- On ground: the Wing Anti-ice System must be ON when OAT is 5°C or below while ground icing conditions exist (as above).
- In flight: the Wing Anti-ice System must be AUTO or ON in icing conditions (as above) or when ICE is annunciated by the ice detection system.
- Do not hold in icing conditions with slats / flaps extended.
- Wing anti-ice must be OFF when above 35.000 ft.
- Wing anti-ice ON is prohibited when TAT is above 15°C.

## Supercooled Large Droplet (SLD) Icing

Continued operation in areas where Supercooled Large Droplet (SLD) icing conditions exist is prohibited. SLD icing conditions are indicated by ice accretion on the flight compartment side windows.

- Wing and Cowl Anti-ice Systems must be ON in SLD icing conditions.
- Leave icing conditions when side window icing occurs.

## Engine Ice Shedding

On ground: Engine ice shedding must be performed when the OAT is 3°C or below and visible moisture in any form is present such as fog, rain, snow, sleet or ice crystals. Engine run-ups must be performed, momentarily, to a minimum of 60% $N_1$, at intervals of not more than 30 minutes. The first run-up must be done as soon as practical, final engine ice shedding within 5 minutes before the take-off.

In flight: When operating for extended periods in icing conditions, engine vibration may be encountered, even with anti-ice on. A momentary increase in thrust up to CLB thrust should stop undesired vibrations.

## Approach and Landing in Icing Conditions

FLAP 5: during an approach and landing in icing conditions, or if there is evidence of ice accretion, $V_{REF}$, $V_{AC}$ and $V_{GA}$ must be increased by 5 kts.

## Potable Water System

If turnaround exceeds 120 minutes with OAT between 0°C to -29°C, potable water system must either be drained or aircraft must remain electrically powered during turnaround and cabin temperature maintained. Below -29°C the aircraft must remain electrically powered during turnaround and cabin temperature maintained.

# Aircraft Systems

### Pressurization

- Maximum positive differential pressure is 8.8 psid.
- Maximum negative differential pressure is -0.5 psid.
- Maximum differential pressure at touchdown is 1.0 psid.

### Minimum oxygen dispatch pressure

- Mountainous region: 950 psi (1300 psi with 3 cockpit crew)
- Other region: 900 psi (1200 psi with 3 cockpit crew)

### Air-conditioning

- Doors should not be closed for more than 15 minutes when passengers are on board and packs are off and no ground ventilation available (except for de-icing).
- When OAT is above 40°C, operation without air conditioning or external cooling is limited to 30 minutes (regardless if doors are open or closed).

### Auto flight

Autopilot (AP)

- Minimum height for autopilot engagement is 400 ft AGL.
- Minimum height for use of autopilot is 80 ft AGL (except for autoland).
- WARNING: autopilot engagement during take-off can result in premature rotation, possibly leading to tail strike, inability to climb or loss of control.

Autothrottle (AT)

- The AT must be disconnected before landing for touch and go procedures.
- During approach and landing with AT engaged, $V_{APP}$ must be increased by 5 kts.

Flight Director (FD)

- During altitude capture (CAP) modes, the flight director guidance / autopilot may not maintain the selected airspeed. Crew intervention may be required.

### Auxiliary Power Unit (APU)

- Honeywell 131-9(C)
- Minimum oil temperature for APU start is -40°C
- Do not perform more than 3 starts (or start attempts) in one hour.
- A two-minute delay must be observed between start attempts.
- The altitude limit for APU bleed air extraction is 23.000 ft.
- APU (or APU BLEED) must be off for take-off when OAT is at or below 15°C.

### Slats / Flaps

- Max altitude with slats / flaps extended is 15.000 ft (-100) or 20.000 ft (-300).

### Flight Spoilers

- Minimum height with flight spoilers extended is 300 ft AGL.
- Minimum speed with flight spoilers extended is $V_{REF}$ + 10 kts.
- Use of MAX spoilers is prohibited except in an emergency as it can cause unexpected loads which can lead to early structural fatigue or damage in combination with severe turbulence and / or excessive g loading.

### Fuel System

- Maximum permissible fuel imbalance between the wing tanks is 454 kg.
- Manual transfer and gravity crossflow must be off for take-off.
- Minimum fuel quantity for go-around is 442 kg per wing and maximum climb attitude of 10 degrees nose-up.
- Max fuel temp for operations is 52.5°C, min fuel temp for take-off is -37°C.
- If either wing tank fuel quantity is greater than 2830 kg:
    a) Operation of the APU in unattended mode is prohibited.
    b) On ground, the left engine must be started first.
    c) A manual transfer must be done to the center tank to reduce wing tank quantity below 2830 kg if center tank fuel quantity is less than 10.670 kg.

# Avionics

## Flight Management System

- FMS navigation north of 73°N and south of 60°S is prohibited.
- Selection of TACAN from the navigation database is prohibited.
- Entry of Place/Bearing/Distance (PBD) using an airport identifier is prohibited, uplinked flight plans must not contain PBD waypoints.
- Use of FPLN UPLINK function in flight is prohibited and any REQUEST PENDING must be cleared before take-off.
- The flight crew must verify any altitude or speed constraint on the last waypoint of any entered STAR procedure and / or first waypoint of any entered approach procedure and correct any altitude or speed constraints as required to agree with current charted procedures.
- Localizer based approaches must be flown with NAV mode selected before the Final Approach Fix (FAF). Use of FMS as primary navigation source beyond the FAF for any localizer-based approach is prohibited.
- Other than automatic temperature compensation, changes to altitudes in the final approach segment (including FAF and step-downs) for any FMS approach is prohibited.
- MAN SPD must be set and used when flying $V_{APP}$ in landing configuration.
- Use of OPT ALT and MAX ALT on FMS PERF – CLB and FMS PERF-CRZ page for flight planning purposes is prohibited.
- True headings must be entered in the xxxT format.

## Displays

- The use of Vertical Situation Display (VSD) is for advisory purpose only.

## Printer

- Do not print Weight & Balance in-flight.
- Do not print en-route charts and NOTAM charts (at any time).

## Altimeters

- Operation using QFE altimeter setting is prohibited.

## Terrain Awareness System (TAWS)

- Airplane must not be navigated based on the terrain display.
- To avoid unwanted alerts, TAWS TERR switch must be selected to INHIB for take-off and landing within 15 nm of an airport not in the TAWS database.

## Take-off and Landing Awareness Function (TLAF)

- When proceeding onto the departure runway do not exceed 35 kts ground speed or 55% $N_1$ until within 10 degrees of the runway heading.

# Engine Limitations

## Engine Type

- PurePower PW1519G (A220-100 only)
- PurePower PW1521G
- PurePower PW1524G
- PurePower PW1525G

## Engine Oil Temperature

- Below -40°C: Engine start prohibited
- From -40°C to -6°C: Idle
- From -6°C to +21°C: 30% $N_1$
- From +21°C to +49°C: 50% $N_1$
- +49°C and above: No limit

## Engine Oil Consumption

- Maximum allowable oil consumption is 0.38 l/hr.

## Engine Operating Limits

- Maximum Take-off thrust (MTO) limited to:
    a) All Engines Operating (AEO): 5 minutes
    b) One Engine inoperative (OEI): 10 minutes
- EGT limitations:
    a) Maximum EGT = 1054°C
    b) Max continuous EGT = 1016°C
- Max $N_1$ / $N_2$ rotational speed = 100%
- Above 29.100 ft, the maximum $N_1$ = 95.2%

## Engine Start

- After three start attempts or 4 minutes of continuous motoring, a 30-minute cooling period is required.
- Do not engage the starter on ground if indicated $N_2$ exceeds 20%.

## Engine Shutdown

- Engines must be operated at or near idle for at least 3 minutes (5 minutes is recommended) before shutdown (does not apply for non-normal shutdowns). Near idle is defined as up to 40% $N_1$. An additional period equal to the time spent above 40% $N_1$ must be accounted for.

## Thrust Reverse

- Thrust reversers are approved for ground use only.
- Activation of reverse thrust in flight is prohibited.
- Use of thrust reverse for backing up (i.e. power back) is prohibited.
- Go-around maneuver and touch and go are prohibited after deployment of the thrust reversers.

OAT between -20°C and -40°C: If turnaround time exceeds 120 minutes, the thrust reversers must be cycled twice before take-off from contaminated runways. Take-off must be initiated within 5 minutes after last thrust reverser cycle, otherwise procedure must be repeated.

OAT below 40°C: If turnaround time exceeds 120 minutes, the thrust reversers must be cycled before take-off from contaminated runways until last cycle time (time from commanding reverse until green REV icon on EICAS) is equal to or less than 2.5 seconds. Take-off must be initiated within 2 minutes after last thrust reverser cycle, otherwise procedure must be repeated.

# Runway & Wind Limitations

### Runway Slope

- Maximum runway slopes approved for take-off and landing are ± 2%.

### Passenger and Cargo Door Operations

- Maximum wind speed for door operation (opening / closing) is 40 kts.
- Maximum wind speed for doors held in open position is 65 kts.

### Tailwind Components

- The maximum tailwind component for take-off and landing is 10 kts.

### Crosswind Components

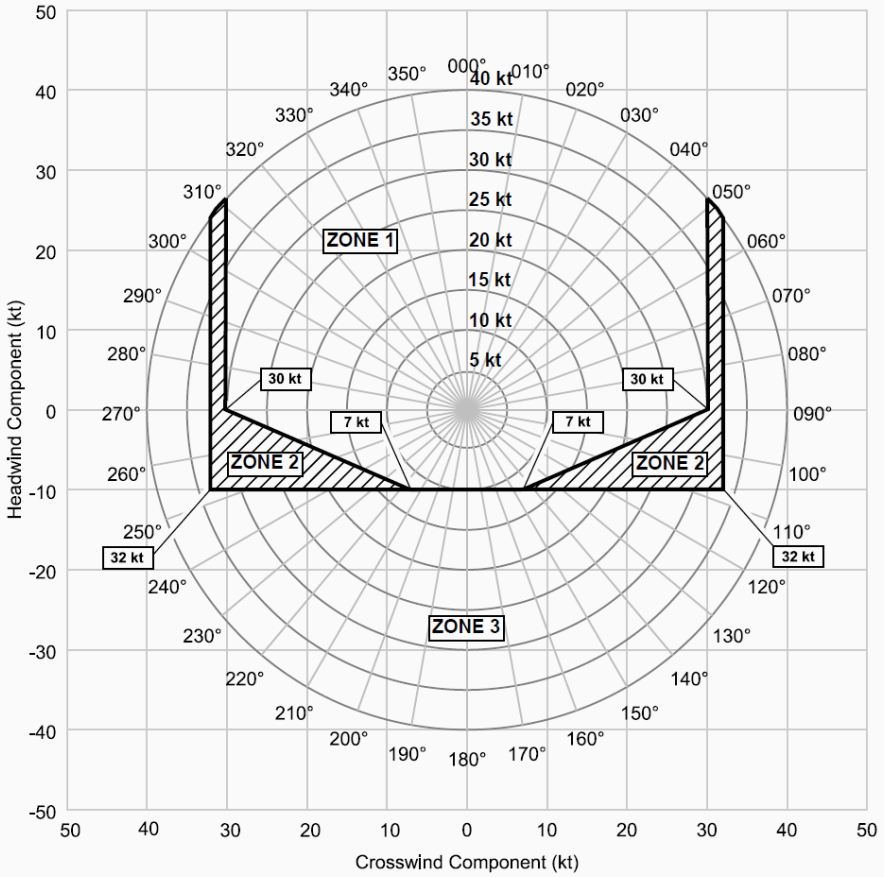
- Maximum crosswind component for take-off is 32 kts (engine limitation).
- Maximum crosswind component for landing is 29 kts (demonstrated).

### Engine Operating Wind Limitations

- For starting and ice shedding: max crosswind = 43 kts, max tailwind = 18 kts
- No limitations for taxi.
- High power run-ups (> 60% $N_1$) must be done with aircraft into the wind.

### Contaminated runway Operations

- During taxi, except for the interest of safety, avoid use of thrust reversers if movement area surfaces are covered with standing water, slush, snow or ice.
- When operating on runways contaminated with ice, both thrust reversers must be operative before dispatch.
- Anti-skid must be operative.




| Engine operating limits due to wind | | |
|---|---|---|
| **Zone** | **Altitude** | **Limitation** |
| 1 | < 7000 ft | Normal take-off technique can be used. |
| | 7000 – 9200 ft | High wind take-off if crosswind exceeds 22 kts. |
| | ≥ 9200 ft | High wind take-off must be used (always). |
| 2 | High wind take-off must be used. | |
| 3 | Take-off prohibited. | |

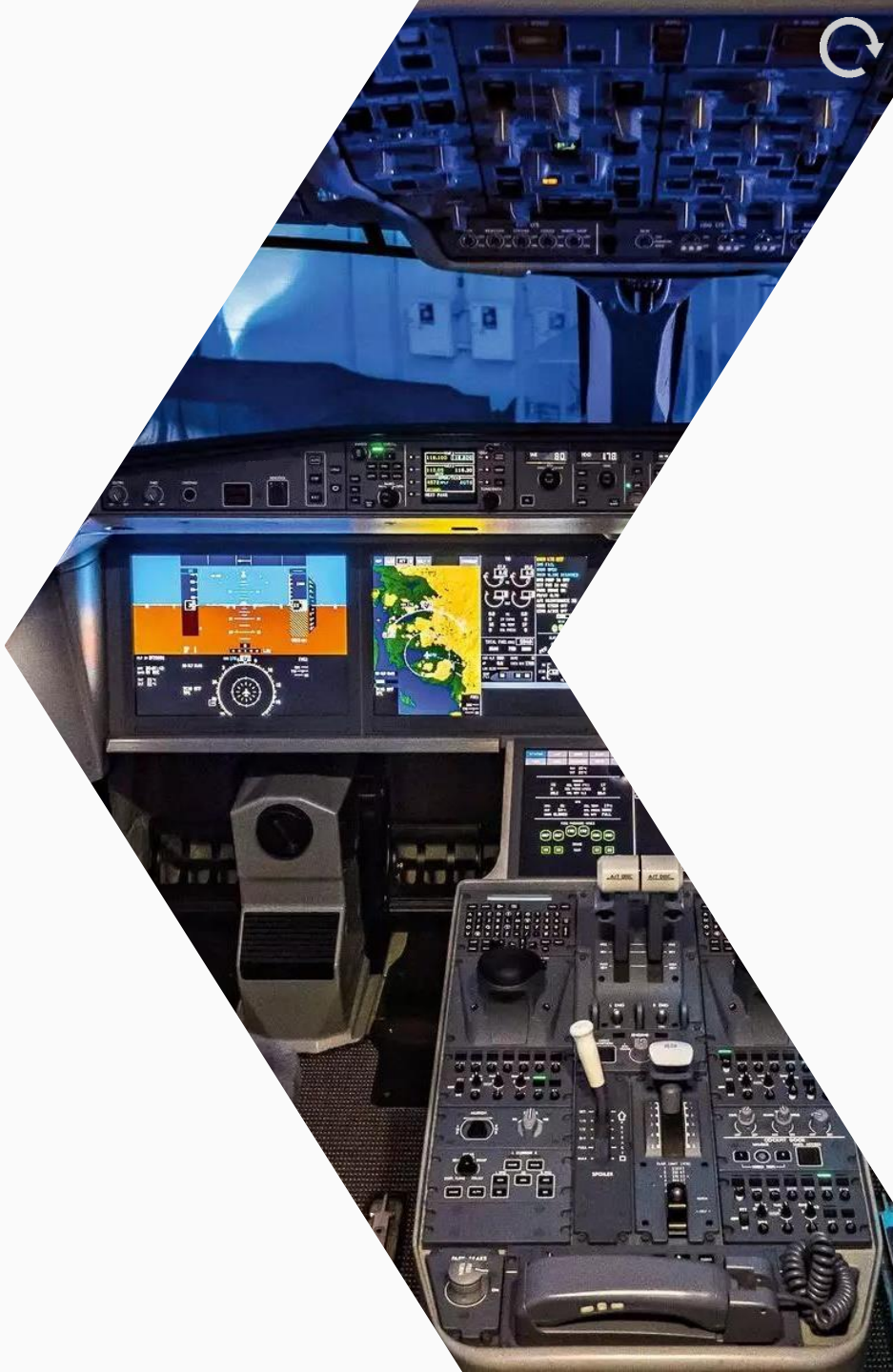
# Runway Condition Assessment Matrix (RCAM)

| Runway Condition Assessment Matrix | | | |
|---|---|---|---|
| **Runway Condition Code (RWYCC)** | **Runway Surface Condition Description** | **Pilot-reported Braking Action** | **Maximum Recommended Crosswind** |
| 6 | Dry | - | 32 kts (take-off)<br>29 kts (landing) |
| 5 | • Frost<br>• Wet (less than 3 mm)<br>3 mm depth or less of:<br>• Slush<br>• Dry snow<br>• Wet Snow | Good | |
| 4 | -15°C and colder:<br>• Compacted snow | Good to Medium | 27 kts |
| 3 | • Wet (“slippery when wet”)<br>• Dry snow or wet snow<br>More than 3 mm depth:<br>• Dry snow<br>• Wet snow<br>Warmer than -15°C:<br>• Compacted snow | Medium | 20 kts |
| 2 | More than 3 mm depth:<br>• Water<br>• Slush | Medium to Poor | 10 kts |
| 1 | Ice | Poor | |
| 0 | - | - | - |

## Other Limitations

Refer to Supplemental Procedures for additional limitations

# Normal Procedures



# Introduction

## Operational Philosophy

A. Clean Cockpit: objects may fall and cause hazards such as damage to equipment or accidentally operate controls or switches. All objects must be stored in their dedicated stowage areas.

B. Dark Cockpit concept: during normal operations, there are no lights on when switches are in their AUTO or normal position. When lights are out, systems are ready and fit for flight.

C. Crew responsibility: it is required that crew members monitor the airplane and systems through periodical checks of the instruments and displays.

D. Checklists: on ground, roles are divided between Right Pilot (CM2) and Left Pilot (CM1), in-flight, for Power-on / Power-off and cockpit preparation (e.g. FMS set-up) roles are divided between Pilot Flying (PF) and Pilot Monitoring (PM). The PF (or CM1 on ground) normally initiates checklists. The other pilot then reads and actions the checklist. There are two types of checklists: **challenge & response** and **read & do**.

- Challenge and Response: the PM (CM2 on ground) reads the challenge, the other pilot verifies the existing configuration and then responds. Some items may require a response from both pilots (e.g. sidestick and oxygen masks checks). If the configuration is not in accordance with the procedure (e.g. dispatch with system inoperative), the response must be modified to reflect the actual situation (the same applies to "As required" items).

  e.g. CM2 reads: "FLAP", CM1 responds: "... SELECTED".

- Read and Do: the PM (CM2 on ground) reads the challenge, verifies the existing configuration, then reads the response. No verification is required from the other pilot. Both challenge and response are read out loud so that the other pilot can follow all actions.

  e.g. PM reads: "EICAS", verifies and calls: "CHECKED".

E. Callouts: all FMA mode changes must be acknowledged, specific callouts for each procedure are shown on the next page.

**Note:** checklists are normally read by CM2 while on ground and performed by the PM while in flight. The Preflight (PM) and Power-on / Power-off (PF) checklists are exceptions. All checklists having "Power" or "After" in the title are read and do checklists.

| Pre-flight | | | In-flight | | | Post-flight | | |
|---|---|---|---|---|---|---|---|---|
| **Title** | **Type** | **Read** | **Title** | **Type** | **Read** | **Title** | **Type** | **Read** |
| Power-on | Read & Do | *PF | After take-off, High altitude climb check | Read & Do | PM | After landing | Read & Do | CM2 |
| Pre-flight | Challenge Response | *PM | Descent & approach, Before landing | Challenge Response | | Shutdown | Challenge Response | |
| Before start, Before taxi, Before take-off | | CM2 | After go-around | Read & Do | | Power-off | Read & Do | *PF |

Responses that are written in uppercase are read as challenge-response-response:

- Lowercase ("Complete") CM2: "TAKE-OFF BRIEFING", CM1 reads: "COMPLETE"
- Uppercase ("OFF") CM2: "L ENG RUN ... OFF", CM1 completes the action: "OFF"

The pilot reading the checklist calls out the title of the checklist, then the phrase "CHECKLIST". For example:

- PM: "AFTER TAKEOFF **CHECKLIST**"

When a procedure is completed, the pilot reading the checklist calls the title of the checklist and the phrase "COMPLETE". For example:

- PM: "AFTER TAKEOFF CHECKLIST **COMPLETE**"

Conditions in checklists ( YES NO in ECL) must be confirmed by both pilots before choosing a direction. For example:

- PM: "FIRST FLIGHT OF THE DAY, **YES or NO**?"
- PF: "YES" or "NO"

Airbus A220 standard callouts and procedural flow on the next pages.

## Standard callouts

| Event | CM1 (ground) or PF (flight) | CM2 (ground) or PM (flight) |
|---|---|---|
| Checklist reading<br><br>Both pilots respond to:<br><br>• EICAS and INFO<br>• Altimeter setting<br>• Approach minima<br>• Sidestick test<br>• Oxygen mask test | "PREFLIGHT CHECKLIST"<br><br>"YES" or "NO"<br><br>Response →<br><br>Interruption required:<br>"STOP / CONTINUE ECL" | "PREFLIGHT CHECKLIST"<br><br>"FIRST FLIGHT OF THE DAY, YES or NO?"<br><br>← Challenge<br><br>"PREFLIGHT CHECKLIST COMPLETE" |
| Flight control check | "FLIGHT CONTROL CHECK"<br><br>"RUDDER" | Ailerons: "FULL LEFT, FULL RIGHT, NEUTRAL"<br>Elevators: "FULL UP, FULL DOWN, NEUTRAL"<br><br>Rudder: "FULL LEFT, FULL RIGHT, NEUTRAL" |
| Transfer of controls or communications | "YOU HAVE CONTROL (ATC)"<br><br>Giving controls or communications:<br>"YOU HAVE CONTROL (ATC)" | Taking controls or communications:<br>"I HAVE CONTROL (ATC)"<br><br>"I HAVE CONTROL (ATC)" |
| Configuration changes | "GEAR UP"<br><br>"FLAP 2" | "POSITIVE CLIMB"<br><br>"GEAR UP"<br><br>"SPEED CHECKED" (speed ≥F or ≤$V_{FE}$), then call: "FLAP 2", or "CHECK SPEED" |
| Flight director, Autopilot or Autothrottle | FMA: "AUTOPILOT ON / OFF"<br><br>Alternatively:<br>"(DE)SELECT AUTOPILOT"<br><br>FMA: "AUTOPILOT ON / OFF" | "CHECKED"<br><br>"AUTOPILOT (DE)SELECTED"<br><br>"CHECKED" |
| FCP selections<br><br>Push = "(DE)SELECT"<br>Turn = "SET"<br><br>AP ON: PF does not call FCP selections, only FMA modes.<br><br>For altitude selections, keep finger on the ALT SEL button until other pilot confirms.<br><br>Include magenta altitude call-out only if more restrictive. | Manual flight (example):<br><br>Modes (push): "SELECT HDG"<br>Call FMA mode change: "HDG"<br><br>Selections (turn): "SET HDG 360"<br>Call heading indication: "HDG 360"<br><br>Autopilot ON (example):<br><br>Call FMA mode changes: "V-FLC"<br><br>Altitudes are called with corresponding colour:<br>e.g. "FL350 BLUE", "3000 FT MAGENTA" | "HDG SELECTED"<br>"CHECKED"<br><br>"HDG 360 SET"<br>"CHECKED"<br><br>"CHECKED"<br><br>"CHECKED" |
| FMA callouts | Active modes: "HDG", "FLC", "FMS 1(2)"<br>Armed modes: "APPR LOC 1(2) WHITE" | Active modes (1st line) without associated color, armed modes are called with associated color. |
| FMS changes<br>(PM below 10.000 ft) | Pilot making the change: "CONFIRM ..."<br>Other pilot after confirming: "EXECUTE" | |

## Specific callouts per flight phase

| Event | CM1 (ground) or PF (flight) | CM2 (ground) or PM (flight) |
|---|---|---|
| Taxi | Crossing or entering taxiways / runways:<br>"CLEAR LEFT SIDE" | "CLEAR RIGHT SIDE" |
| Line-up | "CLEAR LEFT SIDE"<br>"LIGHTS & BELTS" | "CLEAR RIGHT SIDE"<br>"LIGHTS & BELTS ON" |
| Take-off | Pressing AT on FCP:<br>"THRUST WHITE"<br>Advancing thrust to 55% $N_1$:<br>"TAKE-OFF"<br><br>Moving thrust levers forward until AT engages: "THRUST"<br><br>Confirm speed indication:<br>"CHECKED" | High wind take-off at 20 kts groundspeed: "20 KTS"<br><br>At 80 kts:<br>"80 KTS, THRUST SET"<br><br>"V1" (auto-callout)<br>At $V_R$: "ROTATE" |
| Climb or Descent | Above MSA and cleared to FL, Not later than transition altitude:<br>"SET STANDARD"<br><br>Cleared to an altitude, no later than transition level:<br>"SET QNH"<br><br>"LIGHTS & BELTS"<br><br>"CHECKED" | "STANDARD SET"<br><br>"QNH ... SET"<br><br>Passing 10.000 ft:<br>"10.000 FT"<br>"LIGHTS & BELTS ON / OFF"<br><br>1000 ft prior level-off:<br>"1000 TO GO" |
| Approach | "CORRECTING" or<br>"GO-AROUND, FLAP 2 (4)"<br><br>"CHECKED" or<br>"GO-AROUND, FLAP 2 (4)"<br><br>"CHECKED"<br><br>"CONTINUE" or<br>"GO-AROUND, FLAP 2 (4)" | ½ dot vertical deviation, 100 ft low / high for NPA:<br>"GLIDESLOPE" or "VERTICAL DEVIATION"<br><br>½ dot lateral deviation (ILS, LPV, VOR)<br>1 dot for RNP (non-LPV), for NDB 5° :<br>"LOC", "LATERAL DEVIATION", "COURSE"<br><br>Target +10 kts or target -5 kts: "SPEED"<br>VS exceeding 1000 ft/min: "SINKRATE"<br><br>At 1000 ft AAE:<br>"STABILIZED" or<br>"NOT STABILIZED"<br><br>"PLUS-100" (auto-callout)<br><br>"MINIMUMS" (auto-callout) |
| Landing | Latest at 80 ft AGL:<br>"AUTOPILOT OFF"<br><br>Autobrake disconnect:<br>"MANUAL BRAKING" | 30 ft AGL: "RETARD" or "NO RETARD"<br><br>Bank exceeding 7.0° left or right: "BANK"<br>Pitch exceeding 7.5° up or below 0°: "PITCH"<br><br>"REVERSE GREEN" or "NO REVERSE" |

## Power On (PF)

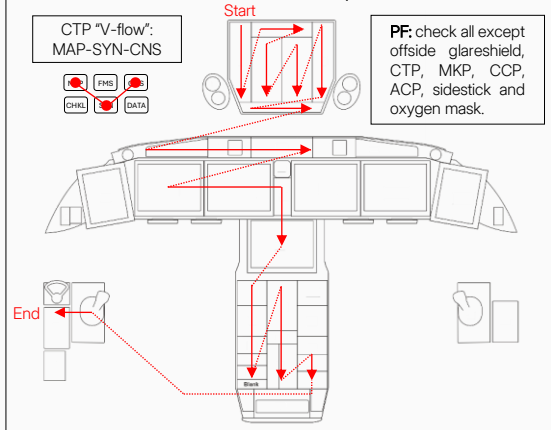
- Steps a. to d. performed by memory:
  a. PARK BRAKE.....ON
  b. BATT 1.....AUTO
  c. BATT 2.....AUTO
  d. ECL.....DU 2
- Complete POWER ON CHECKLIST.
- Establish AC power (GPU or start APU).

## Exterior Inspection (PM)

- Check aircraft technical status.
- NAV (LOGO + WING at night) lights ON.
- Set-up fueling / defueling panel.
- Perform external walkaround.
- Returning from walkaround, perform FDRAS (cockpit door) test (first flight of the day only).
- Exterior lights as required.

## Preflight (PF)

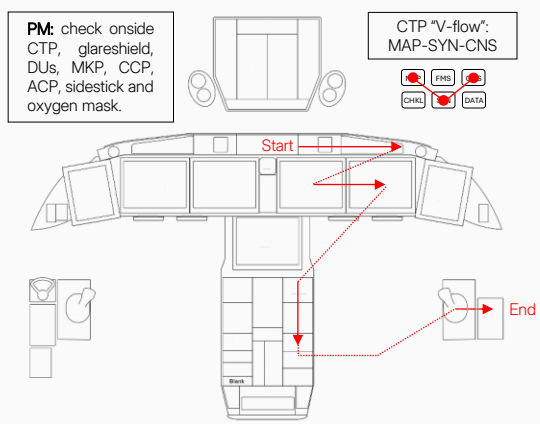
- Panel flow CM1 = PF (example):




- All switches AUTO / middle position, no lights.

## Preflight (PM)

- Panel flow CM2 = PM (example):




- All switches AUTO / middle position, no lights

## Cockpit Preparation (PF)

- ACARS INIT, FPLN UPLINK
- FMS set-up, CTP 1 (NAVs, BRG 1 / 2 etc.)

PF ➞ CM1

## Cockpit Preparation (PM)

- COM frequencies, ATIS, en-route CLX.
- CTP 2 (NAVs, BRG 1 / 2 etc.)

(PM) PREFLIGHT CHECKLIST
Departure briefing

PM ➞ CM2

## Before Start (CM1)

- Read ZFW, CG, PAX from final loadsheet.
- Update performance & VSPEEDS.

Ready for pushback and / or start-up:

## Before Start (CM2)

- Enter figures into FMS FPLN – FUEL.
- Update performance & VSPEEDS.

XPDR ... AUTO

(CM2) BEFORE START CHECKLIST

## Engine Start (CM1)

- Call for engine 1 / 2 start.

Both engines stabilized:

## Engine Start (CM2)

- ENG 1 / 2 RUN SWITCH ... ON

(CM2) BEFORE TAXI CHECKLIST

Turnaround: RESET ALL CHKL, start from POWER ON

Challenge Response

Read & Do

## Taxi (CM1)

- Check left clear, taxi light NARROW or WIDE.

Cabin READY:

CM1 ➞ PF

## Taxi (CM2)

- Check right side clear.

(CM2) BEFORE TAKEOFF CHECKLIST

CM2 ➞ PM

## Take-off (PF)

- Autothrottle ARM, "TAKE-OFF"
- 55% $N_1$ ➞ TO thrust

## Take-off (PM)

- Lights ON and cabin signal given.

Clean config, above GD:

(PM) AFTER TAKEOFF CHECKLIST

## Climb (PF)

- Altimeters standard
- 10.000 ft, lights and seatbelts.

## Climb (PM)

- Altimeters standard
- 10.000 ft, lights OFF and seatbelts AUTO.

Before 35.000', no icing:

(PM) HIGH ALTITUDE CLIMB CHECKLIST

## Cruise (PF)

- FMS, CTP 1 (NAVs, minima), Perf & VSPEEDS.

Before starting descent:

## Cruise (PM)

- ATIS, CTP 2 (NAVs, etc.), Perf & VSPEEDS.

Arrival Briefing
(PM) DESCENT AND APPROACH CHECKLIST

## Descent (PF)

- 10.000 ft, lights and seatbelts.
- Altimeters set to QNH.

## Descent (PM)

- 10.000 ft, lights and seatbelts ON.
- Altimeters set to QNH.

## Approach and Landing (PF)

Landing Configuration:

## Approach and Landing (PM)

Cabin signal given.

(PM) BEFORE LANDING CHECKLIST

## Go-around (PF)

- Press TOGA, "GO-AROUND, FLAP 2 / 4"

PF ➞ CM1

## Go-around (PM)

Clean configuration above green dot speed:

PM ➞ CM2

(PM) AFTER GO AROUND CHECKLIST

or

## After Landing (CM1)

Vacate runway.

## After Landing (CM2)

Start APU, lights, flaps:

(CM2) AFTER LANDING CHECKLIST

## Shutdown (CM1)

PARK BRAKE ... SET

CM1 ➞ PF

## Shutdown (CM2)

XPDR ... STBY

(CM2) SHUTDOWN CHECKLIST

CM2 ➞ PM

or

## Power Off (PF)

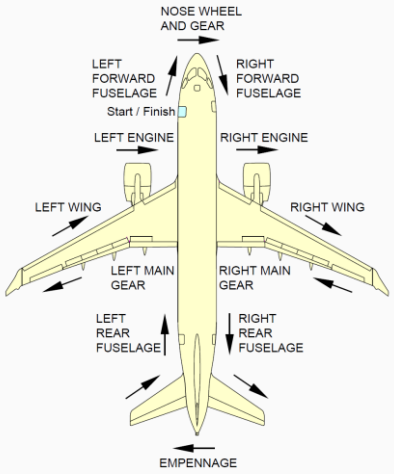
Before leaving aircraft:

## Exterior Inspection (PM)

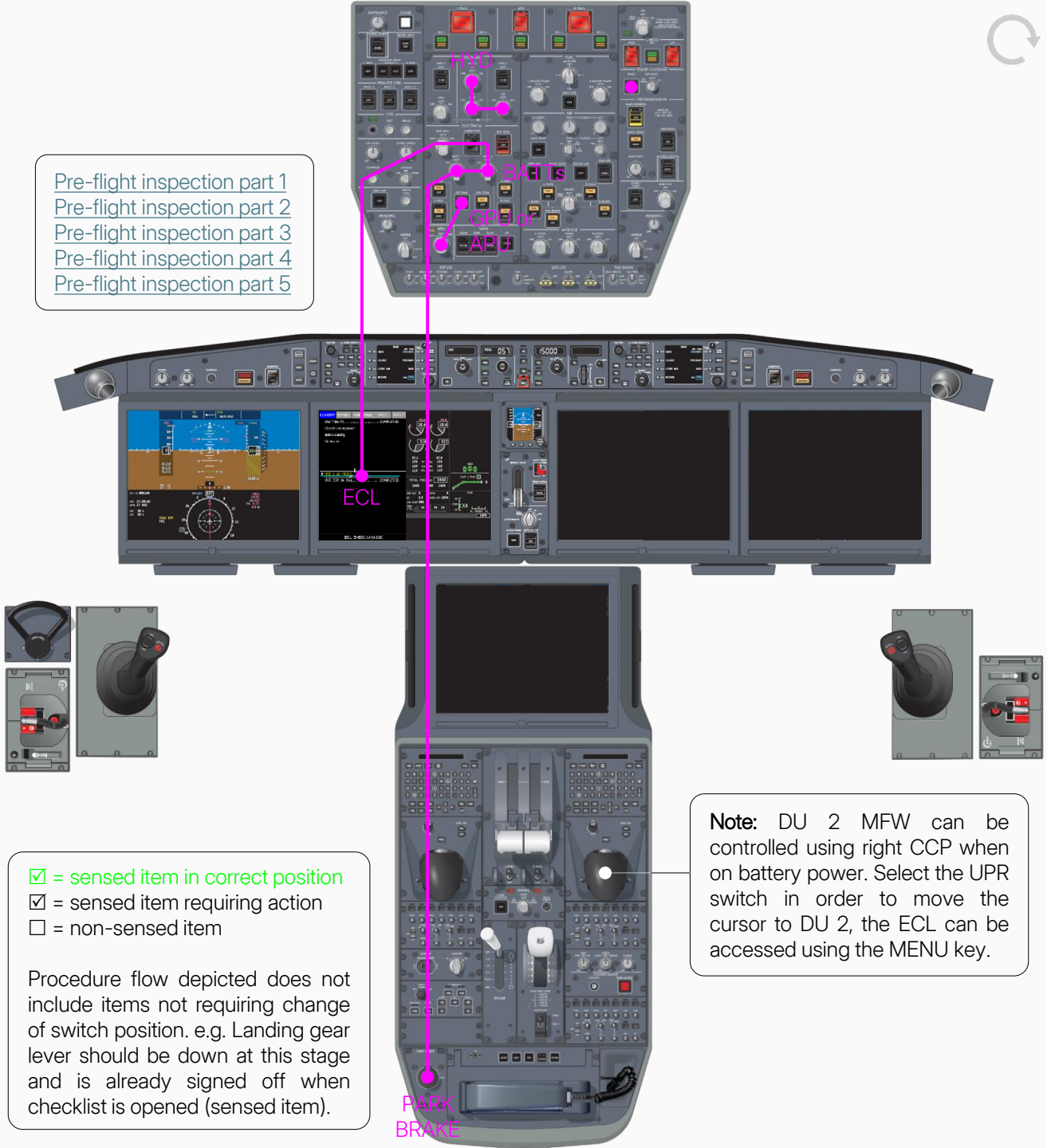
Post flight inspection.

(PF) POWER OFF CHECKLIST

# (PF) Power-on | (PM) Walkaround

| Event | Pilot Flying | Pilot Monitoring |
|---|---|---|
| Power-on Checklist (Read & Do) | These steps must be done if AC electrical power is removed from the airplane.<br><br>Before batteries are ON, the ECL is not available, first 4 items are done by memory:<br><br>☑ PARK BRAKE.....ON<br>☑ BATT 1.....AUTO<br>☑ BATT 2.....AUTO<br>☐ ECL.....DU 2<br><br>*Confirm ECL version before continuing.*<br><br>☑ PTU.....AUTO<br>☑ HYD 3A.....AUTO<br>☑ HYD 3B.....AUTO<br>☑ HYD 2B.....AUTO<br>☑ Landing gear lever.....DN<br>☑ Thrust levers.....IDLE<br>☑ L ENG run.....OFF<br>☑ R ENG run.....OFF<br>☐ APU or external power.....As required<br>☑ EQUIP COOLING, INLET.....Select auto | The airplane and its visible components must be checked as follows:<br><br>• Proximate area is free of potential FOD<br>• Flight control surfaces unobstructed<br>• All vents, ports, intakes and exhausts are unobstructed.<br>• All covers, plugs, pins are removed.<br>• Any evidence of fluid leaks<br>• Condition of skin (visible damage)<br>• Access panels and doors are secured.<br>• Probes for evidence of freezing, severe discoloration, condition and security.<br>• Tire condition and inflation.<br><br>During cold weather operations, ensure that the airplane fuselage, wings and tail surfaces are free from ice, snow or frost.<br><br>a. Aircraft technical status..... Check<br>b. Exterior lights..... As required<br>c. Refueling / defueling panel.....Set<br><br>Start at the forward passenger door, continue in the direction shown and finish back at the forward passenger door.<br><br>  |
| APU start (if no GPU) | a. Select START for a minimum of 3 sec.<br>b. Allow switch to return to RUN position.<br><br>APU IN START status message. When APU is ready to load APU ON is shown.<br><br>Wait for APU GEN to show online on the ELEC synoptic page or until air audible from the air vents before making electrical changes (e.g. disconnecting GPU).<br><br>To avoid interruption of the start cycle, no electrical system configuration change (such as engine shutdown or selecting GPU) should be performed until APU ON.<br><br>If the APU SHUTDOWN message is shown during APU start, the APU switch must be set to OFF before a new APU start cycle can be attempted. | |

Pre-flight inspection part 1
Pre-flight inspection part 2
Pre-flight inspection part 3
Pre-flight inspection part 4
Pre-flight inspection part 5




☑ = sensed item in correct position
☑ = sensed item requiring action
☐ = non-sensed item

Procedure flow depicted does not include items not requiring change of switch position. e.g. Landing gear lever should be down at this stage and is already signed off when checklist is opened (sensed item).

**Note:** DU 2 MFW can be controlled using right CCP when on battery power. Select the UPR switch in order to move the cursor to DU 2, the ECL can be accessed using the MENU key.

# › Pre-flight

Check all switches in the AUTO – middle position, guarded with no lights. Instrument and display lighting set as required.
♦ = first flight of the day only.

| Event | Pilot Flying | Pilot Monitoring |
|---|---|---|
| Flight Compartment | • Airplane documents...On board, checked<br>• Operations Engineering Bulletins<br>• Emergency Equipment.....Checked<br>• Gear pins (3).....On board | • Operations Engineering Bulletins<br>*Review together all applicable OEBs (can be performed as part of departure briefing).* |
| Overhead Panel | • PROBE HEAT.....GND ON, if required<br>*In case of snow or ice on the probes.*<br>• FWD CARGO.....As required<br>*LO or HI HEAT if carrying sensitive cargo*<br>• EMER LTS.....ARM<br>• EXT LTS.....NAV ON<br>*LOGO & WING in low visibility or at night.*<br>• Cabin signs.....As required<br>*Seatbelts ON for boarding after refueling* | |
| CTP Control Tuning Panel | • RANGE.....As required<br>• NAV SRC.....As required<br>• BARO.....Set<br>• MENU.....Set CRS, WXR, PFD/NAV<br>CTP "V-flow" (MAP-SYN-CNS):<br>a. MAP.....As required<br>• OVLY, VSD, SYMBOL (IRS aligned)<br>b. SYN.....Cycle and check<br>• FLT CTRL.....Perform sidestick test<br>♦ AVIONIC.....ICE DETECT test<br>♦ CB.....Circuit breaker status<br>c. CNS.....Set<br>• COMs, NAVs, XPDR 1 SEL, set FLT ID | • RANGE.....As required<br>• NAV SRC.....As required<br>• BARO.....Set<br>• MENU.....Set CRS, WXR, PFD/NAV<br>CTP "V-flow" (MAP-SYN-CNS):<br>a. MAP.....As required<br>• OVLY, VSD, SYMBOL (IRS aligned)<br>b. SYN.....Cycle and check<br>• FLT CTRL – Perform sidestick test<br>*Full range on both axis, release from full deflection (roll axis), soft stop (pitch axis). Confirm full deflection and the blue dot returning to center when releasing sidestick* |
| FCP Flight Control Panel | • SPD.....Set green dot, FMS SPD<br>• XFR.....PF side<br>• ALT.....Set initial altitude<br>*After FMS – PERF – DEP setup completed:*<br>• TOGA + NAV + VNAV..... Select<br>• HDG.....Set RWY HDG | |
| Displays | • All displays configured, no flags.<br>• EICAS and INFO..... Checked<br>• Altimeters.....Set and cross-checked | • All displays configured, no flags.<br>• EICAS and INFO..... Checked<br>• Altimeters.....Set and cross-checked |
| Center Panel | • Autobrake.....RTO | • ISI.....No flags, set BARO |
| Pedestal | • ACP.....As required<br>*Set SPKR and INT on for oxygen mask test.*<br>♦ Oxygen mask.....Checked<br>♦ Rudder trim.....Checked (& centered)<br>♦ FDRAS test.....EMER ACCESS, DENY | • ACP.....As required<br>*Set SPKR and INT on for oxygen mask test.*<br>♦ Oxygen mask.....Checked<br>*When returning from the walkaround:*<br>♦ FDRAS test.....EMER ACCESS, REQ |

CM1 = PF (example)

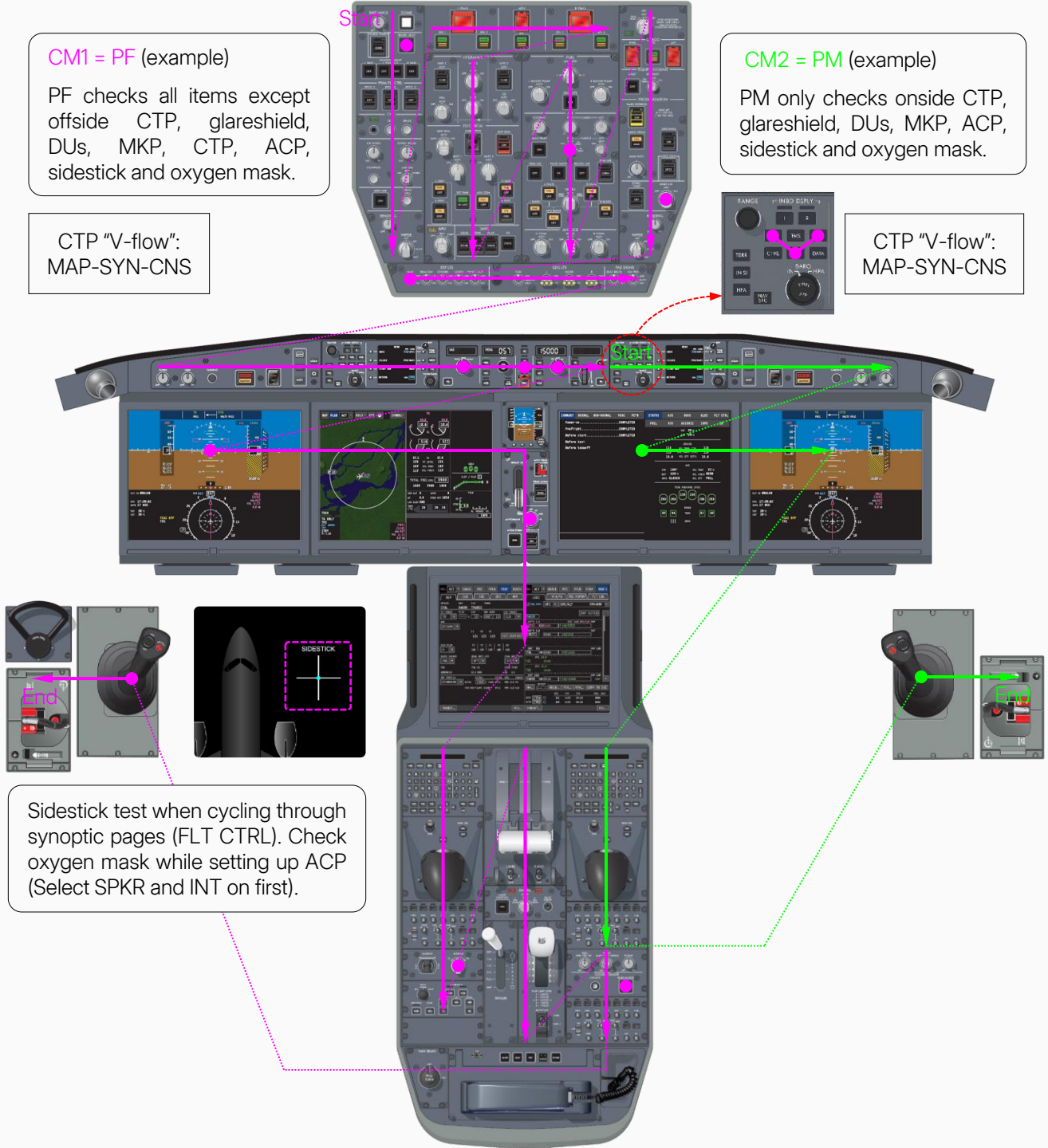
PF checks all items except offside CTP, glareshield, DUs, MKP, CTP, ACP, sidestick and oxygen mask.

CM2 = PM (example)

PM only checks onside CTP, glareshield, DUs, MKP, ACP, sidestick and oxygen mask.

CTP "V-flow": MAP-SYN-CNS

CTP "V-flow": MAP-SYN-CNS




Sidestick test when cycling through synoptic pages (FLT CTRL). Check oxygen mask while setting up ACP (Select SPKR and INT on first).

# Cockpit Preparation

| Event | Pilot Flying | Pilot Monitoring |
|---|---|---|
| ACARS<br>Initialization | CNS – DLK – AOC – PREFLIGHT – INIT:<br>• Select INIT REQ*<br>• Enter ATS FLT ID | |
| ATIS<br>En-route clearance | • Set QNH (including ISI)<br>• Set squawk code, cleared altitude<br>• Take-off performance.....Compute | Datalink (CNS – DLK – ATS) or Voice:<br>• Receive ATIS, set QNH<br>• En-route clearance<br>• Take-off performance.....Compute |
| FMS set-up<br>(next pages) | FMS set-up flow:<br>• DBASE<br>• POS: FMS – IRS – GNSS – VOR/DME<br>• FPLN: INIT – WIND/TEMP – FUEL<br>• PERF: DEP – CLB – CRZ – DES<br>• ROUTE: VIA/TO – LEGS<br>After setting FMS – PERF – DEP (see note):<br>• **Press TOGA + NAV + VNAV**<br>• Set runway heading<br>After FMS set-up, leave onside DU 5 MFW on FMS – PERF – DEP page. | *After PF completes FMS set-up, the PM will check all FMS entries by going through the FMS pages in the same order as the FMS preparation was conducted.*<br>*The PM crosschecks the FMS departure route with departure charts including all speed and altitude constraints, then selects COPY TO SEC to create a back-up.*<br>After FMS set-up, leave onside DU 5 MFW on FMS – ROUTE – LEGS page. |
| CNS<br>CTP | • Set BARO minima to OEI ACCEL ALT<br>• NAV 1: as required or set auto-tuning<br>• BRG 1 / 2 pointers as desired | • COMs: set frequencies as required<br>• NAV 2: as required or set auto-tuning<br>• BRG 1 / 2 pointers as desired |
| Pre-flight<br>Checklist | "PREFLIGHT CHECKLIST"<br>Response → | "PREFLIGHT CHECKLIST"<br>← Challenge<br>"PREFLIGHT CHECKLIST COMPLETE" |
| Departure briefing<br>(next pages) | Step 1. Cockpit door closed – set distraction free environment<br>Step 2B - PLAN<br>• Aircraft technical status & OEBs<br>• Take-off perf (PERF – DEP rundown)<br>• Stopping margin for RTO<br>• OEI procedure, return / diversion<br>• Non-standard operation<br>Step 3B - IDENTIFY THREATS | Step 2A - PLAN<br>• Planned taxi route & hotspots<br>• Take-off runway (intersection)<br>• Standard Instrument Departure<br>• First cleared altitude, MSA / MORA<br>• Extra fuel and endurance<br>Step 3A - IDENTIFY THREATS<br>Step 4 - MITTIGATIONS, Step 5 - MISCELLANEOUS |

**FMS set-up (next page)**

DBASE – check aircraft / engine variant and navigation database validity.
POS – cycle through each tab, check FMS position, RAIM availability, etc.
FPLN:

- INIT: open SEC FPLN, select FPLN UPLINK to uplink a flight plan via ACARS. Enter departure and arrival, other missing data, ACTIVATE SEC, enter FLT NO last.
- WIND/TEMP: upload by selecting FPLN WIND REQ or enter manually. If no FPLN UPLINK, enter the route first before uplink or enter wind.
- FUEL: enter preliminary ZFW, CG, PAX, FOB and TAXI fuel.

PERF:

- DEP: enter thrust, flaps, bleed source, anti-icing, NADP and speeds (last). SET VSPEEDS, **then press TOGA + NAV + VNAV** and set runway heading.
- CLB: climb speed profile, limits.
- CRZ: cruise speed profile.
- DES: descent speed profile, limits.

ROUTE:

- VIA/TO: enter route (or check if uplinked), enter waypoint (TO) first, then airway (VIA).
- LEGS: using FIX page, enter fixes as desired for reference (e.g. OEI departure route).

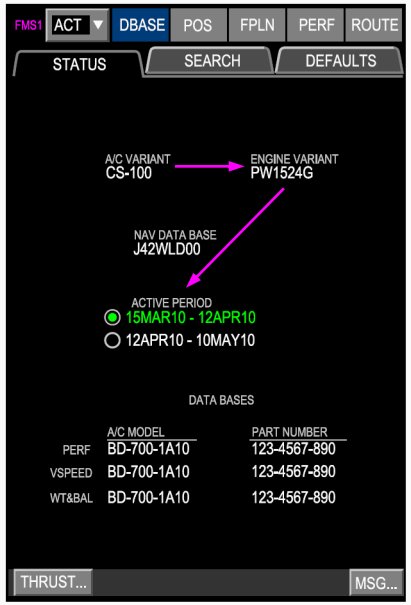
**Note 1:** entering a FOB on the FPLN – FUEL page that differs from the actual FOB (before refueling completed) will result in FMS FUEL message on EICAS. Selecting "SET VSPEEDS" and TOGA before refueling is completed or FPLN – INIT not finalized will cause FMS PERF DEP VSPEEDS to post together with a red FD flag on the PFD. It might be needed to cycle the FOB after refueling.

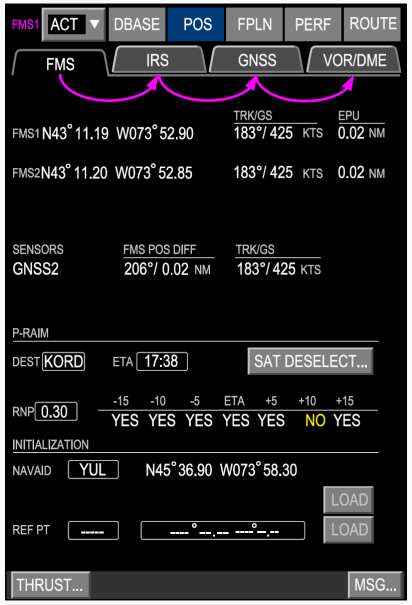
**Note 2:** after MAC ZFW & VSPEEDS are set, any change resulting in an EXEC prompt will cause TO CG and TRIM to disappear on the PERF – DEP page. Reselect V2, then press SET VSPEEDS.

| Requiring response from both pilots | First flight of the day only (♦) |
|---|---|
| • EICAS and INFO<br>• Altimeters<br>• Minimums<br>• Sidestick test<br>• Oxygen mask test | • Ice detector test<br>• Circuit breakers<br>• Rudder trim test<br>• Oxygen mask test<br>• FDRAS (cockpit door) test |

# FMS Preparation

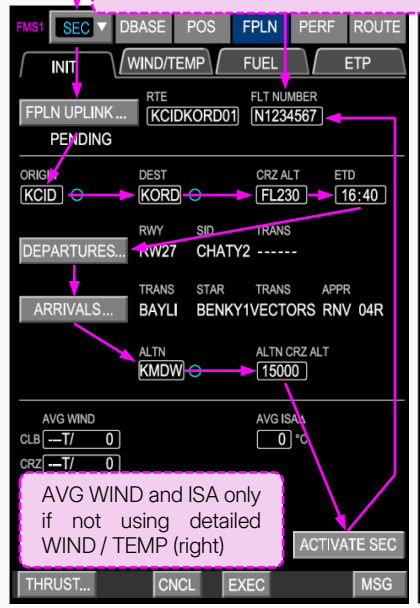
START



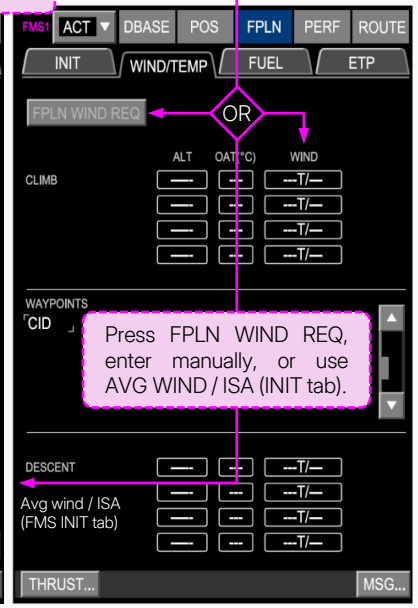




FPLN will be uplinked as SEC FPLN

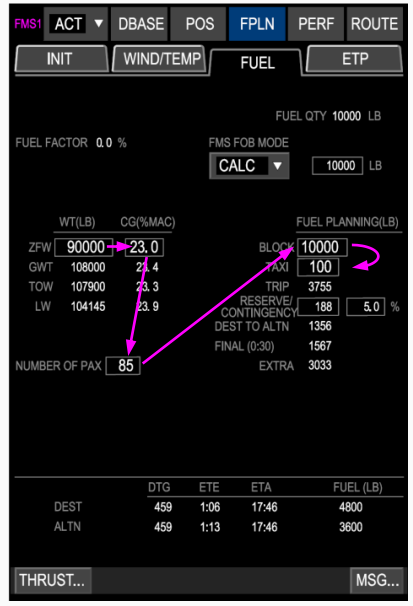
IATA FLT NO. after ACTIVATE SEC

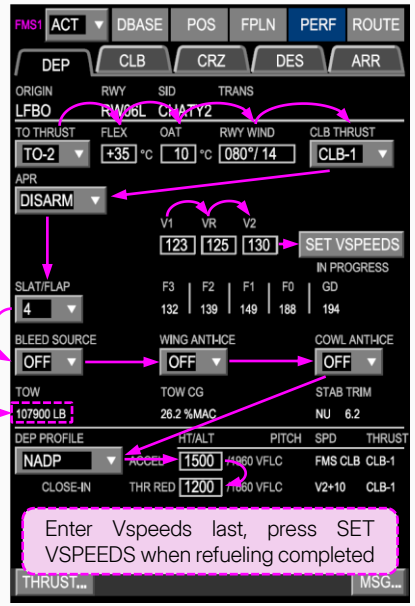



AVG WIND and ISA only if not using detailed WIND / TEMP (right)



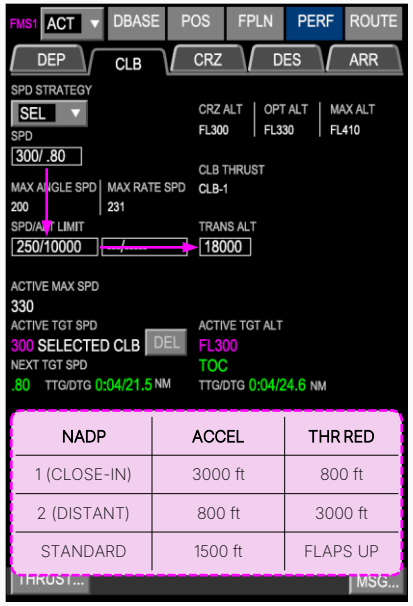

Press FPLN WIND REQ, enter manually, or use AVG WIND / ISA (INIT tab).



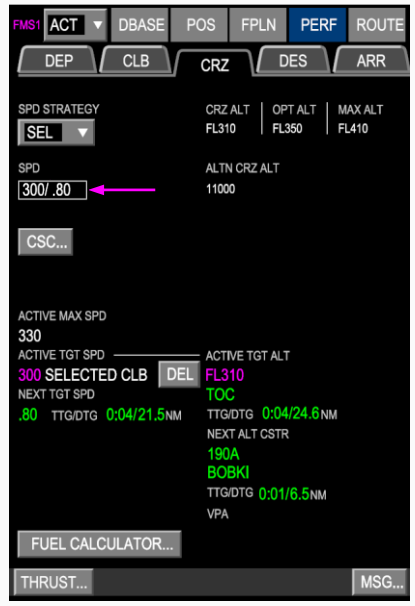



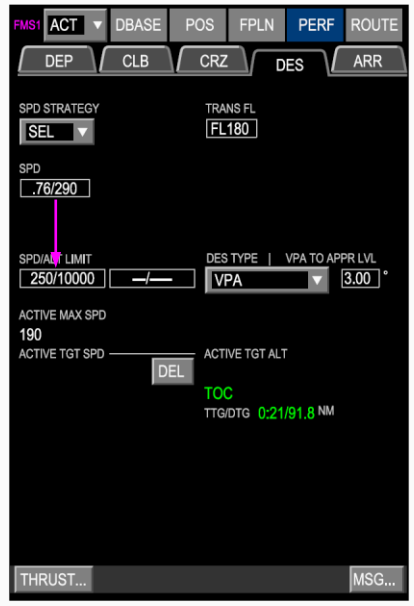

Take-off weight

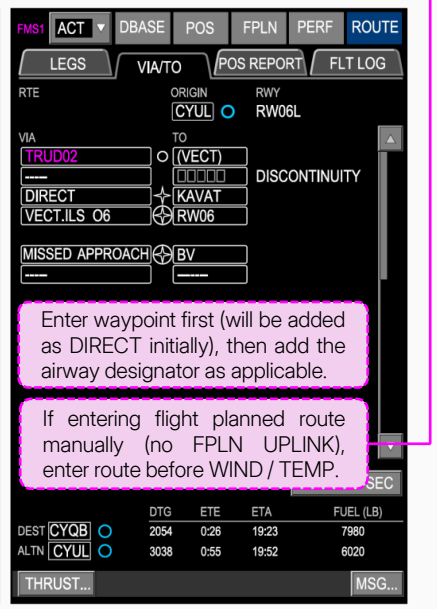
Enter Vspeeds last, press SET VSPEEDS when refueling completed




| NADP | ACCEL | THR RED |
|---|---|---|
| 1 (CLOSE-IN) | 3000 ft | 800 ft |
| 2 (DISTANT) | 800 ft | 3000 ft |
| STANDARD | 1500 ft | FLAPS UP |




Enter waypoint first (will be added as DIRECT initially), then add the airway designator as applicable.

If entering flight planned route manually (no FPLN UPLINK), enter route before WIND / TEMP.

END

# Departure Briefing

**Explanation / Notes:**

1. For the briefing to be of good quality it is important to minimize interruptions. The cockpit door should be closed. It should be the task of the Commander to anticipate potential disturbances and to manage this step. In case of interruption of the briefing it should be resumed at the beginning of the step where the interruption occurred.

2. The PM should start to brief the main items of the PLAN. This ensures that both pilots share the same mental image of the flight trajectory after the FMS preparation (by PF) and check (by PM). The PF then briefs the aircraft technical status including OEBs. He then considers at least the following items:

   - Take-off performance (rundown of FMS – PERF – DEP)
   - Consideration for RTO, including stop margin available
   - The One Engine Inoperative (OEI) departure trajectory
   - Considerations for return or diversion (weather / weight)

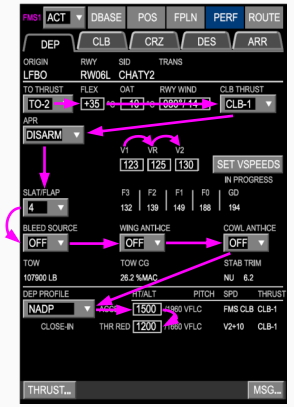
   The PF recalls any Special Operations or Supplementary procedures to be applied. Briefing the PLAN should normally only be a high-level description. It should normally not be a repetition of the detailed setting and checking of the flight trajectory in the FMS performed in the respective SOP items.

3. The PM should brief the THREATS identified throughout the preparation for the flight. The PF will highlight additional threats if required.

4. The PF and PM will discuss and agree on the MITIGATION for the threats identified.

5. MISCELLANEOUS is intended to take into consideration additional items:

   - Low visibility operations
   - Intended use of automation after take-off
   - Observer safety briefing and duties
   - Dangerous Goods on board etc.

| Step | Pilot Flying | Pilot Monitoring |
|---|---|---|
| 1 | Cockpit door closed – set distraction free environment | |
| 2A | | PLAN<br>• Planned taxi route & hotspots<br>• Take-off runway (intersection)<br>• Standard Instrument Departure<br>• First cleared altitude<br>• MSA / MORA for climb trajectory<br>• Extra fuel and endurance |
| 2B | PLAN<br>• Technical status, OEBs<br>• Take-off performance<br>• Stop margin for RTO<br>• OEI departure procedure<br>• Return / diversion considerations<br>• Non-standard operation |  |
| 3A | | IDENTIFY THREATS |
| 3B | IDENTIFY THREATS | |
| 4 | MITIGATIONS | |
| 5 | MISCELLANEOUS | |

# Before Start & Engine Start Procedure

| Event | Pilot Flying / Crewmember 1 | Pilot Monitoring / Crewmember 2 |
|---|---|---|
| APU start | Start APU (if not already) in accordance with local regulations (see Power On for details). | |
| Final Loadsheet | *Read from the final loadsheet:*<br>• ZFW, MAC ZFW, PAX<br>*Crosscheck TOW & TRIM with load sheet.* | *Enter figures in FMS – FPLN – FUEL, read:*<br>• TOW and STAB TRIM<br>**Note:** If APU is running, re-enter taxi fuel before reading TOW. Re-enter $V_2$ for the TO CG and STAB TRIM to display. |
| | Update and crosscheck take-off performance, set TRIM, update and SET VSPEEDS. | |
| Ready for start-up and / or pushback | ✓ All doors closed and slides armed<br>✓ Ground ready, pushback tug attached<br>✓ Pushback and / or start-up clearance | • XPDR.....AUTO<br>• Pushback and/or start-up clearance. |
| Before start Checklist | "BEFORE START CHECKLIST"<br>**A220-FOT-21-63-001:** TRIM AIR – OFF | "BEFORE START CHECKLIST" |
| | ☐ Takeoff briefing.....Complete<br>☐ APU or external power.....As required<br>☑ SEATBELTS.....ON<br>☑ DOORS.....Closed and locked<br>☑ BEACON.....ON<br>☐ PARK BRAKE.....As required | |
| | | "BEFORE START CHKL COMPLETE" |
| Pushback (if required) | • Coordinate with ground crew.<br>• Release parking brake. | |
| Engine Start Procedure<br>**A220-FOT Bowed Rotor Start**<br>If OAT >35ºC and engines were shutdown within the last 3 hours, dry motor (crank) for 45 sec before engine start:<br>• L(R) Thrust Lever.....IDLE<br>• CONT IGNITION..... OFF<br>• ENG START..... L(R) CRANK<br>*Hold for 45 seconds.*<br>• ENG START.....AUTO<br>• L(R) $N_2$.....Wait until 0% | *Monitor AIR SYN page (duct press 38 psi).*<br>"START ENGINE 1(2)"<br>TIMING.....Start<br>Wait for START flag to disappear before starting second engine. Repeat procedure. | *Monitor ELEC SYN page.*<br>ENG 1 (2) RUN switch.....ON<br>"ENGINE 1(2) ON"<br>TIMING.....Start<br>Monitor engine start sequence (≈ 2 min):<br>• Motoring at 12% $N_2$ for 45 seconds*.<br>• Fuel flow and ignition at 18-20% $N_2$.<br>• Starter and ignition cut-out at 51% $N_2$.<br>• START flag out as engine stabilizes.<br>• Abort start if exceedance is imminent. |
| | The FADEC ensures that engine start stays within limits and will abort if the engine starting limits are exceeded. EICAS L ENG START ABORT or R ENG START ABORT will be shown. | |

**A220-FOT Bowed Rotor Start**
High vibrations have been encountered with high OAT, causing compressor rubbing, which may increase internal clearances and then lead to off-idle engine stalls. If OAT is >35ºC and engines were shutdown within last 3 hours, dry motor (crank) for 45 seconds before engine start.

**Note:** motoring will be longer if engine thrust has not been maintained at idle for 3 minutes before last engine shutdown or if OAT is lower than -17°C or above +43°C. ENG START DELAY advisory is displayed if motoring time will exceed 55 seconds.

When starting engines with external power connected (e.g. APU inop), select HYD 3A + 3B OFF until one or both engines are running.

For a crossbleed start, start either engine using an air starter unit and external power. After first engine start, disconnect GPU and air starter.

**Pre-FADEC version 2.12.1:** increase $N_1$ of the donor engine to 30% before starting second engine.

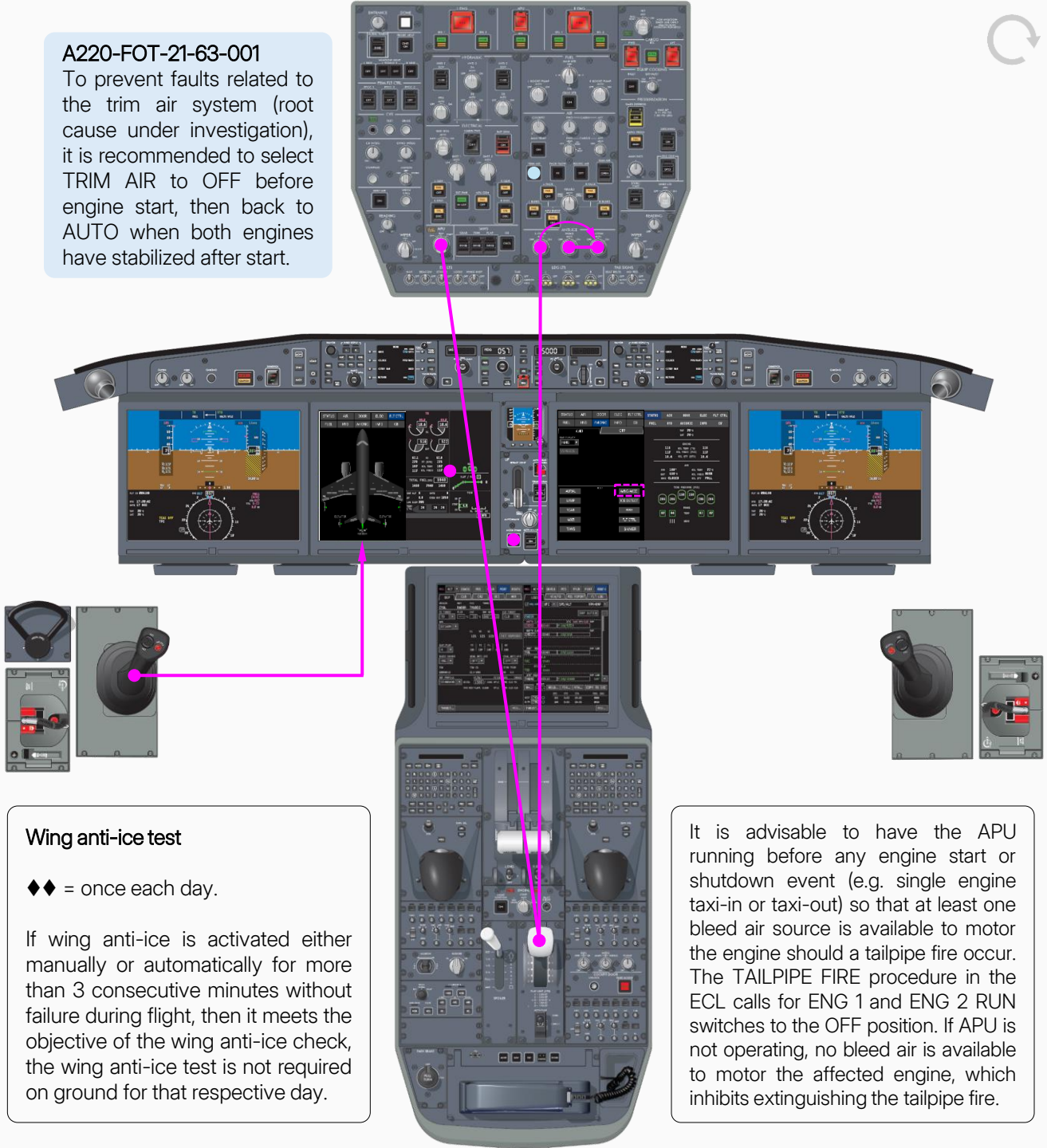
Engine Start Limitations:

- Maximum crosswind = 43 kts.
- Maximum tailwind = 18 kts.
- If either wing tank fuel quantity is greater than 2830 kg, left engine must be started first.
- After 3 start attempts or 4 min of continuous motoring, a 30-minute cooling period is required.
- Do not engage the starter if indicated $N_2$ exceeds 20%.

# Before Taxi

| Event | Crewmember 1 | Crewmember 2 |
|---|---|---|
| Single engine taxi operation<br><br>(not recommended on slippery/contaminated taxiways and aprons) | When the left engine is used to taxi out, flaps should only be deployed while aircraft is stationary to prevent reduced nosewheel steering function. Make sure all three hydraulic systems are pressurized during flight control check.<br><br><u>Before moving slats / flaps or performing flight control checks:</u><br><br>• Left engine used for taxi-out: select HYD 2B – ON until both engines are running.<br>• Right engine used for taxi-out: select PTU – ON until both engines are running. | |
| Before taxi checklist | "BEFORE TAXI CHECKLIST" | "BEFORE TAXI CHECKLIST" |
| | ☐ APU.....As required<br>*If OFF, will start a 1-minute cooldown, then shuts down and closes the inlet door.*<br>☐ FLAP.....( ) selected<br>*Set take-off flaps as entered in the FMS (FMS PERF DEP CONFIG if mismatch).*<br>☐ ♦♦ Wing anti-ice test.....Complete<br>a. At or below 8000 ft, advance $N_1$ to 30% (dual bleed) or 50% (single bleed).<br>b. Above 8000 ft, advance $N_1$ to 35% (dual bleed) or 60% (single bleed).<br>c. ANTI-ICE, WING.....ON<br>d. AVIONIC synoptic page.....Select<br>e. AVIO, TEST – WING A/ICE.....Select<br>f. Confirm PASS is shown.<br>g. Thrust levers.....As required<br>☐ ANTI-ICE, COWL.....As required<br>*ON if OAT is 10°C or below and visible moisture in any form is present.*<br>☐ ANTI-ICE, WING.....As required<br>*ON if OAT is 5°C or below and visible moisture in any form is present.*<br>☐ Flight controls.....Checked<br>a. FLT CTRL synoptic page.....Monitor<br>b. Sidestick.....Checked<br>*Pilot Flying checks flight controls full and correct range of movement.*<br>c. Rudder.....Checked<br>*Pilot Flying ensures full deflection to both sides (wait for flaps in position).*<br>☑ NOSE STEER.....Select on<br>☐ EICAS and INFO.....Checked | |
| | **A220-FOT-21-63-001:** TRIM AIR – AUTO | "BEFORE TAXI CHKL COMPLETE" |
| All clear signal received | • AMM and / or taxi charts displayed.<br>• Confirm taxi instructions.<br><br>"CLEAR LEFT SIDE"<br><br>• PARK BRAKE.....OFF<br>• TAXI LTS.....NARROW or WIDE | • AMM and / or taxi charts displayed.<br>• Obtain taxi clearance.<br><br>"CLEAR RIGHT SIDE" |

**A220-FOT-21-63-001**
To prevent faults related to the trim air system (root cause under investigation), it is recommended to select TRIM AIR to OFF before engine start, then back to AUTO when both engines have stabilized after start.



**Wing anti-ice test**

♦♦ = once each day.

If wing anti-ice is activated either manually or automatically for more than 3 consecutive minutes without failure during flight, then it meets the objective of the wing anti-ice check, the wing anti-ice test is not required on ground for that respective day.

It is advisable to have the APU running before any engine start or shutdown event (e.g. single engine taxi-in or taxi-out) so that at least one bleed air source is available to motor the engine should a tailpipe fire occur. The TAILPIPE FIRE procedure in the ECL calls for ENG 1 and ENG 2 RUN switches to the OFF position. If APU is not operating, no bleed air is available to motor the affected engine, which inhibits extinguishing the tailpipe fire.

# Taxi and Before Take-off

| Event | Crewmember 1 | Crewmember 2 |
|---|---|---|
| During taxi | Crossing or entering taxiways / runways:<br>"CLEAR LEFT SIDE"<br>• STROBES.....ON (active runways)<br><br>Controlling taxi speed:<br>Rather than riding the brakes, let the taxi speed increase to 30 kts groundspeed before braking to 15 kts, then repeat.<br><br>When holding position:<br>• TAXI LTS.....OFF<br>• PARK BRAKE.....As required | "CLEAR RIGHT SIDE"<br><br>*Parking brake must be used if holding time is expected to exceed 1 minute as holding the brakes may overheat the EMAs.* |
| Cabin<br>READY | "BEFORE TAKEOFF CHECKLIST" | "BEFORE TAKEOFF CHECKLIST" |
| | ☐ APU BLEED.....OFF, if OAT is at or below 15°C<br>*If APU is shutdown APU bleed can be left in AUTO.*<br>☐ FMS.....Set<br>☐ Trims.....Set<br>☐ FLAP.....Set<br>☐ Anti-ice.....As required<br>*CAIS if OAT is 10°C or below, WAIS if OAT is 5°C or below with visible moisture.*<br>☐ AUTOBRAKE.....As required<br>☐ EICAS and INFO.....Checked<br>☐ Cabin.....Ready<br>*On EICAS, check CABIN tile is showing READY*<br>☐ Runway.....( ) confirmed | "BEFORE TAKEOFF CHKL COMPLETE" |
| Line-up | "CLEAR LEFT SIDE"<br><br>"LIGHTS & BELTS" | "CLEAR RIGHT SIDE"<br>• STROBES.....ON<br>• TAXI LTS.....WIDE<br>• LDG LTS, L & R.....ON<br>• SEATBELTS.....Cycle OFF, ON<br>"LIGHTS & BELTS ON" |
| | • MAP.....As required<br>*Confirm FMS departure runway (cyan), then zoom out of the AMM.*<br>• DU 5.....Set<br>*Pilot Flying side: FMS, PERF – DEP, Pilot Monitoring side: FMS, ROUTE – LEGS*<br>• WXR.....As required<br>*ON if weather anticipated on the departure route.* | |
| Take-off clearance received | • Confirm take-off clearance.<br>• NOSE LDG LT.....ON | • Readback take-off clearance. |

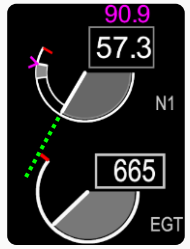
- Airplane geometry for take-off:
  a. Bank angles of more than 15 degrees can cause an engine nacelle strike.
  b. Pitch angles of more than 11 degrees can cause an aft fuselage strike.
- Take-off techniques:
  a. Rolling take-off: take-off thrust is set without holding brakes (standard).
  b. Standing take-off: holding brakes until thrust levers are advanced to take-off setting.
  c. High wind take-off: 55% $N_1$ until 20 kts groundspeed before setting take-off thrust.
- Autothrottle use:
  a. When thrust levers are advanced towards the take-off thrust setting and the autothrottle is selected, the autothrottle engages (THRUST) and sets target take-off $N_1$ as selected in the FMS and limited by the Thrust Limitation at Low speed (TLLS).
  b. For higher-than-normal headwinds (more than 40 kts), the autothrottle may not achieve the take-off $N_1$ value. Consider use of manual thrust for take-off (see point c).
  c. If autothrottle is inoperative or not armed, derated or reduced thrust is not allowed.
  d. During take-off, autothrottle must not be selected or reselected after thrust levers are advanced for take-off until the aircraft is at or above 400 ft AGL (RED OEB).
  e. Failure to advance thrust levers to take-off thrust by 60 kts groundspeed may cause the extension of the spoilers.
- Autothrottle disconnect during take-off:
  a. Max thrust (TO) with or without FLEX:
     - Below 60 kts: abort take-off.
     - Above 60 kts: thrust levers must be set to MAX to continue the take-off.
  b. Derated thrust (TO-1, TO-2 or TO-3) with or without FLEX:
     - Below 60 kts: abort take-off.
     - Above 60 kts: if $N_1$ on both engines matches $N_1$ target, continue take-off. If $N_1$ of either or both engines does not match the $N_1$ target, abort the take-off, thrust levers should not be advanced further except in an emergency.
- Certain combinations of take-off derate and climb derate will result in higher climb thrust than take-off thrust causing a **thrust increase after take-off**. From the tables below, combinations with an X will result in a thrust reduction, other combinations will cause a thrust increase. In the latter case, there is no noise benefit selecting a THR RED altitude below the ACCEL altitude.

| PW1521G | CLB | CLB-1 | CLB-2 |
|---|---|---|---|
| TO | X | X | X |
| TO-1 | * | * | X |
| TO-2 | * | * | * |
| * THR RED = ACCEL ALT (e.g. use 3000/3000) | | | |

| PW1524G | CLB | CLB-1 | CLB-2 |
|---|---|---|---|
| TO | X | X | X |
| TO-1 | X | X | X |
| TO-2 | * | * | X |
| TO-3 | * | * | * |

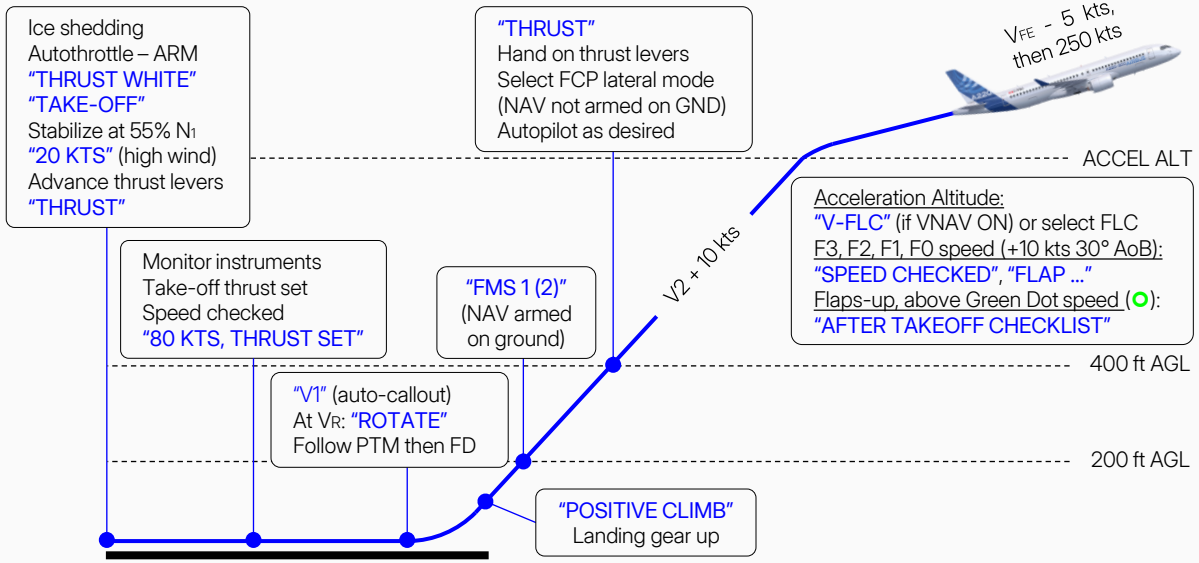
# Take-off Procedure

| Event | Crewmember 1 | Crewmember 2 |
|---|---|---|
| When aligned and CM2 = PF. | "YOU HAVE CONTROLS" | "I HAVE CONTROLS" |

| Event | Pilot Flying | Pilot Monitoring |
|---|---|---|
| Take-off Initiation<br>  <br><br>Placing the thrust command arc in line with the EGT red line gives approximately 55% $N_1$. | • HEADING.....VERIFY<br>*Heading agrees with assigned runway.*<br>• Ice shedding*.....As required<br>• BRAKES.....As required<br>*Depending on take-off technique used.*<br>• Autothrottle.....ARM<br>"THRUST WHITE"<br>"TAKE-OFF"<br>• Thrust levers.....Advance<br>a. Advance to 55% $N_1$ on both engines<br>b. Check $N_1$ stabilized for both engines<br>*If high wind take-off, wait for 20 kts GS*<br>c. Increase thrust until AT engages:<br>"THRUST" | • HEADING.....VERIFY<br>*Heading agrees with assigned runway.*<br>"CHECKED"<br>High wind take-off, at 20 kts groundspeed:<br>"20 KTS"<br>"CHECKED" |
| | At "THRUST" call CM1 takes over the thrust levers. | |
| At 80 kts | • Airspeed.....Check<br>*Both PFD airspeed indications agree.*<br>"CHECKED" | • Flight and engine instruments.....Monitor<br>• Take-off $N_1$.....Check<br>*Check $N_1$ reached $N_1$ target (TRV).*<br>• Airspeed.....Check<br>*Both PFD airspeed indications agree.*<br>"80 KTS, THRUST SET" |
| At $V_1$ | | "V1" (auto-callout) |
| | CM1 removes hand from the thrust levers. | |
| At $V_R$ | Rotate with 3 to 5 degrees / sec towards PTM and transition to FD when available. | "ROTATE" |
| Positive rate of climb indicated | "GEAR UP" | "POSITIVE CLIMB"<br>• Landing gear.....UP<br>"GEAR UP" |
| 200 ft AGL | "FMS 1(2)" (NAV armed on ground) | "CHECKED" |
| 400 ft AGL | • Autothrottle HOLD → THRUST mode:<br>Call: "THRUST", Hand on thrust levers.<br>• NAV or HDG (NAV not armed on GND)<br>• Autopilot.....As desired | "CHECKED" |

**Note:** in case of ice shedding before take-off (OAT 3°C or below with visible moisture), hold the brakes (do not use parking brake), increase thrust to 60% $N_1$ momentarily, then move thrust levers to idle to shed any ice, arm autothrottle and continue as for a normal or high wind take-off.

| Event | Pilot Flying | Pilot Monitoring |
|---|---|---|
| Acceleration Altitude | Confirm FLC (VNAV active) or select FLC:<br>"(V-)FLC"<br>F3, F2, F1, F0 speed (+ 10 kts if 30° bank):<br>"FLAP ..." | "CHECKED"<br>"SPEED CHECKED", "FLAP ..." |
| Flaps up and above Green Dot speed | "AFTER TAKEOFF CHECKLIST"<br>With AP engaged, keep hands on sidestick and thrust levers until at least 2500 ft AGL. | "AFTER TAKEOFF CHECKLIST"<br>☑ Landing gear.....UP<br>☑ FLAP.....0<br>☑ ANTI-ICE, L COWL..... AUTO<br>☑ ANTI-ICE, WING..... AUTO<br>☑ ANTI-ICE, R COWL..... AUTO<br>☑ APU BLEED.....Select auto<br>☐ EICAS.....Checked<br>"AFTER TAKEOFF CHKL COMPLETE" |



## ( Y ) Pitch Axis

Whenever the autopilot is disengaged, a "speed trim" bug ( ◁ ) appears on the speed tape to indicate what speed the aircraft is currently trimmed for. Trim speed is adjusted with the pitch trim rocker switches on the sidestick. Contrary to a "real" Airbus, the A220 has inherent speed stability, pitch control laws are not load-factor based.

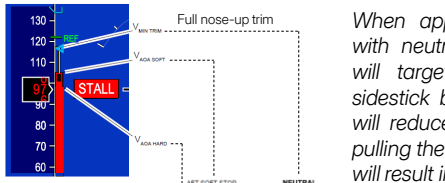
With neutral sidestick, the A220 will return to trim speed, much like a conventional aircraft. The pilot can see the exact speed the aircraft is trimmed for on the speed tape, eliminating the need for endless finetuning. The maximum selectable trim speed is $V_{MAX}$ ($V_{MO}$ / $M_{MO}$ or $V_{FE}$ / $V_{LE}$ when not in clean configuration). The minimum trim speed is $V_{MIN\ TRIM}$ indicated by an amber bar on the speed tape.

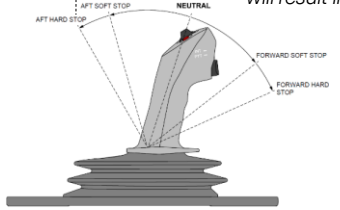
In NORMAL law, the pilot provides elevator and speed trim inputs to the FBW system using the sidestick controller and trim switches. The flight control system moves the elevators accordingly and then proceeds to offload the elevators by adjusting the horizontal stabilizer. In-flight, the pilot has no direct control over the horizontal stabilizer.

When both thrust levers are advanced for take-off, the elevators automatically move 12° down at neutral sidestick until reaching 80 kts. This is done to improve nosewheel steering effectiveness, nose down sidestick input during initial take-off roll is therefore unnecessary.

The A220 features limits for nose-up / down pitch angle, load-factor, high AOA, low / high speed, tail strike and tailplane AOA protection.

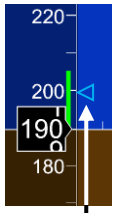
**Note:** the A220 fly-by-wire flight control laws aim to make the aircraft behave conventionally, the aircraft does not promptly fly or return to the trim speed. Instead, it will enter a phugoid around the trim speed without manual input much like a conventional aircraft. Similarly, when holding the sidestick at the aft soft stop, the nose will raise, speed will drop, initially undershooting $V_{AOA\ SOFT}$, the nose will then lower automatically, speed will pick up, overshooting $V_{AOA\ SOFT}$, the nose will raise again and the cycle repeats, much like a "falling leaf".



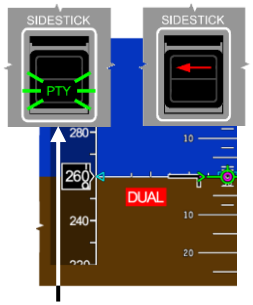

*When applying full nose-up trim with neutral sidestick, the aircraft will target $V_{MIN\ TRIM}$, pulling the sidestick back to the aft soft stop will reduce the target to $V_{AOA\ SOFT}$, pulling the sidestick to the hard stop will result in $V_{AOA\ HARD}$ (max AOA).*

*Min recommended speed in flight is Green Dot or next F-speed if not in clean config (e.g. for FLAP 2, min speed is F3).*




***Tip:** during speed changes while flying manually, keep the speed trim bug between ½ and ⅔ the length of the green speed trend vector to stay in-trim.*

Speed trim

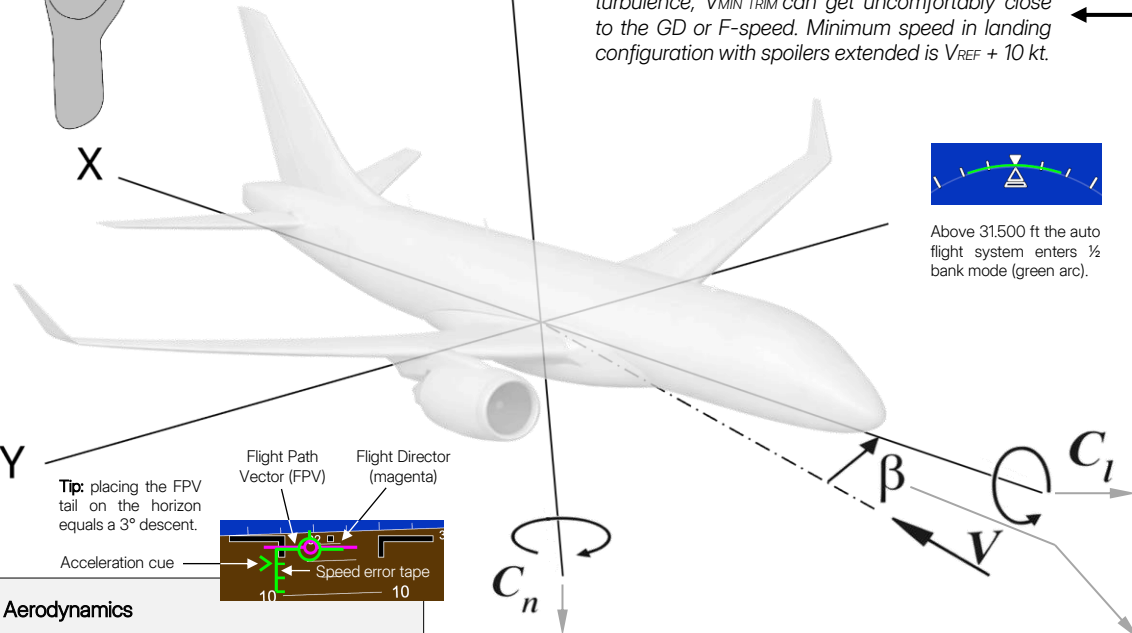



Sidestick Priority

*Manipulating both sidesticks simultaneously will result in a "DUAL INPUT" warning, the two control inputs will be algebraically summed up.*

*Priority can be taken by pressing and holding the AP / PTY pushbutton on the sidestick (momentary priority) or by pressing the SIDESTICK pushbutton on the glareshield (latched priority).*

*This will cause a "PRIORTY LEFT / RIGHT" alert and inhibition of the opposite sidestick controller.*




*When spoilers are extended during turns and/or turbulence, $V_{MIN\ TRIM}$ can get uncomfortably close to the GD or F-speed. Minimum speed in landing configuration with spoilers extended is $V_{REF}$ + 10 kt.*



Above 31.500 ft the auto flight system enters ½ bank mode (green arc).

**Tip:** placing the FPV tail on the horizon equals a 3° descent.




## Aerodynamics

The A220 features a low drag design, making energy management challenging. Speed reduction while maintaining descent path or speed reduction and configuration during approach can pose a real challenge, especially if glide path is steep.

## Autothrottle

The autothrottle system is usually not very responsive, it is advisable to start a slow gradual descent to allow thrust reduction to idle. On approach many pilots prefer using manual thrust, especially when it is gusty.

## ( Z ) Yaw Axis

Rudder is <u>only</u> used to maintain centerline, de-crab during crosswind landing and engine failure or other failures causing yaw.

In case of engine failure or other failure resulting in yaw the pilot needs to take manual control of the rudder using rudder pedals and rudder trim (also when AP is engaged). Partial compensation is provided, limiting the sideslip angle to 5° to assist the pilot.

In case of go-around during autoland after touchdown has already occurred, autopilot will remain engaged and go-around will be automatic, but the pilot needs to take manual control of rudder.

## Introduction

Other than having sidestick controllers, the A220 does not behave fundamentally different from any conventional aircraft. The A220 is not part of the Airbus "tech tree", the A220 fly-by-wire control laws aim to make the aircraft behave conventionally but does feature full flight envelope protection. Consequently, the A220 fly-by-wire offers less augmentation, thus requiring frequent control inputs to maintain trajectory. Unless approaching the boundaries of the flight envelope, fly-by-wire augmentations are subtle.

## Secondary Flight Controls

Multifunction spoilers retract automatically at high AOA or high thrust lever angle. Ground spoilers extend and retract automatically during landing or rejected take-off (no arm position). Effect of thrust, slat / flap extension and spoiler extension is automatically compensated for by the FBW.

## ( X ) Roll Axis

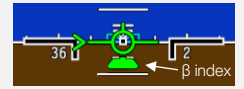
- When the sidestick controller is released to its center position with a bank angle of 30° or less the current bank angle is maintained.
- At higher bank angles, continuous sidestick input is required to maintain the bank, releasing the sidestick will cause the aircraft to roll-out to 30° angle of bank.
- The maximum bank angle is limited to 80° and limited further based on speed and pitch angle.
- Automatic nose-up compensation is provided for bank angles up to 33° to compensate for loss of lift.

**Note:** in case of a disturbance, the aircraft does not restore bank angle promptly. As part of the A220 design philosophy, there is a noticeable lag, manual control input may be necessary to maintain the required trajectory.

## ( β ) Engine failure or other failure causing sideslip

The aircraft features partial compensation in case of engine failure or other failures causing yaw. The aim is to assist the pilot in controlling the aircraft without "masking" the failure.

- Sideslip angle is limited to 5°
- Bank angle is limited to 20°



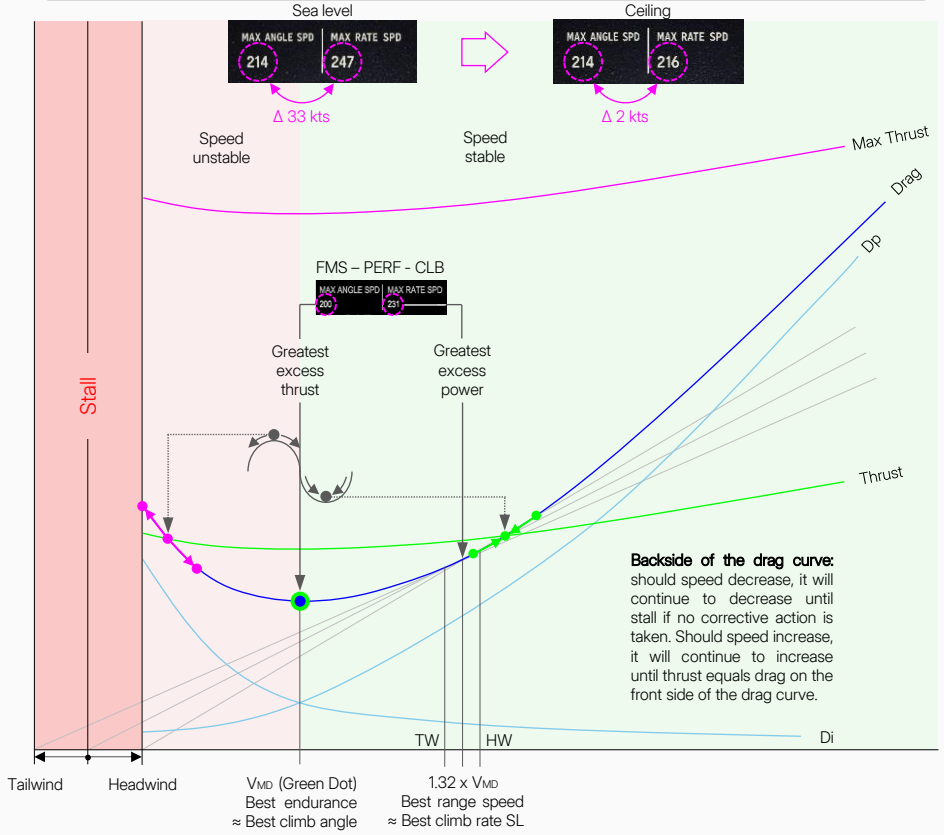

When flaps are extended, the BETA indicator will show as a trapezoid below the FPV. When flaps are retracted, the BETA indicator remains displayed as an open trapezoid.

## Performance

The total drag is comprised of induced drag (Di) and parasitic drag (Dp), the combined drag is high at low speed, reduces, then increases again at higher speed. There are two distinct points:

a. Minimum drag speed ($V_{MD}$): this is the green dot speed, it is the speed for best endurance, it is close to the point of greatest excess thrust, thus the speed for best climb gradient ($V_X$).

b. Best speed - drag ratio (1.32 x $V_{MD}$): this speed achieves maximum range, it can be found by drawing a line from the origin onto the drag curve and is depending on wind component (higher for headwind, lower for tailwind). At sea level, this is close to the point of greatest excess power and therefore the speed for best climb rate ($V_Y$). With increasing altitude (lower thrust available), $V_Y$ decreases until $V_X$ and $V_Y$ will coincide at the ceiling.

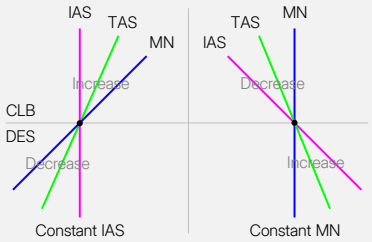



## Speed Stability (back side of the drag curve)

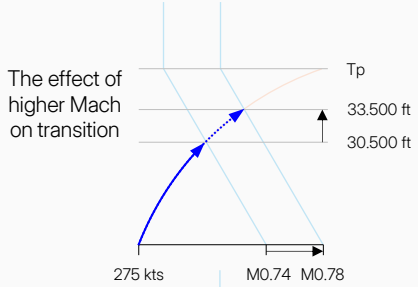
Jet aircraft have a considerably shallower drag curve compared to propeller aircraft, therefore there is less speed stability, also engine spool-up time is slower. Above the green dot, speed is stable, following a disturbance, speed automatically recovers. Below the green dot (or flap maneuvering speed) speed is unstable, when speed changes due to a disturbance, it will continue to diverge if no action is taken. This is especially dangerous at high altitude as there might not be sufficient thrust available to recover without descending. A220 fly-by-wire features inherent speed stability in the normal mode, in direct-mode there is no stability augmentation.

## IAS, TAS and MACH Number

During climb at constant IAS, both TAS and MACH will increase, when keeping MACH constant, both IAS and TAS will decrease (vice versa for descent). The speed of sound decreases with altitude as ambient temperature decreases. At the tropopause ambient temperature and therefore speed of sound will remain constant with further increase in altitude.




As can be seen below, a climb profile with a higher Mach number will result in a higher change-over altitude while a profile with a higher IAS will result in a lower change-over altitude. The highest TAS (therefore highest groundspeed assuming no wind) is reached at the change-over altitude, best fuel economy however generally occurs at higher level.

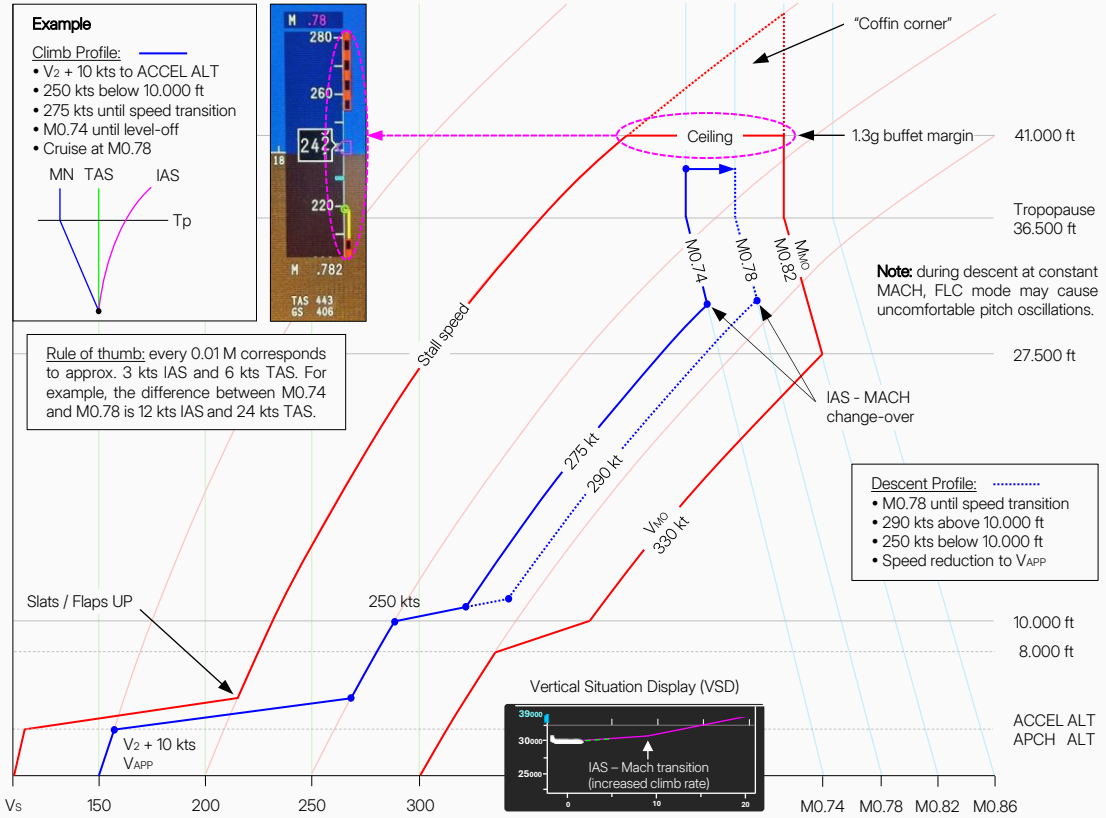



## "Coffin Corner"

As altitude increases, the margin between low-speed stall and critical Mach number decreases. When climbing above MAX ALT, the margin between low-speed and high-speed buffet onset is insufficient for maneuvering or handle atmospheric disturbances. When continuing to climb even further, the stall speed and the critical Mach number will eventually coincide, theoretically the aircraft cannot climb further, slow-down, nor speed-up without stalling. The area where this occurs is known as the "coffin corner".

## Maneuver ("Buffet") Margin

At the aircraft operational ceiling (MAX ALT) a buffet margin of 1.3g is available for maneuvering and provide margin for atmospheric disturbances. This allows for a 40-degree bank angle (30° bank with a 10° overshoot). Remember that the auto-flight system enters ½ bank mode above 31.500 ft (15° bank = 1.05g).



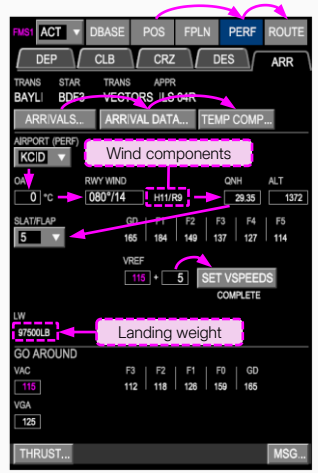

## IAS – MACH Transition

When climbing at constant IAS, Mach number increases. When continuing a constant IAS climb, maximum Mach number ($M_{MO}$) will eventually be exceeded. Similarly, when descending at constant Mach number, the maximum operating speed ($V_{MO}$) will be reached. It is therefore necessary to change from constant IAS to constant Mach during climb and vice versa during descent. Upon change-over from IAS to Mach, the nose will start pitching-up as Mach is held constant at the expense of IAS. Climb rate increases slightly as speed is traded for altitude (vice versa for descent). The change-over from IAS to MACH is automatic when using FMS speed (using speed profile as programmed in the FMS). When using MAN speed, change-over between IAS and MACH occurs at 27.500 ft.

# Climb & Cruise

**Note:** green dot speed (O) provides maximum climb gradient, minimum drag therefore maximum endurance (e.g. holding) and is the minimum recommended speed in clean configuration. Maximum bank angle at green dot is 30° below 31.500 ft or 15° at / above 31.500 ft (25° for OEI).

| Event | Pilot Flying | Pilot Monitoring |
|---|---|---|
| Transition Altitude | Above MSA and cleared to a FL, not later than passing transition altitude:<br>"SET STANDARD"<br>• BARO.....STD | • BARO.....STD<br>"STANDARD SET" |
| Passing 10.000 ft | "LIGHTS & BELTS" | "10.000 FT"<br>• LOGO, WING, TAXI & LDG lights.....OFF<br>• SEATBELTS.....AUTO<br>"LIGHTS & BELTS OFF" |
| Wing anti-ice test (if required) | | Wing anti-ice test can be performed In flight by manual or automatic activation of the wing anti-ice system for 3 consecutive minutes instead of test on ground.<br><br>In case of manual activation, TAT 15°C or below and constant thrust (CLB or IDLE):<br><br>• Dual pack: climb or descent < 31.000 ft<br>• Single pack: during climb < 19.000 ft |
| Before 35.000 ft | "HIGH ALTITUDE CLIMB CHECKLIST"<br><br>**Note:** If cruising level is below 35.000 ft, override the high altitude climb checklist by selecting FCTN tile, then OVERRIDE CHKL.<br><br>**CAUTION:** If still in icing conditions upon reaching 35.000 ft, climb must be stopped. Failure to select WING ANTI-ICE OFF could result in engine nacelle overheating, triggering the engine fire warning. | "HIGH ALTITUDE CLIMB CHECKLIST"<br>☑ ANTI-ICE, WING.....OFF<br>Avoid icing conditions.<br>"HIGH ALTITUDE CLIMB CL COMPLETE" |
| 2000 ft prior level-off | • VS.....Select<br>• Rate.....Maximum 2000 ft/min | |
| 1000 ft prior level-off | • Rate.....Maximum 1000 ft/min<br>*Sterile cockpit until level-off complete.* | "1000 TO GO"<br>*Sterile cockpit until level-off complete.* |
| Level-off | Altitude capture mode:<br>"SPD", "(V-)ALT-S CAP"<br>Level-off complete:<br>"(V-)ALT-S" | "CHECKED"<br><br>"CHECKED" |

| Event | Pilot Flying | Pilot Monitoring |
|---|---|---|
| Approach Preparation | • ISI.....Set QNH<br>• Landing performance.....Compute<br><br>1. POS - FMS: check RAIM availability<br>2. PERF - DES: set transition level<br>3. PERF - ARR:<br><br>a. ARRIVALS: select arrival, runway and approach type, arrival transitions<br>b. ARRIVAL DATA: select RNP or LPV<br>c. TEMP COMP: select if OAT 0°C or below, correct the approach minima<br>d. AIRPORT: select arrival airport<br>e. OAT, RWY WIND, QNH: enter values<br>f. SLAT / FLAP: set landing flap setting<br>g. VREF: add 5 kts if autothrottle is used and ½ the gust (together max 10 kts). *For example: if wind is 36010G20KT, gust is 10 kts (from 10 kts steady to 20 kts), gust factor is ½ the gust = 5 kts.*<br>h. SET VSPEEDS: select<br><br>**Note:** within 50 nm from the origin, the departure airport is selected for return. Select arrival airport from the dropdown and enter ICAO code in ARRIVALS box.<br><br>**Note:** do not manually set ILS or localizer frequency active as this will interfere with the automatic NAV-to-NAV transfer on approach. Set final CRS and ILS / LOC frequency standby only as back-up.<br><br>• CTP / CNS.....Set<br>*Set CRS, Frequencies and Minima (CFM)*<br>• AUTOBRAKE.....Set | • ATIS.....Receive (Voice or DLK)<br>• Landing performance.....Compute<br><br>  <br><br>*After PF completes FMS set-up, the PM will check all FMS entries by going through the FMS pages in the same order as the FMS preparation was conducted.*<br><br>*The PM crosschecks the FMS arrival route, approach and transitions on the ROUTE – LEGS page with arrival / approach charts including speed and altitude constraints.*<br><br>• CTP / CNS.....Set<br>*Set CRS, Frequencies and Minima (CFM)* |
| Arrival briefing (next page) | Step 1. Set distraction free environment<br><br>Step 2B - PLAN<br>• Aircraft technical status<br>• Guidance for approach<br>• Landing flap setting<br>• Use of reverse and autobrakes<br>• Stop-margin, planned runway exit<br>• Planned taxi route & hotspots<br>Step 3B - IDENTIFY THREATS<br><br>Step 2A - PLAN<br>• Minimum altitudes, MSA<br>• Arrival route and transitions<br>• Approach incl. arrival data<br>• Approach minima<br>• Go-around procedure<br>• Extra fuel and endurance<br>Step 3A - IDENTIFY THREATS<br><br>Step 4 - MITTIGATIONS, Step 5 - MISCELLANEOUS | |

# Arrival Briefing

Explanation / Notes:

1. For the briefing to be of good quality it is important to minimize interruptions. Allocating the right time before top of descent mitigates potential disturbances. Anticipate potential disturbances, in case of interruption of the briefing it should be resumed at the beginning of the step where the interruption has occurred.

2. The PM should start to brief the main items of the PLAN. This ensures that both pilots share the same mental image of the flight trajectory after the FMS preparation (by PF) and check (by PM). He considers at least the following:

   - Minimum altitudes
   - STAR, Transition and Approach designators
   - Rundown of FMS – PERF – ARR – ARRIVAL DATA
   - Minima and go-around procedure
   - Extra fuel and time

   The PF briefs his considerations for landing. Briefing the PLAN should normally only be a high-level description. It should not be a repetition of the detailed setting and checking of the flight trajectory in the FMS performed in the respective SOP items.

3. The PM should brief the THREATS they have identified. The PF will highlight additional threats if required.

4. The PF and PM will discuss and agree on the MITIGATION for the threats.

5. MISCELLANEOUS is intended to take into consideration additional items such as circle-to-land, RNP-AR, CAT II / III, autoland, etc.

| Step | Pilot Flying | Pilot Monitoring |
|---|---|---|
| 1 | Set distraction free environment | |
| 2A | | PLAN<br>• Minimum altitudes, MSA<br>• Arrival route and transitions<br>• Approach incl. arrival data<br>• Approach minima<br>• Go-around procedure<br>• Extra fuel and endurance |
| 2B | PLAN<br>• Aircraft technical status<br>• Guidance for approach (see matrix)<br>• Landing flap setting<br>• Use of reverse and autobrakes<br>• Stop-margin & planned runway exit<br>• Planned taxi route & hotspots | |
| 3A | | IDENTIFY THREATS |
| 3B | IDENTIFY THREATS | |
| 4 | MITIGATIONS | |
| 5 | MISCELLANEOUS | |

# Descent Planning

| Event | Pilot Flying | Pilot Monitoring |
|---|---|---|
| Approach briefing completed | "DESCENT & APPROACH CHECKLIST"<br><br>☑ ANTI-ICE, WING (below 35.000 feet).....AUTO<br>☐ FMS.....Set<br>☐ Minima.....Set<br>*MDA and / or DH set.*<br>☐ AUTOBRAKE.....As required<br>☐ EICAS.....Checked<br>*Both pilots check messages.*<br>☐ Approach briefing.....Complete | "DESCENT & APPROACH CHECKLIST"<br><br>"DESCENT & APPROACH CL COMPLETE"<br>or "WING ANTI-ICE TO GO" |
| Descent planning | Determine descent angle, set VPA on FMS – PERF – DES page. | |
| Top of Descent | • VNAV.....As required<br>• ALT SEL.....Set cleared level<br>• Vertical mode.....As required<br>FMA (e.g.): "(V-)FLC", "(V-)ALT-S WHITE" | Obtain descent clearance.<br><br>"CHECKED" |
| Below 35.000 ft | Complete DESCENT & APPROACH CHECKLIST (below 35.000 ft)<br>☑ ANTI-ICE, WING (below 35.000 feet).....AUTO | |
| Passing 10.000 ft | "LIGHTS & BELTS" | "10.000 FT"<br>• LOGO & WING.....As required<br>• TAXI LTS.....WIDE<br>• LDG LTS, L & R..... ON<br>• SEATBELTS.....ON<br>"LIGHTS & BELTS ON" |
| Transition Level | Cleared to an altitude, not later than passing transition level:<br>"SET QNH"<br>• BARO.....Set QNH | • BARO.....Set QNH<br>"QNH ... SET" |
| 2000 ft prior level-off | • VS.....Select<br>• Rate.....Maximum 2000 ft/min | |
| 1000 ft prior level-off | • Rate.....Maximum 1000 ft/min<br>*Sterile cockpit until level-off complete.* | "1000 TO GO"<br>*Sterile cockpit until level-off complete.* |
| Level-off | Altitude capture mode:<br>"SPD", "(V-)ALT-S CAP"<br>Level-off complete:<br>"(V-)ALT-S" | "CHECKED"<br><br>"CHECKED" |

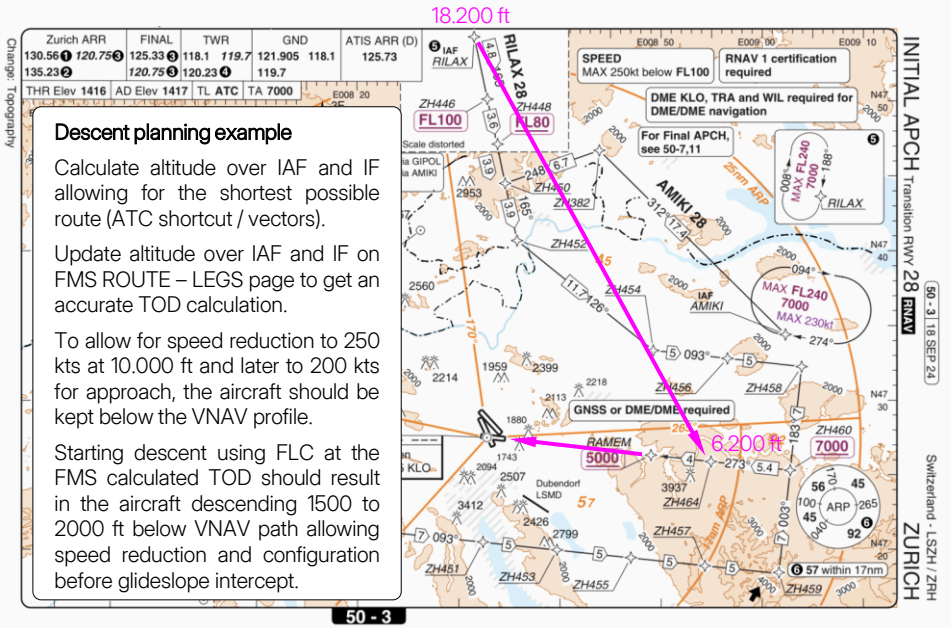
**Note:** flight crew should pay attention to barometric reference (QNH) that significantly differs from the one used for approach preparation (compare with ISI), the flight crew should consider confirming the barometric reference from all available sources.

## Descent Planning

Rules of thumb:

1. Distance and altitude (3 nm for every 1000 ft):
   a. Distance to descent = altitude (in 1000s ft) x 3
      *Example: 10.000 ft = 10(000) x 3 = 30 nm*
   b. Altitude at distance = distance / 3 (in 1000s ft)
      *Example: 30 nm = 30 / 3 = 10(000) ft*
   c. Alternatively: altitude = distance x 300
      *Example: at 5 nm = 5 x 300 = 1500 ft*
2. Speed reduction:
   a. Level flight, speed reduction = 10 kts every 1.0 nm
   b. Using FLC, speed reduction = 10 kts every 1½ nm
3. Effect of speed reduction on a nominal 3° descent path:
   a. 300 kts to 250 kts in FLC = +1000 ft and +3 nm
   b. 250 kts to 200 kts in FLC = +1000 ft and +3 nm
   c. Total speed reduction = +2000 ft and +6 nm

Example descent planning below.




## Idle Descent (table)

The average (idle) descent path from TOD to approach level including all speed reductions can be calculated as follows:

Weight:
- Landing weight 55t = 3.0°
- Each 5t below 55t = +0.2°
- Each 5t above 55t = -0.1°

ISA deviation:
- Each 10 below ISA = +0.1°
- Each 10 above ISA = -0.1°

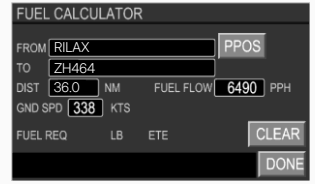
Wind:
- Each 10 kts headwind = +0.1°
- Each 10 kts tailwind = -0.1°

Anti-icing use anticipated = -0.2°

### RAMEM (FAF)
- 5000 ft

### ZH464 (IF)
- RAMEM to ZH464 = 4 nm
- 4 x 300 ft = 1200 ft
- 5000 + 1200 = 6200 ft

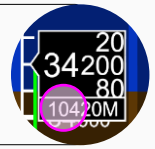
### RILAX (IAF)
- Open FUEL CALCULATOR (FMS – PERF – CRZ)
- RILAX to ZH464 = 36 nm
- 36 / 3 = 12(000) ft
- 12.000 + 6200 = 18.200 ft



# Energy Management

| Approximate VPA at GW 55t | | | | | |
|---|---|---|---|---|---|
| M0.78 | M0.74 | 290 kts | 275 kts | 250 kts | GD |
| 3.1° | 3.0° | 3.5° | 3.3° | 3.0° | 2.5° |

**Tip:** selecting ALT – M on the FCP will show altitude in meters below the altitude window. The first 2 or 3 numbers indicate the remaining distance required for descent to SL




The Airbus A220 does not (currently) have advanced tools for planning and managing the descent. The only available mode is constant Vertical Path Angle (VPA) to approach level. The (average) descent angle is calculated by the pilot and entered on FMS – PERF – DES.

The A220 has a very low drag profile, speed reduction during descent can therefore be challenging. At 250 kts the aircraft will follow roughly a 3.0-degree vertical path angle, around 210 kts the aircraft experiences minimum drag and will descent with a 2.5-degree path angle (± 1000 ft/min). Thus, the higher the speed, the steeper the descent angle.

Plan for the shortest possible route (allowing for vectors / short-cuts) and aim to be 1500 ft (headwind) to 2000 ft (tailwind) below profile (TOD 5 to 6 nm ahead) before starting speed reduction. Use of spoilers is considered standard procedure and sometimes unavoidable.

TOD = 105 nm
VNAV OFF, FLC

35000 ft

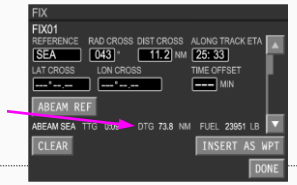
VNAV path
Actual path

M0.78 (0.74)

30.500 ft
(28.500 ft)

Speed transition

To monitor distance to go (DTG) versus altitude, direct distance to any point can be found using the FIX page.




300 kts

Reduce 250 kts

12.000 ft
11.000 ft

10.000 ft

2000 ft below path
TOD 6 nm ahead

250 kts

Reduce 200 kts

200 kts

GS intercept F2
Reducing 180 kts

180 kts

Reduce 160 kts (latest 6 nm)
Reduce VAPP (latest 4 nm)

3000 ft

2000 ft

1000 ft below path
TOD 3 nm ahead

3.0°

160 kts, VAPP

25 nm - 15 nm | GS | 6 nm | THR

250 → 200 kts F1 | 200 → 180 kts F2 | 180 → 160 → VAPP GD, F3, F4(5)

$$3\ x\ \frac{35.000}{1000} = 3\ x\ 35 = 105\ nm$$

**Key points:**

- Plan for the shortest route.
- When above path, maintain high speed. Use flight spoilers when needed and start using them early.
- Aim for 1500 - 2000 ft below VNAV profile before speed reduction.
- Intercept the glidepath decelerating to 180 kts in F2 configuration.
- **For an approach steeper than 3.0 degrees, intercept glidepath in F3 configuration, reducing to 180 kts.**
- Start configuring for landing as a continuous flow, no later than 6 nm.
- Start reducing from 160 kts to VAPP no later than 4 nm.

Selecting FLC at TOD, the actual descent path will roughly follow the VNAV profile until MACH – IAS transition. Then starts to diverge until 1500 to 2000 ft below path by the time speed reduction is required.

**Note:** while speed is in MACH, FLC mode might cause uncomfortable pitch fluctuations, VS or FPA can be used initially to control descent speed while selecting a lower speed target for the autothrottle to keep thrust at idle.

**1. Descent Schedule**

At TOD:

- VNAV OFF, select FLC

Between 12.000 ft and 11.000 ft:

- Start reducing speed to 250 kts

Between 25 nm and 15 nm (latest):

- Start speed reduction to 200 kts
- Select Flap 1

Before glideslope intercept:

- Select Flap 2
- Start speed reduction to 180 kts

Latest at 6 nm / 2000 ft AGL:

- Select gear down, Flap 3, Flap 4 (5)
- Speed 160 kts to 4 nm (latest), VAPP

**2. Vertical Path Angle (VPA)** (table)

Weight:

- Landing weight 55t = 3.0°
- Each 5t below 55t = +0.2°
- Each 5t above 55t = -0.1°

ISA deviation:

- Each 10°C below ISA = +0.1°
- Each 10°C above ISA = -0.1°

Wind correction:

- Each 10 kts of HW = +0.1°
- Each 10 kts of TW = -0.1°

Anti-icing:

- Use anticipated = -0.2°

**Example**

- Cruise at FL350
- Aerodrome elevation 0 ft
- Gross weight = 55t
- 10 kts tailwind, ISA -10 °C
- Resulting average VPA = 3.0°

**3. Distance to Descent**

$$Distance\ nm\ (3°) = 3\ x\ \frac{altitude}{1000}$$

- Each 0.1° above 3.0° = -3.0%
- Each 0.1° below 3.0° = +4.0%

# Approach Operations

## Slat / Flap Extension

The minimum recommended speed is F-speed for the next flap setting. For example, in F1 configuration, F2 is the minimum speed. To minimize thrust lever movement, it is recommended to select the next flap position as the speed reaches the next flap $V_{FE}$ minus 10 kts. At certain weights, the posted F1 speed may be higher than the green dot speed. Maneuvering at green dot but below the posted F1 can be safely performed at F0. As soon as speed reduction below green dot is required, select F1.

If FMS F-speeds are not displayed, the suggested slat / flap extension speeds are:

- FLAP 1 = 220 kts
- FLAP 2 = 200 kts
- FLAP 3 = 180 kts
- FLAP 4 = 160 kts
- FLAP 5 = Slowing to $V_{APP}$

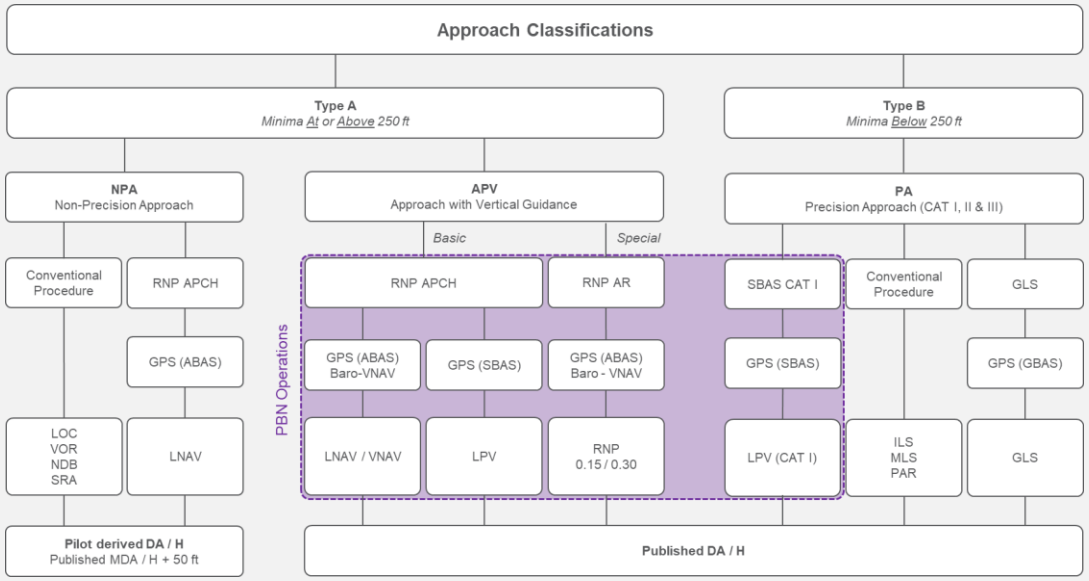
## Stabilized Approach Concept

Approaches must be stabilized latest upon reaching 1000 ft AAE with only small corrections required to maintain the correct speed, lateral and vertical trajectory. Circling approaches must be stabilized by 500 ft AAE with wings level at 300 ft AAE.

| | |
|---|---|
| Lateral | • Maximum 1 dot deviation<br>• Maximum 5º deviation for NDB approach |
| Vertical | • Maximum 1 dot deviation<br>• Maximum 100 ft deviation for 2D approach |
| Speed * | • Maximum $V_{REF}$ + 20 kts<br>• Minimum $V_{REF}$<br>**Note:** a later speed stabilization is acceptable if required by ATC (e.g. 160 kts to 4 DME), speed trend must be negative at 1000 ft AAE and stabilized latest at 500 ft AAE. |
| Rate of descent * | • Maximum 1000 ft/min (unless briefed otherwise, e.g. steep approach) |
| Configuration | • Checklists and briefings completed<br>• Gear down, landing flaps |
| **Note:** ignoring fluctuations due turbulence, momentary speed < $V_{REF}$ acceptable if positive speed trend. | |

## Approach Types

There are many approach categories and types:




Procedurally approaches can be divided into 2 types:

- 2D approach operations – only lateral guidance
- 3D approach operations – both lateral and vertical guidance

Most approaches can be flown as a 3D approach, except for LOC and VOR or NDB approaches if there is no GNSS overlay. If no GNSS overlay, there might be an FMS overlay (REF ONLY APPR). For a "REF ONLY APPR", advisory VNAV may be available. The absence of a path from FAF to the runway on the VSD, GP ANGLE blank in ARRIVAL DATA window or V-MDA coded on LEGS page indicates no advisory VNAV.

All approaches must be flown as Constant Descent Final Approach (CDFA) (without level-offs). For non-precision approaches, add 50 ft to the published minima to convert the Minimum Descent Altitude (MDA) into a (pilot derived) Decision Altitude (DA).

# Approach Guidance

**A220-FOT-22-30-012 –** During initial LOC interception above 2000 ft AGL, the flight guidance relies exclusively on LOC and GS deviation signals, without accounting for wind, drift, or groundspeed. As a result, the aircraft may overshoot the localizer, which is particularly undesirable during parallel approaches. In order to minimize the overshoot, it is recommended to sequence the flight plan correctly (i.e. extend the final approach track when being vectored) and then use FMS guidance for the initial interception. Once cleared for the approach and established on an intercept heading, select NAV (FMS 1(2)). Delay arming approach mode until the lateral deviation is within 1 dot, then select APPR (APPR LOC 1(2)). Even when following these guidelines, some localizer overshoot may occur in crosswind conditions. A single overshoot of approximately ½ dot may be expected with a 20 kt crosswind, increasing to 1 dot with a 50 kt crosswind. Delaying APPR mode selection until the aircraft is fully established using FMS guidance may not eliminate overshoot. Below 2000 ft AGL, when the PFCCs take over approach guidance (APPR LOC), LOC and GS tracking will improve noticeably.

## General Guidelines for Approach

a. Do not manually change NAV SRC for approach, APPR mode will automatically initiate NAV-to-NAV transfer for ILS, LOC and VOR approaches, manually changing NAV SRC or tuning approach frequencies could interfere with NAV-to-NAV transfer back to FMS during a missed approach.
b. Verify that correct QNH is set before commencing the approach, TEMP COMP must be selected when OAT at the arrival airport is 0°C or below.
c. ILS and LPV approach angle are unaffected by temperature, all other approaches use BARO-VNAV, approach angle is affected by temperature.
d. Transition from arrival to final approach segment during LPV (RNP-1 to angular and BARO to GPS altitude) may cause a jump in flight guidance.
e. As soon as APPR 1(2) or LAND 2(3) appears on the ASA (ILS at or below 1500 ft AAE), approach mode can only be canceled by selecting TOGA.
f. For ILS and LOC, intercept the final approach course in NAV (FMS 1(2)), then select APPR within 1 dot localizer deviation to prevent overshoot.
g. LOC approach: select APPR mode to initiate NAV-to-NAV transfer, when established on localizer, select NAV to cancel GS mode (latest at FAF).
h. Step down fixes published on final approach charts are only applicable for non-precision approaches (LNAV ONLY, LOC ONLY, VOR and NDB).
i. Add 50 ft to published minima for all non-precision approaches (LNAV ONLY, LOC ONLY, VOR and NDB), cross-check altitude versus distance.
j. For all 2D operations, use of VNAV beyond the FAF is not permitted. Before FAF, set MDA and deselect VNAV (FPA becomes the active mode).

## Final Approach Track Extension

When on radar vectors, approach waypoints may not sequence properly, the final approach track might not become the active leg (white instead of magenta). To sequence the flight plan properly, the final approach can be extended and made active:

1. Select Direct-To FMS quick access key
2. Select direct-to the FAF, IF or any other waypoint on the final approach track
3. Enter the final approach course under CRS
4. Confirm and EXEC the modification

**Note:** do NOT extend the final approach track from a waypoint within the FAF (e.g. the runway) as this will inhibit the NAV-to-NAV transfer.

Final Approach Track Extension

# Approach Matrix

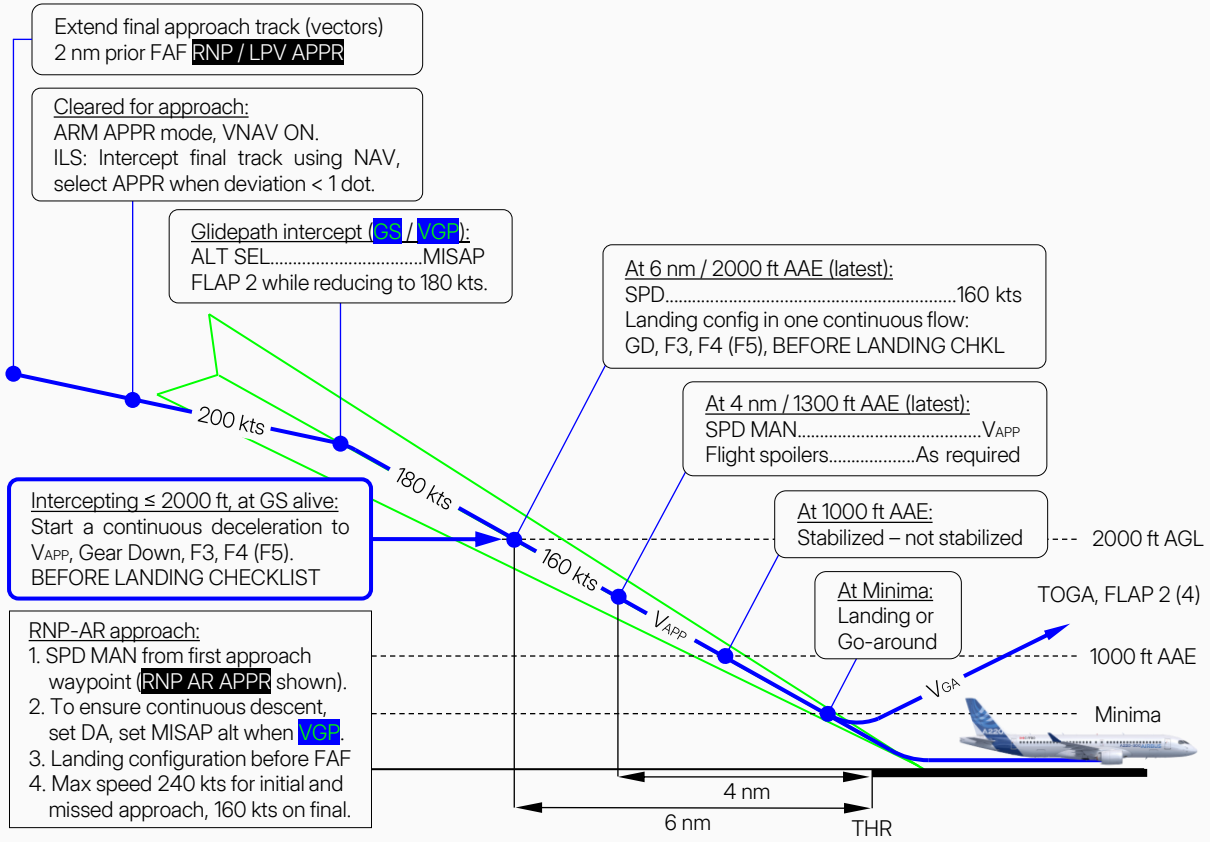
| Operation | | | Approach Category | Approach Type | FCP Mode Selection APPR | FCP Mode Selection VNAV | FCP Mode Selection ALT SEL | FMA Lateral | FMA Vertical | Deviation Scale | HSI or ASA Indications | Approach Minima |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Type B | PA | 3D | Conventional | ILS | A220-FOT-22-30-012<br>NAV, APPR (within 1 dot) | Selected | MISAP altitude when GS mode is active | APPR LOC 1(2), then APPR LOC < 2000 ft | GS | Angular | APPR 1 (2)<br>LAND 2(3) | BARO, DA (CAT I)<br>RADIO, DH (CAT II/III) |
| Type A | APV | 3D | RNP | LPV | Select APPR when cleared for approach | Must be selected to arm VGP mode | MISAP altitude when VGP mode is active<br>For RNP-AR, set DA to ensure continuous descent, set MISAP altitude when VGP is the active mode. | APPR FMS 1(2) | VGP | Angular | LPV ARM<br>LPV APPR | BARO, DA |
| | | | | RNP-AR | | | | | | ± 0.3 nm<br>± 150 ft | RNP AR ARM<br>RNP AR APPR | BARO, DA |
| | | | | LNAV / VNAV | | | | | | | RNP ARM<br>RNP APPR | BARO, DA |
| | NPA | | | LNAV only | | | | | | | | BARO, MDA + 50 ft |
| | | | GNSS | VOR (GNSS overlay) | | | | | | | - | BARO, MDA + 50 ft |
| | | | | NDB (GNSS overlay) | | | | | | | | |
| | | 2D | Conventional | LOC only | A220-FOT-22-30-012<br>NAV, APPR (within 1 dot)<br>NAV again latest at FAF | Use of VNAV beyond FAF is not permitted<br>Deselect VNAV latest at the FAF | Set the MDA before starting final descent, then set the MISAP altitude when FPA is the active mode and below MISAP altitude | APPR LOC 1(2), then LOC 1(2) NAV selected | VPATH before FAF FPA when VNAV mode is deselected (latest at FAF). | Angular<br>No vertical guidance | - | BARO, MDA + 50 ft |
| | | | | VOR (not in database) | Select APPR when cleared for approach | | | APPR VOR 1(2) | | Full scale = 10°<br>No vertical guidance | - | |
| | | | FMS | VOR (FMS overlay) | | | | APPR FMS 1(2) | | ± 1.0 nm<br>± 150 ft<br>(for reference only) | REF APPR<br>APPR FOR REF ONLY<br>(X-check BRG pointers) | |
| | | | | NDB (FMS overlay) | | | | | | | | |

# 3D Approach Operation



| Event | Pilot Flying | Pilot Monitoring |
|---|---|---|
| Before approach | 31 nm prior FAF: "RNP ARM / LPV ARM"<br>2 nm prior FAF: "RNP APPR / LPV APPR" | Extend the final approach track (vectors). |
| Cleared for approach | • APPR.....ARM<br>• VNAV.....ON<br>• ALT SEL.....DA (RNP AR)<br><br>"APPR LOC 1(2), GS WHITE" (ILS)<br>"APPR FMS 1(2), VGP WHITE" (other)<br><br>**A220-FOT-22-30-012:** for ILS approach, it is recommended to use FMS guidance by selecting NAV, delay arming APPR mode until within 1 dot lateral deviation. | **RNP AR Approach**<br>• Check RNP AR ARM 31 nm from the first approach waypoint.<br>• Check RNP AR APPR at the first approach waypoint, select MAN SPD.<br>• Set the Decision Altitude, when VGP is active, select the MISAP altitude.<br>• Landing configuration latest at FAF.<br>• Maximum speed is 240 kts for initial and missed approach, 160 kts on final. |
| Lateral path intercept | "APPR LOC 1(2)" (ILS)<br>"APPR FMS 1(2)" (other) | "LOCALIZER ALIVE"<br><br>"CHECKED" |
| Vertical path Intercept<br><br>Intercepting glidepath from above: select ALT 1000 ft above AAE, VS as required | GP intercept at FLAP 2, reducing to 180 kts<br>"GS" (ILS)<br>"VGP" (other)<br>• ALT SEL.....MISAP<br>"GO-AROUND ALTITUDE ... SET" | "GLIDESLOPE ALIVE"<br><br>"CHECKED"<br><br>"CHECKED" |
| 2500 ft AGL | Hands on sidestick and thrust levers. | "TWENTY-FIVE HUNDRED" (auto-callout) |
| 6 nm / 2000 ft AAE (latest) | • SPD.....160 kts<br>Landing config. in one continuous flow:<br>"GEAR DOWN"<br><br>"FLAP 3, 4 (5)" | "GEAR DOWN"<br>• SEATBELTS.....Cycle OFF, ON<br>"SPEED CHECKED", "FLAP 3, 4 (5)" |
| Landing configuration established | "BEFORE LANDING CHECKLIST" | "BEFORE LANDING CHECKLIST" |
| | ☐ Altimeters..... ( ) Set and cross-checked<br>☑ Landing gear.....DN<br>☐ FLAP..... ( ) indicating<br>☐ Cabin.....Ready | |
| | | "BEFORE LANDING CHKL COMPLETE" |

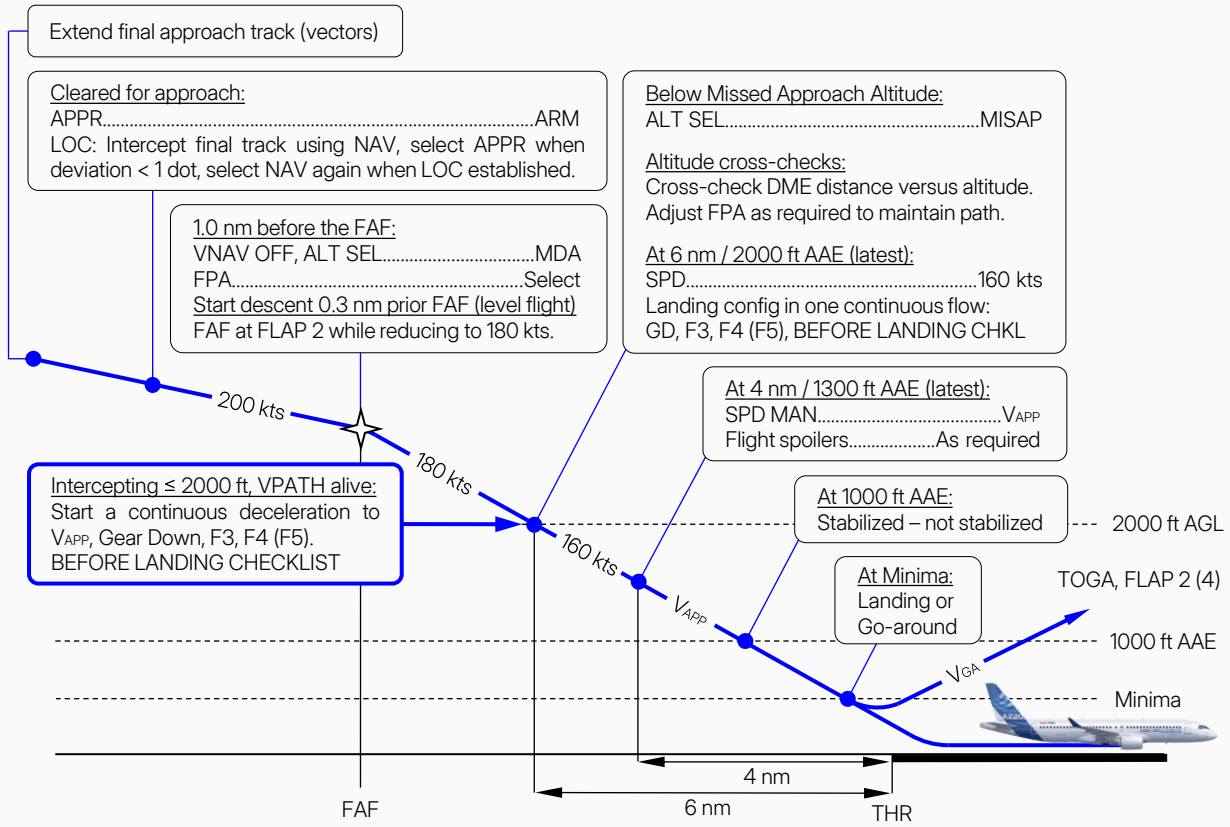
| Event | Pilot Flying | Pilot Monitoring |
|---|---|---|
| 4 nm / 1300 ft AAE (latest) | • SPD MAN.....$V_{APP}$<br>• Flight Spoilers.....As required | At any point, landing clearance received:<br>• NOSE LDG LT.....ON |
| 1000 ft AAE | "CHECKED" or<br>"GO-AROUND, FLAP 2(4)" | "STABILIZED" or<br>"NOT STABILIZED" |
| 100 ft above minima | "CHECKED" | "PLUS-100" (auto-callout) |
| At minima | "CONTINUE" or<br>"GO-AROUND, FLAP 2(4)" | "MINIMUMS" (auto-callout) |



# 2D Approach Operation

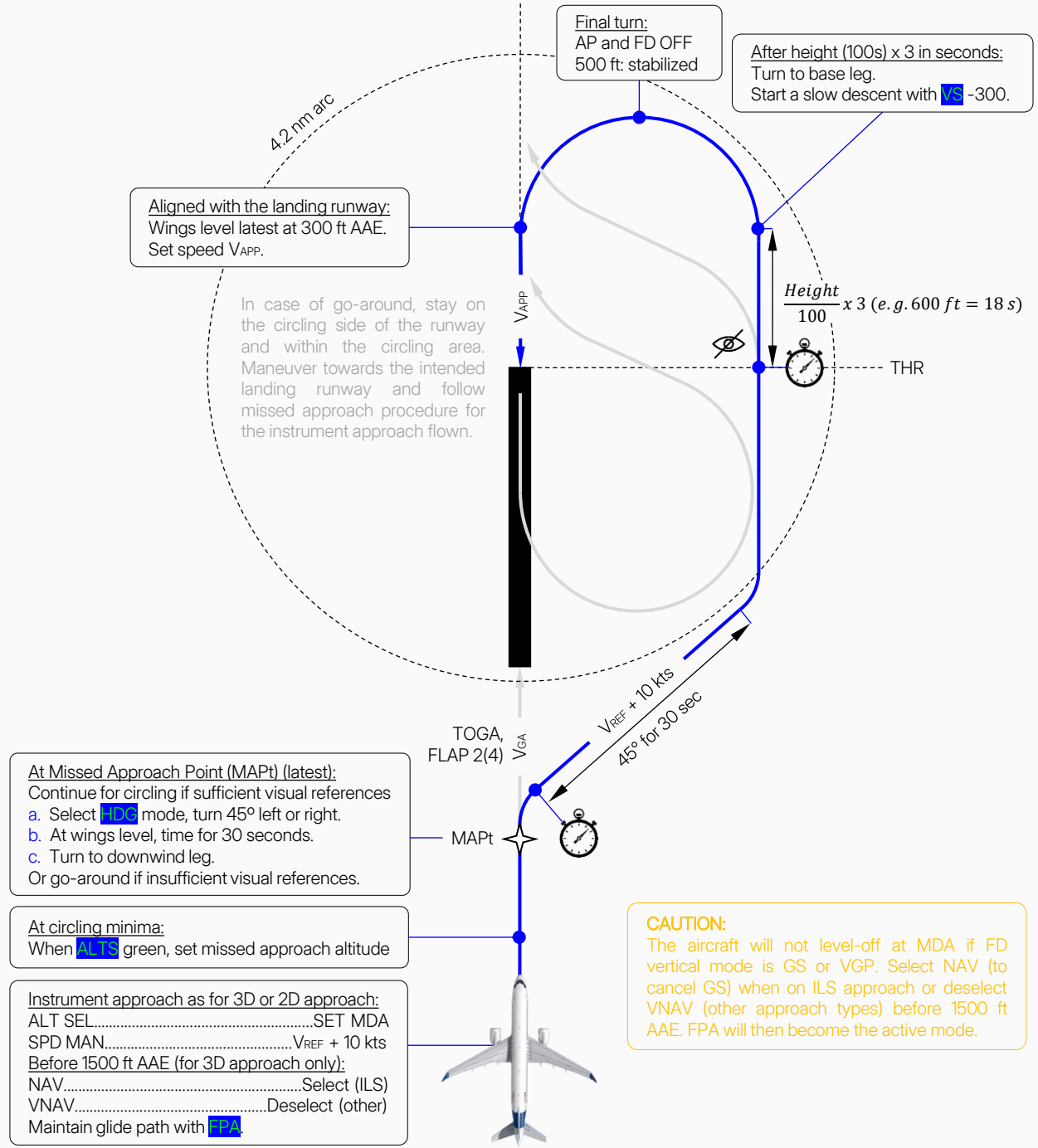
| Event | Pilot Flying | Pilot Monitoring |
|---|---|---|
| Cleared for approach | • APPR.....ARM<br>"APPR LOC 1(2) WHITE" (LOC)<br>"APPR VOR 1(2) WHITE" (VOR)<br>"APPR FMS 1(2) WHITE" (FMS)<br>**A220-FOT-22-30-012:** for LOC approach, it is recommended to use FMS guidance by selecting NAV, delay arming APPR mode until within 1 dot lateral deviation. | Extend final approach track (radar vectors).<br>"CHECKED" |
| Lateral path intercept | "APPR LOC 1(2)" (LOC)<br>"APPR VOR 1(2)" (VOR)<br>"APPR FMS 1(2)" (FMS)<br>• NAV.....Select (LOC)<br>*When established to cancel GS mode.* | "LOCALIZER ALIVE"<br>"CHECKED" |
| 1.0 nm before the FAF | • VNAV.....OFF<br>• ALT SEL.....MDA<br>• FPA.....Select<br>Start descent 0.3 nm prior FAF (level flight)<br>Passing FAF at FLAP 2, reducing to 180 kts<br>"FPA 3.0" | **Note:** the FPA selected on the FCP is the flight path angle from the aircraft, not the path angle towards a fixed point on the ground. Closely monitor altitude versus distance to the runway when using FPA.<br>"CHECKED" |
| Below MISAP altitude | • ALT SEL.....MISAP<br>"GO-AROUND ALTITUDE ... SET" | "CHECKED" |
| Altitude cross-checks | "CHECKED" or "CORRECTING"<br>Adjust FPA as required to maintain path. | "... NM, ... FT CHECKED" |
| 2500 ft AGL | Hands on sidestick and thrust levers. | "TWENTY-FIVE HUNDRED" (auto-callout) |
| 6 nm / 2000 ft AAE (latest) | • SPD.....160 kts<br>Landing config. in one continuous flow:<br>"GEAR DOWN"<br>"FLAP 3, 4 (5)" | "GEAR DOWN"<br>• SEATBELTS.....Cycle OFF, ON<br>"SPEED CHECKED", "FLAP 3, 4 (5)" |
| Landing configuration established | "BEFORE LANDING CHECKLIST"<br>☐ Altimeters..... ( ) Set and cross-checked<br>☑ Landing gear.....DN<br>☐ FLAP..... ( ) indicating<br>☐ Cabin.....Ready | "BEFORE LANDING CHECKLIST"<br>"BEFORE LANDING CHKL COMPLETE" |

| Event | Pilot Flying | Pilot Monitoring |
|---|---|---|
| 4 nm / 1300 ft AAE (latest) | • SPD MAN.....$V_{APP}$<br>• Flight Spoilers.....As required | At any point, landing clearance received:<br>• NOSE LDG LT.....ON |
| 1000 ft AAE | "CHECKED" or<br>"GO-AROUND, FLAP 2(4)" | "STABILIZED" or<br>"NOT STABILIZED" |
| 100 ft above minima | "CHECKED" | "PLUS-100" (auto-callout) |
| At minima | "CONTINUE" or<br>"GO-AROUND, FLAP 2(4)" | "MINIMUMS" (auto-callout) |



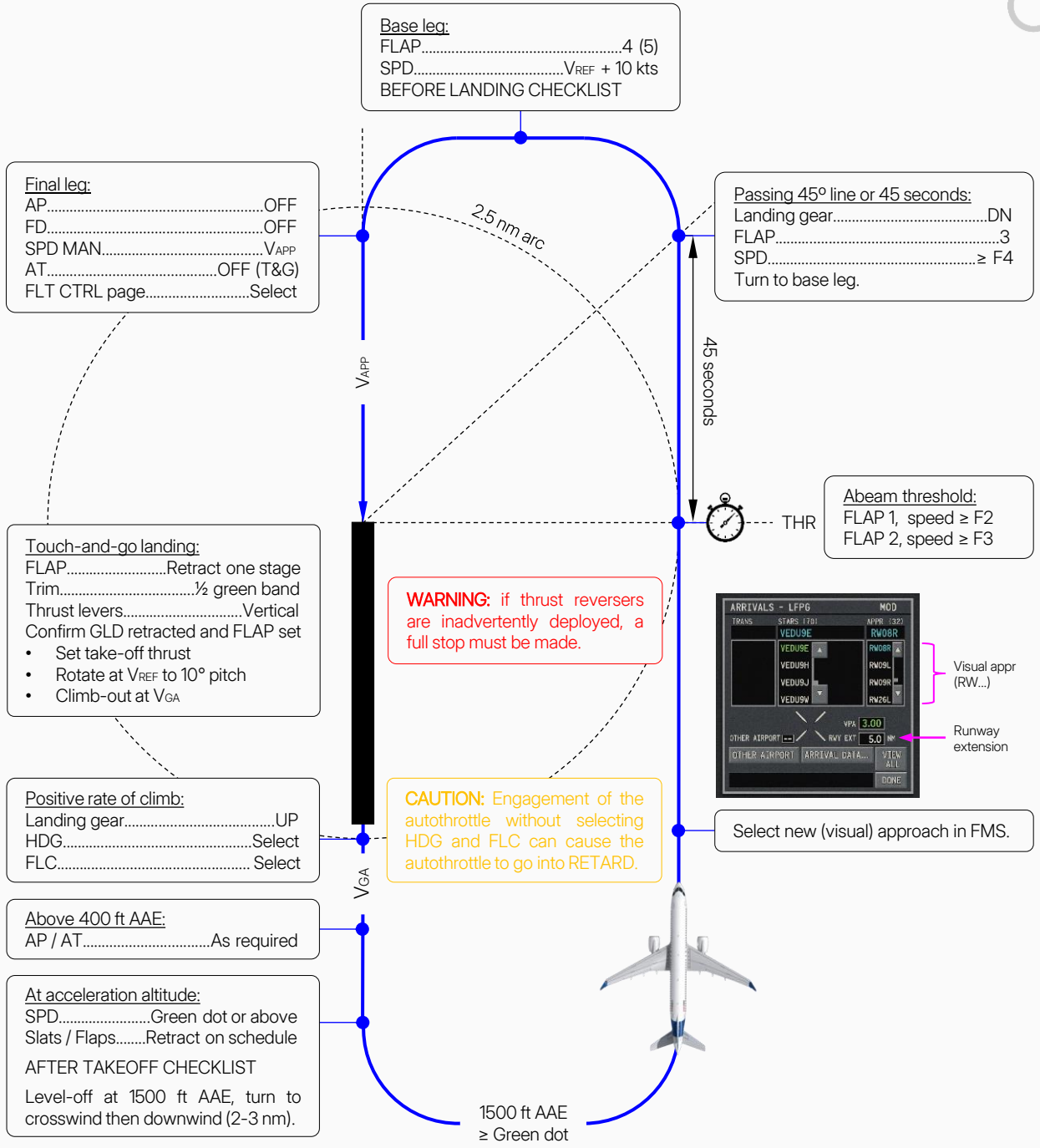
# Circling Approach

| Event | Pilot Flying | Pilot Monitoring |
|---|---|---|
| Approach set-up | • FIX page.....Set<br>*For reference, create the following fixes:*<br>1. Landing RWY extended centerline.<br>2. Landing RWY threshold abeam line.<br>3. 4.2 nm arc surrounding LDG RWY. | CAT-C aircraft (EASA CAT.OP.MPA.110):<br>• Maneuvering speed max. 180 kts<br>• Circling area radius: R = 4.2 nm<br>• Minimum circling height = 600 ft<br>• Minimum VIS 5000 m (realistically) |
| Instrument Approach | As for 2D or 3D approach except as below:<br>• ALT SEL.....MDA<br>• SPD MAN.....$V_{REF}$ + 10 kts<br>*When established in landing configuration.*<br><br>In case of 3D approach before 1500 ft AAE:<br>• NAV.....Select (ILS)<br>• VNAV.....Deselect (other)<br>*To cancel GS / VGP mode (see Caution)*<br>"FPA 3.0" | CAUTION: GS and VGP modes will not level-off at the circling MDA. In case of ILS approach, GS becomes latched below 1500 ft AAE and the only way to cancel would be TOGA. Therefore, GS or VGP must be deselected before 1500 ft AAE.<br><br>"CHECKED" |
| 100 ft above circling minima | "CHECKED" | "PLUS-100" (auto-callout) |
| At circling minima | "CHECKED, ALT-S CAP"<br>"ALT-S" (MDA captured)<br><br>• ALT SEL.....MISAP<br>"GO-AROUND ALTITUDE ... SET" | "MINIMUMS" (auto-callout)<br><br>"CHECKED"<br><br>"CHECKED" |
| Latest at the Missed Approach Point (MAPt) | "CIRCLING" or<br>"GO-AROUND, FLAP 2(4)" | |
| Circling | • HDG.....Select, 45º L or R<br>• Timing.....Start at wings level<br>After 30 seconds have elapsed:<br>• HDG.....Downwind leg | Note: in case of go-around during circling, make an initial turn towards the intended landing runway, then follow the missed approach for instrument approach flown. |
| Base turn | • Timing.....Start abeam threshold<br>After height (100) x 3 (e.g. 600 ft = 18 sec):<br>• HDG.....Base leg<br>• VS.....Select -300 ft/min | At 500 ft AAE (latest):<br>"STABILIZED"<br>(except wings level) or<br>"NOT STABILIZED" |
| Final turn | "AUTOPILOT OFF"<br>• AP / PTY.....Press twice<br>*2nd press to cancels AP disconnect alert.* | • FD.....OFF (PF side)<br>Aligned (wings level) by 300 ft AAE (latest):<br>• SPD MAN.....$V_{APP}$ |



# Visual Approach / Circuit Pattern

| Event | Student Pilot | Instructor Pilot |
|---|---|---|
| Preparation | • FIX page.....Set<br>*For reference, create the following fixes:*<br>1. Landing RWY extended centerline.<br>2. Landing RWY threshold abeam line.<br>3. Landing RWY threshold 45° offset.<br>4. 2.5 nm arc surrounding LDG RWY. | CAUTION: engagement of the autothrottle without selecting HDG and FLC can cause the autothrottle to go into RETARD mode.<br><br>WARNING: if reverse thrust has been initiated, a full stop must be made. |
| Take-off | When positive climb is indicated:<br>• Landing gear.....UP<br>• HDG.....Select<br>• FLC.....Select<br><br>Above 400 ft AAE:<br>• AP / AT.....As required<br><br>At ACCEL ALT:<br>• SPD.....Green dot or above<br>Level-off at 1500 ft AAE, turn to crosswind, then downwind (approx. 2-3 nm spacing). | During touch-and-go landings, the FMS F-speeds are not shown in accordance with the correct take-off / climb profile. The recommended flap retraction speeds are:<br><br>• FLAP 4 to 3 = 170 kts<br>• FLAP 3 to 2 = 180 kts<br>• FLAP 2 to 1 = 190 kts<br>• FLAP 1 to 0 = 200 kts<br><br>• Slats / Flaps.....Retract on schedule<br>"AFTER TAKEOFF CHECKLIST" |
| Downwind | Abeam threshold:<br>• FLAP 1.....Select<br>• SPD.....≥ F2<br>• FLAP 2.....Select<br>• SPD.....≥ F3<br>Passing 45° line / 45 seconds elapsed:<br>• Landing gear.....DN<br>• FLAP 3.....Select<br>• SPD.....≥ F4<br>Turn towards base leg. | After touch-and-go is performed, selection of a new approach in the FMS resets the FMS F-speeds and the appropriate flap extension speeds are shown.<br><br>Alternatively, time 45 seconds abeam THR.<br><br>Note: low thrust lever angles below 1000 ft AGL with flaps set at less than FLAP 4 causes a "TOO LOW FLAP" TAWS alert. |
| Base | • FLAP 4(5).....Select<br>• SPD.....$V_{REF}$ + 10 kts | Note: F5 recommended for touch-and-go.<br>"BEFORE LANDING CHECKLIST" |
| Final | • AP.....OFF<br>• FD.....OFF<br>For touch-and-go landing:<br>• AT.....OFF | When aligned with the runway:<br>• SPD MAN.....$V_{APP}$<br>• FLT CTRL synoptic page.....Select<br>*To monitor ground lift-dump retraction.* |
| Touch-and-go after touchdown | Do not select TOGA, if thrust reversers are inadvertently deployed, perform a full stop.<br>• Thrust levers.....Vertical<br><br>• Thrust levers.....Advance for take-off<br>• Rotate at $V_{REF}$ to 10° pitch, climb at $V_{GA}$ | • Slats / Flaps.....Retract one stage<br>• Trim.....Middle of green band<br>• FLT CTRL page.....Check<br>*Confirm GLD retracted and FLAP set.*<br>• Symmetrical thrust.....Confirm |



# Go-around / Balked Landing

| Event | Pilot Flying | Pilot Monitoring |
|---|---|---|
| Initiating go-around | "GO-AROUND, FLAP 2(4)"<br>• TOGA.....Select<br>*Automatically engages the autothrottle, sets Flight Director to give initial pitch guidance and synchronizes HDG bug.*<br>• Thrust levers.....Go-around thrust<br>*Monitor thrust lever movement to MAX.*<br>• Rotate to follow FD. | In case of FLAP 4 landing:<br>• FLAP.....Select 2<br>In case of FLAP 5 landing:<br>• FLAP..... Select 4<br>"FLAP 2(4)"<br>Bank exceeding 7.0° left or right: "BANK"<br>Pitch less than 10° or above 25°: "PITCH" |
| Positive rate of climb indicated | "GEAR UP"<br><br>"SELECT MAN SPD, SET $V_{GA}$" | "POSITIVE CLIMB"<br>• Landing gear.....UP<br>"GEAR UP"<br><br>• SPD MAN.....Set $V_{GA}$<br>"MAN SPD SELECTED, $V_{GA}$ SET" |
| 400 ft AGL | • Confirm or select lateral mode.<br>"FMS 1(2)"<br><br>• Autopilot.....As desired<br>"AUTOPILOT ON" | "CHECKED"<br><br>"CHECKED" |
| Acceleration altitude<br><br>AEO: 1000 ft AAE<br>OEI: MSA or MISAP alt | • FLC.....Select<br>• MAN SPD.....Green dot or above<br>"(V-)FLC"<br><br>F3, F2, F1, F0 speed (+ 10 kts if 30° bank):<br>"FLAP ..." | "CHECKED"<br><br>"SPEED CHECKED", "FLAP ..." |
| Flaps up and above green dot speed | "AFTER GO-AROUND CHECKLIST"<br><br>With AP engaged, keep hands on sidestick and thrust levers until at least 2500 ft AGL. | "AFTER GO-AROUND CHECKLIST"<br><br>☑ Landing gear.....UP<br>☑ FLAP.....0<br>☐ EICAS.....Checked<br><br>"AFTER GO-AROUND CL COMPLETE" |
| Follow-up | Evaluate fuel state, prepare for new approach or diversion.<br>Continue from "DESCENT AND APPROACH CHECKLIST" | |

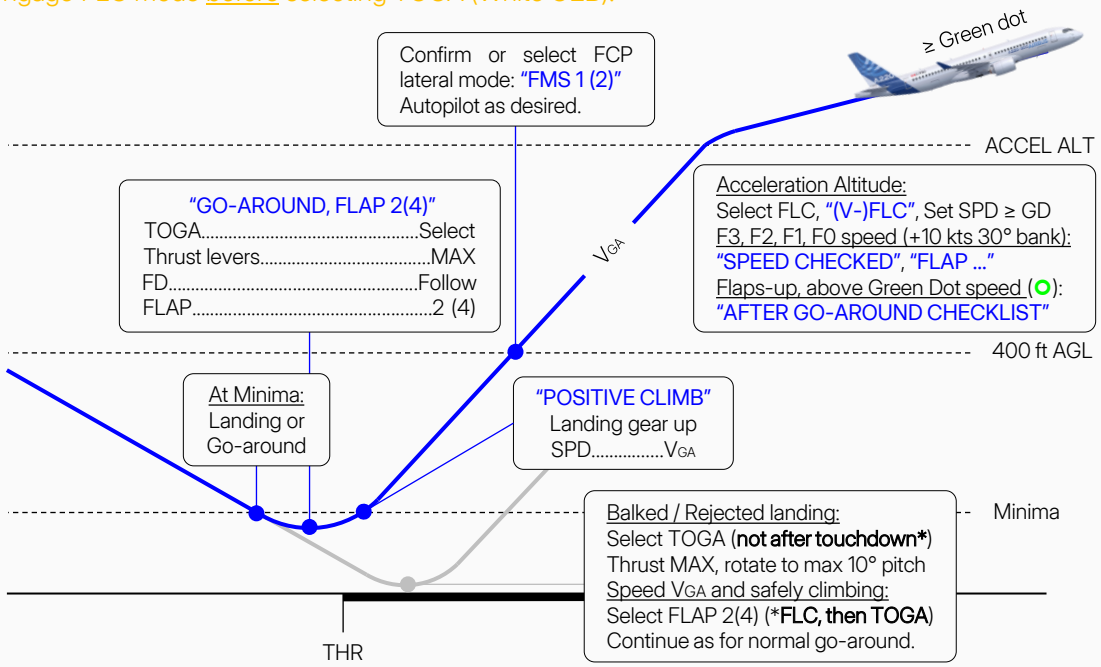
### Balked or Rejected Landing

If there are doubts whether a safe landing can be made, a go-around must be initiated. Go-around can be initiated after touchdown until the deployment of the thrust reversers.

A balked or rejected landing is a missed approach initiated after the aircraft has entered the low-energy regime, either before or after main landing gear contact with the runway. The low energy regime is entered upon initiating the landing flare with idle thrust and negative speed trend. The engines are usually at or close to idle and require several seconds to accelerate to full thrust.

- TOGA.....Select*
  ** In case of GA after touchdown: manual GA, at a safe altitude: press FLC first, then TOGA.*
- Thrust levers.....Go-around thrust
- Rotate to follow FD, don't exceed 10° pitch until no risk of touchdown. Speed $V_{GA}$ and climbing:
- FLAP.....2(4)

**CAUTION:** In case of go-around after main landing gear touchdown has occurred, do not press TOGA, perform a manual go-around, disregard or deselect Flight Directors. At a safe altitude, engage FLC mode before selecting TOGA (White OEB).



# Landing & After Landing Procedure

| Event | Pilot Flying | Pilot Monitoring |
|---|---|---|
| If using AT | "RETARD WHITE" | After the "RETARD WHITE" call, only active Autoland modes are called out by the PM. |
| Latest at 80 ft AGL | "AUTOPILOT OFF"<br>• AP / PTY.....Press twice<br>*2nd press to cancels AP disconnect alert.* | "CHECKED" |
| 30 ft AGL | Start the landing flare.<br><br>In case of no retard:<br>• AT.....Disconnect<br>• Thrust levers.....Idle | "THIRTY" (auto-callout)<br>Autothrottle used for landing:<br>"RETARD" or "NO RETARD"<br>Bank exceeding 7.0° left or right: "BANK"<br>Pitch exceeding 7.5° up or < 0°: "PITCH" |
| Touchdown | • De-rotate, slowly lower the nosewheel.<br>• Thrust levers.....IDLE REV<br>*Wait for the reverser interlock to release.*<br>• Reverse.....As required | "REVERSE GREEN"<br>or "NO REVERSE" |
| 80 kts | "CHECKED"<br>When appropriate for vacating and to minimize RWY occupancy time, disconnect autobrakes (before 30 kts to prevent jerks):<br>"MANUAL BRAKING" | "80 KTS"<br>Call: "AUTOBRAKE OFF" if autobrake does not engage as expected or AB disengages and no manual braking call from the PF. |
| 60 kts | • Thrust levers.....IDLE REV | |
| Taxi speed | • Thrust levers.....Forward IDLE | |

| Event | Crewmember 1 | Crewmember 2 |
|---|---|---|
| If CM2 = PF | "I HAVE CONTROL" | Before vacating: "YOU HAVE CONTROL" |
| Runway vacated | • TIMING.....START<br>"AFTER LANDING CHECKLIST"<br><br>A220-FOT-36-11-001 - Transitioning from engine bleed to APU bleed after landing could cause the PRSOV (engine bleed valve) to fail in the open position, triggering the L (R) BLEED FAIL caution. A design update of PRSOV is required, until then it is recommended to switch APU BLEED OFF before APU start (OAT ≤ 35°C). After both engines are OFF, set APU BLEED - AUTO. | • TIMING.....START<br><br>Procedure flow:<br>A220-FOT-36-11-001: APU BLEED – OFF<br>APU.....As required<br>STROBES.....OFF<br>TAXI LTS.....NARROW or WIDE<br>LANDING LTS (3).....OFF<br>FLAP.....0<br>"AFTER LANDING CHECKLIST"<br>☑ FLAP.....0<br>"AFTER LANDING CHKL COMPLETE" |

A220-FOT-36-11-001 - Transitioning from engine bleed to APU bleed after landing could cause the PRSOV (engine bleed valve) to fail in the open position, triggering L (R) BLEED FAIL caution. A design update of the PRSOV is required, until then it is recommended to switch APU BLEED to OFF before APU start (OAT ≤ 35°C). After both engines are OFF, switch APU BLEED to AUTO.

## Maneuvering

While in the landing configuration with all engines operating at VREF (or corrected VREF if FLAP 5 in icing conditions), maximum bank angle is 25°. For 30-degree bank while in the landing configuration, add 5 kts to VREF (or corrected VREF).

Minimum speed with flight spoilers extended is VREF + 10 kts.

## Approach Speed - VAPP

a. FLAP 5 only: during an approach and landing in icing conditions, or if there is evidence of ice accretion, VREF, VAC and VGA must be increased by 5 kts.
b. If autothrottle is used for approach and landing, add 5 kts to VREF.
c. In case of turbulence or gusty wind conditions are anticipated during the approach and landing, it is recommended to add half of the gust to VREF. (e.g. 36010G20KT, add 5 kts).
d. The autothrottle speed increment and gust factor combined should not exceed 10 kts.
e. The total speed increment should be entered on the FMS – PERF – ARR page.

## Airplane geometry for landing

a. Bank angles of more than 15 degrees can cause an engine nacelle strike.
b. Pitch angles of more than 10 degrees can cause an aft fuselage strike.

## Reverse thrust Operation

Reverse thrust significantly reduces the aircraft stopping distance when used in conjunction with braking, especially on slippery surfaces. Reverse thrust is most effective at high speeds.

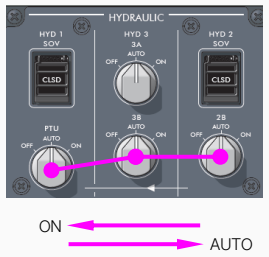
REV MAX must be used when landing on contaminated runways. On wet runways, REV IDLE may only be used in the OLD (unfactored landing distance) for RWYCC 2 is lower than the LDA.

Raise the thrust reverser levers to the interlock / deploy position. After reverser interlock release, pull thrust reverser levers back to the maximum position, if required. The PM monitors engine indications and calls any abnormalities.

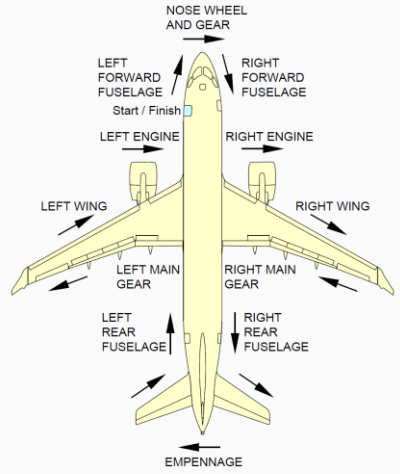
Maintain reverse thrust as required. The thrust reverse auto cutback feature automatically reduces the reverse thrust to reverse idle at 60 knots.

Stow reversers when the aircraft has achieved the desired taxi speed. When autobrake is not selected, do not wait for thrust reverser deployment before applying manual braking.

# Taxi-in, Parking & Shutdown

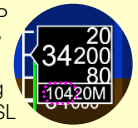
| Event | Crewmember 1 | Crewmember 2 |
|---|---|---|
| Single engine taxi operation<br><br>(not recommended on slippery/contaminated taxiways and aprons) | If flaps must be retracted after one engine is shutdown, it is recommended to use the right engine for single engine taxi-in to prevent loss of nosewheel steering function.<br><br>• Left engine used for taxi-in: select HYD 2B – ON before right engine is shutdown.<br>• Right engine used for taxi-in: select PTU – ON before left engine is shutdown. | |
| $N_1$ < 40% for ≥ 3 min (5 min recommended) | Single engine taxi-in desired:<br>"SHUTDOWN ENGINE 1(2)" | • ENG 1 (2) RUN switch.....OFF<br>"ENGINE 1(2) OFF" |
| During taxi | • Taxi up to 30 kts GS, then brake to 15 kts<br>• STROBES.....ON (active runways)<br>• PARK BRAKE.....ON (holding > 1 min) | |
| APU start | Start APU (if not already) at least 1 minute before arrival at the parking stand unless it is confirmed that ground power will be made available upon arrival (see Power On for details). | |
| Approaching parking stand | "CLEAR LEFT SIDE"<br><br>• TAXI LTS.....OFF | "CLEAR RIGHT SIDE" |
| Shutdown<br><br>  <br> | • PARK BRAKE.....ON<br>"SHUTDOWN CHECKLIST"<br><br>☑ PARK BRAKE.....ON<br>☑ NOSE STEER.....OFF<br>☐ APU and/or external power.....As required<br>*Engines must run 3 min near idle (< 40% $N_1$) prior shutdown ( 5 min recommended).*<br>☑ L ENG run.....OFF<br>☑ R ENG run.....OFF<br>☑ BEACON.....OFF<br>☐ ♦♦ Flight control test.....Complete<br>*Once each day only, do not make electrical changes (e.g. connect GPU) during the test.*<br>a. HYD 2B.....ON<br>b. HYD 3B.....ON<br>c. PTU.....ON<br>d. AVIONIC, TEST – FLT CTRL.....Select<br>e. Confirm PASS is shown<br>f. PTU.....AUTO<br>g. HYD 3B.....AUTO<br>h. HYD 2B.....AUTO<br><br>**A220-FOT-36-11-001:** APU BLEED – AUTO<br><br>Continue with Power-on for subsequent flights or Power-off when leaving the aircraft. | • XPDR.....CODE 2000, STBY<br><br>"SHUTDOWN CHECKLIST"<br><br>"SHUTDOWN CHECKLIST COMPLETE" |

# (PF) Power-off | (PM) Walkaround

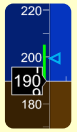
| | Crewmember 1 | Crewmember 2 |
|---|---|---|
| Event | Pilot Flying | Pilot Monitoring |
| Power-off Checklist (Read and Do) | Power-off checklist:<br><br>☐ ECL.....DU2<br>☑ EQUIP COOLING, INLET.....OFF<br>☑ EMER LTS.....OFF<br>☑ HYD 3A.....OFF<br>☑ HYD 3B.....OFF<br>☑ HYD 2B.....OFF<br>☑ APU.....OFF<br><br>*Confirm APU has completed its cooldown and is shut down (RPM = 0%, inlet door = closed) before removing battery power.*<br><br>☐ EXT PWR (if connected).....AVAIL<br>☑ DOME.....Select off<br>☑ BATT 1.....OFF<br>☑ BATT 2.....OFF<br><br>**Note:** after a complete power-off, a minimum of 60 seconds is recommended before electrical power is restored.<br><br> <br><br>Alternatively, a "partial power-off" can be performed. EXT PWR is kept IN USE with BATT 1 & 2 OFF. Keep EQUIP COOLING, INLET in AUTO to prevent bay overheating. | Perform an external inspection similar to the external inspection performed before flight. Check for damages, bird strikes etc.<br><br>The airplane and its visible components must be checked as follows:<br><br>• Proximate area is free of potential FOD<br>• Flight control surfaces unobstructed<br>• All vents, ports, intakes and exhausts are unobstructed.<br>• All covers, plugs, pins are removed.<br>• Any evidence of fluid leaks<br>• Condition of skin (visible damage)<br>• Access panels and doors are secured.<br>• Probes for evidence of freezing, severe discoloration, condition and security.<br>• Tire condition and inflation.<br><br>Start at the forward passenger door, continue in the direction shown and finish back at the forward passenger door.<br><br>  <br> |

# Practical Information (stuff the FCOM does not explain)

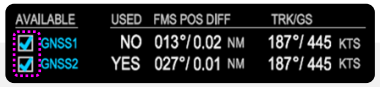
**Tip:** selecting ALT – M on the FCP will show altitude in meters below the altitude window. The first 2 or 3 numbers indicate the remaining distance required for descent to SL

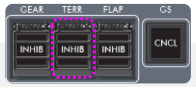



When flying manually, keep the speed trim bug between ½ and ⅔ the length of the green speed trend vector during speed changes to remain in-trim.




Placing the thrust command arc in-line with the EGT redline will give approx. 55% N1




## GNSS Interference

When encountering GPS jamming or entering an area of known GNSS interference: deselect GNSS 1 and 2 on both FMSs. In case of GPS spoofing, in addition to the above, select TAWS TERR to OFF to prevent nuisance alerts. Select auto-tuning, if DME-DME position updating is lost (FMS reversion to IRS position), terrain display is lost (purple screen).

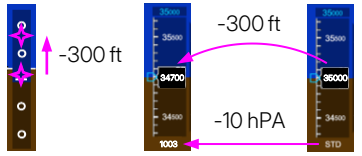
## VNAV (or lack thereof)

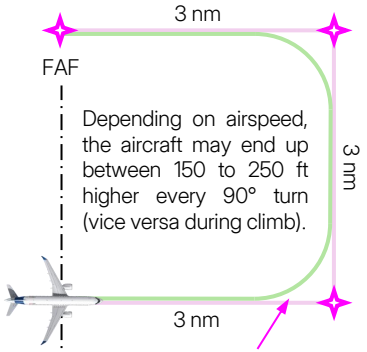
- A220 VNAV is essentially a glorified "FPA keeper". The only descent mode available is Vertical Path Angle (VPA), the FMS does not plan deceleration segments and does not consider aircraft descent capability (e.g. weight, wind, temperature).
- During descent, the VPA does not recalculate when changed by the pilot. Change any altitude constraint (e.g. by 1 ft) to force recalculation.
- FMS speed constraints or speed limits (e.g. 250 kts below 10.000 ft) may cause the FMS speed target to change at inappropriate times (too early or too late, not considering aircraft deceleration capability).
- The FMS does not anticipate BARO change between STD and QNH.

  *Example: when on profile before changing from STD to QNH, after setting a local QNH of 1003 hPA, the aircraft will now suddenly be 300 ft below path. (10 hPA below standard x 30 ft per hPA = -300 ft).*
- The FMS does not anticipate the distance lost by "short-cutting" fly-by waypoints. The higher the speed, the greater the turning radius, therefore the aircraft starts turning early, and "cuts" the corners more.

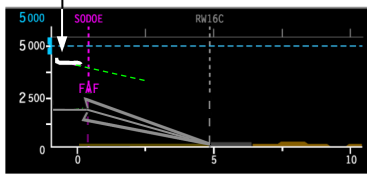
  *Example: for an arrival with downwind, base and final segments, each turn being 90° and each leg being 3 nm. The FMS assumes that 9 nm will be available for descent. However, the actual distance is less and every turn the aircraft will end up further the above profile.*
- When on radar vectors and deviating from the FMS routing, the VSD will show a vertical profile along the aircraft's track rather than along the flight plan. Essentially a vertical cross-section or "side-view".

  *Example: while on downwind, under radar vectors, with the aircraft abeam the FAF, it is projected over FAF on the VSD. It provides no useful information about altitude versus distance to go until on final.*
- The FMS provides a 5 nm level segment prior to FAF. Update the altitude over waypoint directly prior to FAF for a continuous descent.
- If there are any points between FAF and the runway, a level segment is drawn between those points on the VCP. The ILS feather is drawn from the last point before the runway which may cause confusion.

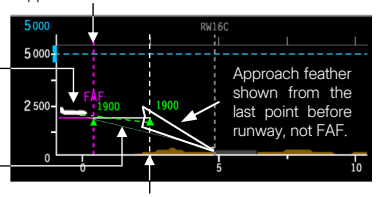



**Note:** the FMS only draws turns on the MAP for RNP-AR approaches.



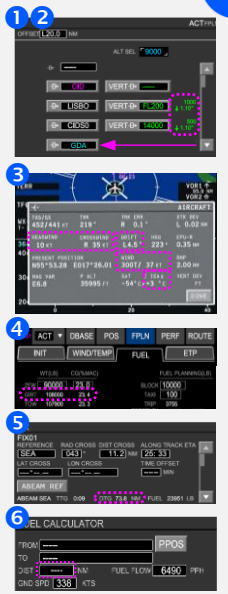

Approach starts here ...

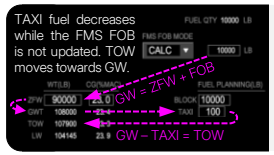



... not here

## Where to find information?

1. The direct-to window provides vertical path angles and the vertical speed information for vertical direct-to waypoints with constraints.
2. Continuing to scroll forward past the last flight plan waypoint will display the history section (cyan), allowing to restore points.
3. Information such as ISA deviation, wind, wind components and drift (among other things) can be found by selecting the aircraft symbol on the MAP, then press INFO.
4. Aircraft Gross Weight (GWT) can be found on the FMS, FPLN-FUEL page or the PERF-ARR page by selecting the departure airport (will be shown as Landing Weight - LW).
5. Direct distance to any point can be found by entering the waypoint to the FIX box.
6. For distance between any two points, use the FUEL CALCULATOR (FMS, PERF-CRZ).




With APU running but engines off, the APU fuel burn is subtracted from the TAXI fuel but the FMS FOB (and therefore GW) is not updated, causing the TOW to increase. Reenter the TAXI fuel before crosschecking TOW with load sheet.

Refueling rate is 400 kg/min for wing tanks (combined) and 400 kg/min for center tank. Thus, refueling rate is 800 kg/min when all tanks are being filled and 400 kg/min thereafter. Full refueling takes approximately 28 min.

Format to create points: **ROBTU 360/10/WPT1**
Waypoint ident - radial / distance / name. (name is optional and saves it as PILOT WPT).

During constant MACH descent, FLC mode might start chasing the speed, causing pitch oscillations. Maintain speed using VS or FPA, select lower speed target to keep thrust at idle.

Windshield heating gets hot on ground, many pilots switch windshield heating to OFF as part of AFTER LANDING flow. Then back to AUTO during the BEFORE TAXI flow.

## 3D Approach Operation:

- ✓ APPR.....ARM*
- ✓ VNAV.....Select
- ✓ ALT SEL.....DA (RNP AR only)

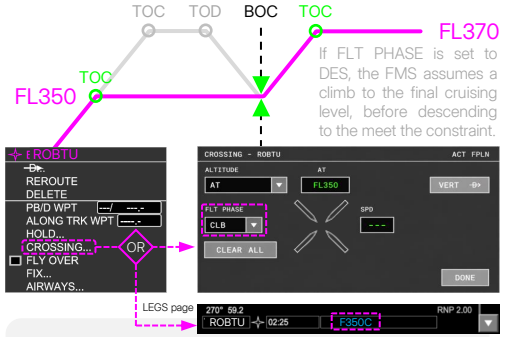
## 2D Approach Operation:

- ✓ APPR.....ARM*
- ✓ VNAV.....Deselect (latest at FAF)
- ✓ ALT SEL.....MDA, descent with FPA

*For ILS and LOC approaches, use FMS guidance for intercept, select APPR within 1 dot lateral deviation. For LOC approach, press NAV when localizer established to cancel GS mode.*

Programming step climb(s):

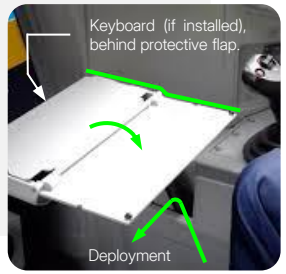
1. Set final cruise level on FPLN – INIT (e.g. FL370).
2. At the waypoint where further climb is expected, (BOC) set the initial cruising level (e.g. FL350).
3. On the CROSSING box, ensure FLT PHASE is set to CLB or add "C" behind the level on legs page.



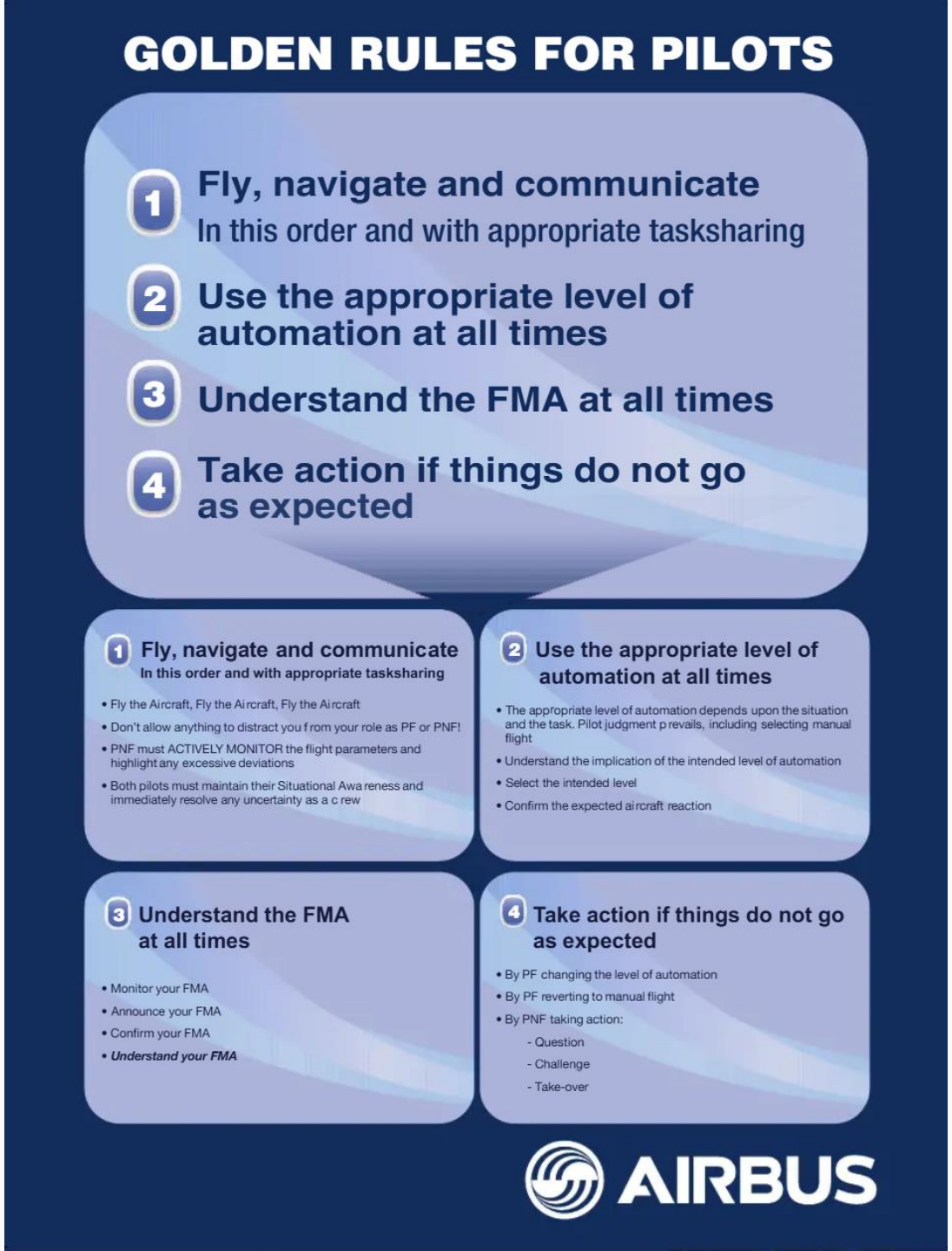

Cockpit tables slide out from the side panels. Aircraft with built-in EFB have a keyboard included in the table. On other aircraft, this compartment is empty (glove compartment ☺).

Pilot tables have some shortcomings. At times, during deployment, the whole assembly may release from its mount.

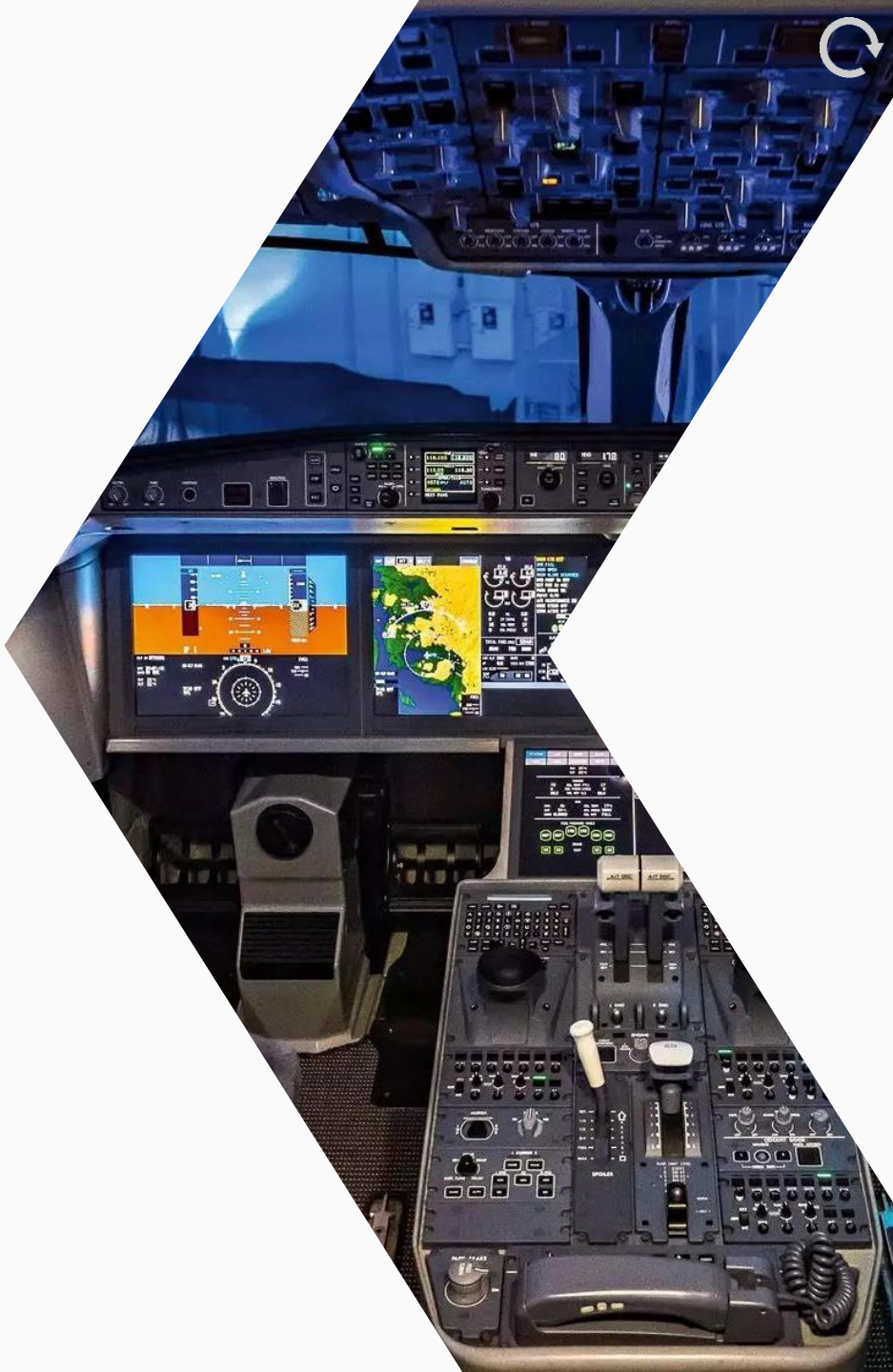
The tables also do not provide a flat surface for meal trays for example.



# › “Golden Rules”



# Supplemental Procedures



# Operation on Contaminated Runways

Runway conditions can be categorized as:

- Dry (RWYCC 6)
- Wet (RWYCC 5)
- Contaminated (RWYCC 4 or less)

Runway contamination causes:

- Additional drag
- Power loss or malfunction due to spray ingestion
- Reduced braking performance
- Directional control problems
- Possibility of structural damage

Requirements for operations on contaminated surfaces:

- During taxi, avoid use of thrust reverse, single engine taxi not recommended.
- Taxi slowly and avoid sharp turns, avoid taxiing closely behind other aircraft.
- Use light brake applications during taxi to heat up brakes before take-off and to prevent brakes from freezing at altitude and locking on landing.
- When operating on runways contaminated with ice, both thrust reversers must be operative before dispatch.
- Anti-skid must be operative.

Downgrade (applicable if depth < 3 mm, if depth ≥ 3 mm - delay the take-off):

- RWYCC 4 - use **Compacted Snow** for take-off performance computation.
- RWYCC 3 – use **10 mm Dry Snow** for take-off performance computation.
- RWYCC 2 – Flight crew should delay take-off (preferred) or compute with **Ice**.
- RWYCC 1 – Flight crew should compute the take-off performance with **Ice**.

Landing:

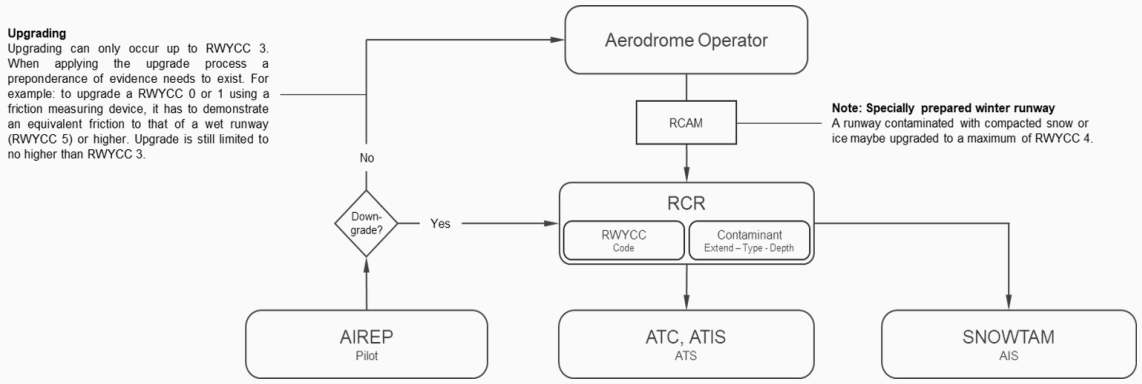
- Make a positive landing to aid wheel spin-up.
- Use reverse thrust, this reduces the use of wheel brakes and chance of skid.
- If loss of directional control occurs, it may be necessary to move thrust levers out of reverse and even select forward idle thrust to recover the centerline.

Hydroplaning:

- Hydroplaning usually does not occur in water depths of less than 5.0 mm, but if hydroplaning has begun it may persist if water depth and groundspeed are less than required for initiation.
- Hydroplaning can occur at all speeds, however, reference speeds for hydroplaning are: 127 kts for main wheels and 110 kts for nose wheels.

Grooved or Porous Friction Course (PFC) runways:

Ground handling characteristics are not as good as can be achieved on dry runways. A grooved or PFC runway is considered wet when there is sufficient moisture on the runway surface causing it to appear partially reflective but without significant areas of standing water (not overflowing).



# Operation on Contaminated Runways

Runway Condition Report (RCR) is distributed:

- Dispatch: SNOWTAM
- In-flight: ATIS or ATC

The Runway Condition Report has 2 parts:

- Performance section: RWYCC, extend, depth and type of contaminant
- Situational awareness: "DRIFTING SNOW", "CHEMICALLY TREATED"

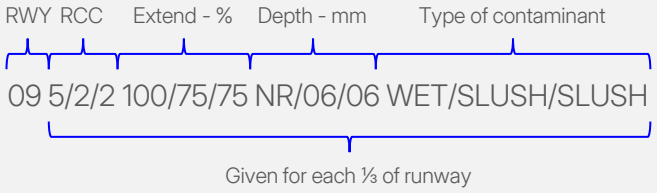
Example of an RCR:

09 5/2/2 100/75/75 NR/06/06 WET/SLUSH/SLUSH

DRIFTING SNOW. RWY 09 LOOSE SAND. CHEMICALLY TREATED. TWY A CLSD.

Decoding:

- Given for the runway end with lowest number
- RWYCC is given for each third of runway
- Extend of the runway covered in % for each third
- Depth of the contaminant in mm for each third (NR = Not Reported)
- Type of contaminant for each third of runway




| Runway Condition Assessment Matrix | | | |
|---|---|---|---|
| **Runway Condition Code (RWYCC)** | **Runway Surface Condition Description** | **Pilot-reported Braking Action** | **Maximum Recommended Crosswind** |
| 6 | Dry | - | 32 kts (take-off)<br>29 kts (landing) |
| 5 | • Frost<br>• Wet (less than 3 mm)<br>3 mm depth or less of:<br>• Slush<br>• Dry snow<br>• Wet Snow | Good | |
| 4 | -15°C and colder:<br>• Compacted snow | Good to Medium | 27 kts |
| 3 | • Wet ("slippery when wet")<br>• Dry snow or wet snow<br>More than 3 mm depth:<br>• Dry snow<br>• Wet snow<br>Warmer than -15°C:<br>• Compacted snow | Medium | 20 kts |
| 2 | More than 3 mm depth:<br>• Water<br>• Slush | Medium to Poor | 10 kts |
| 1 | Ice | Poor | |
| 0 | - | - | - |

# Navigation

## Reduced Vertical Separation Minima (RVSM)

The A220 is certified for RVSM operations. RVSM operations must not be started or continued unless all required equipment, as specified in the list that follows is operational:

- Autopilot
- Altitude alerting system
- Altitude reporting transponder (1)
- Air data computer (2)

## Traffic alert and Collision Avoidance System (TCAS)

Pilots are authorized to deviate from their Air Traffic Control (ATC) clearance in order to comply with a TCAS Resolution Advisory (RA) command.

## Enhanced mode S surveillance

Parameters available:

- Magnetic heading
- Indicated airspeed
- Mach number
- Vertical rate (barometric, not inertial)
- Roll angle
- Track angle rate (at least 1 functioning FMS)
- True track angle (at least 1 functioning FMS)
- Groundspeed (at least 1 functioning FMS)
- Selected altitude
- Barometric pressure setting

## Flight Management System (FMS)

The FMS has been demonstrated capable of, and meeting the requirements for the following operation:

| Operational Capability | FMS | GNSS | IRS |
|---|---|---|---|
| RNP-10 | 2 | 2 in navigation mode | - |
| | | 1 in navigation mode | 1 |
| | | - | 2 (max duration of 1 ½ hrs after loss of both GNSS) |
| RNP-4 | 2 | 2 in navigation mode (2 SBAS for LPV) | - (2 for RNP AR) |
| Oceanic & remote | | | |
| LPV * | | | |
| RNP AR * | | | |
| RNP-2 | 1 | 1 in navigation mode | - |
| RNAV-2 | | | |
| RNP APCH * | | | |
| B-RNAV | 1 | 1 in navigation mode or 1 DME/DME | - |
| RNAV-5 | | | |
| RNP-1 | 1 | 1 in navigation or Precision Approach (PA) mode | - |
| P-RNAV | | | |
| RNAV-1 | | | |
| * LPV APPR, RNP AR APPR, RNP APPR must be shown at the FAF or first point for RNP AR. | | | |

- The FMS employs a spherical Earth model opposed to WGS-84.
- Capable of RF legs if FMS is receiving information from a GNSS sensor and the aircraft uses the flight director or coupled autopilot to operate.
- Baro-VNAV vertical guidance deviation information, when used, pilots must use primary barometric altimeter to ensure compliance with all restrictions.
- LPV approach operations to LPV minima must not be started unless 2 FMS and 2 SBAS GNSS sensors are fully operational.

# Communication & Surveillance

Automatic Dependent Surveillance Broadcast (ADS-B)

ADS-B operations must not be started or continued if any of these caution messages are shown:

- ADS-B 1 OUT FAIL
- ADS-B 2 OUT FAIL
- ADS-B OUT FAIL
- XPDR 1 FAIL
- XPDR 2 FAIL

Aircraft Communications Addressing and Reporting System (ACARS)

The ACARS is approved for transmission and receipt of messages that will not create an unsafe condition if the message is improperly received. For messages such as pre-departure clearance, digital automatic terminal information service, weight and balance, take-off data (speeds, trim settings, runway distances) approved operational procedures must be used to verify that the message received is valid and that the content is not corrupted.

Controller-to-Pilot Data Link Communication (CPDLC)

CPDLC is limited to providing a supplementary means of communication, voice must remain the primary means of communication. Use of CPDLC should be avoided during periods of high pilot workload and critical phases of flight where pilot-ATC communications are time critical.

Emergency Locator Transmitted, Distress Tracking (ELT-DT)

ELT-DT usage has been demonstrated to meet the applicable regulations.

# CPDLC

Reserved for future use

# RNP - Authorization Required (AR)

General

- The A220 is approved for RNP AR approaches with a RNP greater than or equal to 0.3 and missed approaches with 1.0 nm accuracy.
- RNP AR is approved with autopilot engaged or using flight director only.
- MAN SPD must be set and used before the first waypoint of the approach.
- RNP AR APPR replaces RNP AR ARM on the PFD at the first waypoint.
- Flaps must be set to FLAP 4 or FLAP 5 at or before the FAF.

System limitations

RNP AR approach must not be started unless all required equipment is operational (easy rule: if there are more than 2 installed you can lose 1 system except for TOGA sw):

- PFCC (2)
- DU (4)
- FD (2)
- FMS (2)
- GNSS (2)
- IRS (2)
- ADS (3) both primary altimeters showing within 100 ft
- TAWS must be operational
- RAD ALT (2)
- TOGA sw (1)
- Nav data (1) must be valid

The approach can be continued if failure occurs during the approach unless otherwise specified in the non-normal procedures.

Maximum airspeed by approach segment

- Initial and intermediate (IAF to FAF): 240 kts
- Final approach (FAF to DA): 160 kts
- Missed approach (DA to IAF): 240 kts

Approach Briefing

The standard briefing should include RNP AR segment speed restrictions and contingency plans for missed approach from any point during the approach. Consider the following:

- Turn radius of each RF leg, maximum speeds and RNP value
- Equipment failure (including GNSS), loss of VNAV
- Go-around for any failure affecting RNP AR operations
- Hazards such as terrain, wind variations, windshear
- Terrain should be displayed on MFW for terrain clearance (loss of guidance)
- Planned recovery maneuver from a failure during a RF leg
- Use of lateral deviation on the non-failed side or transfer of flight guidance

RNP AR Approach

- Check and call RNP AR ARM within 31 nm of the first approach waypoint
- Do not accept direct to FAF, fix after FAF or fix immediately prior a RF leg
- Cleared for approach:
  a. Select APPR mode
  b. Ensure VNAV selected
  c. Set DA to ensure continuous descent
- At the first approach waypoint:
  a. Check and call RNP AR APPR
  b. Speed MAN mode
- Before or latest at the FAF:
  a. FLAP 4 or 5
  b. When VGP is active, set MISAP altitude
- Discontinue the approach (RNP AR MA will be shown):
  a. Lateral deviation exceeds 1 x RNP value (2 dots)
  b. Vertical deviation exceeds 75 ft (1 dot) after the FAF
  c. NO APPR flag on PFD or loss of navigation capability
- Approach can be continued if GNSS REVERTED shown on 1 side or AP failure
- During missed approach, continue to follow lateral track **including speed constraints**
- Perform terrain escape maneuver (with or without TAWS alert) for loss of navigation
- **Pre-Avionic Built 6.5:** should an underspeed (USPD) occur, VGP will no longer arm. First reselect VNAV, then deselect and reselect APPR. **Post-Avionic Built 6.5:** reselect VNAV.

# CAT II, CAT III & Autoland Operations

CAT II / III and Autoland capabilities

- CAT II – dual or single engine:
    - Decision height not less than 100 ft AGL
- CAT III – dual engine only with autoland:
    - LAND 2 (fail passive) with a decision height not less than 50 ft AGL
    - LAND 3 (fail operational) has no decision height
- Autoland operations – dual engine only on dry or wet runways up to MLW:
    - CAT II can be performed as an autoland if ASA shows at least LAND 2
    - CAT III must be performed as an autoland
    - Using CAT I equipment or CAT II / III with ILS critical area unprotected:
        - The ILS beam quality and effect of terrain on AP/FD is checked
        - Flight crew is aware that LOC or GS beam fluctuations can occur and prepared to immediately disconnect the autopilot.
        - At least LAND 2 capability is displayed on the ASA and flight crew uses CAT II / III approved procedures.
        - Visual references are obtained at the CAT I decision altitude.
    - The alert height is 200 ft AGL. Above the alert height the approach must be discontinued if a required aircraft system or ground system fails during CAT III operations.
    - The use of autobrake is recommended for all autoland operations.
    - Since autothrottle must be used, the approach speed must be increased by 5 kts ($V_{REF}$ + 5 kts).

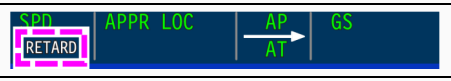
System Limitations

An ILS approach to CAT II or CAT III minima with autoland must not be started unless all required equipment, as specified below is operational. Some equipment failures during approach will cause the system to automatically "down-mode" to a different ASA (e.g. LAND 3 to LAND 2) or to a NO AUTOLAND condition.

| System or Limitation | CAT II – APPR 2 (no autoland) | CAT II or III – LAND 2(3) (autoland) |
|---|---|---|
| AURAL WARN INHIB | Operation is prohibited if selected on | |
| Glideslope angle | 2.5° to 3.5° | 2.5° to 3.25° |
| Runway slope | 2.0% up to 2.0% downslope | 0.8% up to 1.0% downslope |
| Runway width | - | 45 m |
| Engines | Single or Dual | Dual |
| PFCC (normal mode) | 1 | 2 |
| Primary Flight Display | 2 | |
| Flight Director | 2 | |
| Autopilot | 1 (On) | 2 (On) |
| Autopilot minimum height | 80 ft AGL | Not applicable |
| Autothrottle | On or Off | 1 (On) |
| TOGA switches | 1 | |
| IRS | 2 | 3 |
| ADS (smart probes) | 2 | 4 |
| Radio Altimeter | 1 | 2 |
| VHF NAV | 2 | |
| Nosewheel steering | - | Must be operational |
| Multi-function Spoilers | - | 4 pairs |
| Hydraulics | All systems must be operational | |
| Electrics | 2 generators must be on, approach can be continued with 1 | |
| Windshield heat | 2 | |
| Windshield wiper | 2 | |
| RWYCC | - | 6 (dry) or 5 (wet) |
| Max headwind | 25 kts | 25 kts (≥ CAT I = 29 kts) |
| Max crosswind | 15 kts | 15 kts (≥ CAT I = 24 kts) |
| Max tailwind | 10 kts | 10 kts (≥ CAT I = 10 kts) |
| Max pressure altitude | - | 8466 ft |

# CAT II, CAT III & Autoland Operations

| Event | Pilot Flying | Pilot Monitoring |
|---|---|---|
| In addition to a normal 3D Approach Operation | | |
| Below 2000 ft AGL<br>PFCC guidance | "APPR LOC" | "CHECKED" |
| 1500 to 800 ft AAE<br>ASA indication shown | "CHECKED" | "APPR 2 (LAND 2) (LAND 3)" |
| 1000 ft AAE<br>No downgrade | "CHECKED, NO DOWNGRADE" or<br>"GO-AROUND, FLAP 2(4)" | "STABILIZED" or<br>"NOT STABILIZED" |
| Below 400 ft AAE<br>RETARD mode arm | "RETARD WHITE" | After the "RETARD WHITE" call, only active Autoland modes are called out by the PM. |
| 200 ft to 150 ft AAE<br>ALIGN mode active | In case of no ALIGN:<br>"GO-AROUND, FLAP 2(4)" | "ALIGN" or "NO ALIGN" |
| 100 ft above minima<br>(Except for LAND 3) | "CHECKED" | "PLUS-100" (auto-callout) |
| At Decision Height<br>(Except for LAND 3) | If insufficient visual references:<br>"GO-AROUND, FLAP 2(4)" | "MINIMUMS" (auto-callout) |
| Latest at 80 ft AGL<br>(CAT II manual land) | In case of APPR 2:<br>"AUTOPILOT OFF" | "CHECKED" |
| Below 65 ft AGL<br>FLARE mode active | In case of no FLARE:<br>"GO-AROUND, FLAP 2(4)" | "FLARE" or "NO FLARE" |
| Below 20 ft AGL<br>RETARD mode active | In case of no RETARD:<br>Autothrottle disconnect - Thrust levers idle | "RETARD" or "NO RETARD" |
| Nose gear touchdown<br>ROLLOUT mode active | **Do not forget to select reverse thrust**<br>If no ROLLOUT: "AUTOPILOT OFF" | "ROLLOUT" or "NO ROLLOUT"<br>"REVERSE GREEN" or "NO REVERSE" |
| Safe taxi speed<br>AB, AP/FD OFF | "MANUAL BRAKING"<br>"AUTOPILOT OFF" | Select FD (PF side) OFF to prevent FD MODE CHANGE master caution alert. |

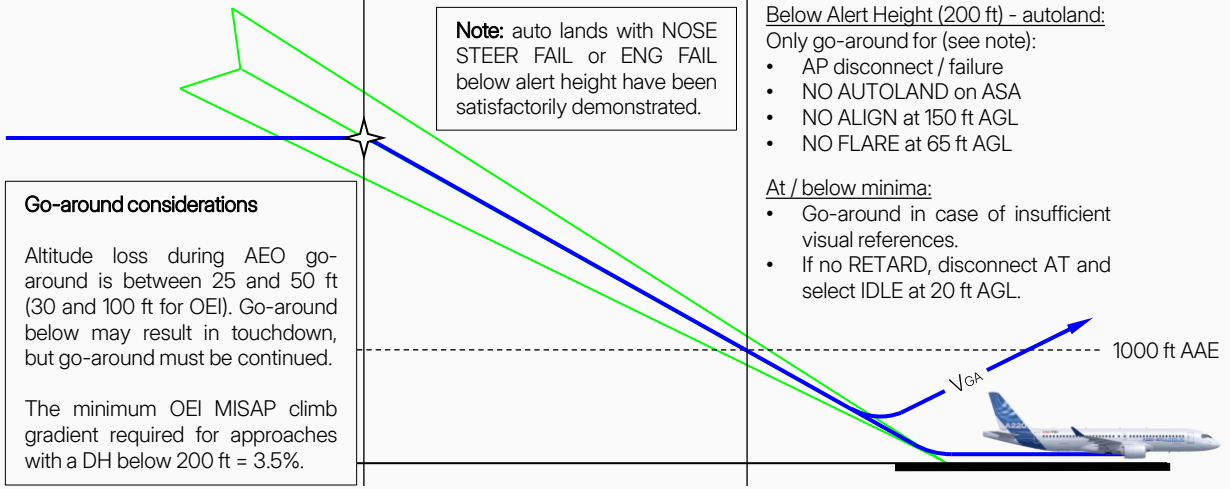
| | | | |
|---|---|---|---|
| Above 2000 ft AGL | | 200 ft 150 ft AAE | |
| Below 2000 ft AGL | | Below 65 ft AGL | |
| 1500 ft 800 ft AAE | | Below 20 ft AGL | |
| Below 400 ft AAE |  | 2s after WOW | |
| 350 ft 250 ft AAE | | Below 30 kts GS | ROLLOUT command bar removed.<br>NWS or rudder input disengages AP. |

| Before FAF | Within FAF, above 1000 ft AAE | Below 1000 ft AAE |
|---|---|---|
| If RVR or visibility is below landing minima, approach may be started and continued to 1000 ft AAE. | If RVR or visibility is below landing minima by 1000 ft AAE, approach shall be discontinued. | If RVR or visibility reduces below landing minima below 1000 ft AAE, approach may be continued to DA/H. |
| Attempt can be made to restore systems by performing non-normal checklists or procedures. All checklists and procedures must be completed prior 1000 ft AAE. | | Any system failure or loss of guidance should normally lead to a go-around. |
| | | In case of smoke, fumes or fire of any kind, the aircraft is committed to land. |
| Change of approach type is allowed if approach set-up and re-briefing can be completed prior crossing the FAF. | A change of approach type is allowed if the only change is the approach minima (e.g. CAT III to CAT II). | Continuation in case of down-mode without a down-grade is allowed. Down-grade shall result in go-around. |
| In case of pilot incapacitation, discontinue the approach and perform pilot incapacitation drill, decide whether to return or divert. | In case of pilot incapacitation, approach can be continued if stabilized and incapacitated pilot is not obstructing the controls. | In case of pilot incapacitation, approach can be continued if stabilized and incapacitated pilot is not obstructing the controls. |
| Effect of aerodrome equipment (e.g. failure of approach lights) on landing minima must be considered. | Effect of aerodrome equipment (e.g. failure of approach lights) on landing minima must be considered. | In case of failure of aerodrome equipment, approach continuation is commander discretion. |
| Autothrottle can be reengaged once, if unsuccessful, autoland not allowed. | Autothrottle can be reengaged once, if unsuccessful, autoland not allowed. | Autothrottle can be reengaged once, if unsuccessful, autoland not allowed. |
| | **Note:** auto lands with NOSE STEER FAIL or ENG FAIL below alert height have been satisfactorily demonstrated. | Below Alert Height (200 ft) - autoland:<br>Only go-around for (see note):<br>• AP disconnect / failure<br>• NO AUTOLAND on ASA<br>• NO ALIGN at 150 ft AGL<br>• NO FLARE at 65 ft AGL |
| | | At / below minima:<br>• Go-around in case of insufficient visual references.<br>• If no RETARD, disconnect AT and select IDLE at 20 ft AGL. |

**Go-around considerations**

Altitude loss during AEO go-around is between 25 and 50 ft (30 and 100 ft for OEI). Go-around below may result in touchdown, but go-around must be continued.

The minimum OEI MISAP climb gradient required for approaches with a DH below 200 ft = 3.5%.




Autoland: after touchdown, without ROLLOUT: TOGA buttons are active.
With ROLLOUT, TOGA buttons are inactive, make a manual go-around.

# Glideslope Angle from 3.5 to 3.8°

This part contains operational guidance to conduct approaches with glidepath angles greater than 3.5 degrees, up to and including 3.8 degrees.

The aircraft is capable of CAT I approaches and approaches with vertical guidance (3D approaches only) with glideslope / glidepath angles greater than 3.5, up to and including 3.8 degrees as below:

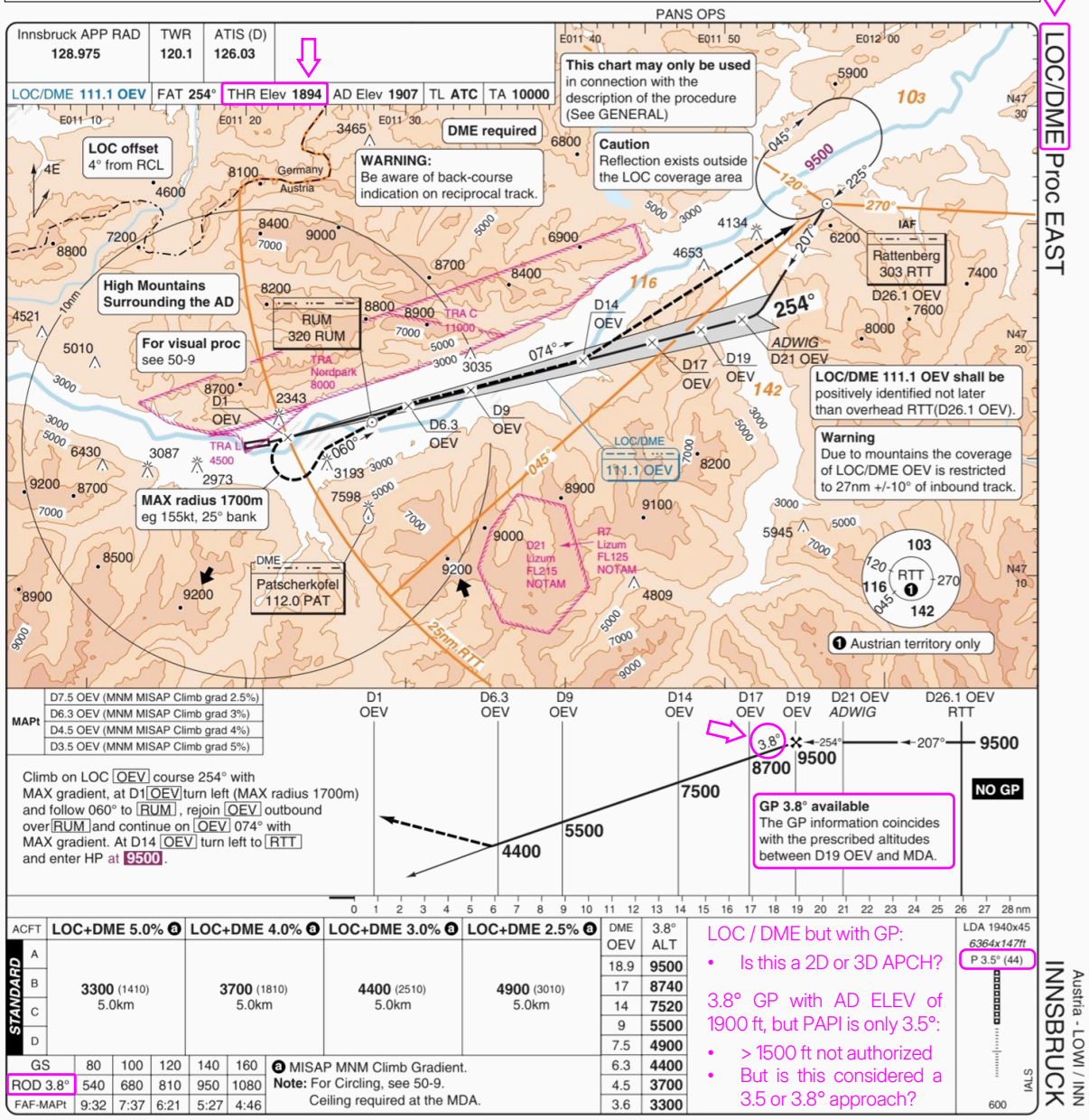
- Maximum airport pressure altitude for landing is 1500 ft.
- Maximum temperature for landing is ISA + 25°C.
- FLAP 4 landings are prohibited, only FLAP 5 landings are permitted.
- The maximum tailwind component is 5 kts.
- Approach with single bleed source is prohibited unless WAI is selected OFF.

An approach with a glidepath greater than 3.5 degrees, up to 3.8 degrees must not be started or a go-around must be accomplished for the conditions below:

- Any failure that prohibits FLAP 5 landings.
- Single bleed operation if icing conditions are forecasted or present.
- CAT II, III and Autoland operations not permitted.

**Note:** consider intercepting the glidepath with at least FLAP 3 and GEAR DOWN, preferably fully configure the aircraft before intercepting the glidepath in order to control the approach speed.

## Suitable or not? (matter of interpretation)



# Derated Thrust and Reduced Thrust Take-off

Engine take-off thrust can be set at less than full take-off thrust (TO) to prolong engine life as follows:

- Derated take-off thrust (TO-1, TO-2 or TO-3) with APR ARMED or DISARMED
- Set a reduced thrust take-off (FLEX) using the assumed temperature method
- A combination of the above **(Avionic Built 6.5 and higher only).**

The derated thrust and/or reduced thrust take-off N1 setting is automatically calculated by the FADEC based on entry in the PERF – DEP FMS page. The reduced N1 is then set when the autothrottle is engaged for take-off.

Assumed temperature should be at least 5°C higher than the OAT, otherwise it could cause the assumed temperature to be rejected. The wing anti-ice test can lead to rejection of the assumed temperature.

The flight crew must confirm:

- The target N1 shown on the EICAS (with engines running, take-off thrust selected and bleeds configured for take-off) is the same as the N1 value from thrust rating tables or performance application (N1 within 1%).
- The derated thrust or assumed temperature (FLEX) shown on the EICAS thrust mode display are the same as used to determine performance.

Derated thrust can be cancelled before take-off by selecting a different take-off thrust rating (e.g. TO, TO-1, TO-2, TO-3), this will also clear the assumed temperature input (FLEX).

Limitations:

Initiation of take-off using derated thrust and/or reduced engine thrust (FLEX) procedures is prohibited:

- With an engine that cannot achieve available full rated thrust
  *Take-off at full rated thrust (TO) or power assurance test is needed every 100 flights, approved trend monitoring can increase this interval to 1000 flights.*
- De-rated thrust can be used on dry, wet, contaminated runways (except ice).
- Reduced thrust take-off (FLEX) can be used on dry and wet runways only.
- When autothrottle is inoperative or not armed for take-off.
- Derated thrust operation with APR DISARM is prohibited.
- When windshear or downdrafts are forecasted.
- When the anti-skid system is inoperative.

Pre-Avionic Built 6.5:

- Reduced thrust (FLEX) from derated take-off thrust (TO-1,2,3) is prohibited.

During de-rated thrust procedure, the take-off thrust setting is considered a limitation for take-off and thrust levers should not be advanced further except in an emergency. A further thrust increase after an engine failure can result in a loss of directional control while on ground.

During a full rated thrust (TO) take-off in combination with reduced thrust (FLEX) operation, FLEX can be cancelled and maximum take-off thrust can be achieved at any time by setting the thrust levers to MAX.

# Low Visibility Operations (LVO)

RVR Reports

- TDZ – the touchdown RVR
- MID – the mid-point RVR
- RO – the rollout RVR

Where only two RVR sensors are available, both are controlling. Where three RVR sensors are available, for landing, the touchdown zone and mid-point are controlling, and rollout is for information only. Only the relevant parts (those used with groundspeed over 60 kts) need to be considered.

RVR requirements

| Category | RVR | DH |
|---|---|---|
| CAT I | 550 m | 200 ft |
| CAT II | 300 m | 100 ft |
| CAT IIIA (LAND 2) | 200 m | 50 ft |
| CAT IIIB (LAND 3) | 75 m | no DH |
| Take-off RVR: all relevant parts minimum 125 m | | |

A take-off alternate is required if the weather is below the minimum for approach, making an immediate return after take-off impossible (assuming engine failure, therefore no CAT III capability). The take-off alternate must be within 60 minutes or less at the one engine inoperative cruise speed in still air.

**Down-mode** is a change in approach capability while a **down-grade** is a change in the planned approach. In case of a down-mode from LAND 3 to LAND 2 while the crew prepared for a CAT IIIB approach, this would be an example of a down-grade. If in the same example, the crew prepared for a CAT IIIA or CAT II approach instead, this would be a down-mode without a down-grade. The decision to continue the approach after a down-grade is 1000 ft AAE. Below 1000 ft AAE a go-around must be completed.

**Note:** AP switch on the FCP is disabled during autoland, the AP DISC switch on the sidestick must be used. TOGA is disabled after landing + 2 seconds (ROLLOUT active).

Departure

- Plan low visibility taxi route to the runway or de-icing pad.
- Use of Airport Moving Map and airport charts, brief hotspots.
- Taxi at reduced speed, stop for all illuminated stop bars.
- All tasks (checklists, FMS changes etc.) when stopped with park brake set.
- Both crew members must be qualified for low visibility operations.
- Start during pushback and single engine taxi is not recommended.
- Use exterior lights (TAXI-WIDE, LOGO, WING INSP lights) even in daylight.
- If unsure of position, vacate any runway, stop the aircraft and inform ATC.

Arrival

- CAT II DH height is 100 ft above the touchdown zone, depending on terrain in front of the runway and slope, published DH can be < 100 ft (which is 100 ft above TDZ).
- Consider use of landing / taxi lights, visual cues required and approach lights.
- Wind limitations for CAT II / III and Autoland, correct seating position (eye reference).
- The effect of MEL dispatch items on approach capability.
- Approach reversion plan in case of autoland down-mode.
- Failures below alert height including autoland and engine failure.
- Brief CAT II / III and autoland specific procedures and call-outs.
- When possible, Autoland is recommended for all CAT II approaches (LAND 2 / 3).
- For a CAT 2 approach with APPR 2, autopilot must be used to fly the approach, but the approach is completed with a manual landing. AP must be disengaged by 80 ft.
- For landing on contaminated runways, it is recommended to disengage the AP after nosewheel touchdown, and a manual rollout be performed.
- If taxi-off is performed while rollout guidance is active (> 30 kts), the coupled (PF) side FD should be removed to prevent FD MODE CHANGE master caution alert.
- After touchdown, if visual references are lost, a go-around must not be attempted.

# Cold Weather Operations

**Ambient Temperature** SAT and TAT do not read accurately on ground, ATIS or other reliable means must be used to determine ambient temperature.

**Cold weather operations** refers to ground handling, take-offs and landings conducted on surface conditions where frozen moisture is present or when conditions are conducive to moisture freezing (OAT at or below 0°C).

**Cold soak** is the effect cold fuel can have on the wing surface resulting in moisture to freeze on the upper and lower wing surface if fuel temperature is at or below 0°C. It is possible to have ice or frost form even with ambient temperature above freezing.

The airplane is considered cold-soaked when it has been at an ambient temperature of -7°C or below for more than 8 consecutive hours without following the recommended procedure for preventing cold soak.

**Critical surfaces** are defined as the wings, horizontal stabilizer, vertical stabilizer, control surfaces and engine inlets.

The fuselage is not considered a critical surface, even so, if the upper surface of the fuselage is contaminated with ice, snow or frost to the extent that surface features and markings cannot be distinguished, the surface must be cleaned or de-iced.

Frost on the upper surface of the wing must be removed, take-off is permitted with frost on the lower surface of the wing to a maximum thickness of 3.0 mm.

The upper surface of the horizontal stabilizer may not be visible from the ground, the upper surface of the wing should be used as a comparative surface.

If de-/anti-icing fluid is allowed to dry on airplane surfaces, the fluid itself can become a contaminant. If left on the airplane surfaces for long periods they may dehydrate and form a gel or dried deposit that will not shear off, even at high speed.

**De-icing and Anti-icing fluid** application is the most common means of ground de-icing and anti-icing protection. These fluids are water/glycol solutions broadly classified as Type I, Type II, Type III (not applicable for A220) and Type IV.

**Type I fluid** contains glycol but no thickening agents (fluid is very thin), it provides very short holdover times and therefore a very limited anti-icing protection. Type I is often used for de-icing purposes or in very light snow or frost conditions. The fluid is orange.

**Type II and Type IV fluids** contain glycol and thickening agents, their high viscosity permits the application of a layer of fluid that is effective for anti-icing purposes and protects for a significant period of time (long holdover time). During take-off, the airstream imparts a shear stress to the fluid causing it to flow off the surface. Type II fluid has a transparent, slightly yellow color while Type IV fluids are green.

**De-icing and Anti-icing Procedures**

De-icing is the removal of snow, ice or frost using mechanical means, hot water or a heated mixture of water and de-/anti-icing fluid. Anti-icing is the application of de-/anti-icing fluid with a useful holdover time to prevent the accumulation of snow, ice or frost on clean or de-iced aircraft surfaces.

- Mechanical removal of loose contamination: if a significant amount of loose snow is present, the expenditure of a large amount of de-icing fluid can be avoided.
- One-step de-/anti-icing: fluid is applied in one step to remove frozen contamination and at the same time provide time-limited ice protection.
- Two-step de-/anti-icing: two fluid applications are made, the first using hot water or a water / fluid mixture (generally Type I). The second step applies de-/anti-icing fluid for anti-icing purposes. The second step must be applied before the first step freezes (typically within 3 minutes).

Benefits of two-step over one-step de-/anti-icing treatment are the removal of contaminants in a separate step which is usually done with cheaper Type I fluid, longer holdover time (holdover time begins from start of the second step) and preventing fluid residue built-up that can dehydrate and form a gel that will not shear off or refreeze.

# Cold Weather Operations

**Holdover time** is the published estimated time that an application of an approved de-icing /anti-icing fluid is effective in preventing frost, ice or snow from adhering to treated surfaces. Holdover time begins when the final application of de-/anti-icing fluid starts. It is considered to be no longer effective when it's ability to absorb more precipitation has been exceeded. (Generic) Holdover timetables can be found under Quick Reference.

**Notes:**

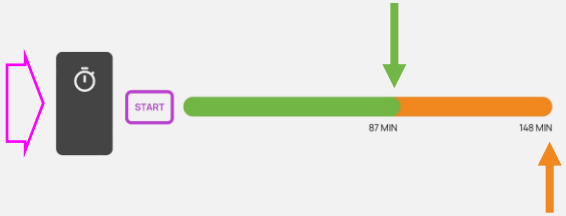
- FAA Holdover Time Guidelines are published every summer for the next season.
- Fluids used during ground de-/anti-icing do not provide in-flight icing protection.
- The time of protection will be shortened in heavy weather conditions, heavy precipitation rates or high moisture content.
- High wind velocity, jet blast or blowing snow may reduce holdover time below the lowest time stated in the range.
- The holdover time may be reduced when the aircraft skin temperature is below OAT.
- Whenever frost or ice occurs on the lower surface of the wing in the area of the fuel tank (cold soaked wing), the 50/50 dilution of Type II and IV is not to be used for the anti-icing step because fluid freezing may occur.
- The use of RVR is not permitted for determining snowfall intensities.
- ATIS / METAR reported snowfall intensity (e.g. –SN) is not to be used for determining holdover times, use "Snowfall intensities as a function of prevailing visibility" table.
- Lowest Operational Use Temperature (LOUT) must be respected, the fluid must provide a buffer above freezing point of at least 10°C (Type I) or 7°C (Type II / IV).
- Fluid must be heated to 60°C at the nozzle, a clean aircraft may be anti-iced with unheated fluid, the second step (anti-icing) of a two-step procedure may be unheated unless Type I fluid is used.
- **Standard holdover times only apply if the aircraft is de-/anti-iced with slats and flaps retracted and slats / flaps are deployed as close as possible to the departure. In all other cases, adjusted holdover times must be used, reduce the holdover time to 76%.**

**Factors affecting Holdover Time (HOT)**

Conservative approach would be to use lowest time.

Less favorable conditions:
- Low end of temp range
- Low end of VIS range
- Lower skin temp
- High wind speed

Considered in determining HOT
- Ambient temperature
- Fluid type & concentration
- Precipitation type and rate




- Aircraft skin temperature
- Wind velocity and direction
- Taxi speed
- Jet blast

More favorable conditions:
- High end of temp range
- High end of VIS range
- Higher skin temp
- Low or no wind

**Checks before and after de-/anti-icing treatment**

A. Contamination check during walkaround to determine the need for de-/anti-icing.
B. Post de-/anti-icing check is a visual check after the treatment has been completed, the de-icing operator will conduct a close visual check to ensure that critical surfaces remain free of contamination.
C. Pre-takeoff check is performed shortly prior to take-off. The flight crew shall assess whether the applied holdover time is still appropriate. The purpose of the check is to verify that frost, ice and snow have not accumulated on critical surfaces. The check is passed if review confirms beyond any doubt, that critical surfaces are free from frost, ice and snow. Three methods are acceptable:
    1. Mental review of conditions since treatment and HOT is valid.
    2. Check of a representative surface and HOT has not expired.
    3. Check of the wings (from the cabin) and HOT has not expired.
D. Pre-takeoff contamination check is to be performed when the condition of critical surfaces of the aircraft cannot be effectively assessed during pre-takeoff check or when holdover time has been exceeded, the check is performed from outside the aircraft. Another way to comply would be to undergo a new de-/anti-icing treatment

# Cold Weather Operations

### Contamination Check

To determine whether the aircraft needs a de-/anti-icing treatment, a "contamination check" is performed as part of the walkaround inspection, check the following areas:

- Check protective covers removed
- ADS probes, AOA vanes, windshields and wipers
  *Be aware of ice ridges forming on the nose*
- Aircraft critical surfaces free of frost, ice and snow
- Nosewheel and main landing gear areas
- APU and air conditioning intake and exhaust areas
- Engine inlet, spinner, fan blades and cowlings
  *Including back side of fan blades*
- Fuel tanks and hydraulic components
- Water system and drain masts

### Cabin preparation

In case of a cold-soaked aircraft, it is recommended that the cabin interior is warmed up before dispatch to ensure proper operation of all exits.

- Warm up cabin to a comfortable level (ground air or APU)
- Check all doors

CAUTION: failure to disconnect ground conditioned air before closing the passenger doors can result in rapid pressurization of the aircraft.

### Preparation for de-/anti-icing treatment

Avoid direct impingement on engine intakes, APU intake, flight crew emergency exit hatch, windshields, cabin windows, AOA vanes, probes, wheels and brakes. Spray must not be directed at trailing edges to avoid partially melted contaminants being forced into hinge mechanisms. Aircraft that has been previously anti-iced (but not flown) must not receive Type II or IV fluid on top of the existing film, it must be de-iced first.

**Clean Aircraft Concept** the (aerodynamic) clean aircraft concept prohibits take-off when frost, ice, snow or other contaminants are present on the aircraft critical surfaces. The adverse effects of airframe contamination:

- Decreased thrust
- Decreased lift
- Increased drag
- Increased stall speed
- Trim changes
- Altered stall characteristics

The removal procedure should be done at the latest possible time before take-off to maximize the time that de-/anti-icing fluids are able to provide protection.

These general precautions must be observed:

- Do not assume that dry or loose snow will be blown off during take-off roll.
- A thorough inspection of critical surfaces must be done to determine the presence of contamination and need for de-/anti-icing treatment before take-off.
- Holdover time begins when the final application of de-/anti-icing fluid starts.
- If holdover time is exceeded, or at any time contamination is suspected, a new inspection must be done and if required, de-/anti-icing treatment must be repeated.

CAUTION:

Do not use de-/anti-icing fluid to remove frozen contaminants from the spinner, fan or other engine components. De-/anti-icing fluids that are ingested into the engine can have a corrosive and contaminating effect.

De-/anti-icing fluid must not be sprayed directly into or applied on the engine inlet, APU inlet or exhaust vents. Before engine start, make sure that there is no ice on the fan blades. Ice on the blades can cause rotor imbalance and damage to the engine.

# Cold Weather Operations

## De-icing / Anti-icing Phraseology

| Event | Ground Crew | Flight Crew |
|---|---|---|
| Prior to de/anti-icing | *"C-FFDO taxi lane … , move forward to the stop bar, confirm brakes set and aircraft is configured."* | *"Brakes set and aircraft configured, request two-step procedure for wings and stabilizers, C-FFDO."* |
| Starting de/anti-icing | *"C-FFDO, hold position, de-icing starts now."* | *"Hold position, C-FFDO."* |
| De/anti-icing completed<br><br>(anti-icing code) | *"C-FFDO, de-icing completed, Type II, Clariant Safe Wing MP II, 100%, start time HH:MM (local time), post de / anti-icing check performed, all clear signal on your left/right."* | |

## Post de-/anti-icing check

An aircraft shall not be dispatched after a de-icing / anti-icing procedure until the aircraft has received a visual check by qualified staff. Once the treatment has been completed, the de-icing operator will conduct a close visual check of the surface where the treatment commenced, to ensure that it has remained free of contamination.

## Contents of the "Anti-icing Code":

- The anti-icing fluid type (Type I, Type II or Type IV)
- The brand of the anti-icing fluid
- Volume concentration (e.g. 75%)
- The local time at the beginning of the final step
- Post de/anti-icing check completed

# De-icing / Anti-icing Procedure

## Remote De-icing / Anti-icing

**Before fluid is applied:**

**(1) Engine(s) is/are operating**

→ **Yes – Go to (2)**
→ **No – Go to (35)**

**(2) Engine(s) is/are operating:**

**Before taxi, before de-icing:**

Do not perform (override) the BEFORE TAXI checklist.

(3) APU..... As required
(4) ♦♦ Wing anti-ice test..... Complete (Refer to Before Taxi)
(5) ANTI-ICE. COWL..... AUTO or ON
(6) ANTI-ICE WING..... AUTO or ON
(7) EICAS and INFO..... Checked

**When aircraft in position / ready for de-icing:**

(8) PARK BRAKE..... ON
(9) NOSE STEER..... OFF
(10) Thrust levers..... IDLE for duration of the operation

*If the APU is running, make sure the personnel that does the fluid application are aware of the location of the APU air intake and have been instructed to avoid fluid spray that can be ingested by the APU.*

(11) APU.....As required

*It is recommended to turn APU OFF. Confirm door is closed if APU is OFF.*

(12) L PACK..... OFF
(13) R PACK..... OFF
(14) L BLEED..... OFF
(15) APU BLEED.....OFF
(16) R BLEED.....OFF
(17) EQUIP COOLING, INLET..... OFF

CAUTION: stabilizer should remain in current position for de-icing if there are signs of ice around the visor fairing or the horizontal stabilizer/elevator-to-aft-fuselage interface.

(18) Stabilizer trim..... Set to 5 units

CAUTION: if accumulation of frost, ice or snow is suspected while slats/flaps are extended, leave slats/flaps at current position for de-icing.

(19) FLAP.....0

*To prevent damage, the pilot must advise the de-icing operator to avoid direct spraying of anti-icing fluid on the following:*

- *Windshields, side windows, and cabin windows*
- *ADS (to prevent ingestion) and TAT probes*
- *AOA vanes*
- *Engine cowls intake (to prevent ingestion) and exhaust*
- *APU inlet / exhaust*
- *Antennas, and*
- *Wheels and brakes*

**After de-icing/anti-icing spraying:**

*The PIC must ensure that an inspection is done to confirm that critical surfaces are clear of contamination.*

(20) APU..... As required
(21) FLAP.....( ) selected

*Set take-off flaps as entered in FMS*

(22) Stabilizer trim.....( ) set
(23) Flight controls..... Checked

a) FLT CTRL synoptic page.....Monitor
b) Sidesticks..... Checked

*One pilot checks the flight control for full and correct range of movement.*

c) Rudder.....Checked

*Nosewheel steering off or use PEDAL DISC switch on tiller. Ensure full deflection (monitor both sides for correct movement).*

(24) NOSE STEER..... Select on
(25) PARK BRAKE..... As required

**One minute after de-icing / anti-icing is complete:**

(26) L BLEED..... Select auto
(27) APU BLEED..... Select auto
(28) R BLEED..... Select auto
(29) EQUIP COOLING, INLET..... Select auto
(30) ANTI-ICE, COWL..... AUTO or ON
(31) ANTI-ICE, WING..... AUTO or ON

**One minute after bleeds are set to auto:**

(32) L PACK.....Select auto
(33) R PACK..... Select auto
(34) Before take-off procedure..... Accomplish

**- COMPLETE -**

## De-icing / Anti-icing on Parking Stand

**(35) Engine(s) is/are not operating:**

**(36) APU is OFF:**

→ **Yes – Go to (37)**
→ **No – Go to (54)**

**(37) APU is OFF:**

**Before fluid is applied:**

(38) Doors..... closed
(39) External power..... As required
(40) EQUIP COOLING, INLET..... OFF

*To prevent damage, the pilot must advise the de-icing operator to avoid direct spraying of anti-icing fluid on the following:*

- *Windshields, side windows, and cabin windows*
- *ADS (to prevent ingestion) and TAT probes*
- *AOA vanes*
- *Engine cowls intake (to prevent ingestion) and exhaust*
- *APU inlet / exhaust*
- *Antennas, and*
- *Wheels and brakes*

**After de-icing/anti-icing spraying:**

*The PIC must ensure that an inspection is done to confirm that critical surfaces are clear of contamination.*

**Before APU is started or external air is established:**

(41) L PACK..... OFF
(42) R PACK.....OFF
(43) L BLEED..... OFF
(44) APU BLEED..... OFF
(45) R BLEED..... OFF
(46) APU or external air..... START / Establish

**One minute after de-icing / anti-icing is complete:**

(47) L BLEED..... Select auto
(48) APU BLEED..... Select auto
(49) R BLEED..... Select auto
(50) EQUIP COOLING, INLET..... Select auto

**One minute after bleeds are set to auto:**

(51) L PACK.....Select auto
(52) R PACK..... Select auto
(53) Before start procedure.....Accomplish

**- COMPLETE -**

**(54) APU is ON:**

**Before fluid is applied:**

(55) Doors..... closed
(56) External power..... As required
(57) L PACK..... OFF
(58) R PACK..... OFF
(59) L BLEED..... OFF
(60) APU BLEED..... OFF
(61) R BLEED.....OFF
(62) EQUIP COOLING, INLET.....OFF

*To prevent damage, the pilot must advise the de-icing operator to avoid direct spraying of anti-icing fluid on the following:*

- *Windshields, side windows, and cabin windows*
- *ADS (to prevent ingestion) and TAT probes*
- *AOA vanes*
- *Engine cowls intake (to prevent ingestion) and exhaust*
- *APU inlet / exhaust*
- *Antennas, and*
- *Wheels and brakes*

**After de-icing/anti-icing spraying:**

*The PIC must ensure that an inspection is done to confirm that critical surfaces are clear of contamination.*

**One minute after de-icing / anti-icing is complete:**

(63) L BLEED..... Select auto
(64) APU BLEED..... Select auto
(65) R BLEED..... Select auto
(66) EQUIP COOLING, INLET..... Select auto

**One minute after bleeds are set to auto:**

(67) L PACK..... Select auto
(68) R PACK..... Select auto
(69) Before start procedure..... Accomplish

**- COMPLETE -**

# Cold Weather Operations

### Pushback and Engine Start

- It is recommended to delay engine start until pushback or towing is completed. The flight crew must be aware that on slippery surfaces, the parking brake may not be enough to prevent forward motion of the airplane at idle thrust.
- During cold weather starts, initial oil pressure response may be slow and the oil pressure indication may be higher than normal.

### Taxi

- Avoid large and rapid nosewheel steering inputs on slippery surfaces.
- Maintain greater than normal distance between airplanes.
- Use moderate brake application to warm the brakes before take-off to eliminate moisture built-up and prevent possible brake freezing.

### Ice Shedding

- Engine ice shedding must be performed when the OAT is 3°C or below and visible moisture in any form is present such as fog, rain, snow, sleet or ice crystals. Engine run-ups must be performed, momentarily, to a minimum of 60% $N_1$ and at intervals not exceeding 30 minutes. Final run-up must be done within 5 min before take-off.
- In heavy snow conditions (visibility ≤ 400m), more frequency engine run-ups to 60% $N_1$ are recommended with a minimum 10-second hold.
- If final engine ice shedding is required before take-off, advance thrust levers to 60% $N_1$, then set thrust levers to IDLE, continue with normal take-off procedure.
- The following may be observed during ice shedding: vibration may increase during ice shedding, sharp metallic noise as ice impacts the nacelle, airframe vibration and change in engine noise may be observed.

### Take-off

- Consider delaying gear retraction if take-off from slush or snow-covered runway.

### In flight

When operating for extended periods in icing conditions, engine vibration may be encountered, even with the anti-ice on. A momentary increase in thrust, up to CLB thrust, should stop undesired vibration.

### Descent and Approach

- Anticipate wing anti-ice requirements during descent, this may increase the idle thrust and result in lower descent rate.
- Avoid holding in icing conditions for longer than necessary.
- Ensure the cleared runway width is sufficient and take note of braking action reports.
- Consider diversion during extreme weather conditions (e.g. freezing rain), when braking action is reported to be poor and for strong crosswinds on slippery runways.

### Landing

- Accomplish a positive landing to ensure initial wheel spin up, free up frozen brakes and reduce the risk of hydroplaning.
- Lower the nosewheel immediately.
- Use autobrake, when using manual braking, apply brakes normally with steadily increasing pressure to allow the anti-skid system to modulate brake pressure.
- Use maximum reverse thrust as soon as possible after touchdown, thrust reversers are most effective at high speed.
- Expect skidding and hydroplaning and be prepared to make necessary corrections.
- If a loss of directional control occurs, reduce reverse thrust to idle reverse, if necessary, return to forward idle thrust to regain centerline.

### Taxi-in and Parking

- Do not retract flaps in order to prevent damage by crushing ice that formed in the slots of surfaces. At the stand, perform a visual inspection before retracting flaps.
- Expect movement areas to be slippery, single engine taxi-in is not recommended.

# Cold Weather Operations

**Leaving the Aircraft**

Based on the forecasted conditions, the crew should assess if the airplane will become cold-soaked or not and either prevent or prepare the aircraft for cold soak conditions.

Cold soak is not likely

- Park the aircraft on a clear, sanded spot
- Park into the wind if possible

The flight crew must ensure that ground personnel provide adequate servicing of the aircraft, appropriate for the expected weather, especially for prolonged overnight stops.

- Ensure wheel chocks are in place
- Switch PARK BRAKE off
  *To prevent brake freezing*
- Protective covers are installed
- Engine covers are installed
- Water, lavatory and waste as required
  *Drain if interior heating not provided*

Cold soak prevention

The intent is to minimize heat loss from the aircraft and minimize the risk of critical equipment reaching the point of cold soak, this to ensure fast preparation for next flight.

In addition to that for "Cold soak not likely":

- While servicing the aircraft, open time of external doors should be minimized except for the mid equipment bay door which is kept open to prevent aircraft pressurization
- Keep lavatory doors open and connect external power
  *Electrical equipment kept warm by own consumption*

CAUTION: If the turnaround time is expected to exceed 120 minutes with OAT at or below 0°C, it is recommended to use an external source of air-conditioning to maintain a minimum cabin temperature to minimize freezing conditions in the water system.

- Connect external low-pressure air
- Monitor APU and engine oil temperatures, before the oil temperature reaches the lower limits for ground starting (-40°C):

  a) Start the APU and / or engines
  b) Run the APU and / or engines long enough to warm up the oil
  c) Warm up the cabin using bleed and packs

Cold soak preparation

In addition to that for "Cold soak not likely":

- Drain waste and water systems
- Remove aircraft batteries and store at room temperature
  *If OAT is below -15°C, especially if needed for APU start*
- For next flight, continue with "Preparation after cold soak"

Preparation after cold soak

- Use a heater before applying electrical power (if below -40°C)
  *Maintenance procedure*
- Connect external low pressure ground air to warm up the cabin
- If APU oil temperature is too low for APU start, preheat the APU
  *Maintenance procedure*
- If engine oil temperature is too low for start, preheat the engines
  *Maintenance procedure*
- Install aircraft batteries (if removed)
- Connect external power
- Apply power to the aircraft
- Disconnect heaters and external air
- Start APU to warm up cabin
- Switch HYD 3A and HYD 3B on (if cold soaked below -40°C)
  *Hydraulic reservoir temperature warmed up to at least -6°C*
- Proceed with cold weather aircraft pre-flight preparation

# Temperature Compensation

All altitudes depicted on approach charts are based on ISA conditions. The true altitude will be lower when temperature is lower than ISA and the altimeter error may become significant at extremely low temperatures. Temperature error is zero at the aerodrome elevation and increases with height, therefore the correction is only to be applied over the height (not the altitude).

When the temperature at the aerodrome is at or below 0 °C, it is the responsibility of the Commander to consider the effect of temperature on the minima and reference altitudes. Correct the MSA first, then all altitudes below the temperature corrected MSA must be corrected for low temperature as well such as:

- DA, MDA, circling minima
- Initial and intermediate approach altitudes
- Missed approach altitude
- Engine-out acceleration altitude
- Step-down fixes / checking point altitudes

These altitudes are corrected using Method 1, Method 2 or Method 3.

**Method 1:** Approximate Correction (at or above -15°C OAT only):

The approximate correction to be applied is 4 ft per 1°C below ISA for every 1000 ft height above the altimeter setting source or alternatively, add 4% of the height above the altimeter setting source for every 10°C below ISA. This approximate method can be used as long as the temperature is at or above -15°C OAT.

**Method 2:** Tabulated Correction (any temperature):

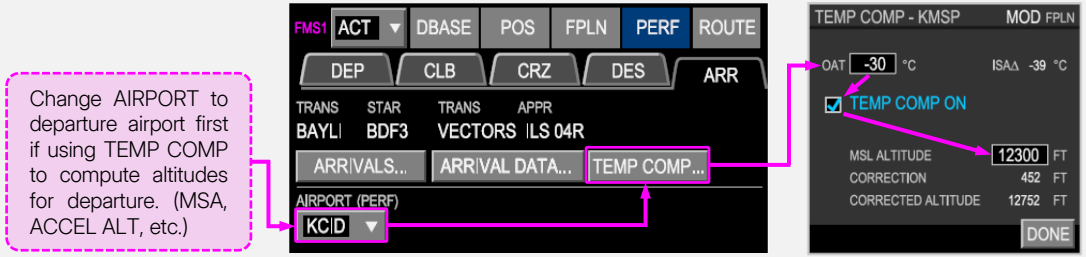
| AD °C | Height above the elevation of the altimeter setting source (ft) | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 200 | 300 | 400 | 500 | 600 | 700 | 800 | 900 | 1000 | 1500 | 2000 | 3000 | 4000 | 5000 |
| 0 | 20 | 20 | 30 | 30 | 40 | 40 | 50 | 50 | 60 | 90 | 120 | 170 | 230 | 280 |
| -10 | 20 | 30 | 40 | 50 | 60 | 70 | 80 | 90 | 100 | 150 | 200 | 290 | 390 | 490 |
| -20 | 30 | 50 | 60 | 70 | 90 | 100 | 120 | 130 | 140 | 210 | 280 | 420 | 570 | 710 |
| -30 | 40 | 60 | 80 | 100 | 120 | 140 | 150 | 170 | 190 | 280 | 380 | 570 | 760 | 950 |
| -40 | 50 | 80 | 100 | 120 | 150 | 170 | 190 | 220 | 240 | 360 | 480 | 720 | 970 | 1210 |
| -50 | 60 | 90 | 120 | 150 | 180 | 210 | 240 | 270 | 300 | 450 | 590 | 890 | 1190 | 1500 |

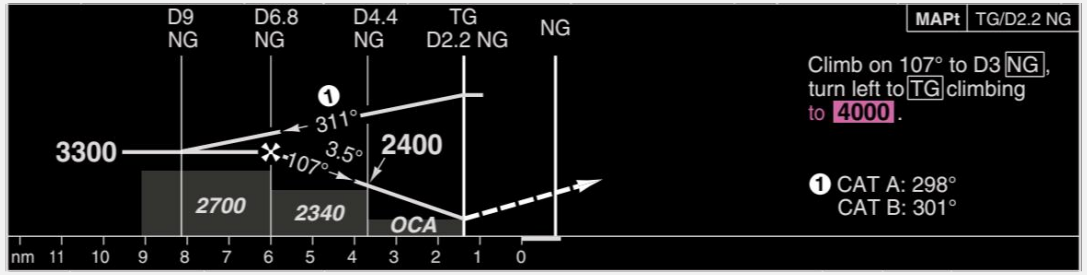
**Method 3:** FMS Temperature Compensation (any temperature):

The FMS temperature compensation function is accessible via the FMS – PERF – ARR page and can be used to automatically correct all FMS coded arrival and approach altitudes (indicated on the LEGS page by a ©). It can also be used to temperature correct any non-FMS coded altitude:

- Navigate to FMS – PERF – ARR, open TEMP COMP dialog box.
- Enter the aerodrome OAT, select TEMP COMP ON checkbox then hit EXEC.
- Enter any other altitude to be corrected for approach such as: MSA, DA/MDA, step downs or MISAP altitude into the MSL ALTITUDE box, note the CORRECTED ALTITUDE, set DA/MDA.

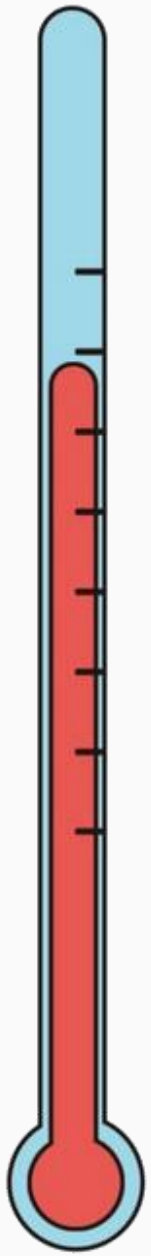
**Note:** the FMS TEMP COMP function is for approach only and cannot be used to correct FMS coded altitudes for departure. However, TEMP COMP can be used as a tool to manually compute temperature corrections for departure (MSA, OEI ACCEL ALT, etc.). Change AIRPORT on the PERF – ARR page to the departure airport before use, otherwise computations are invalid.




Example: highest MSA = 4000 ft, AD ELEV = 1027 ft, DA = 1660 ft, ISA = -20°C.

- To simplify calculations: AD ELEV → 1000 ft | 4% for each 10°C below ISA = 8% → 10%.
- Corrected MSA = MSA – AD ELEV = 3000 ft → 10% of 3000 ft = 300 ft + MSA = **4300 ft.**
- Correct everything at / below the corrected MSA: Initial approach = 3530 ft (3600 ft), Minima = 1726 ft (1730 ft), Missed approach = 4300 ft, Step-downs of 2870 ft and 2540 ft.



| | |
|---|---|
| +15°C | Above 15°C, wing anti-ice ON prohibited |
| | At and below 15°C, APU (or APU bleed) must be OFF for take-off |
| +10°C | Cowl anti-ice must be selected ON before take-off if visible moisture present |
| | In flight, +10°C TAT to -40°C SAT in visible moisture is considered icing conditions |
| +5°C | Wing anti-ice must be selected ON before take-off if visible moisture present |
| +3°C | Engine ice shedding must be performed to 60% N1 at intervals not exceeding 30 minutes |
| 0°C | Ground operations, take-offs and landings with OAT < 0°C is considered cold weather operations |
| | If turnaround is expected to exceed 120 min, cabin temperature must be maintained during turnaround |
| -7°C | Or below for more than 8 consecutive hour is considered cold soak |
| -15°C | Recommended to remove aircraft batteries and store at room temperature |
| -18°C | Engine operation is prohibited in ground fog conditions |
| -20°C | If turnaround exceeds 120 min, thrust reversers must be cycled twice before take-off from contaminated runway, take-off must be initiated within 5 minutes, otherwise repeat procedure |
| -37°C | Minimum fuel temperature for start of flight |
| -40°C | Minimum oil temperature for APU and engine start |
| | Oil temp -40 to -6°C = idle only, -6 to +21°C = max 30% N1, 21 to 49°C = max 50% N1, above 49°C = no limit |
| | If turnaround exceeds 120 min, thrust reversers must be cycled until cycle time is less than 2 ½ seconds before take-off from contaminated runway, take-off must be initiated within 2 minutes, otherwise repeat procedure |
| | Before applying electrical power, use heater to warm up equipment (maintenance action) |
| | ACMP 3A and 3B must be ON to heat hydraulic system 3 fluid temperature to at least -6°C before take-off |
| -47°C | JET A-1 freezing point |
| -54°C | Minimum temperature for take-off or landing |
| -70.5°C | Minimum operating temperature at tropopause (ISA -14°C) |

# Super-cooled Large Droplet (SLD) Icing

Super-cooled Large Droplets (SLD) exist in liquid form at temperature below 0 °C. If SLD are large enough, the mass prevents the pressure wave travelling ahead of an airfoil from deflecting it. When this occurs, the droplets impact further aft than a typical cloud-sized droplet and form clear ice (possibly beyond the protected area).

The presence of SLD can only be determined by observation of the resulting ice accumulation on unprotected surfaces. The indicator for differentiating SLD ice is observation of ice accumulation on the flight compartment side windows. Any ice accumulation on the side windows should be taken as an indication that SLD icing conditions are present.

Operations in SLD icing conditions is prohibited. When SLD is recognized (observation of side window icing), ensure the wing and cowl anti-ice systems are operating and leave SLD icing conditions immediately. After leaving SLD icing conditions, the wings should be monitored for signs of ice formation. If ice is observed, then the Ice dispersal procedure should be accomplished.

During the Ice dispersal procedure, speed is increased to $V_{MO}$ / $M_{MO}$ (if possible) to shed any ice formations and increase Total Air Temperature (TAT). If ice remains, landing speed and operational landing distance must be adjusted.

# Hot Weather Operations

The main focus is passenger and crew comfort and the significant decrease in airplane performance when ground temperatures are high.

In order to keep the aircraft as cool as possible, all doors should be kept closed as much as possible. Cargo bay doors should not be left open.

**If a ground air source is available:**

- Connect low pressure ground air
- Connect external power

**If APU is the only air source available:**

- L PACK and R PACK auto
- RECIRC AIR and TRIM AIR auto
- CKPT and CABIN temperature as required
- AIR CARGO AFT and FWD set to VENT
- EQUIP COOLING, EXHAUST set auto
- WSHLD and WINDOW HEAT off
- All air outlets and gaspers open
- Window shades closed

If cooling air is available from an external source, the supply should be connected immediately after engine shutdown and should not be removed until just before engine start.

CAUTION: failure to disconnect ground conditioned air before closing the passenger doors can result in rapid pressurization of the aircraft.

**Departure**

Excessive use and riding of brakes should be avoided. The recommended technique is to allow the airplane to accelerate, then brake to a very slow taxi speed and release the brakes completely.

High speed short radius turns should be avoided because hot temperatures will tend to wear tires at higher rate.

**Arrival**

Runway conditions, particularly at the touchdown area could be slick due to heavy deposits of rubber and oil which have melted. The possibility of hydroplaning is not remote.

This condition also holds true at the opposite end of the runway in use, therefore, slow down to a manageable speed before turning off from the runway to preclude skidding and departing from the intended path.

Always opt for the longest runway available for landing, if possible.

# Windshear

Windshear is a sudden change in wind speed and / or direction over a short distance. Severe windshear is indicated by sudden airspeed changes greater than 15 kts or vertical speed changes greater than 500 ft/min. Most windshear occurs horizontally, under certain conditions (mountains, thunderstorms, strong frontal systems) windshear can occur vertically, examples of this are downbursts and microbursts.

Indications of windshear are characterized by uncontrolled changes in excess of the following (separate or combination of):

- 15 kts indicated airspeed
- 500 ft/min vertical speed
- 5 degrees in pitch attitude
- 1 dot displacement from glideslope
- 10-degree variation from nominal heading
- Unusual thrust lever position for significant period of time

The following conditions can identify the existence of windshear:

- Thunderstorms with heavy rain
- The presence of virga (rain evaporating before reaching the ground)
- Strong frontal activity
- Low level Jetstream
- Temperature inversions
- A combination of the following conditions:

    a. Extreme variations in wind speed and / or direction in a short time
    b. Evidence of a gust front such as dust blowing over the airport
    c. Surface temperature higher than 30°C
    d. Dew point spread of 4°C or more

Strong microburst can also occur without precipitation, windshear can be caused by strong surface winds in the vicinity of hills or buildings or by sea breeze fronts. Avoidance is the best course of action if windshear is known or suspected. Stay clear of thunderstorm cells, heavy precipitation, virga and areas of known windshear.

- Monitor windshear pilot reports
- Use weather radar to locate thunderstorm cells

**At departure airport:**

- Delay take-off until conditions are favorable
- Use the longest and most favorable runway
- Select minimum flap setting acceptable
- Use full take-off thrust (no derate or FLEX)
- Monitor IAS during take-off roll
- Obstacles permitting, increase initial climb speed

**At arrival airport:**

- Delay the approach until conditions are favorable or divert
- Use the longest and most favorable runway available
- Add an appropriate wind correction to final approach speed
- Fly a normal approach, use autopilot and autothrottle
  *To allow for more monitoring and recognition time*
- Do not make large thrust reductions or trim changes
  *Speed could vary from increase to decrease in short time*
- If significant change in rate of descent, consider go-around

The Predictive Windshear System (PWS) uses the weather radar to predict windshear conditions and is active even in the TEST or STBY mode. During take-off, the PWS function is active below 2400 ft radio altitude, alerts are inhibited from 80 kts IAS until 50 ft AGL. During approach, PWS is active from 2300 ft radio altitude until 50 ft.

There are two alert levels:

- CAUTION – 3 nm prior encounter: "MONITOR RADAR DISPLAY"
- WARNING – 1 ½ nm prior encounter: "(GO-AROUND) WINDSHEAR AHEAD" x2

Accompanied by a yellow or red "WSHR – AHEAD" on the PFD.

The Reactive Windshear Detection and Escape Guidance System is part of the TAWS and detects (negative) windshear conditions between 10 ft and 1500 ft radio altitude. It does not detect windshear on the runway; therefore, it is crucial to monitor airspeed during take-off.

- WINDSHEAR x3 aural alert and a red "WINDSHEAR" on the PFD (see Windshear Alerts)

# Flight in Turbulence

Flight through turbulence should be avoided as much as possible. Areas with known severe turbulence should be avoided, if possible.

Delay of take-off or approach may be required.

Definitions:

| Intensity Level | Aircraft Reaction | Inside Cabin |
|---|---|---|
| Light | • Momentary changes in aircraft altitude and attitude. | • Occupants may feel a slight strain against seatbelt.<br>• Unsecured objects may displace slightly. |
| Moderate | • Changes in aircraft altitude and attitude.<br>• The aircraft remains under positive control at all times.<br>• Usually causes variations in airspeed. | • Occupants feel a definite strain against seatbelts.<br>• Unsecured objects are dislodged. |
| Severe | • Causes large abrupt changes in altitude and/or attitude.<br>• Large variations in indicated airspeed.<br>• Aircraft may be temporarily out of control. | • Occupants are forced violently against seatbelts or shoulder straps.<br>• Unsecured objects are tossed about. |
| Extreme | The aircraft is violently tossed about and is practically impossible to control.<br>May cause structural damage. | |

Turbulence frequency:

- Occasional: less than ⅓ of the time
- Intermittent: ⅓ to ⅔ of the time
- Continuous: more than ⅔ of the time

Whenever possible, inform the cabin crew of anticipated turbulence so that they can plan their activities accordingly.

Before entering an area of known turbulence:

- Secure all loose objects in flight compartment and cabin
- Shoulder harness should be worn by flight crew
- When turbulence is encountered:

Light turbulence:

- Select SEATBELTS to ON
- Advise passengers over the PA system of turbulence and request that they fasten their seatbelts.

Moderate to Severe turbulence:

- Select SEATBELTS to ON
- Advise passengers over the PA system of turbulence and request that they fasten their seatbelts.
- Advise cabin crew to be seated (*"CABIN CREW BE SEATED"*)

When clear of turbulence:

- SEATBELTS as required
- Inspect cabin for damage if severe or extreme turbulence was encountered
- Crew must report hard landings, severe turbulence and aircraft upsets

The autopilot and autothrottle should be used when flying through turbulence, however, some flight conditions might require manual control.

Large altitude variations are possible, do not make abrupt control inputs to maintain altitude, allow altitude to vary provided there is adequate terrain clearance. Inform ATC if unable to maintain accurate altitude in RVSM airspace. If large pitch changes occur, hold wings level and smoothly adjust pitch back to the required attitude.

**Best turbulent air penetration speed is: 265 kts below 10.000 ft, 280 kts / M0.75 above.**

# Operation in Volcanic Ash / Dust

Refer to Volcanic Ash Encounters.

# Overweight Landings

The aircraft is built for an immediate return after take-off, up to the maximum take-off weight (MTOW). Overweight landings up to MTOW can be safely accomplished when normal landing procedures and techniques are used, there are no adverse handling characteristics associated with heavier than normal landing weights (autoland not permitted).

- Check landing distance, use longest runway, consider wind and slope.
- Avoid landing with a tailwind, down slope and reduced braking action.
- FLAP 4 or 5 landings will not exceed brake energy levels, but use of thrust reversers should be maximized to reduce chance of wheel fuse plug melting.
- Do not carry extra speed on final, fly a normal profile and rate of descent. A smooth landing is desirable but avoid tendency to flare long.
- If a long landing occurs, perform a go-around.
- After touchdown apply maximum reverse thrust, use all of the available runway for stopping to minimize brake temperatures.
- Overweight landings must be recorded in the technical log.

# Hard Landings

A hard landing is described as:

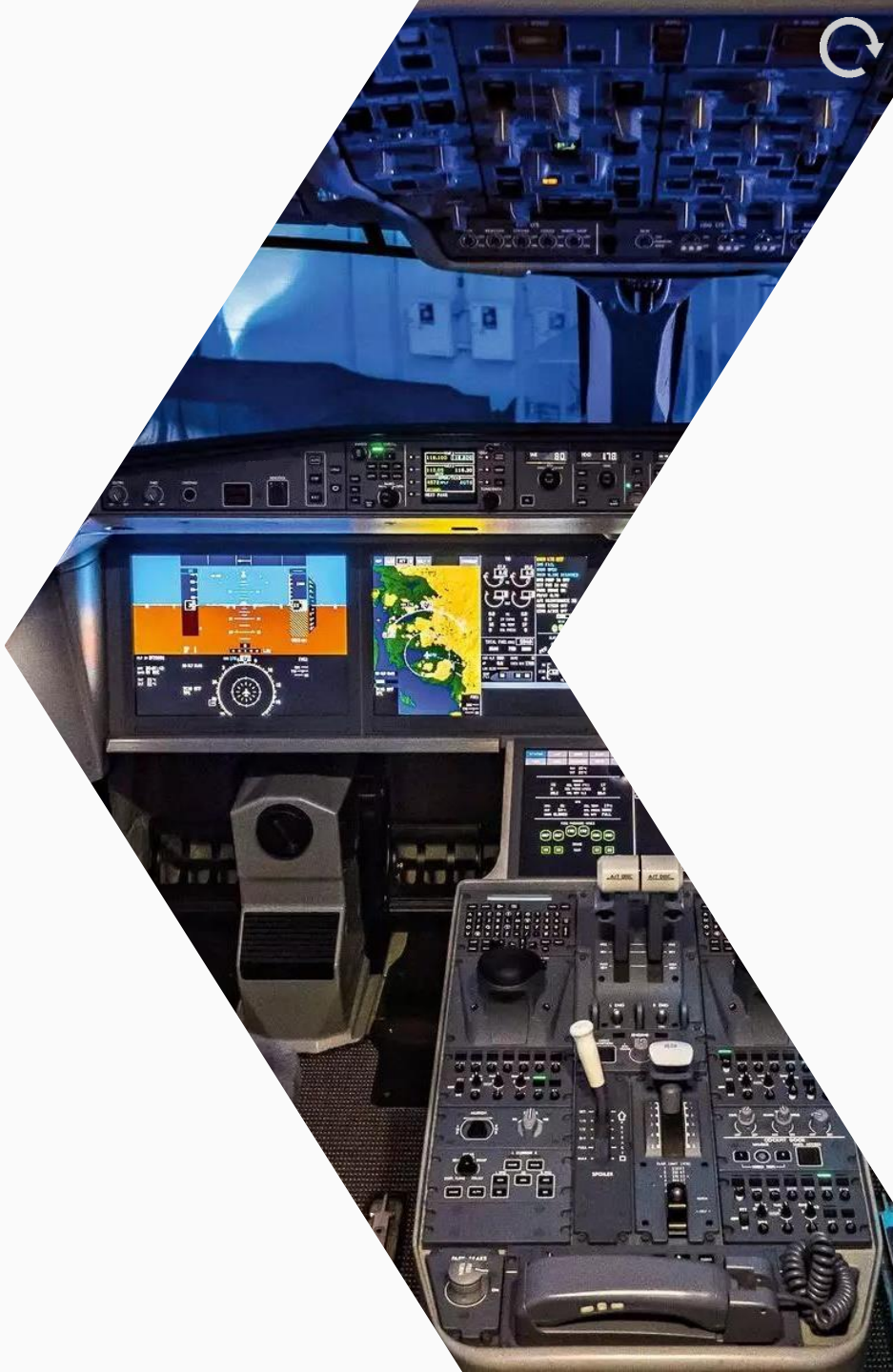
- A landing with a vertical rate of more than 600 ft/min when the aircraft gross weight is less than or equal to the Maximum Landing Weight (MLW)
- A landing with a vertical descent rate of more than 360 ft/min when the aircraft gross weight is more than the MLW (i.e. overweight landing)
- Leaving the prepared surface (i.e. runway excursion)
- A landing where the nose landing gear touches the ground first
- A touchdown that causes a tire burst
- A tail, wing tip or nacelle strike
- A hard de-rotation (abnormally hard nosewheel touchdown)

After a hard landing or de-rotation at any weight, it is possible that damage was caused to the aircraft structure and systems, damage can be visible or hidden.

The aircraft must be inspected before next flight. Hard landings are recorded in the Aircraft Health Monitoring System (AHMS) and Flight Data Recorder (FDR).

Crews must at all times report landings of abnormal severity or attitude, hard landings, severe turbulence encounters or aircraft upsets.

# Non-normal Procedures



# Introduction

## General

The abnormal and emergency procedures address situations involving failure conditions, in which the use of the normal or alternate systems can be expected to maintain an acceptable level of airworthiness. During abnormal / emergency situations, it is imperative that the crew understand and complete the applicable procedure and ensure that effective communication and coordination is maintained.

It is essential that the Commander, in coordination with the First Officer, assess the situation and clearly determines the task distribution for the various phases of flight. The commander could decide to take over controls for landing in Emergency and Abnormal situations, especially for those failures affecting aircraft handling. Abnormal and Emergency operations are non-routine, and all actions in abnormal procedures are to be announced before performing them. Under no circumstance may control of the aircraft be compromised.

### a) Definitions

Emergency situation:

*A condition of being threatened by serious and / or imminent danger and requiring immediate assistance.*

Usually triggered by MASTER WARNING + tripple chime + red alert on EICAS and refer to an emergency checklist (red) or priority procedure (non-EICAS event).

Abnormal situation:

*A condition involving an aircraft or other vehicle's safety, or some person onboard or insight, not requiring immediate assistance.*

Usually triggered by MASTER CAUTION + single Chime + amber alert on EICAS and refer to a non-normal checklist (amber) or non-normal procedure (non-EICAS event). If no immediate action is required, PF may delay crew actions or checklist if necessary.

### b) Priority Levels

During normal operations, all lights are extinguished.

| | | |
|---|---|---|
| • Red | Warning | Emergency checklist |
| • Amber | Caution | Non-normal checklist |
| • Cyan | Advisory | No in-flight action |
| • White | Status | Information only |

Situations that are not detected by EICAS (e.g. pilot incapacitation or windshear) may have associated procedures that are either performed by memory (e.g. windshear escape maneuver) or by using the ECL PROC tile. Note the terminology used, EICAS alerts have associated "Checklists" non-EICAS events are called "Procedures".

| | |
|---|---|
| Checklists priority levels: | Emergency checklist and non-normal checklist |
| Procedure priority levels: | Priority procedures and non-normal procedures |

### c) Procedure Initiation

| On Ground | In Flight |
|---|---|
| • CM1 initiates procedures<br>• Checklists are challenge and response (Read by CM2, response by CM1)<br>• Irreversible items don't need confirmation | • PF initiates procedures<br>• Checklists are read and performed by the PM<br>• Irreversible items need confirmation |

No action shall be taken:

- Until flight path is stabilized.
- Below 400 ft AGL (if not time critical, delay until above MSA).

### d) Basic Failure Management

The Basic Failure Management (BFM) concept serves as a step-by-step guide and should enable the crew to deal with any malfunction or abnormal situation, from minor to severe, in a structured manner. Not all steps always apply or not always precisely in the order given. BFM should never replace common sense and good airmanship.

# Emergency Communications

## a) Emergency Transponder Codes

Emergency transponder codes:

- EMERGENCY - Mode A Code 7700
- COMM FAILURE - Mode A Code 7600
- UNLAWFUL INTERFERENCE - Mode A Code 7500

Aircraft already receiving an air traffic service, and transmitting a code, should retain the code in use. Aircraft in an emergency, who are not receiving an air traffic service should set the transponder to EMERGENCY (Mode A Code 7700) as part of their initial actions. A pilot may select Mode A Code 7700 whenever he has specific reason to believe that this would be the best course of action.

## b) Emergency Radio telephony

Abbreviated (short) message:

- **"MAYDAY"** (x3) or **"PAN PAN"** (x3)
- Name of the station addressed
- Aircraft identification (callsign)
- Nature of urgency / emergency
- The phrase: **"STANDBY"**

**Note:** whenever workload permits, crew should pass the remaining information from the full emergency message below such as persons on board, endurance and intentions.

Full message:

- **"MAYDAY"** (x3) or **"PAN PAN"** (x3)
- Name of the station addressed
- Aircraft identification (callsign)
- Nature of urgency / emergency
- Intention of the pilot in command
- Present or last know position, flight level / altitude and heading (if not under radar control)
- Other useful information such as endurance, POB, aircraft color / markings and survival aids

## c) PA announcements

| |
|---|
| **Turbulence Encounter**<br>• Before entering turbulence: *"CABIN CREW BE SEATED (reason)"*<br>• Encountering unexpected turbulence: *"CABIN CREW BE SEATED" (self-explanatory)* |
| **Announcing emergency during any phase of flight**<br>• Time permitting: Cabin call button (3 seconds) or, *"ATTENTION – SENIOR TO FLIGHT DECK"*<br>• No time available (e.g. fire on short final): *"ATTENTION – CREW AT STATIONS"*<br>• Cancellation of emergency: *"NORMAL OPERATIONS"* |
| **Emergency on ground (e.g. rejected take-off)**<br>• After stopping: *"ATTENTION - CREW AT STATIONS"*<br>• Evacuation not required: *"CABIN CREW & PASSENGERS REMAIN SEATED"*<br>• Evacuation required: *"EVACUATE, EVACUATE"* press the EVAC CMD switch |
| **Rapid disembarkation** *"RAPID DISEMBARKATION"* |
| **Unprepared emergency landing**<br>• When need for emergency landing is apparent: *"ATTENTION – CREW AT STATIONS"*<br>• 30 seconds before impact (approx. 500 ft AGL): *"BRACE FOR IMPACT"* |
| **Prepared emergency landing**<br>• 2 minutes before impact (approx. 2000 ft AGL): *"FINISH PREPARATIONS"*<br>• 30 seconds before impact (approx. 500 ft AGL): *"BRACE FOR IMPACT"* |
| **Emergency descent**<br>• Initiating emergency descent: *"DESCENT, DESCENT"*<br>• Descent completed and safe altitude: *"SAFE TO REMOVE MASKS"* |

## d) NITS – Briefing

Time permitting, cabin crew shall be briefed about the abnormal or emergency situation. Preferably, NITS briefing is done in person by requesting the Senior to come to the flight deck. If not possible, briefing can be performed via interphone. If no time is available (e.g. engine fire on short final), cabin crew should be made aware of a developing situation by calling on PA: *"ATTENTION – CREW AT STATIONS"*.

- **N**ature of abnormal / emergency situation
- **I**ntentions of the commander
- **T**ime before landing (synchronize watches)
- **S**pecial instructions, e.g. prepared emergency landing

Cabin crew shall read back the NITS briefing to verify information has been passed on correctly.

# Emergency Communications

### Emergency / Urgency Message

- **"MAYDAY"** x3 or **"PAN PAN"** x3
- Name ATSU
- Callsign
- Nature of emergency or urgency
- Intentions
- Position, altitude / level, heading (as applicable)
- Other relevant info. (endurance, POB etc.)

*"**MAYDAY MAYDAY MAYDAY**, Schiphol Tower, FastJet 123, engine failure, continuing runway track, climbing to altitude 2000 ft, endurance 2½ hrs, 100 POB"*

### Abreviated Urgency / Emergency Message

- **"MAYDAY"** x3 or **"PAN PAN"** x3
- Name ATSU
- Callsign
- Nature of emergency
- **"STANDBY"**

**Note:** crew should pass more details when workload permits.

*"**MAYDAY MAYDAY MAYDAY**, Shiphol Tower, FastJet 123, engine failure, **STANDBY**"*

### Cancelling Distress

- Name ATSU
- Callsign
- **"CANCEL MAYDAY"**
- Other relevant information

*"Schiphol Tower, FastJet 123, **CANCEL MAYDAY**, engine restarted"*

### Rejected Take-off

- Callsign
- **"STOPPING"**

*"**FastJet 123, STOP IMMEDIATELY! I SAY AGAIN STOP IMMEDIATELY! ACKNOWLEDGE!**"*

*"**STOPPING,** FastJet 123"*

### Communication Failure

- **"TRANSMITTING BLIND"**
- Name ATSU
- Callsign & aircraft type
- Intentions
- Position, altitude / level, heading (as applicable)
- Other relevant info. (endurance, POB etc.)
- Time of next transmission
- **"I SAY AGAIN ..."**

*"**TRANSMITTING BLIND,** Shiphol Tower, Fastjet 123, Airbus A220, continuing BLUFA 1A arrival to ARTIP to join the published hold, FL100, starting approach at time ... , Endurance 1½ hr, 100 POB, next transmission at time ... , **I SAY AGAIN ...** (repeat entire message)"*

### TCAS

- Callsign
- **"TCAS RA"**

*"FastJet 123, **TCAS RA**"*

When "clear of conflict":

- Callsign
- **"CLEAR OF CONFLICT"**
- **"RETURNING ..." / "... RESUMED"**

*"FastJet 123, **CLEAR OF CONFLICT, RETURNING FL150 / FL150 RESUMED**"*

# Basic Failure Management

| Step | Action | Description |
|---|---|---|
| 1 | Identify and call-out | The first pilot to recognize a non-normal situation should call this to the attention of the other pilot:<br>(e.g. "ENGINE OUT / FIRE, WINDSHEAR, STALL, UPSET, MASTER CAUTION / WARNING …" )<br>Cancel MASTER CAUTION / WARNING flasher. |
| 2A | System check (time available) | Where applicable, check the following (C-S-I):<br>• Control: system correctly configured<br>• Supply: system source, check circuit breakers (but do not reset)<br>• Indication: compare flight instruments, check EICAS and synoptic pages |
| 2B | Recovery Action (time critical) | If applicable, perform relevant recovery action by memory:<br>• Emergency descent<br>• Windshear escape<br>• Terrain escape<br>• Upset recovery<br>• Stall recovery |
| 3 | Short-term plan | This step is meant to initially keep the aircraft on a safe trajectory away from terrain / obstacles while dealing with the problem, or towards a place where a safe landing can be made if the situation is time-critical (e.g. SMOKE, FIRE, BOMB on board).<br>• Level-off and set course to the nearest suitable airport<br>• Following engine out departure procedure<br>• Drift down or depressurization procedure |
| 4 | ECL actions | Communications are transferred to the PF while the PM confirms the checklist title and then reads the checklists out loud in the order given (as applicable):<br>1. WARNING<br>2. Priority procedure<br>3. NORMAL<br>4. CAUTION<br>5. Non-normal procedure<br>Confirmation is required in-flight for "irreversible" actions:<br>• Engines: thrust levers, engine run switches<br>• All guarded switches<br>• Checklist conditions: ○YES ○NO (read as a YES or NO question)<br>Read LIMITATIONS (lower part of SUMMARY page) and preview DEFERRED ITEMS. |
| 5 | Long-term plan | After completing the applicable failure procedure and collecting all available data (e.g. weather), decide on a long-term solution.<br>• Decide using FORDEC (return, continue, divert)<br>• Communicate the plan to ATC, cabin crew (NITS) and passengers<br>• Execute the plan and revise if needed (redo FORDEC) |

Non-annunciated (non-EICAS) checklists are called "procedures". There are two categories of procedures: *priority* and *non-normal* procedures.

Although certain maneuvers are carried out by memory (see step - 3 recovery action), the A220 does not have memory or immediate action items as part of checklists. All checklists are read and do.

# Step 1 – Identify and call-out failure or non-normal situation

### a) Annunciated failures

The EICAS provides flight crew with operational support for the management of normal, abnormal and emergency situations. The EICAS generates MASTER WARNING and MASTER CAUTION alerts when a non-normal condition is detected.

The source of the non-normal condition is identified using the EICAS together with local lights (e.g. FAIL light on overhead panel switches) and aircraft system synoptic pages.

The non-normal condition shall be called out before cancelling the MASTER WARNING or MASTER CAUTION flasher, for example:

- PM: "MASTER CAUTION - L PACK FAIL" - cancels MASTER CAUTION.

In the case of associated alerts, for example, in case of an engine failure, multiple alerts may be triggered, call:

- PM: "ASSOCIATED" - cancels MASTER WARNING / CAUTION.

| Level | Definition | Indication |
|---|---|---|
| Warning | Immediate awareness, action or crew decision required. | Master Warning + triple chime + flashing red message on EICAS + voice alert for some warnings (e.g. fire) |
| Caution | Immediate awareness, crew action required before end of the flight. | Master caution + single chime + flashing amber message on EICAS |
| Advisory | Crew awareness, no associated in-flight procedures. More info might be available on INFO synoptic page. | Cyan EICAS message flashing 5 seconds, then steady. |
| Status | Selected system states (e.g. SEATBELT ON). | White EICAS message |
| INFO messages are non-alerting-type messages with no in-flight associated procedures. INFO messages are used for dispatch and trouble shooting. They are displayed on the INFO synoptic page and require acknowledgement to remove the EICAS flag (will remain displayed on ground). | | |

### b) Non-annunciated Failures

Some failures are not annunciated, or not directly annunciated. For example, a compressor stall or surge could cause erroneous faults to appear, such as generators dropping offline and engine exceedances. However, there is no EICAS alert directly indicating a compressor stall or surge.

Other conditions will not generate an alert whatsoever and depend on vigilance of the crew in order to be detected timely. Examples of such conditions are pilot incapacitation or smoke / fumes in cockpit or cabin.

Non-exhaustive list of non-EICAS events:

- Pilot incapacitation
- Forced landing or ditching
- Unlawful interference, bomb on board
- Rejected take-off, emergency evacuation, rapid disembarkation
- Air-conditioning smoke, smoke or fumes in the cockpit or cabin
- Instrument failures and unreliable airspeed (disagree is indicated)
- Cracked, shattered or delaminated windows or windshields
- Display Unit failure, engine tailpipe fire, tire burst
- Rudder pedal, slat-flap lever or spoiler lever jammed
- Windshear or wake turbulence encounters, volcanic ash
- TAWS alert, TCAS alert, stall or aircraft upset

These need to be called out as clearly as possible to the attention of the other pilot.

For example:

- Either pilot: "RUDDER PEDAL JAM", "STALL", "UPSET", "WINDSHEAR".

## Step 2A – System check

System checks are performed after identifying the problem (Step 1). When the situation is time critical, this step usually does not apply, or only in an abbreviated form. Continue with Step 2B.

For example:

- With an engine fire indication, the problem is obvious and there is no need for an in-depth analysis of the problem.
- For TCAS, TAWS or windshear alerts, there is no applicable system check.
- In case of aircraft upset, compare aircraft attitude with standby instrument and offside PFD before applying recovery action (very short system check).

Where applicable, system check is performed by the PM (CM2 on ground) as follows (C-S-I):

- Control: check switches and levers – is the system correctly configured?
- Supply: check the system source, is it available? e.g. a pack fault could be caused by a bleed problem. Check circuit breakers (but do not reset).
- Indication: check EICAS, local alerts and synoptic pages – is the indication in accordance with the system configuration?

Under certain conditions, system check is part of a procedure or common sense, system checks are not specifically called for, examples:

- Comparing flight instruments in case of disagree or unreliable airspeed.
- Identifying the source of smoke or fumes.

## Step 2B – Recovery Action

Although the A220 does not have official memory or immediate action items as part of checklists, some situations are time critical and require recovery action in the form of maneuvers to be executed from memory without delay, non-exhaustive list of such events:

- Rejected take-off
- Engine failure after take-off
- Go-around or balked / rejected landing
- Windshear escape maneuver (predictive or reactive)
- TCAS Resolution Advisories (RAs)
- Terrain escape maneuver
- Aircraft upset recovery
- Emergency descent
- Stall recovery

Here the flight crew has to react quickly to resolve the initial crisis. After recovery, when the flight path is stable and time is available; trouble shooting can start regarding the root cause of event, required checklists and follow-up.

Although the situation and actions required may appear obvious, avoid rushing into conclusions. In case of an aircraft upset for example, perform a quick instrument cross-check to ensure it is not just an isolated instrument problem before applying upset recovery techniques.

# Step 3 – Short term plan

The short-term plan aims to place the aircraft on an initial safe trajectory away from conflicting terrain or obstacles. This could simply mean continuing the current flight path, starting a drift down in case of engine failure over high terrain, etc. It's important to remember: Aviate – Navigate – Communicate, in that order.

### a) Critical Phases of Flight

The short-term plan is usually prepared and briefed before any critical phase of flight. For example, flying the engine out procedure in case of engine failure after take-off and then return for landing or diversion to take-off alternate is discussed during the departure briefing. Go-around and missed-approach procedure, whether the OEI MISAP climb gradient is assured or not, etc. is discussed during the arrival briefing.

### b) Time Critical Events

In case of a time critical event (e.g. SMOKE or FIRE of any kind, BOMB onboard, structural damage, etc.) set course to the nearest suitable airport and place the aircraft on a trajectory from where a safe landing can be made whilst dealing with the emergency. In case of fire onboard the time available before the situation becomes unsurvivable is statistically between 15 and 20 minutes, but in severe cases less.

### c) Communications

When time permits, ATC could be informed with an abbreviated MAYDAY or PAN PAN call during this stage. Cabin crew can be informed about the nature of the situation during this stage as well to mentally prepare without providing any details yet. The official NITS briefing is part of the last failure management step after all data has been collected and a plan formulated.

### d) Procedure Initiation

The PF (CM1 on ground) calls for "ECL ACTIONS", before any checklist is read, there must be crew consensus regarding checklist priority. The PM (CM2 on ground) reads the checklist while communications are transferred to the PF (CM1 on ground).

# Step 4 – ECL actions

### a) Checklist Priority

| | |
|---|---|
| 1. EMERGENCY | • WARNING - emergency checklist<br>• Priority procedure (non-EICAS) |
| 2. NORMAL | • NORMAL - normal checklist |
| 3. NON-NORMAL | • CAUTION - non-normal checklist<br>• Non-normal procedure (non-EICAS) |

### b) Checklist Reading

In-flight checklists are considered READ and DO, while on ground checklists are considered CHALLENGE and RESPONSE (except for **emergency evacuation** and **rapid disembarkation** which are READ and DO by CM2). The READ-RESPOND-RESPOND principle is used where the PM (CM2 on ground) reads the challenge and the response, then he himself or CM1 on ground performs the action and repeats the response:

- In flight: PM: "APU GEN...OFF", switches APU GEN – OFF, responds "OFF".
- On ground: CM2: "APU GEN...OFF", CM1 switches APU GEN OFF, responds "OFF".

### c) Conditions

Certain checklist items are considered "conditions" ( ○YES ○NO in ECL). These items require crew evaluation and shall be read as a "YES or NO" question. The "question" must be answered by the other pilot before proceeding with the checklist. Example:

- PM (CM2 on ground): "APU GEN FAIL caution message stays out: YES or NO?"
- PF (CM1 on ground): "YES" or "NO"

### d) Irreversible items

The following items are considered as "irreversible" actions. In-flight, these must be confirmed by the other pilot: thrust levers, engine run switches, all guarded switches and checklist conditions (O) in ECL (read as a YES or NO question):

- PF: "CONFIRM LEFT THRUST LEVER?", PM: "LEFT THRUST LEVER CONFIRMED"
- "Mental-two-count" – even after confirmation from the other pilot, thrust lever angle should be reduced by 50% for 2 seconds, note engine response, then set flight idle.

# Step 5 – Long term plan

During this final step, all information is gathered and a plan formulated. The time available dictates to what level and in what detail a plan is prepared. In case of time critical emergency, the decision is relatively straight forward, namely, land at the nearest suitable airport. The short-term plan in such case is also the long-term plan.

First, all facts must be collected about the technical state of the aircraft, available landing options and weather conditions. Consider consequences of related failure of systems, the possibility to return after take-off, continue to destination, diverting to (take-off) alternate or the nearest suitable airport, flight time, approach aids, runway length, emergency services, aircraft performance, weather conditions, fuel status, maintenance availability and services available for passengers (non-exhaustive list).

FORDEC model:

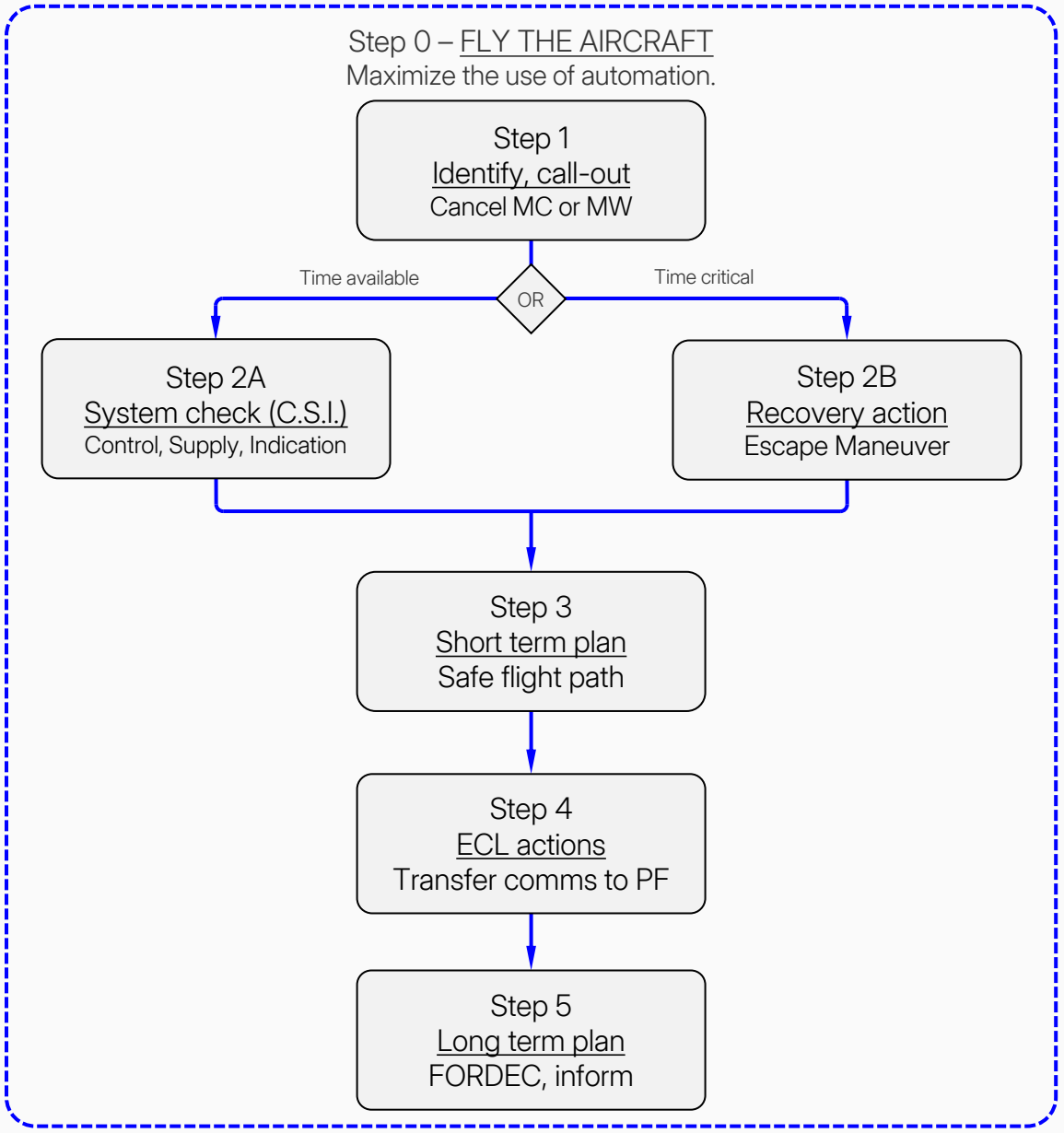
- Facts – What are the facts: airports, technical, limitations, weather, etc.
- Options – What options do we have? Continue, return, divert?
- Risks & Benefits – What are the risks and benefits of each option?
- Decide – Which option is best suited, has most benefits and least risks?
- Execute – Who will do what, when and how? Communicate the plan.
- Check – Are the assumptions still correct? If not, start from facts.

After a decision has been made, the plan must be communicated to everyone involved:

- Air Traffic Control
- Cabin Crew – NITS
- Passengers

Some checklists may include “deferred items”, such items are deferred to a later flight phase such as descent and approach or before landing checklist. Deferred items may include limitations or affect landing performance, it is therefore advisable to review these as part of the long-term plan, especially if landing distance is known to be critical.

For failures where a $\Delta V_{REF}$ is given, this increment must be added to $V_{REF}$, $V_{AC}$ and $V_{GA}$. In the event of multiple failures, use the highest $\Delta V_{REF}$. In case of multiple OLD factors, use the highest for related failures or multiply factors for unrelated failures.



# Basic Failure Management Example

| Event | Pilot Flying | Pilot Monitoring |
|---|---|---|
| Step 1<br>Identify & Call-out | If the PM is preoccupied call: "IDENTIFY" | "MASTER CAUTION, L PACK FAIL"<br>• Press to cancel MASTER CAUTION<br>• Open CHKL on ECL SUMMARY tab |
| Step 2A<br>System Check | | • Control: PACK switch correct position.<br>• Supply: BLEED available, check CBs.<br>• Indication: check synoptic pages, lights |
| Step 2B<br>Recovery Action | Not applicable. | |
| Step 3<br>Short term plan | Safe flight path, above MSA. | |
| Step 4<br>ECL Actions<br><br>1. WARNING<br>2. Priority procedures<br>3. Normal checklists<br>4. CAUTION<br>5. Non-normal procedures | "ECL ACTIONS, I HAVE ATC"<br><br>"YES" or "NO"<br><br>"CONFIRM"<br><br>"YOU HAVE ATC" | "YOU HAVE ATC"<br>• Title of procedure.....ANNOUNCE "L PACK FAIL"<br>• OEB / ECL.....PERFORM<br>READ-RESPOND-RESPOND:<br>"L PACK ... OFF", perform action, "OFF"<br>CHECKLIST CONDITIONS:<br>"L PACK FAIL CAUTION MESSAGE GOES OUT, YES or NO?"<br>COMPLETE CHECKLIST:<br>"COMPLETE" or "COMPLETE W/DFRD"<br>• COMPLETE (W/DFRD).....SELECT<br>Repeat the above procedure for all other checklists in order of priority (use pointer). After completing all non-normal checklists and procedures on the SUMMARY page:<br>• LIMITATIONS..... READ<br>• DEFFERED ITEMS.....PREVIEW<br>"ECL ACTIONS COMPLETED"<br>"I HAVE ATC" |
| Step 5<br>Long term plan | a. Collect facts, limitations, how will it affect dispatch if no maintenance at destination?<br>b. Decide on course of action using FORDEC decision model (return, continue, divert?)<br>c. Communicate plan (if required) to ATC, cabin crew (NITS briefing) and passengers.<br>d. Execute the plan, revise plan as new facts are becoming available (redo FORDEC). | |

# Rejected Take-off

To assist with the decision-making process, the take-off inhibit function inhibits alerts that are non-essential between 80 kts and 400 ft AGL. Therefore, any warnings triggered during this period must be considered as significant.

The decision to reject the take-off is the responsibility of the Commander. Therefore, the Commander should keep his/her hand on the thrust levers until the aircraft reaches $V_1$, regardless whether he is Pilot Flying (PF) or Pilot Monitoring (PM).

The PM shall closely monitor essential instruments during the take-off roll and will immediately call out as clearly as possible abnormalities, or any adverse condition significantly affecting safety of flight. e.g. "ENGINE EXCEEDANCE" or e.g. "MASTER WARNING ...".

The decision to reject the take-off must be made timely, not later than $V_1$. If the decision is to reject the take-off, the Commander must clearly announce, **"STOP".** This announcement both confirms the decision to abort the take-off and transfer of controls to CM1. It is the only time that hand-over of control is not accompanied by the phrase "MY CONTROLS". If a malfunction occurs and the Commander's decision is to continue the take-off, he/she should announce "CONTINUE".

Below 100 kts the take-off is in the low energy regime and can be aborted safely without significant risk. Take-off is aborted in case of any failure or adverse condition.

Between 100 kts and $V_1$ the take-off is in the high energy regime, take-off is aborted in case of major malfunction or adverse conditions only (non-exhaustive list):

- Collision hazard
- Master warning
- Obvious pilot incapacitation
- Smoke or fire of any kind
- Engine related problems
- Windshear encounter
- Major or multiple bird strike event
- Problems controlling the aircraft
- Aircraft unsafe or unable to fly

| Event | Crewmember 1 | Crewmember 2 |
|---|---|---|
| Malfunction or adverse condition | Either pilot (whoever notices first), calling out the condition as clearly as possible:<br>"ENGINE FAILURE / FIRE", "BIRDS", "MASTER CAUTION / WARNING ..." | |
| Decision making | Any condition below 100 kts or major conditions between 100 kts and $V_1$: "STOP"<br><br>In other cases: "CONTINUE" | |
| Rejected take-off | Immediately and simultaneously:<br>• Thrust levers.....REV IDLE<br>• Brakes.....MAX<br>*Manual braking if no autobrake.*<br>• Thrust levers.....MAX REV<br>"CHECKED" | "REVERSE GREEN" or "NO REVERSE"<br>At 80 kts: "80 KTS" |
| Aircraft stopped | • Thrust levers.....IDLE<br>• PARK BRAKE..... SET<br>• CABIN CREW (PA).....NOTIFY<br>"ATTENTION - CREW AT STATIONS" | • ATC.....NOTIFY<br>"... STOPPING" |
| Situation assessment | • SITUATION ASSESSMENT..... PERFORM<br>• ECL ACTIONS.....PERFORM | |
| | If evacuation required:<br>CM1: "EMERGENCY EVACUATION PROCEDURE"<br>• CHKL.....DU2 and DU3<br>• EMERGENCY EVACUATION PRIORITY PROCEDURE.....READ & DO (CM2) | |
| | • CABIN CREW (PA).....NOTIFY<br>"EVACUATE, EVACUATE" | • ATC.....NOTIFY<br>"MAYDAY x3, ...." |
| | If evacuation not required:<br>CM1: "REJECTED TAKEOFF PROCEDURE"<br>• REJECTED TAKEOFF PRIORITY PROCEDURE.....PERFORM | |
| | • CABIN CREW (PA).....NOTIFY<br>"CABIN CREW & PAX REMAIN SEATED" | • ATC.....NOTIFY<br>"... , REQUEST TAXI" |
| | **Note:** before attempting another take-off:<br><br>• Monitor brake temperature. It could take up to 10 minutes for max brake temperature to be reached at the sensor location. If BRAKE OVHT (warning) appears, no dispatch.<br>• If take-off is rejected above 30 kts groundspeed, consider new de/anti-icing treatment.<br>• If rejected take-off due to autothrottle disengagement, consider standing take-off (wind permitting) until take-off thrust achieved or use manual (maximum TO) thrust.<br>• Set-up for a new departure, continue from BEFORE TAXI CHECKLIST.<br>• Check fuel on board versus minimum fuel required for take-off. | |

# Single Engine Operation

The causes of engine flameout can generally be divided into two categories:

- External causes such as icing, volcanic ash, severe turbulence, fuel mismanagement. These causes, which can affect both engines simultaneously can be determined easily and an immediate relight can be attempted.
- Internal causes such as engine stalls or failures, usually affect a single engine. These causes are not easily determined. In these cases, the engine is shut down first, then the cause of the flameout is investigated.
- If the cause of the flameout cannot be determined, the need for engine restart should be evaluated against the risk of further engine damage or fire that may result from a restart attempt.

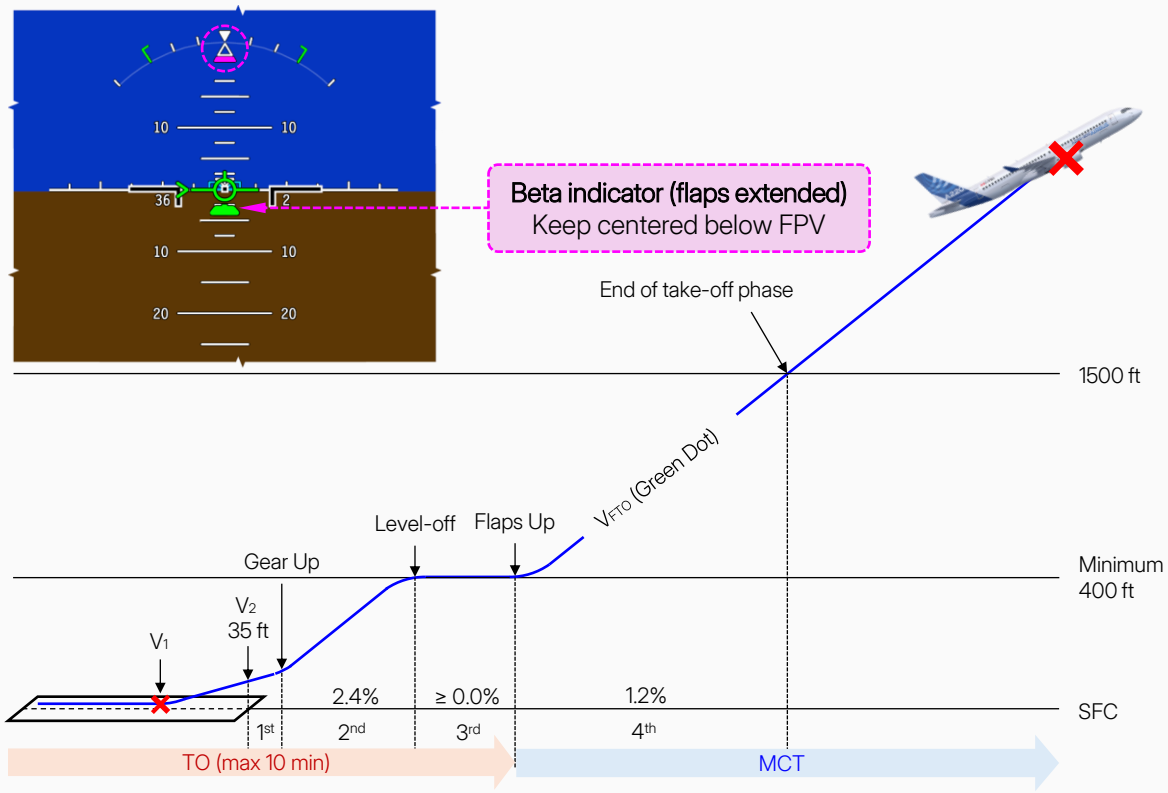
Automatic Power Reserve (APR) provides automatic, additional thrust when an OELT condition exists during take-off (up to 400 ft AGL). APR is automatically armed during a derated or FLEX take-off (can be disarmed in the FMS). When an OELT condition occurs, FADEC will increase the $N_1$ to the next thrust level (e.g. TO-2 to TO-1).

One-engine Inoperative (OEI) guidance mode is active during take-off and go-around with an engine failure or Opposite Engine Low Thrust (OELT) event until slats / flaps are up. Guidance provides optimum rate of climb (beta indicator). When slats / flaps are up, the magenta slip indicator becomes white again, the green filled trapezoid displayed under the FPV remains displayed as an open trapezoid.

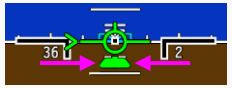
The FBW (in normal mode) provides partial compensation in case of engine failure or other failures causing yaw. Sideslip angle is limited to 5°, bank angle is limited to 20°. The pilot must take manual control of the rudder, using rudder pedals and rudder trim, even with autopilot engaged.

| Bank Angle Limitations | | | | | |
|---|---|---|---|---|---|
| **Configuration** | **TKOF** | **CLB** | **Green Dot** | **APCH** | **GA** |
| One Engine Inoperative | 15° ($V_2$)<br>30° ($V_2$ + 10) | 15° (F) | 25° | 25° ($V_{REF}$)<br>30° ($V_{REF}$ + 5) | 20° ($V_{AC}$) |
| **Note:** until $V_2$ + 10 kts or green dot (FLAP 0) is reached, HDG mode must be used in the ½ BANK mode. | | | | | |

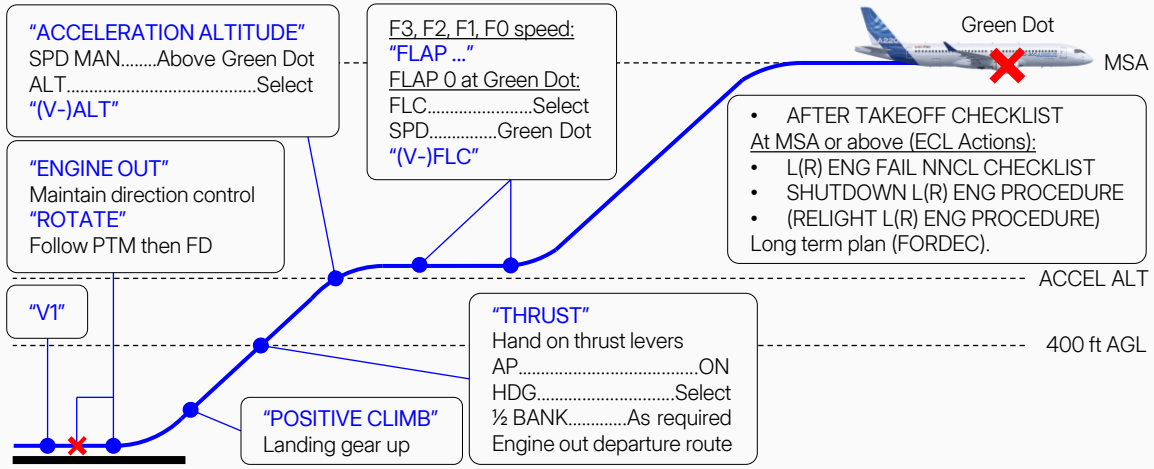
| Segment | Gradient | Speed | Thrust | Start | End |
|---|---|---|---|---|---|
| 1st | > 0.0% | $V_2$ (min) | TO | 35 ft AGL | GEAR UP |
| 2nd | 2.4% | $V_2$ to $V_2$ +10 kts | TO | GEAR UP | ACCEL ALT |
| 3rd | ≥ 0.0% | $V_2$ to $V_{FTO}$ | TO | ACCEL ALT | FLAPS UP |
| 4th | 1.2% | $V_{FTO}$ (GD) | MCT | FLAPS UP | 1500 ft AGL |



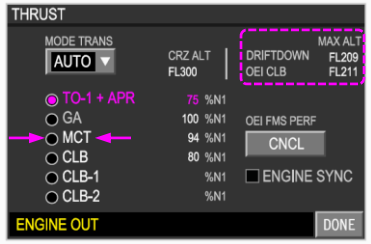
# Engine Failure at Take-off

| Event | Pilot Flying | Pilot Monitoring |
|---|---|---|
| At $V_1$ | CM1 removes hand from thrust levers. | "V1" (auto-callout) |
| Engine flameout | Keep directional control.<br>"CHECKED" | "ENGINE FAILURE" |
| At $V_R$ | Rotate with 3 to 5 degrees / sec towards PTM and transition to FD when available. | "ROTATE" |
| Positive rate of climb indicated | "GEAR UP" | "POSITIVE CLIMB"<br>• Landing gear.....UP<br>"GEAR UP" |
| 400 ft AGL | • Autothrottle HOLD ➡ THRUST mode: Call: "THRUST", hand on thrust levers<br>• Autopilot.....As desired<br>• HDG + ½ bank.....Select | Any EICAS warnings or cautions associated to the engine failure, press to cancel MASTER CAUTION / WARNING:<br>"ASSOCIATED" |
| Step 3 - short term plan (engine out procedure) | Follow engine out departure procedure.<br>If turns are required below ACCEL ALT, HDG ½ BANK must be used until $V_2$ +10 kts | Time permitting, short "MAYDAY" call. |
| Acceleration Altitude (OEI)<br>← Green Dot<br> <br>ACCEL ALT → | At acceleration altitude:<br>• SPD MAN.....Above green dot<br>*To prevent thrust roll-back in SPD mode.*<br>• ALT.....Select<br>"(V-)ALT"<br>At F3, F2, F1, F0 speed:<br>"FLAP ..."<br>FLAP 0 and green dot:<br>• FLC.....Select<br>• SPD MAN.....Adjust to green dot<br>"(V-)FLC" | "ACCELERATION ALTITUDE"<br>"CHECKED"<br>"SPEED CHECKED", "FLAP ..."<br>"CHECKED" |
| Flaps up and above green dot speed | "AFTER TAKEOFF CHECKLIST"<br>Hands on sidestick and thrust levers until at least 2500 ft AGL. Use rudder trim to keep the beta indicator centered below the FPV.<br>  | "AFTER TAKEOFF CHECKLIST"<br>☑ Landing gear.....UP<br>☑ FLAP.....0<br>☑ ANTI-ICE, L COWL.....AUTO<br>☑ ANTI-ICE, WING.....AUTO<br>☑ ANTI-ICE, R COWL.....AUTO<br>☑ APU BLEED.....Select auto<br>☐ EICAS.....Checked<br>"AFTER TAKEOFF CHKL COMPLETE" |

| Event | Pilot Flying | Pilot Monitoring |
|---|---|---|
| ECL Actions<br><br>Preferably climb to MSA before continuing.<br><br>~~1. WARNING~~<br>~~2. Priority procedures~~<br>~~3. Normal checklists~~<br>4. CAUTION<br>5. Non-normal procedures<br><br>Do NOT relight engine if:<br><br>• Abnormal vibrations<br>• Loss of oil QTY/PRESS<br>• No rotational speeds ($N_1$ or $N_2$ at 0% RPM)<br>• Surges, exceedances<br>• Shutdown due to fire or otherwise damaged (e.g. bird strikes). | "ECL ACTIONS, I HAVE ATC"<br><br>"CONFIRM"<br><br>"L(R) ENG RUN CONFIRMED"<br><br>Check indications: "YES" or "NO"<br><br>"CONFIRM"<br><br>"YES" or "NO"<br><br>"YOU HAVE ATC" | "YOU HAVE ATC"<br>Title: "L(R) ENG FAIL"<br>Shutdown – Left (Right) engine procedure.....Accomplish<br>"COMPLETE (W/DFRD)"<br><br>Title: "SHUTDOWN L(R) ENG"<br>Confirm AT / AP, thrust levers (mental-2-count), engine run switches and checklist conditions:<br>e.g. "CONFIRM L(R) ENG RUN?"<br><br>Select ENG run OFF, respond: "OFF"<br>"ENGINE DAMAGE SUSPECTED OR PRECAUTIONARY SHUTDOWN YES or NO?<br><br>If NO: Relight – Left (Right) engine Procedure.....Accomplish<br>"COMPLETE (W/DFRD)"<br><br>Title: "RELIGHT L(R) ENG"<br>Perform non-normal procedure.<br>"ENGINE RELIGHTS YES or NO?"<br><br>Read limitations and preview deferred items.<br><br>"ECL ACTIONS COMPLETED"<br>"I HAVE ATC" |
| Step 5<br>Long term plan | a. Facts: limitations, weather, weight, consider deferred items (landing performance).<br>b. Decide on course of action using FORDEC decision model (return, take-off alternate?)<br>c. Communicate the plan to ATC, cabin crew (NITS briefing) and passengers.<br>d. Execute plan, revise plan as new facts are becoming available (redo FORDEC). | |



# Engine Failure in Flight

| Event | Pilot Flying | Pilot Monitoring |
|---|---|---|
| Step 1<br>Identify & Call-out | Either pilot: "ENGINE FAILURE"<br>Apply rudder input, then trim the rudder. | Any associated CAUTIONs / WARNINGs:<br>"ASSOCIATED" |
| Step 2A - System Check | Not applicable. | |
| Step 2B - Recovery Action | Not applicable. | |
| Step 3<br>Short term plan<br>(if required) | Driftdown procedure:<br>• Thrust.....MCT<br>• ALT SEL.....DRIFTDOWN<br>• HDG.....As required<br>At green dot speed:<br>• AT.....Disengage<br>*To prevent thrust reduction to idle in FLC*<br>• SPD MAN.....Green dot<br>• FLC.....Select<br>At driftdown altitude:<br>• AT.....As required | "MAYDAY x3, ..."<br>  |
| Step 4<br>ECL Actions<br><br>~~1. WARNING~~<br>~~2. Priority procedures~~<br>~~3. Normal checklists~~<br>4. CAUTION<br>5. Non-normal procedures<br><br>Do NOT relight engine if:<br><br>• Abnormal vibrations<br>• Loss of oil QTY/PRESS<br>• No rotational speeds ($N_1$ or $N_2$ at 0% RPM)<br>• Surges, exceedances<br>• Shutdown due to fire or otherwise damaged (e.g. bird strikes). | "ECL ACTIONS, I HAVE ATC"<br><br>"CONFIRM"<br><br>"CONFIRM"<br><br>"CONFIRM"<br><br>"YOU HAVE ATC" | "YOU HAVE ATC"<br><br>Title: "L(R) ENG FAIL"<br>Shutdown – Left (Right) engine procedure .....Accomplish<br>"COMPLETE (W/DFRD)"<br><br>Title: "SHUTDOWN L(R) ENG"<br>Confirm AT / AP, thrust levers (mental-2-count), engine run switches and conditions.<br><br>Relight Left (Right) engine procedure ..... Accomplish if applicable<br>"COMPLETE (W/DFRD)"<br><br>Title: "RELIGHT L(R) ENG"<br>Perform non-normal procedure.<br>"COMPLETE (W/DFRD)"<br><br>Read limitations, preview deferred items.<br><br>"ECL ACTIONS COMPLETED"<br>"I HAVE ATC" |
| Step 5<br>Long term plan | a. Facts: limitations, weather, weight, consider deferred items (landing performance).<br>b. Decide on course of action using FORDEC decision model (return, continue, divert?)<br>c. Communicate the plan to ATC, cabin crew (NITS briefing) and passengers.<br>d. Execute plan, revise plan as new facts are becoming available (redo FORDEC). | |

# Single Engine Go-around

| Event | Pilot Flying | Pilot Monitoring |
|---|---|---|
| Initiating<br>go-around | "GO-AROUND, FLAP 2(4)"<br><br>• TOGA..... Select<br>• Thrust levers.....Go-around thrust<br>• Maintain directional control.<br>• Rotate to follow FD. | In case of FLAP 4 landing:<br>• FLAP.....Select 2<br>In case of FLAP 5 landing:<br>• FLAP..... Select 4<br>"FLAP 2(4)" |
| Engine<br>flameout | Keep directional control.<br>"CHECKED" | "ENGINE FAILURE" |
| Positive rate of<br>climb indicated | "GEAR UP"<br><br>"SELECT MAN SPD, SET $V_{AC}$" | "POSITIVE CLIMB"<br><br>• Landing gear.....UP<br>"GEAR UP"<br><br>• SPD MAN.....Set $V_{AC}$<br>"MAN SPD SELECTED, $V_{AC}$ SET" |
| 400 ft AGL | • Confirm or select lateral mode:<br>"FMS 1(2)"<br>• Autopilot.....As desired<br>"AUTOPILOT ON" | Time permitting, short "MAYDAY" call.<br>"CHECKED"<br><br>"CHECKED" |
| Acceleration altitude:<br>MSA or MISAP ALT | At acceleration altitude:<br>• FLC.....Select<br>"(V-)FLC"<br>• SPD MAN.....Green dot<br>At F3, F2, F1, F0 speed:<br>"FLAP ..." | "ACCELERATION ALTITUDE"<br><br>"CHECKED"<br><br>"SPEED CHECKED", "FLAP ..." |
| Flaps up and above<br>green dot speed | "AFTER GO-AROUND CHECKLIST"<br><br>Hands on sidestick and thrust levers until at least 2500 ft AGL. Use rudder trim to keep the beta indicator centered below the FPV. | "AFTER GO-AROUND CHECKLIST"<br>☑ Landing gear.....UP<br>☑ FLAP.....0<br>☐ EICAS.....Checked<br>"AFTER GO-AROUND CL COMPLETE" |
| Step 4<br>ECL Actions | • L(R) ENG FAIL NON-NORMAL CHECKLIST<br>• SHUTDOWN L(R) ENG NON-NORMAL PROCEDURE<br>• RELIGHT L(R) ENG NON-NORMAL PROCEDURE (if applicable) | |
| Step 5<br>Long term plan | a. Facts: limitations (no CAT III), weather, weight, deferred items (landing performance).<br>b. Decide on course of action using FORDEC decision model (return or divert?).<br>c. Communicate the plan to ATC, cabin crew (NITS briefing) and passengers.<br>d. Execute plan, revise plan as new facts are becoming available (redo FORDEC). | |

# Engine Fire on Ground

| Event | Crewmember 1 | Crewmember 2 |
|---|---|---|
| Step 1<br>Identify & Call-out | "CHECKED" | "MASTER WARNING, L(R) ENG FIRE"<br>• Press to cancel MASTER WARNING<br>• Open CHKL on ECL SUMMARY tab |
| Step 2A<br>System Check | Engine fire is recognized by MASTER WARNING + L(R) ENG FIRE on EICAS + aural "LEFT (RIGHT) ENGINE FIRE" + FIRE flag above N1 indicator + FIRE light below ENG run switch. | |
| Step 2B<br>Recovery Action | • Thrust levers.....IDLE<br>• PARK BRAKE.....SET<br>• CABIN CREW (PA).....NOTIFY<br>"ATTENTION - CREW AT STATIONS" | • ATC.....NOTIFY<br>"MAYDAY x3, ..." |
| Step 3<br>Short term plan | Not applicable. | |
| Step 4<br>ECL Actions<br><br>1. WARNING<br>2. Priority procedures<br>~~3. Normal checklists~~<br>~~4. CAUTION~~<br>~~5. Non-normal procedures~~ | "ECL ACTIONS, I HAVE ATC"<br><br>"NO"<br><br>Set parking brake, respond: "ON"<br><br>Check thrust levers idle, respond: "IDLE"<br><br>No confirmation required for non-reversible actions on ground.<br><br>"CONFIRM"<br><br>"YOU HAVE ATC" | "YOU HAVE ATC"<br><br>Title: "L(R) ENG FIRE"<br>"IN FLIGHT, YES or NO?"<br><br>Challenge – Response – Response:<br>Example, read: "PARK BRAKE...ON"<br><br>Read: "THRUST LEVERS...IDLE"<br><br>No confirmation required for non-reversible actions on ground.<br><br>"COMPLETE"<br><br>"ECL ACTIONS COMPLETED"<br>"I HAVE ATC" |
| | <u>If evacuation required:</u><br>CM1: "EMERGENCY EVACUATION PROCEDURE"<br>• CHKL.....DU2 <u>and</u> DU3<br>• EMERGENCY EVACUATION PRIORITY PROCEDURE.....READ & DO (CM2) | |
| | • CABIN CREW (PA).....NOTIFY<br>"EVACUATE, EVACUATE" | • ATC.....NOTIFY<br>"... , EVACUATION IN PROGRESS" |
| | <u>If evacuation not required:</u> | |
| | • CABIN CREW (PA).....NOTIFY<br>"CABIN CREW & PAX REMAIN SEATED" | • ATC.....NOTIFY<br>"... , REQUEST TAXI" |
| Step 5<br>Long term plan | Return to ramp. | |

# Engine Fire at Take-off

| Event | Pilot Flying | Pilot Monitoring |
|---|---|---|
| At $V_1$ | | "V1" (auto-callout) |
| | CM1 removes hand from thrust levers. | |
| Engine fire | "CHECKED" | "MASTER WARNING, L(R) ENG FIRE"<br>Press to cancel MASTER WARNING |
| At $V_R$ | Rotate with 3 to 5 degrees / sec towards PTM and transition to FD when available. | "ROTATE" |
| Positive rate of climb indicated | "GEAR UP" | "POSITIVE CLIMB"<br>• Landing gear.....UP<br>"GEAR UP" |
| 400 ft AGL | • Autopilot.....As desired<br>• HDG + ½ bank.....Select | |
| Step 3<br>Short term plan<br><br>Fire fighting can be started when above 400 ft AGL. | Follow engine out departure procedure. If turns are required below ACCEL ALT, HDG ½ BANK must be used until $V_2$ +10 kts.<br><br>"ECL ACTIONS, I HAVE ATC"<br><br>"YES"<br><br>At ACCEL ALT: "STOP ECL" | "YOU HAVE ATC"<br><br>Title: "L(R) ENG FIRE"<br>"IN FLIGHT, YES or NO?"<br><br>Confirm AT / AP, thrust levers, engine run switches, guarded switches, conditions. |
| Acceleration Altitude (OEI) | <u>At acceleration altitude:</u><br>• SPD MAN.....Above green dot<br>• ALT.....Select<br>"(V-)ALT"<br><u>At F3, F2, F1, F0 speed:</u><br>"FLAP ..."<br><u>FLAP 0 and green dot:</u><br>• FLC.....Select<br>• SPD MAN.....Adjust to green dot<br>"(V-)FLC" | "ACCELERATION ALTITUDE"<br><br>"CHECKED"<br><br>"FLAP ..."<br><br>"CHECKED" |
| Step 4<br>ECL Actions | "CONTINUE ECL" | Continue the L(R) ENG FIRE checklist. |
| | • AFTER TAKEOFF CHECKLIST<br>• SHUTDOWN L(R) ENG NON-NORMAL PROCEDURE | |
| Step 5<br>Long term plan | In case of an unextinguishable ENG fire, land immediately at the nearest suitable airport. | |

# Smoke, Fire or Fumes on Board

| Event | Pilot Flying | Pilot Monitoring |
|---|---|---|
| Step 1<br>Identify & Call-out | Smoke or fumes in cockpit or cabin (without EICAS alert).<br>Either pilot call out the condition (whoever notices first). | |
| Step 2A<br>System Check | Determine the source of smoke or fumes: ask cabin crew whether there is smoke or fumes in the cabin, if so, from where. Determine if source is air-conditioning, electrical or cabin. | |
| Step 2B<br>Recovery Action | If required:<br>• OXYGEN MASK.....DON, 100% and EMERGENCY<br>• Cabin and crew communication.....Establish<br>**WARNING:** during smoke / fire / fumes emergencies, prepare to land the aircraft without delay while conducting fire suppression and / or smoke evacuation procedures. Any time smoke / fumes becomes the greatest threat, perform smoke or fumes removal procedure.<br>**CAUTION:** passenger masks should not be deployed for smoke or fire procedures. Continuing to destination is not recommended. If the decision to continue is made, the flight crew must acknowledge that the threat has been positively identified, confirmed to be extinguished and the smoke / fumes have dissipated. The crew should consider an overweight landing, tailwind landing, ditching or off-airport landing. | |
| Step 3<br>Short term plan | If climbing, level-off:<br>• ALT.....Select<br>• FMS Direct-to QAK NEAREST ARPTS list<br>• Set course to the nearest suitable airport.<br>• Start descent if required. | • ATC.....Inform<br>"MAYDAY x3, ..." |
| Step 4<br>ECL Actions | "ECL ACTIONS, I HAVE ATC"<br><br>If smoke / fumes becomes greatest threat:<br>"STOP CHECKLIST - SMOKE OR FUMES REMOVAL PROCEDURE"<br><br>"YOU HAVE ATC" | "YOU HAVE ATC"<br><br>Title: "SMOKE / FIRE / FUMES"<br><br>**Note:** Smoke / fire / fumes procedure is not available from ECL, refer to (paper) QRH.<br><br>a. Time does not permit and source cannot be quickly identified, or<br>b. Source is air-conditioning, or<br>c. Source is electrical, or<br>d. Source is cabin<br><br>"ECL ACTIONS COMPLETED"<br>"I HAVE ATC" |
| Step 5<br>Long term plan | • Short term plan is also the long-term plan. Land at the nearest suitable airport.<br>• Do NOT delay landing in order to complete checklists or procedures.<br>• Selecting approach in the FMS (or MAP display) will tune CRS and frequency.<br>• Inform Cabin Crew (NITS) and Passengers, detailed information to ATC (persons and fuel on board, dangerous goods locations (if carried) etc.). | |

## Fire and Smoke Drills

Time is critical during smoke / fire emergencies, whether on ground or in the air. The flight crew shall consider an immediate landing or evacuation anytime the situation cannot be controlled. Statistically there is between 15 and 20 minutes before a fire onboard can become unsurvivable, in severe cases even less.

To continue to destination, the flight crew must acknowledge that the threat has been positively identified, confirmed to be extinguished and the smoke / fumes have dissipated (for example: lavatory bin fire, PED or oven fire).

In case of fire, smoke or fumes:

- Identify source and type of smoke or fire.
- Switch off any electrical supplied involved.
- Use appropriate firefighting equipment.
- Follow appropriate procedures according to QRH.

The Smoke / Fire / Fumes procedure is not available from the ECL, see next pages.

## Oven Fire

Switch off the electrical supply.

- Open oven door slightly and discharge a short burst of BCF into the oven.
- Close the oven door.
- Leave for a few minutes, then with the BCF extinguisher ready, open the oven door and check fire extinguished.
- Should re-ignition occur, repeat the process.
- Put the smoke hood on and repeat the process if required.

Following the discharge of the BCF extinguisher the oven door must be closed and the area cleared of personnel until smoke and fumes disappear.

# Smoke / Fire / Fumes Priority Procedure (not in ECL)

1) Oxygen masks.....If required, on, 100% and EMERGENCY
2) Cabin and crew communication.....Establish
3) Plan to land at the nearest suitable airport.

**WARNING:** during smoke/fire/fumes emergencies, prepare to land the aircraft without delay while conducting fire suppression and/or smoke evacuation procedures.

**CAUTION:** passenger masks should not be deployed when performing smoke or fire procedures.

4) EQUIP COOLING, INLET..... OFF
5) CABIN PWR.....OFF
6) RECIRC AIR..... OFF

**WARNING:** any time smoke/fumes becomes the greatest threat accomplish the Smoke or fumes removal procedure.

**CAUTION:** continuing to destination is not recommended. If the decision to continue is made, the flight crew must acknowledge that the threat has been positively identified, confirmed to be extinguished, and the smoke/fumes have dissipated.

7) Time permits and source of smoke/fire/fumes can be quickly identified and eliminated:

→ **Yes – Go to (11)**
→ **No – Go to (8)**

**8) Time does not permit and source of smoke/fire/fumes cannot be quickly identified:**

**CAUTION:** the crew should consider an overweight landing, tailwind landing, ditching or off-airport landing.

9) Land immediately at the nearest suitable airport.
10) Smoke or fumes removal procedure.....Accomplish if required. Refer to smoke, fire and fumes – Smoke or fumes removal.

**- COMPLETE -**

**11) Time permits and source of smoke/fire/fumes can be quickly identified and eliminated:**
12) Select appropriate procedure for source of smoke/fire/fumes:

→ **Air-conditioning – Got to (13)**
→ **Electrical – Go to (33)**
→ **Cabin – Go to (107)**

**13) Source is identified as coming from air-conditioning:**

**NOTE:** smoke may take considerable amount of time to dissipate.

14) XBLEED..... MAN CLSD
15) APU BLEED..... OFF
16) L PACK.....OFF
17) Altitude.....Not above 31000 feet
18) Smoke/fire/fumes stops:

→ **Yes – Go to (19)**
→ **No – Go to (22)**

**19) Smoke/fire/fumes stops:**
20) Land immediately at the nearest suitable airport.
21) Smoke or fumes removal procedure.....Accomplish if required. Refer to smoke, fire and fumes – Smoke or fumes removal.

**- COMPLETE -**

**22) Smoke/fire/fumes does not stop:**
23) L PACK.....Select auto
24) R PACK.....OFF
25) Smoke/fire/fumes stops:

→ **Yes – Go to (26)**
→ **No – Go to (29)**

**26) Smoke/fire/fumes stops:**
27) Land immediately at the nearest suitable airport.
28) Smoke or fumes removal procedure.....Accomplish if required. Refer to smoke, fire and fumes – Smoke or fumes removal.

**- COMPLETE -**

**29) Smoke/fire/fumes does not stop:**
30) R PACK.....Select auto

**CAUTION:** the crew should consider an overweight landing, tailwind landing, ditching or off-airport landing.

31) Land immediately at the nearest suitable airport.
32) Smoke or fumes removal procedure.....Accomplish if required. Refer to smoke, fire and fumes – Smoke or fumes removal.

**- COMPLETE -**

**33) Source is identified as coming from electrical:**

**NOTE:** smoke may take a considerable amount of time to dissipate.

The series of steps that follow isolate each channel of the electrical system, one at a time.

34) ECL.....DU 5
35) ELEC synoptic page.....Select

**Channel 1 isolation:**

**NOTE:** during this procedure DU 1, DU 3, AC BUS 1, DC BUS 1, and DC ESS BUS 1 will not be powered.

36) BUS ISOL.....MAIN
37) Affected generator.....OFF

**NOTE:** the affected generator supplies power to AC BUS 1 on the ELEC synoptic page.

38) BATT 1.....OFF
39) CTP, CPDR/TCAS.....Select XPDR 2
40) FD.....Couple to left side

**NOTE:** The use of EICAS, REV compresses both PFD and allows two additional MFW. To recover the use of autopilot, ensure FD is coupled to left side and re-engage autopilot if necessary.

41) Smoke/fire/fumes stops:

→ **Yes – Go to (42)**
→ **No – Go to (47)**

**42) Smoke/fire/fumes stops:**

**CAUTION:** Do not land on contaminated runway.

Significant systems affected:

| | | |
|---|---|---|
| APU start | Autopilot (FD right) | Nose landing light |
| APU door actuator | ADSP 1 | Left landing light |
| PTU | Autothrottle | Left taxi light |
| HYD pump 2B | XPDR 1 | L Map + reading lts |
| HYD 1 press ind | TCAS | Cockpit door lock |
| HYD 2 quantity ind | TAWS | Fwd lav smoke det |
| Crew oxy press ind | DME 1 | Toilet flush |
| Rudder trim | VHF 3 | Navigation lights |
| PAX oxy auto dply | HF 1 | Seat belt signs |
| Attendant call lights | SATCOM | L + R pwr outlets |
| Flaps slow | HUD | Fuel inerting |
| Reduced braking | RAD ALT 3 | Trim air |
| Left thrust reverse | DU 1 and DU 3 | TRU 1 |
| L wiper & heating | Tire press ind | ACP 3 |
| Left fuel pump | MKP 1 | |

43) Land immediately at the nearest suitable airport.
44) Reversion Switch Panel, L or R PFD.....Select ADS until ADS source data indication is not amber on both PFDs.

**NOTE:** RNP AR approaches are prohibited.

**On approach:**

45) FMS, PERF – ARR – VREF.....VREF
46) OLD factor.....Multiply by 1.70

**- COMPLETE -**

**47) Smoke/fire/fumes does not stop:**

**Channel 2 isolation:**

**NOTE:** During this procedure DU 4, DU 5, AC BUS 2, DC BUS 2, and DC ESS BUS 2 will not be powered.

48) BUS ISOL.....Confirm MAIN
49) BATT 1..... AUTO
50) Affected generator..... Select ON

**NOTE:** Before Channel 1 isolation, the affected generator supplied power to AC BUS 1 on the ELEC synoptic page.

**After 30 seconds:**

51) ECL.....DU 2
52) Affected generator.....OFF

**NOTE:** The affected generator supplied power to AC BUS 2 on the ELEC synoptic page.

53) BATT 2 .....OFF
54) CTP, XPDR/TCAS.....Select XPDR 1
55) CTP 2.....Select off
56) CTP 1.....ATC on VHF 1, if required
57) Smoke/fire/fumes stops:

→ **Yes – Go to (58)**
→ **No – Go to (70)**

**58) Smoke/fire/fumes stops:**

**CAUTION:** Do not land on contaminated runway.

Significant systems affected:

| | | |
|---|---|---|
| HYD pumps 3A/3B | TCAS | Reduced braking |
| HYD pump 2B | WXR | Right thrust reverse |
| HYD 2 press ind | Wing anti-ice | Right landing light |
| HYD 3 quantity ind | ADSP 2 & ADSP 4 | Nose taxi light |
| Right fuel pump | HF 2 | Right taxi light |
| VHF 2 | FMS 2 | R Map + reading lts |
| CTP 2 | GNSS 2 | Right window heat |
| MKP 2 | APPR 1 | R wiper & heating |
| CCP 2 | DME 2 | Outflow valve MAN |
| PAX oxy auto dply | NAV 2 | Aft lav smoke det |
| Slats | RAD ALT 2 | Recirculation fan |
| Flaps slow | DU 4 and DU 5 | TRU 2 |
| XPDR 2 | R HUD | |

# Smoke / Fire / Fumes Priority Procedure (not in ECL)

59) Land immediately at the nearest suitable airport.
60) Reversion Switch Panel, L or R PFD.....Select ADS until ADS source data indication is not amber on both PFDs.
61) Use AVIONIC, CTP tab for R CTP functions.

**NOTE:** RNP AR approaches are prohibited.

**On approach:**

62) Check SLAT position:

→ **SLAT position IN – Go to (63)**
→ **SLAT position OUT/MID/FULL – Go to (67)**

**63) SLAT position IN:**
64) Maximum landing weight.....Use the table to determine the value with corrections as applicable.

| Landing weight (kg) limited by max brake energy & tire speed (VREF + 15 kts) | | | | | | |
|---|---|---|---|---|---|---|
| OAT °C | Pressure Altitude (ft) | | | | | |
| | 0 | 2000 | 4000 | 6000 | 8000 | 10000 |
| ≤-20 | 86990 | 84410 | 81690 | 79100 | 76430 | 73520 |
| 0 | 84050 | 81370 | 78740 | 76110 | 73480 | 70570 |
| 20 | 81280 | 78690 | 76060 | 73430 | 70480 | 65540 |
| 30 | 80010 | 77380 | 74790 | 72160 | 68170 | 63140 |
| ≥-40 | 78780 | 76150 | 73570 | 70710 | 65770 | 60910 |

**Corrections:**
- Tailwind: decrease maximum landing weight by 18% per 10 kts of tailwind.
- Downhill: decrease maximum landing weight by 2% per 1% downhill slope.
- Decrease maximum landing weight by 14% for 10 kts $V_{APP}$ increase.

65) FMS, PERF – ARR – VREF..... $V_{REF}$ + 15
66) OLD factor.....Multiply by 2.05

**- COMPLETE -**

**67) SLAT position OUT/MID/FULL:**
68) FMS, PERF – ARR – VREF..... $V_{REF}$ + 5
69) OLD factor.....Multiply by 1.85

**- COMPLETE -**

**70) Smoke/fire/fumes does not stop:**

**Channel 3 isolation:**

**NOTE:** During this procedure DU 2, AC ESS BUS, and DC ESS BUS 3 will not be powered.

71) HYD 3A.....OFF
72) BATT 2.....AUTO
73) Affected generator.....Select on

**NOTE:** Before Channel 2 isolation, the affected generator supplied power to AC BUS 2 on the ELEC synoptic page.

74) CTP 2.....Select on

**After 30 seconds:**

75) ECL..... DU 5
76) BUS ISOL..... ESS
77) CTP, XPDR/TCAS..... Select XPDR 2
78) CTP 1..... Select off
79) CTP 2.....ATC on VHF 2, if required
80) Smoke/fire/fumes stops:

→ **Yes – Go to (81)**
→ **No – Go to (88)**

**81) Smoke/fire/fumes stops:**

Significant systems affected:

| | | |
|---|---|---|
| HYD 1 quantity ind | CCP 1 | GNSS 1 (no LPV) |
| HYD 3 press ind | REV switch panel | NAV 1 |
| ISI (ATT only) | DU 2 | RAD ALT 1 |
| Rudder degraded | L HUD | Altn gear extension |
| L fuel pump | Slats / flaps slow | Ram air |
| Gravity transfer | MFS (1 pair) | Equip bay cooling |
| ADSP 3 | Right stick shaker | Left window heat |
| VHF 1 | Wing anti-ice | CDPLC |
| XPDR 1 | PFCC 3 | TRU 3 |
| CTP 1 | FMS 1 (no LPV) | OMS |

82) Land immediately at the nearest suitable airport.
83) FD.....Couple to right side
84) Reversion Switch Panel, L or R PFD.....Select ADS until ADS source data indication is not amber on both PFDs.
85) Use AVIONIC, CTP tab for L CTP functions.

**NOTE:** RNP AR approaches are prohibited.

**On approach:**

86) FMS, PERF – ARR – VREF..... $V_{REF}$
87) OLD factor.....Multiply by 1.20

**- COMPLETE -**

**88) Smoke/fire/fumes does not stop:**

**These steps will deploy the RAT:**

**NOTE:** During this procedure DU 3, DU 4, DU 5, AC BUS 1, AC BUS 2, DC BUS 1, and DC BUS 2 will not be powered.

89) BUS ISOL..... AUTO
90) CTP 1.....Select on, ATC on VHF 1
91) ECL..... DU 2
92) RAT GEN..... ON
93) CTP, XPDR/TCAS..... Select XPDR 1
94) FD..... Couple to left side

**CAUTION:** The **EMER PWR ONLY** warning message will be shown after all generators are selected off. The steps are included in this procedure and do not need to be done separately.

95) L GEN..... OFF
96) R GEN..... OFF
97) APU GEN.....OFF
98) L PACK..... Confirm auto
99) R PACK..... Confirm auto
100) EQUIP COOLING, EXHAUST.....ON
101) Land immediately at the nearest suitable airport
102) Leave/avoid icing conditions.

**CAUTION:** Below 148 KIAS the RAT generator may be inoperative and the airplane will be operating on battery power only. Brakes are not available when batteries are depleted. Do not land on contaminated runway.

103) Airspeed.....Not less than 155 KIAS until landing assured.

**NOTE:** RNP AR approaches are prohibited.

Significant systems affected when operating on RAT power:

| | | |
|---|---|---|
| ACMP 2B, 3A, 3B | L+R thrust reverser | GNSS 2 (no LPV) |
| PTU | Exterior lighting | DME 1 and 2 |
| Rudder trim | L+R ICE detectors | NAV 2 and 3 |
| XPDR 2 | ADS-B | RAD ALT 2 and 3 |
| TCAS | VHF DLK, CPDLC | L, R WSHLD heat |
| Right fuel pump | VHF 2 | R window heating |
| ADSP 1, ADSP 2 | HUD | Left, right wipers |
| Autothrottle | DU 3, DU 4, DU 5 | L+R map & read lts |
| NAV to NAV XFR | CCP 2 (DU 2 only) | Recirculation fan |
| IRU 2 | TAWS | Fuel inerting |
| TRU 1, TRU 2 | WXR | Toilet flush |
| Attendant call lights | HF 2 | Seat belt signs |
| Slats/flaps slow | SATCOM | MKP 2 |

104) Airspeed.....Not less than 155 KIAS until landing assured.

**CAUTION:** Below 148 KIAS the batteries may be the only source of electrical power, and last for at least 5 minutes. Batteries will recharge above 148 KIAS if go-around is required. Brakes are not available when batteries are depleted.

105) Approach speed.....VREF

**NOTE:** FMS VSPEEDS not available.

106) OLD factor.....Multiply by 1.35

**- COMPLETE -**

**107) Source is identified as coming from the cabin:**
108) Designated crew member.....Advise

a. To isolate and extinguish source of smoke or fire, and to secure the area.
b. To turn off all electrical sources via CMS.

**CAUTION:** continuing to destination is not recommended. If the decision to continue is made, the flight crew must acknowledge that the threat has been positively identified, confirmed to be extinguished, and the smoke/fumes have dissipated.

109) Land immediately at the nearest suitable airport.
110) Smoke or fumes removal procedure.....Accomplish if required. Refer to smoke, fire and fumes – Smoke or fumes removal.

**- COMPLETE -**

# Emergency Descent

| Event | Pilot Flying | Pilot Monitoring |
|---|---|---|
| Step 1<br>Identify & Call-out | "CHECKED"<br><br>"CHECKED" | "MASTER CAUTION, CABIN ALT"<br><br>If situation worsens:<br>"MASTER WARNING, CABIN ALT" |
| Step 2A<br>System Check | Check cabin ALT, RATE and DIFF. Rapid or explosive decompression is recognized by physical sensation (ears, sinuses etc.) and may be accompanied by mist in the cabin and loud noise and vibration. These cases usually indicate structural damage. In case of slow decompression, cabin pressure might be restored using the appropriate procedure. | |
| Step 2B<br>Recovery Action | If required:<br>• OXYGEN MASK.....DON, 100%<br>• Crew communication.....Establish<br><br>"EMERGENCY DESCENT"<br>Above 25.000 ft:<br>• EDM.....Select on (if not automatic)<br>Emergency descent:<br>• ALT.....10.000 ft or MSA if higher<br>• FLC.....Select (if no EDM)<br>• HDG.....As required<br>Thrust levers at IDLE:<br>• SPOILER lever.....FULL then MAX<br>• Airspeed.....As required<br>*If structural damage, do not exceed speed at which the decompression occurred. In other cases, do not exceed VMO / MMO.* | • SEAT BELTS.....ON<br>• LDG LTS.....ON<br>• Cabin.....Advise<br>"DESCENT, DESCENT"<br>• PAX OXY (if required).....DPLY |
| Step 3<br>Short term plan | Follow depressurization procedure, escape route to stay clear of terrain (if applicable). | • XPDR.....7700<br>• ATC.....Advise |
| Step 4<br>ECL Actions | • CABIN ALT (Caution) or CABIN ALT (Warning) NON-NORMAL CHECKLIST<br>• EDM: EMERGENCY DESCENT (Caution) or EMERGENCY DESCENT (warning) NNCL<br>• No EDM: EMERGENCY DESCENT NON-NORMAL PROCEDURE<br>• UNPRESSURIZED FLIGHT NON-NORMAL PROCEDURE (after level-off)<br><br>At a safe altitude:<br>• SPOILER lever.....RET<br>• Oxygen masks.....As required<br>*Close container doors, press TEST/RESET.*<br>• EDM.....Select OFF<br>• FCP modes.....As required | • Oxygen masks.....As required<br>*Close container doors, press TEST/RESET.*<br>• Cabin.....Advise<br>"SAFE TO REMOVE MASKS" |
| Step 5<br>Long term plan | • Determine the cause of decompression, ask cabin crew to check for structural damage.<br>• Use FORDEC and decide on a plan (consider structural damage, medical conditions).<br>• Communicate the plan to ATC, cabin crew (NITS) and passengers. | |

The emergency descent maneuver is designed to descent to a safe cabin altitude in the minimum amount of time with the least possible passenger discomfort. If the descent is performed because of rapid loss of cabin pressure, oxygen masks are donned and crew communication is established at the FIRST indication of a pressurization problem. At 40.000 ft, the Time of Useful Consciousness (TUC) is between 15 and 20 seconds.

Passenger oxygen masks will deploy automatically when cabin altitude is above 14.500 ft, seatbelts will switch ON and an announcement will play on the PA. After descent has been initiated, PM will squawk 7700, notify ATC and declare emergency. Also, request the local QNH and MSA if required and watch for conflicting traffic both visually and by using TCAS.

Level-off altitude should be at or above Grid Minima, MSA or 10.000 ft whichever is higher. The crew oxygen masks should not be removed if the cabin altitude remains above 10.000 ft. A situation assessment should be made based on weather, oxygen available, fuel remaining and suitable / available airports.

When the situation is under control and aircraft is established at 10.000 ft or below, the oxygen masks can be removed. To regain boom set/microphone function, close the left oxygen mask container door. The mask does not need to be placed back inside the container, then press the test and reset button and release it, the white flag disappears. Now the boom set/microphone function is recovered.

When conditions permit, the Commander calls the cabin to give a NITS briefing and verify passenger physical condition.

### Emergency Descent Mode (EDM)

The Emergency Descent Mode activates automatically when above 25.000 ft if the cabin altitude exceeds 14.500 ft. It can also be manually selected above 25.000 ft. EDM provides the following functions:

- "EMERGENCY DESCENT" aural alert sounds.
- AP and AT engage (if not already engaged), thrust idle.
- Altitude is preselected to 15.000 ft (can be changed).
- HDG mode engages, and present heading is maintained (can be changed).
- Descent speed is set to VMO -10 kts or MMO – 0.02M.
- Transponder is set to 7700 and seatbelts come on (only for automatic EDM activation).

However, speed brakes do NOT automatically extend.

# TCAS Alerts

| Event | Pilot Flying | Pilot Monitoring |
|---|---|---|
| Traffic Advisory (TA) "TRAFFIC, TRAFFIC" aural alert, ● on MAP | Announce: "TCAS, I HAVE CONTROL"<br><br>Ready to disconnect AP and maneuver in accordance with the Resolution Advisory. | "CHECKED"<br>• SEAT BELTS.....ON<br>• Exterior lights..... ON |
| | Attempt to establish visual contact with the intruder and call:<br>"TRAFFIC IN SIGHT, ... 'O CLOCK, HIGH (LEVEL) (LOW)" | |
| Resolution Advisory "CLIMB / DESCENT" aural alert, ■ on MAP | • AP / PTY.....Press twice<br>*2nd press to cancel AP disconnect warning*<br>• FLIGHT PATH.....AMEND<br>*Maneuver into the green "FLY-TO" zone.*<br>*Initial RA: respond within 5 sec and 0.25 g.*<br>*Increase / reversal: within 2½ sec & 0.35 g.* | • FD..... (both sides) OFF<br>*To force autothrottle into SPD mode.*<br>• FLIGHT PATH.....MONITOR<br>• ATC..... NOTIFY<br>"... , TCAS RA" |
| End of RA, "CLEAR OF CONFLICT" aural alert. | Return promptly and smoothly to the last assigned ATC clearance.<br>• FCP modes..... As required<br>• AP.....ON | • ATC..... NOTIFY<br>"CLEAR OF CONFLICT, RETURNING TO FL ... / FL ... RESUMED" |
| | **Reselect MAP – WX overlay.** | |

**Notes:**
- TCAS Resolution Advisories have priority over ATC clearances, call: "UNABLE, TCAS RA".
- Most Traffic Advisories do not result in Resolution Advisories and most RAs only require a minor altitude change.
- RAs do NOT require evasive maneuvering, g-loading is similar to a 30° bank turn or 45° for increase and reversals.
- To prevent nuisance alerts, reduce vertical rate to 2000 fpm - 2000 ft to go and 1000 fpm - 1000 ft before level off.

| RA | Required Vertical Rate | Aural Alert |
|---|---|---|
| Climb / descent | 1500 ft/min | "CLIMB, CLIMB" (or descent) |
| Crossing climb / descent | 1500 ft/min | "CLIMB, CROSSING CLIMB" x2 (or descent) |
| Maintain climb / descent | 1500 to 4400 ft/min | "MAINTAIN VERTICAL SPEED, MAINTAIN" |
| Maintain crossing climb / descent | 1500 to 4400 ft/min | "MAINTAIN VERTICAL SPEED, CROSSING MAINTAIN" |
| Level-off | 0 ft/min | "LEVEL OFF, LEVEL OFF" |
| Reversal climb / descent | 1500 ft/min | "CLIMB, CLIMB NOW" x2 (or descent) |
| Increase climb / descent | 2500 ft/min | "INCREASE CLIMB, INCREASE CLIMB" (or descent) |
| Preventive RA | No change | "MONITOR VERTICAL SPEED" |
| RA removed | - | "CLEAR OF CONFLICT" |

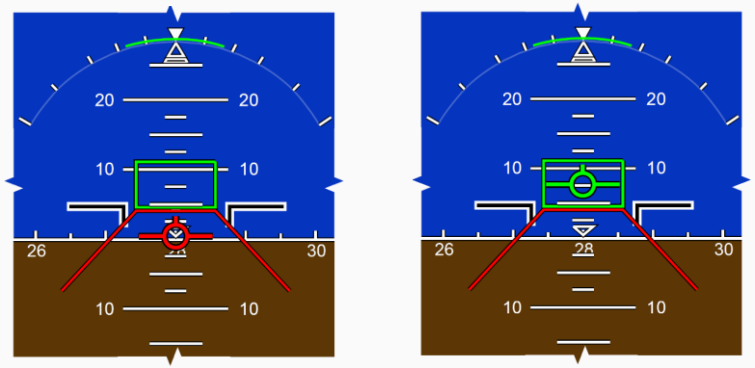
There are two types of TCAS alerts: Traffic Advisory (TA) and Resolution Advisory (RA).

## Traffic Advisory (TA) ●

Traffic Advisory informs the pilot of any surrounding traffic. The TA display shows the intruding aircraft's relative position and altitude with the trend arrow indicating if it is climbing or descending at a rate greater than 500 ft/min.

## Resolution Advisory (RA) ■

Resolution Advisory warns the pilot on the vertical maneuver to carry out to avoid collision with the threat aircraft. Red and green areas are displayed on the PFD, or limitation for climb or descent to avoid a possible collision. Maneuver the FPV into the green "FLY-TO" zone.




1. The TCAS system can only generate Resolution Advisories for intruders equipped with operative mode S or mode C transponders (providing valid intruder's altitude information).
2. Traffic Advisories can only be generated for intruders equipped with operative mode S, C or A transponder (the TCAS system provides no indication of aircraft without operative transponders).

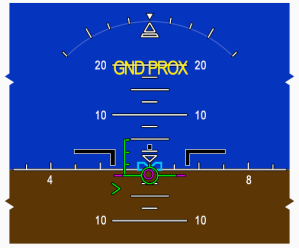
CAUTION: The TCAS equipment is viewed as a supplement to the flight crew who, with the aid of the ATC, have the primary responsibility for avoiding mid-air collisions. In any case, TCAS RAs have priority over ATC instructions.

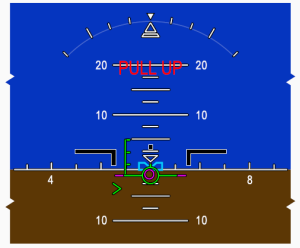
In the case of TCAS resolution advisory, ATC is not responsible for aircraft separation until resuming the initial clearance. Normally, the PF will remain the PF throughout the TA and subsequent RA, however negotiating TAs or RAs is the responsibility of the Commander.

# TAWS Alerts

| Event | Pilot Flying | Pilot Monitoring |
|---|---|---|
| Terrain Caution<br>GND PROX on PFD<br>And aural alerts | Adjust flight path until alert ceases. | |
| Terrain warning<br>PULL UP on PFD<br>Aural "PULL UP" | "PULL UP, MAX THRUST"<br>Rapid and simultaneous action:<br>• AP / PTY..... Press<br>• Sidestick.....SOFT STOP<br>*Sidestick at SOFT STOP initially to trade speed for altitude, respecting stick shaker.*<br>• Thrust levers.....MAX<br>*A momentary EGT exceedance can occur.*<br>• Spoilers..... RET<br>• Configuration.....MAINTAIN<br><br>Approaching $V_{MIN\ TRIM}$:<br>• Sidestick..... As required<br>*Ease sidestick slightly off the SOFT STOP to prevent aircraft from undershooting AoA SOFT, which will cause FBW to lower nose.* | Monitor PF actions, call out omissions.<br><br>Callout radio altitude:<br>"300 FT", "200 FT CLOSING" etc.<br><br>Monitor MAP display and VSD:<br>The terrain display provides situational awareness only, and may not provide the accuracy and/or fidelity upon which to solely base terrain avoidance maneuvering. |
| When flightpath is safe<br>and TAWS alert ceases | • THRUST..... MCT<br>• SPEED..... GREEN DOT<br>CONTINUE MAX GRADIENT CLIMB TO MSA, THEN RESUME NORMAL CLIMB. | |

| Mode | Caution / Advisory | Warning |
|---|---|---|
| 1 | "SINK RATE" | "PULL UP" |
| 2 | "TERRAIN TERRAIN" | "TERRAIN TERRAIN - PULL UP" |
| 3 | "DON'T SINK" | - |
| 4 | "TOO LOW TERRAIN / GEAR / FLAPS" | - |
| 5 | "GLIDE SLOPE"<br>(softer, less frequent) | "GLIDE SLOPE"<br>(louder, more frequent) |
| 6 | "BANK ANGLE"<br>"2500", "1000", "500", "100", "50", "40", "30", "20", "10", "PLUS 100", MINIMUMS" | - |
| 7 | - | "WINDSHEAR" x3 |
| PDA | "TOO LOW TERRAIN" | - |
| FLTA | "TERRAIN / OBSTACLE AHEAD" | "TERRAIN / OBSTACLE AHEAD - PULL UP" |




In case of emergency conditions, or when landing at an airport not in the terrain database, TERR may be switched to INHIB, refer to ditching and forced landing procedure.

In case of landing with landing gear up or unsafe, GEAR may be switched to INHIB, refer to the landing with gear unsafe procedure.

In case of landing with abnormal flaps setting (other than 4 or 5), FLAP may be selected to INHIB position. Refer to the reduced flap landing procedure.

When flying under daylight VMC conditions, should a warning threshold be deliberately exceeded or encountered due to known specific terrain at certain locations, the warning can be regarded as cautionary and the approach can be continued. A go-around shall be initiated in case warning cause cannot be identified immediately.

"SINK RATE"
Adjust pitch attitude and thrust to silence the warning.

"DON'T SINK"
Adjust pitch attitude and thrust to maintain level or climbing flight.

"TOO LOW GEAR" – "TOO LOW FLAPS" – "TOO LOW TERRAIN"
Perform a go-around.

"GLIDE SLOPE" – "GLIDE SLOPE"
Establish the aircraft on the glide slope or go-around.

"BANK ANGLE"
Adjust roll attitude to correct bank angle.

"TERRAIN TERRAIN" – "TERRAIN / OBSTACLE AHEAD" – "TOO LOW TERRAIN"
Verify flight path, if in doubt, perform a maximum performance climb until alert ceases.

"PULL UP" – "TERRAIN TERRAIN - PULL UP" – "TERRAIN / OBSTACLE AHEAD - PULL UP"
Perform TERRAIN ESCAPE maneuver.

"2500", "1000", "500", "100", "50", "40", "30", "20", "10", "PLUS 100", "MINIMUMS"
Situational awareness only.

# Windshear Alerts

| Event | Pilot Flying | Pilot Monitoring |
|---|---|---|
| Predictive windshear<br>WSHR-AHEAD on PFD<br>Aural: "MONITOR RADAR DISPLAY" | Predictive windshear is active below 2400 ft AGL, including on ground (inhibited between 80 kts and 50 ft AGL). Predictive windshear creates caution alerts 3 nm prior entering convective weather, assess situation using MAP WX display, adjust flight path as required. | |
| Predictive windshear<br>WSHR-AHEAD on PFD<br>Aural: "(GO-AROUND) WINDSHEAR AHEAD" | Predictive windshear creates warnings 1 ½ nm prior entering convective weather, assess the situation using the MAP weather display, adjust flight path as required. Be ready to perform the WINDSHEAR ESCAPE maneuver upon entering reactive windshear alert. | |
| Reactive windshear<br>WINDSHEAR on PFD<br>Aural: "WINDSHEAR" x3<br><br>The FD cue is set taking into consideration pitch and low speed limits for the current configuration. Use of FD windshear guidance provides the most effective recovery from windshear encounter. | Either pilot:<br>"WINDSHEAR"<br><br>"WINDSHEAR, MAX THRUST"<br>Rapid and simultaneous action:<br>• AP / PTY.....Press<br>• Thrust levers.....MAX<br>• TOGA.....Select<br>• FD guidance.....Follow<br>*Follow escape guidance (if available) or pull sidestick to the aft SOFT STOP while respecting the stick shaker if no guidance.*<br>• SPOILER lever.....RET<br>• Configuration.....MAINTAIN | Monitor PF actions, call out omissions.<br><br>Callout radio altitude:<br><br>"300 FT DESCENDING"<br>"200 FT LEVEL"<br>"250 FT CLIMBING" etc.<br><br>Monitor wind speed and direction on MAP. |
| When flightpath is safe and out of windshear | • Autopilot.....As required<br>• Autothrottle.....As required<br>• Landing gear.....UP<br>• FD modes.....As required<br>*TO/WSHR or GA/WSHR will remain the active mode until another mode selection is made, the aircraft will not level of at preselected altitude while the WSHR mode is active.*<br>• FLAP.....Retract on schedule<br>• ATC.....Advise | |

There is no sure method of predicting when and where windshear will occur. However the presence of thunderstorms accompanied with heavy rain, presence of virga, strong frontal activity, low level jetstreams, extreme variations of wind speed and/or direction, evidence of gust fronts such as blowing dust over the airport surface, etc. can be identify the existence of windshear.

Strong microbursts can also occur without precipitation, even during sunny weather conditions, or be caused by strong surface winds in the vicinity of small hills or large buildings or sea breezes. Avoidance is the best course of action if windshear is known or suspected in the area. Stay clear of thunderstorm cells, heavy precipitation, virga and areas of known windshear.

When windshear activity is known or suspected:

<u>At departure airport:</u>

- Delay the takeoff until conditions are favorable.
- Use the longest and most favorable runway available.
- Select the minimum flap setting acceptable for the field length available.
- Use full takeoff thrust (no derate or FLEX).
- Monitor the IAS during the takeoff roll.
- Obstacle clearance permitting, consider using an increased initial climb speed.

<u>At arrival airport:</u>

- Delay the approach until conditions are favorable or divert to another airport.
- Use the longest and most favorable runway available.
- Add an appropriate wind correction to the final approach speed.
- Fly a normal approach and consider using the autopilot and autothrottle to allow for more monitoring and recognition time.
- Do not make large thrust reductions or trim changes to correct rapid increases in airspeed as they could vary from an increase to a decrease in a very short time.
- If there is a significant reduction or increase in rate of descent, consider a go-around.

The aircraft is equipped with a Predictive Windshear System (PWS) relying on the weather radar and reactive windshear as part of the TAWS system that alerts for negative windshear (loss of performance) encounters. PWS is active from ground to 2400 ft AGL (inhibited between 80 kts and 50 ft AGL). The reactive windshear system operates between 10 ft AGL and 1500 ft AGL.

**WARNING:** during take-off, closely monitor airspeed for detection of windshear.

# Upset Recovery

| Event | Pilot Flying | Pilot Monitoring |
|---|---|---|
| Step 1 - Identify & Call-out | Either pilot: "UPSET" | |
| Step 2A - System Check | Confirm it is an upset and not instrument failure, compare attitude with ISI and offside PFD. | |
| Step 2B<br>Recovery Action | "UPSET, I HAVE CONTROL"<br><u>Regain situational awareness:</u><br>• Pitch attitude, i.e. nose high or low<br>• Bank angle, airspeed<br>• AP and AT..... OFF<br><u>If nose low:</u><br>• **Unload:** in case of STALL or bank angle exceeding 90°<br>• **Pitch & Roll:** roll towards wings level, pitch input according to bank angle:<br>a. AoB > 90° → PUSH<br>b. AoB 60-90° → NEUTRAL<br>c. AoB < 60° → PULL UP<br>• **Rudder:** neutral if sufficient roll authority<br>• **Thrust:** reduce, spoilers as required<br><u>If nose high:</u><br>• **Unload:** push as much as needed<br>• **Pitch & Roll:** roll to assist in lowering nose, wings level approaching horizon<br>• **Rudder:** neutral if sufficient roll authority<br>• **Thrust:** increase, spoilers retracted | Throughout the recovery, verify all required actions have been taken and call out omissions.<br><br>**NOTE:** reducing thrust may assist in achieving a nose down pitch rate. During recovery, control pitch attitude in a smooth steady manner, respecting stick shaker.<br><br>**CAUTION:** upset recovery is performed referencing the PFD airplane symbol, do not us the Flight Path Vector for recovery.<br><br>**WARNING:** avoid excessive use of pitch trim or rudder which can result in high structural loads. |
| Step 3 - Short term plan | Stabilize the trajectory, climb back to MSA and inform ATC. | |
| Step 4 – ECL Actions | UPSET NON-NORMAL PROCEDURE, relevant checklists depending on cause of upset. | |
| Step 5 - Long term plan | Did the aircraft exceed limits or sustain structural damage? Are passengers or crew hurt? | |

| Actions (U-P-R-T) | Nose High | Nose Low |
|---|---|---|
| Unload | Push the sidestick as much as needed to break the STALL (if applicable) and to lower the nose. | Unload only in case of STALL or if bank angle exceeds 90° (elevators working in reverse). |
| Pitch & Roll | Roll to assist elevators in obtaining nose down or roll the lift vector off if insufficient pitch authority is available (e.g. elevator or stabilizer jam). Roll wings level upon approaching the horizon. | Roll to wings level (lift vector up), pitch as below:<br>• When bank angle > 90° → PUSH<br>• When bank angle 60°-90° → NEUTRAL<br>• When bank angle < 60° → PULL UP |
| Rudder | <u>Use of rudder is not recommended.</u> If control provided by ailerons and spoilers is ineffective, rudder input may be required to induce roll for recovery. A small amount of rudder input is usually sufficient. Excessive rudder applied too quickly or held too long can result in loss of lateral and directional control. | |
| Thrust (drag) | Increase, check spoilers retracted. | Reduce, spoilers extend if required. |

An airplane upset is an undesired aircraft state characterized by unintentional divergences from parameters normally experienced during operations. An airplane upset may involve pitch and/or bank angle divergences as well as inappropriate airspeeds for the conditions. Generally, unintentional flight outside the following parameters is considered an aircraft upset:

- Pitch greater than 25° nose-up
- Pitch greater than 10° nose-down
- Bank angle greater than 45°
- Within these parameters, but at airspeeds inappropriate for conditions

Deviations from the desired airplane state will become greater until action is taken to stop the divergence. Return to the desired airplane state can be achieved through auto-flight system response or pilot intervention. The aircraft is protected against aircraft upset in NORMAL mode.

Active monitoring, in the context of undesired aircraft state, means to proactively keep track of the environment, the aircraft energy state and flight path trajectory. Active monitoring enhances flight crew engagement and combats complacency, ensures awareness and avoidance of undesired aircraft states and provides a strong countermeasure against startle.

Each pilot should:

- Know and understand the expected airplane state for the situation
- Communicate expectations and keep track of current airplane state
- Detect and communicate deviations from expectations
- Assess risk and decide on a response
- Update and communicate understanding
- Take timely corrective actions

Generally, the causes for airplane upsets can be divided into 3 categories (non-exhaustive list):

<u>Pilot induced:</u>

- Poor Instrument cross-check or misinterpretation.
- Incorrect adjustment of pitch, bank and power e.g. inputs being reactionary, unplanned or excessive.
- Inattention and complacency, e.g. neglect of active monitoring or fixation on certain instruments.
- Distraction from primary duties e.g. letting external or internal distractions from controlling the airplane first.
- Misuse of airplane automation or over-relying on automation.

<u>Environmentally Induced:</u>

- Wake Turbulence
- Windshear

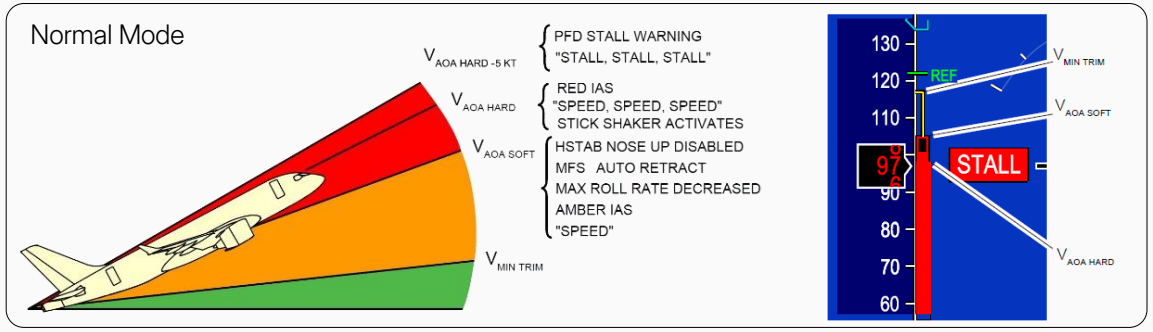
<u>System Induced:</u>

- Flight control DIRECT mode
- Failures inducing yaw and/or roll
- Instrument failure

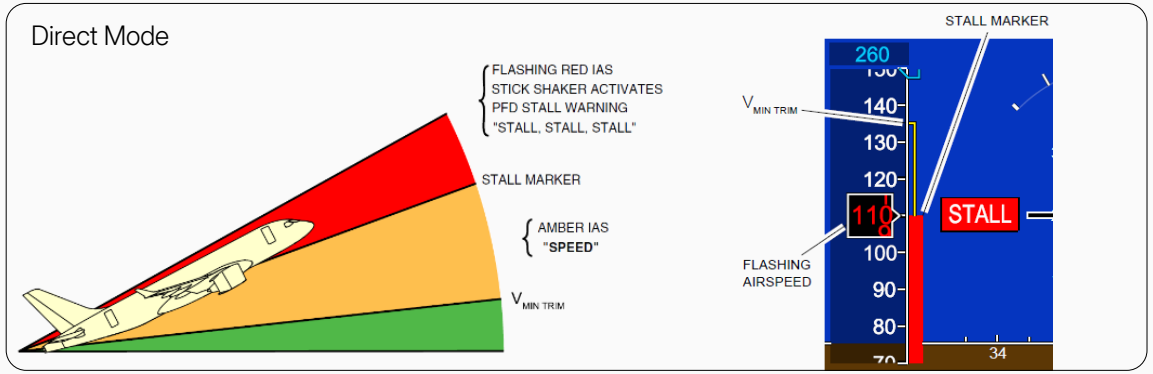
# Stall Recovery

| Event | Pilot Flying | Pilot Monitoring |
|---|---|---|
| Step 1 - Identify & Call-out | Either pilot: "STALL" | |
| Step 2A - System Check | Not applicable. | |
| Step 2B<br>Recovery Action | "STALL, I HAVE CONTROL"<br>• Autopilot..... Disengage<br>• Autothrottle..... Select off<br>• Pitch attitude..... Lower nose<br>• Roll attitude..... Wings level<br>• Thrust Levers..... As required<br>• SPOILER lever.....RET<br>• Return to desired flight path. | Throughout the recovery, verify all required actions have been performed and call-out omissions.<br><br>**Note:** during recovery, control pitch attitude in a smooth steady manner, respecting stick shaker. |
| Step 3 - Short term plan | Stabilize the trajectory, climb back to MSA and inform ATC. | |
| Step 4 – ECL Actions | STALL NON-NORMAL PROCEDURE, relevant checklists depending on cause of stall. | |
| Step 5 - Long term plan | Did the aircraft exceed limits or sustain structural damage? Are passengers or crew hurt? | |

Normal Mode




Direct Mode




Aerodynamic stall can occur at any altitude, thrust setting or airspeed and depends on aircraft configuration, weight and maneuvering load factor. For swept-wing turbojet aircraft the stall angle generally reduces as altitude increases, due to the Mach effect.

The normal load factor protection limits the load based on aircraft configuration and sidestick position (soft or hard stop). Turbulence can cause an abrupt increase in AOA, due to the change in direction of airflow relative to the wing.

FBW normal mode provides flight envelope protection to help maintain the aircraft within the operational flight envelope. The soft limit (soft stop) is used to provide a tactile cue at the edge of the operational envelope. The High AOA Protection (HAP) ensures the aircraft does not exceed:

- Operational AOA envelope at the sidestick (soft stop)
- Maximum AOA at full aft sidestick (hard stop).

The high angle-of-attack protection (HAP) function adjusts the protection AOA:

- When in icing conditions and the wing anti-ice system is not activated or has failed
- During abnormal slat/flap configurations
- When sideslip angles are greater than 5 degrees

Other flight envelope protection functions in the normal mode:

- When the AOA at $V_{MINTRIM}$ is exceeded, the stabilizer nose-up trim is disabled.
- When the AOA is between $V_{MINTRIM}$ and the soft stop, spoilers retract and roll rate is reduced.
- When the AOA is greater than the AOA at the soft stop, the stick shaker is activated.
- At the hard stop, the AOA is limited to below the stall and the maximum roll angle is decreased.

In FBW direct mode, these protections are compromised and not available.

Stall recovery is not a precision maneuver. The objective in recovery is to restore the aircraft to normal flight, this is done by reducing the AOA. After recovery, crews should evaluate why the stall occurred. High altitude recovery typically involves substantial loss of altitude to regain airspeed.

A stall is characterized by any or all of the following indications:

| See | Feel | Hear |
|---|---|---|
| • Amber or red airspeed indication<br>• Red STALL flag on PFD<br>• Inability to arrest the descent | • Aerodynamic buffeting<br>• Stick shaker activation<br>• A lack of pitch and roll authority | • SPEED aural alert<br>• STALL aural alert<br>• Buffeting and/or stick shaker |

# Volcanic Ash Encounters

Flight operations in areas of known volcanic activity should be avoided. This consideration is most important during hours of darkness or daytime instrument meteorological conditions when volcanic ash/dust clouds may not be visible.

Flight planning considerations should include the review of pertinent NOTAMs, PIREPs, and Volcanic Ash Advisories when a flight is planned into areas of possible volcanic activity. When volcanic activity is currently reported, stay well clear of the area or, if possible, stay on the upwind side of the volcanic ash.

The airplane weather radar is not capable of detecting volcanic ash/dust clouds and is therefore not reliable under these circumstances. The presence of volcanic ash/dust can be indicated by:

- Smoke or dust appearing in the flight compartment
- An acrid odor similar to electrical smoke
- Multiple engine malfunctions such as power loss, fluctuating N1, compressor stall, increasing EGTs, flameouts, etc.
- At night, static discharges (also known as Saint Elmo's fire) can be observed around the windshield and/or windows, accompanied by a orange glow in the engine inlets

Flight into volcanic ash/dust clouds can result in the degradation of airplane and engine performance. the adverse effects of volcanic ash/dust encounter are as follows:

- Rapid erosion and damage to the internal components of the engines
- Ash/dust build-up and blockage of the guide vanes and inlet cowls, which can cause surge, loss of thrust and/or high EGT
- Ash/dust blockage of the smart probes, which can cause erratic airspeed indications
- The abrasive properties of volcanic material can cause serious damage to the engines, wing and tail leading edge surfaces and windshields
- Volcanic ash/dust can also cause the windshield to become translucent, effectively reducing visibility

The best course of action is avoidance. However, volcanic ash/dust clouds may sometimes extend for hundreds of miles and an encounter could be unavoidable. The need to exit the area as quickly as possible cannot be overemphasized.

If an inadvertent encounter is experienced, this procedure is recommended:

1. Thrust ..... Reduce towards IDLE

*If altitude permits, engine thrust should be reduced towards idle to maximize the engine stall margin and lower the EGT. This action would also reduce the build-up of volcanic material on the turbine vanes.*

2. Reverse course if possible.

*Clear air must be regained as soon as possible. Terrain permitting, a descending 180-degree turn is likely to be the fastest way out of the contaminated air.*

3. ANTI-ICE, L COWL ..... ON
4. ANTI-ICE, WING ..... ON
5. ANTI-ICE, R COWL ..... ON

*These actions (if thrust is at IDLE) increase bleed air extraction from the engines and further improve the engine stall margin.*

6. EGT ..... Monitor

*Closely monitor the EGT and ensure that the limits are not exceeded.*

- If it becomes necessary to shut down an engine to prevent exceeding EGT limits, an engine restart can be attempted once the engine has cooled down.
- If the engine fails to start, repeated attempts should be done immediately.

Note:

A successful start may not be possible until the airplane is clear of the volcanic ash/dust, and the airspeed and altitude are within the airstart envelope.

After exiting the area of volcanic ash/dust cloud and the engine(s) are restarted, systems should be restored to normal operation. ATC must be advised of the encounter.

# Pilot Incapacitation

There are two types of pilot incapacitation:

- Obvious
- Subtle

Failure to respond to two subsequent calls, pilot incapacitation must be assumed.

In the event of illness, the flight crew have absolute authority to divert or take all necessary action to achieve medical attention.

The cabin crew (if carried) will normally undertake the safe securing of an incapacitated flight crew member, and may also assist with cockpit duties, if this is required.

**Remaining Pilot:**

- Sidestick priority.....LATCH
- Autopilot.....ON
- FCP coupled side.....REMAINING PILOT
- Maintain safe flight path
- Pilot Incapacitation Priority Procedure.....PERFORM
- Flight attendant.....CALL TO FLIGHT DECK
  *Press CABIN CALL button for 3 seconds or PA: "ATTENTION – SENIOR TO FLIGHT DECK". Cabin crew will render first aid, remove the pilot from the flight deck or if not possible, secure the incapacitated pilot and assist with cockpit duties if this is required.*
- Perform situation assessment, decide on course of action.
- Inform ATC, cabin crew (NITS) and passengers.

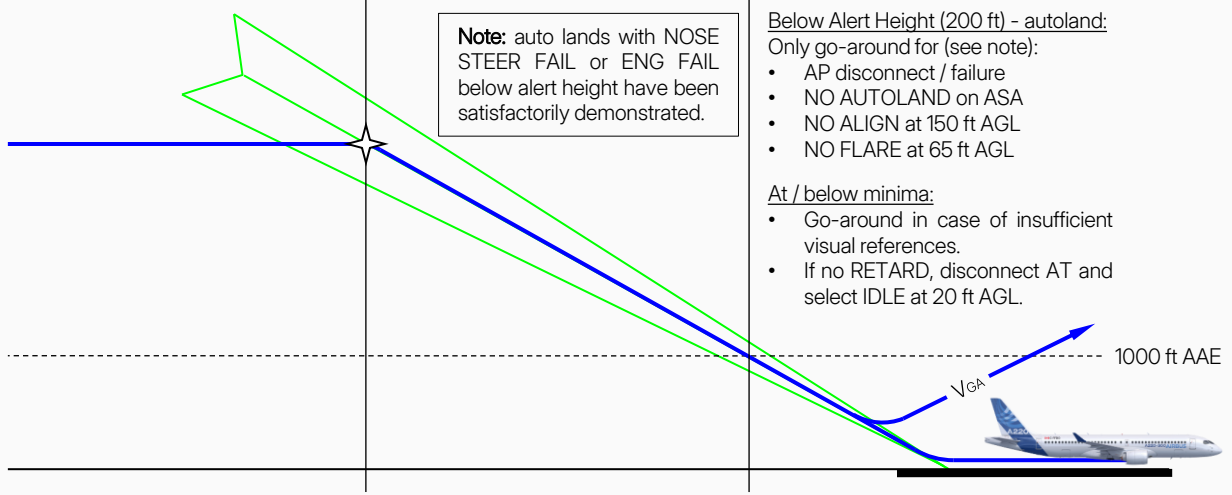
The remaining pilot must land as soon as possible at a suitable airport, taking into account the incapacitated pilot state of health, airport equipment (prefer airport with 3D approach), weather and runway conditions, knowledge of airport by the remaining pilot etc. Request medical assistance upon landing, right pilot should consider lack of nosewheel steering control (no tiller).

The remaining pilot must:

- Perform both PF and PM tasks
- Verify and call out all actions
- Perform all checklists out loud

# Failures during Approach

| Before FAF | Within FAF, above 1000 ft AAE | Below 1000 ft AAE |
|---|---|---|
| If RVR or visibility is below landing minima, approach may be started and continued to 1000 ft AAE. | If RVR or visibility is below landing minima by 1000 ft AAE, approach shall be discontinued. | If RVR or visibility reduces below landing minima below 1000 ft AAE, approach may be continued to DA/H. |
| Attempt can be made to restore systems by performing non-normal checklists or procedures. All checklists and procedures must be completed prior 1000 ft AAE. | | Any system failure or loss of guidance should normally lead to a go-around. |
| | | In case of smoke, fumes or fire of any kind, the aircraft is committed to land. |
| Change of approach type is allowed if approach set-up and re-briefing can be completed prior crossing the FAF. | A change of approach type is allowed if the only change is the approach minima (e.g. CAT III to CAT II). | Continuation in case of down-mode without a down-grade is allowed. Down-grade shall result in go-around. |
| In case of pilot incapacitation, discontinue the approach and perform pilot incapacitation drill, decide whether to return or divert. | In case of pilot incapacitation, approach can be continued if stabilized and incapacitated pilot is not obstructing the controls. | In case of pilot incapacitation, approach can be continued if stabilized and incapacitated pilot is not obstructing the controls. |
| Effect of aerodrome equipment (e.g. failure of approach lights) on landing minima must be considered. | Effect of aerodrome equipment (e.g. failure of approach lights) on landing minima must be considered. | In case of failure of aerodrome equipment, approach continuation is commander discretion. |
| Autothrottle can be reengaged once, if unsuccessful, autoland not allowed. | Autothrottle can be reengaged once, if unsuccessful, autoland not allowed. | Autothrottle can be reengaged once, if unsuccessful, autoland not allowed. |
| | **Note:** auto lands with NOSE STEER FAIL or ENG FAIL below alert height have been satisfactorily demonstrated. | Below Alert Height (200 ft) - autoland:<br>Only go-around for (see note):<br>• AP disconnect / failure<br>• NO AUTOLAND on ASA<br>• NO ALIGN at 150 ft AGL<br>• NO FLARE at 65 ft AGL<br>At / below minima:<br>• Go-around in case of insufficient visual references.<br>• If no RETARD, disconnect AT and select IDLE at 20 ft AGL. |




Autoland: after touchdown, without ROLLOUT: TOGA buttons are active.
With ROLLOUT, TOGA buttons are inactive, make a manual go-around.

# Emergency Landing

There are two main forms of emergency landing

- Unprepared – those that take place without warning.
- Prepared – where there is ample warning to pre-plan the landing and evacuation.

It is normally not considered necessary to prepare the cabin for an emergency landing during the following situations:

- Single engine landings
- Flaps up landings (All or Partial), if the runway is sufficiently long

Even though an emergency situation is declared during a single engine operation, it is expected that, under normal circumstances, a safe landing will be made.

Should it become necessary to perform an emergency landing, the occupants safety is dependent upon three main factors, namely: crash protection, evacuation and survival. The most important duties for the cockpit crew are:

- To alert the ground rescue organization as soon as possible
- To select a suitable landing place and perform the landing in a manner to avoid excessive impact and deceleration forces and minimize fire hazard
- To facilitate a quick evacuation after the landing

If extra cockpit crew members are carried, they shall be utilized in the cockpit as long as required. Thereafter, they shall be ordered to proceed to the cabin to assist the cabin crew.

Unprepared emergency landing

This can arise during and/or immediately after take-off or immediately prior to or after landing. It is the most critical case, and a successful evacuation will depend entirely upon the competence and initiative of the crew. In these circumstances, there may not be time to carry out the full preparation procedures and the following should be applied:

After rejected take-off or as soon as the need for an emergency landing becomes apparent, the PM should call on PA:

- "ATTENTION - CREW AT STATIONS"

this call serves to warn and to mentally prepare the cabin crew that an emergency is developing, and subsequent emergency landing and evacuation may be imminent. Then 30 seconds before impact (approximately 500 ft):

- "BRACE FOR IMPACT"

If the PA is unserviceable, the signals will be given from the flight deck by repeated flashing of the SEAT BELT sign.

Prepared emergency landing

Action will be initiated by the flight crew by pressing CABIN CALL for 3 seconds or by PA:

- "ATTENTION - SENIOR TO FLIGHT DECK"

These are the ALERT signals and will be used in an emergency only, thus triggering an immediate response from the Senior, and enabling the remaining crew to disengage "quietly" from their cabin service duties and go to their designated emergency station for briefing.

Upon arrival to the flight deck the Senior will be briefed by the Commander. NITS must be repeated by the Senior to ensure information has been correctly received and understood.

The commander will assign who does the passenger briefing.

Once the cabin crew and passengers have been briefed, and all preparations completed, the Senior will report to the Commander:

- "CABIN SECURED"

Two minutes before impact (approx. 2000 ft AGL), the flight crew will call:

- "FINISH PREPARATIONS"

30 seconds before impact (approx. 500 ft AGL) the flight crew will call:

- "BRACE FOR IMPACT"

If the PA is unserviceable, the signals will be given from the flight deck by repeated flashing of the SEAT BELT sign.

# Ditching

Ditching should only be considered in the most extreme situations, such as major fuel leak, inextinguishable fire, severe damage or controllability issue. Besides the preparations in accordance with a "normal" emergency landing, pilots should take note of the following:

Wind Direction may be assessed by observing the waves, which move and break downwind, spray from wave tops is also a reliable indication. The following can be used for wind speed:

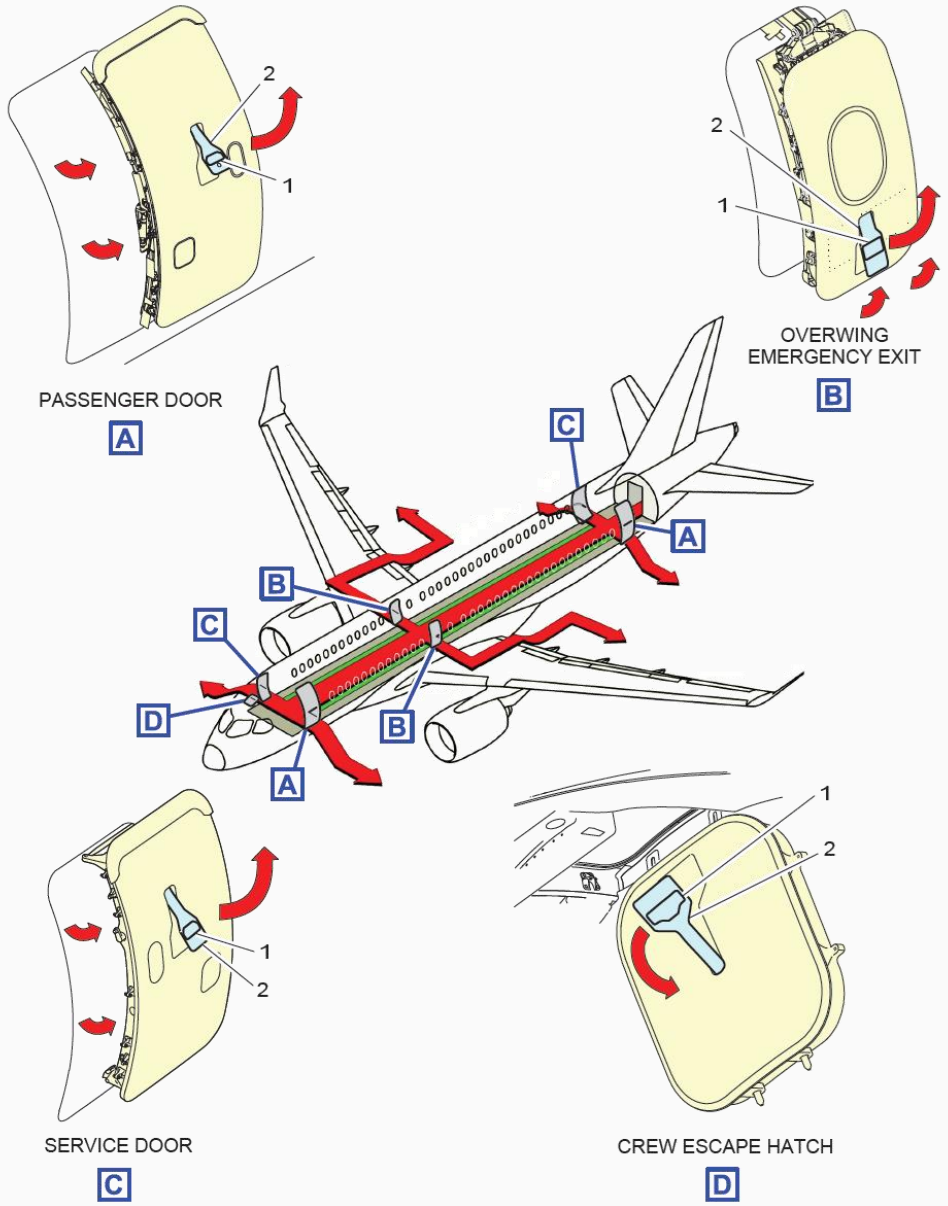
- A few white crests 8 – 17 kt
- Many white crests 17 – 26 kt
- Streaks of foam along the water 23 – 35 kt
- Spray from wave tops 35 – 43 kt

This is better assessed from a height of 500 to 1000 ft, particularly the direction of the swell, which may not be obvious when seen from a lower height. When there is no swell, align the landing direction into the wind. In the presence of a swell and provided that drift does not exceed 10°, land parallel to the swell and as nearly into wind as possible. If the drift exceeds 10°, land into wind. The presence of drift on landing in itself is not dangerous but every effort should be made to minimize roll.

Whenever a landing on water is expected, all occupants shall put on their life vest before landing. Cabin Crew shows the passengers how to don life vests and instruct them not to inflate them inside the aircraft. Before a landing at night in darkness, passengers should be instructed to pull out the life vest battery plugs. The light will then automatically come on when the battery comes into contact with water. Life vests on small children should be inflated inside the aircraft before the impact or as soon as the aircraft stops. The adult life vest should be inflated as soon as the person leaves the aircraft.

For easy and quick identification during the evacuation, the passengers should be advised that all crew members will wear red life vests. As the passengers leave the aircraft, remind them to inflate their life vest and swim away to clear the jumping area.

- For evacuation, use only those doors that are not under the water line.
- Escape slides are not designed as life rafts and are not stable enough to evacuate on, escape slides should be released from the aircraft after inflation and used to hold on to only.
- Overwing escape slides will not inflate.
- A "life-line" is available inside the overwing emergency exit door sill and can be attached to the (yellow) mooring hook on the wing to assist in evacuating onto the wings.



# Emergency Evacuation | Rapid Disembarkation

The passenger evacuation procedure includes the steps required to prepare for passenger evacuation and is initiated when a condition potentially endangering life or physical well-being of passengers and crew exists.

Time permitting, brief flight attendant(s):

- Nature of emergency
- Evacuation signals
- Specific directions regarding exits to be used.

The commander, or in his absence the first officer, will order an evacuation using the PA. If required, the exits to be used may be indicated. Only in cases that are clearly catastrophic (ditching, explosive fire, and structural damage) should cabin crew act on their own initiation. To indicate the flight crew is conscious and actively working the situation (standby for evacuation):

- "ATTENTION - CREW AT STATIONS"

If no evacuation signal and no other form of communication is established, but it is unmistakably apparent that the evacuation must be executed, after 30 seconds, cabin crew shall give the command to evacuate.

The EMERGENCY EVACUATION PRIORITY PROCEDURE includes the steps required to secure the aircraft to prevent movement, shutdown engines and prepare the aircraft for safe evacuation, the procedure is considered READ and DO by CM2 and shall not be conducted from memory. DITCHING and FORCED LANDING PRIORTY PROCEDURES already include the steps required to prepare for evacuation.

If the Commander chooses to evacuate the passengers by a particular exit, the Commander should state the exit first, followed by the evacuation command, e.g.

- "LEFT (RIGHT) SIDE, EVACUATE, EVACUATE!"

Evacuation is considered complete when all passengers and crew have been evacuated from the aircraft. Some post evacuation actions should be carried out as a standard procedure:

- FOLLOW ORDERS from the Commander.
- COMMAND PASSENGERS away (up wind) from the aircraft because of risk of fire.
- COUNT DOWN the passengers.
- COMMUNICATE with the passengers as required.
- CARE for the injured.
- COORDINATE actions to overcome the first few hours.

Rapid Disembarkation is the expedited exit of passengers and crew from a parked aircraft utilizing the boarding entrance(s) and the associated airport infrastructure (air-bridge, jetway or boarding stairs).

Rapid disembarkation can be initiated only when the aircraft is on-stand with doors open and the air-bridge connected or boarding stairs in place.

The decision to initiate rapid disembarkation is likely to be taken in response to an occurrence involving some aspect of ground handling activity. During ground support activities, accidents occur, machinery malfunctions, and situations sometimes arise which pose a potential risk to the passengers in the process of boarding. These risks might include, but are not limited to:

- Fuel spills.
- Smoke or fumes, from a variety of potential sources, entering the aircraft cabin.
- A fire in proximity to the aircraft caused by malfunctioning ground support equipment.
- An accident, either directly involving the aircraft, on an adjacent stand or ground support vehicles in proximity to the aircraft.
- A bomb threat.

The Commander or designated individual will first determine the immediacy of the threat and then decide on the most appropriate course of action, be it an Emergency Evacuation or a Rapid Disembarkation. Perform the RAPID DISEMBARKATION PRIORITY PROCEDURE to prepare the aircraft. Time permitting, call Senior to the flight deck for a NITS briefing, otherwise call on PA:

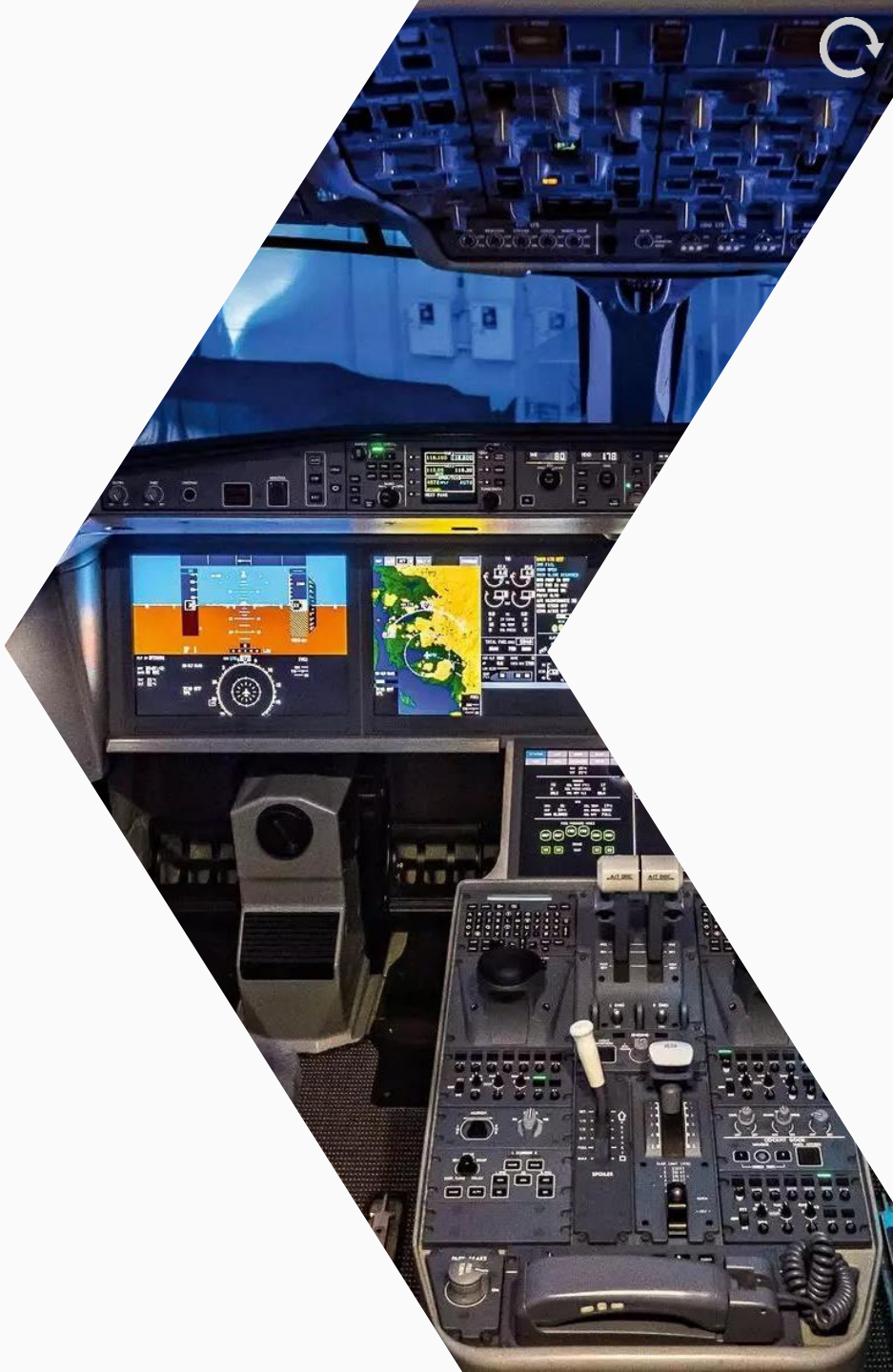
- "RAPID DISEMBARKATION"

Upon hearing the execution phrase for Rapid Disembarkation, the cabin crew and ground staff will, if necessary, first reverse the passenger flow on the air-bridge or boarding stairs, so passengers about to embark move away from the aircraft.

At the same time, the flight attendant in charge will make a public address system announcement advising the on-board passengers of the need for immediate deplanement.

In the event that electrical power is interrupted, a megaphone will be used or shouted instructions will be provided. It is important to highlight, that the already started rapid disembarkation procedure may turn into an evacuation situation, and all crew members must be ready to change procedure.

# EICAS Message List



# Crew Alerting

To avoid distraction during take-off and landing phases, most caution and advisory messages are inhibited if no immediate crew action is required:

- Take-off inhibit: from 80 kts until 400 ft AGL.
- Landing inhibit: from 400 ft AGL until speed is less than 50 kts.

Messages that are not inhibited during the take-off phase:

Air Conditioning, bleed air and pressurization:
- EQUIP BAY OVHT
- CABIN ALT
- CABIN DIFF PRESS

Auto flight:
- EMERGENCY DESCENT
- EMERGENCY DESCENT

Auxiliary Power System:
- APU FIRE
- APU OVERSPEED

Doors:
- COCKPIT DOOR EMER ACCESS

Electrical:
- EMER PWR ONLY

Fire and overheat protection:
- FWD (AFT) CARGO FIRE
- APU FIRE, L(R) ENG FIRE
- EQUIP BAY (LAV) SMOKE
- MLG BAY OVHT

Flight controls:
- L(R) ELEVATOR FAIL
- RUDDER FAIL
- L(R) SIDESTICK FAIL

Fuel:
- FUEL COLLECTOR LO LVL
- FUEL LEAK SUSPECT

Ice and rain protection:
- L(R) WING A/ICE FAIL
- ICE

Landing gear:
- GEAR DISAGREE

Navigation:
- TCAS OFF

Powerplant:
- DUAL ENG FAIL
- L(R) ENG FAIL
- L(R) ENG START ABORT

All STATUS (white) messages

Take-off configuration warning is triggered when TLA is above 23° and indicated $N_1$ is above 55% on both engines in the following cases:

| Message | Voice Alert |
|---|---|
| CONFIG AP | "CONFIG AUTOPILOT" |
| CONFIG BRAKE | "CONFIG BRAKE" |
| CONFIG FLAP | "CONFIG FLAP" |
| CONFIG RUDDER TRIM | "CONFIG TRIM" |
| CONFIG SIDESTICK | "CONFIG SIDESTICK" |
| CONFIG SPOILER | "CONFIG SPOILER" |
| CONFIG STAB TRIM | "CONFIG TRIM" |

Aural alerts:

- A-chord: overspeed
- Double beep: pitch or rudder trim running time limit (3 seconds)
- Tripple chime: warning message
- Single chime: caution message
- Cavalry charge: autopilot disconnect
- Single C-chord: altitude alert
- Double C-chord: VTA - Vertical Track Alert
- Voice messages: as below (for TCAS, TAWS, see respective chapter)

| Type | Condition / Message | Voice Alert |
|---|---|---|
| EICAS | CABIN ALT<br>CABIN DIFF PRESS<br>EMERGENCY DESCENT<br>EMERGENCY DESCENT<br>CKPT DOOR EMER ACCESS<br>FWD (AFT) CARGO FIRE<br>APU FIRE<br>EQUIP BAY (LAV) SMOKE<br>L(R) ENG FIRE<br>MLG BAY OVHT<br>GEAR<br>CONFIG ... | "CABIN ALTITUDE"<br>"CABIN PRESSURE"<br>"EMERGENCY DESCENT"<br>"EMERGENCY DESCENT"<br>"COCKPIT DOOR"<br>"CARGO FIRE"<br>"APU FIRE"<br>"SMOKE"<br>"LEFT (RIGHT) ENGINE FIRE"<br>"GEAR BAY OVERHEAT"<br>"GEAR"<br>"CONFIG ..." |
| Windshear | WSHR-AHEAD (PFD)<br>WSHR-AHEAD (PFD)<br>WINDSHEAR (PFD) | "MONITOR RADAR DISPLAY"<br>"(GO AROUND) WINSHEAR AHEAD"<br>"WINDSHEAR WINDSHEAR WINDSHEAR" |
| Engine start | Thrust lever not idle | "THRUST LEVER" |
| SMS<br>TLAF | Mode 1, Mode 4 - RUNWAY (PFD)<br>Mode 3 - RUNWAY (PFD) | "NOT A RUNWAY, NOT A RUNWAY"<br>"RUNWAY DISAGREE, RUNWAY DISAGREE" |
| Autothrottle | Disconnect | "AUTOTHROTTLE" (2 sec or continuous) |
| Sidestick | Priority<br>Dual input | "PRIORITY LEFT (RIGHT)"<br>"DUAL INPUT" |
| Speed | Overspeed<br>Underspeed<br>Stall | "OVERSPEED" (single or continuous)<br>"SPEED" (single or continuous)<br>"STALL" (continuous) |
| Test | TCAS<br>Aural alert test | "TCAS SYSTEM TEST OK (FAIL)"<br>"AURAL WARNING TEST 1 (2)" |

# EICAS Message List

| Bleed Air System | |
|---|---|
| Warning messages | Description |
| None | - |
| Caution messages | Description |
| AIR SYS ESS CTLR FAIL | IASC 1 failure due to channel B and C failure and IASC 2 failure due to channel B and C failure or not received by DMC. |
| APU BLEED LEAK | APU bleed leak detected. |
| ENG BLEED MISCONFIG | XBLEED valve selected open and both engine bleed selected on or low flow on APU and L/R ENG bleed off. |
| L AIR SYS CTLR FAIL | IASC 1 failure due to channel A and B failure or not received by DMC. |
| L BLEED FAIL | Left engine bleed failure, loss of L BLEED leak detection, or loss of both IASC 1 channels. |
| L BLEED LEAK | Left bleed leak detected. |
| L BLEED OVHT | Left engine bleed overtemperature detected. |
| L PACK LEAK | Left PACK leak detected. |
| LEAK DET FAIL | Entire leak detection system failed. |
| R AIR SYS CTLR FAIL | IASC 2 failure due to channel A and B failure or not received by DMC. |
| R BLEED FAIL | Right engine bleed failure, loss of R BLEED leak detection, or loss of both IASC 2 channels. |
| R BLEED LEAK | Right bleed leak detected. |
| R BLEED OVHT | Right engine bleed overtemperature detected. |
| R PACK LEAK | Right PACK leak detected. |
| XBLEED FAIL | XBLEED valve failed in position in AUTO or MAN mode. |
| Advisory messages | Description |
| AIR SYSTEM FAULT | Loss of redundant or non-critical function for the air system (bleed air or trim air) or Emergency Ram Air Valve (ERAV) inoperative. |
| LEAK DET FAULT | Single loop leak detection failure. |
| Status messages | Description |
| APU BLEED OFF | APU bleed selected off. |
| L BLEED OFF | Left engine bleed selected off. |
| L BLEED OFF | Left engine bleed selected off. |
| R BLEED OFF | Right engine bleed selected off. |
| XBLEED MAN CLSD | Cross-bleed valve manually selected and confirmed closed. |
| XBLEED MAN OPEN | Cross-bleed valve manually selected and confirmed opened. |

| Air Conditioning System | |
|---|---|
| Warning messages | Description |
| EQUIP BAY OVHT | Forward equipment bay or mid equipment bay overtemperature detected (more than 67 °C (152 °F)). |
| Caution messages | Description |
| AIR SYS ESS CTRL FAIL | IASC 1 failed, channel B and C failed and IASC 2 channel B and C failed. |
| EQUIP BAY COOL FAIL | Loss of both extraction FANS confirmed by IASC channel (safety or control), or both fans are operating, or total loss of one zone temperature measurement. |
| FWD CARGO HEAT FAIL | LO HEAT AND HI HEAT mode not available. |
| FWD CARGO LO TEMP | Low temperature in FWD CARGO when FWD CARGO selected to LO/HI HEAT. |
| L AIR SYS CTRL FAIL | IASC 1 failed, channel A and channel B failed or not received by DMC. |
| L PACK FAIL | Left pack inoperative, or loss of left pack leak detection, or loss of both IASC 1 channels. |
| L PACK OVHT | Left compressor discharge temperature overheat or left pack discharge temperature overheat. |
| R AIR SYS CTRL FAIL | IASC 2 failed, channel A and channel B failed or not received by DMC. |
| R PACK FAIL | Right pack inoperative, or loss of right pack leak detection, or loss of both IASC 2 channels. |
| R PACK OVHT | Right compressor discharge temperature overheat or right pack discharge temperature overheat. |
| RAM AIR FAIL | RAM AIR switch selected OPEN and Emergency Ram Air Valve (ERAV) not detected open, or RAM AIR switch selected to CLOSE and ERAV not detected closed. |
| RECIRC AIR FAIL | RFAN internal failure detected or communication failure. |
| TRIM AIR FAIL | Trim air failure or loss of trim air leak detection or loss of both IASC 1 channels. |
| TRIM AIR LEAK | Trim air leak detected. |

# EICAS Message List

| Advisory messages | Description |
|---|---|
| AVIONIC FAULT | Loss of redundant or non-critical function for the avionics system. |
| AVIONIC FAN FAULT | Loss of fan functionality within avionics LRUs. |
| EQUIP BAY COOL FAULT | Loss of redundant or non-critical function for the air-conditioning system. |
| PACK FAULT | Loss of redundant or non-critical function for the air-conditioning system. |
| Status messages | Description |
| AFT CARGO AIR OFF | AFT CARGO selected and confirmed OFF. |
| FWD CARGO AIR OFF | FWD CARGO selected and confirmed OFF. |
| EXHAUST AIR VLV ONLY | Exhaust valves open manually. |
| EXHAUST AIR ON | Exhaust air system manually selected on (supplemental fan turns on and valves open). |
| INLET AIR OFF | Inlet air system (skin heat exchanger) selected off (fans turn off and valves close) or aft galley and aft lavatory extraction selected OFF (no supplementary avionic cooling). |
| MAN TEMP ON | Temperature control selected manually. |
| PACK FLOW HI | Cabin airflow selected to HI. |
| L PACK OFF | Left pack selected to OFF. |
| R PACK OFF | Right pack selected to OFF. |
| RAM AIR OPEN | Ram air selected to OPEN. |
| TRIM AIR OFF | Trim air selected to OFF. |

| **Pressurization System** | |
|---|---|
| Warning messages | Description |
| CABIN ALT | Cabin altitude more than 10000 feet or up to 15000 feet High Altitude Airfield Operation (HAAO). |
| CABIN DIFF PRESS | Positive differential pressure is more than the pneumatic relief setting. |
| Caution messages | Description |
| AUTO PRESS FAIL | Both cabin pressure control auto functions are failed. |
| CABIN ALT | Cabin altitude between 8500 feet and 10000 feet, or up to 15000 feet (HAAO), or above 10000 feet for 30 minutes (HAAO). |
| DITCHING MISCONFIG | DITCHING guarded switch selected to ON and RAM AIR switch selected OPEN. |
| EMER DEPRESS ON | EMER DEPRESS guarded switch selected to ON confirmed by either one auto mode or one manual mode. |
| LDG ELEV MISCONFIG | High airport elevation landing is computed and HAAO option is not available on the aircraft. |
| Advisory messages | Description |
| CABIN ALT LEVEL HI | Cabin altitude caution and warning thresholds higher due to high altitude landing field elevation mode. |
| PRESSURIZATION FAULT | Loss of redundant or non-critical function for the pressurization system. |
| Status messages | Description |
| CABIN PRESS MAN | Manual pressurization mode selected. |
| DITCHING ON | DITCHING switch selected to ON and sequence completed. |

# EICAS Message List

| Automatic Flight Control System | |
|---|---|
| Warning messages | Description |
| CONFIG AP | Autopilot (AP) engaged on takeoff. |
| EMERGENCY DESCENT | Emergency Descent Mode (EDM) automatically activated. |
| Caution messages | Description |
| AT RETARD INHIBIT | Autothrottle (AT) system retard mode inoperative due to radio altimeter failure. |
| EMERGENCY DESCENT | Emergency Descent Mode (EDM) is active and it has been manually engaged. |
| FD FAIL | AFCS coupled left and FD channels 1A and 2A failed. |
| FD MODE CHANGE | Uncommanded flight director mode changed to basic mode. |
| STEEP NOT AVAIL | Steep approach automatic flight mode not available and detected while mode is armed. |
| Advisory messages | Description |
| APPR1 NOT AVAIL | ILS CAT I not supported due to sensor failure. |
| APPR2 NOT AVAIL | ILS CAT II not supported due to sensor failure. |
| AT FAIL | Both autothrottle channels failed. |
| AUTO FLIGHT FAULT | Loss of redundant or non-critical function for the auto flight systems. |
| L ENG A/T OFF | Left engine AT command is off. |
| LAND 2 NOT AVAIL | LAND 2 (CAT III fail passive) capability not supported due to sensor failure. |
| LAND 3 NOT AVAIL | LAND 3 (CAT III fail operational) capability not supported due to sensor failure. |
| R ENG A/T OFF | Right engine AT command is off. |
| Status messages | Description |
| None | - |

| Auxiliary Power Unit | |
|---|---|
| Warning messages | Description |
| APU FIRE | APU fire detected. |
| APU OVERSPEED | APU has experienced an overspeed condition (above 106% and has not shut down. |
| Caution messages | Description |
| APU | APU failure in flight that does not lead to automatic shutdown. |
| APU BLEED FAIL | Failure of the APU to provide bleed air when requested or bleed leak detection fails. |
| APU DOOR OPEN | APU inlet door failed to close when commanded. |
| APU FIRE DET FAIL | Failure of the APU fire detection system. |
| APU OIL LO TEMP | APU low oil temperature (below −35 °C (−31 °F)) detected |
| Advisory messages | Description |
| APU FAULT | Loss of redundant or non-critical function for the APU. |
| APU OIL LO QTY | APU low oil quantity detected on ground while APU is not running. |
| APU SHUTDOWN | Failures detected that will cause the APU to shut down. |
| Status messages | Description |
| APU IN START | APU is performing the startup sequence. |
| APU ON | APU is running. |

# EICAS Message List

| Communication | |
|---|---|
| Warning messages | Description |
| None | - |
| Caution messages | Description |
| AURAL WARN FAIL | Channel B on both RIUs failed. |
| KU BAND ON | The INLET switch is selected to OFF and the Ku-band or Ka-band system (if installed) is powered or status is unknown while on the ground. |
| L CTP TUNING FAIL | L RIU channel B failure leading to L CTP not capable of tuning left side radio. |
| L-R RADIO TUNING FAIL | Four channels of RIU failed, leading to only emergency audio available. |
| R CTP TUNING FAIL | R RIU channel B failure leading to R CTP not capable of tuning right side radio. |
| Advisory messages | Description |
| AUDIO PNL 1 FAIL | ACP1 failed. |
| AUDIO PNL 2 FAIL | ACP2 failed. |
| DATALINK FAIL | Datalink system (CMU) failed. |
| DATALINK STATUS | No communication currently available to send/receive data. |
| CABIN COM FAULT | Loss of either PA/intercom through audio control panel or loss of PA/intercom through handset. |
| Status messages | Description |
| AURAL WARN INHIBIT | Aural warning system has been disabled by the flight crew. |
| CTP OVERRIDE | L or R CTP override through AVIONIC synoptic page. |
| DSPL TUNE INHIB | Left or right display tune inhibited. |
| VHF COM 121.5 ENABLE | VHF operating in emergency mode. |
| VHF3 IN VOICE | VHF3 selected in voice mode by flight crew. |

| Doors | |
|---|---|
| Warning messages | Description |
| CKPT DOOR EMER ACCESS | Emergency request from cabin. |
| Caution messages | Description |
| AFT DOOR | Aft passenger or service door failed or opened with at least one engine running. |
| AFT SLIDE | Aft passenger or service door slide is faulty with at least one engine running. |
| CARGO DOOR | Forward or aft cargo compartment door failed or opened with at least one engine running. |
| CKPT DOOR LOCK FAIL | Failure of the FDRAS to lock door or loss of communication to DMC. |
| DOOR SLIDE DISARMED | Any access door slide is still disarmed in flight or with one engine running. |
| EQUIP BAY DOOR | Forward, mid, or aft equipment bay door failed or open with at least one engine running. |
| FWD DOOR | Forward passenger or service door failed or open with at least one engine running. |
| FWD SLIDE | Forward passenger or service door evacuation slide is faulty with at least one engine running. |
| OVERWING DOOR | Left or right overwing emergency exit door failed or open with at least one engine running. |
| WING SLIDE | Left or right overwing escape slide faulty or deployed. |
| Advisory messages | Description |
| CKPT DOOR OPEN | Flight compartment door open. |
| DOOR FAULT | Any electrical or mechanical failure of any door while on ground and no engine running. |
| DOOR OPEN | Any door open while on ground and no engine running. |
| DOOR SLIDE DISARMED | Any door slide disarmed while on ground and no engine running. |
| DOOR SLIDE FAULT | Any electrical or mechanical failure of any door slide while on ground and no engine running. |
| Status messages | Description |
| CKPT DOOR LOCK OFF | Flight compartment door system is in maintenance mode or when associated SSPC is the PULL and LOCK condition. |

# EICAS Message List

| Electrical | |
|---|---|
| Warning messages | Description |
| EMER PWR ONLY | Loss of the main power generation in flight. |
| Caution messages | Description |
| AC BUS 1 | AC BUS 1 not powered with at least one AC primary power source available. |
| AC BUS 2 | AC BUS 2 not powered with at least one AC primary power source available. |
| AC ESS BUS | AC ESS bus not powered with at least one AC primary power source available or in emergency mode. |
| APU GEN FAIL | APU GEN not online, with APU running and APU GEN switch selected to AUTO, or total loss of Time-Triggered Protocol (TTP) communication from the AGCU. |
| BATT 1 FAIL | Battery 1 not connected to DC ESS 1 bus when required, or battery fault from battery charger. |
| BATT 2 FAIL | Battery 2 not connected to DC ESS 2 bus when required, or battery fault from battery charger. |
| BATT 1 OVERTEMP | Battery 1 temperature greater than or equal to 71 °C (160 °F) and appears as long as it is above 65 °C (149 °F). |
| BATT 2 OVERTEMP | Battery 2 temperature greater than or equal to 71 °C (160 °F) and appears as long as it is above 65 °C (149 °F). |
| BATT DISCHARGING | Battery 1 or 2 (or both) discharging for 5 minutes or more on ground. |
| DC BUS 1 | DC BUS 1 not powered with at least one AC primary power source available. |
| DC BUS 2 | DC BUS 2 not powered with at least one AC primary power source available. |
| DC EMER BUS | DC EMER bus not powered. |
| DC ESS BUS 1 | DC ESS 1 bus not powered. |
| DC ESS BUS 2 | DC ESS 2 bus not powered. |
| DC ESS BUS 3 | DC ESS 3 bus not powered. |
| L GEN FAIL | L GEN not online and control switch set to AUTO, or total loss of TTP communication from left GCU. |
| R GEN FAIL | R GEN not online and control switch set to AUTO, or total loss of TTP communication from right GCU. |
| L GEN OIL | L GEN low oil pressure or high oil temperature detected. |
| R GEN OIL | R GEN low oil pressure or high oil temperature detected. |
| RAT GEN FAIL | RAT generator not online in flight when RAT is deployed, or fault detected during BIT or RAT is not supplying power as demanded. |
| Advisory messages | Description |
| BATT CHARGER FAULT | Battery charger 1 or 2 inoperative. |
| BATT EMER PWR ON | Loss of all AC power sources in flight and the only power source is the battery system. |
| CB TRIP | Thermal CB, SSPC, or FUSE open and not acknowledged. |
| ELECTRICAL FAULT | Loss of redundant or non-critical function for the electrical system. |
| EXT PWR AVAIL | External power connected and ready to be used. |
| LOAD SHED | Automatic load shed commanded. |
| RAT GEN ON | RAT generator online. |
| RAT DPLY | RAT deployed in flight or on ground. |
| TRU FAULT | TRU1, TRU2 or TRU3 failed or not connected to their respective DC BUS. |
| Status messages | Description |
| APU GEN OFF | APU generator is off while APU is operating with APU GEN switch selected to OFF. |
| BATT 1 OFF | BATT 1 selected and confirmed OFF. |
| BATT 2 OFF | BATT 2 selected and confirmed OFF. |
| BATT PWR CONFIG | ELEC switch on maintenance panel selected to DC ESS 3. |
| CABIN PWR OFF | CABIN PWR switch selected to OFF on ELECTRICAL panel. |
| ELEC BUS ISOL ESS | AC ESS and DC ESS 3 busses cross feeding manually inhibited. |
| ELEC BUS ISOL MAIN | AC BUS 1, AC BUS 2, DC BUS 1, DC BUS 2, DC ESS 1, DC ESS 2 busses cross feeding manually inhibited. |
| EXT PWR IN USE | External power connected and online. |
| L GEN DISC | L GEN disconnected manually. |
| R GEN DISC | R GEN disconnected manually. |
| L GEN OFF | L GEN OFF while left engine running with L GEN switch selected to OFF. |
| R GEN OFF | R GEN OFF while right engine running with R GEN switch selected to OFF. |

# EICAS Message List

| Electronic Display System | |
|---|---|
| Warning messages | Description |
| CONFIG AP | Autopilot engaged on takeoff |
| CONFIG BRAKE | Parking brake set |
| CONFIG FLAP | Slat/Flap not configured for takeoff |
| CONFIG RUDDER TRIM | Rudder trim position not in T/O range (out of the green arc) |
| CONFIG SIDESTICK | One sidestick priority latched at takeoff |
| CONFIG SPOILER | Spoiler not stowed |
| CONFIG STAB TRIM | Stabilizer trim position not in T/O range (out of the green arc) |
| Caution messages | Description |
| ADS 3 FAIL | Loss of data or internal failure of Air Data System No. 3. |
| ADS 1 SLIPCOMP FAIL | Sideslip Compensation capability (cross and diagonal) is lost on ADSP 1. To avoid dual EICAS caution message, inhibited if reversion to Estimated Beta is due to cross-side PROBE HEAT FAIL. |
| ADS 2 SLIPCOMP FAIL | Sideslip Compensation capability is lost on ADSP 2. To avoid dual EICAS caution message, inhibited if reversion to Estimated Beta is due to cross-side PROBE HEAT FAIL. |
| ADS 3 SLIPCOMP FAIL | Sideslip Compensation capability is lost on ADSP 3. To avoid dual EICAS caution message, inhibited if reversion to Estimated Beta is due to cross-side PROBE HEAT FAIL. |
| ADS 4 SLIPCOMP FAIL | Sideslip Compensation capability is lost on ADSP 4. To avoid dual EICAS caution message, inhibited if reversion to Estimated Beta is due to cross-side PROBE HEAT FAIL. |
| ADS-B OUT FAIL | ADS-B function unavailable due to loss of GPS input at both transponder level. |
| ADS-B 1 OUT FAIL | XPDR 1 is the selected XPDR and has declared an ADS-B out function failed (both GPS input invalid). |
| ADS-B 2 OUT FAIL | XPDR 2 is the selected XPDR and has declared an ADS-B out function failed (both GPS input invalid). |
| ADS ISI PROBE FAIL | ADSP 3 and ADSP 4 combined heater failure. |
| ADS ISI SLIPCOMP FAIL | Sideslip Compensation capability is lost on ADSP 3 and ADSP 4. |
| AURAL WARN FAIL | Aural warning alerts failed. |
| DMC 1 FAIL | Data Module Cabinet (DMC) channel 1A and 1B failed. |
| DMC 2 FAIL | DMC channel 2A and 2B failed. |
| DUAL ADS FAIL | Two ADS sources are failed, or DMC channels 1B and 2B have failed. |
| EFIS COMPARATOR FAIL | Electronic Flight Instrument System (EFIS) comparator has failed. |
| EFIS MISCOMPARE | Data source parameters miscompare between the PFDs. |
| ENG DSPL MISCOMPARE | Engine parameters do not agree between the Primary Flight Displays (PFDs). |
| IPC 1 FAIL | Integrated Processing Cabinet (IPC) 1 has failed. |
| IPC 2 FAIL | IPC 2 has failed. |
| IPC 3 FAIL | IPC 3 has failed. |
| IPC 4 FAIL | IPC 4 has failed. |
| RAD ALT FAIL | Lost of radio altimeter function. |
| Advisory messages | Description |
| ADS FAULT | Lost of redundant or non-critical function for the Air Data System (ADS) system. |
| ADS 1 DEGRADED | Static Source Error Connection (SSEC) correction lost and based on default input value(s) for ADS 1 – Includes loss of Angle Of Attack (AOA) offset. |
| ADS 2 DEGRADED | SSEC correction lost and based on default input value(s) for ADS 2 – Includes loss of AOA offset. |
| ADS 3 DEGRADED | SSEC correction lost and based on default input value(s) for ADS 3 – Includes loss of AOA offset. |
| ADS 4 DEGRADED | SSEC correction lost and based on default input value(s) for ADS 4 – Includes loss of AOA offset. |
| ADS 1 FAIL | Loss of ADS 1 channel. |
| ADS 2 FAIL | Loss of ADS 2 channel. |
| ADS 4 FAIL | Loss of ADS 4 channel. |
| ADS MAINT MODE ACTIVE | Any of the ADS is in maintenance mode. |
| CAS MISCOMPARE | EICAS processing logic for one particular message in disagreement between Data Module Cabinet (DMC) channels. (Miscompare is done for warning, caution, advisory, caution and aurals) |
| DMC 1A FAIL | DMC channel 1A failure. |

# EICAS Message List

| | |
|---|---|
| DMC 1B FAIL | DMC channel 1B failure. |
| DMC 2A FAIL | DMC channel 2A failure. |
| DMC 2B FAIL | DMC channel 2B failure. |
| L PFD-ISI SAME ADS | The left PFD and the ISI have the same source of air data. |
| R PFD-ISI SAME ADS | The right PFD and the ISI have the same source of air data. |
| RAD ALT 1 FAIL | Radio altimeter system 1 failure. |
| RAD ALT 2 FAIL | Radio altimeter system 2 failure. |
| RAD ALT 3 FAIL | Radio altimeter system 3 failure. |
| Status messages | Description |
| A/C MAINTENANCE SW | Maintenance switch in UPLOAD or MAINT position. |
| AURAL WARN INHIB | Aural system has been disabled manually. |
| CTP OVERRIDE | Left or right Control Tuning Panel (CTP) override through avionics EICAS page. |
| CURSOR INHIB | Left or right cursor is inhibited. |

| Fire and Overheat Protection | |
|---|---|
| Warning messages | Description |
| AFT CARGO FIRE | Aft cargo compartment smoke detected. |
| APU FIRE | APU fire detected. |
| EQUIP BAY SMOKE | Smoke is detected in either equipment bay. |
| FWD CARGO FIRE | Forward cargo compartment smoke detected. |
| LAV SMOKE | Lavatory smoke detected. |
| L ENG FIRE | Left engine fire detected. |
| R ENG FIRE | Right engine fire detected. |
| MLG BAY OVHT | Main landing gear bay overheat detected. |
| Caution messages | Description |
| AFT CARGO BTL FAIL | Failure (aft cargo squibs inoperative) of either of the cargo compartment fire extinguishing bottles (HRD or LRD). |
| AFT CARGO SMOKE FAIL | Failure of the aft cargo smoke detection system. |
| APU BTL FAIL | Failure (low pressure or squib inoperative) of APU fire-extinguishing bottle. |
| APU FIRE DET FAIL | Failure of the APU fire detection system. |
| CARGO BTL FAIL | Failure (low pressure or both squibs inoperative) of either of the cargo compartment fire-extinguishing bottles (HRD or LRD). |
| EQUIP BAY SMOKE FAIL | Failure of both smoke detectors in either equipment bay. |
| FIRE SYSTEM FAIL | Total failure of FIDEX system, including loss of both channels of the controller unit. |
| FWD CARGO BTL FAIL | Failure (forward cargo squibs inoperative) of either of the cargo compartment fire-extinguishing bottles (HRD or LRD). |
| FWD CARGO SMOKE FAIL | Failure of the forward cargo smoke detection system. |
| LAV SMOKE FAIL | Failure of the lavatory smoke detection system. |
| L–R ENG BTL FAIL | Failure (abnormally low pressure in both bottles or all four squibs inoperative) of engine fire-extinguishing bottles. |
| L ENG BTL FAIL | Failure (both squibs inoperative left engine – bottles 1 and 2) of fire extinguishing bottles. |
| R ENG BTL FAIL | Failure (both squibs inoperative right engine – bottles 1 and 2) of fire extinguishing bottles. |
| L ENG FIRE DET FAIL | Failure of the left engine fire detection system. |
| R ENG FIRE DET FAIL | Failure of the right engine fire detection system. |
| MLG BAY OVHT DET FAIL | Failure of the main landing gear bay overheat detection system. |
| Advisory messages | Description |
| APU BTL LO | Low pressure condition found on APU fire-extinguishing bottle (normal or abnormal). |
| CARGO BTL LO | Low pressure condition found on either cargo compartment fire-extinguishing bottles – HRD or LRD (normal). |
| ENG BTL 1 LO | Low pressure condition found on engine BTL 1 (normal or abnormal). |
| ENG BTL 2 LO | Low pressure condition found on engine BTL 2 (normal or abnormal). |
| FIRE SYSTEM FAULT | Loss of redundant or non-critical function of the FIDEX system. |
| L ENG BTL FAULT | Loss of left squib on BTL 1 or BTL 2. |
| R ENG BTL FAULT | Loss of right squib on BTL 1 or BTL 2. |
| Status messages | Description |
| None | - |

# EICAS Message List

| Fly-By-Wire System | |
|---|---|
| Warning messages | Description |
| FLT CTRL DIRECT | Aircraft in AFCU direct mode. |
| Caution messages | Description |
| ADS DEGRADED | PFCC is reporting errors from the air data system that could impact the input to the CLAWS. |
| ALPHA LIMIT | Reduction of upper aircraft pitch angle limit (stall protection limit) due to failures. |
| FLT CTRL DIRECT | Aircraft in direct mode. |
| FLT CTRL DIRECT ADS | Aircraft in direct mode due to air data input failure. |
| FLT CTRL DIRECT IRS | Aircraft in direct mode due to inertial data input failure. |
| Advisory messages | Description |
| FLT CTRL FAULT | Loss of redundant or non-critical function for the Primary Flight Control Systems (PFCSs). |
| FLT CTRL IN TEST | Flight control FBW system in automated test. |
| FLT CTRL TEST REQ | Power-up Built-In Test (PBIT) test interval exceeded. |
| PFCC 1 FAIL | Loss of PFCC 1 (loss of redundancy). |
| PFCC 2 FAIL | Loss of PFCC 2 (loss of redundancy). |
| PFCC 3 FAIL | Loss of PFCC 3 (loss of redundancy). |
| Status messages | Description |
| PFCC 1 OFF | PFCC 1 selected OFF. |
| PFCC 2 OFF | PFCC 2 selected OFF. |
| PFCC 3 OFF | PFCC 2 selected OFF. |

| Primary flight controls | |
|---|---|
| Warning messages | Description |
| CONFIG RUDDER TRIM | Rudder trim position out of range for takeoff. |
| L ELEVATOR FAIL | Left elevator failed below 400 ft. |
| R ELEVATOR FAIL | Right elevator failed below 400 ft. |
| RUDDER FAIL | All three rudder PCUs failed. |
| Caution messages | Description |
| AILERON FAIL | Either left or right aileron failed. |
| L ELEVATOR FAIL | Left elevator failed above 400 ft. |
| R ELEVATOR FAIL | Right elevator failed above 400 ft. |
| L SIDESTICK | Left sidestick failed. |
| R SIDESTICK | Right sidestick failed. |
| RUDDER DEGRADED | Two of the three rudder PCUs failed. |
| PITCH AUTHORITY | Left or right elevator command near maximum operational authority. |
| ROLL AUTHORITY | Left or right aileron command near maximum operational authority. |
| YAW AUTHORITY | Rudder command near maximum operational authority. |
| Advisory messages | Description |
| L PITCH TRIM SW FAIL | Pilot pitch trim switch has failed. |
| R PITCH TRIM SW FAIL | Copilot pitch trim switch has failed. |
| Status messages | Description |
| None | - |

# EICAS Message List

| Secondary flight controls | |
|---|---|
| Warning messages | Description |
| CONFIG SPOILER | Spoiler (MFS or GS) out at takeoff. |
| CONFIG STAB TRIM | Stab trim position out of range for takeoff. |
| Caution messages | Description |
| GND LIFT DUMP FAIL | Automatic ground lift dump function failed. |
| GND SPOILER FAIL | Ground spoilers failed retracted (Panel). |
| SPOILER DEGRADED | Two pairs of MFS failed. |
| SPOILER DPLY | Flight spoiler deployed below 300 ft. |
| SPOILER FAIL | Three or more MFS failed. |
| SPOILER LEVER FAIL | Spoiler lever manual control failed. |
| STAB DEGRADED | Selected stab position is invalid. |
| STAB TRIM FAIL | Stab trim function failed. |
| Advisory messages | Description |
| SPOILER MISMATCH | Spoiler auto retract function engaged. |
| SPOILER DPLY | Flight spoiler deployed with either gear deployed or flaps 4 or 5. |
| Status messages | Description |
| SPOILER OUT | Indicates that the flight spoilers have been commanded. |
| SPOILER MAX | Spoilers are deployed at their maximum position. |

| High Lift System | |
|---|---|
| Warning messages | Description |
| CONFIG FLAP | Slat/Flap not configured for takeoff. |
| Caution messages | Description |
| FLAP FAIL | Any failure that results in the flap system being inoperable. Both flap channels unavailable due to monitor failure. |
| FLAP SLOW | Flap speed less than 30% of dual channel nominal speed. |
| SLAT FAIL | Any failure that results in the slat system being inoperable. Both slat channels unavailable due to monitor failure. |
| SLAT SLOW | Slat speed less than 30% of dual channel nominal speed. |
| SLAT SKEW | Slat skew condition detected or slat asymmetry. |
| SLAT-FLAP FAIL | Any failure that results in the slat and flap systems being inoperable. All channels unavailable due to monitor failure. |
| SLAT-FLAP LEVER FAIL | Failure to command flaps through the flap lever. |
| Advisory messages | Description |
| FLAP FAULT | Loss of redundancy or loss of non-critical functions within the flap system. |
| FLAP SLOW | Only one flap channel available to operate system at approximately half speed (30% – 60%). |
| SLAT FAULT | Loss of redundancy or loss of non-critical functions within the slat system. |
| SLAT SLOW | Only one slat channel available to operate system at approximately half speed (30% – 60%). |
| SLAT-FLAP SLOW | One flap and one slat channel available to operate system at slow speed (RAT deployment) (30% – 60%). |
| Status messages | Description |
| ALTN FLAP DPLY | Alternate flap setting deployed selected (setting 3). |

# EICAS Message List

| Fuel | |
|---|---|
| Warning messages | Description |
| None | - |
| Caution messages | Description |
| APU FUEL SOV FAIL | APU fuel Shutoff Valve (SOV) failed to close/open correctly. |
| FUEL COLLECTOR LO LVL | Fuel in the collector is low. The respective scavenge ejector pump may be inoperative for the left wing tank or right wing tank. |
| FUEL CTR XFR FAIL | Fuel transfer out of center tank has failed, center tank fuel may become unusable. |
| FUEL IMBALANCE | Fuel imbalance between left and right wing tank is greater than 454 kg (1000 lb). |
| FUEL LEAK SUSPECT | Potential fuel leak suspected. |
| FUEL MAN XFR FAIL | Manual transfer failed to perform the commanded operation or to stop the transfer. |
| FUEL TANK HI TEMP | Fuel temperature above the operating limit. |
| FUEL TANK LO TEMP | Fuel temperature is almost at Jet A fuel freezing point of –37 °C (–35 °F). |
| L ENG FUEL LO PRESS | Left engine fuel feed pressure is low. |
| L ENG FUEL SOV FAIL | Left engine fuel SOV failed to close/open correctly. |
| R ENG FUEL LO PRESS | Right engine fuel feed pressure is low. |
| R ENG FUEL SOV FAIL | Right engine fuel SOV failed to close/open correctly. |
| L FUEL LO QTY | Low fuel level detected in left wing tank. |
| R FUEL LO QTY | Low fuel level detected in right wing tank. |
| Advisory messages | Description |
| APU FUEL SOV CLSD | APU fuel SOV is closed after the APU FIRE switch is pressed. |
| FUEL COMPUTER FAIL | Fuel Quantity Computer (FQC) has failed or loss of both ARINC 429 channels from FQC to DMC. |
| FUEL CTR XFR FAULT | Fuel transfer from center tank to the left or right wing tank has failed. |
| FUEL FAULT | Loss of redundant or non-critical function of the fuel system. |
| FUEL GRAV XFR FAIL | Fuel gravity SOV failed to open or close. |
| FUEL INERTING FAULT | Fault detected in the Fuel Tank Inerting System (FTIS). |
| FUEL MAN XFR COMPLETE | Wing tank to wing tank target fuel transfer of 180 kg (400 lb) or wing tank to center tank target fuel transfer of 360 kg (800 lb) has been completed. |
| L BOOST PUMP FAIL | Left AC boost pump has failed to indicate pressure when it is turned on manually or automatically. |
| R BOOST PUMP FAIL | Right AC boost pump has failed to indicate pressure when it is turned on manually or automatically. |
| L ENG FUEL SOV CLSD | Left fuel SOV is closed after the L ENG FIRE switch is pressed. |
| R ENG FUEL SOV CLSD | Right fuel SOV is closed after R ENG FIRE switch is pressed. |
| L FUEL EJECTOR FAIL | Left engine motive flow is faulty. |
| R FUEL EJECTOR FAIL | Right engine motive flow is faulty. |
| Status messages | Description |
| FUEL GRAV XFR ON | Gravity SOV has been commanded open on the FUEL panel. |
| FUEL MAN XFR TO CTR | MAN XFR switch selected to CTR (center tank) and either L or R BOOST PUMP switch has been selected ON. |
| FUEL MAN XFR TO L | MAN XFR switch selected to L (left wing tank) on the FUEL panel. |
| FUEL MAN XFR TO R | MAN XFR switch selected to R (right wing tank) on the FUEL panel. |
| FUEL XFR CTR READY | MAN XFR switch selected to CTR (center tank) but no AC boost pump has been selected ON on the FUEL panel. |
| L BOOST PUMP OFF | Left AC boost pump is turned off from the FUEL panel. |
| R BOOST PUMP OFF | Right AC boost pump is turned off from the FUEL panel. |
| L BOOST PUMP ON | Left AC boost pump is turned on from the FUEL panel. |
| R BOOST PUMP ON | Right AC boost pump is turned on from the FUEL panel. |

# EICAS Message List

| Hydraulics | |
|---|---|
| Warning messages | Description |
| None | - |
| Caution messages | Description |
| HYD 1 HI TEMP | The system No. 1 reservoir temperature is more than 107 °C (225 °F) or 30 seconds after temperature switch indicates high temperature. |
| HYD 2 HI TEMP | The system No. 2 reservoir temperature is more than 107 °C (225 °F) or 30 seconds after the reservoir temperature switch indicates high temperature. |
| HYD 3 HI TEMP | The system No. 3 reservoir temperature is more than 107 °C (225 °F) or 30 seconds after the reservoir temperature switch indicates high temperature. |
| HYD 1 LO PRESS | Low pressure is detected in system No. 1 by:<br>• Engine Driven Pump (EDP 1A) pressure switch, and<br>• Power Transfer Unit (PTU) pressure switch, or<br>• System No. 1 pressure transducer. |
| HYD 2 LO PRESS | Low pressure is detected in system No. 2 by:<br>• EDP 2A pressure switch, and<br>• ACMP 2B pressure switch, or<br>• System No. 2 pressure transducer. |
| HYD 3 LO PRESS | Low pressure is detected in system No. 3 by:<br>• ACMP 3A pressure switch, and<br>• ACMP 3B pressure switch, or<br>• System No. 3 pressure transducer. |
| HYD 1-2 LO PRESS | Low pressure detected in system No. 1 and system No. 2. |
| HYD 1-3 LO PRESS | Low pressure detected in system No. 1 and system No. 3. |
| HYD 2-3 LO PRESS | Low pressure detected in system No. 2 and system No. 3. |
| HYD 1 SOV FAIL | System No. 1 Shutoff Valve (SOV) failed to close when commanded or open when supposed to be closed automatically. |
| HYD 2 SOV FAIL | System No. 2 SOV failed to close when commanded or open when supposed to be closed automatically |
| HYD EDP 1A FAIL | Low pressure in system No. 1 when system No. 1 SOV is open. |
| HYD EDP 2A FAIL | Low pressure in system No. 2 when system No. 2 SOV is open. |
| HYD PTU FAIL | PTU low pressure detected when PTU commanded to be online. |
| HYD PUMP 2B FAIL | ACMP 2B low pressure detected when commanded to be online. |
| HYD PUMP 3A FAIL | ACMP 3A low pressure detected when commanded to be online. |
| HYD PUMP 3B FAIL | ACMP 3B low pressure detected when commanded to be online. |
| HYD RAT PUMP FAIL | Ram Air Turbine (RAT) is deployed and system No. 3 pressure is less than 1800 psi. |
| Advisory messages | Description |
| HYD 1 LO QTY | Aircraft on ground: Reservoir quantity less than 18%. Aircraft in flight: Reservoir quantity less than 13%. |
| HYD 2 LO QTY | Aircraft on ground: Reservoir quantity less than 16%. Aircraft in flight: Reservoir quantity less than 13%. |
| HYD 3 LO QTY | Aircraft on ground: Reservoir quantity less than 14%. Aircraft in flight: Reservoir quantity less than 11%. |
| HYD 1 SOV CLSD | System No. 1 SOV is automatically closed with the L ENG FIRE switch on the engine and APU fire panel or by the overheat relay. |
| HYD 2 SOV CLSD | System No. 2 SOV is automatically closed with the R ENG FIRE switch on the engine and APU fire panel or by the overheat relay. |
| HYDRAULIC FAULT | Loss of non-critical functions or loss of redundancy in the hydraulic systems. |
| Status messages | Description |
| HYD 1 SOV CLSD | System No. 1 SOV manually closed with the HYD 1 SOV guarded switch on the HYDRAULIC panel. |
| HYD 2 SOV CLSD | System No. 2 SOV manually closed with the HYD 2 SOV guarded switch on the HYDRAULIC panel. |
| HYD PTU OFF | PTU manually selected OFF. |
| HYD PTU ON | PTU manually selected ON. |
| HYD PUMP 2B OFF | ACMP 2B manually selected OFF. |
| HYD PUMP 2B ON | ACMP 2B manually selected ON. |
| HYD PUMP 3A OFF | ACMP 3A selected OFF. |
| HYD PUMP 3A ON | ACMP 3A selected ON. |
| HYD PUMP 3B OFF | ACMP 3B selected OFF. |
| HYD PUMP 3B ON | ACMP 3B selected ON. |

# EICAS Message List

| Ice and Rain Protection | |
|---|---|
| Warning messages | Description |
| L WING A/ICE FAIL | Left Wing Anti-Ice Valve (WAIV) not closed in any condition on ground, out of test sequence, overpressure, or low pressure when WAI is not inhibited for at least 15 seconds. |
| R WING A/ICE FAIL | Right Wing Anti-Ice Valve (WAIV) not closed in any condition on ground, out of test sequence, overpressure, or low pressure when WAI is not inhibited for at least 15 seconds. |
| Caution messages | Description |
| ADS ISI PROBE HEAT | ADS 3 – 4 combined heaters failed. |
| ADS 1 PROBE HEAT FAIL | ADS 1 probe not heated per expectations. |
| ADS 2 PROBE HEAT FAIL | ADS 2 probe not heated per expectations. |
| ADS 3 PROBE HEAT FAIL | ADS 3 probe not heated per expectations. |
| ADS 4 PROBE HEAT FAIL | ADS 4 probe not heated per expectations. |
| COWL A/ICE ON | Left or right Cowl Anti-Ice System (CAIS) manually set to ON while the Outside Air Temperature (OAT) is more than approximately 15 °C ( 59 °F). |
| ICE | Ice detected and Wing Anti-Ice System (WAIS) or Cowl Anti-Ice System (CAIS) are failed or off. During the ice detection test (approximately 10 seconds) it is displayed on the EICAS page. |
| L COWL A/ICE FAIL | Left cowl anti-ice system failed (valves closed). This message is inhibited when ENG OPER DEGRADED EICAS caution message is displayed. |
| L COWL A/ICE FAIL ON | Left cowl anti-ice system failed (valves open). This message is inhibited when ENG NACELLE OVHT EICAS caution message is displayed. |
| L ICE DET FAIL | Left ice detector has failed. |
| L SIDE WDW HEAT FAIL | Heater for left side window failed. |
| L WING A/ICE LO HEAT | Combination of low pressure and temperature for left Wing Anti-Ice System (WAIS) (also function of altitude) when correctly configured. |
| L WING A/ICE OVHT | Left wing bleed overtemperature detected. |
| L WSHLD HEAT FAIL | Heater for left windshield failed. |
| R COWL A/ICE FAIL | Right cowl anti-ice system failed (valves closed). This message is inhibited when ENG OPER DEGRADED EICAS caution message is displayed. |
| R COWL A/ICE FAIL ON | Right cowl anti-ice system failed (valves open). This message is inhibited when ENG NACELLE OVHT EICAS caution message is displayed. |
| R ICE DET FAIL | Right ice detector has failed. |
| R SIDE WDW HEAT FAIL | Heater for right side window failed. |
| R WING A/ICE LO HEAT | Combination of low pressure and temperature for right Wing Anti-Ice System (WAIS) (also function of altitude) when correctly configured. |
| R WING A/ICE OVHT | Right wing bleed overtemperature detected. |
| R WSHLD HEAT FAIL | Heater for right windshield failed. |
| WING A/ICE FAIL | Left or right Wing Ant-Ice System (WAIS) failed. |
| WING A/ICE LEAK | Left or right wing anti-ice leak detected. |
| WING A/ICE MISCONFIG | Wing Anti-Ice System (WAIS) set to ON or Wing Anti-Ice System (WAIS) set to AUTO and ice detected with the engine bleed OFF for both engines, or WAIS inhibited in high altitude due to single bleed operation. |
| WING A/ICE ON | Wing Anti-Ice System (WAIS) set to ON and Total Air Temperature (TAT) more than 15 °C (59 °F). |
| Advisory messages | Description |
| ICE | Ice detected, Wing Anti-Ice System (WAIS) and Cowl Anti-Ice System (CAIS) are working correctly. |
| L SIDE WDW HT FAIL ON | Heater of the left side window failed on. |
| R SIDE WDW HT FAIL ON | Heater of the right side window failed on. |
| L WSHLD HEAT FAIL ON | Heater of the left windshield failed on. |
| R WSHLD HEAT FAIL ON | Heater of the right windshield failed on. |
| WING A/ICE FAULT | Loss of redundant or non-critical function for the WAIS, or WAIV high leakage. |
| Status messages | Description |
| ADS PROBE HEAT GND ON | Probe heaters manually commanded ON and the Air Data Smart Probes (ADSP) are heating the Multifunction Probes (MFPs). |
| L COWL A/ICE OFF | Left cowl anti-ice manually set to OFF. |
| R COWL A/ICE OFF | Right cowl anti-ice manually set to OFF. |
| L-R COWL A/ICE OFF | Left and right cowl anti-ice manually set to OFF. |
| L COWL A/ICE ON | Left cowl anti-ice manually set to ON. |

# EICAS Message List

| | |
|---|---|
| R COWL A/ICE ON | Right cowl anti-ice manually set to ON. |
| L-R COWL A/ICE ON | L COWL and R COWL anti-ice switches manually set to ON. |
| L SIDE WDW HEAT OFF | Heater for left side window set to OFF. |
| R SIDE WDW HEAT OFF | Heater for right side window set to OFF. |
| L WSHLD HEAT OFF | Heater for left windshield set to OFF. |
| R WSHLD HEAT OFF | Heater for right windshield set to OFF. |
| WING A/ICE OFF | Wing Anti-Ice System (WAIS) set to OFF. |
| WING A/ICE ON | Wing Anti-Ice System (WAIS) set to ON. |

| **Landing Gear System** | |
|---|---|
| Warning messages | Description |
| GEAR | Attempt to land is detected and one (or more) of the landing gear is not down and locked. |
| Caution messages | Description |
| GEAR DISAGREE | Discrepancy between landing gear position and commanded position of the landing gear lever after time delay. |
| GEAR FAIL | Landing gear normal extension/retraction system failed or no communication is available. |
| WOW FAIL | Weight-On-Wheels (WOW) functionality system failed or no communication is available. |
| Advisory messages | Description |
| ALTN GEAR DN | Alternate extension dormancy check is done on the uplocks. |
| GEAR FAULT | Landing gear extension/retraction system redundancy is lost or dormancy test has detected a failure of the alternate extension. |
| WOW FAULT | Weight-On-Wheels (WOW) functionality system redundancy is lost. |
| Status messages | Description |
| ALTN GEAR DN | Alternate extension switch is detected in down position. |
| GEAR AURAL CNCL | Gear aural warning inhibited. |
| PEDAL STEER DISC | Pedal steering has been disconnected. |

| **Nosewheel Steering System** | |
|---|---|
| Warning messages | Description |
| None | - |
| Caution messages | Description |
| NOSE STEER FAIL | Nosewheel steering system failed OR no communication is available from both LGSCUs. |
| NOSE STEER MISALIGN | Nose wheel angle is beyond active steering range. |
| Advisory messages | Description |
| NOSE STEER FAULT | Loss of redundancy of nosewheel steering system, or reduced functionality. |
| Status messages | Description |
| NOSE STEER OFF | Steering disabled (castor mode) through the NOSE STEER switch. |
| PEDAL STEER DISC | Rudder pedal steering off is evaluated, and reported by one of the Landing Gear and Steering Control Units (LGSCUs). |

| **Wheels, Tires and Brakes** | |
|---|---|
| Warning messages | Description |
| BRAKE OVHT | Any brake temperature reaches the temperature range of 15 or more. |
| CONFIG BRAKE | Parking brake is applied when in takeoff mode. |
| Caution messages | Description |
| AUTOBRAKE FAIL | Failure of the autobrake system after being armed or autobrake function cannot be armed. |
| BRAKE FAIL | Total loss of braking in the normal mode (NORM BRAKE FAIL) and the alternate mode (ALTN BRAKE FAIL). |
| BRAKE ON | Detection of uncommanded brake force greater than 200 lb or parking brake is on in flight. |
| CPLT BRAKE PEDAL FAIL | Normal and alternate modes for the copilot pedal braking are not available on the left and/or right copilot brake pedals. |
| L BRAKE FAIL | Three or more EMAs on left landing gear are failed, or two EMAs adjacent to each other on left landing gear have failed, or at least one wheel speed transducer (both channels) on left landing gear wheels has failed. |

# EICAS Message List

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|---|---|
| NORM BRAKE FAIL | Total loss of braking in the normal mode or communication failure from Brake Data Concentrator Units (BDCUs). |
| NOSE TIRE LO PRESS | Nose tire pressure below 70% of nominal pressure in flight. |
| PARK BRAKE FAIL | Failure to set parking brake or insufficient parking force available after commanding the function. |
| PLT BRAKE PEDAL FAIL | Normal and alternate modes for the pilot pedal braking are not available on the left and/or right pilot brake pedals. |
| R BRAKE FAIL | Three or more EMAs on right landing gear are failed, or two EMAs adjacent to each other on right landing gear have failed, or at least one wheel speed transducer (both channels) on right landing gear wheels has failed. |
| Advisory messages | Description |
| ALTN BRAKE FAIL | Failure of the alternate brake function or loss of a pilot and copilot brake pedal. |
| BRAKE FAULT | Loss of redundant or non-critical function for the brake control system. |
| BRAKE TEMP | Any brake temperature range reaches 7 or greater. |
| L BRAKE DEGRADED | Partial loss of functionality on left brake. |
| R BRAKE DEGRADED | Partial loss of functionality on right brake. |
| TIRE LO PRESS | On ground, any tire < 90% of nominal pressure. In flight, main tire < 70% of tire pressure. |
| TIRE PRESS FAULT | One or more sensors failed on tire (info collector). |
| Status messages | Description |
| ALTN BRAKE ON | ALTN BRAKE switch has been selected ON and confirmed on. |
| AUTOBRAKE HI | AUTOBRAKE switch is in the HI position and arming conditions are acceptable. |
| AUTOBRAKE LO | AUTOBRAKE switch is in the LO position and arming conditions are acceptable. |
| AUTOBRAKE MED | AUTOBRAKE switch is in the MED position and arming conditions are acceptable. |
| AUTOBRAKE RTO | AUTOBRAKE switch is in the RTO position and arming conditions are acceptable. |
| PARK BRAKE ON | PARK BRAKE switch has been selected to ON and parking brake on. |

| **Lighting** | |
|---|---|
| Warning messages | Description |
| None | - |
| Caution messages | Description |
| EMER LTS OFF | The emergency light switch is selected to OFF. |
| Advisory messages | Description |
| None | - |
| Status messages | Description |
| CABIN PWR OFF | The cabin power switch is selected to OFF on the electrical control panel. |
| EMER LTS ON | The emergency lights are on. |
| NO PED | The NO PED signs are activated (auto or manual) in the cabin. |
| SEAT BELTS | The seat belt signs are activated (auto or manual) in the cabin. |

| **Navigation** | |
|---|---|
| Warning messages | Description |
| None | - |
| Caution messages | Description |
| GNSS NOT AVAIL | Displayed when the FMS does not use Global Navigation Satellite System (GNSS) position data as part of its calculations to determine position. Displayed if either of the conditions that follow is true:<br>• All GNSS sensors are not available and at least one sensor is enabled, or<br>• At least one sensor is disabled and one sensor is enabled and no enabled sensor is available. |
| IRS SAME SOURCE | Two of the three IRS failed. |
| IRS SET HEADING | Align-In-Motion (AIM) occurs but not in AIM attitude mode or excessive motion in reversion attitude mode. |
| TAWS FAIL | Terrain Awareness and Warning System (TAWS) failed. |
| TCAS FAIL | Traffic Alert and Collision Avoidance System (TCAS) failed. |
| TCAS OFF | TCAS in the standby mode. One or more display is reporting TCAS in standby and aircraft is in-air and TCAS is not failed. |

# EICAS Message List

| | |
|---|---|
| WXR ON | Weather radar is transmitting while aircraft is on ground and no engine is running. |
| XPDR 1 FAIL | Transponder 1 is active and has failed. |
| XPDR 2 FAIL | Transponder 2 is active and has failed. |
| Advisory messages | Description |
| IRS 1 FAIL | IRS 1 failure reported. |
| IRS 2 FAIL | IRS 2 failure reported. |
| IRS 3 FAIL | IRS 3 failure reported. |
| IRS 1 PWR FAULT | The IRS is operating on auxiliary power, or auxiliary power is not available to the IRS. |
| TAWS GPWS FAIL | TAWS mode 1–6 not available and the other TAWS functions are working. |
| TAWS MAP FAIL | Absolute terrain map function of TAWS not available and the other TAWS functions are working. |
| TAWS TERR FAIL | Terrain awareness function inoperative or not available and the other TAWS functions are working. |
| TAWS WINDSHEAR FAIL | TAWS mode 7 not available and the other TAWS functions are working. |
| WXR AUTO FAULT | WXR is reporting an auto tilt mode fault and other WXR functions are working. |
| WXR CTRL FAULT | Both WXR controls have faults. |
| WXR FAIL | WXR transceiver inoperative. |
| WXR FAULT | WXR minor failure. |
| WXR PWS FAIL | WXR is reporting a PWS failure and other WXR functions are working. |
| WXR TURB FAULT | WXR is reporting a turbulence mode fault and other WXR functions are working. |
| Status messages | Description |
| CTP OVERRIDE | Left or right Control Tuning Panel (CTP) override through AVIONIC synoptic page (CTP tab). |
| CURSOR INHIB | Left or right Cursor Control Panel (CCP) cursor inhibited. |
| TAWS FLAP INHIB | TAWS flap alerts inhibited by the flight crew. |
| TAWS GEAR INHIB | TAWS gear alerts inhibited by the flight crew. |
| TAWS GS CNCL | TAWS Glideslope (GS) alerts cancelled by the flight crew. |
| TAWS TERR INHIB | TAWS terrain alerts inhibited by the flight crew. |

| Oxygen and Emergency Equipment | |
|---|---|
| Warning messages | Description |
| None | - |
| Caution messages | Description |
| CREW OXY LO PRESS | Flight crew oxygen low pressure. |
| ELT ON | ELT is transmitting. |
| Advisory messages | Description |
| None | - |
| Status messages | Description |
| PAX OXY DPLY | Cabin oxygen masks deployed automatically or manually. |

| Power Plant | |
|---|---|
| Warning messages | Description |
| DUAL ENG FAIL | Dual engine failure. |
| L ENG OIL PRESS | Left engine oil pressure below or above normal range. |
| R ENG OIL PRESS | Right engine oil pressure below or above normal range. |
| Caution messages | Description |
| L COWL A/ICE FAIL | Left cowl anti-ice system failed (valves closed). This message is inhibited when the EICAS caution message L ENG OPER DEGRADED is displayed. |
| R COWL A/ICE FAIL | Right cowl anti-ice system failed (valves closed). This message is inhibited when the EICAS caution message R ENG OPER DEGRADED is displayed. |
| L COWL A/ICE FAIL ON | Left cowl anti-ice system failed (valves open). This message is inhibited when the EICAS caution message L ENG NACELLE OVHT is displayed. |
| R COWL A/ICE FAIL ON | Right cowl anti-ice system failed (valves open). This message is inhibited when the EICAS message R ENG NACELLE OVHT is displayed. |
| COWL A/ICE ON | Left or right cowl anti-ice manually selected ON while the OAT is above approximately 15 °C. |
| ENG OIL LO TEMP | Oil of either engine is too cold to allow a high thrust to be set. |

# EICAS Message List

| | |
|---|---|
| ENG SETTING MISMATCH | Thrust management data received/fed back by FADEC does not match (APR setting, N1 target, etc.). |
| ENG VIBRATION | Left or right engine vibration (either fan, N1 or N2). |
| L ENG EXCEEDANCE | Left N1 or left N2 or left ITT or left oil temperature above threshold. |
| R ENG EXCEEDANCE | Right N1 or right N2 or right ITT or right oil temperature above threshold. |
| L ENG FAIL | Left engine sub-idle. |
| R ENG FAIL | Right engine sub-idle. |
| L ENG NACELLE OVHT | Left burst cowl anti-ice duct or left HPC valve failed open or buffer air shutoff valve failed open. |
| R ENG NACELLE OVHT | Right burst cowl anti-ice duct or right HPC valve failed open or buffer air shutoff valve failed open. |
| L-R ENG FUEL FILTER | Both engine fuel filters are bypassed (clogged filters). Likely fuel contamination. |
| L ENG OIL FILTER | Left engine oil filter is bypassed (clogged filter). |
| R ENG OIL FILTER | Right engine oil filter is bypassed (clogged filter). |
| L ENG OPER DEGRADED | Uncertainty on T2/Total Air Temperature (TAT) (degraded or default value), or HPC stator vane actuator not tracking as it should. |
| R ENG OPER DEGRADED | Uncertainty on T2/Total Air Temperature (TAT) (degraded or default value), or HPC stator vane actuator not tracking as it should. |
| L ENG START ABORT | Left engine starting procedure has aborted. |
| R ENG START ABORT | Right engine starting procedure has aborted. |
| L ENG STARTER FAIL ON | Air Turbine Starter (ATS) is not disengaging or the left Starter Air Valve (SAV) failed open. |
| R ENG STARTER FAIL ON | Air Turbine Starter (ATS) is no disengaging or the right Starter Air Valve (SAV) failed open. |
| L REVERSER FAIL | Left thrust reverser failed or not available. Icon only shown if thrust reversers have been commanded. |
| R REVERSER FAIL | Right thrust reverser failed or not available. Icon only shown if thrust reversers have been commanded. |
| L REVERSER UNLOCK | Multiple left reverser locks unlocked. |
| R REVERSER UNLOCK | Multiple right reverser locks unlocked. |
| L THROTTLE FAIL | Left thrust lever position not recognized. |
| R THROTTLE FAIL | Left cowl anti-ice system failed (valves closed). This message is inhibited when the EICAS caution message L ENG OPER DEGRADED is displayed. |
| THROTTLE IN REVERSE | Right cowl anti-ice system failed (valves closed). This message is inhibited when the EICAS caution message R ENG OPER DEGRADED is displayed. |
| Advisory messages | Description |
| L ENGINE FAULT | Loss of redundant or non-critical function for the left engine. |
| R ENGINE FAULT | Loss of redundant or non-critical function for the right engine. |
| L ENG FUEL FILTER | Left engine fuel filter is in impending bypass or is bypassed. |
| R ENG FUEL FILTER | Right engine fuel filter is in impending bypass or is bypassed. |
| ENG START DELAY | FADEC detects a risk of bowed rotor. |
| L FUEL FLOW DEGRADED | FADEC supplies synthesized fuel flow instead of measured fuel flow. Accuracy of the displayed left fuel flow is degraded. |
| R FUEL FLOW DEGRADED | FADEC supplies synthesized fuel flow instead of measured fuel flow. Accuracy of the displayed right fuel flow is degraded. |
| L ENG OIL LO QTY | The oil quantity in the left engine is below 9.8 quarts. |
| R ENG OIL LO QTY | The oil quantity in the right engine is below 9.8 quarts. |
| L ENG PCE DOOR OPEN | Left Pre-cooler Exhaust (PCE) door commanded open (due to pre-cooler overtemperature condition). |
| R ENG PCE DOOR OPEN | Right Pre-cooler Exhaust (PCE) door commanded open (due to pre-cooler overtemperature condition). |
| L ENG STARTER OVHT | Left starter usage does not fulfill the duty cycle criteria: 30 minutes cool down time for 4 minutes (low N2) or 30 seconds (high N2). |
| R ENG STARTER OVHT | Right starter usage does not fulfill the duty cycle criteria: 30 minutes cool down time for 4 minutes (low N2) or 30 seconds (high N2). |
| Status messages | Description |
| APR DISARM | Automatic Power Reserve (APR) is disarmed in the Flight Management System (FMS) and derated takeoff is being used. |
| ENG CONT IGNITION ON | At least one FADEC receives request from flight deck switch to have continuous ignition. |
| L COWL A/ICE ON | Left cowl anti-ice manually selected ON. |
| R COWL A/ICE ON | Right cowl anti-ice manually selected ON. |

# EICAS Message List

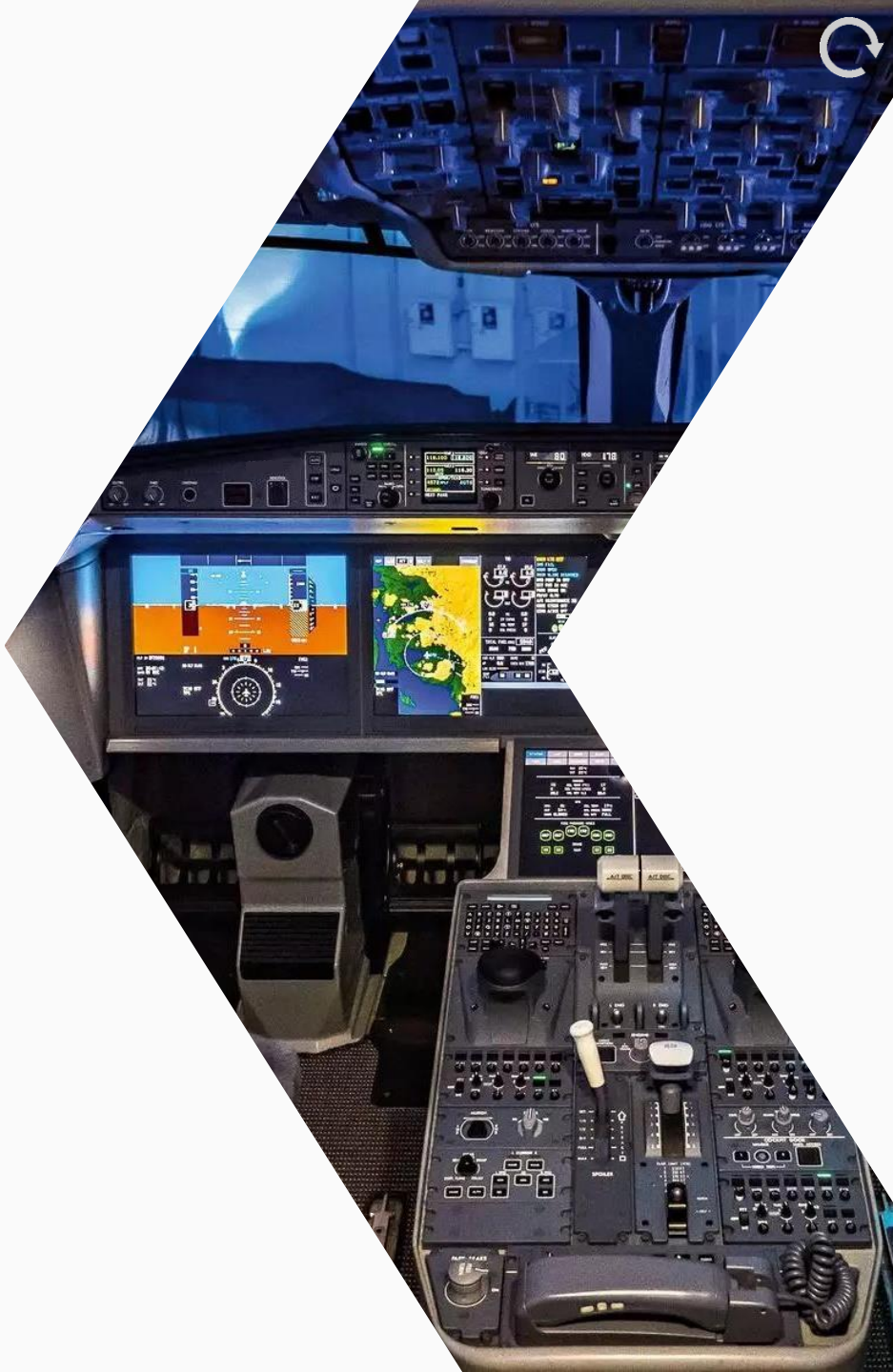
| | |
|---|---|
| L–R COWL A/ICE ON | Left and right cowl anti-ice manually selected ON. |
| L COWL A/ICE OFF | Left cowl anti-ice manually selected OFF. |
| R COWL A/ICE OFF | Right cowl anti-ice manually selected OFF. |
| L–R COWL A/ICE OFF | Left and right cowl anti-ice manually selected OFF. |
| L ENG SHUTDOWN | In-flight pilot commanded engine shutdown (L ENG switch to OFF). |
| R ENG SHUTDOWN | In-flight pilot commanded engine shutdown (R ENG switch to OFF). |
| L REVERSER INHIBIT | Left thrust reverser manually inhibited, as per maintenance procedure. |
| R REVERSER INHIBIT | Right thrust reverser manually inhibited, as per maintenance procedure. |

| **Recording** | |
|---|---|
| Warning messages | Description |
| None | - |
| Caution messages | Description |
| None | - |
| Advisory messages | Description |
| CVR FAIL | CVR failure. |
| FDR FAIL | FDR failure. |
| HEALTH MGMT FAULT | Generic fault detected in the HMU. |
| FDR ACCEL FAIL | Accelerometer declared failed if signal is not reading a reasonable acceleration. |
| Status messages | Description |
| A/C MAINTENANCE SW | Maintenance switch in UPLOAD or MAINT position. |
| PILOT EVENT | Momentary status message for confirmation of PILOT EVENT switch selection. |
| BATT PWR CONFIG | ELEC switch on the maintenance panel selected to DC ESS 3. |

| **Flight Management System** | |
|---|---|
| Caution messages | Description |
| APPROACH NOT AVAIL | Shows when a Global Navigation Satellite System (GNSS) approach is selected, the aircraft is in the arrival terminal area, the FMS navigation mode is not GNSS base or the SBAS service provider is not available (for an LPV only approach), or the GNSS accuracy is less than required for the approach. |
| FMS 1 FAIL | Flight Management System (FMS) 1 has detected a failure. |
| FMS 2 FAIL | FMS 2 has detected a failure. |
| FMS FUEL | Shows when FMS message CHECK FUEL AT ALTN or CHECK FUEL AT DEST or CHECK FUEL PLAN or CHECK BLOCK FUEL is posted. |
| FMS OEI PERF ACTIVE | One Engine Inoperative (OEI) performance data active and can be canceled on the THRUST window. |
| FMS PERF DEP CONFIG | Mismatch between sensed aircraft configuration state (bleeds, ANTI-ICE switches, SLAT/FLAP lever position) and FMS entered configuration. |
| FMS PERF DEP VSPEEDS | FMS Gross Speed check has found Vspeeds values not entered between minimum/maximum operational allowed values. |
| FMS POSITION | Shows when one of the FMS messages that follow is shown: INITIALIZE POSITION or RE-ENTER SET POS or RESET INITIAL POS. |
| LPV NOT AVAIL | Shows when a GNSS LPV approach is selected, the aircraft is in the arrival terminal area, and the FMS navigation mode is not GNSS based, or the SBAS service provider is not available, or the GNSS accuracy is less than required for the approach. |
| UNABLE RNP | Shows when loss of integrity condition exists. The aircraft may not be able to maintain the required RNP accuracy when this message shows. |

| **Surface Management System** | |
|---|---|
| Advisory messages | Description |
| SMS FAIL | SMS aural and visual alerts not available. |
| SMS RWY N/A | SMS runway not present in the FMS data. |
| Status messages | Description |
| SMS RUNWAY INHIB | SMS aural and visual alerts inhibited by the flight crew. |

# Nuisance Guidance



# Nuisance Guidance

| Nuisance | Indications | Background Information | Recommendations |
|---|---|---|---|
| Equipment bay cooling fault | EQUIP BAY COOL FAULT (advisory) | EQUIP COOLING, INLET must be switched OFF before aircraft power is removed, otherwise this advisory will post next power-up. | When power is restored, cycle EQUIP COOLING INLET switch OFF, back to AUTO with at least one door open. |
| Recirculation fan noise on ground and in flight | - | The recirculation fan can be noisy during initial pull-up or pull-down temperature sequences. | To reduce noise levels, the flight crew can select RECIRC AIR to OFF without consequence. |
| N1 Sync function deactivates | Green SYNC disappearing or cycling on / off on EICAS | The FADEC synchronizes N1 speeds between the two engines to minimize acoustic beat noise by adjusting the TLA via the autothrottle (similar to a propeller synchronizer). When there is a mismatch between autothrottle channels, synchronization mode disengages (green SYNC disappears). | No crew action required. |
| Loss of autothrottle RETARD mode on approach over mountainous or irregular terrain | No RETARD (white) on the FMA<br><br>Thrust levers not moving to idle thrust during flare | The autothrottle depends on the landing phase to be active. The landing phase is triggered when the radio altimeter detects an altitude below 400 ft AGL and will remain active for a defined period of time. During approaches over mountainous terrain the radio altimeter might detect an altitude of 400 ft AGL and trigger the landing phase timer early. If landing occurs after timer has run out, RETARD will not arm. This has been observed for RWY 28 at RKTH (Pohang, Korea) and RWY 34 at LGTS (Thessaloniki, Greece). | It is recommended to disconnect the autothrottle and manually select IDLE. |
| No autothrottle upon manually activating Emergency Descent Mode (EDM) | No AT green on FMA,<br><br>Thrust levers not moving to idle thrust | The PFCC engages the AP, the EICAS engages the AT. These work at different frequencies. As a result, if the flight crew presses the EDM pushbutton for less than 100 milliseconds, the AP activates but not the AT and thrust levers are not retarded to idle. | To manually activate EDM, it is recommended to press and hold the EDM button for 1 second. |
| Unexpected autothrottle behavior in flight | RETARD (white) or RETARD (green) while not on approach<br><br>GA thrust mode on EICAS or thrust mode toggling between CLB and GA every 2 seconds | Reports of unexpected changes in the autothrottle mode in flight:<br><br>• RETARD mode armed in cruise or early on the approach<br>• RETARD activating in cruise or early on approach, commanding idle thrust<br>• RETARD arming during climb, CLB thrust changing to GA thrust, causing an increase in rate of climb or thrust mode toggling between CLB and GA thrust every 2 seconds with slight TLA movement and thrust fluctuations | Flight crew should be vigilant to possible uncommanded autothrottle changes to RETARD mode (armed or active) and disconnect the autothrottle as required. Also uncommanded changes in thrust from CLB to GA (on EICAS) while the aircraft is not in the landing configuration should be overridden with the FMS THRUST dialog box. |
| Electrical fault due to ground power unit disconnect or off | ELECTRICAL FAULT (adv)<br>24 ELECTRICAL FAULT – GND CART INOP (info) | The Ground Power Unit has either been disconnected or turned off while it was IN USE. To prevent this, EXT PWR switch must be selected to AVAIL before it is disconnected or switched OFF. | To clear the message, cycle the EXT PWR switch, if the messages are still displayed, a full aircraft power reset must be performed. |

# Nuisance Guidance

| Nuisance | Indications | Background Information | Recommendations |
|---|---|---|---|
| Electrical fault due to electrical power being restored too quickly | ELECTRICAL FAULT (adv)<br>24 ELECTRICAL FAULT – EPC 3 DEGRADED (info) | When the aircraft is powered on within 7 seconds after power-off, these messages may be shown approximately 30 seconds after power is restored. | To prevent this, there must be a minimum time of 15 seconds between aircraft power-off and power-on, a full aircraft power reset must be done to clear these messages. |
| ADS degraded due to crosswind on ground | ADS 1 (2)(3)(4) DEGRADED (advisory) | In a location with reported crosswinds and during taxi ADS 1(2)(3)(4) DEGRADED can occur and latch. | The message should clear above 50 to 80 kts. If not, maintenance action is required. |
| Electronic FCS test (PBIT) | During test (advisory):<br>FLT CTRL IN TEST<br>ADS 1 (2)(3)(4) DEGRADED<br>NOSE STEER FAULT<br><br>After test (advisory):<br>FLT CTRL FAULT | PBIT is the electronic power-on test for the system, designed to ensure integrity of the electrical unit protections and identify dormancies in the system. PBIT is an automated test, no crew action is required.<br><br>The PBIT test sequence is initiated automatically at PFCC power-up when all three hydraulic systems are unpressurized approximately 20 seconds after the first PFCC completed initialization or when the system completes 20 electrical hours uninterrupted. It takes approximately 80 secs to complete. | During PBIT sequence: FLT CTRL IN TEST and ADS 1 (2)(3)(4) DEGRADED are shown, these messages clear after PBIT sequence.<br><br>NOSE STEER FAULT can post momentarily after PBIT completes. If no FLT CTRL FAULT is shown, the test is considered a PASS. |
| Actuation FCS test (ABIT) | Before test (advisory):<br>FLT CTRL TEST REQ<br><br>During test (advisory):<br>FLT CTRL IN TEST<br><br>After test (advisory):<br>FLT CTRL FAULT | The ABIT is a test designed to ensure integrity of the mechanical protections related to actuators and identify dormancies in the system. There should not be surface motion, under some failure conditions, surface motion is part of the failure indication.<br><br>The ABIT must be done once a day or when FLT CTRL TEST REQ advisory is shown after the aircraft completed 20 flight hours. The flight crew initiates the test via the AVIONIC synoptic page (AVIO, TEST – FLT CTRL TEST) when all three hydraulic systems are pressurized. The ABIT takes approximately 50 seconds to complete. | See Parking & Shutdown for ABIT test procedure.<br><br>Wait until both engines spool down (N2 = 0%) before performing AVIO, TEST – FLT CTRL TEST on the AVIONIC synoptic page.<br><br>Phase / EICAS / AVIO, TEST page<br>In progress / FLT CTRL IN TEST / PROG<br>Test failed / FLT CTRL FAULT / FAULT<br>Successful / - / PASS |
| Flight control input on ground | FLT CTRL IN TEST (advisory) | While FLT CTRL IN TEST (advisory) is shown, do not make aircraft configuration changes such as APU start / shutdown, GPU connection / disconnection, engine start, switch hydraulic pumps or switch PFCCs.<br>Do NOT move sidesticks, rudder pedals, spoiler or flap levers. | It is not recommended to make flight control inputs during electrical power supply changes (APU start / shutdown, external power connection / disconnection) or when FLT CTRL IN TEST is shown on EICAS. |
| Flight control test required | FLT CTRL TEST REQ (advisory) | Message can only be shown when the airplane is on ground, a flight control test must be completed before flight. | See Actuation FCS test (ABIT) above. |

# Nuisance Guidance

| Nuisance | Indications | Background Information | Recommendations |
|---|---|---|---|
| Equipment bay smoke failure indication while on ground | EQUIP BAY SMOKE FAIL (caution) | Electrical power transients during engine start or shutdown can cause one or two extraction fans to cycle, the FIDEX control unit will temporarily detect a low flow condition. | These messages can show for up to 2 minutes after engine start or shutdown. If they stay on after 2 minutes, maintenance action is required. |
| Fuel collector low level during single engine operations while on ground | FUEL COLLECTOR LO LVL (caution) | Might appear during single engine operation on ground when the fuel quantity is less than 850 kg on the side where the engine is not operating and will extinguish when the second engine is started. | No crew action required. |
| Fuel manual transfer caution message after fuel transfer completed | FUEL MAN XFR FAIL (caution) | After a successful transfer, the FUEL MAN XFR COMPLETE (advisory) is shown, if the crew does not select MAN XFR switch to OFF for a period of time, FUEL MAN XFR FAIL messages appears and will latch. | On ground: reset L FUEL QTY (CDC 1-6-3) and R FUEL QTY (CDC 2-8-3) on the CB synoptic page. In-flight: select MAN XFR switch to OFF, then to the desired position. |
| Wing anti-ice test while in single bleed configuration | - | Make sure that the engine with the operating bleed is at or above 50% (or 60% above 8000 ft) N1 before the wing anti-ice test is started. This will prevent EGT exceedance and possible damage to the engine. Coordination with ATC might be required to ensure a safe engine blast zone behind the aircraft (apron, runway, other). | See Before Taxi for wing anti-ice test procedure. Above 15°C OAT, if the wing anti-ice system is ON for more than 3 minutes, the WING A/ICE ON caution appears. |
| Gear fault after selection of landing gear lever to the up or down position | GEAR FAULT (advisory) | After selection of the landing gear lever to the UP or DN position, the GEAR FAULT advisory message can be shown and latch due to malfunction during the automatic dormancy check of the alternate gear extension system together with INFO message:<br>• 32 GEAR FAULT – L GEAR UPLK MOTOR FAULT<br>• 32 GEAR FAULT – R GEAR UPLK MOTOR FAULT<br>• 32 GEAR FAULT – NOSE GEAR UPLK MOTOR FAULT | The landing gear system continues to operate correctly. After landing, the latched GEAR FAULT advisory message (and INFO message if any) can be cleared by selecting the ALTN GEAR switch to the DN position, wait 10 seconds, and then select the switch back to NORM position.<br>If messages do not clear, maintenance action is required. |
| Weight-on-wheel fail due to center of gravity movement | WOW FAIL (caution) | During boarding / deboarding operations, the center of gravity can move aft and result In light weight condition on the nose landing gear shock strut causing the WOW FAIL (weight-on-wheel fail) caution message. | The aircraft must be loaded and unloaded in accordance with the WBM loading instructions. To clear the caution, center of gravity must move towards the front of aircraft. |
| Nose landing gear shock strut assembly leakage | - | The tiller deflection rates on the nose landing gear steering must be managed to help reduce the potential of leakage from the nose landing gear shock strut assembly. Avoid large steering commands when the aircraft is static, forward movement is preferred before steering. | To reduce side loads, crew should be careful with sharp turn angles, when possible crew should use a smooth and gradual movement of the tiller and avoid sharp and quick inputs to reach maximum tiller deflection. |

# Nuisance Guidance

| Nuisance | Indications | Background Information | Recommendations |
|---|---|---|---|
| Take-off and Landing Awareness Function (TLAF) nuisance alerts | "NOT A RUNWAY" aural alert<br><br>"RUNWAY" shown on PFD | There have been reports of TLAF nuisance alerts during taxi and take-off phases, these nuisance alerts are caused by:<br><br>• Spurious GPS spike data used by the TLAF<br>• TLAF using non-synchronized FMS navigation database cycle<br>• Database errors (e.g. incorrect runway magnetic heading)<br>• Inaccurate runway data in FMS navigation database | To prevent nuisance TLAF alerts (other than spurious GPS spike data), perform a complete power-off after the first selection of a new ACTIVE PERIOD FMS navigation database. If power-off cannot be done, inhibit TLAF during taxi and take-off phases. |
| Inoperative STBY WXR ON switch (range knob) on the Control Tuning Panel (CTP) | - | The STBY WXR ON switch on the CTP is not active (by design). | - |
| Engine oscillations during crossbleed engine start | Fluctuations in EGT, N2, FF and to a lesser extend N1 | During a ground start, the starter valve cycles to maintain a low N2 (13%) at the beginning of the start phase. When the opposite engine is used as a bleed source, the engines continuously adjusts itself to the bleed demand (fluctuations in EGT, N2 and FF). | This behavior is considered normal, as long as parameters stay within limits. |
| Engine fault advisory during crossbleed engine start | L (R) ENGINE FAULT (adv)<br>73 L(R) ENGINE FAULT – FUEL EQUAL INOP (info) | May be displayed if the second engine start is commanded too quickly after the first engine reaches IDLE or in strong crosswind and tailwind conditions. | It is recommended to wait at least 1 minute after the first engine reaches idle and position the aircraft into the wind during strong wind conditions. |
| Engine shutdown procedure and cooling requirements<br><br>High power engine shutdown | - | The engine must operate at or near idle for at least 3 minutes (cumulative) before shutdown (recommended 5 minutes). Taxi times may be counted when thrust is at or near idle.<br><br>This does not apply to non-normal engine shutdowns. Near idle is defined as up to 40% N1. An additional period equal to or greater than the time spend above 40% N1 must be accounted for.<br><br>CAUTION: engine shutdown after high power operation (above 40% N1) without adequate cooldown can cause accelerated deterioration. In case of a non-normal shutdown engine shutdown at high power (greater than 40% N1), the engine must not be restarted until cleared by maintenance. | For an inadvertent high-power shutdown, the following procedure is required. Perform the tasks, in order:<br><br>1. If restart is possible, restart the engine within 2 minutes and stabilize at IDLE for 10 minutes before shutdown.<br>2. If restart is not possible, but cranking is possible, dry crank the engine immediately. Dry motor for 4 minutes, then pause for a 60-second period starting when N2 speed is 0%. This should be repeated two more times.<br>3. If neither restart nor cranking is possible, it is recommended to only attempt engine restart after 8 hours (no restriction on emergency in-flight restarts). |

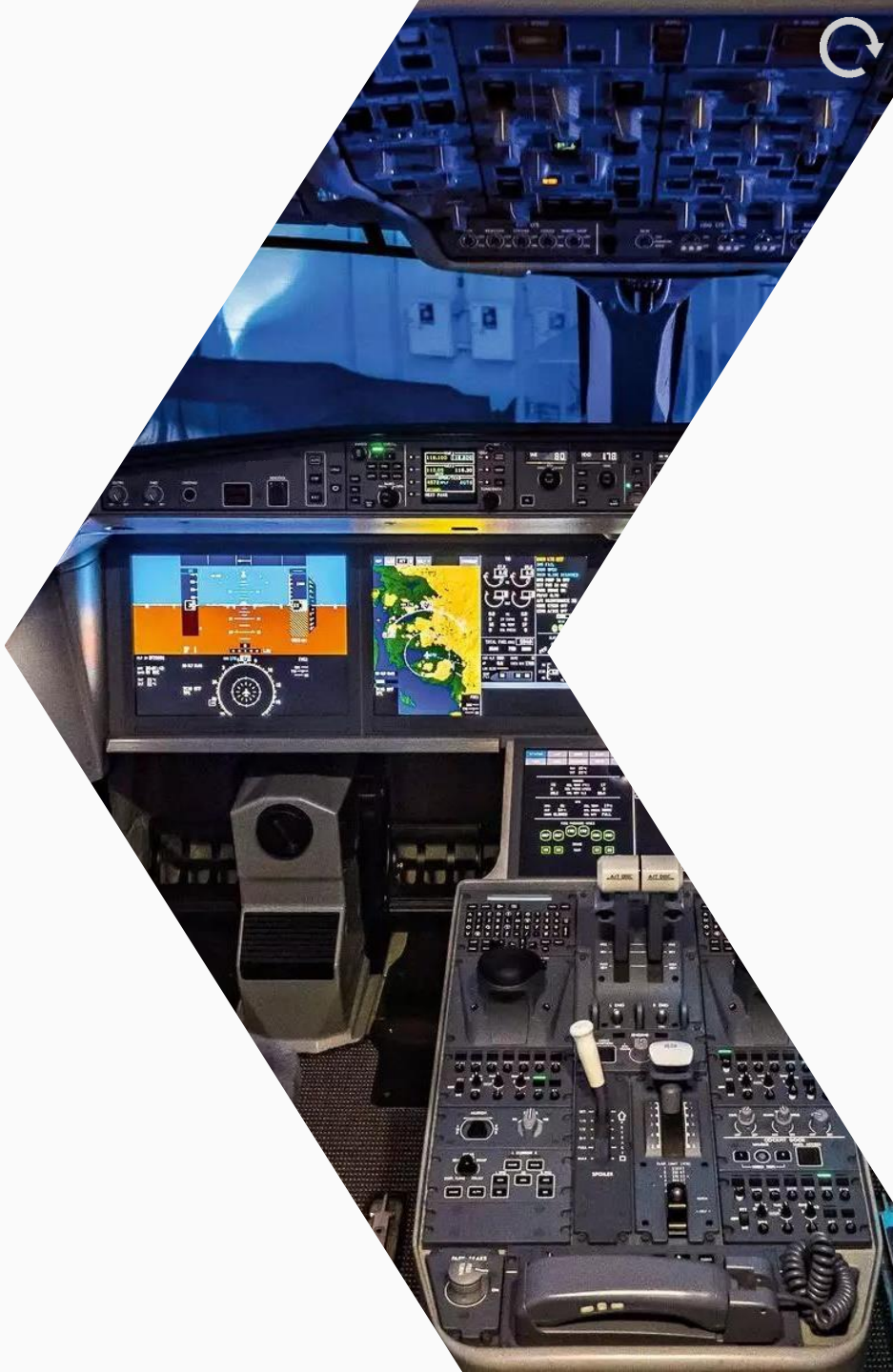
# Nuisance Guidance

| Nuisance | Indications | Background Information | Recommendations |
|---|---|---|---|
| Flight Management System (FMS) resets | FMS 1 FAIL or FMS 2 FAIL (caution)<br><br>FMS FAIL flag on the associated PFD<br><br>FMS MAP shown in amber on the associated map<br><br>FMS NOT AVAILABLE shown on the FMS window<br><br>FMS speed target indicates magenta dashes above speed tape | There are several known instances of FMS reset, flight crew must be aware of conditions that can lead to such resets so they can be prevented.<br><br>CAUTION: on FMS recovery after reset, the FMS speed target may revert to approach speed, the FMS will not automatically sequence the missed approach procedure if TOGA is selected within 2 minutes after a reset.<br><br>Conditions that can lead to a FMS reset:<br><br>• When an out-of-range number of passengers is entered on the FPLN – FUEL page.<br>• When simultaneous flight plan edits are made – a FMS reset can occur, or a modification done to a waypoint different from the one that was planned when a flight plan is edited at the same time as a waypoint is automatically sequenced (passed) or during another simultaneous crew action. This can occur on the following entries:<br><br>a. Direct-to dialog box: entry of an altitude in the vertical direct field during sequencing of a waypoint, selection of vertical direct-to soft switch during automatic sequencing of a waypoint.<br>b. FPLN – WIND/TEMP: entry of ALT, OAT or WIND values if the entry occurs at the same time as the flight plan is cleared.<br>c. ROUTE – LEGS: entry or deletion of speed or altitude values in the SPD / ALT field can cause the entry to be applied to the wrong leg or an FMS reset can occur if selection occurs during sequencing of a waypoint. | It is recommended to avoid the steps that could lead to known FMS reset scenarios.<br><br>If a reset occurs, use MAN SPD. In case of TOGA within 2 minutes after FMS reset, the crew must select the SEQ AUTO checkbox on the ROUTE – LEGS page manually. |
| Block fuel locked out | BLOCK FUEL field appear greyed out | In certain cases, during FMS programming on DU 5, the required BLOCK FUEL field can appear greyed out on either FMS 1 or FMS 2 or both, prohibiting data entry. | In such cases, the crew should use DU 2 (FMS 1) or DU 3 (FMS 2) to enter the block fuel. |
| Departure performance locked-out | All entries on the PERF – DEP tab appear greyed out | In certain cases, after all required entries are made on the FPLN – FUEL tab, the PERF – DEP tab can appear to be locked out, where entries or changes are not allowed, the stab trim may start flashing. | In such cases, to reset the PERF – DEP tab, the crew must re-enter the ZFW – WT and CG values on the FPLN – FUEL tab. |

# Nuisance Guidance

| Nuisance | Indications | Background Information | Recommendations |
|---|---|---|---|
| Flight plan uplink soft switch locked-out | The FPLN UPLINK soft switch appear greyed out | Sometimes the FPLN UPLINK soft switch may appear to be greyed out and cannot be accessed. | Switching to any other FMS page (for example FPLN – WIND/TEMP), then back to FPLN – INIT will unfreeze the FPLN UPLINK soft switch. |
| Predictive Windshear fail shown intermittently in flight | WXR PWS FAIL (advisory) | The weather radar may report a nuisance failure of the Predictive Windshear System (PWS). When this condition occurs, the WXR PWS FAIL advisory is shown. This nuisance message normally self-clears. | If the WXR PWS FAIL advisory message is no longer shown after landing, no further action is required. If it stays on after landing, maintenance action is required. |
| Predictive Windshear false alerts | Take-off:<br><br>Aural: "WINDSHEAR AHEAD"<br>Red WSHR-AHEAD on PFD<br><br>Approach:<br><br>Aural: "GO-AROUND, WINDSHEAR AHEAD"<br>WSHR-AHEAD on PFD | There have been reports of suspected Predictive Windshear System (PWS) false alerts at certain airports, example EHAM – Amsterdam while on approach for RWY 27 or 18C.<br><br>The PWS can be susceptible to false alerts at runways considered to be in a clutter-rich environment. A clutter-rich environment is defined as a runway with an adjacent area which has expressways, high volume streets and / or concentrated urban clutter with highways that have random vehicle traffic patterns.<br><br>CAUTION: Reactive Windshear System (RWS) is not affected by these conditions, alerts are valid. | Take-off:<br><br>Before take-off, flight crew should evaluate possible windshear conditions. If a predictive windshear warning is triggered during the take-off phase (less than 80 kts):<br><br>• Aural: "WINDSHEAR AHEAD" x2<br>• Red WSHR-AHEAD on PFD<br><br>The take-off can be continued if flight crew evaluates that there are no signs of windshear conditions and the reactive windshear system is operational.<br><br>Approach:<br><br>When a predictive windshear warning is triggered on approach:<br><br>• Aural: "GO-AROUND, WINDSHEAR AHEAD" x2<br>• Red WSHR-AHEAD on PFD<br><br>Flight crew should evaluate possible erroneous PWS alerts. If the flight crew evaluates that there are no signs of windshear conditions present and if the reactive windshear system is operational, the approach can be continued. |

# MEL / CDL



# Minimum Equipment List (MEL)

With various levels of redundancy designed into an aircraft, serviceability of every system or installed component may not be necessary when the remaining operative components can provide the required level of safety.

The Minimum Equipment List (MEL) lists those items that may be inoperative and yet maintain the required level of safety by following appropriate conditions and limitations. It does not list obviously required items such as wings, flaps and rudders. It is important that repairs be accomplished at the earliest opportunity. When operating with multiple inoperative items, the interrelationships between those items and the effect on aircraft operation and crew workload must be considered. Supplemental procedures the MEL might refer to are found on next pages.

The MEL applies until the aircraft starts moving under its own power, for example, the MEL applies should a defect be discovered during pushback, but not during taxi. After the aircraft starts moving under its own power, the non-normal checklist will specify whether return to ramp is required or not. It is good practice to consult the MEL nonetheless, for example when leaving a maintenance base for an airport without maintenance with a defect containing (M) item.

The MEL consists of 2 sections:

- Section 1 – LRU / component oriented: requires fault isolation through maintenance.
- Section 2 – CAS Message oriented: based on indicated alert without fault isolation.

Dispatch under Section 2 (no fault isolation) will generally be more restrictive, without fault isolation the worst-case failure condition for the posted CAS message is assumed.

(M) Specific maintenance procedure which must be accomplished prior to operation.
(O) Specific operational procedure for planning and/or operating with the listed item inoperative.

Repair intervals:

- Category A: as specified in the MEL (no standard interval)
- Category B: shall be repaired within 3 consecutive days*
- Category C: shall be repaired within 10 consecutive days*
- Category D: shall be repaired within 120 consecutive days*

* Excluding the day of discovery, for example if discovered at 10:00 on the 26th, the time interval begins 00:01 on the 27th and ends at 23:59 on the 29th for a Category B item (3 days).

Master Minimum Equipment List (MMEL) available from Transport Canada website.

## Section 1 – LRU / Component Orientated MEL Relief (example)

| System & Sequence Nº | Item | 1. | 2. Number Installed | 3. Number Required For Dispatch | 4. Remarks or Exceptions |
|---|---|---|---|---|---|
| 21 – AIR CONDITIONING | | | | | |
| 51-01 Air Conditioning Packs | | | | | |
| 2) One pack inoperative | C | 2 | 1 | (M)(O) | Except for extended operations, may be inoperative provided:<br>(a) Affected Air Conditioning Pack is selected OFF,<br>(b) Flight is conducted at or below FL310,<br>(c) 26 FIRE SYSTEM FAULT – EQUIP BAY SMOKE DET REDUND LOSS is not displayed,<br>(d) Operations are conducted in accordance with AFM supplement 5 (Operations with Airplane Systems Inoperative), and<br>(e) Operations with Steep Approach are not conducted. |

(M) Maintenance action required, the aircraft cannot dispatch if no maintenance available

## Section 2 – CAS Message Orientated MEL Relief (example)

| CAS Message Indication | 1. | 2. Remarks and Exceptions |
|---|---|---|
| 21-00-133-01<br>**L PACK FAIL**<br>(CAUTION)<br>21 L PACK FAIL – L PACK INOP | C | (O) Except for extended operations, may be displayed provided:<br>(a) None of the below INFO messages is displayed:<br>26 FIRE SYSTEM FAULT – EQUIP BAY SMOKE DET REDUND LOSS<br>21 PACK FAULT – R BYPASS VLV INOP<br>21 PACK FAULT – R PACK DISCH PRESS SNSR INOP<br>21 R PACK FAULT – R PACK TEMP SNSR INOP<br>21 R PACK OVHT – R PACK INOP<br>(b) Left air conditioning pack is selected OFF,<br>(c) Flight is conducted at or below FL 310,<br>(d) Operations are conducted in accordance with Airplane Flight manual (AFM) Supplement 5 (Operations with Airplane Systems Inoperative), and<br>(e) Operations with steep approach are not conducted. |

No maintenance action required for the same fault if dispatched under section 2 (but may sometimes be more restrictive)

# Configuration Deviation List (CDL)

The Configuration Deviation List (CDL) is part of the Aircraft Flight Manual (AFM) and contains additional limitations for operation without secondary airframe and/or nacelle parts such as seals, panels, static dischargers and gear doors.

Performance penalties can either be applied using the penalties provided or by using the CAFM by entering the CDL index. If several CDL items are missing, the user must add the various CDL index values, the minimum and maximum values are 0 and 30 respectively.

The CDL index has no impact on the stall speed, maneuvering capability and thrust settings and is not considered for take-off speed or distance calculations (drag increment has a negligible effect). Similarly, CDL index is not considered during landing speed or landing distance calculations.

Not more than 1 part for any system can be missing unless specific combinations are given. Unless otherwise specified, parts from different systems can be missing at the same time. The performance penalties are cumulative unless specific penalties for combinations of missing parts are given.

No more than 3 missing parts having a “negligible” effect on performance can be missing for take-off without applying a performance penalty. When more than 3 such parts are missing, the following performance penalties must be applied for each subsequent missing part:

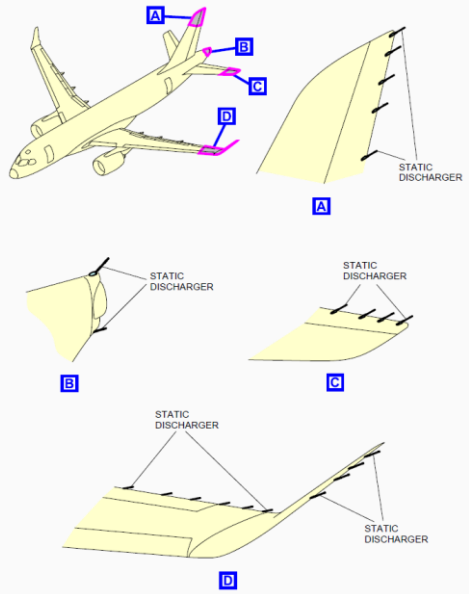
- MTOW or MLW reduced by 61 kgs for take-off and landing for each part.
- Max. weight en-route (driftdown) reduced by 195 kgs for each missing part.
- Fuel consumption must be increased by 0.54% for each missing part.
- Using CAFM, a CDL index of 1.5 must be applied for each missing part.

Any number of items identified as having “no impact” on aircraft performance can be missing at the same time without performance penalty.

## Example 1: CDL item with no impact on aircraft performance

**SYSTEM 23 COMMUNICATIONS**

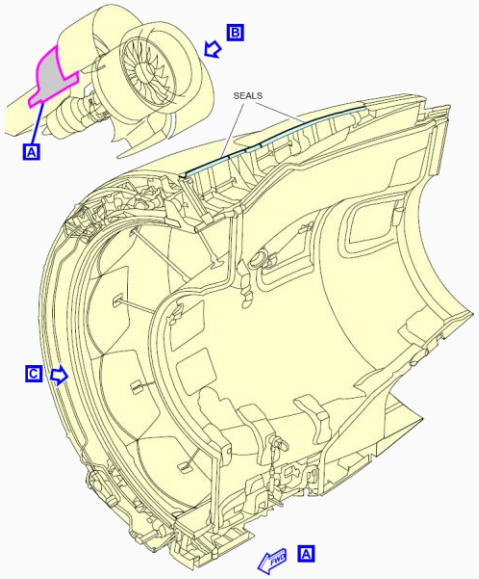
| SUB-SYSTEM ITEM | 1 Number required for all flight conditions except as provided in column 2 | 2 Remarks and/or exceptions |
|---|---|---|
| 23–01 Static discharger | 35 | One or two can be missing with no impact on aircraft performance. |




## Example 2: CDL item with impact on aircraft performance

**SYSTEM 54 NACELLES/PYLONS**

| SUB-SYSTEM ITEM | 1 Number required for all flight conditions except as provided in column 2 | 2 Remarks and/or exceptions |
|---|---|---|
| 54–09 (B) Nacelle seal – External/environmental[2]<br>[2]Part number:<br>• 999–3584–1<br>• 999–3585–1<br>• 999–3585–2<br>• 999–3841–3<br>• 999–3841–4<br>• 999–3842–1<br>• 999–3842–2<br>• 999–3843–3<br>• 999–3843–4<br>• 999–3844–1<br>• 999–3844–2 | 36 | Any number or combination can be missing if:<br>• The performance limited weights are reduced by:<br>**Takeoff weight**: 20 kg/seal \| 44 lb/seal<br>**Enroute climb**: 66 kg/seal \| 144 lb/seal<br>**Landing weight**: 20 kg/seal \| 44 lb/seal<br>CAFM/CIFP: Add 0.5 to the CDL index for each missing seal.<br>• The mission fuel requirements are increased by:<br>**Fuel consumption**: +0.18% on fuel used/seal |



# Operation with Systems Inoperative

Ground spoiler system or multi-function spoiler system 1 inoperative:

- Minimum landing runway width is 45 m if a pedal or rudder jam is encountered.
- ASDR x 1.20 and LDR x 1.30

Landing gear locked down:

- Flight in known or forecasted icing conditions is prohibited.

Brake system Electric Motor Actuator (EMA) or Electric Motor Actuator Controller (EMAC) inoperative:

- Minimum landing runway width is 45 m if a pedal or rudder jam is encountered.
- 1 or 2 EMA inoperative or 1 EMAC inoperative: ASDR x 1.04, LDR x 1.03
- 3 or 4 EMA inoperative or 2 EMAC inoperative: ASDR x 1.11, LDR x 1.10

Airconditioning or bleed source inoperative:

- FLAP 4 landings are prohibited in icing conditions during single bleed operation with all engines operating (AEO) due to high idle thrust level affecting descent capability. Increase VREF by 10 kts to improve descent capability. Additional speed for icing, autothrottle or wind gusts not required.
- Reduced thrust (FLEX) operations are prohibited with air conditioning pack or bleed source inoperative and bleed source is engine (allowed if APU is the bleed source for take-off which is not permitted if OAT is at or below 15°C).

Thrust reverser inoperative:

- Dispatch with one thrust reverser inoperative: the maximum crosswind component for take-off and landing on a wet runway is limited to 25 kts.

TOGA switches inoperative (thrust levers):

- Take-off: use of autothrottle and FMS speeds during take-off phase is prohibited (use manual, max TO thrust).
- At VR, rotate towards the target pitch angle determined from the table that follows and adjust pitch to capture V2 (OEI) or a minimum of V2+10 kts (AEO).

| Target pitch angle at take-off | | | | | | |
|---|---|---|---|---|---|---|
| ΔV = (V2-VR) | FLAP 2 | | FLAP 3 | | FLAP 4 | |
| | AEO | OEI | AEO | OEI | AEO | OEI |
| 0 | 12 | 8 | 12 | 8 | 11 | 8 |
| 1 | 14 | 10 | 13 | 8 | 12 | 8 |
| 2 | 16 | 11 | 14 | 10 | 13 | 8 |
| 3 | 17 | 12 | 15 | 11 | 14 | 9 |
| 4 | 18 | 12 | 16 | 11 | 15 | 11 |
| 5 | 19 | 13 | 17 | 12 | 16 | 11 |
| 6 | 20 | 13 | 18 | 12 | 17 | 11 |
| 7 | 21 | 14 | 19 | 12 | 18 | 12 |
| 8 | 22 | 14 | 20 | 13 | 19 | 12 |
| 9 | 23 | 14 | 21 | 13 | 20 | 12 |
| 10 and above | 23 | 15 | 22 | 13 | 21 | 13 |

- Go-around: use of autothrottle and FMS speeds during go-around phase is prohibited. Rotate towards the target pitch angle below (do not exceed 22°):
- OEI: rotate towards 8 degrees and adjust pitch to capture VAC.
- AEO: rotate towards 11 degrees and adjust pitch to capture VGA.

Flap or Slat channel inoperative:

Slats / flaps will operate at half speed, performance limited take-off weight:

- If final segment climb is not limiting: -5%
- If final segment climb is limiting: -6%

# Operation with Systems Inoperative

BTMS (Brake Temperature Measuring System) inoperative:

After landing or rejected take-off, the minimum brake cooling time at the ramp must be observed to ensure enough brake energy capacity before the next take-off, minimum cooling time is determined as follows:

1. Determine the reference brake energy: the reference brake energy measures the energy absorbed by the brakes in million foot-pounds (mfp). The value for brake energy is at the intersection of the brakes-on speed, airport pressure altitude and the weight, airport outside air temperature.

    - Wind correction: enter the table with the brakes-on speed minus 0.5 times the headwind or plus 2 times the tailwind.
    - Slope correction: decrease brake energy by 3% for each percent of upslope, increase brake energy by 3% for each percent of downslope.
    - Ground speed can be used for brakes-on speed by entering the table at sea level with OAT 15°C.

2. Use the reference brake energy to determine the brake cooling time: to determine the brake cooling time, select the appropriate table based on thrust reverser use during the last braking event. Enter the table with the value determined for brake energy and brake level applied during the last braking event. Two specific zones are identified in the tables. If value is in these zones, these procedures must be done:

    - Caution zone (shaded cells): the wheel fuse plugs can melt because of cumulative braking energy absorbed during taxi after rejected take-off event.
    - Fuse plug melt zone: clear runway immediately. Unless required, do not set the parking brake. Do not approach the gear or try to taxi.
    - A tire pressure check must be done after 40 minutes.

Example: landing weight 50t, landing speed 140 kts, OAT 25°C at pressure altitude 0 ft. Reference brake energy = 29.9 mfp. Landing with autobrake medium without using reverse (reverse idle) = 67 minutes of brake cooling required before next take-off.

**Reference Brake Energy – MFP per Brake**

| Brakes on speed (kts) | Pressure altitude (1000 ft) | 40000 kg OAT - °C | | | 50000 kg OAT - °C | | | 60000 kg OAT - °C | | | 70000 kg & above OAT - °C | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 0 | 25 | 50 | 0 | 25 | 50 | 0 | 25 | 50 | 0 | 25 | 50 |
| 80 | 0 | 11.1 | 11.7 | 12.6 | 12.3 | 13.0 | 14.0 | 13.3 | 14.0 | 15.1 | 14.2 | 14.9 | 16.1 |
| | 2 | 11.8 | 12.4 | 13.3 | 13.1 | 13.8 | 14.8 | 14.1 | 14.9 | 16.0 | 15.1 | 15.8 | 17.0 |
| | 5 | 12.8 | 13.4 | 14.5 | 14.2 | 14.9 | 16.1 | 15.3 | 16.1 | 17.3 | 16.3 | 17.2 | 18.5 |
| | 10 | 14.4 | 15.2 | 16.3 | 16.0 | 16.9 | 18.1 | 17.3 | 18.2 | 19.6 | 18.5 | 19.4 | 20.9 |
| 100 | 0 | 15.2 | 16.0 | 17.2 | 17.6 | 18.5 | 19.9 | 19.9 | 20.9 | 22.5 | 22.2 | 23.4 | 25.1 |
| | 2 | 16.2 | 17.0 | 18.3 | 18.7 | 19.6 | 21.1 | 21.0 | 22.2 | 23.8 | 23.5 | 24.8 | 26.6 |
| | 5 | 17.5 | 18.4 | 19.8 | 20.2 | 21.3 | 22.9 | 22.8 | 24.0 | 25.8 | 25.5 | 26.9 | 28.9 |
| | 10 | 19.8 | 20.9 | 22.4 | 22.9 | 24.1 | 25.9 | 25.8 | 27.2 | 29.2 | 28.8 | 30.4 | 32.6 |
| 120 | 0 | 19.4 | 20.4 | 22.0 | 22.9 | 24.1 | 25.9 | 26.4 | 27.8 | 29.8 | 30.2 | 31.7 | 34.1 |
| | 2 | 20.6 | 21.7 | 23.3 | 24.3 | 25.6 | 27.5 | 28.0 | 29.4 | 31.6 | 32.0 | 33.7 | 36.2 |
| | 5 | 22.3 | 23.5 | 25.3 | 26.3 | 27.7 | 29.8 | 30.3 | 31.9 | 34.3 | 34.7 | 36.5 | 39.2 |
| | 10 | 25.2 | 26.6 | 28.5 | 29.8 | 31.3 | 33.7 | 34.3 | 36.1 | 38.8 | 39.2 | 41.3 | 44.4 |
| 140 | 0 | 23.6 | 24.8 | 26.7 | 28.4 | 29.9 | 32.2 | 33.2 | 35.0 | 37.6 | 38.4 | 40.5 | 43.5 |
| | 2 | 25.0 | 26.3 | 28.3 | 30.1 | 31.7 | 34.1 | 35.2 | 37.1 | 39.9 | 40.8 | 42.9 | 46.1 |
| | 5 | 27.1 | 28.5 | 30.7 | 32.7 | 34.4 | 37.0 | 38.2 | 40.2 | 43.3 | 44.2 | 46.5 | 50.0 |
| | 10 | 30.7 | 32.3 | 34.7 | 37.0 | 38.9 | 41.8 | 43.2 | 45.5 | 48.9 | 50.0 | 52.6 | 56.6 |
| 160 | 0 | 27.8 | 29.2 | 31.4 | 34.1 | 35.9 | 38.6 | 40.5 | 42.6 | 45.8 | 47.3 | 49.7 | 53.5 |
| | 2 | 29.4 | 31.0 | 33.3 | 36.2 | 38.1 | 40.9 | 42.9 | 45.2 | 48.5 | 50.1 | 52.7 | 56.7 |
| | 5 | 31.9 | 33.6 | 36.1 | 39.2 | 41.3 | 44.4 | 46.5 | 49.0 | 52.7 | 54.3 | 57.2 | 61.5 |
| | 10 | 36.1 | 38.0 | 40.8 | 44.3 | 46.7 | 50.2 | 52.6 | 55.4 | 59.5 | 61.4 | 64.7 | 69.5 |

| Reference brake energy (mfp) | One or No Thrust Reverser – Cooling Time (min) | | | | Two Thrust Reversers – Cooling Time (min) | | | |
|---|---|---|---|---|---|---|---|---|
| | Rejected take-off | Landing | | | Rejected take-off | Landing | | |
| | Manual or AB RTO | Manual or AB High | Autobrake Medium | Autobrake Low | Manual or AB RTO | Manual or AB High | Autobrake Medium | Autobrake Low |
| 10 | 30 | No special procedure required – dispatch allowed without further time delay. | | | 32 | No special procedure required – dispatch allowed without further time delay. | | |
| 12 | 36 | | | | 37 | | | |
| 14 | 41 | 30 | | | 42 | 27 | | |
| 16 | 46 | 36 | 31 | 28 | 46 | 33 | | |
| 18 | 51 | 42 | 37 | 33 | 51 | 38 | | |
| 20 | 56 | 47 | 42 | 39 | 56 | 44 | 31 | |
| 22 | 61 | 53 | 47 | 44 | 60 | 49 | 35 | |
| 24 | 66 | 58 | 53 | 48 | 64 | 54 | 39 | |
| 26 | 70 | 63 | 57 | 53 | 68 | 59 | 43 | |
| 28 | 75 | 68 | 62 | 58 | 72 | 63 | 47 | 30 |
| 30 | 79 | 73 | 67 | 62 | 76 | 68 | 50 | 33 |
| 32 | 83 | 78 | 71 | 67 | 80 | 73 | 54 | 35 |
| 34 | | 82 | 76 | 71 | 84 | 77 | 57 | 38 |
| 36 | | 87 | 80 | 75 | | 81 | 61 | 41 |
| 38 | | | 84 | 79 | | | 64 | 43 |
| 40 | | | | 83 | | | 68 | 46 |
| 42 | | | | | | | 71 | 48 |
| 44 | | | | | | | 74 | 51 |
| 46 | | Fuse Plug Melt Zone | | | | Fuse Plug Melt Zone | 77 | 53 |
| 48 | | | | | | | 80 | 56 |
| 50 | | | | | | | 83 | 58 |
| 52 & Above | | | | | | | | 90 |

# Operation with Systems Inoperative

RECIRC AIR selected OFF:

- For operations with CARGO FWD selected to OFF or VENT, the impact of RECIRC AIR selected to OFF is negligible.
- Multiply all fuel flows with the AIR CARGO FWD selected to LO HEAT or HI HEAT and RECIRC AIR selected to OFF by 1.005.

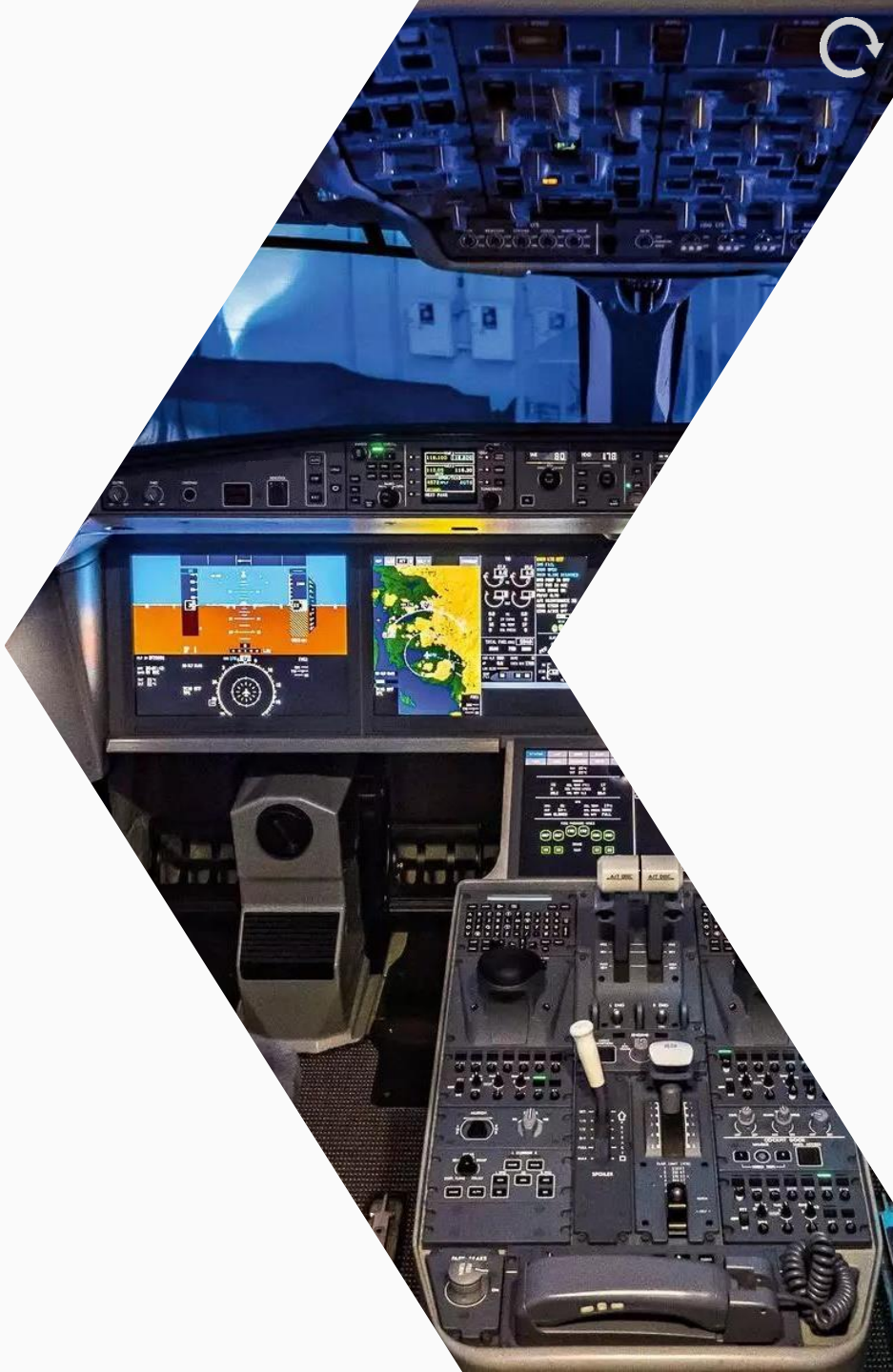
Active Clearance Control (ACC) valve failed in the closed position:

- For loss of active clearance control function of the left and / or right engine, multiply all fuel flows by 1.03.

Pre-cooler Exit (PCE) doors in open position:

- Multiply the one engine inoperative take-off distance and take-off run, the all-engines operating take-off distance, the take-off run and the accelerate-stop distance by 1.05.
- Reduce the take-off weight limited by climb requirements by 544 kg.
- If one or more obstacles are present in the take-off flight path, multiply the distance of the obstacle from the reference point by a factor of 0.95 (obstacles moved closed to the reference point). Additionally, reduce the take-off weight limited by obstacle clearance by 680 kg.
- Reduce the net ceiling by 800 ft. Alternatively, decrease the weight limited by net ceiling by 1225 kg.
- Decrease the cruise altitude limited by the climb ceiling or the maximum cruise speed capability by 800 feet. Alternatively, decrease the weight limited by climb ceiling by 2268 kg.
- Reduce landing weight limited by climb requirements by 544 kg.
- Multiply all fuel flows by 1.02.

# Performance (A220 – 300)

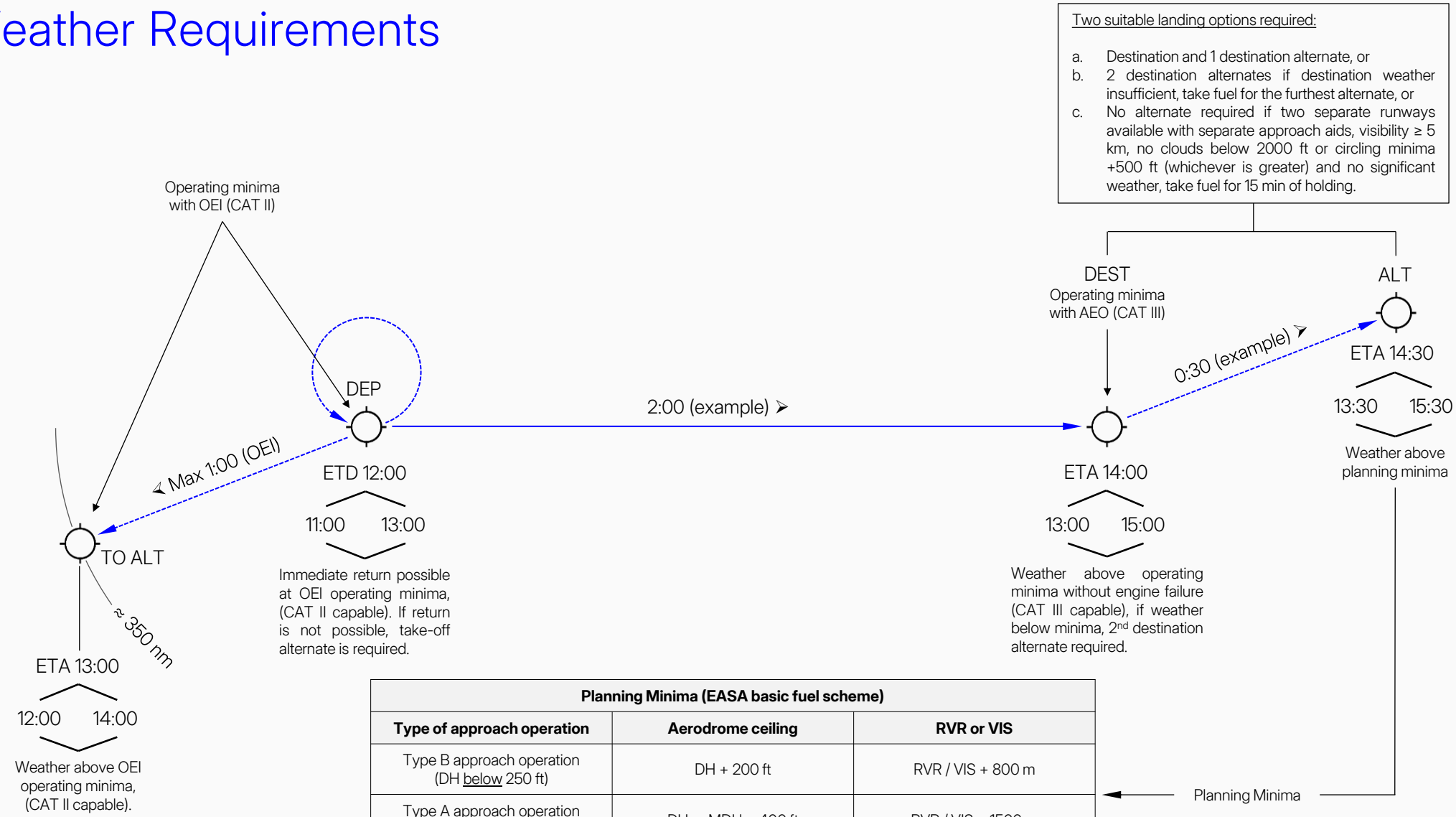


# Flight Planning Structure

| Step | Description | Details |
|---|---|---|
| 1 | Route and aerodromes | Area, route and aerodrome qualification, self-briefing, limitations and required experience. |
| 2 | Aircraft status | Open defects (MEL / CDL), operational impact and deadlines.<br>Active Operational Engineering Bulletins (OEBs). |
| 3 | NOTAM / SNOWTAM | For departure, en-route, destination and alternate(s):<br>Approach aid serviceability, runway condition report, runway / taxiway closures etc. |
| 4 | Weather | For departure, en-route, destination and alternate(s):<br>Check weather at ETD or ETA ± 1 hour, within limits:<br>• Ceiling, visibility, wind<br>• Significant weather<br>En-route weather: SIGMETs, AIRMETs, SWC. |
| 5 | Performance | Maximum (limiting) weight per flight phase:<br>• Take-off, climb<br>• En-route (drift down)<br>• Landing and<br>• Go-around (OEI) |
| 6 | Flight planning | Fuel required for:<br>• Taxi and APU, trip fuel (departure to destination)<br>• Contingency fuel (5% of trip fuel, minimum 5 minutes of flying)<br>• Alternate fuel (destination to alternate, 15 min holding if no alternate required)<br>• Final reserve fuel (30 minutes of holding at 1500 ft above the alternate)<br>• Additional, extra and discretionary fuel as required |
| 7 | Weight & balance | Determine most limiting weight for take-off (lowest of a, b, c):<br>a. Maximum take-off weight<br>b. Maximum landing weight + trip fuel<br>c. Maximum zero fuel weight + take-off fuel<br>Prepare load sheet. |

Might affect weather limits:
special aerodrome limitations, aircraft defects (e.g. no CAT III capability), approach aid availability, runway conditions (for crosswind limits), etc.

# Weather Requirements




| Planning Minima (EASA basic fuel scheme) | | |
|---|---|---|
| **Type of approach operation** | **Aerodrome ceiling** | **RVR or VIS** |
| Type B approach operation (DH below 250 ft) | DH + 200 ft | RVR / VIS + 800 m |
| Type A approach operation (MDH / DH at or above 250 ft) | DH or MDH + 400 ft | RVR / VIS + 1500 m |
| Circling approach operations | MDH + 400 ft | VIS + 1500 m |
| Wind limitations should be applied considering the runway condition (dry, wet, contaminated). | | |

# Weather Minima (EASA Basic Fuel Scheme)

- **Departure** weather and performance should allow for return after take-off within ETD ± 1 hr, using operating minima but taking engine failure into consideration (CAT II capable).
- **Take-off Alternate** (max 1 hour in still air, ISA at OEI cruising speed ≈ 350 nm) allowing for landing within ETA ± 1 hr, at operating minima considering engine failure (CAT II capable).
- **Destination** weather should allow for a landing within ETA ± 1 hr using operating minima, engine failure does not need to be considered (CAT III capable). If weather is below landing minima, two destination alternates are required.
- **Destination Alternate** weather allows for landing within ETA ± 1 hr using planning minima:
  - Type A approach operation: MDH/DH +400 ft, required visibility +1500 m.
  - Type B approach operation: DH +200 ft, required visibility +800 m.

  (Type A means MDH/DH is At or Above 250 ft, Type B if DH is Below 250 ft)
- Other variations such as no destination alternate, isolated aerodrome, en-route alternate.

- Performance (dispatch)

# Take-off Performance

Example:
PW1521, OAT = 20°C
Field length available = 2000 m

Result:
MTOW = 62000 kg

**Note:** 67.585 kg Max Structural Take-off Weight option is used for take-off performance.

| PW1521G Field Length Required (m) | | | | |
|---|---|---|---|---|
| Weight (kg) | OAT (°C) – A/I | | | |
| | 5°C ALL | 10°C CAI | 20°C OFF | 30°C OFF |
| 46000 | 1110 | 1130 | 1150 | 1220 |
| 48000 | 1180 | 1200 | 1225 | 1300 |
| 50000 | 1250 | 1270 | 1295 | 1380 |
| 52000 | 1340 | 1360 | 1385 | 1490 |
| 54000 | 1430 | 1450 | 1475 | 1600 |
| 56000 | 1530 | 1555 | 1575 | 1730 |
| 58000 | 1650 | 1675 | 1690 | 1885 |
| 60000 | 1765 | 1790 | 1800 | 2035 |
| 62000 | 1930 | 1955 | 1970 | |
| 64000 | 2170 | 2200 | 2220 | |
| 66000 | Climb limited | | | |
| 67585 | | | | |

Conditions:
- Sea level, QNH 1013.25 hPa
- Dry runway, wind calm
- Optimum take-off thrust and flap setting
- Non-limiting runway, no limiting obstacles
- Rolling take-off, APR armed
- APU OFF, bleeds and packs ON
- BTMS level 6 or below

| PW1524G Field Length Required (m) | | | | |
|---|---|---|---|---|
| Weight (kg) | OAT (°C) – A/I | | | |
| | 5°C ALL | 10°C CAI | 20°C OFF | 30°C OFF |
| 46000 | 1085 | 1095 | 1125 | 1170 |
| 48000 | 1135 | 1145 | 1175 | 1235 |
| 50000 | 1180 | 1195 | 1220 | 1300 |
| 52000 | 1255 | 1270 | 1300 | 1385 |
| 54000 | 1325 | 1345 | 1375 | 1465 |
| 56000 | 1405 | 1425 | 1455 | 1560 |
| 58000 | 1490 | 1510 | 1540 | 1665 |
| 60000 | 1570 | 1595 | 1620 | 1765 |
| 62000 | 1680 | 1705 | 1730 | 1905 |
| 64000 | 1790 | 1815 | 1835 | 2040 |
| 66000 | 1880 | 1900 | 1915 | 2160 |
| 67585 | 2015 | 2040 | 2035 | 2360 |

Conditions:
- Sea level, QNH 1013.25 hPa
- Dry runway, wind calm
- Optimum take-off thrust and flap setting
- Non-limiting runway, no limiting obstacles
- Rolling take-off, APR armed
- APU OFF, bleeds and packs ON
- BTMS level 6 or below

# Climb (All Engines Operating)

| | CLB | (CLB-2) |
|---|---|---|
| PW1521 | 7.3 | (6.3) |
| PW1524 | 7.6 | (6.5) |

Example:

PW1521
CLB-2
ISA + 20°C
TOW 60t

Result:

Average climb gradient from SL to 10.000 ft is 7.3%.

**Average AEO Gross Climb Gradient to 10000 ft - % CLB (CLB-2)**

| Weight (kg) | ISA Deviation (°C) | | | | | | |
|---|---|---|---|---|---|---|---|
| | ≤ +10°C | +15°C | +20°C | +25°C | +30°C | +35°C | +40°C |
| 70900 | 8.1 (7.0)<br>8.4 (7.3) | 7.3 (6.3)<br>7.6 (6.5) | 6.5 (5.5)<br>6.7 (5.7) | 5.6 (4.7)<br>5.9 (4.9) | — (4.1)<br>5.1 (4.2) | — (—)<br>4.4 (3.5) | — |
| 68000 | 8.6 (7.5)<br>9.1 (7.9) | 7.8 (6.8)<br>8.2 (7.0) | 6.9 (5.9)<br>7.2 (6.2) | 6.1 (5.2)<br>6.3 (5.4) | 5.3 (4.4)<br>5.5 (4.6) | — (—)<br>4.8 (3.9) | — (—)<br>4.1 (3.3) |
| 65000 | 9.2 (8.1)<br>9.7 (8.5) | 8.4 (7.3)<br>8.8 (7.6) | 7.5 (6.4)<br>7.8 (6.7) | 6.6 (5.6)<br>6.8 (5.8) | 5.7 (4.8)<br>6.0 (5.0) | 5.0 (4.1)<br>5.2 (4.3) | — (—)<br>4.5 (3.6) |
| 60000 | 10.4 (9.2)<br>11.0 (9.7) | 9.5 (8.3)<br>9.9 (8.7) | 8.4 (7.3)<br>8.8 (7.6) | 7.5 (6.4)<br>7.8 (6.7) | 6.6 (5.6)<br>6.8 (5.8) | 5.8 (4.8)<br>6.0 (5.0) | 5.0 (4.1)<br>5.2 (4.3) |
| 55000 | 11.8 (10.5)<br>12.5 (11.0) | 10.8 (9.5)<br>11.3 (9.9) | 9.6 (8.4)<br>10.1 (8.8) | 8.5 (7.4)<br>8.9 (7.7) | 7.5 (6.4)<br>7.8 (6.7) | 6.6 (5.6)<br>6.9 (5.8) | 5.9 (4.9)<br>6.1 (5.0) |
| 50000 | 13.6 (12.0)<br>14.3 (12.6) | 12.4 (10.9)<br>13.0 (11.4) | 11.1 (9.7)<br>11.6 (10.1) | 9.8 (8.6)<br>10.3 (8.9) | 8.7 (7.5)<br>9.1 (7.8) | 7.7 (6.6)<br>8.0 (6.8) | 6.8 (5.7)<br>7.1 (5.9) |
| 45000 | 15.8 (14.0)<br>16.6 (14.7) | 14.4 (12.7)<br>15.1 (13.3) | 12.9 (11.3)<br>13.5 (11.8) | 11.5 (10.0)<br>12.0 (10.5) | 10.2 (8.8)<br>10.4 (9.2) | 9.0 (7.8)<br>8.8 (8.1) | 8.0 (6.8)<br>8.3 (7.0) |
| 40000 | 18.8 (16.6)<br>19.7 (17.4) | 17.1 (15.1)<br>17.9 (15.8) | 15.4 (13.5)<br>16.1 (14.1) | 13.7 (11.9)<br>14.3 (12.4) | 12.1 (10.5)<br>12.6 (10.9) | 10.7 (9.3)<br>11.2 (9.7) | 9.5 (8.1)<br>9.9 (8.5) |

- Normal take-off
- TO thrust until FLAP 0
- Flap retraction at 1500 ft AAE
- Climbing at FLAP 0 at 250 kts

PW1521G Corrections:
- Each 1000 ft above SL = -0.8
- ECS ON = -0.1
- ALL A/I + ECS ON = -1.6

PW1524G Corrections:
- Each 1000 ft above SL = -0.9
- ECS ON = -0.1
- ALL A/I + ECS ON = -1.9

# Enroute

| | |
|---|---|
| PW1521 | 41.600 |
| PW1524 | (41.700) |

Example:

PW1521
ISA + 10°C
Highest en-route MSA = 18000 ft

Result:

Maximum weight at most critical point en-route is 58.5t. Add fuel burn from take-off to most critical point to find maximum take-off weight limited by terrain clearance.

Or conservatively, assume 58.5t to be the maximum take-off weight.

**OEI net level off weight - PW1521G (PW1524G)**

| Bleed Source | Weight (kgs) | | | |
|---|---|---|---|---|
| | Altitude (ft) | ISA and below | ISA +10°C | ISA +20°C |
| ENG | 28000 | 41.600<br>(41.700) | 40.300<br>(40.400) | — |
| | 26000 | 45.000<br>(45.100) | 43.000<br>(43.000) | 40.500<br>(40.500) |
| | 24000 | 48.500<br>(48.600) | 46.000<br>(45.900) | 43.300<br>(43.100) |
| APU | 22000 | 52.300<br>(52.300) | 52.200<br>(52.300) | 49.300<br>(49.400) |
| | 20000 | 56.100<br>(56.100) | 56.000<br>(56.100) | 52.800<br>(52.800) |
| | 18000 | 58.500<br>(60.200) | 58.500<br>(60.100) | 55.700<br>(56.400) |
| | 16000 | 60.000<br>(64.300) | 59.900<br>(63.900) | 57.800<br>(60.100) |
| | 14000 | 62.800<br>(68.000) | 62.700<br>(67.800) | 60.300<br>(63.000) |
| | 12000 | 66.200<br>(71.000) | 66.100<br>(71.000) | 63.000<br>(66.100) |
| | 10000 | 69.200<br>(71.000) | 69.000<br>(71.000) | 65.700<br>(68.900) |

Conditions:
- Thrust MCT
- Clean configuration
- Engine bleed above 23.000 ft

# Landing PW1521G

Shaded areas limited by maximum structural landing weight (not performance limiting).

- F4 dry runway (F4 wet runway)
- F5 dry runway (F5 wet runway)

| Step 1 – Wind Corrected Runway Length | | | | | |
|---|---|---|---|---|---|
| Runway Length (m) | Wind (kts) | | | | |
| | Tailwind | Calm | Headwind | | |
| | –10 | 0 | +10 | +20 | +30 |
| 1500 | 1140 (1100)<br>1180 (1140) | 1500 (1500)<br>1500 (1500) | 1580 (1590)<br>1580 (1590) | 1670 (1690)<br>1670 (1680) | 1760 (1770)<br>1760 (1760) |
| 1600 | 1260 (1200)<br>1260 (1230) | 1600 (1600)<br>1600 (1600) | 1690 (1690)<br>1690 (1690) | 1780 (1790)<br>1780 (1790) | 1870 (1880)<br>1860 (1870) |
| 1700 | 1350 (1290)<br>1370 (1340) | 1700 (1700)<br>1700 (1700) | 1800 (1800)<br>1800 (1800) | 1890 (1900)<br>1900 (1900) | 1980 (1990)<br>1990 (2000) |
| 1800 | 1460 (1420)<br>1460 (1420) | 1800 (1800)<br>1800 (1800) | 1910 (1900)<br>1900 (1900) | 2010 (2000)<br>2000 (2000) | 2110 (2100)<br>2100 (2100) |
| 1900 | 1550 (1500)<br>1550 (1510) | 1900 (1900)<br>1900 (1900) | 2000 (2010)<br>2000 (2010) | 2110 (2110)<br>2110 (2120) | 2210 (2220)<br>2210 (2220) |
| 2000 | 1640 (1600)<br>1640 (1620) | 2000 (2000)<br>2000 (2000) | 2110 (2120)<br>2110 (2120) | 2210 (2230)<br>2210 (2230) | 2320 (2340)<br>2320 (2340) |
| 2100 | 1740 (1710)<br>1740 (1710) | 2100 (2100)<br>2100 (2100) | 2210 (2220)<br>2210 (2220) | 2320 (2330)<br>2320 (2330) | 2420 (2460)<br>2420 (2440) |
| 2200 | 1830 (1800)<br>1830 (1800) | 2200 (2200)<br>2200 (2200) | 2310 (2320)<br>2310 (2320) | 2420 (2440)<br>2420 (2440) | 2530 (2560)<br>2530 (2560) |
| 2300 | 1920 (1890)<br>1910 (1890) | 2300 (2300)<br>2300 (2300) | 2410 (2420)<br>2410 (2420) | 2530 (2550)<br>2530 (2550) | 2640 (2670)<br>2640 (2660) |
| 2400 | 2010 (1980)<br>2000 (1980) | 2400 (2400)<br>2400 (2400) | 2520 (2520)<br>2520 (2520) | 2630 (2650)<br>2650 (2650) | 2750 (2770)<br>2750 (2770) |
| 2500 | 2060 (2080)<br>2090 (2080) | 2500 (2500)<br>2500 (2500) | 2610 (2630)<br>2620 (2630) | 2730 (2760)<br>2740 (2760) | 2850 (2880)<br>2860 (2880) |
| 2600 | 2160 (2170)<br>2180 (2150) | 2600 (2600)<br>2600 (2600) | 2720 (2730)<br>2720 (2730) | 2840 (2860)<br>2850 (2860) | 2960 (2990)<br>2980 (2990) |

Example:

FLAP 4 landing, runway dry
10 kts headwind, runway length 2000 m
Pressure altitude 1000 ft

Result:

Corrected runway length = 2110 m
Performance limited landing weight = 71.2t (limited by max structural landing weight of 58.74t)

| Corrected RWY Length (m) | Step 2 - Runway limited weight (1000 kgs) | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | Pressure Altitude (ft) | | | | | | | |
| | 0 | 1000 | 2000 | 3000 | 4000 | 6000 | 8000 | 10000 |
| 1200 | — (—)<br>43.8 (—) | — (—)<br>42.6 (—) | — (—)<br>41.4 (—) | — | — | — | — | — |
| 1300 | 43.5 (—)<br>48.7 (—) | 42.4 (—)<br>47.4 (—) | 41.2 (—)<br>46.1 (—) | 40.1 (—)<br>44.9 (—) | — (—)<br>43.6 (—) | — (—)<br>41.1 (—) | — | — |
| 1400 | 48.1 (39.7)<br>54.0 (44.6) | 46.8 (—)<br>52.5 (43.4) | 45.5 (—)<br>51.0 (42.2) | 44.1 (—)<br>49.6 (39.1) | 42.9 (—)<br>48.1 (—) | 40.5 (—)<br>45.4 (—) | — (—)<br>42.8 (—) | — |
| 1500 | 52.8 (43.7)<br>59.2 (48.9) | 51.3 (42.6)<br>57.5 (47.6) | 49.9 (41.4)<br>56.0 (46.3) | 48.4 (40.2)<br>54.4 (45.1) | 47.0 (39.1)<br>52.8 (43.8) | 44.3 (—)<br>49.8 (41.3) | 41.6 (—)<br>46.9 (—) | — (—)<br>43.8 (—) |
| 1600 | 57.3 (47.7)<br>65.6 (53.5) | 55.6 (46.4)<br>63.6 (52.1) | 54.0 (45.1)<br>61.5 (50.6) | 52.4 (43.8)<br>59.5 (49.2) | 50.9 (42.6)<br>57.5 (47.7) | 48.0 (40.2)<br>54.0 (45.0) | 45.2 (—)<br>50.9 (42.4) | 42.2 (—)<br>47.6 (—) |
| 1700 | 62.1 (51.8)<br>67.1 (58.1) | 60.1 (50.4)<br>66.1 (56.4) | 58.2 (49.0)<br>65.1 (54.9) | 56.3 (47.6)<br>64.1 (53.3) | 54.6 (46.2)<br>62.3 (51.8) | 51.2 (43.5)<br>58.3 (48.9) | 48.2 (40.9)<br>54.5 (46.0) | 45.1 (—)<br>50.8 (43.0) |
| 1800 | 66.9 (55.6)<br>67.1 (63.4) | 64.8 (54.1)<br>66.1 (61.4) | 62.7 (52.6)<br>65.1 (59.5) | 60.6 (51.1)<br>64.1 (57.5) | 58.6 (49.6)<br>63.0 (55.8) | 55.0 (46.7)<br>60.5 (52.6) | 51.5 (44.0)<br>57.6 (49.6) | 47.9 (41.0)<br>54.4 (46.3) |
| 1900 | 71.7 (59.8)<br>67.1 (67.1) | 69.5 (57.9)<br>66.1 (66.1) | 67.3 (56.1)<br>65.1 (64.2) | 65.0 (54.4)<br>64.1 (62.1) | 62.9 (52.7)<br>63.0 (60.1) | 58.8 (49.7)<br>60.5 (56.2) | 55.0 (46.8)<br>57.6 (52.7) | 51.2 (43.7)<br>55.0 (49.3) |
| 2000 | 72.3 (63.9)<br>67.1 (67.1) | 71.2 (61.9)<br>66.1 (66.1) | 70.1 (60.0)<br>65.1 (65.1) | 69.0 (58.0)<br>64.1 (64.1) | 67.2 (56.1)<br>63.0 (63.0) | 62.8 (52.7)<br>60.5 (60.0) | 58.4 (49.4)<br>57.6 (56.1) | 54.5 (46.2)<br>55.0 (52.2) |
| 2100 | 72.3 (68.1)<br>67.1 (67.1) | 71.2 (66.0)<br>66.1 (66.1) | 70.1 (63.9)<br>65.1 (65.1) | 69.0 (61.8)<br>64.1 (64.1) | 67.9 (59.7)<br>63.0 (63.0) | 65.2 (55.9)<br>60.5 (60.5) | 62.4 (52.4)<br>57.6 (57.6) | 57.8 (48.7)<br>55.0 (55.0) |
| 2200 | 72.3 (72.3)<br>67.1 (67.1) | 71.2 (70.1)<br>66.1 (66.1) | 70.1 (67.9)<br>65.1 (65.1) | 69.0 (65.6)<br>64.1 (64.1) | 67.9 (63.5)<br>63.0 (63.0) | 65.2 (59.3)<br>60.5 (60.5) | 62.4 (55.5)<br>57.6 (57.6) | 59.8 (51.6)<br>55.0 (55.0) |
| 2300 | 72.3 (72.3)<br>67.1 (67.1) | 71.2 (71.2)<br>66.1 (66.1) | 70.1 (70.1)<br>65.1 (65.1) | 69.0 (69.0)<br>64.1 (64.1) | 67.9 (67.2)<br>63.0 (63.0) | 65.2 (62.8)<br>60.5 (60.5) | 62.4 (58.6)<br>57.6 (57.6) | 59.8 (54.5)<br>55.0 (55.0) |
| 2400 | 72.3 (72.3)<br>67.1 (67.1) | 71.2 (71.2)<br>66.1 (66.1) | 70.1 (70.1)<br>65.1 (65.1) | 69.0 (69.0)<br>64.1 (64.1) | 67.9 (67.9)<br>63.0 (63.0) | 65.2 (65.2)<br>60.5 (60.5) | 62.4 (61.9)<br>57.6 (57.6) | 59.8 (57.3)<br>55.0 (55.0) |
| 2500 | 72.3 (72.3)<br>67.1 (67.1) | 71.2 (71.2)<br>66.1 (66.1) | 70.1 (70.1)<br>65.1 (65.1) | 69.0 (69.0)<br>64.1 (64.1) | 67.9 (67.9)<br>63.0 (63.0) | 65.2 (65.2)<br>60.5 (60.5) | 62.4 (62.4)<br>57.6 (57.6) | 59.8 (59.8)<br>55.0 (55.0) |
| 2600 | 72.3 (72.3)<br>67.1 (67.1) | 71.2 (71.2)<br>66.1 (66.1) | 70.1 (70.1)<br>65.1 (65.1) | 69.0 (69.0)<br>64.1 (64.1) | 67.9 (67.9)<br>63.0 (63.0) | 65.2 (65.2)<br>60.5 (60.5) | 62.4 (62.4)<br>57.6 (57.6) | 59.8 (59.8)<br>55.0 (55.0) |

# Landing PW1524G

Shaded areas limited by maximum structural landing weight (not performance limiting).

- F4 dry runway (F4 wet runway)
- F5 dry runway (F5 wet runway)

| Step 1 – Wind Corrected Runway Length | | | | | |
|---|---|---|---|---|---|
| Runway Length (m) | Wind (kts) | | | | |
| | Tailwind | Calm | Headwind | | |
| | –10 | 0 | +10 | +20 | +30 |
| 1500 | 1140 (1100)<br>1180 (1140) | 1500 (1500)<br>1500 (1500) | 1580 (1590)<br>1580 (1590) | 1670 (1690)<br>1670 (1680) | 1760 (1770)<br>1760 (1760) |
| 1600 | 1260 (1200)<br>1260 (1230) | 1600 (1600)<br>1600 (1600) | 1690 (1690)<br>1690 (1690) | 1780 (1790)<br>1780 (1790) | 1870 (1880)<br>1860 (1870) |
| 1700 | 1350 (1290)<br>1370 (1340) | 1700 (1700)<br>1700 (1700) | 1800 (1800)<br>1800 (1800) | 1890 (1900)<br>1900 (1900) | 1980 (1990)<br>1990 (2000) |
| 1800 | 1460 (1420)<br>1460 (1420) | 1800 (1800)<br>1800 (1800) | 1910 (1900)<br>1900 (1900) | 2010 (2000)<br>2000 (2000) | 2110 (2100)<br>2100 (2100) |
| 1900 | 1550 (1500)<br>1550 (1510) | 1900 (1900)<br>1900 (1900) | 2000 (2010)<br>2000 (2010) | 2110 (2110)<br>2110 (2120) | 2210 (2220)<br>2210 (2220) |
| 2000 | 1640 (1600)<br>1640 (1620) | 2000 (2000)<br>2000 (2000) | 2110 (2120)<br>2110 (2120) | 2210 (2230)<br>2210 (2230) | 2320 (2340)<br>2320 (2340) |
| 2100 | 1740 (1710)<br>1740 (1710) | 2100 (2100)<br>2100 (2100) | 2210 (2220)<br>2210 (2220) | 2320 (2330)<br>2320 (2330) | 2420 (2460)<br>2420 (2440) |
| 2200 | 1830 (1800)<br>1830 (1800) | 2200 (2200)<br>2200 (2200) | 2310 (2320)<br>2310 (2320) | 2420 (2440)<br>2420 (2440) | 2530 (2560)<br>2530 (2560) |
| 2300 | 1920 (1890)<br>1910 (1890) | 2300 (2300)<br>2300 (2300) | 2410 (2420)<br>2410 (2420) | 2530 (2550)<br>2530 (2550) | 2640 (2670)<br>2640 (2660) |
| 2400 | 2010 (1980)<br>2000 (1980) | 2400 (2400)<br>2400 (2400) | 2520 (2520)<br>2520 (2520) | 2630 (2650)<br>2650 (2650) | 2750 (2770)<br>2750 (2770) |
| 2500 | 2060 (2080)<br>2090 (2080) | 2500 (2500)<br>2500 (2500) | 2610 (2630)<br>2620 (2630) | 2730 (2760)<br>2740 (2760) | 2850 (2880)<br>2860 (2880) |
| 2600 | 2160 (2170)<br>2180 (2150) | 2600 (2600)<br>2600 (2600) | 2720 (2730)<br>2720 (2730) | 2840 (2860)<br>2850 (2860) | 2960 (2990)<br>2980 (2990) |

| Corrected RWY Length (m) | Step 2 - Runway limited weight (1000 kgs) | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | Pressure Altitude (ft) | | | | | | | |
| | 0 | 1000 | 2000 | 3000 | 4000 | 6000 | 8000 | 10000 |
| 1200 | — (—)<br>43.8 (—) | — (—)<br>42.6 (—) | — (—)<br>41.4 (—) | — | — | — | — | — |
| 1300 | 43.5 (—)<br>48.7 (—) | 42.4 (—)<br>47.4 (—) | 41.2 (—)<br>46.1 (—) | 40.1 (—)<br>44.9 (—) | — (—)<br>43.6 (—) | — (—)<br>41.1 (—) | — | — |
| 1400 | 48.1 (39.7)<br>54.0 (44.6) | 46.8 (—)<br>52.5 (43.4) | 45.5 (—)<br>51.0 (42.2) | 44.1 (—)<br>49.6 (39.1) | 42.9 (—)<br>48.1 (—) | 40.5 (—)<br>45.4 (—) | — (—)<br>42.8 (—) | — |
| 1500 | 52.8 (43.7)<br>59.2 (48.9) | 51.3 (42.6)<br>57.5 (47.6) | 49.9 (41.4)<br>56.0 (46.3) | 48.4 (40.2)<br>54.4 (45.1) | 47.0 (39.1)<br>52.8 (43.8) | 44.3 (—)<br>49.8 (41.3) | 41.6 (—)<br>46.9 (—) | — (—)<br>43.8 (—) |
| 1600 | 57.3 (47.7)<br>65.6 (53.5) | 55.6 (46.4)<br>63.6 (52.1) | 54.0 (45.1)<br>61.5 (50.6) | 52.4 (43.8)<br>59.5 (49.2) | 50.9 (42.6)<br>57.5 (47.7) | 48.0 (40.2)<br>54.0 (45.0) | 45.2 (—)<br>50.9 (42.4) | 42.2 (—)<br>47.6 (—) |
| 1700 | 62.1 (51.8)<br>71.1 (58.1) | 60.1 (50.4)<br>68.9 (56.4) | 58.2 (49.0)<br>66.7 (54.9) | 56.3 (47.6)<br>64.5 (53.3) | 54.6 (46.2)<br>62.3 (51.8) | 51.2 (43.5)<br>58.3 (48.9) | 48.2 (40.9)<br>54.5 (46.0) | 45.1 (—)<br>50.8 (43.0) |
| 1800 | 66.9 (55.6)<br>72.6 (63.4) | 64.8 (54.1)<br>72.6 (61.4) | 62.7 (52.6)<br>71.8 (59.5) | 60.6 (51.1)<br>69.5 (57.5) | 58.6 (49.6)<br>67.2 (55.8) | 55.0 (46.7)<br>62.7 (52.6) | 51.5 (44.0)<br>58.6 (49.6) | 47.9 (41.0)<br>54.4 (46.3) |
| 1900 | 71.7 (59.8)<br>72.6 (68.5) | 69.5 (57.9)<br>72.6 (66.4) | 67.3 (56.1)<br>71.9 (64.2) | 65.0 (54.4)<br>70.7 (62.1) | 62.9 (52.7)<br>69.5 (60.1) | 58.8 (49.7)<br>66.5 (56.2) | 55.0 (46.8)<br>62.8 (52.7) | 51.2 (43.7)<br>58.2 (49.3) |
| 2000 | 72.6 (63.9)<br>72.6 (72.6) | 72.6 (61.9)<br>72.6 (71.0) | 71.8 (60.0)<br>71.9 (68.7) | 69.5 (58.0)<br>70.7 (66.4) | 67.2 (56.1)<br>69.5 (64.2) | 62.8 (52.7)<br>66.5 (60.0) | 58.6 (49.4)<br>63.3 (56.1) | 54.5 (46.2)<br>60.3 (52.2) |
| 2100 | 72.6 (68.1)<br>72.6 (72.6) | 72.6 (66.0)<br>72.6 (72.6) | 72.6 (63.9)<br>71.9 (71.9) | 72.6 (61.8)<br>70.7 (70.7) | 71.5 (59.7)<br>69.5 (68.4) | 66.8 (55.9)<br>66.5 (63.9) | 62.4 (52.4)<br>63.3 (59.7) | 57.8 (48.7)<br>60.3 (55.4) |
| 2200 | 72.6 (72.3)<br>72.6 (72.6) | 72.6 (70.1)<br>72.6 (72.6) | 72.6 (67.9)<br>71.9 (71.9) | 72.6 (65.6)<br>70.7 (70.7) | 72.6 (63.5)<br>69.5 (69.5) | 70.9 (59.3)<br>66.5 (66.5) | 66.1 (55.5)<br>63.3 (63.3) | 61.3 (51.6)<br>60.3 (58.7) |
| 2300 | 72.6 (72.6)<br>72.6 (72.6) | 72.6 (72.6)<br>72.6 (72.6) | 72.6 (71.8)<br>71.9 (71.9) | 72.6 (69.5)<br>70.7 (70.7) | 72.6 (67.2)<br>69.5 (69.5) | 71.7 (62.8)<br>66.5 (66.5) | 68.4 (58.6)<br>63.3 (63.3) | 64.8 (54.5)<br>60.3 (60.3) |
| 2400 | 72.6 (72.6)<br>72.6 (72.6) | 72.6 (72.6)<br>72.6 (72.6) | 72.6 (72.6)<br>71.9 (71.9) | 72.6 (72.6)<br>70.7 (70.7) | 72.6 (70.9)<br>69.5 (69.5) | 71.7 (66.3)<br>66.5 (66.5) | 68.4 (61.9)<br>63.3 (63.3) | 65.2 (57.3)<br>60.3 (60.3) |
| 2500 | 72.6 (72.6)<br>72.6 (72.6) | 72.6 (72.6)<br>72.6 (72.6) | 72.6 (72.6)<br>71.9 (71.9) | 72.6 (72.6)<br>70.7 (70.7) | 72.6 (72.6)<br>69.5 (69.5) | 71.7 (69.8)<br>66.5 (66.5) | 68.4 (65.1)<br>63.3 (63.3) | 65.2 (60.4)<br>60.3 (60.3) |
| 2600 | 72.6 (72.6)<br>72.6 (72.6) | 72.6 (72.6)<br>72.6 (72.6) | 72.6 (72.6)<br>71.9 (71.9) | 72.6 (72.6)<br>70.7 (70.7) | 72.6 (72.6)<br>69.5 (69.5) | 71.7 (71.7)<br>66.5 (66.5) | 68.4 (68.4)<br>63.3 (63.3) | 65.2 (63.4)<br>60.3 (60.3) |

# Approach and Landing Climb (One Engine Inoperative)

| | F4 | F5 |
|---|---|---|
| PW1521 | 71.0 | (66.6) |
| PW1524 | 71.0 | (71.0) |

Example:

PW1521
FLAP 4
ISA + 10°C
Missed approach (pressure) altitude 3000 ft

Result:

68.7t (limited by max structural landing weight of 58.74t)

Repeat for alternate(s).

**Approach and Landing Climb Limited Weight (1000 kgs)**

| Pressure ALT (ft) | Temperature (°C) | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | -40°C | -30°C | -20°C | -10°C | 0°C | 10°C | 20°C | 30°C | 40°C |
| 0 | 71.0 (67.1)<br>71.0 (71.0) | 71.0 (67.1)<br>71.0 (71.0) | 71.0 (67.1)<br>71.0 (71.0) | 71.0 (67.1)<br>71.0 (71.0) | 71.0 (67.0)<br>71.0 (71.0) | 71.0 (66.9)<br>71.0 (71.0) | 71.0 (66.8)<br>71.0 (71.0) | 71.0 (66.6)<br>71.0 (71.0) | 64.4 (59.7)<br>70.8 (65.9) |
| 1000 | 71.0 (66.1)<br>71.0 (71.0) | 71.0 (66.1)<br>71.0 (71.0) | 71.0 (66.1)<br>71.0 (71.0) | 71.0 (66.1)<br>71.0 (71.0) | 71.0 (66.0)<br>71.0 (71.0) | 70.9 (65.9)<br>71.0 (71.0) | 70.8 (65.8)<br>71.0 (71.0) | 69.2 (64.3)<br>71.0 (71.0) | 62.2 (57.6)<br>68.4 (63.6) |
| 2000 | 70.1 (65.1)<br>71.0 (71.0) | 70.1 (65.1)<br>71.0 (71.0) | 70.0 (65.1)<br>71.0 (71.0) | 70.0 (65.1)<br>71.0 (71.0) | 69.9 (65.0)<br>71.0 (71.0) | 69.8 (64.9)<br>71.0 (71.0) | 69.7 (64.8)<br>71.0 (71.0) | 66.9 (62.1)<br>71.0 (71.0) | 60.1 (55.7)<br>66.1 (61.4) |
| 3000 | 68.9 (64.0)<br>71.0 (70.7) | 68.9 (64.1)<br>71.0 (70.7) | 68.9 (64.0)<br>71.0 (70.7) | 68.9 (64.0)<br>71.0 (70.6) | 68.8 (63.9)<br>71.0 (70.6) | 68.7 (63.8)<br>71.0 (70.4) | 68.5 (63.7)<br>71.0 (70.3) | 64.5 (59.9)<br>70.9 (66.1) | 58.0 (53.7)<br>63.8 (59.2) |
| 4000 | 67.7 (63.0)<br>71.0 (69.4) | 67.7 (63.0)<br>71.0 (69.5) | 67.7 (62.9)<br>71.0 (69.5) | 67.7 (62.9)<br>71.0 (69.4) | 67.6 (62.8)<br>71.0 (69.3) | 67.5 (62.7)<br>71.0 (69.2) | 67.4 (62.6)<br>71.0 (69.1) | 62.3 (57.7)<br>68.4 (63.8) | 55.9 (51.8)<br>61.5 (57.0) |
| 6000 | 65.0 (60.4)<br>71.0 (66.4) | 65.0 (60.4)<br>71.0 (66.4) | 65.0 (60.4)<br>71.0 (66.4) | 65.0 (60.3)<br>71.0 (66.4) | 64.9 (60.3)<br>71.0 (66.3) | 64.8 (60.2)<br>71.0 (66.2) | 63.7 (59.1)<br>70.2 (65.1) | 57.8 (53.5)<br>63.5 (59.1) | 51.8 (48.0)<br>57.0 (52.9) |
| 8000 | 62.2 (57.5)<br>68.0 (63.1) | 62.2 (57.5)<br>68.1 (63.1) | 62.2 (57.5)<br>68.1 (63.1) | 62.1 (57.5)<br>68.1 (63.1) | 62.1 (57.4)<br>68.0 (63.0) | 62.0 (57.3)<br>68.0 (62.9) | 59.0 (54.6)<br>65.1 (60.2) | 53.4 (49.4)<br>58.7 (54.5) | 47.7 (44.2)<br>52.5 (48.6) |
| 10000 | 59.6 (54.9)<br>64.8 (60.0) | 59.6 (54.9)<br>64.8 (60.1) | 59.6 (54.9)<br>64.9 (60.1) | 59.5 (54.9)<br>64.9 (60.1) | 59.5 (54.8)<br>64.8 (60.0) | 59.4 (54.7)<br>64.4 (59.3) | 54.6 (50.5)<br>60.3 (55.6) | 49.1 (45.4)<br>53.9 (50.0) | 43.7 (40.5)<br>47.9 (44.4) |

Shaded areas limited by maximum structural landing weight (not performance limiting).

Correction – no ice accumulation PW1521 (PW1524)

FLAP 4
- COWL A/I ON: -7.0 (-7.1)
- ALL A/I ON: -11.5 (-12.1)

FLAP 5
- COWL A/I ON: -6.4 (-6.6)
- ALL A/I ON: -10.7 (-11.1)

Correction – with ice accumulation PW1521 (PW1524)

FLAP 4
- COWL A/I ON: -11.2 (-11.6)
- ALL A/I ON: -15.2 (-16.2)

FLAP 5
- COWL A/I ON: -9.7 (-10.2)
- ALL A/I ON: -13.6 (-14.3)

- Flight Planning

# Flight Planning

| Step | Description | Details |
|---|---|---|
| 1 | Extra fuel<br>Additional fuel<br>Discretionary fuel | • Extra fuel is the fuel required for delays that can be foreseen.<br>• Additional fuel is fuel required to cover for known defects (increased drag, APU running during flight etc.) or failure during the most critical point along the route (engine failure or pressurization loss), drift down followed by escape route and diversion if this is not otherwise covered.<br>• Discretionary fuel is the "extra" fuel carried on discretion of the commander based on weather, delays anticipated, etc.<br>• A conservative 35 kg/min can be used to determine the extra fuel and discretionary fuel. |
| 2 | "Start" weight | <u>If payload is known:</u><br>• Zero Fuel Weight = Dry Operating Weight + the weight of passengers, baggage and cargo.<br>• Then add the extra, additional and discretionary fuel (STEP 1) to have a "start" weight for fuel planning purposes.<br>• Ensure none of the limiting masses as determined during PERFORMANCE are exceeded.<br><u>If payload is not known:</u><br>• A conservative fuel calculation can be performed using the maximum landing mass at the alternate aerodrome.<br>• Ensure none of the limiting masses as determined during PERFORMANCE are exceeded. |
| 3 | Final reserve fuel (30 min) | • Enter the holding fuel table with the "start" weight as determined in STEP 2 and aerodrome elevation + 1500 ft (holding at 1500 ft AAE).<br>• Multiply the total fuel flow (kg/min) x 30 to find the final reserve fuel. |
| 4 | Alternate fuel | • **Landing weight at alternate = "start" weight (STEP 2) + final reserve fuel (STEP 3).**<br>• Enter the "Flight Planning to Alternate" table with the landing weight at alternate and the ground distance to find the average TAS.<br>• Enter the corrected air distance table with the ground distance and average TAS and read the corrected air distance.<br>• Enter the "Flight Planning to Alternate" table with landing weight at alternate and corrected air distance, find the alternate fuel required. |
| 5 | Trip fuel | • **Landing weight at destination = landing weight at alternate (STEP 3) + alternate fuel (STEP 4).**<br>• Enter the "Flight Planning to Destination" table with the landing weight at destination and the ground distance to find the average TAS.<br>• Enter the corrected air distance table with the ground distance and average TAS and read the corrected air distance.<br>• Enter the "Recommended Altitude" table with corrected air distance and rough estimate for the take-off weight to find cruise level.<br>• Enter the "Flight Planning to Destination" table with landing weight, corrected air distance, and cruise level in order to find trip fuel. |
| 6 | Contingency fuel | • Contingency fuel = 5% of the trip fuel or 5 minutes of flying at 1500 ft AAE at holding consumption at destination (whichever is greater).<br>• 5 minutes at holding consumption can be calculated using the holding fuel table or by using a conservative 35 kg/min. |
| 7 | Taxi fuel and APU | • Taxi fuel = expected taxi time x 10 kg/min.<br>• APU fuel is determined using APU fuel consumption of 136 kg/hour. |
| 8 | Block fuel<br>Take-off fuel | • Block fuel = extra fuel + additional fuel + discretionary fuel + final reserve fuel + alternate fuel + trip fuel + contingency fuel + taxi and APU.<br>• Take-off fuel = block fuel – taxi and APU fuel.<br>• <u>Minimum block and take-off fuel as above – discretionary fuel.</u> |

# Holding

| | |
|---|---|
| Holding speed - kts | 240 |
| Total fuel flow – kg/min | 33 |

Example:

Alternate airport elevation = 3500 ft
Holding altitude = 1500 ft AAE
3500 ft + 1500 ft = 5000 ft AMSL
Maximum landing weight = 58.74t

Result:

Holding speed = 219 kts
Total fuel flow = 30 kg/min
Final reserve fuel = 30 min

30 kg/min x 30 min = 900 kg

**Note:** the most conservative number for holding consumption is 35 kg/min.

**Holding Speed (IAS) and Total Fuel Flow (kg/min)**

| Pressure ALT (ft) | Gross Weight (kg) | | | | | | |
|---|---|---|---|---|---|---|---|
| | 40000 | 45000 | 50000 | 55000 | 60000 | 65000 | 70000 |
| 1500 | 176<br>21 | 186<br>23 | 197<br>25 | 207<br>28 | 217<br>30 | 227<br>33 | 236<br>35 |
| 5000 | 176<br>21 | 187<br>23 | 198<br>25 | 209<br>27 | 219<br>30 | 228<br>32 | 237<br>35 |
| 10000 | 177<br>20 | 189<br>23 | 200<br>25 | 210<br>27 | 220<br>30 | 230<br>32 | 239<br>34 |
| 15000 | 179<br>20 | 191<br>22 | 202<br>24 | 212<br>26 | 222<br>28 | 231<br>30 | 240<br>33 |
| 20000 | 181<br>19 | 192<br>21 | 203<br>23 | 213<br>26 | 223<br>28 | 232<br>30 | 241<br>33 |
| 25000 | 182<br>19 | 193<br>21 | 204<br>23 | 215<br>26 | 224<br>28 | 233<br>31 | 242<br>33 |
| 30000 | 183<br>19 | 195<br>21 | 205<br>23 | 215<br>26 | 226<br>29 | 236<br>32 | 245<br>35 |

Conditions:
- Green dot
- All Engines Operating
- Clean configuration
- Engine bleed – packs on
- ISA conditions
- CG 27% MAC

Effect on total fuel flow

| No ice accumulation | With ice accumulation |
|---|---|
| • COWL A/I ON: +6%<br>• ALL A/I ON: +14% | • COWL A/I ON: +17%<br>• ALL A/I ON: +24% |

# Ground to Air Distance Conversion Table

| Wind kts | | Avg TAS | Ground Distance 50 | 100 | 200 | 400 | 600 | 800 | 1000 | 1200 | 1400 | 1600 | 1800 | 2000 | 2200 | 2400 | 2600 | 2800 | 3000 | 3200 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Headwind | -150 | 200 kts | 200 | 400 | 800 | 1600 | 2400 | 3200 | - | - | - | - | - | - | - | - | - | - | - | - |
| | | 300 kts | 100 | 200 | 400 | 800 | 1200 | 1600 | 2000 | 2400 | 2800 | 3200 | - | - | - | - | - | - | - | - |
| | | 400 kts | 80 | 160 | 320 | 640 | 960 | 1280 | 1600 | 1920 | 2240 | 2560 | 2880 | 3200 | - | - | - | - | - | - |
| | | 500 kts | 71 | 143 | 286 | 571 | 857 | 1143 | 1429 | 1714 | 2000 | 2286 | 2571 | 2857 | 3143 | - | - | - | - | - |
| | -100 | 200 kts | 100 | 200 | 400 | 800 | 1200 | 1600 | 2000 | 2400 | 2800 | 3200 | - | - | - | - | - | - | - | - |
| | | 300 kts | 75 | 150 | 300 | 600 | 900 | 1200 | 1500 | 1800 | 2100 | 2400 | 2700 | 3000 | - | - | - | - | - | - |
| | | 400 kts | 67 | 133 | 267 | 533 | 800 | 1067 | 1333 | 1600 | 1867 | 2133 | 2400 | 2667 | 2933 | 3200 | - | - | - | - |
| | | 500 kts | 63 | 125 | 250 | 500 | 750 | 1000 | 1250 | 1500 | 1750 | 2000 | 2250 | 2500 | 2750 | 3000 | - | - | - | - |
| | -80 | 200 kts | 83 | 167 | 333 | 667 | 1000 | 1333 | 1667 | 2000 | 2333 | 2667 | 3000 | - | - | - | - | - | - | - |
| | | 300 kts | 68 | 136 | 273 | 545 | 818 | 1091 | 1364 | 1636 | 1909 | 2182 | 2455 | 2727 | 3000 | - | - | - | - | - |
| | | 400 kts | 63 | 125 | 250 | 500 | 750 | 1000 | 1250 | 1500 | 1750 | 2000 | 2250 | 2500 | 2750 | 3000 | - | - | - | - |
| | | 500 kts | 60 | 119 | 238 | 476 | 714 | 952 | 1190 | 1429 | 1667 | 1905 | 2143 | 2381 | 2619 | 2857 | 3095 | - | - | - |
| | -60 | 200 kts | 71 | 143 | 286 | 571 | 857 | 1143 | 1429 | 1714 | 2000 | 2286 | 2571 | 2857 | 3143 | - | - | - | - | - |
| | | 300 kts | 63 | 125 | 250 | 500 | 750 | 1000 | 1250 | 1500 | 1750 | 2000 | 2250 | 2500 | 2750 | 3000 | - | - | - | - |
| | | 400 kts | 59 | 118 | 235 | 471 | 706 | 941 | 1176 | 1412 | 1647 | 1882 | 2118 | 2353 | 2588 | 2824 | 3059 | - | - | - |
| | | 500 kts | 57 | 114 | 227 | 455 | 682 | 909 | 1136 | 1364 | 1591 | 1818 | 2045 | 2273 | 2500 | 2727 | 2955 | 3182 | - | - |
| | -40 | 200 kts | 63 | 125 | 250 | 500 | 750 | 1000 | 1250 | 1500 | 1750 | 2000 | 2250 | 2500 | 2750 | 3000 | - | - | - | - |
| | | 300 kts | 58 | 115 | 231 | 462 | 692 | 923 | 1154 | 1385 | 1615 | 1846 | 2077 | 2308 | 2538 | 2769 | 3000 | - | - | - |
| | | 400 kts | 56 | 111 | 222 | 444 | 667 | 889 | 1111 | 1333 | 1556 | 1778 | 2000 | 2222 | 2444 | 2667 | 2889 | 3111 | - | - |
| | | 500 kts | 54 | 109 | 217 | 435 | 652 | 870 | 1087 | 1304 | 1522 | 1739 | 1957 | 2174 | 2391 | 2609 | 2826 | 3043 | - | - |
| | -20 | 200 kts | 56 | 111 | 222 | 444 | 667 | 889 | 1111 | 1333 | 1556 | 1778 | 2000 | 2222 | 2444 | 2667 | 2889 | 3111 | - | - |
| | | 300 kts | 54 | 107 | 214 | 429 | 643 | 857 | 1071 | 1286 | 1500 | 1714 | 1929 | 2143 | 2357 | 2571 | 2786 | 3000 | - | - |
| | | 400 kts | 53 | 105 | 211 | 421 | 632 | 842 | 1053 | 1263 | 1474 | 1684 | 1895 | 2105 | 2316 | 2526 | 2737 | 2947 | 3158 | - |
| | | 500 kts | 52 | 104 | 208 | 417 | 625 | 833 | 1042 | 1250 | 1458 | 1667 | 1875 | 2083 | 2292 | 2500 | 2708 | 2917 | 3125 | - |
| CALM | 0 | n/a | 50 | 100 | 200 | 400 | 600 | 800 | 1000 | 1200 | 1400 | 1600 | 1800 | 2000 | 2200 | 2400 | 2600 | 2800 | 3000 | 3200 |
| Tailwind | 20 | 200 kts | 45 | 91 | 182 | 364 | 545 | 727 | 909 | 1091 | 1273 | 1455 | 1636 | 1818 | 2000 | 2182 | 2364 | 2545 | 2727 | 2909 |
| | | 300 kts | 47 | 94 | 188 | 375 | 563 | 750 | 938 | 1125 | 1313 | 1500 | 1688 | 1875 | 2063 | 2250 | 2438 | 2625 | 2813 | 3000 |
| | | 400 kts | 48 | 95 | 190 | 381 | 571 | 762 | 952 | 1143 | 1333 | 1524 | 1714 | 1905 | 2095 | 2286 | 2476 | 2667 | 2857 | 3048 |
| | | 500 kts | 48 | 96 | 192 | 385 | 577 | 769 | 962 | 1154 | 1346 | 1538 | 1731 | 1923 | 2115 | 2308 | 2500 | 2692 | 2885 | 3077 |
| | 40 | 200 kts | 42 | 83 | 167 | 333 | 500 | 667 | 833 | 1000 | 1167 | 1333 | 1500 | 1667 | 1833 | 2000 | 2167 | 2333 | 2500 | 2667 |
| | | 300 kts | 44 | 88 | 176 | 353 | 529 | 706 | 882 | 1059 | 1235 | 1412 | 1588 | 1765 | 1941 | 2118 | 2294 | 2471 | 2647 | 2824 |
| | | 400 kts | 45 | 91 | 182 | 364 | 545 | 727 | 909 | 1091 | 1273 | 1455 | 1636 | 1818 | 2000 | 2182 | 2364 | 2545 | 2727 | 2909 |
| | | 500 kts | 46 | 93 | 185 | 370 | 556 | 741 | 926 | 1111 | 1296 | 1481 | 1667 | 1852 | 2037 | 2222 | 2407 | 2593 | 2778 | 2963 |
| | 60 | 200 kts | 38 | 77 | 154 | 308 | 462 | 615 | 769 | 923 | 1077 | 1231 | 1385 | 1538 | 1692 | 1846 | 2000 | 2154 | 2308 | 2462 |
| | | 300 kts | 42 | 83 | 167 | 333 | 500 | 667 | 833 | 1000 | 1167 | 1333 | 1500 | 1667 | 1833 | 2000 | 2167 | 2333 | 2500 | 2667 |
| | | 400 kts | 43 | 87 | 174 | 348 | 522 | 696 | 870 | 1043 | 1217 | 1391 | 1565 | 1739 | 19131 | 2087 | 2261 | 2435 | 2609 | 2783 |
| | | 500 kts | 45 | 89 | 179 | 357 | 536 | 714 | 893 | 1071 | 1250 | 1429 | 1607 | 1786 | 964 | 2143 | 2321 | 2500 | 2679 | 2857 |
| | 80 | 200 kts | 36 | 71 | 143 | 286 | 429 | 571 | 714 | 857 | 1000 | 1143 | 1286 | 1429 | 1571 | 1714 | 1857 | 2000 | 2143 | 2286 |
| | | 300 kts | 39 | 79 | 158 | 316 | 474 | 632 | 789 | 947 | 1105 | 1263 | 1421 | 1579 | 1737 | 1895 | 2053 | 2211 | 2368 | 2526 |
| | | 400 kts | 42 | 83 | 167 | 333 | 500 | 667 | 833 | 1000 | 1167 | 1333 | 1500 | 1667 | 1833 | 2000 | 2167 | 2333 | 2500 | 2667 |
| | | 500 kts | 43 | 86 | 172 | 345 | 517 | 690 | 862 | 1034 | 1207 | 1379 | 1552 | 1724 | 1897 | 2069 | 2241 | 2414 | 2586 | 2759 |
| | 100 | 200 kts | 33 | 67 | 133 | 267 | 400 | 533 | 667 | 800 | 933 | 1067 | 1200 | 1333 | 1467 | 1600 | 1733 | 1867 | 2000 | 2133 |
| | | 300 kts | 38 | 75 | 150 | 300 | 450 | 600 | 750 | 900 | 1050 | 1200 | 1350 | 1500 | 1650 | 1800 | 1950 | 2100 | 2250 | 2400 |
| | | 400 kts | 40 | 80 | 160 | 320 | 480 | 640 | 800 | 960 | 1120 | 1280 | 1440 | 1600 | 1760 | 1920 | 2080 | 2240 | 2400 | 2560 |
| | | 500 kts | 42 | 83 | 167 | 333 | 500 | 667 | 833 | 1000 | 1167 | 1333 | 1500 | 1667 | 1833 | 2000 | 2167 | 2333 | 2500 | 2667 |
| | 150 | 200 kts | 29 | 57 | 114 | 229 | 343 | 457 | 571 | 686 | 800 | 914 | 1029 | 1143 | 1257 | 1371 | 1486 | 1600 | 1714 | 1829 |
| | | 300 kts | 33 | 67 | 133 | 267 | 400 | 533 | 667 | 800 | 933 | 1067 | 1200 | 1333 | 1467 | 1600 | 1733 | 1867 | 2000 | 2133 |
| | | 400 kts | 36 | 73 | 145 | 291 | 436 | 582 | 727 | 873 | 1018 | 1164 | 1309 | 1455 | 1600 | 1745 | 1891 | 2036 | 2182 | 2327 |
| | | 500 kts | 38 | 77 | 154 | 308 | 462 | 615 | 769 | 923 | 1077 | 1231 | 1385 | 1538 | 1692 | 1846 | 2000 | 2154 | 2308 | 2462 |

Example:

Headwind:
40 kts
Ground distance:
1400 nm
Average TAS:
400 kts

Result:

Air distance:
1556 nm

# Flight Planning to Alternate

| Flight Planning to Alternate | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Landing Weight (kgs) | | Corrected Air Distance (nm) | | | | | | | | | | | | | |
| | | 50 | 75 | 100 | 125 | 150 | 175 | 200 | 225 | 250 | 275 | 300 | 500 | 700 | 900 | 1100 |
| 40000 | Fuel (kg) | 460 | 590 | 708 | 818 | 925 | 1025 | 1119 | 1209 | 1291 | 1371 | 1450 | 2078 | 2711 | 3350 | 3994 |
| | Flight level | 120 | 180 | 230 | 270 | 310 | 330 | 360 | 390 | 400 | 410 | 410 | 410 | 410 | 410 | 410 |
| | Time (min) | 16 | 20 | 24 | 24 | 31 | 35 | 41 | 41 | 44 | 48 | 51 | 80 | 109 | 138 | 166 |
| | TAS (kts) | 188 | 225 | 250 | 268 | 290 | 300 | 316 | 329 | 341 | 344 | 353 | 375 | 385 | 391 | 398 |
| 45000 | Fuel (kg) | 479 | 615 | 740 | 854 | 966 | 1071 | 1178 | 1273 | 1366 | 1454 | 1541 | 2218 | 2908 | 3602 | 4302 |
| | Flight level | 110 | 170 | 220 | 250 | 290 | 310 | 340 | 360 | 380 | 390 | 400 | 410 | 400 | 400 | 400 |
| | Time (min) | 16 | 20 | 24 | 28 | 32 | 35 | 39 | 42 | 45 | 49 | 52 | 80 | 110 | 139 | 168 |
| | TAS (kts) | 188 | 225 | 250 | 268 | 281 | 300 | 316 | 321 | 333 | 344 | 346 | 375 | 382 | 388 | 393 |
| 50000 | Fuel (kg) | 499 | 641 | 771 | 894 | 1010 | 1123 | 1232 | 1340 | 1440 | 1536 | 1630 | 2370 | 3115 | 3881 | 4647 |
| | Flight level | 100 | 160 | 200 | 240 | 270 | 300 | 320 | 340 | 360 | 370 | 380 | 390 | 390 | 380 | 380 |
| | Time (min) | 16 | 20 | 25 | 29 | 32 | 36 | 39 | 43 | 46 | 49 | 52 | 81 | 110 | 140 | 169 |
| | TAS (kts) | 188 | 225 | 240 | 259 | 281 | 292 | 308 | 321 | 326 | 337 | 346 | 370 | 382 | 386 | 391 |
| 55000 | Fuel (kg) | 520 | 669 | 807 | 935 | 1057 | 1177 | 1291 | 1401 | 1514 | 1619 | 1722 | 2523 | 3333 | 4167 | 5000 |
| | Flight level | 100 | 150 | 190 | 230 | 250 | 290 | 310 | 320 | 340 | 360 | 370 | 370 | 370 | 360 | 360 |
| | Time (min) | 16 | 21 | 24 | 29 | 33 | 36 | 39 | 43 | 46 | 49 | 53 | 81 | 110 | 140 | 169 |
| | TAS (kts) | 188 | 214 | 240 | 259 | 281 | 292 | 308 | 314 | 326 | 337 | 340 | 370 | 382 | 386 | 391 |
| 59000 | Fuel (kg) | 540 | 693 | 836 | 969 | 1096 | 1220 | 1340 | 1459 | 1573 | 1684 | 1793 | 2645 | 3506 | 4393 | 5278 |
| | Flight level | 100 | 150 | 190 | 220 | 250 | 280 | 300 | 320 | 330 | 340 | 350 | 360 | 360 | 350 | 350 |
| | Time (min) | 16 | 21 | 25 | 29 | 33 | 36 | 40 | 43 | 46 | 50 | 53 | 82 | 110 | 139 | 168 |
| | TAS (kts) | 188 | 214 | 240 | 259 | 273 | 292 | 300 | 314 | 326 | 330 | 340 | 366 | 382 | 388 | 393 |

Conditions:

- ISA, CG 27%, AEO, clean
- Climb: 250, 275, M0.78
- Long Range Cruise
- Descent: M0.78, 275, 250

Corrections:

Up to 150 nm corrected air distance:

- 10°C below ISA or lower: fuel -2%, time +2%
- 10°C above ISA: fuel +2%, time 0%
- 20°C above ISA: fuel +4%, time +1%

Between 175 and 275 nm corrected air distance:

- 10°C below ISA or lower: fuel -2%, time +2%
- 10°C above ISA: fuel +3%, time 0%
- 20°C above ISA: fuel +6%, time +1%

300 nm and above corrected air distance:

- 10°C below ISA or lower: fuel -1%, time +2%
- 10°C above ISA: fuel +3%, time 0%
- 20°C above ISA: fuel +7%, time +1%

Example:

Landing weight 50t, Corrected Air Distance = 200 nm

Result:

Alternate fuel 1232 kgs, FL320, time 39 min, avg TAS 308 kts.

# Recommended Cruise Altitude

| M0.78 | LRC |
|---|---|
| 320 | (360) |

Example:

Take-off weight around 58t
Corrected air distance >300 nm

Result:

Cruise at MACH 78: FL370
Long Range Cruise: FL390

| Recommended Altitude (Flight Level) | | | | | | | |
|---|---|---|---|---|---|---|---|
| Take-off Weight (kg) | Corrected Air Distance | | | | | | |
| | 50 | 100 | 150 | 200 | 250 | 300 | > 300 |
| 40000 | 220 (140) | 260 (260) | 290 (340) | 320 (400) | 370 (410) | 410 | 410 |
| 42000 | 220 (140) | 260 (250) | 290 (330) | 310 (390) | 370 (410) | 400 (410) | 410 |
| 44000 | 220 (130) | 250 (240) | 290 (320) | 310 (380) | 360 (410) | 390 (410) | 410 |
| 46000 | 220 (130) | 250 (240) | 280 (310) | 310 (370) | 360 (410) | 390 (410) | 410 |
| 48000 | 210 (130) | 240 (230) | 270 (300) | 300 (360) | 350 (400) | 380 (410) | 400 (410) |
| 50000 | 200 (120) | 230 (220) | 270 (300) | 290 (350) | 340 (390) | 380 (410) | 390 (410) |
| 52000 | 200 (120) | 230 (220) | 260 (290) | 290 (340) | 340 (380) | 380 (400) | 390 (400) |
| 54000 | 200 (110) | 230 (210) | 260 (280) | 290 (330) | 330 (380) | 370 (400) | 380 (400) |
| 56000 | 200 (110) | 230 (210) | 260 (280) | 280 (330) | 330 (370) | 360 (390) | 370 (390) |
| 58000 | ~~200 (110)~~ | ~~230 (200)~~ | ~~260 (270)~~ | ~~280 (320)~~ | ~~330 (360)~~ | ~~360 (390)~~ | 370 (390) |
| 60000 | 190 (110) | 220 (200) | 250 (260) | 270 (310) | 320 (350) | 350 (380) | 360 (380) |
| 62000 | 180 (100) | 210 (190) | 240 (260) | 260 (310) | 310 (350) | 340 (370) | 360 (370) |
| 64000 | 170 (100) | 200 (190) | 240 (250) | 260 (300) | 310 (340) | 340 (370) | 360 (370) |
| 66000 | 180 (100) | 210 (180) | 240 (240) | 260 (290) | 310 (340) | 330 (370) | 350 (370) |
| 68000 | — | 200 (170) | 230 (230) | 250 (280) | 300 (330) | 330 (370) | 340 (370) |
| 70000 | — | 210 (170) | 230 (230) | 250 (270) | 290 (320) | 320 (360) | 330 (360) |
| 70900 | — | 210 (170) | 230 (220) | 250 (270) | 290 (320) | 320 (360) | 320 (360) |

Conditions:
- All Engine Operating (AEO)
- Clean configuration
- ISA conditions

Profile:
- Climb: 250 kts, 275 kts, M0.78 (LRC)
- Cruise: 275 kts / M0.78 (LRC)
- Descent: M0.78, 275 kts, 250 kts

# Flight Planning to Destination

| Landing Weight (kg) | Flight Level | | Corrected Air Distance (nm) | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 200 | 400 | 600 | 800 | 1000 | 1200 | 1400 | 1600 | 1800 | 2000 | 2200 | 2400 | 2600 | 2800 | 3000 | 3200 |
| 44000 | 290 | Fuel (kg) | 1258 | 2219 | 3183 | 4151 | 5122 | 6097 | 7075 | 8057 | 9044 | 10034 | 11027 | 12025 | 13026 | 14031 | 15041 | 16055 |
| | | Time (min) | 38 | 64 | 90 | 116 | 143 | 169 | 195 | 221 | 247 | 273 | 299 | 325 | 351 | 377 | 403 | 430 |
| | | TAS (kts) | 316 | 375 | 400 | 414 | 420 | 426 | 431 | 434 | 437 | 440 | 441 | 443 | 444 | 446 | 447 | 447 |
| | 350 | Fuel (kg) | 1219 | 2010 | 2804 | 3602 | 4404 | 5209 | 6019 | 6832 | 7650 | 8472 | 9298 | 10128 | 10963 | 11801 | 12645 | 13492 |
| | | Time (min) | 39 | 65 | 92 | 119 | 146 | 172 | 199 | 226 | 253 | 279 | 306 | 333 | 360 | 386 | 413 | 440 |
| | | TAS (kts) | 308 | 369 | 391 | 403 | 411 | 419 | 422 | 425 | 427 | 430 | 431 | 432 | 433 | 435 | 436 | 436 |
| | 390 | Fuel (kg) | | 1932 | 2643 | 3357 | 4076 | 4799 | 5528 | 6262 | 7002 | 7749 | 8501 | 9261 | 10028 | 10802 | 11584 | 12374 |
| | | Time (min) | — | 65 | 92 | 119 | 146 | 173 | 200 | 227 | 254 | 281 | 307 | 334 | 361 | 388 | 415 | 442 |
| | | TAS (kts) | | 369 | 391 | 403 | 411 | 416 | 420 | 423 | 425 | 427 | 430 | 431 | 432 | 433 | 434 | 434 |
| 48000 | 290 | Fuel (kg) | 1292 | 2268 | 3248 | 4230 | 5217 | 6208 | 7202 | 8200 | 9202 | 10208 | 11219 | 12233 | 13253 | 14276 | 15305 | 16338 |
| | | Time (min) | 39 | 65 | 91 | 117 | 143 | 169 | 195 | 221 | 247 | 273 | 300 | 326 | 352 | 378 | 404 | 430 |
| | | TAS (kts) | 308 | 369 | 396 | 410 | 420 | 426 | 431 | 434 | 437 | 440 | 440 | 442 | 443 | 444 | 446 | 447 |
| | 350 | Fuel (kg) | 1267 | 2077 | 2890 | 3708 | 4530 | 5356 | 6187 | 7022 | 7861 | 8705 | 9553 | 10406 | 11265 | 12130 | 13001 | 13880 |
| | | Time (min) | 39 | 66 | 92 | 119 | 146 | 173 | 200 | 226 | 253 | 280 | 307 | 333 | 360 | 387 | 414 | 441 |
| | | TAS (kts) | 308 | 364 | 391 | 403 | 411 | 416 | 420 | 425 | 427 | 429 | 430 | 432 | 433 | 434 | 435 | 435 |
| | 390 | Fuel (kg) | | 2013 | 2752 | 3496 | 4246 | 5004 | 5768 | 6540 | 7319 | 8106 | 8901 | 9704 | 10517 | | | |
| | | Time (min) | — | 66 | 93 | 120 | 147 | 173 | 200 | 227 | 254 | 281 | 308 | 335 | 362 | — | — | — |
| | | TAS (kts) | | 364 | 387 | 400 | 408 | 416 | 420 | 423 | 425 | 427 | 429 | 430 | 431 | | | |
| 52000 | 290 | Fuel (kg) | 1332 | 2323 | 3318 | 4317 | 5319 | 6325 | 7336 | 8352 | 9371 | 10396 | 11425 | 12459 | 13498 | 14542 | 15591 | 16646 |
| | | Time (min) | 39 | 65 | 91 | 117 | 143 | 170 | 196 | 222 | 248 | 274 | 300 | 326 | 352 | 378 | 405 | 431 |
| | | TAS (kts) | 308 | 369 | 396 | 410 | 420 | 424 | 429 | 432 | 435 | 438 | 440 | 442 | 443 | 444 | 444 | 445 |
| | 350 | Fuel (kg) | | 2150 | 2985 | 3824 | 4667 | 5515 | 6368 | 7227 | 8091 | 8962 | 9841 | 10727 | 11621 | 12523 | 13435 | 14355 |
| | | Time (min) | — | 66 | 93 | 120 | 146 | 173 | 200 | 227 | 254 | 280 | 307 | 334 | 361 | 387 | 414 | 441 |
| | | TAS (kts) | | 364 | 387 | 400 | 411 | 416 | 420 | 423 | 425 | 429 | 430 | 431 | 432 | 434 | 435 | 435 |
| | 390 | Fuel (kg) | | 2109 | 2884 | 3666 | 4457 | 5255 | 6063 | 6880 | | | | | | | | |
| | | Time (min) | — | 66 | 93 | 120 | 147 | 174 | 201 | 228 | — | — | — | — | — | — | — | — |
| | | TAS (kts) | | 364 | 387 | 400 | 408 | 414 | 418 | 421 | | | | | | | | |
| 56000 | 290 | Fuel (kg) | 1376 | 2383 | 3394 | 4410 | 5430 | 6454 | 7483 | 8517 | 9557 | 10601 | 11651 | 12705 | 13766 | 14831 | | |
| | | Time (min) | 40 | 66 | 92 | 118 | 144 | 170 | 196 | 222 | 248 | 275 | 301 | 327 | 353 | 379 | — | — |
| | | TAS (kts) | 300 | 364 | 391 | 407 | 417 | 424 | 429 | 432 | 435 | 436 | 439 | 440 | 442 | 443 | | |
| | 350 | Fuel (kg) | | 2228 | 3086 | 3950 | 4820 | 5697 | 6582 | 7475 | 8376 | 9285 | 10204 | 11131 | 12068 | 13014 | 13970 | |
| | | Time (min) | — | 67 | 93 | 120 | 147 | 174 | 200 | 227 | 254 | 281 | 308 | 334 | 361 | 388 | 415 | — |
| | | TAS (kts) | | 358 | 387 | 400 | 408 | 414 | 420 | 423 | 425 | 427 | 429 | 431 | 432 | 433 | 434 | |
| | 390 | Fuel (kg) | | 2212 | 3031 | | | | | | | | | | | | | |
| | | Time (min) | — | 67 | 94 | — | — | — | — | — | — | — | — | — | — | — | — | — |
| | | TAS (kts) | | 358 | 383 | | | | | | | | | | | | | |
| 60000 | 290 | Fuel (kg) | 1436 | 2465 | 3499 | 4539 | 5583 | 6632 | 7686 | 8746 | 9812 | | | | | | | |
| | | Time (min) | 40 | 66 | 92 | 119 | 145 | 171 | 197 | 223 | 249 | — | — | — | — | — | — | — |
| | | TAS (kts) | 300 | 364 | 391 | 403 | 414 | 421 | 426 | 430 | 434 | | | | | | | |
| | 350 | Fuel (kg) | | 2343 | 3243 | 4151 | 5068 | 5994 | 6929 | 7875 | 8829 | 9794 | | | | | | |
| | | Time (min) | — | 67 | 94 | 121 | 148 | 174 | 201 | 228 | 255 | 282 | — | — | — | — | — | — |
| | | TAS (kts) | | 358 | 383 | 397 | 405 | 414 | 418 | 421 | 424 | 426 | | | | | | |

Conditions:
- ISA
- CG 27% MAC
- All engines operating

Profile:
- Climb: 250 kts, 275 kts, M0.78
- Cruise: M0.78
- Descent: M0.78, 275 kts, 250 kts

Corrections:
- 10°C below ISA or lower: fuel -1%, time + 2%
- Every 5% CG variation from 27% increase (forward) or decrease (aft) trip fuel by 2.5%
- Per 10°C above ISA: fuel +3%, time 0%
- Increase total trip fuel by 2% with cowl anti-ice on and 6% with wing and cowl anti-ice on

Example:
- Corrected air distance 1600 nm
- Landing weight 52000 kg, FL350
- Trip fuel = 7227 kgs, trip time = 3h47m

- Weight & Balance

## Section 1 – Maximum Weights & Under Load

Allowed weight for take-off is the lowest of:

a. Max Zero Fuel Weight + take-off fuel
b. Maximum Landing Weight + trip fuel
c. Maximum Take-off Weight

MLW and MTOW in this context are either the maximum structural weight or performance limited weight if lower. "Under Load" is the additional load the aircraft is able to carry.

Catering requirements vary for short, medium and long-haul operations. Typical values are 250 kgs for short flights and 500 kgs for medium to long flights. If needed, DOW and DOI can be determined from empty weight by using the data provided on the load sheet.

## Section 2 – PAX and LOAD distribution

Passengers are subdivided into Ad(ult), Ch(ild) and In(fant). Note that infants are only included in the total passenger number (usually written as "+"), there are no weight or index corrections applicable to infants.

Weight for Adults and Childs can be read directly from the "PAX WEIGHT" table and added together to get total passenger weight.

Passengers are divided between the 4 cabin zones. In this example (148 seat configuration with front left wardrobe installed):

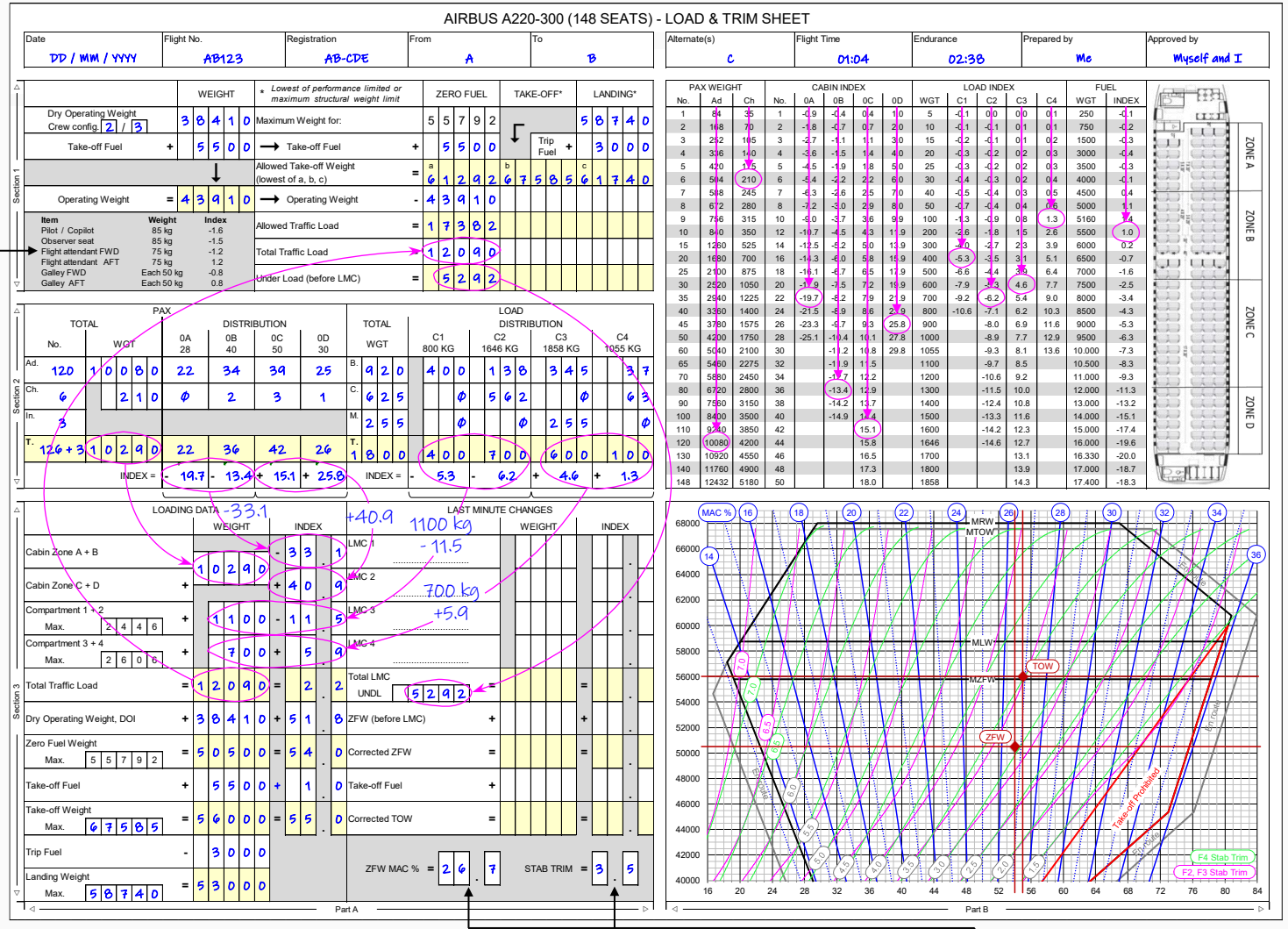
- **Zone 0A** from row 1 to 6 (28 seats)
- **Zone 0B** from row 7 to 14 (40 seats)
- **Zone 0C** from row 15 to 25 (50 seats)
- **Zone 0D** from row 26 to 31 (30 seats)

B(agage), C(argo) and M(ail) is divided between forward and aft cargo holds. Each hold has 2 compartments separated by cargo nets. There are individual weight limits for each compartment, note that the combined weight limit for compartment 3 and 4 is lower than the sum of the individual limits (2606 kg).

After establishing the total PAX per zone and weight per compartment, the index for each zone and compartment can be read directly from the "CABIN" and "LOAD" index tables. 0A, 0B, C1 and C2 have negative index values while 0C, 0D, C3 and C4 are positive.

## Dry Operating Weight (DOW) & Dry Operating Index (DOI)

DOW is the weight of an aircraft ready for operation but excluding fuel. It includes the empty weight of the aircraft, crew and their baggage, catering and other service-related equipment. The DOW and corresponding index (DOI) are usually provided by the operator in tabulated form and account for different crew and catering configurations (for example, flights with additional crew and different catering options for different routes and operations). A typical DOW and Index for an A220-300 with a standard crew configuration of 2 flight crew and 3 cabin crew (2 / 3) but without catering: DOW = 38000 kg, DOI = 50.0 units.




## Section 3 – Weight & Index computation

Combine all negative and positive indexes (0A+0B, 0C+0D, C1+C2 and C3+C4), add weights and indexes to determine the total traffic load. With the total traffic load, return to section 1 and calculate the underload. Compute the rest of the weights / indexes, take-off fuel index is read from the FUEL table.

The ZFW and ZFW CG (MAC) are entered into the FMS – INIT – FUEL page, the FMS calculated TOW and STAB TRIM is then crosschecked with the load sheet. If APU is running, there might be a TOW mismatch, re-enter the TAXI fuel to reset APU fuel used before crosschecking. V2 needs to be re-entered to show STAB TRIM after every EXEC prompt.

## Center of Gravity (CG) Envelope

The ZFW, TOW with corresponding indexes are plotted onto the Center of Gravity (CG) envelope to determine the following:

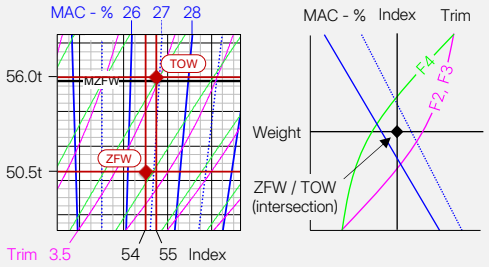
- Whether the TOW is within the take-off / landing envelope (black outline) while out of the take-off prohibited zone (red outline).
- Whether the ZFW lies within the en route envelope (dark gray line).
- ZFW CG in % Mean Aerodynamic Chord (MAC)
- Take-off STAB TRIM setting

On the CG graph the following is depicted:

- Take-off / landing envelope (**thick black outline**)
- En-route envelope (thick dark gray outline)
- Take-off prohibited zone (**thick red outline**)
- Index lines (straight up), index values (bottom)
- Weight lines (horizonal), weight values (left side)
- Mean aerodynamic chord (MAC) lines with MAC values at the top (blue and blue dotted lines)
- Take-off stab trim lines with trim values at the bottom (for FLAP 2, FLAP 3 and FLAP 4 take-off)

Plotting the ZFW and TOW onto the CG graph:

a. Find the ZFW and TOW index values at bottom (54 and 55 respectively), draw lines straight up.
b. Find the ZFW and TOW weight on the left (50.5t, 56.0t respectively), draw horizontal lines.
c. The ZFW and TOW are at the intersection of those horizontal and vertical lines, ensure the ZFW and TOW are within the envelope.
d. Find and note the ZFW MAC by referencing the ZFW plot (point) with the blue MAC % lines.
e. Find the take-off STAB TRIM by referencing the TOW plot with either the magenta (F2, F3) or green (F4) lines or use table on next page.




**Note:** the FMS calculated TOW CG and STAB TRIM might slightly differ or change due to fuel distribution between main (wing) tanks and center tank.

# Stabilizer Trim

Example:

Take-off weight = 56t
TOW MAC - % = 26.7%

Result:

FLAP 2 and 3 ≈ 3.4 units
(FLAP 4 ≈ 3.4 units)

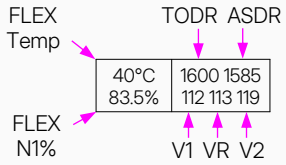
**Stabilizer Trim (units) FLAP 2 and FLAP 3 (FLAP 4)**

| Weight (kg) | CG Position M.A.C. (%) | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 |
| 67585 | Forward CG Limit | | | | | 6.1 (7.0) | 5.9 (6.7) | 5.6 (6.4) | 5.3 (6.1) | 5.0 (5.9) | 4.7 (5.6) | 4.4 (5.4) | 4.1 (5.1) | 3.9 (4.8) | 3.6 (4.5) | 3.3 (4.3) | 3.0 (4.0) | Aft CG Limit | | | | |
| 67000 | | | | | | 6.1 (6.9) | 5.8 (6.6) | 5.6 (6.3) | 5.3 (6.1) | 5.0 (5.8) | 4.7 (5.5) | 4.4 (5.3) | 4.1 (5.0) | 3.8 (4.8) | 3.6 (4.5) | 3.3 (4.2) | 3.0 (3.9) | 2.7 (3.6) | | | | |
| 66000 | | | | | 6.4 (7.0) | 6.1 (6.8) | 5.8 (6.5) | 5.5 (6.2) | 5.2 (6.0) | 5.0 (5.7) | 4.7 (5.4) | 4.4 (5.2) | 4.1 (4.9) | 3.8 (4.6) | 3.6 (4.3) | 3.3 (4.1) | 3.0 (3.8) | 2.7 (3.5) | | | | |
| 65000 | | | | | 6.3 (6.9) | 6.1 (6.6) | 5.8 (6.4) | 5.5 (6.1) | 5.2 (5.8) | 4.9 (5.6) | 4.6 (5.3) | 4.4 (5.0) | 4.1 (4.8) | 3.8 (4.5) | 3.5 (4.2) | 3.2 (3.9) | 3.0 (3.7) | 2.7 (3.4) | 2.4 (3.1) | | | |
| 64000 | | | | 6.6 (7.1) | 6.3 (6.8) | 6.0 (6.5) | 5.7 (6.2) | 5.5 (6.0) | 5.2 (5.7) | 4.9 (5.4) | 4.6 (5.2) | 4.3 (4.9) | 4.1 (4.6) | 3.8 (4.4) | 3.5 (4.1) | 3.2 (3.8) | 2.9 (3.6) | 2.7 (3.3) | 2.4 (3.0) | 2.1 (2.7) | | |
| 63000 | | | | 6.5 (6.9) | 6.3 (6.7) | 6.0 (6.4) | 5.7 (6.1) | 5.4 (5.8) | 5.1 (5.6) | 4.9 (5.3) | 4.6 (5.0) | 4.3 (4.8) | 4.0 (4.5) | 3.8 (4.2) | 3.5 (4.0) | 3.2 (3.7) | 2.9 (3.4) | 2.6 (3.2) | 2.4 (2.9) | 2.1 (2.6) | 1.8 (2.3) | |
| 62000 | | | 6.8 (7.1) | 6.5 (6.8) | 6.2 (6.5) | 6.0 (6.3) | 5.7 (6.0) | 5.4 (5.7) | 5.1 (5.4) | 4.8 (5.2) | 4.6 (4.9) | 4.3 (4.6) | 4.0 (4.4) | 3.7 (4.1) | 3.5 (3.9) | 3.2 (3.6) | 2.9 (3.3) | 2.6 (3.0) | 2.4 (2.8) | 2.1 (2.5) | 1.8 (2.2) | |
| 61000 | | | 6.7 (6.9) | 6.4 (6.6) | 6.2 (6.3) | 5.9 (6.1) | 5.6 (5.8) | 5.3 (5.6) | 5.1 (5.3) | 4.8 (5.0) | 4.5 (4.8) | 4.2 (4.5) | 4.0 (4.3) | 3.7 (4.0) | 3.4 (3.7) | 3.2 (3.4) | 2.9 (3.2) | 2.6 (2.9) | 2.3 (2.6) | 2.1 (2.4) | 1.8 (2.1) | 1.5 (1.8) |
| 60000 | | 6.9 (6.9) | 6.6 (6.6) | 6.4 (6.4) | 6.1 (6.1) | 5.8 (5.9) | 5.6 (5.7) | 5.3 (5.4) | 5.0 (5.2) | 4.7 (4.9) | 4.5 (4.7) | 4.2 (4.4) | 3.9 (4.1) | 3.7 (3.9) | 3.4 (3.6) | 3.1 (3.3) | 2.9 (3.1) | 2.6 (2.8) | 2.3 (2.5) | 2.1 (2.2) | 1.8 (2.0) | 1.5 (1.7) |
| 59000 | | 6.8 (6.6) | 6.5 (6.4) | 6.3 (6.2) | 6.0 (5.9) | 5.7 (5.8) | 5.5 (5.6) | 5.2 (5.4) | 4.9 (5.1) | 4.7 (4.9) | 4.4 (4.6) | 4.1 (4.3) | 3.9 (4.0) | 3.6 (3.7) | 3.4 (3.5) | 3.1 (3.2) | 2.8 (2.9) | 2.6 (2.7) | 2.3 (2.4) | 2.0 (2.1) | 1.8 (1.8) | 1.5 (1.6) |
| 58000 | 6.9 (6.7) | 6.7 (6.4) | 6.4 (6.1) | 6.2 (5.9) | 5.9 (5.8) | 5.6 (5.6) | 5.4 (5.5) | 5.1 (5.3) | 4.9 (5.1) | 4.6 (4.8) | 4.3 (4.6) | 4.1 (4.2) | 3.8 (3.9) | 3.6 (3.6) | 3.3 (3.3) | 3.0 (3.1) | 2.8 (2.8) | 2.5 (2.5) | 2.2 (2.3) | 2.0 (2.0) | 1.7 (1.7) | 1.5 (1.5) |
| 57000 | 6.9 (6.6) | 6.6 (6.3) | 6.3 (6.1) | 6.1 (5.9) | 5.8 (5.7) | 5.6 (5.6) | 5.3 (5.4) | 5.0 (5.2) | 4.8 (5.0) | 4.5 (4.8) | 4.2 (4.5) | 4.0 (4.1) | 3.7 (3.8) | 3.5 (3.5) | 3.2 (3.2) | 2.9 (3.0) | 2.7 (2.7) | 2.4 (2.4) | 2.2 (2.2) | 1.9 (1.9) | 1.6 (1.6) | 1.4 (1.4) |
| 56000 | 6.8 (6.5) | 6.5 (6.3) | 6.3 (6.0) | 6.0 (5.9) | 5.7 (5.7) | 5.5 (5.5) | 5.2 (5.3) | 5.0 (5.1) | 4.7 (5.0) | 4.4 (4.7) | 4.2 (4.4) | 3.9 (4.0) | 3.6 (3.7) | 3.4 (3.4) | 3.1 (3.2) | 2.9 (2.9) | 2.6 (2.6) | 2.3 (2.3) | 2.1 (2.1) | 1.8 (1.8) | 1.5 (1.5) | 1.3 (1.3) |
| 55000 | 6.7 (6.5) | 6.4 (6.2) | 6.2 (6.0) | 5.9 (5.8) | 5.7 (5.7) | 5.4 (5.4) | 5.1 (5.2) | 4.9 (5.1) | 4.6 (4.9) | 4.3 (4.6) | 4.1 (4.3) | 3.8 (3.9) | 3.6 (3.6) | 3.3 (3.3) | 3.0 (3.1) | 2.8 (2.8) | 2.5 (2.5) | 2.2 (2.3) | 2.0 (2.0) | 1.7 (1.7) | 1.5 (1.4) | 1.2 (1.2) |
| 54000 | 6.6 (6.4) | 6.4 (6.2) | 6.1 (5.9) | 5.8 (5.8) | 5.6 (5.6) | 5.3 (5.4) | 5.1 (5.1) | 4.8 (5.0) | 4.5 (4.8) | 4.3 (4.5) | 4.0 (4.2) | 3.7 (3.9) | 3.5 (3.5) | 3.2 (3.2) | 3.0 (3.0) | 2.7 (2.7) | 2.4 (2.4) | 2.2 (2.2) | 1.9 (1.9) | 1.6 (1.6) | 1.4 (1.4) | |
| 53000 | 6.5 (6.4) | 6.3 (6.1) | 6.0 (5.9) | 5.8 (5.7) | 5.5 (5.5) | 5.2 (5.3) | 5.0 (5.0) | 4.7 (4.9) | 4.4 (4.7) | 4.2 (4.4) | 3.9 (4.1) | 3.7 (3.8) | 3.4 (3.4) | 3.1 (3.1) | 2.9 (2.9) | 2.6 (2.6) | 2.3 (2.3) | 2.1 (2.1) | 1.8 (1.8) | 1.5 (1.5) | 1.3 (1.3) | |
| 52000 | 6.5 (6.3) | 6.2 (6.1) | 5.9 (5.8) | 5.7 (5.7) | 5.4 (5.5) | 5.2 (5.2) | 4.9 (4.9) | 4.6 (4.8) | 4.4 (4.7) | 4.1 (4.4) | 3.8 (4.0) | 3.6 (3.7) | 3.3 (3.3) | 3.0 (3.1) | 2.8 (2.8) | 2.5 (2.5) | 2.2 (2.2) | 2.0 (2.0) | 1.7 (1.7) | 1.5 (1.4) | | |
| 51000 | 6.4 (6.2) | 6.1 (6.0) | 5.9 (5.8) | 5.6 (5.6) | 5.3 (5.4) | 5.1 (5.1) | 4.8 (4.8) | 4.5 (4.7) | 4.3 (4.6) | 4.0 (4.3) | 3.7 (4.0) | 3.5 (3.6) | 3.2 (3.2) | 3.0 (3.0) | 2.7 (2.7) | 2.4 (2.4) | 2.2 (2.1) | 1.9 (1.9) | 1.6 (1.6) | 1.4 (1.3) | | |
| 50000 | 6.3 (6.2) | 6.0 (6.0) | 5.8 (5.7) | 5.5 (5.5) | 5.3 (5.3) | 5.0 (5.0) | 4.7 (4.7) | 4.5 (4.6) | 4.2 (4.5) | 3.9 (4.2) | 3.7 (3.9) | 3.4 (3.5) | 3.1 (3.1) | 2.9 (2.9) | 2.6 (2.6) | 2.3 (2.3) | 2.1 (2.1) | 1.8 (1.8) | 1.5 (1.5) | 1.3 (1.3) | | |
| 49000 | 6.2 (6.1) | 6.0 (5.9) | 5.7 (5.7) | 5.4 (5.4) | 5.2 (5.2) | 4.9 (4.9) | 4.6 (4.7) | 4.4 (4.5) | 4.1 (4.3) | 3.8 (4.1) | 3.6 (3.8) | 3.3 (3.4) | 3.1 (3.1) | 2.8 (2.8) | 2.5 (2.5) | 2.2 (2.2) | 2.0 (2.0) | 1.7 (1.7) | 1.4 (1.4) | | | |
| 48000 | 6.1 (6.1) | 5.9 (5.8) | 5.6 (5.6) | 5.4 (5.4) | 5.1 (5.1) | 4.8 (4.8) | 4.6 (4.6) | 4.3 (4.4) | 4 .0(4.2) | 3.7 (3.9) | 3.5 (3.7) | 3.2 (3.3) | 3.0 (3.0) | 2.7 (2.7) | 2.4 (2.4) | 2.2 (2.1) | 1.9 (1.9) | 1.6 (1.6) | 1.4 (1.3) | | | |
| 47000 | 6.1 (6.0) | 5.8 (5.8) | 5.5 (5.5) | 5.3 (5.3) | 5.0 (5.0) | 4.7 (4.7) | 4.5 (4.5) | 4.2 (4.3) | 3.9 (4.1) | 3.7 (3.8) | 3.4 (3.6) | 3.1 (3.2) | 2.9 (2.9) | 2.6 (2.6) | 2.3 (2.3) | 2.1 (2.1) | 1.8 (1.8) | 1.5 (1.5) | 1.3 (1.3) | | | |
| 46000 | 6.0 (6.0) | 5.7 (5.7) | 5.5 (5.5) | 5.2 (5.2) | 4.9 (4.9) | 4.7 (4.6) | 4.4 (4.4) | 4.1 (4.2) | 3.8 (3.9) | 3.6 (3.7) | 3.3 (3.5) | 3.0 (3.1) | 2.8 (2.8) | 2.5 (2.5) | 2.2 (2.2) | 2.0 (2.0) | 1.7 (1.7) | 1.4 (1.4) | | | | |
| 45000 | 5.8 (5.8) | 5.5 (5.5) | 5.3 (5.3) | 5.0 (5.0) | 4.8 (4.8) | 4.5 (4.5) | 4.2 (4.2) | 4.0 (4.0) | 3.7 (3.7) | 3.4 (3.4) | 3.2 (3.2) | 2.9 (2.9) | 2.6 (2.6) | 2.4 (2.4) | 2.1 (2.1) | 1.8 (1.8) | 1.6 (1.6) | 1.3 (1.3) | Take-off Prohibited | | | |
| 44000 | 5.6 (5.6) | 5.4 (5.4) | 5.1 (5.1) | 4.8 (4.8) | 4.6 (4.6) | 4.3 (4.3) | 4.1 (4.1) | 3.8 (3.8) | 3.6 (3.6) | 3.3 (3.3) | 3.0 (3.0) | 2.8 (2.8) | 2.5 (2.5) | 2.2 (2.2) | 2.0 (2.0) | 1.7 (1.7) | 1.5 (1.5) | | | | | |
| 43000 | 5.4 (5.4) | 5.1 (5.1) | 4.9 (4.9) | 4.6 (4.6) | 4.4 (4.4) | 4.1 (4.1) | 3.9 (3.9) | 3.6 (3.6) | 3.4 (3.4) | 3.1 (3.1) | 2.9 (2.9) | 2.6 (2.6) | 2.4 (2.4) | 2.1 (2.1) | 1.8 (1.8) | 1.6 (1.6) | 1.3 (1.3) | | | | | |
| 42000 | 5.2 (5.2) | 4.9 (4.9) | 4.7 (4.7) | 4.4 (4.4) | 4.2 (4.2) | 3.9 (3.9) | 3.7 (3.7) | 3.5 (3.5) | 3.2 (3.2) | 3.0 (3.0) | 2.7 (2.7) | 2.5 (2.5) | 2.2 (2.2) | 2.0 (2.0) | 1.7 (1.7) | 1.5 (1.5) | 1.2 (1.2) | | | | | |
| 41000 | 5.0 (5.0) | 4.7 (4.7) | 4.5 (4.5) | 4.2 (4.2) | 4.0 (4.0) | 3.8 (3.8) | 3.5 (3.5) | 3.3 (3.3) | 3.0 (3.0) | 2.8 (2.8) | 2.5 (2.5) | 2.3 (2.3) | 2.0 (2.0) | 1.8 (1.8) | 1.5 (1.5) | 1.3 (1.3) | | | | | | |
| 40000 | 4.8 (4.8) | 4.5 (4.5) | 4.3 (4.3) | 4.1 (4.1) | 3.8 (3.8) | 3.6 (3.6) | 3.3 (3.3) | 3.1 (3.1) | 2.9 (2.8) | 2.6 (2.6) | 2.4 (2.4) | 2.1 (2.1) | 1.9 (1.9) | 1.6 (1.6) | 1.4 (1.4) | 1.1 (1.1) | | | | | | |

- Performance (in-flight)

# Take-off PW1521G

**Note:** 67.585 kg Maximum Structural Take-off Weight option is used for take-off performance.




FLEX in blue if N1% lower than fixed derate.

## Take-off Performance and V-speeds

TAKE-OFF WEIGHT - kgs

| OAT - °C / A/I | | | 5°C & Below ALL | | | | | 10°C CAI | | | | | 20°C OFF | | | | | 30°C OFF | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| THRUST N1% | | | TO 85.4% | TO-1 82.3% | TO-2 79.2% | TO + FLEX | | TO 86.2% | TO-1 83.0% | TO-2 79.9% | TO + FLEX | | TO 87.7% | TO-1 84.5% | TO-2 81.3% | TO + FLEX | | TO 87.6% | TO-1 84.3% | TO-2 81.1% | TO + FLEX | |
| 45000 | FLAP 2 | | 1210 1225<br>109 115 125 | 1270 1285<br>110 115 125 | 1390 1405<br>113 118 125 | 35°C<br>78.4% | 1625 1640<br>117 121 124 | 1225 1240<br>109 115 125 | 1290 1305<br>110 115 125 | 1405 1425<br>113 118 125 | 43°C<br>77.3% | 1705 1720<br>117 121 124 | 1250 1265<br>109 115 125 | 1295 1310<br>109 115 125 | 1390 1405<br>112 117 125 | 47°C<br>79.3% | 1640 1655<br>115 120 124 | 1335 1350<br>110 116 125 | 1385 1400<br>111 116 125 | 1495 1510<br>113 118 124 | 47°C<br>80.6% | 1640 1655<br>115 120 124 |
| | FLAP 3 | | 1125 1140<br>104 109 120 | 1180 1200<br>105 109 120 | 1280 1295<br>108 111 119 | 33°C<br>79.3% | 1460 1475<br>112 114 118 | 1140 1155<br>104 109 120 | 1195 1210<br>105 109 120 | 1295 1310<br>108 111 119 | 41°C<br>78.2% | 1530 1545<br>112 114 118 | 1160 1175<br>104 109 120 | 1200 1215<br>104 109 120 | 1285 1300<br>107 110 119 | 47°C<br>79.3% | 1520 1535<br>110 113 118 | 1235 1250<br>105 110 119 | 1280 1295<br>106 110 119 | 1375 1390<br>108 111 119 | 47°C<br>80.6% | 1520 1535<br>110 113 118 |
| | FLAP 4 | | 1060 1075<br>100 103 114 | 1110 1125<br>101 103 114 | 1200 1215<br>103 105 114 | 30°C<br>80.6% | 1325 1325<br>106 107 113 | 1075 1090<br>100 103 114 | 1125 1140<br>101 103 114 | 1215 1230<br>103 105 114 | 38°C<br>79.6% | 1400 1385<br>106 107 113 | 1095 1110<br>100 103 114 | 1130 1145<br>100 103 114 | 1200 1215<br>102 104 114 | 47°C<br>79.3% | 1450 1440<br>106 107 113 | 1160 1175<br>101 104 114 | 1200 1215<br>101 104 114 | 1285 1300<br>104 105 113 | 47°C<br>80.6% | 1450 1440<br>106 107 113 |
| 50000 | FLAP 2 | | 1435 1450<br>118 124 131 | 1525 1540<br>118 124 131 | 1675 1690<br>121 126 131 | 28°C<br>81.5% | 1755 1770<br>123 127 131 | 1455 1470<br>118 124 131 | 1545 1560<br>118 124 131 | 1700 1715<br>121 126 131 | 37°C<br>80.0% | 1890 1905<br>124 127 131 | 1485 1500<br>117 124 131 | 1545 1565<br>118 124 131 | 1675 1690<br>120 125 131 | 46°C<br>79.6% | 1990 2005<br>124 127 131 | 1590 1605<br>118 125 131 | 1665 1680<br>119 125 131 | 1810 1825<br>121 126 131 | 46°C<br>81.0% | 1990 2005<br>124 127 131 |
| | FLAP 3 | | 1325 1340<br>112 117 125 | 1405 1420<br>113 117 126 | 1550 1565<br>116 119 125 | 26°C<br>82.4% | 1575 1590<br>117 120 125 | 1345 1360<br>112 117 125 | 1425 1440<br>113 117 126 | 1570 1585<br>116 119 125 | 35°C<br>80.9% | 1700 1715<br>118 120 125 | 1370 1385<br>112 117 125 | 1425 1440<br>112 117 125 | 1540 1885<br>115 118 125 | 40°C<br>80.4% | 1780 1795<br>118 120 124 | 1465 1480<br>113 118 125 | 1530 1545<br>114 118 125 | 1670 1685<br>116 119 125 | 44°C<br>81.8% | 1780 1795<br>118 120 124 |
| | FLAP 4 | | 1235 1250<br>108 110 120 | 1305 1320<br>108 110 120 | 1450 1460<br>112 112 119 | 23°C<br>83.9% | 1415 1430<br>111 112 119 | 1255 1270<br>108 110 120 | 1325 1340<br>108 110 120 | 1470 1960<br>112 112 119 | 31°C<br>82.6% | 1530 1520<br>112 113 119 | 1280 1295<br>107 110 120 | 1330 1345<br>108 110 120 | 1435 1450<br>110 112 119 | 44°C<br>82.1% | 1600 1585<br>112 113 119 | 1365 1380<br>108 111 119 | 1425 1440<br>109 111 119 | 1575 1580<br>112 112 119 | 40°C<br>83.5% | 1600 1585<br>112 113 119 |
| 55000 | FLAP 2 | | 1690 1705<br>126 132 137 | 1810 1825<br>126 132 137 | 2030 2050<br>130 133 137 | 22°C<br>84.4% | 1905 1920<br>129 133 137 | 1715 1730<br>126 132 137 | 1835 1850<br>126 132 137 | 2060 2075<br>130 133 137 | 30°C<br>83.0% | 2040 2060<br>130 134 137 | 1745 1760<br>125 131 137 | 1830 1850<br>126 131 137 | 2000 2015<br>128 133 137 | 38°C<br>82.9% | 2120 2140<br>130 134 137 | 1880 1895<br>127 132 137 | 2000 1995<br>127 132 137 | 2195 2215<br>130 134 137 | 38°C<br>84.3% | 2120 2135<br>130 134 137 |
| | FLAP 3 | | 1560 1575<br>120 124 131 | 1670 1685<br>121 125 131 | 1885 1900<br>125 127 131 | 21°C<br>85.0% | 1740 1755<br>123 126 131 | 1580 1595<br>120 124 131 | 1695 1710<br>121 125 131 | 1910 1925<br>125 127 131 | 27°C<br>84.2% | 1805 1820<br>124 126 131 | 1610 1625<br>120 124 131 | 1685 1700<br>120 124 131 | 1855 1870<br>123 126 131 | 36°C<br>83.7% | 1900 1915<br>124 126 130 | 1735 1750<br>121 125 131 | 1825 1840<br>122 125 131 | 2035 2050<br>125 127 131 | 36°C<br>85.1% | 1900 1915<br>124 126 130 |
| | FLAP 4 | | 1455 1470<br>115 117 125 | 1550 1565<br>116 117 125 | 1875 1725<br>119 119 124 | — | | 1480 1495<br>115 117 125 | 1575 1590<br>116 117 125 | 1900 1750<br>119 119 124 | 23°C<br>86.0% | 1605 1620<br>118 118 125 | 1500 1515<br>115 117 125 | 1570 1585<br>115 117 125 | 1780 1735<br>118 119 124 | 32°C<br>85.3% | 1700 1705<br>118 118 124 | 1635 1650<br>117 118 125 | 1715 1730<br>117 118 125 | — | — | |
| 60000 | FLAP 2 | | 1980 1990<br>133 139 143 | 2135 2150<br>134 139 143 | — | — | | 2005 2020<br>133 139 143 | 2165 2180<br>134 139 143 | — | — | | 2035 2050<br>133 139 143 | 2150 2165<br>133 139 143 | 2410 2425<br>137 140 143 | 31°C<br>85.7% | 2255 2270<br>136 140 143 | 2220 2235<br>135 139 143 | 2355 2370<br>135 139 143 | — | — | |
| | FLAP 3 | | 1835 1850<br>128 131 136 | 1990 2005<br>129 132 137 | — | — | | 1860 1875<br>128 131 136 | 2020 2035<br>129 132 137 | — | — | | 1885 1900<br>127 131 136 | 1995 2010<br>128 131 136 | 2240 2255<br>132 133 137 | 29°C<br>86.5% | 2025 2040<br>129 132 136 | 2055 2070<br>130 132 136 | 2180 2195<br>130 132 136 | — | — | |
| | FLAP 4 | | 1765 1735<br>123 124 130 | 1900 1855<br>124 124 130 | — | — | | 1790 1760<br>123 124 130 | 1925 1880<br>124 124 130 | — | — | | 1790 1800<br>123 124 130 | 1890 1560<br>123 124 130 | — | — | | 2035 1925<br>124 124 130 | 2145 2020<br>124 124 130 | — | — | |
| 65000 | FLAP 2 | | 2320 2335<br>142 145 148 | 2540 2560<br>143 146 149 | — | — | | 2355 2370<br>142 145 148 | 2580 2595<br>143 146 149 | — | — | | 2380 2400<br>141 145 148 | 2535 2550<br>141 145 148 | — | — | | — | — | — | — | |
| | FLAP 3 | | 2155 2170<br>137 137 141 | 2360 2375<br>138 139 142 | — | — | | 2185 2200<br>136 137 141 | 2395 2410<br>138 139 142 | — | — | | 2210 2220<br>136 137 141 | 2355 2370<br>136 138 142 | — | — | | — | — | — | — | |
| | FLAP 4 | | — | — | — | — | | — | — | — | — | | — | — | — | — | | — | — | — | — | |
| 67585 | FLAP 2 | | — | — | — | — | | — | — | — | — | | 2580 2595<br>145 148 151 | 2760 2775<br>146 148 151 | — | — | | — | — | — | — | |
| | FLAP 3 | | — | — | — | — | | — | — | — | — | | — | — | — | — | | — | — | — | — | |
| | FLAP 4 | | — | — | — | — | | — | — | — | — | | — | — | — | — | | — | — | — | — | |

Conditions:
- Sea level, QNH 1013.25 hPa
- Dry runway, wind calm
- Non-limiting runway, no limiting obstacles

- Rolling take-off
- APR armed
- APU OFF, bleeds and packs ON
- BTMS level 6 or below

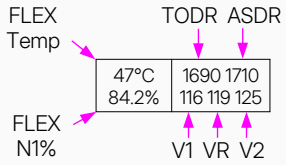
Example:
- Take-off weight 55t
- OAT 20°C
- Runway length 2000 m

Best (climb) performance:
- FLAP 2, no derate (TO), N1 = 87.7%
- TODR = 1745 m, ASDR = 1760 m (stop margin 240 m)
- V1 = 125 kts, VR = 131 kts, V2 = 137 kts

Lowest engine power (reduced wear):
- FLAP 3, derate TO-2, N1 = 81.3%
- TODR = 1855 m, ASDR = 1870 m (stop margin 130 m)
- V1 = 123 kts, VR = 126 kts, V2 = 131 kts

# Take-off PW1524G

**Note:** 67.585 kg Maximum Structural Take-off Weight option is used for take-off performance.




FLEX in blue if N1% lower than fixed derate.

## Take-off Performance and V-speeds

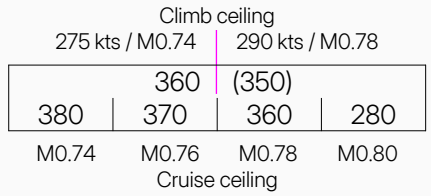
TAKE-OFF WEIGHT - kgs

| OAT - °C A/I | | | 5°C & Below ALL | | | | | | 10°C CAI | | | | | | 20°C OFF | | | | | | 30°C OFF | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| THRUST N1% | | | TO 89.2% | TO-1 85.4% | TO-2 82.3% | TO-3 79.2% | TO + FLEX | | TO 90.0% | TO-1 86.2% | TO-2 83.0% | TO-3 79.9% | TO + FLEX | | TO 91.6% | TO-1 87.7% | TO-2 84.5% | TO-3 81.3% | TO + FLEX | | TO 91.4% | TO-1 87.6% | TO-2 84.3% | TO-3 81.1% | TO + FLEX | |
| 45000 | FLAP 2 | | 1135 1150<br>106 113 126 | 1175 1190<br>107 114 126 | 1270 1285<br>110 115 125 | 1390 1405<br>113 118 125 | 41°C<br>78.8% | 1650 1780<br>117 121 124 | 1150 1165<br>106 114 126 | 1190 1205<br>107 114 126 | 1290 1305<br>110 115 125 | 1405 1425<br>113 118 125 | 47°C<br>79.3% | 1635 1650<br>115 120 124 | 1170 1185<br>106 113 126 | 1210 1225<br>107 113 126 | 1295 1310<br>109 115 125 | 1390 1405<br>112 117 125 | 47°C<br>82.8% | 1520 1535<br>113 118 124 | 1250 1265<br>107 115 125 | 1295 1310<br>108 115 125 | 1385 1400<br>111 116 125 | 1495 1510<br>113 118 124 | 47°C<br>84.2% | 1520 1535<br>113 118 124 |
| | FLAP 3 | | 1060 1075<br>101 107 120 | 1095 1110<br>102 107 120 | 1180 1195<br>105 109 120 | 1280 1295<br>108 111 119 | 40°C<br>79.4% | 1510 1525<br>112 114 118 | 1075 1090<br>101 107 120 | 1110 1125<br>102 107 120 | 1195 1210<br>105 109 120 | 1295 1310<br>108 111 119 | 47°C<br>79.3% | 1515 1530<br>111 113 118 | 1095 1110<br>101 107 120 | 1130 1145<br>102 107 120 | 1200 1215<br>104 109 120 | 1285 1300<br>107 110 119 | 47°C<br>82.8% | 1400 1415<br>108 112 119 | 1165 1180<br>102 108 120 | 1205 1220<br>103 108 120 | 1280 1295<br>106 110 119 | 1375 1390<br>108 111 119 | 47°C<br>84.2% | 1400 1415<br>108 112 119 |
| | FLAP 4 | | 1010 1060<br>100 103 116 | 1055 1070<br>100 103 116 | 1110 1125<br>101 103 114 | 1200 1215<br>103 105 114 | 36°C<br>81.1% | 1345 1355<br>106 107 113 | 1025 1070<br>100 103 116 | 1070 1085<br>100 103 116 | 1125 1140<br>101 103 114 | 1215 1230<br>103 105 114 | 46°C<br>79.7% | 1420 1420<br>106 107 113 | 1045 1100<br>100 103 116 | 1090 1105<br>100 103 116 | 1130 1145<br>100 103 114 | 1200 1215<br>102 104 114 | 47°C<br>82.8% | 1305 1320<br>103 105 113 | 1110 1135<br>100 103 115 | 1150 1165<br>100 103 115 | 1200 1215<br>101 104 114 | 1285 1300<br>104 105 113 | 47°C<br>84.2% | 1305 1320<br>103 105 113 |
| 50000 | FLAP 2 | | 1335 2090<br>115 122 132 | 1395 1410<br>116 122 132 | 1525 1540<br>118 124 131 | 1675 1690<br>121 126 131 | 34°C<br>82.0% | 1790 1810<br>123 127 131 | 1355 1370<br>115 122 132 | 1420 1430<br>116 122 132 | 1545 1560<br>118 124 131 | 1700 1715<br>121 126 131 | 44°C<br>80.5% | 1910 1925<br>123 127 131 | 1385 1400<br>114 122 132 | 1440 1455<br>115 122 132 | 1545 1565<br>118 124 131 | 1675 1690<br>120 125 131 | 47°C<br>82.8% | 1825 1840<br>121 126 131 | 1480 1500<br>116 123 131 | 1545 1560<br>116 123 131 | 1665 1680<br>119 125 131 | 1810 1825<br>121 126 131 | 47°C<br>84.2% | 1825 1840<br>121 126 131 |
| | FLAP 3 | | 1240 1255<br>110 115 126 | 1290 2140<br>110 115 126 | 1405 1420<br>113 117 126 | 1550 1565<br>116 119 125 | 32°C<br>82.9% | 1610 1625<br>117 120 125 | 1260 1275<br>110 115 126 | 1310 1325<br>111 115 126 | 1425 1440<br>113 117 126 | 1570 1585<br>116 119 125 | 43°C<br>81.0% | 1745 1760<br>118 120 124 | 1285 1300<br>109 115 126 | 1330 1345<br>110 115 126 | 1425 1440<br>112 117 125 | 1540 1885<br>115 118 125 | 47°C<br>82.8% | 1690 1710<br>116 119 125 | 1375 1390<br>111 116 126 | 1425 1440<br>111 116 126 | 1530 1545<br>114 118 125 | 1670 1685<br>116 119 125 | 47°C<br>84.2% | 1690 1710<br>116 119 125 |
| | FLAP 4 | | 1165 1180<br>105 109 120 | 1210 1225<br>106 109 120 | 1305 1320<br>108 110 120 | 1450 1980<br>112 112 119 | 29°C<br>84.3% | 1440 1455<br>111 112 119 | 1180 1195<br>105 109 120 | 1225 1240<br>106 109 120 | 1325 1340<br>108 110 120 | 1470 1960<br>112 112 119 | 39°C<br>82.9% | 1565 1560<br>112 113 119 | 1205 1220<br>105 109 120 | 1250 1265<br>106 109 120 | 1330 1345<br>108 110 120 | 1435 1450<br>110 112 119 | 47°C<br>82.8% | 1610 1615<br>112 113 119 | 1285 1300<br>106 110 120 | 1330 1345<br>107 110 120 | 1425 1440<br>109 111 119 | 1575 1580<br>112 112 119 | 47°C<br>84.2% | 1610 1615<br>112 113 119 |
| 55000 | FLAP 2 | | 1565 1580<br>123 130 138 | 1640 1655<br>124 130 138 | 1810 1825<br>126 132 137 | 2030 2050<br>130 133 137 | 28°C<br>84.8% | 1940 1955<br>129 133 137 | 1585 1600<br>123 130 138 | 1665 1680<br>124 130 138 | 1835 1850<br>126 132 137 | 2060 2075<br>130 133 137 | 37°C<br>83.7% | 2065 1365<br>130 134 137 | 1615 1630<br>122 130 138 | 1690 1705<br>123 130 138 | 1830 1850<br>126 131 137 | 2000 2015<br>128 133 137 | 46°C<br>83.2% | 2180 2195<br>130 134 137 | 1740 1755<br>124 131 137 | 1825 1840<br>124 131 137 | 2000 1995<br>127 132 137 | 2195 2215<br>130 134 137 | 46°C<br>84.6% | 2180 2195<br>130 134 137 |
| | FLAP 3 | | 1445 1460<br>117 123 131 | 1510 1525<br>118 123 131 | 1670 1685<br>121 125 131 | 1885 1900<br>125 127 131 | 26°C<br>85.7% | 1735 1750<br>123 126 131 | 1465 1480<br>117 123 131 | 1535 1550<br>118 123 131 | 1695 1710<br>121 125 131 | 1910 1925<br>125 127 131 | 35°C<br>84.5% | 1850 1865<br>123 126 131 | 1495 1510<br>117 123 131 | 1560 1575<br>117 123 131 | 1685 1700<br>120 124 131 | 1855 1870<br>123 126 131 | 43°C<br>84.5% | 1920 1935<br>123 126 130 | 1600 1615<br>118 124 131 | 1675 1690<br>119 124 131 | 1825 1840<br>122 125 131 | 2035 2050<br>125 127 131 | 43°C<br>85.9% | 1920 1935<br>123 126 130 |
| | FLAP 4 | | 1345 1360<br>112 116 125 | 1405 2020<br>113 116 125 | 1550 1565<br>116 117 125 | 1875 1725<br>119 119 124 | 23°C<br>87.1% | 1555 1570<br>117 118 125 | 1365 1380<br>112 116 125 | 1425 1440<br>113 116 125 | 1575 1590<br>116 117 125 | 1900 1750<br>119 119 124 | 32°C<br>85.8% | 1670 1685<br>118 118 124 | 1395 1410<br>112 116 125 | 1450 1465<br>113 116 125 | 1570 1585<br>115 117 125 | 1780 1735<br>118 119 124 | 40°C<br>85.8% | 1740 1750<br>118 119 124 | 1490 1505<br>113 117 125 | 1555 1570<br>114 117 125 | 1715 1730<br>117 118 125 | — | 40°C<br>87.2% | 1740 1750<br>118 119 124 |
| 60000 | FLAP 2 | | 1815 1830<br>130 137 143 | 1915 1930<br>131 137 143 | 2135 2150<br>134 139 143 | — | 23°C<br>87.1% | 2095 2115<br>135 139 143 | 1840 1855<br>130 137 143 | 1940 1485<br>131 137 143 | 2165 2180<br>134 139 143 | — | 31°C<br>86.3% | 2215 2225<br>135 140 143 | 1875 1890<br>130 137 143 | 1970 1985<br>130 137 143 | 2150 2165<br>133 139 143 | 2410 2425<br>137 140 143 | 39°C<br>86.2% | 2310 2325<br>136 140 143 | 2020 2035<br>132 138 143 | 2130 2145<br>132 138 143 | 2355 2370<br>135 139 143 | — | 39°C<br>87.6% | 2310 2325<br>136 140 143 |
| | FLAP 3 | | 1670 1685<br>125 130 137 | 1760 1775<br>125 130 137 | 1990 2005<br>129 132 137 | — | 21°C<br>88.0% | 1880 1895<br>128 131 136 | 1695 1710<br>125 130 137 | 1785 1800<br>125 130 137 | 2020 2035<br>129 132 137 | — | 29°C<br>87.1% | 1985 2000<br>129 132 136 | 1725 1740<br>124 129 137 | 1810 1830<br>125 130 137 | 1995 2010<br>128 131 136 | 2240 2255<br>132 133 137 | 37°C<br>87.0% | 2075 2090<br>129 132 136 | 1870 1885<br>126 131 136 | 1965 1980<br>126 131 136 | 2180 2195<br>130 132 136 | — | 37°C<br>88.4% | 2075 2090<br>129 132 136 |
| | FLAP 4 | | 1555 1570<br>120 122 130 | 1640 1655<br>120 122 130 | 1900 1855<br>124 124 130 | — | — | | 1580 1595<br>120 122 130 | 1660 1680<br>120 123 130 | 1925 1880<br>124 124 130 | — | 25°C<br>88.9% | 1760 1775<br>122 124 130 | 1605 1620<br>119 122 130 | 1685 1700<br>119 122 130 | 1890 1560<br>123 124 130 | — | 33°C<br>88.6% | 1845 1860<br>123 124 130 | 1750 1765<br>121 123 130 | 1835 1855<br>122 123 130 | 2145 2020<br>124 124 130 | — | — | |
| 65000 | FLAP 2 | | 2080 2095<br>138 144 148 | 2210 2225<br>138 144 148 | 2540 2560<br>143 146 149 | — | — | | 2110 2125<br>138 144 148 | 2245 2260<br>138 144 148 | 2580 2595<br>143 146 149 | — | 25°C<br>88.9% | 2335 2350<br>140 145 148 | 2145 2160<br>137 144 148 | 2275 2290<br>138 144 148 | 2535 2550<br>141 145 148 | — | 32°C<br>89.0% | 2420 2435<br>140 145 148 | 2340 2355<br>139 145 148 | 2485 2500<br>140 145 148 | — | — | — | |
| | FLAP 3 | | 1930 1945<br>132 136 141 | 2050 2050<br>133 136 142 | 2360 2375<br>138 139 142 | — | — | | 1955 1970<br>132 136 141 | 2080 2100<br>133 136 142 | 2395 2410<br>138 139 142 | — | 23°C<br>89.7% | 2100 2115<br>134 137 141 | 1985 2000<br>131 136 141 | 2105 2120<br>132 136 142 | 2355 2370<br>136 138 142 | — | 30°C<br>89.9% | 2170 2185<br>134 137 141 | 2170 2185<br>134 137 141 | 2305 2320<br>134 137 141 | — | — | — | |
| | FLAP 4 | | 1825 1845<br>128 128 135 | 1940 1940<br>128 128 135 | — | — | — | | 1850 1865<br>128 128 135 | 1970 1970<br>128 128 135 | — | — | — | | 1870 1885<br>127 128 135 | 1975 1990<br>127 128 135 | — | — | — | | 2105 2045<br>129 129 134 | 2230 2150<br>129 129 134 | — | — | — | |
| 67585 | FLAP 2 | | 2245 2260<br>142 147 151 | 2395 2410<br>142 147 151 | — | — | — | | 2275 2290<br>142 147 151 | 2430 2445<br>142 147 151 | — | — | — | | 2305 2320<br>141 147 151 | 2450 2465<br>141 147 151 | 2760 2775<br>146 148 151 | — | — | | 2530 2545<br>144 148 151 | 2695 2710<br>144 148 151 | — | — | — | |
| | FLAP 3 | | 2080 2095<br>136 139 144 | 2220 2235<br>137 140 144 | — | — | — | | 2110 2125<br>136 139 144 | 2255 2270<br>137 140 144 | — | — | — | | 2135 2155<br>135 139 144 | 2280 2295<br>136 140 144 | — | — | — | | 2345 2360<br>138 140 144 | 2500 2515<br>139 140 144 | — | — | — | |
| | FLAP 4 | | 2015 1960<br>131 131 137 | 2150 2065<br>131 131 137 | — | — | — | | 2040 1990<br>131 131 137 | 2180 2100<br>131 131 137 | — | — | — | | 2035 2035<br>131 131 137 | 2160 2140<br>131 131 137 | — | — | — | | — | — | — | — | — | |

Conditions:

- Sea level, QNH 1013.25 hPa
- Dry runway, wind calm
- Non-limiting runway, no limiting obstacles
- Rolling take-off
- APR armed
- APU OFF, bleeds and packs ON
- BTMS level 6 or below

# Maximum Cruise Altitude




Example:

Top of Climb weight = 56000
Cruise speed = MACH 0.78
Temperature = ISA +10°C

Result:

Climb ceiling (minimum rate of climb 300 ft/min):

- FL370 at 275 kts / M0.74
- FL360 at 290 kts / M0.78

Cruise ceiling (minimum rate of climb 100 ft/min) at M0.78:

- FL390

**Maximum Altitude (Flight Level)**

| Gross Weight (kg) | | Climb at 250 kts, 275 kts, M0.74 (Climb at 250 kts, 290 kts, M0.78) | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | ISA & Below | | | | ISA +10°C | | | | ISA +20°C | | | |
| | | 0.74 | 0.76 | 0.78 | 0.80 | 0.74 | 0.76 | 0.78 | 0.80 | 0.74 | 0.76 | 0.78 | 0.80 |
| 44000 | Climb | | 410 (410) | | | | 410 (410) | | | | 410 (380) | | |
| | Cruise | 410 | 410 | 410 | 410 | 410 | 410 | 410 | 410 | 410 | 410 | 410 | 410 |
| 46000 | Climb | | 410 (410) | | | | 410 (400) | | | | 390 (370) | | |
| | Cruise | 410 | 410 | 410 | 410 | 410 | 410 | 410 | 410 | 410 | 410 | 410 | 400 |
| 48000 | Climb | | 410 (410) | | | | 410 (390) | | | | 390 (370) | | |
| | Cruise | 410 | 410 | 410 | 410 | 410 | 410 | 410 | 410 | 410 | 410 | 400 | 390 |
| 50000 | Climb | | 410 (400) | | | | 400 (390) | | | | 380 (360) | | |
| | Cruise | 410 | 410 | 410 | 410 | 410 | 410 | 410 | 410 | 410 | 400 | 400 | 380 |
| 52000 | Climb | | 400 (400) | | | | 390 (380) | | | | 370 (350) | | |
| | Cruise | 410 | 410 | 410 | 410 | 410 | 410 | 410 | 400 | 400 | 400 | 390 | 380 |
| 54000 | Climb | | 400 (390) | | | | 380 (370) | | | | 360 (330) | | |
| | Cruise | 410 | 410 | 410 | 410 | 410 | 400 | 400 | 390 | 390 | 390 | 380 | 370 |
| 56000 | Climb | | 390 (380) | | | | 370 (360) | | | | 360 (320) | | |
| | Cruise | 410 | 410 | 400 | 400 | 400 | 400 | 390 | 380 | 390 | 380 | 370 | 360 |
| 58000 | Climb | | 380 (370) | | | | 370 (360) | | | | 350 (300) | | |
| | Cruise | 400 | 400 | 400 | 390 | 390 | 390 | 390 | 380 | 380 | 380 | 370 | 360 |
| 60000 | Climb | | 380 (370) | | | | 360 (350) | | | | 340 (300) | | |
| | Cruise | 400 | 390 | 390 | 390 | 390 | 380 | 380 | 370 | 370 | 370 | 360 | 340 |
| 62000 | Climb | | 370x (360) | | | | 360 (340) | | | | 330 (290) | | |
| | Cruise | 390 | 390 | 380 | 380 | 380 | 380 | 370 | 370 | 370 | 360 | 350 | 330 |
| 64000 | Climb | | 360 (360) | | | | 350 (330) | | | | 320 (290) | | |
| | Cruise | 390 | 380 | 380 | 370 | 380 | 370 | 370 | 360 | 360 | 350 | 340 | 310 |
| 66000 | Climb | | 360 (350) | | | | 340 (320) | | | | 310 (290) | | |
| | Cruise | 380 | 380 | 370 | 370 | 370 | 370 | 360 | 350 | 350 | 350 | 330 | 290 |
| 68000 | Climb | | 350 (340) | | | | 330 (300) | | | | 290 (280) | | |
| | Cruise | 370 | 370 | 370 | 360 | 360 | 360 | 350 | 340 | 350 | 330 | 290 | — |
| 70000 | Climb | | 340 (330) | | | | 320 (290) | | | | 290 (280) | | |
| | Cruise | 370 | 360 | 360 | 360 | 360 | 350 | 350 | 330 | 340 | 330 | 310 | — |

Conditions:
- All Engines Operating (AEO)
- Engine bleeds – packs ON
- A/I OFF, APU OFF, CG 27% MAC

Climb ceiling:
- Minimum rate of climb = 300 ft/min

Cruise ceiling:
- Minimum rate of climb = 100 ft/min

# Green Dot and LRC Speed

| | |
|---|---|
| Green Dot – AEO | 201 |
| Green Dot – OEI | 198 |
| Long Range Cruise – OEI | 200 |

Example:

Gross weight = 55000
Cruise altitude = 30000 ft

Result:

- GD – AEO = 218 kts
- GD – OEI = 216 kts
- LRC – OEI = 226 kts

**Green Dot – AEO, Green Dot – OEI, LRC – OEI (kts)**

| Gross Weight (kg) | Pressure Altitude (ft) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 0 | 5000 | 10000 | 15000 | 20000 | 25000 | 30000 | 35000 | 40000 | 41000 |
| 40000 | 178<br>169<br>229 | 178<br>170<br>232 | 179<br>169<br>220 | 181<br>172<br>216 | 183<br>174<br>219 | 185<br>178<br>210 | 186<br>181<br>203 | 186<br>183<br>201 | 188<br>185<br>199 | 189<br>186<br>198 |
| 45000 | 188<br>181<br>236 | 189<br>182<br>234 | 192<br>181<br>224 | 193<br>184<br>218 | 194<br>186<br>224 | 196<br>190<br>214 | 197<br>194<br>205 | 198<br>195<br>203 | 200<br>197<br>201 | 201<br>198<br>200 |
| 50000 | 198<br>192<br>253 | 200<br>194<br>240 | 202<br>192<br>237 | 204<br>195<br>234 | 205<br>198<br>231 | 206<br>202<br>227 | 208<br>205<br>225 | 209<br>207<br>223 | 211<br>209<br>221 | 209<br>210<br>220 |
| 55000 | 208<br>203<br>257 | 211<br>205<br>247 | 212<br>203<br>240 | 214<br>206<br>244 | 215<br>209<br>237 | 217<br>213<br>227 | 218<br>216<br>226 | 220<br>218<br>225 | 219<br>220<br>224 | 214<br>221<br>223 |
| 60000 | 218<br>213<br>262 | 220<br>217<br>259 | 222<br>213<br>253 | 224<br>216<br>253 | 225<br>220<br>251 | 226<br>222<br>251 | 228<br>225<br>249 | 230<br>228<br>248 | 224<br>230<br>247 | 219<br>231<br>246 |
| 65000 | 228<br>222<br>266 | 230<br>224<br>261 | 232<br>222<br>264 | 233<br>225<br>257 | 234<br>229<br>254 | 235<br>232<br>252 | 238<br>234<br>251 | 238<br>236<br>250 | 229<br>238<br>249 | 229<br>239<br>248 |
| 70000 | 237<br>231<br>269 | 239<br>233<br>263 | 240<br>231<br>268 | 242<br>234<br>259 | 243<br>238<br>254 | 244<br>240<br>252 | 248<br>243<br>251 | 243<br>245<br>250 | 238<br>247<br>249 | 234<br>248<br>248 |

Conditions:

- All Engines Operating (Green Dot – AEO)
- One Engine Inoperative (Green Dot / LRC – OEI)
- Clean configuration, Engine bleed – Packs ON
- A/I OFF, APU OFF

# Idle Descent

Example:

- LW = 52t, CRZ ALT = 32000 ft
- Distance = 99 nm, VPA = 3.05º

| Approximate VPA at GW 55t | | | | | |
|---|---|---|---|---|---|
| M0.78 | M0.74 | 290 kts | 275 kts | 250 kts | GD |
| 3.1º | 3.0º | 3.5º | 3.3º | 3.0º | 2.5º |

**Distance and Vertical Path Angle for Idle Descent to SL**

| CRZ ALT | Landing Weight (kg) | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 45000 | 46000 | 47000 | 48000 | 49000 | 50000 | 51000 | 52000 | 53000 | 54000 | 55000 | 56000 | 57000 | 58000 | 58740 |
| 41000 | 112 nm 3.44º | 114 nm 3.39º | 115 nm 3.35º | 117 nm 3.30º | 118 nm 3.26º | 120 nm 3.22º | 121 nm 3.19º | 122 nm 3.16º | 124 nm 3.13º | 125 nm 3.10º | 126 nm 3.07º | 127 nm 3.04º | 128 nm 3.02º | 129 nm 2.99º | 130 nm 2.98º |
| 40000 | 110 nm 3.44º | 111 nm 3.39º | 113 nm 3.34º | 114 nm 3.30º | 116 nm 3.26º | 117 nm 3.22º | 118 nm 3.19º | 119 nm 3.15º | 121 nm 3.12º | 122 nm 3.09º | 123 nm 3.06º | 124 nm 3.04º | 125 nm 3.01º | 126 nm 2.99º | 127 nm 2.97º |
| 38000 | 104 nm 3.43º | 106 nm 3.38º | 107 nm 3.34º | 109 nm 3.29º | 110 nm 3.25º | 111 nm 3.21º | 113 nm 3.18º | 114 nm 3.14º | 115 nm 3.11º | 116 nm 3.08º | 117 nm 3.05º | 118 nm 3.02º | 119 nm 3.00º | 120 nm 2.97º | 121 nm 2.95º |
| 36000 | 100 nm 3.40º | 101 nm 3.36º | 102 nm 3.31º | 104 nm 3.27º | 105 nm 3.23º | 106 nm 3.19º | 108 nm 3.15º | 109 nm 3.12º | 110 nm 3.09º | 111 nm 3.06º | 112 nm 3.03º | 113 nm 3.00º | 114 nm 2.97º | 115 nm 2.95º | 116 nm 2.93º |
| 34000 | 95 nm 3.36º | 97 nm 3.31º | 98 nm 3.27º | 99 nm 3.23º | 101 nm 3.19º | 102 nm 3.15º | 103 nm 3.11º | 104 nm 3.08º | 105 nm 3.05º | 106 nm 3.02º | 107 nm 2.99º | 108 nm 2.96º | 109 nm 2.94º | 110 nm 2.91º | 111 nm 2.89º |
| 32000 | 91 nm 3.32º | 92 nm 3.28º | 93 nm 3.23º | 94 nm 3.19º | 96 nm 3.16º | 97 nm 3.12º | 98 nm 3.08º | 99 nm 3.05º | 100 nm 3.02º | 101 nm 2.99º | 102 nm 2.96º | 103 nm 2.94º | 104 nm 2.91º | 104 nm 2.89º | 105 nm 2.87º |
| 30000 | 85 nm 3.31º | 87 nm 3.26º | 88 nm 3.22º | 89 nm 3.18º | 90 nm 3.14º | 91 nm 3.11º | 92 nm 3.07º | 93 nm 3.04º | 94 nm 3.01º | 95 nm 2.98º | 96 nm 2.95º | 97 nm 2.93º | 97 nm 2.90º | 98 nm 2.88º | 99 nm 2.86º |
| 28000 | 80 nm 3.29º | 81 nm 3.24º | 82 nm 3.20º | 83 nm 3.16º | 84 nm 3.12º | 85 nm 3.09º | 86 nm 3.05º | 87 nm 3.02º | 88 nm 2.99º | 89 nm 2.96º | 90 nm 2.94º | 91 nm 2.91º | 91 nm 2.89º | 92 nm 2.86º | 93 nm 2.85º |
| 26000 | 75 nm 3.26º | 76 nm 3.21º | 77 nm 3.17º | 78 nm 3.13º | 79 nm 3.10º | 80 nm 3.06º | 81 nm 3.03º | 82 nm 3.00º | 82 nm 2.97º | 83 nm 2.94º | 84 nm 2.92º | 85 nm 2.89º | 85 nm 2.87º | 86 nm 2.85º | 87 nm 2.83º |
| 24000 | 70 nm 3.22º | 71 nm 3.18º | 72 nm 3.14º | 73 nm 3.10º | 74 nm 3.06º | 75 nm 3.03º | 75 nm 3.00º | 76 nm 2.97º | 77 nm 2.94º | 78 nm 2.92º | 78 nm 2.89º | 79 nm 2.87º | 80 nm 2.84º | 80 nm 2.82º | 81 nm 2.80º |
| 22000 | 65 nm 3.17º | 66 nm 3.13º | 67 nm 3.09º | 68 nm 3.06º | 69 nm 3.02º | 69 nm 2.99º | 70 nm 2.96º | 71 nm 2.93º | 71 nm 2.90º | 72 nm 2.88º | 73 nm 2.85º | 73 nm 2.83º | 74 nm 2.81º | 74 nm 2.79º | 75 nm 2.77º |
| 20000 | 60 nm 3.12º | 61 nm 3.08º | 62 nm 3.04º | 63 nm 3.01º | 63 nm 2.97º | 64 nm 2.94º | 65 nm 2.91º | 65 nm 2.88º | 66 nm 2.86º | 66 nm 2.84º | 67 nm 2.81º | 68 nm 2.79º | 68 nm 2.77º | 69 nm 2.75º | 69 nm 2.74º |
| 15000 | 48 nm 2.94º | 49 nm 2.90º | 49 nm 2.87º | 50 nm 2.84º | 50 nm 2.82º | 51 nm 2.79º | 51 nm 2.76º | 52 nm 2.73º | 52 nm 2.71º | 52 nm 2.69º | 53 nm 2.68º | 53 nm 2.66º | 54 nm 2.64º | 54 nm 2.62º | 54 nm 2.61º |
| 10000 | 34 nm 2.80º | 34 nm 2.77º | 34 nm 2.74º | 35 nm 2.72º | 35 nm 2.69º | 35 nm 2.67º | 36 nm 2.65º | 36 nm 2.64º | 36 nm 2.62º | 36 nm 2.60º | 36 nm 2.59º | 37 nm 2.58º | 37 nm 2.56º | 37 nm 2.55º | 37 nm 2.54º |

Wind :
- Each 10 kts of headwind = +0.1º
- Each 10 kts of tailwind = -0.1º

Temperature :
- Each 10ºC below ISA = +0.1º
- Each 10ºC above ISA = -0.1º

Anti-icing:
- Use anticipated = -0.2º

Distance:
- Each 0.1º below 3.0º = -3 %
- Each 0.1º above 3.0º = +4 %

# VREF, VAC, VGA

FLAP 4 - VREF, VAC, VGA PW1521 (VGA PW1524) | 115 121 139 (157)
FLAP 5 - VREF, VAC, VGA PW1521 (VGA PW1524) | 110 110 120 (132)

Example:

Landing weight = 54t
Pressure alt = 4000 ft
PW1524G, FLAP 4

Result:

- VREF = 135 kts
- VAC = 141 kts
- VGA = 151 kts

| Landing Speeds VREF – VAC - VGA | | | | | | |
|---|---|---|---|---|---|---|
| Landing Weight (kg) | Pressure Altitude (ft) | | | | | |
| | 0 | 2000 | 4000 | 6000 | 8000 | 10000 |
| 40000 | 115 121 139 (157)<br>110 110 120 (132) | 115 121 134 (149)<br>110 110 120 (127) | 115 121 132 (142)<br>110 110 120 (121) | 116 122 132 (135)<br>110 110 120 (120) | 116 122 132 (132)<br>110 110 120 (120) | 116 122 132 (132)<br>110 110 120 (120) |
| 42000 | 118 124 137 (151)<br>112 112 121 (129) | 118 124 135 (145)<br>112 112 121 (124) | 118 124 135 (139)<br>112 112 121 (121) | 119 125 135 (135)<br>112 112 122 (122) | 119 125 135 (135)<br>112 112 122 (122) | 119 125 135 (135)<br>112 112 122 (122) |
| 44000 | 121 127 137 (146)<br>114 114 124 (129) | 121 127 137 (142)<br>114 114 124 (124) | 121 127 138 (138)<br>114 114 124 (121) | 122 128 138 (138)<br>114 114 124 (122) | 122 128 138 (138)<br>114 114 124 (122) | 122 128 138 (138)<br>114 114 124 (122) |
| 46000 | 124 130 140 (143)<br>117 117 127 (127) | 124 130 140 (141)<br>117 117 127 (127) | 124 130 140 (140)<br>117 117 127 (127) | 125 131 141 (141)<br>117 117 127 (127) | 125 131 141 (141)<br>117 117 127 (127) | 125 131 141 (141)<br>117 117 127 (127) |
| 48000 | 126 132 143 (143)<br>119 119 130 (130) | 127 133 143 (143)<br>119 119 130 (130) | 127 133 143 (143)<br>119 119 130 (130) | 128 134 144 (144)<br>119 119 130 (130) | 128 134 144 (144)<br>119 119 130 (130) | 128 134 144 (144)<br>119 119 130 (130) |
| 50000 | 129 135 145 (145)<br>122 122 132 (132) | 130 136 146 (146)<br>122 122 132 (132) | 130 136 146 (146)<br>122 122 132 (132) | 130 136 146 (146)<br>122 122 132 (132) | 131 137 147 (147)<br>122 122 132 (132) | 131 137 147 (147)<br>122 122 132 (132) |
| 52000 | 132 138 148 (148)<br>124 124 134 (134) | 132 138 149 (149)<br>124 124 134 (134) | 132 138 149 (149)<br>124 124 134 (134) | 133 139 149 (149)<br>125 125 135 (135) | 134 140 150 (150)<br>125 125 135 (135) | 134 140 150 (150)<br>125 125 135 (135) |
| 54000 | 135 141 151 (151)<br>126 126 137 (137) | 135 141 151 (151)<br>127 127 136 (136) | 135 141 151 (151)<br>127 127 136 (136) | 136 142 152 (152)<br>127 127 137 (137) | 136 142 152 (152)<br>128 128 137 (137) | 137 143 153 (153)<br>128 128 137 (137) |
| 56000 | 137 143 154 (154)<br>128 128 139 (139) | 138 144 154 (154)<br>129 129 139 (139) | 138 144 154 (154)<br>129 129 139 (139) | 139 145 155 (155)<br>130 130 139 (139) | 139 145 155 (155)<br>130 130 140 (140) | 139 145 156 (156)<br>130 130 140 (140) |
| 58000 | 140 146 156 (156)<br>131 131 141 (141) | 140 146 156 (156)<br>131 131 141 (141) | 141 147 156 (156)<br>131 131 141 (141) | 141 147 157 (157)<br>132 132 141 (141) | 142 148 157 (157)<br>132 132 142 (142) | 142 148 158 (158)<br>132 132 142 (142) |
| 58740 - MSLW | 141 147 157 (157)<br>132 132 142 (142) | 141 147 157 (157)<br>132 132 142 (142) | 142 148 157 (157)<br>132 132 142 (142) | 142 148 158 (158)<br>133 133 142 (142) | 143 149 158 (158)<br>133 133 143 (143) | 143 149 159 (159)<br>133 133 143 (143) |
| 60000 | 142 148 158 (158)<br>133 133 143 (143) | 143 149 159 (159)<br>133 133 143 (143) | 143 149 159 (159)<br>133 133 143 (143) | 144 150 160 (160)<br>134 134 144 (144) | 144 150 160 (160)<br>134 134 144 (144) | 144 150 161 (161)<br>134 134 145 (145) |
| 62000 | 144 150 160 (160)<br>135 135 145 (145) | 145 151 161 (161)<br>136 136 145 (145) | 146 152 161 (161)<br>136 136 145 (145) | 146 152 162 (162)<br>136 136 146 (146) | 147 153 162 (162)<br>137 137 146 (146) | 147 153 163 (163)<br>137 137 147 (147) |
| 64000 | 146 152 163 (163)<br>137 137 147 (147) | 147 153 163 (163)<br>138 138 147 (147) | 148 154 164 (164)<br>138 138 148 (148) | 149 155 164 (164)<br>139 139 148 (148) | 149 155 165 (165)<br>139 139 149 (149) | 149 155 165 (165)<br>139 139 149 (149) |
| 66000 | 149 155 165 (165)<br>140 140 149 (149) | 150 156 165 (165)<br>140 140 149 (149) | 150 156 166 (166)<br>140 140 150 (150) | 151 157 166 (166)<br>141 141 150 (150) | 151 157 167 (167)<br>141 141 151 (151) | 152 158 167 (167)<br>142 142 151 (151) |
| 68000 | 151 157 168 (168)<br>142 142 152 (152) | 152 158 168 (168)<br>142 142 152 (152) | 152 158 169 (169)<br>142 142 153 (153) | 153 159 169 (169)<br>143 143 153 (153) | 154 160 170 (170)<br>143 143 154 (154) | 154 160 170 (170)<br>144 144 154 (154) |
| 70000 | 153 159 170 (170)<br>144 144 154 (154) | 154 160 170 (170)<br>144 144 154 (154) | 154 160 171 (171)<br>144 144 155 (155) | 155 161 171 (171)<br>145 145 155 (155) | 156 162 172 (172)<br>145 145 156 (156) | 156 162 172 (172)<br>146 146 156 (156) |
| 70900 - MTOW | 154 160 170 (170)<br>144 144 154 (154) | 155 161 171 (171)<br>145 145 154 (154) | 155 161 171 (171)<br>145 145 155 (155) | 156 162 172 (172)<br>146 146 155 (155) | 157 163 172 (172)<br>146 146 156 (156) | 157 163 173 (173)<br>146 146 156 (156) |

Overweight landing zone

# FLAP 4 - Operational Landing Distance (OLD)

Example:
- Landing weight = 55t
- LDA = 2000 m, wet
- Pressure alt = 2000 ft

Result (including 15% operational factor):
- Autobrake HIGH = 1440 x 1.15 = 1656 m
- Autobrake MED = 1645 x 1.15 = 1892 m
- Autobrake LOW = 2105 x 1.15 = 2421 m

**FLAP 4 - Operational Landing Distance (m)**

| Weight (kg) | Auto Brake | Pressure Altitude (ft) – Runway Condition Code (RWYCC) | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | SL | | | | | | 2000 | | | | | | 4000 | | | | | | 6000 | | | | | | 8000 | | | | | | 10000 | | | | | |
| | | Dry | Wet | Contaminated | | | | Dry | Wet | Contaminated | | | | Dry | Wet | Contaminated | | | | Dry | Wet | Contaminated | | | | Dry | Wet | Contaminated | | | | Dry | Wet | Contaminated | | | |
| | | 6 | 5 | 4 | 3 | 2 | 1 | 6 | 5 | 4 | 3 | 2 | 1 | 6 | 5 | 4 | 3 | 2 | 1 | 6 | 5 | 4 | 3 | 2 | 1 | 6 | 5 | 4 | 3 | 2 | 1 | 6 | 5 | 4 | 3 | 2 | 1 |
| 40000 | HIGH | 920 | 1045 | 1190 | 1270 | 1365 | 1575 | 955 | 1095 | 1240 | 1330 | 1435 | 1650 | 1000 | 1150 | 1295 | 1390 | 1510 | 1735 | 1045 | 1210 | 1355 | 1455 | 1595 | 1825 | 1095 | 1275 | 1420 | 1530 | 1690 | 1925 | 1150 | 1350 | 1495 | 1610 | 1800 | 2035 |
| | MED | 1215 | 1215 | 1275 | 1320 | 1365 | 1570 | 1270 | 1275 | 1335 | 1380 | 1435 | 1650 | 1330 | 1335 | 1400 | 1445 | 1510 | 1735 | 1395 | 1400 | 1465 | 1515 | 1590 | 1825 | 1465 | 1470 | 1540 | 1590 | 1680 | 1925 | 1545 | 1550 | 1620 | 1680 | 1790 | 2035 |
| | LOW | 1530 | 1530 | 1530 | 1530 | 1535 | 1695 | 1605 | 1605 | 1605 | 1605 | 1610 | 1780 | 1685 | 1685 | 1685 | 1690 | 1690 | 1875 | 1770 | 1770 | 1770 | 1775 | 1780 | 1970 | 1865 | 1865 | 1865 | 1870 | 1880 | 2080 | 1975 | 1975 | 1975 | 1980 | 1995 | 2205 |
| 45000 | HIGH | 1005 | 1150 | 1295 | 1505 | 1505 | 1725 | 1050 | 1205 | 1350 | 1450 | 1585 | 1810 | 1095 | 1270 | 1410 | 1515 | 1675 | 1905 | 1150 | 1340 | 1480 | 1595 | 1780 | 2005 | 1205 | 1420 | 1555 | 1680 | 1890 | 2120 | 1270 | 1505 | 1640 | 1770 | 2015 | 2245 |
| | MED | 1330 | 1335 | 1395 | 1440 | 1500 | 1720 | 1395 | 1395 | 1460 | 1505 | 1580 | 1810 | 1460 | 1465 | 1530 | 1580 | 1665 | 1900 | 1535 | 1540 | 1610 | 1660 | 1770 | 2005 | 1615 | 1625 | 1695 | 1750 | 1885 | 2120 | 1705 | 1715 | 1790 | 1850 | 2005 | 2245 |
| | LOW | 1685 | 1685 | 1685 | 1690 | 1690 | 1860 | 1605 | 1605 | 1605 | 1605 | 1610 | 1780 | 1685 | 1685 | 1685 | 1690 | 1690 | 1875 | 1770 | 1770 | 1770 | 1775 | 1780 | 1970 | 1865 | 1865 | 1865 | 1870 | 1880 | 2080 | 1975 | 1975 | 1975 | 1980 | 1995 | 2205 |
| 50000 | HIGH | 1090 | 1250 | 1395 | 1500 | 1650 | 1870 | 1140 | 1320 | 1460 | 1570 | 1745 | 1970 | 1195 | 1395 | 1530 | 1650 | 1855 | 2075 | 1255 | 1475 | 1610 | 1735 | 1970 | 2195 | 1320 | 1565 | 1690 | 1825 | 2095 | 2320 | 1385 | 1665 | 1785 | 1930 | 2235 | 2460 |
| | MED | 1445 | 1450 | 1515 | 1560 | 1645 | 1870 | 1515 | 1520 | 1585 | 1635 | 1740 | 1970 | 1595 | 1600 | 1670 | 1720 | 1845 | 2075 | 1680 | 1685 | 1755 | 1810 | 1960 | 2195 | 1770 | 1775 | 1850 | 1910 | 2085 | 2320 | 1865 | 1875 | 1955 | 2015 | 2225 | 2460 |
| | LOW | 1840 | 1840 | 1840 | 1845 | 1850 | 2025 | 1935 | 1935 | 1935 | 1940 | 1950 | 2130 | 2040 | 2040 | 2040 | 2045 | 2060 | 2250 | 2155 | 2155 | 2155 | 2160 | 2175 | 2380 | 2280 | 2280 | 2280 | 2285 | 2305 | 2525 | 2415 | 2415 | 2415 | 2420 | 2445 | 2675 |
| 55000 | HIGH | 1175 | 1360 | 1500 | 1615 | 1805 | 2025 | 1230 | 1440 | 1575 | 1695 | 1915 | 2135 | 1290 | 1520 | 1650 | 1780 | 2030 | 2250 | 1350 | 1615 | 1735 | 1875 | 2160 | 2380 | 1420 | 1715 | 1830 | 1975 | 2300 | 2520 | 1495 | 1825 | 1925 | 2085 | 2455 | 2670 |
| | MED | 1560 | 1565 | 1635 | 1680 | 1800 | 2025 | 1640 | 1645 | 1715 | 1765 | 1905 | 2135 | 1725 | 1735 | 1805 | 1860 | 2025 | 2250 | 1820 | 1830 | 1905 | 1960 | 2155 | 2380 | 1920 | 1930 | 2005 | 2070 | 2295 | 2520 | 2025 | 2040 | 2120 | 2185 | 2445 | 2675 |
| | LOW | 2000 | 2000 | 2000 | 2005 | 2015 | 2190 | 2105 | 2105 | 2105 | 2110 | 2125 | 2315 | 2225 | 2225 | 2225 | 2230 | 2245 | 2445 | 2350 | 2350 | 2350 | 2360 | 2375 | 2590 | 2490 | 2490 | 2490 | 2495 | 2515 | 2745 | 2640 | 2640 | 2640 | 2645 | 2670 | 2915 |
| 58740 MSLW | HIGH | 1235 | 1445 | 1580 | 1700 | 1925 | 2140 | 1295 | 1530 | 1655 | 1785 | 2040 | 2255 | 1355 | 1620 | 1740 | 1880 | 2165 | 2380 | 1420 | 1720 | 1830 | 1980 | 2305 | 2520 | 1495 | 1830 | 1925 | 2085 | 2455 | 2670 | 1570 | 1950 | 2030 | 2205 | 2620 | 2830 |
| | MED | 1650 | 1655 | 1720 | 1770 | 1915 | 2140 | 1730 | 1740 | 1810 | 1860 | 2030 | 2255 | 1825 | 1830 | 1905 | 1960 | 2160 | 2380 | 1920 | 1930 | 2010 | 2070 | 2295 | 2520 | 2030 | 2040 | 2120 | 2185 | 2445 | 2670 | 2145 | 2155 | 2240 | 2310 | 2610 | 2835 |
| | LOW | 2115 | 2115 | 2115 | 2120 | 2130 | 2315 | 2230 | 2230 | 2230 | 2235 | 2250 | 2445 | 2360 | 2360 | 2360 | 2365 | 2380 | 2585 | 2495 | 2495 | 2495 | 2500 | 2520 | 2740 | 2640 | 2640 | 2640 | 2645 | 2670 | 2910 | 2800 | 2800 | 2800 | 2805 | 2830 | 3085 |
| 60000 | HIGH | 1255 | 1470 | 1605 | 1730 | 1960 | 2175 | 1315 | 1555 | 1685 | 1815 | 2080 | 2295 | 1375 | 1650 | 1770 | 1910 | 2210 | 2425 | 1445 | 1750 | 1860 | 2010 | 2350 | 2565 | 1520 | 1865 | 1960 | 2125 | 2505 | 2715 | 1600 | 1990 | 2065 | 2240 | 2675 | 2880 |
| | MED | 1675 | 1680 | 1750 | 1800 | 1955 | 2175 | 1760 | 1770 | 1840 | 1895 | 2075 | 2295 | 1855 | 1865 | 1940 | 1995 | 2200 | 2425 | 1955 | 1965 | 2045 | 2105 | 2345 | 2565 | 2065 | 2075 | 2155 | 2225 | 2500 | 2720 | 2180 | 2195 | 2280 | 2350 | 2665 | 2885 |
| | LOW | 2155 | 2155 | 2155 | 2160 | 2170 | 2355 | 2275 | 2275 | 2275 | 2280 | 2290 | 2490 | 2400 | 2400 | 2400 | 2405 | 2425 | 2635 | 2540 | 2540 | 2540 | 2545 | 2565 | 2790 | 2690 | 2690 | 2690 | 2695 | 2720 | 2965 | 2855 | 2855 | 2855 | 2860 | 2885 | 3140 |
| 65000 | HIGH | 1330 | 1580 | 1705 | 1840 | 2115 | 2325 | 1390 | 1675 | 1790 | 1935 | 2245 | 2455 | 1460 | 1775 | 1880 | 2035 | 2385 | 2595 | 1535 | 1890 | 1980 | 2145 | 2540 | 2745 | 1615 | 2010 | 2085 | 2265 | 2705 | 2910 | 1610 | 2150 | 2200 | 2395 | 2895 | 3090 |
| | MED | 1785 | 1795 | 1865 | 1915 | 2105 | 2330 | 1880 | 1885 | 1960 | 2015 | 2235 | 2455 | 1980 | 1990 | 2065 | 2125 | 2375 | 2595 | 2085 | 2100 | 2180 | 2245 | 2530 | 2750 | 2205 | 2215 | 2300 | 2370 | 2700 | 2915 | 2330 | 2345 | 2430 | 2510 | 2885 | 3095 |
| | LOW | 2305 | 2305 | 2305 | 2310 | 2325 | 2515 | 2435 | 2435 | 2435 | 2440 | 2455 | 2660 | 2570 | 2570 | 2570 | 2580 | 2595 | 2815 | 2720 | 2720 | 2720 | 2730 | 2750 | 2985 | 2885 | 2885 | 2885 | 2890 | 2915 | 3165 | 3060 | 3060 | 3060 | 3070 | 3095 | 3365 |
| 70000 | HIGH | 1385 | 1670 | 1785 | 1925 | 2235 | 2445 | 1455 | 1770 | 1875 | 2025 | 2375 | 2580 | 1525 | 1880 | 1970 | 2135 | 2525 | 2730 | 1605 | 2000 | 2075 | 2250 | 2690 | 2890 | 1690 | 2130 | 2190 | 2375 | 2870 | 3065 | 1780 | 2280 | 2310 | 2515 | 3070 | 3255 |
| | MED | 1870 | 1880 | 1955 | 2010 | 2230 | 2445 | 1970 | 1980 | 2055 | 2115 | 2370 | 2585 | 2075 | 2090 | 2165 | 2230 | 2520 | 2730 | 2190 | 2205 | 2285 | 2355 | 2685 | 2895 | 2315 | 2330 | 2415 | 2490 | 2865 | 3070 | 2450 | 2470 | 2550 | 2635 | 3065 | 3260 |
| | LOW | 2425 | 2425 | 2425 | 2430 | 2445 | 2645 | 2560 | 2560 | 2560 | 2565 | 2580 | 2795 | 2705 | 2705 | 2705 | 2715 | 2730 | 2960 | 2865 | 2865 | 2865 | 2870 | 2895 | 3140 | 3040 | 3040 | 3040 | 3045 | 3070 | 3330 | 3225 | 3225 | 3225 | 3230 | 3265 | 3545 |
| 70900 MTOW | HIGH | 1400 | 1690 | 1805 | 1950 | 2270 | 2475 | 1470 | 1795 | 1895 | 2050 | 2410 | 2615 | 1545 | 1905 | 1995 | 2160 | 2560 | 2765 | 1620 | 2025 | 2100 | 2280 | 2730 | 2925 | 1705 | 2160 | 2215 | 2405 | 2915 | 3105 | 1800 | 2310 | 2335 | 2545 | 3115 | 3295 |
| | MED | 1895 | 1905 | 1975 | 2030 | 2260 | 2475 | 1995 | 2005 | 2080 | 2140 | 2400 | 2615 | 2100 | 2115 | 2190 | 2255 | 2555 | 2765 | 2215 | 2230 | 2310 | 2385 | 2720 | 2930 | 2340 | 2360 | 2440 | 2520 | 2905 | 3110 | 2480 | 2500 | 2580 | 2670 | 3110 | 3300 |
| | LOW | 2455 | 2455 | 2455 | 2460 | 2475 | 2675 | 2590 | 2590 | 2590 | 2600 | 2615 | 2830 | 2740 | 2740 | 2740 | 2745 | 2765 | 2995 | 2900 | 2900 | 2900 | 2910 | 2930 | 3180 | 3075 | 3075 | 3075 | 3085 | 3110 | 3375 | 3265 | 3265 | 3265 | 3275 | 3305 | 3590 |

Overweight landing zone

Manual landing:
- FLAP 4
- VREF
- Manual Landing
- 2 Thrust reversers
- ISA, wind calm
- Zero slope
- Not including operational factor (15%)

1. Corrections (m) – manual and autoland:

| RWYCC | VREF 5 kts | Tailwind 10 kts | Slope -1% | Slope +1% | Reversers 1 inop | Reversers 2 inop | ISA Deviation -10°C | ISA Deviation +10°C |
|---|---|---|---|---|---|---|---|---|
| 6 | 220 | 620 | 50 | 0 | 150 | 190 | -20 | 140 |
| 5 | 220 | 650 | 160 | 0 | 310 | 780 | -20 | 150 |
| 4 | 220 | 620 | 100 | 0 | 200 | 460 | -20 | 140 |
| 3 | 220 | 660 | 150 | 0 | 280 | 680 | -30 | 140 |
| 2 | 240 | 1170 | 480 | 0 | 740 | 2060 | -30 | 240 |
| 1 | 230 | 1100 | 580 | -40 | 740 | 2020 | -40 | 180 |

Autoland:
- FLAP 4
- VREF + 5 kts
- Autoland
- 2 Thrust reversers
- ISA, wind calm, zero slope
- ILS GS 3.00°, TCH 50 ft
- Not including operational factor (15%)

2. Corrections (m) – autoland specific:

| Autobrake (add to distance for man landing) | | ILS TCH and GS | | |
|---|---|---|---|---|
| | | TCH 50' | GS 3.00° | |
| | | +1 ft | -0.1° | +0.1° |
| HIGH | 530 | +7 | +14 | -8 |
| MED | 540 | | | |
| LOW | 620 | | | |

# FLAP 5 - Operational Landing Distance (OLD)

**FLAP 5 - Operational Landing Distance (m)**

| Weight (kg) | Auto Brake | Pressure Altitude (ft) – Runway Condition Code (RWYCC) | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | SL | | | | | | 2000 | | | | | | 4000 | | | | | | 6000 | | | | | | 8000 | | | | | | 10000 | | | | | |
| | | Dry | Wet | Contaminated | | | | Dry | Wet | Contaminated | | | | Dry | Wet | Contaminated | | | | Dry | Wet | Contaminated | | | | Dry | Wet | Contaminated | | | | Dry | Wet | Contaminated | | | |
| | | 6 | 5 | 4 | 3 | 2 | 1 | 6 | 5 | 4 | 3 | 2 | 1 | 6 | 5 | 4 | 3 | 2 | 1 | 6 | 5 | 4 | 3 | 2 | 1 | 6 | 5 | 4 | 3 | 2 | 1 | 6 | 5 | 4 | 3 | 2 | 1 |
| 40000 | HIGH | 860 | 970 | 1115 | 1190 | 1270 | 1475 | 895 | 1015 | 1165 | 1245 | 1335 | 1545 | 935 | 1065 | 1215 | 1300 | 1405 | 1620 | 975 | 1120 | 1270 | 1365 | 1480 | 1705 | 1020 | 1180 | 1330 | 1430 | 1560 | 1795 | 1070 | 1245 | 1390 | 1500 | 1650 | 1890 |
| | MED | 1130 | 1130 | 1190 | 1230 | 1270 | 1470 | 1180 | 1180 | 1240 | 1285 | 1335 | 1545 | 1235 | 1235 | 1300 | 1345 | 1400 | 1620 | 1295 | 1295 | 1360 | 1410 | 1475 | 1705 | 1355 | 1360 | 1425 | 1480 | 1555 | 1795 | 1425 | 1425 | 1495 | 1550 | 1645 | 1890 |
| | LOW | 1415 | 1415 | 1415 | 1415 | 1415 | 1570 | 1480 | 1480 | 1480 | 1485 | 1485 | 1650 | 1555 | 1555 | 1555 | 1560 | 1560 | 1735 | 1635 | 1635 | 1635 | 1640 | 1640 | 1825 | 1720 | 1720 | 1720 | 1720 | 1730 | 1925 | 1810 | 1810 | 1810 | 1815 | 1820 | 2030 |
| 45000 | HIGH | 920 | 1040 | 1190 | 1275 | 1370 | 1585 | 960 | 1095 | 1245 | 1335 | 1445 | 1665 | 1000 | 1150 | 1300 | 1395 | 1520 | 1750 | 1050 | 1210 | 1360 | 1460 | 1605 | 1840 | 1100 | 1275 | 1425 | 1535 | 1700 | 1940 | 1100 | 1275 | 1425 | 1535 | 1700 | 1940 |
| | MED | 1210 | 1210 | 1270 | 1315 | 1370 | 1585 | 1265 | 1265 | 1330 | 1375 | 1440 | 1665 | 1325 | 1325 | 1395 | 1440 | 1520 | 1750 | 1390 | 1390 | 1460 | 1510 | 1600 | 1840 | 1455 | 1460 | 1530 | 1585 | 1690 | 1940 | 1530 | 1535 | 1610 | 1665 | 1795 | 2045 |
| | LOW | 1520 | 1520 | 1520 | 1525 | 1525 | 1690 | 1595 | 1595 | 1595 | 1600 | 1600 | 1775 | 1675 | 1675 | 1675 | 1680 | 1685 | 1870 | 1760 | 1760 | 1760 | 1765 | 1770 | 1965 | 1855 | 1855 | 1855 | 1860 | 1865 | 2075 | 1955 | 1955 | 1955 | 1960 | 1975 | 2190 |
| 50000 | HIGH | 995 | 1135 | 1285 | 1375 | 1495 | 1720 | 1040 | 1195 | 1340 | 1440 | 1575 | 1805 | 1085 | 1255 | 1405 | 1510 | 1665 | 1900 | 1135 | 1320 | 1470 | 1580 | 1765 | 2000 | 1195 | 1395 | 1540 | 1665 | 1875 | 2110 | 1255 | 1485 | 1620 | 1755 | 2000 | 2235 |
| | MED | 1310 | 1315 | 1375 | 1420 | 1495 | 1720 | 1375 | 1375 | 1440 | 1490 | 1575 | 1805 | 1440 | 1440 | 1510 | 1560 | 1660 | 1900 | 1510 | 1510 | 1585 | 1635 | 1755 | 2000 | 1585 | 1590 | 1665 | 1720 | 1865 | 2110 | 1675 | 1680 | 1755 | 1815 | 1990 | 2235 |
| | LOW | 1660 | 1660 | 1660 | 1660 | 1665 | 1835 | 1740 | 1740 | 1740 | 1745 | 1750 | 1930 | 1830 | 1830 | 1830 | 1835 | 1840 | 2035 | 1925 | 1925 | 1925 | 1930 | 1945 | 2140 | 2030 | 2030 | 2030 | 2035 | 2050 | 2260 | 2150 | 2150 | 2150 | 2155 | 2175 | 2400 |
| 55000 | HIGH | 1070 | 1230 | 1375 | 1480 | 1625 | 1855 | 1120 | 1290 | 1440 | 1550 | 1720 | 1950 | 1170 | 1360 | 1505 | 1620 | 1820 | 2050 | 1230 | 1440 | 1580 | 1705 | 1935 | 2165 | 1290 | 1530 | 1665 | 1795 | 2060 | 2290 | 1355 | 1625 | 1750 | 1895 | 2195 | 2430 |
| | MED | 1415 | 1415 | 1480 | 1525 | 1625 | 1855 | 1480 | 1485 | 1550 | 1600 | 1710 | 1950 | 1550 | 1555 | 1625 | 1675 | 1810 | 2050 | 1630 | 1635 | 1710 | 1765 | 1925 | 2165 | 1720 | 1725 | 1805 | 1865 | 2050 | 2290 | 1815 | 1825 | 1905 | 1970 | 2190 | 2430 |
| | LOW | 1795 | 1795 | 1795 | 1800 | 1805 | 1980 | 1885 | 1885 | 1885 | 1890 | 1900 | 2085 | 1985 | 1985 | 1985 | 1985 | 2000 | 2195 | 2095 | 2095 | 2095 | 2095 | 2115 | 2320 | 2215 | 2215 | 2215 | 2220 | 2240 | 2460 | 2345 | 2345 | 2345 | 2350 | 2375 | 2610 |
| 58740 MSLW | HIGH | 1125 | 1300 | 1445 | 1555 | 1725 | 1955 | 1175 | 1365 | 1510 | 1625 | 1825 | 2055 | 1230 | 1445 | 1585 | 1710 | 1935 | 2165 | 1290 | 1530 | 1665 | 1800 | 2060 | 2290 | 1355 | 1625 | 1750 | 1895 | 2195 | 2425 | 1425 | 1735 | 1845 | 2000 | 2345 | 2570 |
| | MED | 1485 | 1490 | 1555 | 1605 | 1720 | 1955 | 1555 | 1560 | 1630 | 1680 | 1820 | 2055 | 1635 | 1640 | 1710 | 1765 | 1930 | 2165 | 1720 | 1730 | 1805 | 1860 | 2055 | 2290 | 1815 | 1825 | 1900 | 1965 | 2190 | 2425 | 1920 | 1930 | 2010 | 2080 | 2335 | 2575 |
| | LOW | 1895 | 1895 | 1895 | 1900 | 1910 | 2085 | 1990 | 1990 | 1990 | 1995 | 2005 | 2195 | 2100 | 2100 | 2100 | 2105 | 2115 | 2320 | 2220 | 2220 | 2220 | 2220 | 2240 | 2455 | 2350 | 2350 | 2350 | 2355 | 2375 | 2600 | 2490 | 2490 | 2490 | 2495 | 2520 | 2760 |
| 60000 | HIGH | 1145 | 1320 | 1465 | 1575 | 1760 | 1985 | 1195 | 1390 | 1530 | 1650 | 1860 | 2090 | 1250 | 1470 | 1610 | 1735 | 1975 | 2205 | 1310 | 1560 | 1690 | 1830 | 2105 | 2330 | 1380 | 1660 | 1780 | 1930 | 2240 | 2470 | 1450 | 1770 | 1880 | 2035 | 2390 | 2615 |
| | MED | 1510 | 1515 | 1580 | 1630 | 1755 | 1985 | 1580 | 1585 | 1655 | 1705 | 1855 | 2090 | 1660 | 1670 | 1740 | 1795 | 1970 | 2205 | 1750 | 1760 | 1835 | 1895 | 2095 | 2330 | 1850 | 1855 | 1935 | 2000 | 2235 | 2470 | 1955 | 1965 | 2045 | 2115 | 2385 | 2620 |
| | LOW | 1925 | 1925 | 1925 | 1930 | 1940 | 2120 | 2025 | 2025 | 2025 | 2030 | 2040 | 2235 | 2135 | 2135 | 2135 | 2140 | 2155 | 2360 | 2260 | 2260 | 2260 | 2265 | 2280 | 2500 | 2390 | 2390 | 2390 | 2400 | 2420 | 2650 | 2535 | 2535 | 2535 | 2540 | 2565 | 2815 |
| 65000 | HIGH | 1210 | 1410 | 1550 | 1670 | 1890 | 2115 | 1265 | 1490 | 1625 | 1755 | 2005 | 2230 | 1325 | 1580 | 1710 | 1850 | 2130 | 2355 | 1390 | 1680 | 1800 | 1950 | 2270 | 2490 | 1465 | 1790 | 1895 | 2055 | 2420 | 2640 | 1540 | 1905 | 2000 | 2170 | 2585 | 2800 |
| | MED | 1600 | 1605 | 1675 | 1725 | 1885 | 2115 | 1680 | 1690 | 1760 | 1815 | 2000 | 2230 | 1770 | 1780 | 1855 | 1915 | 2125 | 2355 | 1870 | 1875 | 1955 | 2020 | 2260 | 2495 | 1970 | 1985 | 2065 | 2130 | 2415 | 2645 | 2085 | 2095 | 2180 | 2255 | 2580 | 2805 |
| | LOW | 2050 | 2050 | 2050 | 2055 | 2070 | 2255 | 2165 | 2165 | 2165 | 2170 | 2180 | 2380 | 2285 | 2285 | 2285 | 2290 | 2310 | 2520 | 2420 | 2420 | 2420 | 2425 | 2445 | 2670 | 2565 | 2565 | 2565 | 2570 | 2590 | 2835 | 2720 | 2720 | 2720 | 2725 | 2755 | 3010 |
| 70000 | HIGH | 1260 | 1485 | 1620 | 1750 | 2000 | 2220 | 1320 | 1575 | 1700 | 1840 | 2120 | 2345 | 1385 | 1670 | 1790 | 1935 | 2255 | 2475 | 1455 | 1775 | 1885 | 2040 | 2405 | 2620 | 1530 | 1890 | 1985 | 2155 | 2565 | 2780 | 1610 | 2020 | 2095 | 2280 | 2740 | 2950 |
| | MED | 1675 | 1685 | 1755 | 1805 | 1990 | 2220 | 1765 | 1770 | 1845 | 1905 | 2115 | 2345 | 1860 | 1870 | 1945 | 2005 | 2250 | 2480 | 1960 | 1970 | 2050 | 2115 | 2395 | 2625 | 2070 | 2085 | 2165 | 2240 | 2560 | 2780 | 2190 | 2205 | 2290 | 2370 | 2735 | 2955 |
| | LOW | 2155 | 2155 | 2155 | 2160 | 2175 | 2365 | 2275 | 2275 | 2275 | 2280 | 2295 | 2505 | 2405 | 2405 | 2405 | 2410 | 2430 | 2650 | 2545 | 2545 | 2545 | 2555 | 2570 | 2810 | 2700 | 2700 | 2700 | 2705 | 2730 | 2980 | 2865 | 2865 | 2865 | 2870 | 2900 | 3170 |
| 70900 MTOW | HIGH | 1275 | 1505 | 1640 | 1770 | 2025 | 2245 | 1335 | 1595 | 1720 | 1860 | 2150 | 2370 | 1400 | 1695 | 1810 | 1960 | 2290 | 2505 | 1470 | 1800 | 1905 | 2065 | 2440 | 2655 | 1545 | 1920 | 2010 | 2180 | 2600 | 2815 | 1630 | 2050 | 2120 | 2310 | 2780 | 2985 |
| | MED | 1695 | 1700 | 1775 | 1830 | 2020 | 2250 | 1785 | 1795 | 1865 | 1925 | 2145 | 2375 | 1880 | 1890 | 1965 | 2030 | 2280 | 2510 | 1985 | 1995 | 2075 | 2140 | 2430 | 2655 | 2095 | 2110 | 2190 | 2265 | 2595 | 2815 | 2215 | 2235 | 2315 | 2400 | 2775 | 2990 |
| | LOW | 2180 | 2180 | 2180 | 2185 | 2200 | 2395 | 2305 | 2305 | 2305 | 2310 | 2325 | 2535 | 2435 | 2435 | 2435 | 2440 | 2460 | 2680 | 2580 | 2580 | 2580 | 2585 | 2605 | 2845 | 2735 | 2735 | 2735 | 2740 | 2765 | 3020 | 2900 | 2900 | 2900 | 2910 | 2940 | 3210 |

Overweight landing zone

Manual landing:

- FLAP 5
- VREF
- Manual Landing
- 2 Thrust reversers
- ISA, wind calm
- Zero slope
- Not including operational factor (15%)

1. Corrections (m) – manual and autoland:

| RWYCC | VREF 5 kts | Tailwind 10 kts | Slope -1% | Slope +1% | Reversers 1 inop | Reversers 2 inop | ISA Deviation -10°C | ISA Deviation +10°C |
|---|---|---|---|---|---|---|---|---|
| 6 | 210 | 590 | 40 | 0 | 120 | 120 | -20 | 120 |
| 5 | 210 | 590 | 120 | 0 | 250 | 610 | -20 | 130 |
| 4 | 210 | 590 | 90 | 0 | 170 | 390 | -20 | 120 |
| 3 | 210 | 630 | 140 | 0 | 240 | 570 | -30 | 140 |
| 2 | 240 | 1080 | 400 | 0 | 610 | 1720 | -30 | 220 |
| 1 | 210 | 1050 | 530 | -30 | 650 | 1790 | -30 | 180 |

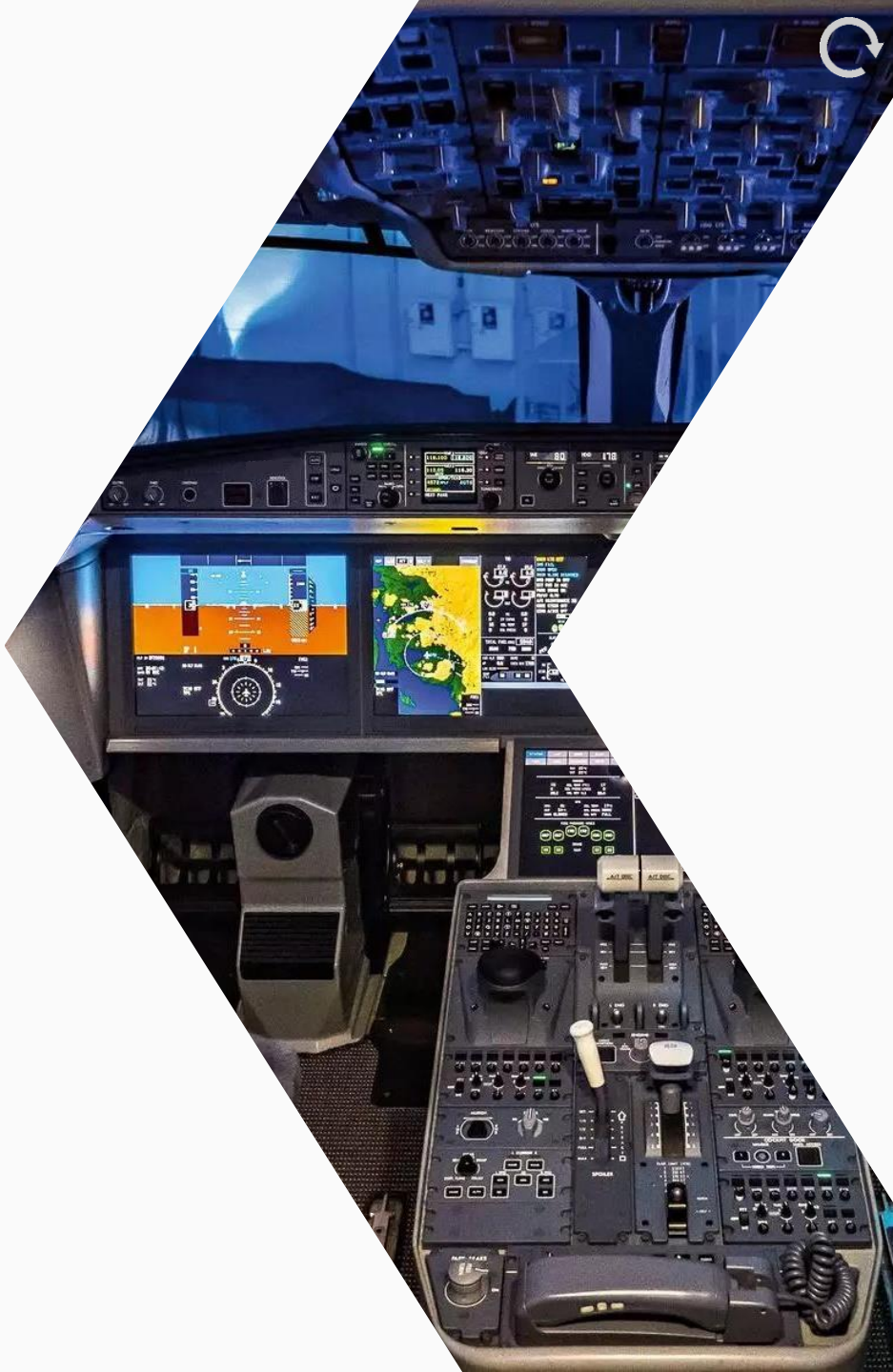
Autoland:

- FLAP 5
- VREF + 5 kts
- Autoland
- 2 Thrust reversers
- ISA, wind calm, zero slope
- ILS GS 3.00°, TCH 50 ft
- Not including operational factor (15%)

2. Corrections (m) – autoland specific:

| Autobrake (add to distance for man landing) | | ILS TCH and GS | | |
|---|---|---|---|---|
| | | TCH 50' | GS 3.00° | |
| | | +1 ft | -0.1° | +0.1° |
| HIGH | 490 | | | |
| MED | 390 | +7 | +14 | -8 |
| LOW | 590 | | | |

# Quick Reference



# Appendix 1 – Abbreviated Normal Procedure

### Power-on

- Steps a. to d. done by memory:
  - a. PARK BRAKE..... ON
  - b. BATT 1..... AUTO
  - c. BATT 2..... AUTO
  - d. ECL..... DU 2
- Complete POWER ON CHECKLIST.
- Establish AC power (GPU or start APU).

### Exterior Inspection

- NAV (LOGO + WING at night).....ON
- Set-up fueling / defueling panel, perform walkaround.
- FDRAS (cockpit door) test when returning to cockpit (♦).
- Set exterior lights as required.

### Preflight

- Documents, OEBs, emergency equipment, gear pins (3)
- Perform panel flow (all switches auto – middle position):

Overhead panel:

- PROBE HEAT.....GND ON (if required)
- FWD CARGO.....LO or HI HEAT (sensitive cargo)
- EMER LTS..... ARM
- EXT LTS.....NAV (LOGO, WING) ON
- Cabin signs.....As required

Control Tuning Panel:

- RANGE, NAV SRC, BARO.....Set
- "V-flow" (MAP – SYN – CNS):
  - a. MAP: set OVLY, select VSD, SYMBOLs
  - b. SYN: cycle and check synoptic pages:
    1. FLT CTRL: perform sidestick test
    2. AVIONIC: perform ICE DETECT test (♦)
    3. CB: check Circuit Breaker status (♦)
  - c. CNS: COMs, NAVs, XPDR 1 SEL, set flight ID.

Flight Control Panel:

- SPD..... Set green dot, FMS SPD
- XFR..... PF side
- ALT.....Initial altitude

Displays:

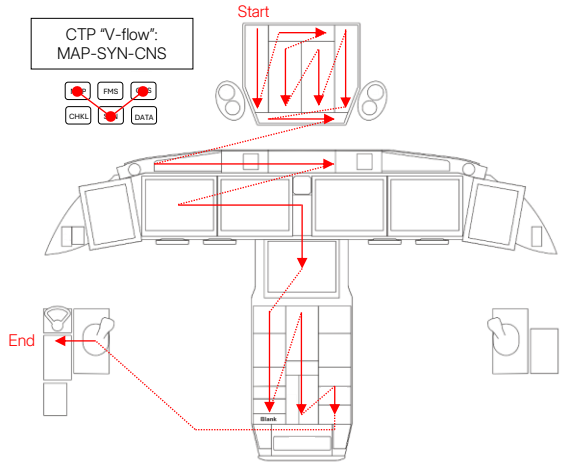
- No flags
- EICAS and INFO..... Checked
- Altimeters.....( ) Set and crosschecked

Center Panel:

- Autobrake..... RTO

Pedestal:

- ACP.....As required
- Set SPKR and INT on, perform oxygen mask test (♦)
- Rudder trim checked (♦) & centered, FDRAS test (♦)




### Cockpit Preparation

ACARS, ATIS and CLX:

- ACARS INIT: CNS – DLK – AOC – PREFLIGHT – INIT
- ATIS and en-route clearance (using datalink or voice):
  - a. Set QNH, squawk code, cleared altitude.
  - b. Take-off performance calculation.

FMS set-up flow:

- DBASE: check database version and validity.
- POS – FMS, IRS, GNSS, VOR/DME: check sensor status.
- FPLN:
  - a. INIT: open SEC FPLN, FPLN UPLINK, enter empty fields, ACTIVATE SEC enter FLT NO.
  - b. WIND/TEMP: upload by selecting WIND REQ or enter manually (enter route first if no uplink)
  - c. FUEL: ZFW, CG, PAX, FOB and TAXI fuel.
- PERF:
  - a. DEP: thrust, flaps, bleed, anti-icing, NADP, enter speeds and select SET VSPEEDS last. **Press TOGA, NAV, VNAV and set RWY HDG.**
  - b. CLB, CRZ, DES: speed profile and limits.
- ROUTE:
  - a. VIA/TO: enter route (if no FPLN UPLINK).
  - b. LEGS: enter fixes for reference (e.g. OEI).

CNS and CTP:

- Set BARO minima to OEI ACCEL ALT.
- Set COM and NAV frequencies, BRG 1/2 pointers.

---

PREFLIGHT CHECKLIST and DEPARTURE BRIEFING

---

### Before Start

- Start APU in accordance with local regulations.
- Set final ZFW, MAC, PAX and crosscheck with load sheet.
- Recalculate take-off performance and SET VSPEEDS.

Doors closed, slides armed, ground ready, tug connected:

- XPDR.....AUTO
- Request pushback and / or start-up clearance

---

BEFORE START CHECKLIST

---

- Pushback (if required), release parking brake.

### Engine Start

- ENG 1 (2) RUN switch.....ON
- Start timing, monitor start sequence.
- Start next engine when START flag disappears.

### Before Taxi

When both engines stabilized:

---

BEFORE TAXI CHECKLIST
Wing anti-ice test if required (♦♦)
Flight control check (full range)

---

Request taxi clearance:

- Left and right-side clear
- PARK BRAKE.....OFF
- TAXI LIGHT.....NARROW or WIDE

### Taxi

- Let taxi speed increase to 30 kts GS, then brake to 15 kts.
- PARK BRAKE ON if holding is expected to exceed 1 min.

When cabin READY:

---

BEFORE TAKE-OFF CHECKLIST

---

### Before Take-off

Cleared for line-up:

- Left and right-side clear
- STROBE, TAXI, LDG LTS L & R..... ON / WIDE
- SEATBELTS..... Cycle OFF, ON
- WXR..... ON (if required)

Take-off clearance received:

- NOSE LDG LT..... ON

### Take-off

When lined-up:

- Heading..... VERIFY
- Ice shedding..... As required
- Brakes.....As required
- Autothrottle..... ARM

Initiating take-off:

- Advance thrust levers, stabilize at 55% N1
- Wait until 20 kts GS (high wind take-off)
- Advance thrust levers for take-off

Take-off roll:

- Check take-off thrust set and speed at 80 kts
- Hand off thrust levers at $V_1$ (auto-callout)

Rotate at $V_R$:

- Rotate with 3 to 5° per second towards the PTM and transition to the FD when it becomes available.
- Retract gear when positive rate of climb.

At 200 ft AGL:

- FMS 1(2) – if NAV armed on ground.

At 400 ft AGL:

- Select lateral mode – if NAV not armed on ground.
- Autopilot.....As desired

At acceleration altitude:

- Confirm VFLC (if VNAV armed on ground) or select FLC.
- Retract flaps on schedule.

When flaps are up and above green dot:

---

AFTER TAKEOFF CHECKLIST

---

### Climb

***Note:*** *wing anti-ice check could be performed in flight instead of BEFORE TAXI (once each day) below 31.000 ft and TAT below 15°C by selecting ANTI-ICE, WING to ON for 3 minutes.*

At transition altitude (latest):

- BARO.....STD

Passing 10.000 ft:

- LOGO, WING, TAXI, LDG LTS (3).....OFF
- SEATBELTS..... AUTO

Before 35.000 ft:

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HIGH ALTITUDE CLIMB CHECKLIST

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Approaching cleared level:

- Vertical rate max 2000 ft/min, 2000 ft prior level-off.
- Vertical rate max 1000 ft/min, 1000 ft prior level-off.

# Appendix 1 – Abbreviated Normal Procedure

## Cruise

ATIS (voice or datalink):
- ISI..... Set QNH
- Landing performance.....Compute

FMS set-up flow:
- POS – FMS: check RAIM availability.
- PERF – DES: set transition level.
- PERF – ARR:
  a. ARRIVALS: select arrival, runway and approach type, arrival transitions.
  b. ARRIVAL DATA: select RNP or LPV.
  c. TEMP COMP: set OAT if 0°C or below, correct approach and arrival altitudes.
  d. OAT, RWY WIND, QNH: enter values.
  e. SLAT / FLAP: set landing flap position.
  f. VREF: add 5 kts if using autothrottle for landing + ½ gust, combined max 10 kts.
  g. SET VSPEEDS: select.

CTP (C-F-M):
- Course: set final approach course
- Frequencies: set (but do NOT set ILS frequency active)
- Minima: set (temperature corrected) DA or MDA + 50 ft

---

ARRIVAL BRIEFING, then
DESCENT & APPROACH CHECKLIST

---

## Descent Planning

Determine VPA for descent:

a. Weight - landing weight 55t = 3.0°:
  Each 5t below 55t = +0.2°
  Each 5t above 55t = -0.1°
b. ISA deviation:
  Each 10°C below ISA = +0.1°
  Each 10°C above ISA = -0.1°
c. Wind (average):
  Each 10 kts headwind = +0.1°
  Each 10 kts tailwind = -0.1°
d. Anti-icing use (if anticipated) = -0.2°

Descent Planning:

a. On the PERF – DES page, enter the calculated VPA.
b. Plan for the shortest possible route (ATC shortcuts).
c. Determine direct distances using FUEL CALCULATOR (can be found on the on the PERF – CRZ page).
d. On the ROUTE – LEGS page, update altitudes for first arrival point / IAF / IF to get an accurate TOD prediction.

## Descent

Approaching TOD:
- VNAV..... OFF
- ALT SEL..... Set cleared level
- FLC..... Select

Below 35.000 ft:

---

Complete DESCENT & APPROACH CHECKLIST
ANTI-ICE, WING.....AUTO

---

Passing 10.000 ft:
- LOGO & WING.....As required
- TAXI, LDG LTS L & R.....WIDE, ON
- SEATBELTS.....ON

At transition level (latest):
- BARO..... QNH

Approaching cleared altitude:
- Vertical rate max 2000 ft/min, 2000 ft prior level-off.
- Vertical rate max 1000 ft/min, 1000 ft prior level-off.

## Energy Management

At TOD:
- Using FLC at the TOD, the aircraft will more or less follow the VNAV profile until MACH – IAS transition point.
- The aircraft will then descent below path until 2000 ft below (TOD 6 nm ahead) before starting speed reduction.

Between 11.000 ft and 12.000 ft:
- Start speed reduction to 250 kts.
- The aircraft will now be 1000 ft below path (TOD in 3 nm).

Between 25 nm and 15 nm (latest):
- Start speed reduction to 200 kts
- Select FLAP 1 latest at F1-speed

Before glidepath intercept:
- Select FLAP 2 latest at F2-speed
- Start speed reduction to 180 kts
- **FLAP 3 if glidepath angle > 3.0°**

Latest at 6 nm (2000 ft AAE):
- Start speed reduction to 160 kts
- Select GEAR DOWN, FLAP 3, FLAP 4 (5)

---

BEFORE LANDING CHECKLIST

---

Latest at 4 nm (1300 ft AAE):
- Start speed reduction to $V_{APP}$
- Use flight spoilers as required
- Stabilized at 1000 ft AAE (latest)

## 3D Approach

Before approach
- 31 nm prior FAF (or first approach waypoint for RNP-AR): Check RNP ARM, RNP AR ARM or LPV ARM.
- 2 nm prior FAF (at first approach waypoint for RNP-AR): Check RNP APPR, RNP AR APPR or LPV APPR.
- Extend the final approach course if on radar vectors.

Cleared for approach:
- APPR.....ARM
  *ILS approach: use NAV, select APPR when within 1 dot.*
- VNAV..... ON
- ALT SEL..... DA (RNP AR)

Glide path intercept:
- Intercept at FLAP 2 while reducing to 180 kts
- GS or VGP, select MISAP altitude on the FCP
- RNP AR: landing configuration latest at FAF

Latest at 6 nm (2000 ft AAE):
- SPD.....160 kts
- Establish landing configuration in one continuous flow: GEAR DOWN, FLAP 3, FLAP 4 (5)
- SEATBELTS.....Cycle OFF, ON

---

BEFORE LANDING CHECKLIST

---

## 2D Approach

Before approach
- Extend the final approach course if on radar vectors.

Cleared for approach:
- APPR.....ARM
  *LOC approach: use NAV to intercept LOC, select APPR when within 1 dot, select NAV again when established.*

1.0 nm before the FAF:
- VNAV.....OFF
- ALT SEL..... MDA
- FPA.....Select

Start descent 0.3 nm before FAF (level flight)
- FAF at FLAP 2 while reducing to 180 kts
- Below MISAP alt, set MISAP alt on FCP

Latest at 6 nm (2000 ft AAE):
- SPD.....160 kts
- Establish landing configuration in one continuous flow.
- SEATBELTS..... Cycle OFF, ON

---

BEFORE LANDING CHECKLIST

---

## Final Approach & Landing

Latest at 4 nm (1300 ft AAE):
- SPD.....$V_{APP}$
- Spoilers.....As required

At 1000 ft AAE:
- Stabilized or go-around

At DA / MDA:
- Landing or go-around

Latest at 80 ft AGL (except for autoland):
- Autopilot.....OFF

At 30 ft AGL ("30" auto call-out):
- Start the landing flare
- Check RETARD mode or select thrust levers IDLE.

At touchdown:
- De-rotate, slowly lower the nosewheel onto the runway.
- Thrust levers.....IDLE REV
- Select reversers as required when interlock releases.

At 60 kts:
- Thrust levers.....IDLE REV
- Disconnect autobrakes by pressing brake pedals.
- Select thrust levers forward idle at safe taxi speed.

## After Landing

When runway vacated:
- Start timing (3 minutes at near idle before engine shutdown).
- APU.....START before reaching stand
- STROBE, LDG LTS (3).....OFF
- TAXI LTS..... NARROW or WIDE
- FLAP.....0

---

AFTER LANDING CHECKLIST

---

## Shutdown & Parking

- TAXI LIGHT..... OFF
- PARK BRAKE.....ON
- XPDR.....CODE 2000, STBY

---

SHUTDOWN CHECKLIST
Flight control test - ABIT (♦♦)

---

Power-off or start from Power-on for subsequent flights.

---

POWER OFF CHECKLIST

---

# Appendix 2 – Normal Checklist

### POWER ON

| | |
|---|---|
| ☑ PARK BRAKE | ON |
| ☑ BATT 1 | AUTO |
| ☑ BATT 2 | AUTO |
| ☐ ECL | DU 2 |
| ☑ PTU | AUTO |
| ☑ HYD 3A | AUTO |
| ☑ HYD 3B | AUTO |
| ☑ HYD 2B | AUTO |
| ☑ Landing gear lever | DN |
| ☑ Thrust levers | IDLE |
| ☑ L ENG run | OFF |
| ☑ R ENG run | OFF |
| ☐ APU or external power | As required |
| ☑ EQUIP COOLING, INLET | Select auto |

### PRE FLIGHT

| | |
|---|---|
| ☐ Airplane documents | On board and checked |
| ☐ Operations Engineering Bulletins (OEB) | Checked |
| ☐ Preliminary takeoff performance | Computed and cross-checked |
| ☐ Emergency equipment | Checked |
| ☐ Gear pins | On board |
| ☐ Overhead panel | Checked |
| ☐ Glareshield | Checked |
| ☐ Displays | Checked |
| ☐ ♦ Ice detector test | Complete |
| ☐ ♦ Circuit breakers | Complete |
| ☐ EICAS and INFO | Checked |
| ☐ Altimeters | ( ) Set and cross-checked |
| ☐ Center panel | Checked |
| ☐ Pedestal | Checked |
| ☐ ♦ Rudder trim | Checked |
| ☐ Sidesticks | Checked |
| ☐ ♦ Oxygen masks | Checked |
| ☐ FDRAS test | Complete |

### BEFORE START

| | |
|---|---|
| ☐ Takeoff briefing | Complete |
| ☐ APU or external power | As required |
| ☑ SEATBELTS | ON |
| ☑ DOORS | Closed and locked |
| ☑ BEACON | ON |
| ☐ PARK BRAKE | As required |

### BEFORE TAXI

| | |
|---|---|
| ☐ APU | As required |
| ☐ FLAP | ( ) selected |
| ☐ ♦♦ Wing anti-ice test | Complete |
| ☐ ANTI-ICE, COWL | As required |
| ☐ ANTI-ICE, WING | As required |
| ☐ Flight controls | Checked |
| ☑ NOSE STEER | Select on |
| ☐ EICAS and INFO | Checked |

### BEFORE TAKE OFF

| | |
|---|---|
| ☐ APU BLEED | OFF, if OAT is at or below 15°C |
| ☐ FMS | Set |
| ☐ Trims | Set |
| ☐ FLAP | Set |
| ☐ Anti-ice | As required |
| ☐ AUTOBRAKE | As required |
| ☐ EICAS and INFO | Checked |
| ☐ Cabin | Ready |
| ☐ Runway | ( ) confirmed |

### AFTER TAKE OFF

| | |
|---|---|
| ☑ Landing gear | UP |
| ☑ FLAP | 0 |
| ☑ ANTI-ICE, L COWL | AUTO |
| ☑ ANTI-ICE, WING | AUTO |
| ☑ ANTI-ICE, R COWL | AUTO |
| ☑ APU BLEED | Select auto |
| ☐ EICAS | Checked |

### HIGH ALTITUDE CLIMB

☑ ANTI-ICE, WING ..... OFF

Avoid icing conditions.

### DESCENT & APPROACH

| | |
|---|---|
| ☑ ANTI-ICE, WING (below 35.000 feet) | AUTO |
| ☐ FMS | Set |
| ☐ Minima | Set |
| ☐ AUTOBRAKE | As required |
| ☐ EICAS | Checked |
| ☐ Approach briefing | Complete |

### BEFORE LANDING

| | |
|---|---|
| ☐ Altimeters | ( ) Set and cross-checked |
| ☑ Landing gear | DN |
| ☐ FLAP | ( ) indicating |
| ☐ Cabin | Ready |

☑ = Sensed item in the correct position for phase of flight
☑ = Sensed item requiring action
☐ = Non-sensed item

### AFTER GO AROUND

| | |
|---|---|
| ☑ Landing gear | UP |
| ☑ FLAP | 0 |
| ☐ EICAS | Checked |

### AFTER LANDING

☑ FLAP ..... 0

### SHUTDOWN

| | |
|---|---|
| ☑ PARK BRAKE | ON |
| ☑ NOSE STEER | OFF |
| ☐ APU and/or external power | As required |
| ☑ L ENG run | OFF |
| ☑ R ENG run | OFF |
| ☑ BEACON | OFF |
| ☐ ♦♦ Flight control test | Complete |

### POWER OFF

| | |
|---|---|
| ☐ ECL | DU2 |
| ☑ EQUIP COOLING, INLET | OFF |
| ☑ EMER LTS | OFF |
| ☑ HYD 3A | OFF |
| ☑ HYD 3B | OFF |
| ☑ HYD 2B | OFF |
| ☑ APU | OFF |
| ☐ EXT PWR (if connected) | AVAIL |
| ☑ DOME | Select off |
| ☑ BATT 1 | OFF |
| ☑ BATT 2 | OFF |

# Appendix 3 – Quick Reference Sheet

### Descent Planning

**Descent Angle**

Weight:
- LW 55t = 3.0°
- Each 5t below 55t = +0.2°
- Each 5t above 55t = -0.1°

Anti-icing:
- Use anticipated = -0.2°

Temperature:
- Each 10°C above ISA = -0.1°
- Each 10°C below ISA = +0.1°

Wind Correction:
- Each 10 kts of headwind = +0.1°
- Each 10 kts of tailwind = -0.1°

**Distance to descent**

- Each 0.1° above 3.0° = -3.0 %
- Each 0.1° below 3.0° = +4.0 %

| RWYCC<br>FC - BA | Crosswind | Thrust | Limitations | | | |
|---|---|---|---|---|---|---|
| 6<br>NIL<br>DRY | 32 kts (TKOF)<br>29 kts (LDG) | MAX or<br>FLEX or<br>DERATE | RWYCC 5: REV MAX except if OLD for RWYCC 2 is less than LDA | | | |
| 5<br>≥ 40<br>GOOD | | | | | | |
| 4<br>36-39<br>MED – GOOD | 27 kts | MAX or<br>DERATE | ≤ RWYCC 4 | | | |
| 3<br>30-35<br>MEDIUM | 20 kts | | ≤ RWYCC 3 | | (Not applicable from BCS 4.0):<br>Choose wide runway with min XWC<br>Autobrake set to LOW | Anti-skid must be operative<br>Manual rollout after Autoland<br>REV MAX |
| 2<br>26-29<br>MED – POOR | 10 kts | | ≤ RWYCC 2 | Manual landing recommended | | |
| 1<br>21-25<br>POOR | 10 kts | MAX only | Both thrust reversers operative | | | |
| 0<br>≤ 20<br>NIL | NIL | - | - | - | - | - |

| Configuration | TKOF | CLB | Green Dot | APCH | GA |
|---|---|---|---|---|---|
| All Engine Operating | 30° ($V_2$ + 10) | 25° (F-speed)<br>30° (F+10 kts) | 30° < 31.500 ft<br>15° > 31.500 ft | 25° ($V_{REF}$) | 30° ($V_{GA}$) |
| One Engine Inoperative | 15° ($V_2$) | 15° (F-speed) | 25° | 30° ($V_{REF}$ + 5) | 20° ($V_{AC}$) |

| Preselected fuel quantity | Automatic Refuel Sequence - kgs | | |
|---|---|---|---|
| | Left Main | Center | Right Main |
| 2000 kg | 1000 | 0<br>(Empty) | 1000 |
| 3000 kg | 1500 | | 1500 |
| 4000 kg | 2000 | | 2000 |
| 5000 kg | 2500 | | 2500 |
| 5200 kg | 2600 (85.6%) | 0 | 2600 (85.6%) |
| 6000 kg | 2200 (73.0%) | 1600 | 2200 (73.0%) |
| 7000 kg | 2600<br>(85.6%) | 1800 | 2600<br>(85.6%) |
| 8000 kg | | 2800 | |
| 9000 kg | | 3800 | |
| 10.000 kg | | 4800 | |
| 11.000 kg | | 5800 | |
| 12.000 kg | | 6800 | |
| 13.000 kg | | 7800 | |
| 14.000 kg | | 8800 | |
| 15.000 kg | | 9800 | |
| 16.000 kg | | 10800 | |
| 16.500 kg | 2600 | 11300 | 2600 |
| 17.000 kg | 2850 | 11300<br>(Full) | 2850 |
| 17.400 kg | 3050 (Full) | | 3050 (Full) |

| Start of FDP at reference time | 1-2 Sectors | 3 Sectors | 4 sectors |
|---|---|---|---|
| 06:00 – 13:29 | 13:00 | 12:30 | 12:00 |
| 13:30 – 13:59 | 12:45 | 12:15 | 11:45 |
| 14:00 – 14:29 | 12:30 | 12:00 | 11:30 |
| 14:30 – 14:59 | 12:15 | 11:45 | 11:15 |
| 15:00 – 15:29 | 12:00 | 11:30 | 11:00 |
| 15:30 – 15:59 | 11:45 | 11:15 | 10:45 |
| 16:00 – 16:29 | 11:30 | 11:00 | 10:30 |
| 16:30 – 16:59 | 11:15 | 10:45 | 10:15 |
| 17:00 – 04:59 | 11:00 | 10:30 | 10:00 |
| 05:00 – 05:14 | 12:00 | 11:30 | 11:00 |
| 05:15 – 05:29 | 12:15 | 11:45 | 11:15 |
| 05:30 – 05:44 | 12:30 | 12:00 | 11:30 |
| 05:45 – 05:59 | 12:45 | 12:15 | 11:45 |

| AD Temp °C | Height above the elevation of the altimeter setting source (ft) | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 200 | 300 | 400 | 500 | 600 | 700 | 800 | 900 | 1000 | 1500 | 2000 | 3000 | 4000 | 5000 |
| 0 | 20 | 20 | 30 | 30 | 40 | 40 | 50 | 50 | 60 | 90 | 120 | 170 | 230 | 280 |
| -5 | 20 | 30 | 40 | 40 | 50 | 60 | 70 | 80 | 80 | 120 | 160 | 230 | 310 | 390 |
| -10 | 20 | 30 | 40 | 50 | 60 | 70 | 80 | 90 | 100 | 150 | 200 | 290 | 390 | 490 |
| -15 | 30 | 40 | 50 | 70 | 80 | 90 | 100 | 110 | 130 | 180 | 240 | 360 | 480 | 600 |
| -20 | 30 | 50 | 60 | 70 | 90 | 100 | 120 | 130 | 140 | 210 | 280 | 420 | 570 | 710 |
| -30 | 40 | 60 | 80 | 100 | 120 | 140 | 150 | 170 | 190 | 280 | 380 | 570 | 760 | 950 |
| -40 | 50 | 80 | 100 | 120 | 150 | 170 | 190 | 220 | 240 | 360 | 480 | 720 | 970 | 1210 |
| -50 | 60 | 90 | 120 | 150 | 180 | 210 | 240 | 270 | 300 | 450 | 590 | 890 | 1190 | 1500 |

| Height ft | Display 1 | | | Display 2 | | |
|---|---|---|---|---|---|---|
| | Range | Tilt | Gain | Range | Tilt | Gain |
| Ground | 10 nm | +5.0° | +2 | 20 nm | +5.0° | +2 |
| 5000 | 20 nm | +3.0° | +2 | 40 nm | +3.0° | +2 |
| 10.000 | 40 nm | +2.0° | +2 | 80 nm | +2.5° | +3 |
| 20.000 | 40 nm | -0.5° | +2 | 80 nm | +1.2° | +3 |
| 30.000 | 40 nm | -2.5° | +3 | 80 nm | +0.5° | +3 |
| 40.000 | 40 nm | -4.0° | +3 | 80 nm | -0.7° | +3 |

| PW1521 | CLB | CLB-1 | CLB-2 |
|---|---|---|---|
| TO | X | X | X |
| TO-1 | ... | ... | X |
| TO-2 | ... | ... | ... |

X = Thrust reduction from TO to CLB.

| PW1524 | CLB | CLB-1 | CLB-2 |
|---|---|---|---|
| TO | X | X | X |
| TO-1 | X | X | X |
| TO-2 | ... | ... | X |
| TO-3 | ... | ... | ... |

X = Thrust reduction from TO to CLB.

| Planning Minima (EASA basic fuel scheme) | | |
|---|---|---|
| Type of Approach | Ceiling | RVR / VIS |
| Type B approach operation<br>(DH below 250 ft) | DA/H + 200 ft | RVR/VIS + 800 m |
| Type A approach operation<br>(MDH / DH at or above 250 ft) | DA/H + 400 ft | RVR/VIS + 1500 m |
| Circling approach operation | DA/H + 400 ft | RVR/VIS + 1500 m |
| Wind limitations should be applied, considering the runway condition (crosswind & tailwind limits). | | |

| CAT-D (Upper)<br>Medium behind: | | Landing (distance) | Take-off (time) | |
|---|---|---|---|---|
| | | | FULL | INT |
| ICAO | S - Super | 7.0 nm | 3m00s | 4m00s |
| | H – Heavy | 5.0 nm | 2m00s | 3m00s |
| RECAT EU | CAT-A Super Heavy | 5.0 nm | 2m20s | 3m20s |
| | CAT-B Upper Heavy | 4.0 nm | 1m40s | 2m40s |
| | CAT-C Lower Heavy | 3.0 nm | 1m20s | 2m20s |
| | Other | MRS | - | - |

| Aircraft Category | $V_{AT}$ | Initial Approach | Final Approach | Max Circling | Max missed approach | |
|---|---|---|---|---|---|---|
| | | | | | Initial | Final |
| C | 121 - 140 | 160 - 240 | 115 - 160 | 180 | 160 | 240 |

| ICAO Max Holding Speed (* Turbulence) | |
|---|---|
| Altitude | Speed |
| ≤ 14.000 ft | 230 kts<br>280 kts* |
| ≤ 20.000 ft | 240 kts<br>280 / M0.8* |
| ≤ 34.000 ft | 265 kts<br>280 / M0.8* |
| > 34.000 ft | M0.83 |
| ≤ 14.000 ft = 1m00s<br>> 14.000 ft = 1m30s | |

| Minima | | |
|---|---|---|
| Category | DH | RVR |
| CAT I | 200 ft | 550 m |
| CAT II | 100 ft | 300 m |
| CAT IIIA | 50 ft | 200 m |
| CAT IIIB | NIL | 75 m |

Take-off RVR

- All relevant parts (min 125 m)

Landing RVR

- TDZ: not below landing minima
- MID: 125 m (75 m rollout mode)
- END: for information only

| GS kts | Turning Radius – nm<br>Angle of Bank | | | |
|---|---|---|---|---|
| | 15° | 20° | 25° | 30° |
| 120 | 0.78 | 0.58 | 0.45 | 0.36 |
| 140 | 1.07 | 0.79 | 0.61 | 0.50 |
| 160 | 1.39 | 1.03 | 0.80 | 0.65 |
| 180 | 1.76 | 1.30 | 1.01 | 0.82 |
| 200 | 2.18 | 1.60 | 1.25 | 1.01 |
| 250 | 3.40 | 2.50 | 1.95 | 1.58 |
| 300 | 4.90 | 3.61 | 2.81 | 2.27 |
| 350 | 6.67 | 4.91 | 3.83 | 3.09 |

| Height above ground | 0 ft | 5000 ft | 10.000 ft | 15.000 ft | 20.000 ft | 25.000 ft | 30.000 ft | 35.000 ft | 40.000 ft | 41.000 ft |
|---|---|---|---|---|---|---|---|---|---|---|
| VHF radio range | 0 nm | 87 nm | 123 nm | 151 nm | 174 nm | 194 nm | 213 nm | 230 nm | 246 nm | 249 nm |

| Unreliable IAS | Pitch | Thrust | Configuration |
|---|---|---|---|
| Below safe alt | 12.5° | MAX | Gear up, Spoilers retract, Slat / flap maintain |
| Above safe alt | 5.0° | CLB | Slat / flap retract in stages |

# Appendix 4 – Manual Load & Trim Sheet

## AIRBUS A220-300 (148 SEATS) - LOAD & TRIM SHEET

| Date | Flight No. | Registration | From | To |
|---|---|---|---|---|
| | | | | |

| Alternate(s) | Flight Time | Endurance | Prepared by | Approved by |
|---|---|---|---|---|
| | | | | |

### Section 1

| | WEIGHT | * Lowest of performance limited or maximum structural weight limit | ZERO FUEL | TAKE-OFF* | LANDING* |
|---|---|---|---|---|---|
| Dry Operating Weight<br>Crew config. ___ / ___ | | Maximum Weight for: | 5 5 7 9 2 | | |
| Take-off Fuel + | | → Take-off Fuel + | | Trip Fuel + | |
| | | Allowed Take-off Weight (lowest of a, b, c) = | a | b | c |
| Operating Weight = | | → Operating Weight - | | | |
| | | Allowed Traffic Load = | | | |
| | | Total Traffic Load - | | | |
| | | Under Load (before LMC) = | | | |

| Item | Weight | Index |
|---|---|---|
| Pilot / Copilot | 85 kg | -1.6 |
| Observer seat | 85 kg | -1.5 |
| Flight attendant FWD | 75 kg | -1.2 |
| Flight attendant AFT | 75 kg | 1.2 |
| Galley FWD | Each 50 kg | -0.8 |
| Galley AFT | Each 50 kg | 0.8 |

### Section 2

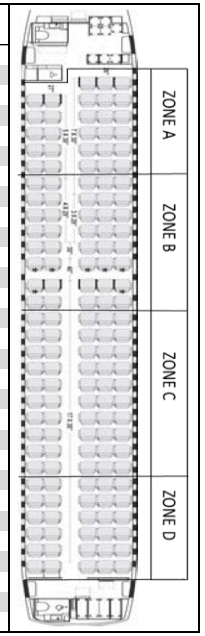
| PAX | TOTAL | | DISTRIBUTION | | | | LOAD | TOTAL | DISTRIBUTION | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | No. | WGT | 0A 28 | 0B 40 | 0C 50 | 0D 30 | | WGT | C1 800 KG | C2 1646 KG | C3 1858 KG | C4 1055 KG |
| Ad. | | | | | | | B. | | | | | |
| Ch. | | | | | | | C. | | | | | |
| In. | | | | | | | M. | | | | | |
| T. | | | | | | | T. | | | | | |
| INDEX = | | | - | - | + | + | INDEX = | | - | - | + | + |

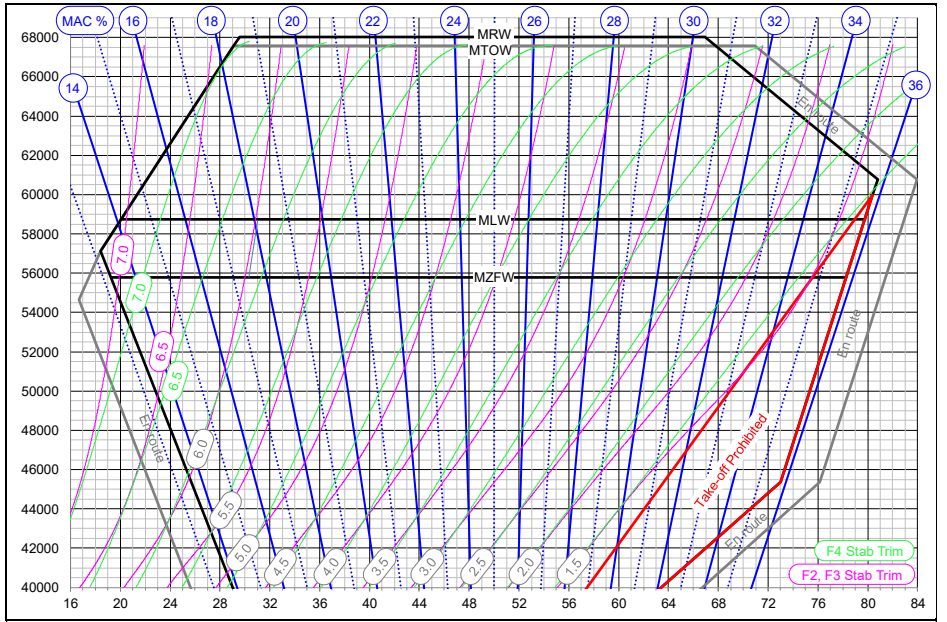
### Section 3

| LOADING DATA | WEIGHT | INDEX | LAST MINUTE CHANGES | WEIGHT | INDEX |
|---|---|---|---|---|---|
| Cabin Zone A + B | | - . | LMC 1 ..... | | . |
| Cabin Zone C + D | | + . | LMC 2 ..... | | . |
| Compartment 1 + 2 Max. 2 4 4 6 | | - . | LMC 3 ..... | | . |
| Compartment 3 + 4 Max. 2 6 0 6 | | + . | LMC 4 ..... | | . |
| Total Traffic Load = | | = . | Total LMC UNDL ___ = | | = . |
| Dry Operating Weight, DOI + | | + . | ZFW (before LMC) + | | + . |
| Zero Fuel Weight Max. 5 5 7 9 2 = | | = . | Corrected ZFW = | | = . |
| Take-off Fuel + | | . | Take-off Fuel + | | . |
| Take-off Weight Max. ___ = | | = . | Corrected TOW = | | = . |
| Trip Fuel - | | | | | |
| Landing Weight Max. ___ = | | | ZFW MAC % = ___ . ___ | STAB TRIM = ___ . ___ | |

Part A

| PAX No. | CABIN INDEX No. | 0A | 0B | 0C | 0D | LOAD INDEX WGT | C1 | C2 | C3 | C4 | FUEL WGT | INDEX |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | -0.9 | -0.4 | 0.4 | 1.0 | 5 | -0.1 | 0.0 | 0.0 | 0.1 | 250 | -0.1 |
| 2 | 2 | -1.8 | -0.7 | 0.7 | 2.0 | 10 | -0.1 | -0.1 | 0.1 | 0.1 | 750 | -0.2 |
| 3 | 3 | -2.7 | -1.1 | 1.1 | 3.0 | 15 | -0.2 | -0.1 | 0.1 | 0.2 | 1500 | -0.3 |
| 4 | 4 | -3.6 | -1.5 | 1.4 | 4.0 | 20 | -0.3 | -0.2 | 0.2 | 0.3 | 3000 | -0.4 |
| 5 | 5 | -4.5 | -1.9 | 1.8 | 5.0 | 25 | -0.3 | -0.2 | 0.2 | 0.3 | 3500 | -0.3 |
| 6 | 6 | -5.4 | -2.2 | 2.2 | 6.0 | 30 | -0.4 | -0.3 | 0.2 | 0.4 | 4000 | -0.1 |
| 7 | 7 | -6.3 | -2.6 | 2.5 | 7.0 | 40 | -0.5 | -0.4 | 0.3 | 0.5 | 4500 | 0.4 |
| 8 | 8 | -7.2 | -3.0 | 2.9 | 8.0 | 50 | -0.7 | -0.4 | 0.4 | 0.6 | 5000 | 1.1 |
| 9 | 9 | -9.0 | -3.7 | 3.6 | 9.9 | 100 | -1.3 | -0.9 | 0.8 | 1.3 | 5160 | 1.4 |
| 10 | 10 | -10.7 | -4.5 | 4.3 | 11.9 | 200 | -2.6 | -1.8 | 1.5 | 2.6 | 5500 | 1.0 |
| 15 | 12 | -12.5 | -5.2 | 5.0 | 13.9 | 300 | -4.0 | -2.7 | 2.3 | 3.9 | 6000 | 0.2 |
| 20 | 14 | -14.3 | -6.0 | 5.8 | 15.9 | 400 | -5.3 | -3.5 | 3.1 | 5.1 | 6500 | -0.7 |
| 25 | 16 | -16.1 | -6.7 | 6.5 | 17.9 | 500 | -6.6 | -4.4 | 3.9 | 6.4 | 7000 | -1.6 |
| 30 | 18 | -17.9 | -7.5 | 7.2 | 19.9 | 600 | -7.9 | -5.3 | 4.6 | 7.7 | 7500 | -2.5 |
| 35 | 20 | -19.7 | -8.2 | 7.9 | 21.9 | 700 | -9.2 | -6.2 | 5.4 | 9.0 | 8000 | -3.4 |
| 40 | 22 | -21.5 | -8.9 | 8.6 | 23.9 | 800 | -10.6 | -7.1 | 6.2 | 10.3 | 8500 | -4.3 |
| 45 | 24 | -23.3 | -9.7 | 9.3 | 25.8 | 900 | | -8.0 | 6.9 | 11.6 | 9000 | -5.3 |
| 50 | 26 | -25.1 | -10.4 | 10.1 | 27.8 | 1000 | | -8.9 | 7.7 | 12.9 | 9500 | -6.3 |
| 60 | 28 | | -11.2 | 10.8 | 29.8 | 1055 | | -9.3 | 8.1 | 13.6 | 10.000 | -7.3 |
| 65 | 30 | | -11.9 | 11.5 | | 1100 | | -9.7 | 8.5 | | 10.500 | -8.3 |
| 70 | 32 | | -12.7 | 12.2 | | 1200 | | -10.6 | 9.2 | | 11.000 | -9.3 |
| 80 | 34 | | -13.4 | 12.9 | | 1300 | | -11.5 | 10.0 | | 12.000 | -11.3 |
| 90 | 36 | | -14.2 | 13.7 | | 1400 | | -12.4 | 10.8 | | 13.000 | -13.2 |
| 100 | 38 | | -14.9 | 14.4 | | 1500 | | -13.3 | 11.6 | | 14.000 | -15.1 |
| 110 | 40 | | | 15.1 | | 1600 | | -14.2 | 12.3 | | 15.000 | -17.4 |
| 120 | 42 | | | 15.8 | | 1646 | | -14.6 | 12.7 | | 16.000 | -19.6 |
| 130 | 44 | | | 16.5 | | 1700 | | | 13.1 | | 16.330 | -20.0 |
| 140 | 46 | | | 17.3 | | 1800 | | | 13.9 | | 17.000 | -18.7 |
| 148 | 48 | | | 18.0 | | 1858 | | | 14.3 | | 17.400 | -18.3 |

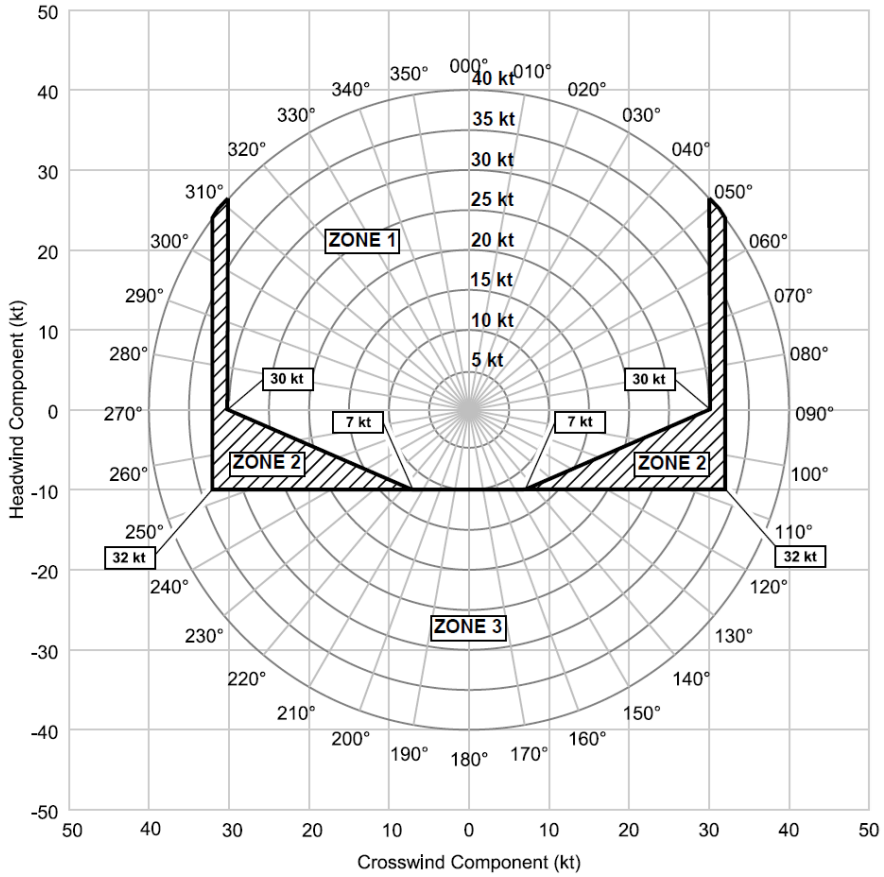



Part B

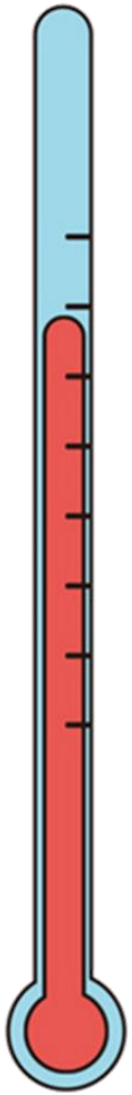
# Appendix 5 – RCAM & Wind Limitations

| Runway Condition Assessment Matrix | | | |
|---|---|---|---|
| **Runway Condition Code (RWYCC)** | **Runway Surface Condition Description** | **Pilot-reported Braking Action** | **Maximum Recommended Crosswind** |
| 6 | Dry | - | 32 kts (take-off)<br>29 kts (landing) |
| 5 | • Frost<br>• Wet (less than 3 mm)<br>3 mm depth or less of:<br>• Slush<br>• Dry snow<br>• Wet Snow | Good | |
| 4 | -15°C and colder:<br>• Compacted snow | Good to Medium | 27 kts |
| 3 | • Wet ("slippery when wet")<br>• Dry snow or wet snow<br>More than 3 mm depth:<br>• Dry snow<br>• Wet snow<br>Warmer than -15°C:<br>• Compacted snow | Medium | 20 kts |
| 2 | More than 3 mm depth:<br>• Water<br>• Slush | Medium to Poor | 10 kts |
| 1 | Ice | Poor | |
| 0 | - | - | - |




| Engine operating limits due to wind | | |
|---|---|---|
| **Zone** | **Altitude** | **Limitation** |
| 1 | < 7000 ft | Normal take-off technique can be used. |
| | 7000 – 9200 ft | High wind take-off if crosswind exceeds 22 kts. |
| | ≥ 9200 ft | High wind take-off must be used (always). |
| 2 | High wind take-off must be used. | |
| 3 | Take-off prohibited. | |

# Appendix 6 – Cold Weather Operations



+15°C
- Above 15°C, wing anti-ice ON prohibited
- At and below 15°C, APU (or APU bleed) must be OFF for take-off

+10°C
- Cowl anti-ice must be selected ON before take-off if visible moisture present
- In flight, +10°C TAT to -40°C SAT in visible moisture is considered icing conditions

+5°C — Wing anti-ice must be selected ON before take-off if visible moisture present

+3°C — Engine ice shedding must be performed to 60% N1 at intervals not exceeding 30 minutes

0°C
- Ground operations, take-offs and landings with OAT < 0°C is considered cold weather operations
- If turnaround is expected to exceed 120 min, cabin temperature must be maintained during turnaround

-7°C — Or below for more than 8 consecutive hour is considered cold soak

-15°C — Recommended to remove aircraft batteries and store at room temperature

-18°C — Engine operation is prohibited in ground fog conditions

-20°C — If turnaround exceeds 120 min, thrust reversers must be cycled twice before take-off from contaminated runway, take-off must be initiated within 5 minutes, otherwise repeat procedure

-37°C — Minimum fuel temperature for start of flight

-40°C
- Minimum oil temperature for APU and engine start
- Oil temp -40 to -6°C = idle only, -6 to +21°C = max 30% N1, 21 to 49°C = max 50% N1, above 49°C = no limit
- If turnaround exceeds 120 min, thrust reversers must be cycled until cycle time is less than 2 ½ seconds before take-off from contaminated runway, take-off must be initiated within 2 minutes, otherwise repeat procedure
- Before applying electrical power, use heater to warm up equipment (maintenance action)
- ACMP 3A and 3B must be ON to heat hydraulic system 3 fluid temperature to at least -6°C before take-off

-47°C — JET A-1 freezing point

-54°C — Minimum temperature for take-off or landing

-70.5°C — Minimum operating temperature at tropopause (ISA -14°C)

# Appendix 7 – De-icing / Anti-icing Procedure

Remote De-icing / Anti-icing

**Before fluid is applied:**

**(1) Engine(s) is/are operating**

→ **Yes – Go to (2)**
→ **No – Go to (35)**

**(2) Engine(s) is/are operating:**

**Before taxi, before de-icing:**

Do not perform (override) the BEFORE TAXI checklist.

(3) APU..... As required
(4) ♦♦ Wing anti-ice test..... Complete (Refer to Before Taxi)
(5) ANTI-ICE. COWL..... AUTO or ON
(6) ANTI-ICE WING..... AUTO or ON
(7) EICAS and INFO..... Checked

**When aircraft in position / ready for de-icing:**

(8) PARK BRAKE..... ON
(9) NOSE STEER..... OFF
(10) Thrust levers..... IDLE for duration of the operation

*If the APU is running, make sure the personnel that does the fluid application are aware of the location of the APU air intake and have been instructed to avoid fluid spray that can be ingested by the APU.*

(11) APU.....As required

*It is recommended to turn APU OFF. Confirm door is closed if APU is OFF.*

(12) L PACK..... OFF
(13) R PACK..... OFF
(14) L BLEED..... OFF
(15) APU BLEED.....OFF
(16) R BLEED.....OFF
(17) EQUIP COOLING, INLET..... OFF

CAUTION: stabilizer should remain in current position for de-icing if there are signs of ice around the visor fairing or the horizontal stabilizer/elevator-to-aft-fuselage interface.

(18) Stabilizer trim..... Set to 5 units

CAUTION: if accumulation of frost, ice or snow is suspected while slats/flaps are extended, leave slats/flaps at current position for de-icing.

(19) FLAP.....0

*To prevent damage, the pilot must advise the de-icing operator to avoid direct spraying of anti-icing fluid on the following:*

- *Windshields, side windows, and cabin windows*
- *ADS (to prevent ingestion) and TAT probes*
- *AOA vanes*
- *Engine cowls intake (to prevent ingestion) and exhaust*
- *APU inlet / exhaust*
- *Antennas, and*
- *Wheels and brakes*

**After de-icing/anti-icing spraying:**

*The PIC must ensure that an inspection is done to confirm that critical surfaces are clear of contamination.*

(20) APU..... As required
(21) FLAP.....( ) selected

*Set take-off flaps as entered in FMS*

(22) Stabilizer trim.....( ) set
(23) Flight controls..... Checked

a) FLT CTRL synoptic page.....Monitor
b) Sidesticks..... Checked

*One pilot checks the flight control for full and correct range of movement.*

c) Rudder.....Checked

*Nosewheel steering off or use PEDAL DISC switch on tiller. Ensure full deflection (monitor both sides for correct movement).*

(24) NOSE STEER..... Select on
(25) PARK BRAKE..... As required

**One minute after de-icing / anti-icing is complete:**

(26) L BLEED..... Select auto
(27) APU BLEED..... Select auto
(28) R BLEED..... Select auto
(29) EQUIP COOLING, INLET..... Select auto
(30) ANTI-ICE, COWL..... AUTO or ON
(31) ANTI-ICE, WING.....AUTO or ON

**One minute after bleeds are set to auto:**

(32) L PACK.....Select auto
(33) R PACK..... Select auto
(34) Before take-off procedure..... Accomplish

**- COMPLETE -**

De-icing / Anti-icing on Parking Stand

**(35) Engine(s) is/are not operating:**

**(36) APU is OFF:**

→ **Yes – Go to (37)**
→ **No – Go to (54)**

**(37) APU is OFF:**

**Before fluid is applied:**

(38) Doors..... closed
(39) External power..... As required
(40) EQUIP COOLING, INLET..... OFF

*To prevent damage, the pilot must advise the de-icing operator to avoid direct spraying of anti-icing fluid on the following:*

- *Windshields, side windows, and cabin windows*
- *ADS (to prevent ingestion) and TAT probes*
- *AOA vanes*
- *Engine cowls intake (to prevent ingestion) and exhaust*
- *APU inlet / exhaust*
- *Antennas, and*
- *Wheels and brakes*

**After de-icing/anti-icing spraying:**

*The PIC must ensure that an inspection is done to confirm that critical surfaces are clear of contamination.*

**Before APU is started or external air is established:**

(41) L PACK..... OFF
(42) R PACK.....OFF
(43) L BLEED..... OFF
(44) APU BLEED..... OFF
(45) R BLEED..... OFF
(46) APU or external air..... START / Establish

**One minute after de-icing / anti-icing is complete:**

(47) L BLEED..... Select auto
(48) APU BLEED..... Select auto
(49) R BLEED..... Select auto
(50) EQUIP COOLING, INLET..... Select auto

**One minute after bleeds are set to auto:**

(51) L PACK.....Select auto
(52) R PACK..... Select auto
(53) Before start procedure.....Accomplish

**- COMPLETE -**

**(54) APU is ON:**

**Before fluid is applied:**

(55) Doors..... closed
(56) External power..... As required
(57) L PACK..... OFF
(58) R PACK..... OFF
(59) L BLEED..... OFF
(60) APU BLEED..... OFF
(61) R BLEED.....OFF
(62) EQUIP COOLING, INLET.....OFF

*To prevent damage, the pilot must advise the de-icing operator to avoid direct spraying of anti-icing fluid on the following:*

- *Windshields, side windows, and cabin windows*
- *ADS (to prevent ingestion) and TAT probes*
- *AOA vanes*
- *Engine cowls intake (to prevent ingestion) and exhaust*
- *APU inlet / exhaust*
- *Antennas, and*
- *Wheels and brakes*

**After de-icing/anti-icing spraying:**

*The PIC must ensure that an inspection is done to confirm that critical surfaces are clear of contamination.*

**One minute after de-icing / anti-icing is complete:**

(63) L BLEED..... Select auto
(64) APU BLEED..... Select auto
(65) R BLEED..... Select auto
(66) EQUIP COOLING, INLET..... Select auto

**One minute after bleeds are set to auto:**

(67) L PACK..... Select auto
(68) R PACK..... Select auto
(69) Before start procedure..... Accomplish

**- COMPLETE -**

# Appendix 8 – Active Frost Holdover Times (use composite)

## TABLE 1: ACTIVE FROST HOLDOVER TIMES FOR SAE TYPE I, TYPE II, TYPE III, AND TYPE IV FLUIDS[1]

| Outside Air Temperature[2,3,4] | Type I Aluminum | Type I Composite |
|---|---|---|
| -1 °C and above (30 °F and above) | 0:45 | 0:35 |
| below -1 to -3 °C (below 30 to 27 °F) | | |
| below -3 to -10 °C (below 27 to 14 °F) | | |
| below -10 to -14 °C (below 14 to 7 °F) | | |
| below -14 to -21 °C (below 7 to -6 °F) | | |
| below -21 to -25 °C (below -6 to -13 °F) | | |
| below -25 °C (below -13 °F) | | |

| Outside Air Temperature[3,4] | Concentration Fluid/Water By % Volume | Type II | Type III[5] | Type IV |
|---|---|---|---|---|
| -1 °C and above (30 °F and above) | 100/0 | 8:00 | 2:00 | 12:00 |
| | 75/25 | 5:00 | 1:00 | 5:00 |
| | 50/50 | 2:00 | 0:30 | 3:00 |
| below -1 to -3 °C (below 30 to 27 °F) | 100/0 | 8:00 | 2:00 | 12:00 |
| | 75/25 | 5:00 | 1:00 | 5:00 |
| | 50/50 | 1:30 | 0:30 | 3:00 |
| below -3 to -10 °C (below 27 to 14 °F) | 100/0 | 8:00 | 2:00 | 10:00 |
| | 75/25 | 4:00 | 1:00 | 5:00 |
| below -10 to -14 °C (below 14 to 7 °F) | 100/0 | 6:00 | 2:00 | 6:00 |
| | 75/25 | 1:00 | 1:00 | 1:00 |
| below -14 to -21 °C (below 7 to -6 °F) | 100/0 | 3:00 | 2:00 | 6:00 |
| below -21 to -25 °C (below -6 to -13 °F) | 100/0 | 2:00 | 2:00 | 4:00 |
| below -25 °C (below -13 °F) | 100/0 | No Holdover Time Guidelines Exist | | |

**NOTES**

1 To use the HOTs in this table, ensure that the fluid and dilution being used is listed in the List of Qualified Fluids Tested for Anti-Icing Performance and Aerodynamic Acceptance table (Table 55 - Table 58). Any restrictions on the use of the fluid have to be identified and applied.
2 Type I Fluid / Water Mixture must be selected so that the freezing point of the mixture is at least 10 °C (18 °F) below outside air temperature.
3 Ensure that the lowest operational use temperature (LOUT) is respected.
4 Changes in outside air temperature (OAT) over the course of longer frost events can be significant; the appropriate holdover time to use is the one provided for the coldest OAT that has occurred in the time between the de/anti-icing fluid application and takeoff.
5 To use the Type III fluid frost holdover times, the fluid brand being used must be known. AllClear AeroClear MAX must be applied unheated.

**CAUTIONS**

- The cautions that apply to the holdover times in the table above can be found on page 9.

# Appendix 8 – Snowfall intensities as a function of prevailing visibility

**TABLE 57: SNOWFALL INTENSITIES AS A FUNCTION OF PREVAILING VISIBILITY**

| Visibility | | Day | | Night | |
|---|---|---|---|---|---|
| Statute Miles | Meters | -1 °C and below 30 °F and below | Above -1 °C Above 30 °F | -1 °C and Below 30 °F and below | Above -1 °C Above 30 °F |
| **≤1/4** (≤3/8) | **≤400** (≤600) | Heavy | Heavy | Heavy | Heavy |
| **1/2** (>3/8 to ≤5/8) | **800** (>600 to ≤1000) | Moderate | Heavy | Heavy | Heavy |
| **3/4** (>5/8 to ≤7/8) | **1200** (>1000 to ≤1400) | Moderate | Moderate | Moderate | Heavy |
| **1** (>7/8 to ≤1 1/8 ) | **1600** (>1400 to ≤1800) | Light | Light | Moderate | Moderate |
| **1 ¼** (>1 1/8 to ≤1 3/8) | **2000** (>1800 to ≤2200) | Light | Light | Moderate | Moderate |
| **1 ½** (>1 3/8 to ≤1 5/8) | **2400** (>2200 to ≤2600) | Light | Light | Moderate | Moderate |
| **1 ¾** (>1 5/8 to ≤1 7/8) | **2800** (>2600 to ≤3000) | Very Light | Light | Light | Light |
| **2** (>1 7/8 to ≤2 ¼) | **3200** (>3000 to ≤3600) | Very Light | Very Light | Light | Light |
| **2 ½** (>2 ¼ to ≤2 ¾) | **4000** (>3600 to ≤4400) | Very Light | Very Light | Light | Light |
| **3** (>2 ¾ to ≤3 ¼) | **4800** (>4400 to ≤5200) | Very Light | Very Light | Very Light | Light |
| **≥3 ½** (>3 ¼) | **≥5600** (>5200) | Very Light | Very Light | Very Light | Very Light |

**NOTES**

- The METAR/SPECI reported visibility or flight crew observed visibility will be used with this visibility table to establish snowfall intensity for Type I, II, III and IV holdover time guidelines, during snow, snow grain, or snow pellet precipitation conditions. This visibility table will also be used when snow, snow grains, or snow pellets are accompanied by blowing or drifting snow, or when snow is mixed with ice crystals, freezing fog, rain or drizzle in the METAR/SPECI.
- The use of Runway Visual Range (RVR) is not permitted for determining visibility used with the holdover tables.
- Some METARs contain tower visibility as well as surface visibility. Whenever surface visibility is available from an official source, such as a METAR, in either the main body of the METAR or in the Remarks ("RMK") section, the preferred action is to use the surface visibility value.
- If the visibility is being reduced by snow along with form(s) of obscuration such as fog, haze, smoke, etc., use of the table above may overestimate the actual snowfall intensity. However, use of the snowfall intensity being reported by the weather observer or automated surface observing system (ASOS), from the FMH-1 Table, may underestimate the actual snowfall intensity as it does not directly correlate to the snowfall intensities used when determining holdover times. Use of the visibility table in all snow conditions with or without obscurations is recommended.

Example for how to read and use the table: `CYVO 160200Z 15011G17KT 1SM -SN DRSN OVC009 M06/M08 A2948`

*In the above METAR the snowfall intensity is reported as light. However, based upon the "Snowfall Intensities as a Function of Prevailing Visibility" table, with a visibility of 1 statute mile, at night and a temperature of -6 °C, the snowfall intensity is classified as moderate. The snowfall intensity of moderate - not the METAR reported intensity of light - will be used to determine which holdover time guideline value is appropriate for the fluid in use.*

# Appendix 8 – Type I Holdover Times (Composites)

## TABLE 3: HOLDOVER TIMES FOR SAE TYPE I FLUID ON CRITICAL AIRCRAFT SURFACES COMPOSED PREDOMINANTLY OF COMPOSITES

| Outside Air Temperature[1,2] | Freezing Fog, Freezing Mist[3], or Ice Crystals[4] | Snow mixed with Freezing Fog[5] | Very Light Snow, Snow Grains or Snow Pellets[6,7] | Light Snow, Snow Grains or Snow Pellets[6,7] | Moderate Snow, Snow Grains or Snow Pellets[6,7] | Freezing Drizzle[8] | Light Freezing Rain | Moderate Snow mixed with Rain[9,10] | Rain on Cold-Soaked Wing[10] | Other[11] |
|---|---|---|---|---|---|---|---|---|---|---|
| -3 °C and above (27 °F and above) | 0:09 - 0:16 | 0:03 - 0:06 | 0:12 - 0:15 | 0:06 - 0:12 | 0:03 - 0:06 | 0:08 - 0:13 | 0:02 - 0:05 | 0:01 - 0:01 | 0:01 - 0:05 | |
| below -3 to -6 °C (below 27 to 21 °F) | 0:06 - 0:08 | 0:02 - 0:05 | 0:11 - 0:13 | 0:05 - 0:11 | 0:02 - 0:05 | 0:05 - 0:09 | 0:02 - 0:05 | CAUTION: No holdover time guidelines exist | | |
| below -6 to -10 °C (below 21 to 14 °F) | 0:04 - 0:08 | 0:02 - 0:05 | 0:09 - 0:12 | 0:05 - 0:09 | 0:02 - 0:05 | 0:04 - 0:07 | 0:02 - 0:05 | | | |
| below -10 °C (below 14 °F) | 0:04 - 0:07 | 0:02 - 0:04 | 0:07 - 0:08 | 0:04 - 0:07 | 0:02 - 0:04 | | | | | |

**NOTES**

1 Type I fluid / water mixture must be selected so that the freezing point of the mixture is at least 10 °C (18 °F) below outside air temperature.
2 Ensure that the lowest operational use temperature (LOUT) is respected.
3 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0 °C (32 °F) and below.
4 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
5 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity is no greater than “moderate”. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
6 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
7 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
8 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
9 These holdover times apply to conditions of “moderate” precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
10 No holdover time guidelines exist for this condition for 0 °C (32 °F) and below.
11 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail.

**CAUTIONS**

- The cautions that apply to the holdover times in the table above can be found on page 11.

# Appendix 8 – Generic Type II Holdover Times

## TABLE 4: GENERIC HOLDOVER TIMES FOR SAE TYPE II FLUIDS[1]

| Outside Air Temperature[2] | Fluid Concentration Fluid/Water By % Volume | Freezing Fog, Freezing Mist[3], or Ice Crystals[4] | Snow mixed with Freezing Fog[5] | Snow, Snow Grains or Snow Pellets[6,7] | Freezing Drizzle[8] | Light Freezing Rain | Moderate Snow mixed with Rain[9,10] | Rain on Cold-Soaked Wing[10] | Other[11] |
|---|---|---|---|---|---|---|---|---|---|
| -3 °C and above (27 °F and above) | 100/0 | 0:55 - 1:50 | 0:20 - 0:40 | 0:30 - 0:55 | 0:35 - 1:05 | 0:25 - 0:35 | 0:05 - 0:05 | 0:07 - 0:45 | |
| | 75/25 | 0:40 - 1:10 | 0:15 - 0:25 | 0:15 - 0:30 | 0:25 - 0:40 | 0:15 - 0:25 | 0:03 - 0:03 | 0:04 - 0:25 | |
| | 50/50 | 0:15 - 0:30 | 0:05 - 0:10 | 0:07 - 0:15 | 0:09 - 0:15 | 0:06 - 0:09 | | | |
| below -3 to -8 °C (below 27 to 18 °F) | 100/0 | 0:30 - 0:45 | 0:15 - 0:30 | 0:20 - 0:40 | 0:20 - 0:45 | 0:15 - 0:20 | | | |
| | 75/25 | 0:25 - 0:55 | 0:09 - 0:15 | 0:10 - 0:25 | 0:15 - 0:30 | 0:09 - 0:20 | | | |
| below -8 to -14 °C (below 18 to 7 °F) | 100/0 | 0:30 - 0:45 | 0:10 - 0:25 | 0:15 - 0:30 | 0:20 - 0:45[12] | 0:15 - 0:20[12] | | | |
| | 75/25 | 0:25 - 0:55 | 0:07 - 0:15 | 0:09 - 0:20 | 0:15 - 0:30[12] | 0:09 - 0:20[12] | | | |
| below -14 to -18 °C (below 7 to 0 °F) | 100/0 | 0:15 - 0:20 | 0:01 - 0:05 | 0:02 - 0:07 | | | | | |
| below -18 to -25 °C (below 0 to -13 °F) | 100/0 | 0:15 - 0:20 | 0:00 - 0:02 | 0:01 - 0:03 | | | | | |
| below -25 °C (below -13 °F) | 100/0 | 0:15 - 0:20 | 0:00 - 0:00 | 0:00 - 0:01 | | | | | |

CAUTION: No holdover time guidelines exist

**NOTES**

1 To use the HOTs in this table, ensure that the fluid and dilution being used is listed in the Type II Fluids Tested for Anti-Icing Performance and Aerodynamic Acceptance table (Table 56). Any restrictions on the use of the fluid have to be identified and applied.
2 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type II fluid cannot be used.
3 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0 °C (32 °F) and below.
4 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
5 The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity is no greater than “moderate”. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
6 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required.
7 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
8 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
9 These holdover times apply to conditions of “moderate” precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 54) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
10 No holdover time guidelines exist for this condition for 0 °C (32 °F) and below.
11 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail.
12 No holdover time guidelines exist for this condition below -10 °C (14 °F).

**CAUTIONS**

- The cautions that apply to the holdover times in the table above can be found on page 14.

# Appendix 8 – Generic Type IV Holdover Times

## TABLE 20: GENERIC HOLDOVER TIMES FOR SAE TYPE IV FLUIDS[1]

| Outside Air Temperature[2] | Fluid Concentration Fluid/Water By % Volume | Freezing Fog, Freezing Mist[3], or Ice Crystals[4] | Snow mixed with Freezing Fog[5] | Very Light Snow, Snow Grains or Snow Pellets[5,6] | Light Snow, Snow Grains or Snow Pellets[5,6] | Moderate Snow, Snow Grains or Snow Pellets[5,6] | Freezing Drizzle[7] | Light Freezing Rain | Moderate Snow mixed with Rain[8,9] | Rain on Cold-Soaked Wing[9] | Other[10] |
|---|---|---|---|---|---|---|---|---|---|---|---|
| -3 °C and above (27 °F and above) | 100/0 | 1:15 - 2:15 | 0:20 - 0:40 | 1:50 - 2:15 | 0:55 - 1:50 | 0:25 - 0:55 | 0:35 - 1:10 | 0:15 - 0:30 | 0:06 - 0:06 | 0:08 - 1:05 | |
| | 75/25 | 1:25 - 2:40 | 0:30 - 0:55 | 2:05 - 2:25 | 1:15 - 2:05 | 0:40 - 1:15 | 1:00 - 1:20 | 0:30 - 0:50 | 0:07 - 0:07 | 0:09 - 1:20 | |
| | 50/50 | 0:30 - 0:55 | 0:07 - 0:20 | 1:00 - 1:10 | 0:25 - 1:00 | 0:10 - 0:25 | 0:15 - 0:40 | 0:09 - 0:20 | | | |
| below -3 to -8 °C (below 27 to 18 °F) | 100/0 | 0:15 - 0:35 | 0:20 - 0:35 | 1:35 - 2:00 | 0:45 - 1:35 | 0:25 - 0:45 | 0:25 - 1:00 | 0:20 - 0:25 | | | |
| | 75/25 | 0:40 - 1:20 | 0:25 - 0:50 | 1:50 - 2:10 | 1:05 - 1:50 | 0:30 - 1:05 | 0:20 - 1:05 | 0:15 - 0:25 | | | |
| below -8 to -14 °C (below 18 to 7 °F) | 100/0 | 0:15 - 0:35 | 0:15 - 0:30 | 1:25 - 1:50 | 0:45 - 1:25 | 0:20 - 0:45 | 0:25 - 1:00[11] | 0:15 - 0:25[11] | | | |
| | 75/25 | 0:40 - 1:20 | 0:20 - 0:45 | 1:45 - 2:00 | 0:55 - 1:45 | 0:25 - 0:55 | 0:20 - 1:05[11] | 0:15 - 0:25[11] | | | |
| below -14 to -18 °C (below 7 to 0 °F) | 100/0 | 0:15 - 0:25 | 0:01 - 0:06 | 0:30 - 0:45 | 0:09 - 0:30 | 0:02 - 0:09 | | | | | |
| below -18 to -25 °C (below 0 to -13 °F) | 100/0 | 0:15 - 0:25 | 0:00 - 0:02 | 0:10 - 0:20 | 0:03 - 0:10 | 0:01 - 0:03 | | | | | |
| below -25 °C (below -13 °F) | 100/0 | 0:15 - 0:25 | 0:00 - 0:01 | 0:07 - 0:10 | 0:02 - 0:07 | 0:00 - 0:02 | | | | | |

CAUTION: No holdover time guidelines exist

**NOTES**

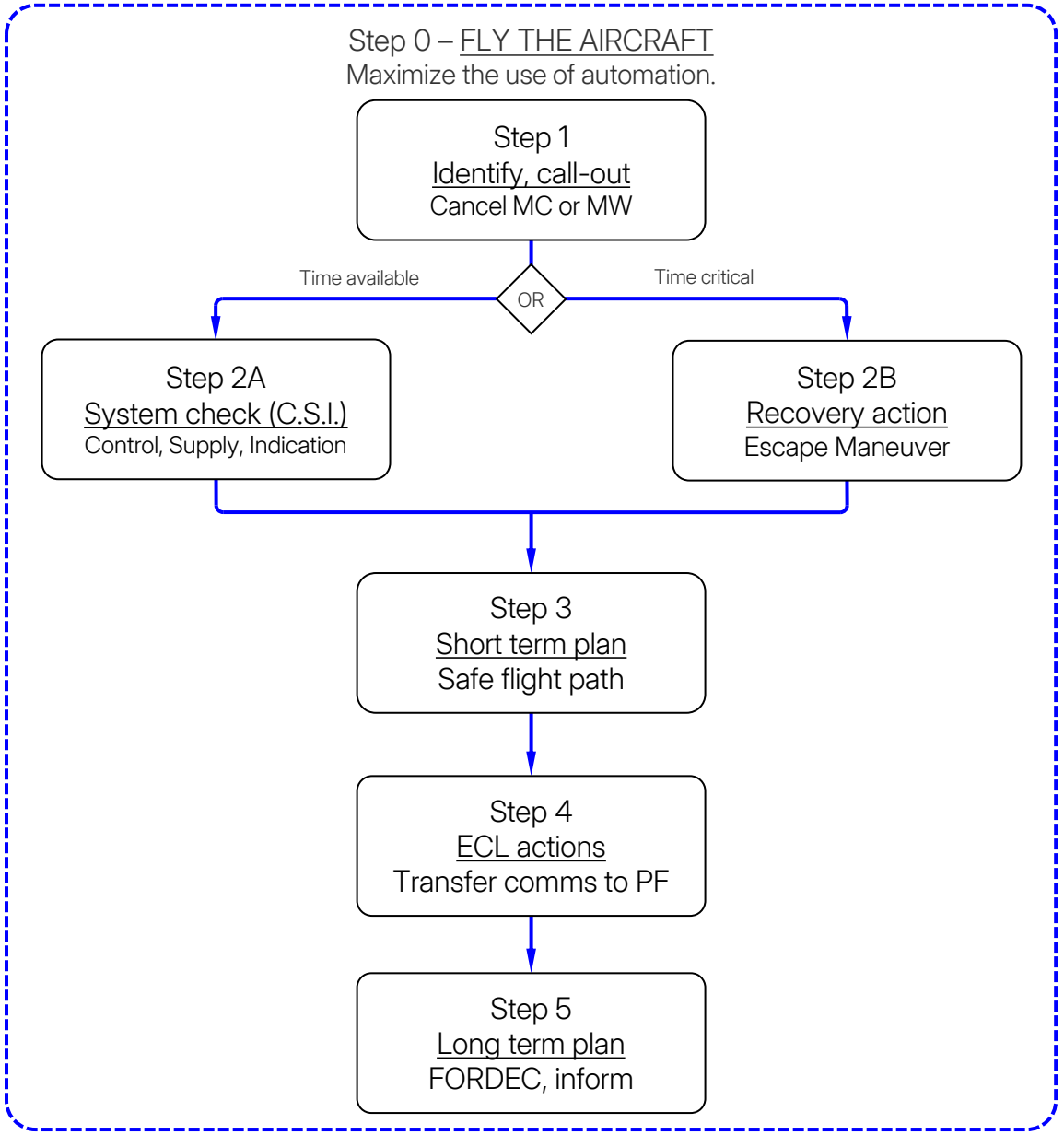
1 To use the holdover times in this table, ensure that the fluid and dilution being used is listed in the Type IV Fluids Tested for Anti-Icing Performance and Aerodynamic Acceptance table (Table 61). Any restrictions on the use of the fluid have to be identified and applied.
2 Ensure that the lowest operational use temperature (LOUT) is respected. Consider use of Type I fluid when Type IV fluid cannot be used.
3 Freezing mist is best confirmed by observation. It is never reported by METAR; however, it can occur when mist is present at 0 °C (32 °F) and below.
4 Use freezing fog holdover times in conditions of ice crystals mixed with freezing fog or freezing mist.
5 To determine snowfall intensity, the Snowfall Intensities as a Function of Prevailing Visibility table (Table 57) is required. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
6 Use snow holdover times in conditions of very light, light, or moderate snow mixed with ice crystals.
7 Includes light, moderate and heavy freezing drizzle. Use light freezing rain holdover times if positive identification of freezing drizzle is not possible.
8 These holdover times apply to conditions of “moderate” precipitation intensity. In cases of very light or light snow mixed with light rain or drizzle, use light freezing rain holdover times. The Snowfall Intensities as a Function of Prevailing Visibility table (Table 57) is required to confirm the precipitation intensity. No holdover times exist if the reported visibility correlates to a “heavy” precipitation intensity.
9 No holdover time guidelines exist for this condition for 0 °C (32 °F) and below.
10 Heavy snow, ice pellets, moderate and heavy freezing rain, small hail and hail (Table 55 and Table 56 provides allowance times for Type IV EG and Type IV PG fluids in ice pellets and small hail. To use these allowance times, ensure that the fluid being used is listed as validated in Table 53).
11 No holdover time guidelines exist for this condition below -10 °C (14 °F).

**CAUTIONS**

- The cautions that apply to the holdover times in the table above can be found on page 32.

# › Appendix 9 – Basic Failure Management

| Step | Action | Description |
|---|---|---|
| 1 | Identify and call-out | Call non-standard condition to the attention of the other pilot:<br>"ENG OUT, WINDSHEAR, UPSET, MASTER CAUTION / WARNING …"<br>Cancel MASTER CAUTION / WARNING flasher. |
| 2A | System Check (time avail) | Where applicable, check the following (C-S-I):<br>• Control: system correctly configured<br>• Supply: system source, check circuit breakers (but do not reset)<br>• Indication: compare flight instruments, EICAS and synoptic pages |
| 2B | Recovery Action (time critical) | If applicable, perform relevant recovery action by memory:<br>• Emergency descent<br>• Windshear / Terrain escape<br>• Upset / Stall recovery |
| 3 | Short-term plan | Safe trajectory away from terrain / obstacles. Turn towards the nearest suitable airport if time-critical (e.g. SMOKE, FIRE, BOMB on board).<br>• Level-off and set course to the nearest suitable airport<br>• Following engine out departure procedure<br>• Drift down or depressurization procedure |
| 4 | ECL actions | Transfer communications to the PF while the PM reads the checklists out loud in the order given (as applicable):<br>1. WARNING<br>2. Priority procedure<br>3. NORMAL<br>4. CAUTION<br>5. Non-normal procedure<br>Confirmation is required **in-flight** for "irreversible" actions:<br>• Engines: Thrust levers, Engine run switches<br>• All guarded switches<br>• Checklist conditions: ○YES ○NO (read as a YES or NO question)<br>Read LIMITATIONS and preview DEFERRED ITEMS |
| 5 | Long-term plan | After completing the applicable failure procedure and collecting all available data (e.g. weather), decide on a long-term solution.<br>• Decide using FORDEC (return, continue, divert)<br>• Communicate the plan to ATC, cabin crew (NITS) and passengers<br>• Execute the plan and revise if needed (redo FORDEC) |



# Appendix 10 – Smoke / Fire / Fumes Priority Procedure (not in ECL)

1) Oxygen masks.....If required, on, 100% and EMERGENCY
2) Cabin and crew communication.....Establish
3) Plan to land at the nearest suitable airport.

**WARNING:** during smoke/fire/fumes emergencies, prepare to land the aircraft without delay while conducting fire suppression and/or smoke evacuation procedures.

**CAUTION:** passenger masks should not be deployed when performing smoke or fire procedures.

4) EQUIP COOLING, INLET..... OFF
5) CABIN PWR.....OFF
6) RECIRC AIR..... OFF

**WARNING:** any time smoke/fumes becomes the greatest threat accomplish the Smoke or fumes removal procedure.

**CAUTION:** continuing to destination is not recommended. If the decision to continue is made, the flight crew must acknowledge that the threat has been positively identified, confirmed to be extinguished, and the smoke/fumes have dissipated.

7) Time permits and source of smoke/fire/fumes can be quickly identified and eliminated:

→ **Yes – Go to (11)**
→ **No – Go to (8)**

**8) Time does not permit and source of smoke/fire/fumes cannot be quickly identified:**

**CAUTION:** the crew should consider an overweight landing, tailwind landing, ditching or off-airport landing.

9) Land immediately at the nearest suitable airport.
10) Smoke or fumes removal procedure.....Accomplish if required. Refer to smoke, fire and fumes – Smoke or fumes removal.

**- COMPLETE -**

**11) Time permits and source of smoke/fire/fumes can be quickly identified and eliminated:**
12) Select appropriate procedure for source of smoke/fire/fumes:

→ **Air-conditioning – Got to (13)**
→ **Electrical – Go to (33)**
→ **Cabin – Go to (107)**

**13) Source is identified as coming from air-conditioning:**

**NOTE:** smoke may take considerable amount of time to dissipate.

14) XBLEED..... MAN CLSD
15) APU BLEED..... OFF
16) L PACK.....OFF
17) Altitude.....Not above 31000 feet
18) Smoke/fire/fumes stops:

→ **Yes – Go to (19)**
→ **No – Go to (22)**

**19) Smoke/fire/fumes stops:**
20) Land immediately at the nearest suitable airport.
21) Smoke or fumes removal procedure.....Accomplish if required. Refer to smoke, fire and fumes – Smoke or fumes removal.

**- COMPLETE -**

**22) Smoke/fire/fumes does not stop:**
23) L PACK.....Select auto
24) R PACK.....OFF
25) Smoke/fire/fumes stops:

→ **Yes – Go to (26)**
→ **No – Go to (29)**

**26) Smoke/fire/fumes stops:**
27) Land immediately at the nearest suitable airport.
28) Smoke or fumes removal procedure.....Accomplish if required. Refer to smoke, fire and fumes – Smoke or fumes removal.

**- COMPLETE -**

**29) Smoke/fire/fumes does not stop:**
30) R PACK.....Select auto

**CAUTION:** the crew should consider an overweight landing, tailwind landing, ditching or off-airport landing.

31) Land immediately at the nearest suitable airport.
32) Smoke or fumes removal procedure.....Accomplish if required. Refer to smoke, fire and fumes – Smoke or fumes removal.

**- COMPLETE -**

**33) Source is identified as coming from electrical:**

**NOTE:** smoke may take a considerable amount of time to dissipate.

The series of steps that follow isolate each channel of the electrical system, one at a time.

34) ECL.....DU 5
35) ELEC synoptic page.....Select

**Channel 1 isolation:**

**NOTE:** during this procedure DU 1, DU 3, AC BUS 1, DC BUS 1, and DC ESS BUS 1 will not be powered.

36) BUS ISOL.....MAIN
37) Affected generator.....OFF

**NOTE:** the affected generator supplies power to AC BUS 1 on the ELEC synoptic page.

38) BATT 1.....OFF
39) CTP, CPDR/TCAS.....Select XPDR 2
40) FD.....Couple to left side

**NOTE:** The use of EICAS, REV compresses both PFD and allows two additional MFW. To recover the use of autopilot, ensure FD is coupled to left side and re-engage autopilot if necessary.

41) Smoke/fire/fumes stops:

→ **Yes – Go to (42)**
→ **No – Go to (47)**

**42) Smoke/fire/fumes stops:**

**CAUTION:** Do not land on contaminated runway.

Significant systems affected:

| | | |
|---|---|---|
| APU start | Autopilot (FD right) | Nose landing light |
| APU door actuator | ADSP 1 | Left landing light |
| PTU | Autothrottle | Left taxi light |
| HYD pump 2B | XPDR 1 | L Map + reading lts |
| HYD 1 press ind | TCAS | Cockpit door lock |
| HYD 2 quantity ind | TAWS | Fwd lav smoke det |
| Crew oxy press ind | DME 1 | Toilet flush |
| Rudder trim | VHF 3 | Navigation lights |
| PAX oxy auto dply | HF 1 | Seat belt signs |
| Attendant call lights | SATCOM | L + R pwr outlets |
| Flaps slow | HUD | Fuel inerting |
| Reduced braking | RAD ALT 3 | Trim air |
| Left thrust reverse | DU 1 and DU 3 | TRU 1 |
| L wiper & heating | Tire press ind | ACP 3 |
| Left fuel pump | MKP 1 | |

43) Land immediately at the nearest suitable airport.
44) Reversion Switch Panel, L or R PFD.....Select ADS until ADS source data indication is not amber on both PFDs.

**NOTE:** RNP AR approaches are prohibited.

**On approach:**

45) FMS, PERF – ARR – VREF.....VREF
46) OLD factor.....Multiply by 1.70

**- COMPLETE -**

**47) Smoke/fire/fumes does not stop:**

**Channel 2 isolation:**

**NOTE:** During this procedure DU 4, DU 5, AC BUS 2, DC BUS 2, and DC ESS BUS 2 will not be powered.

48) BUS ISOL.....Confirm MAIN
49) BATT 1..... AUTO
50) Affected generator..... Select ON

**NOTE:** Before Channel 1 isolation, the affected generator supplied power to AC BUS 1 on the ELEC synoptic page.

**After 30 seconds:**

51) ECL.....DU 2
52) Affected generator.....OFF

**NOTE:** The affected generator supplied power to AC BUS 2 on the ELEC synoptic page.

53) BATT 2 .....OFF
54) CTP, XPDR/TCAS.....Select XPDR 1
55) CTP 2.....Select off
56) CTP 1.....ATC on VHF 1, if required
57) Smoke/fire/fumes stops:

→ **Yes – Go to (58)**
→ **No – Go to (70)**

**58) Smoke/fire/fumes stops:**

**CAUTION:** Do not land on contaminated runway.

Significant systems affected:

| | | |
|---|---|---|
| HYD pumps 3A/3B | TCAS | Reduced braking |
| HYD pump 2B | WXR | Right thrust reverse |
| HYD 2 press ind | Wing anti-ice | Right landing light |
| HYD 3 quantity ind | ADSP 2 & ADSP 4 | Nose taxi light |
| Right fuel pump | HF 2 | Right taxi light |
| VHF 2 | FMS 2 | R Map + reading lts |
| CTP 2 | GNSS 2 | Right window heat |
| MKP 2 | APPR 1 | R wiper & heating |
| CCP 2 | DME 2 | Outflow valve MAN |
| PAX oxy auto dply | NAV 2 | Aft lav smoke det |
| Slats | RAD ALT 2 | Recirculation fan |
| Flaps slow | DU 4 and DU 5 | TRU 2 |
| XPDR 2 | R HUD | |

# Appendix 10 – Smoke / Fire / Fumes Priority Procedure (not in ECL)

59) Land immediately at the nearest suitable airport.
60) Reversion Switch Panel, L or R PFD.....Select ADS until ADS source data indication is not amber on both PFDs.
61) Use AVIONIC, CTP tab for R CTP functions.

**NOTE:** RNP AR approaches are prohibited.

**On approach:**

62) Check SLAT position:

→ **SLAT position IN – Go to (63)**
→ **SLAT position OUT/MID/FULL – Go to (67)**

**63) SLAT position IN:**
64) Maximum landing weight.....Use the table to determine the value with corrections as applicable.

| Landing weight (kg) limited by max brake energy & tire speed (VREF + 15 kts) | | | | | | |
|---|---|---|---|---|---|---|
| OAT °C | Pressure Altitude (ft) | | | | | |
| | 0 | 2000 | 4000 | 6000 | 8000 | 10000 |
| ≤-20 | 86990 | 84410 | 81690 | 79100 | 76430 | 73520 |
| 0 | 84050 | 81370 | 78740 | 76110 | 73480 | 70570 |
| 20 | 81280 | 78690 | 76060 | 73430 | 70480 | 65540 |
| 30 | 80010 | 77380 | 74790 | 72160 | 68170 | 63140 |
| ≥-40 | 78780 | 76150 | 73570 | 70710 | 65770 | 60910 |

**Corrections:**
- Tailwind: decrease maximum landing weight by 18% per 10 kts of tailwind.
- Downhill: decrease maximum landing weight by 2% per 1% downhill slope.
- Decrease maximum landing weight by 14% for 10 kts $V_{APP}$ increase.

65) FMS, PERF – ARR – VREF..... $V_{REF}$ + 15
66) OLD factor.....Multiply by 2.05

**- COMPLETE -**

**67) SLAT position OUT/MID/FULL:**
68) FMS, PERF – ARR – VREF..... $V_{REF}$ + 5
69) OLD factor.....Multiply by 1.85

**- COMPLETE -**

**70) Smoke/fire/fumes does not stop:**

**Channel 3 isolation:**

**NOTE:** During this procedure DU 2, AC ESS BUS, and DC ESS BUS 3 will not be powered.

71) HYD 3A.....OFF
72) BATT 2.....AUTO
73) Affected generator.....Select on

**NOTE:** Before Channel 2 isolation, the affected generator supplied power to AC BUS 2 on the ELEC synoptic page.

74) CTP 2.....Select on

**After 30 seconds:**

75) ECL..... DU 5
76) BUS ISOL..... ESS
77) CTP, XPDR/TCAS..... Select XPDR 2
78) CTP 1..... Select off
79) CTP 2.....ATC on VHF 2, if required
80) Smoke/fire/fumes stops:

→ **Yes – Go to (81)**
→ **No – Go to (88)**

**81) Smoke/fire/fumes stops:**

Significant systems affected:

| | | |
|---|---|---|
| HYD 1 quantity ind | CCP 1 | GNSS 1 (no LPV) |
| HYD 3 press ind | REV switch panel | NAV 1 |
| ISI (ATT only) | DU 2 | RAD ALT 1 |
| Rudder degraded | L HUD | Altn gear extension |
| L fuel pump | Slats / flaps slow | Ram air |
| Gravity transfer | MFS (1 pair) | Equip bay cooling |
| ADSP 3 | Right stick shaker | Left window heat |
| VHF 1 | Wing anti-ice | CDPLC |
| XPDR 1 | PFCC 3 | TRU 3 |
| CTP 1 | FMS 1 (no LPV) | OMS |

82) Land immediately at the nearest suitable airport.
83) FD.....Couple to right side
84) Reversion Switch Panel, L or R PFD.....Select ADS until ADS source data indication is not amber on both PFDs.
85) Use AVIONIC, CTP tab for L CTP functions.

**NOTE:** RNP AR approaches are prohibited.

**On approach:**

86) FMS, PERF – ARR – VREF..... $V_{REF}$
87) OLD factor.....Multiply by 1.20

**- COMPLETE -**

**88) Smoke/fire/fumes does not stop:**

**These steps will deploy the RAT:**

**NOTE:** During this procedure DU 3, DU 4, DU 5, AC BUS 1, AC BUS 2, DC BUS 1, and DC BUS 2 will not be powered.

89) BUS ISOL..... AUTO
90) CTP 1.....Select on, ATC on VHF 1
91) ECL..... DU 2
92) RAT GEN..... ON
93) CTP, XPDR/TCAS..... Select XPDR 1
94) FD..... Couple to left side

**CAUTION:** The **EMER PWR ONLY** warning message will be shown after all generators are selected off. The steps are included in this procedure and do not need to be done separately.

95) L GEN..... OFF
96) R GEN..... OFF
97) APU GEN.....OFF
98) L PACK..... Confirm auto
99) R PACK..... Confirm auto
100) EQUIP COOLING, EXHAUST.....ON
101) Land immediately at the nearest suitable airport
102) Leave/avoid icing conditions.

**CAUTION:** Below 148 KIAS the RAT generator may be inoperative and the airplane will be operating on battery power only. Brakes are not available when batteries are depleted. Do not land on contaminated runway.

103) Airspeed.....Not less than 155 KIAS until landing assured.

**NOTE:** RNP AR approaches are prohibited.

Significant systems affected when operating on RAT power:

| | | |
|---|---|---|
| ACMP 2B, 3A, 3B | L+R thrust reverser | GNSS 2 (no LPV) |
| PTU | Exterior lighting | DME 1 and 2 |
| Rudder trim | L+R ICE detectors | NAV 2 and 3 |
| XPDR 2 | ADS-B | RAD ALT 2 and 3 |
| TCAS | VHF DLK, CPDLC | L, R WSHLD heat |
| Right fuel pump | VHF 2 | R window heating |
| ADSP 1, ADSP 2 | HUD | Left, right wipers |
| Autothrottle | DU 3, DU 4, DU 5 | L+R map & read lts |
| NAV to NAV XFR | CCP 2 (DU 2 only) | Recirculation fan |
| IRU 2 | TAWS | Fuel inerting |
| TRU 1, TRU 2 | WXR | Toilet flush |
| Attendant call lights | HF 2 | Seat belt signs |
| Slats/flaps slow | SATCOM | MKP 2 |

104) Airspeed.....Not less than 155 KIAS until landing assured.

**CAUTION:** Below 148 KIAS the batteries may be the only source of electrical power, and last for at least 5 minutes. Batteries will recharge above 148 KIAS if go-around is required. Brakes are not available when batteries are depleted.

105) Approach speed.....VREF

**NOTE:** FMS VSPEEDS not available.

106) OLD factor.....Multiply by 1.35

**- COMPLETE -**

**107) Source is identified as coming from the cabin:**
108) Designated crew member.....Advise

a. To isolate and extinguish source of smoke or fire, and to secure the area.
b. To turn off all electrical sources via CMS.

**CAUTION:** continuing to destination is not recommended. If the decision to continue is made, the flight crew must acknowledge that the threat has been positively identified, confirmed to be extinguished, and the smoke/fumes have dissipated.

109) Land immediately at the nearest suitable airport.
110) Smoke or fumes removal procedure.....Accomplish if required. Refer to smoke, fire and fumes – Smoke or fumes removal.

**- COMPLETE -**

# Safe flights!

# AIRBUS A220